THE GHISSU MODEL ANSWER SERIES

With a foreword by BHARGAVA V

IRMS, UPSC CSE 2024 · FORMER ASST. COMMISSIONER, APPSC GROUP-I 2023

APPSC GROUP-I MAINS

Paper V

Science, Technology and Environmental Issues

MODEL ANSWERS TO PREVIOUS YEAR QUESTIONS

2008 *to* 2025

292 questions · in handwriting style

SRI CHARAN VEGIREDDI

FOUNDER · GHISSU.COM

FOREWORD

When I sat for APPSC Group-I Mains, I noticed that most aspirants in the hall knew the content. What separated the final list was how each answer met the exact demand of the question within the time and the pages available. That is why understanding the syllabus, analysing previous year questions (PYQs) and following the trends in the pattern matter so much.

During my own preparation, I worked extensively with the APPSC Mains PYQs. I answered each question myself, working on how I presented the answer and on the value additions I could bring to it. This improved the presentation of my content, and with it, my confidence. Technology gave me an extra edge as well: I used AI to refine my answers.

This book brings together previous year questions with model answers written to the demand of the exam. It analyses the examination from every angle, which makes the aspirant's journey simpler and more focused. Attempt each question yourself before you read the model answer, and then compare. You can also get your written answer evaluated on ghissu.com. Practicing these questions year by year will sharpen your sense of the demand, the structure and the depth that score marks.

I hope this book helps aspirants prepare with clarity and confidence, and succeed in realising their goals.

All the best,

BHARGAVA V
Indian Railway Management Service (IRMS)
UPSC Civil Services Examination 2024
Former Assistant Commissioner of State Tax, Andhra Pradesh
(APPSC Group-I, 2023)

A FREE RESOURCE FROM GHISSU

Preface

APPSC Group-I Mains is a writing exam. The syllabus tells you what to study. The previous papers show you what the commission actually asks, how it words each question and how much it expects for ten marks. This book gives you every question from eight sittings, 2008 to 2025, each with an answer written the way an aspirant can write it in the exam hall.

How this book is arranged

Each paper is a separate book. Inside it, the questions run year by year, newest first, with the same order and numbering as the official question papers. The arrangement follows the latest APPSC syllabus. Questions from the older sittings, when the papers were organised differently, are placed in the paper they belong to today, and their label shows where they were first printed. Every answer fits in three pages, the space the 44-page answer booklet allows per question. The answers are drafted with Ghissu's Model Answer engine, then checked and edited, and set in a handwriting style so you can see exactly what three pages in the exam hall look like.

Why this book

After graduating in Mathematics and Computing Engineering from DTU, New Delhi, I spent years as an aspirant myself. I wrote two UPSC Civil Services Mains, two APPSC Group-I Mains and one APPSC Group-II Mains. I am still a learner, walking the same path as you.

Then, for nearly two years, I worked on the other side as Mentor-cum-Head Reviewer at Vajiram & Ravi, New Delhi, where I headed the evaluation of its Mains Test Series. I reviewed more than 10,000 answer copies, trained over 600 evaluators and mentored more than 100 aspirants. One lesson stayed with me: marks are rarely lost because of too little reading. They are lost when the demand is only half answered, the structure wanders or the conclusion says nothing. These are habits, and habits change only through writing and honest feedback.

This book is half of the method

A model answer shows you where to reach. It cannot show you how far your own answer is from it. That is what ghissu.com is for. There you will find the same questions arranged subject-wise, topic-wise and sub-topic-wise. Alongside them are UPSC questions from 1979 to 2026 that match the APPSC syllabus, and more practice questions that cover each topic fully. You can write an answer, get it evaluated within minutes, and compare it with the exam-hall thinking, toppers' answers and value additions for that question.

> **To use the topic-wise PYQs and resources:** sign in at **ghissu.com** and join the waitlist. Requests are usually approved within one or two days. The next pages explain how to use this book and the website together.

An apology, in advance

Every question and every answer has been checked with care. Even so, some errors may remain. If you find anything that could be clearer or more accurate, please write to **hello@ghissu.com**. Every correction makes the book better for the next reader.

With best wishes,

SRI CHARAN VEGIREDDI

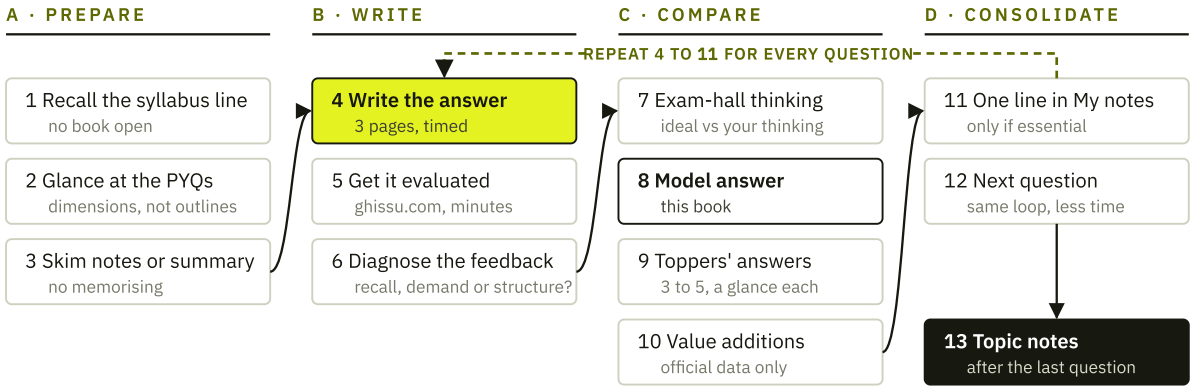
Founder, ghissu.com

READ THIS BEFORE THE FIRST QUESTION

How to use this book with ghissu.com

Keeping a book of model answers open in front of you feels like preparation, but it is passive reading, and passive reading fades. The method below keeps you active at every stage: you recall before you read, you write before you compare, and you compare in order to find the half mark you left behind in each part of the answer, from the introduction to the conclusion.

It assumes you have covered the syllabus once from a standard book or a coaching programme. Work topic by topic, using the topic-wise view of these questions on ghissu.com, and keep a time limit for every step.



Where the website comes in. Every answer page in this book carries a QR code at the bottom right, and every question in the index has a Practice link. Both open that exact question on ghissu.com, where you can upload a handwritten answer, type it against a running timer, or dictate it by voice.

SUGGESTED TIME FOR THE FIRST QUESTIONS OF A TOPIC	MINUTES
Recall, glance at the PYQs and skim your notes (steps 1 to 3, once per topic)	10 to 15
Write the answer at exam pace (step 4)	12
Read the feedback and the exam-hall thinking (steps 5 to 7)	8 to 10
Model answer, toppers' answers and value additions (steps 8 to 10)	8 to 10

A beginner may need 30 to 45 minutes for each of the first one to three questions of a sub-topic. That is expected. The time falls with every question, and what you gain is the confidence to attempt any question from the topic, at any time.

PHASE A Prepare, without opening the answers

1 **Recall the syllabus line.** Read the line from the syllabus and brainstorm everything you remember about it, on paper.

2 **Glance at the previous questions.** See which questions were asked and along which dimensions. Do not outline or think deeply yet. This shows the gaps in what you recalled and where the marks in this topic usually lie.

3 **Skim your notes.** If you have notes, skim them without trying to memorise. If not, read the summary, strategy, value addition and toppers' answers for the topic on ghissu.com, just enough to attempt the first question with confidence.

PHASE B Write and get feedback

4 **Write the first question.** Do not wait for the perfect moment; there is none. Answer writing is the skill the examiner tests most, more than reading or revision. If the topic has few APPSC questions, ghissu.com adds the related UPSC questions and practice questions.

5 **Get it evaluated.** Submit a full answer, or only a quick outline, on ghissu.com. Feedback returns in two to three minutes. Upload a handwritten page, type with the timer running, or dictate, whichever suits your day.

6 **Diagnose before you read further.** Find the one major issue. If it is only recall, move on; it disappears as you finish the topic. If it is a missed demand or a weak structure, slow down and understand it with the exam-hall thinking for that question. Ask Ghissu and the Ghissu bull can explain further, in your own language.

PHASE C Compare with the ideal

7 **Exam-hall thinking.** See how the question is broken down: its demand and structure, how to open the introduction, how to generate points, which frameworks and value additions to use and how to conclude. Measure the distance between your thinking and the ideal. By the end of a topic, your own thinking will often go further.

8 **The model answer in this book.** Note new terms, data and value additions, and see how a complete answer fits three pages. Do not memorise it.

9 **Toppers' answers.** Glance through three to five toppers' answers on ghissu.com, drawn from more than 1.5 lakh answers, mostly by UPSC toppers. The APPSC syllabus overlaps heavily with UPSC, so they apply directly. Spend little time on each. Your mind will start to notice how they earn the extra half mark in each part, and their keywords will come back to you in the hall.

10 **Value additions.** For the latest figures and data, use the value additions on the question or the topic. They are taken from official government sources, ready to quote in an answer.

PHASE D Consolidate

11 **Hold back the notes.** Do not write short notes after a single question; you will end up copying everything. If something is essential, add it to My notes on that question. Your notes collect topic by topic on their own, so nothing is written twice.

12 **Next question, same loop.** Repeat steps 4 to 11 for every question in the topic, keeping to the time limits so that no step runs away with your day.

13 **Write your topic notes last.** When every question is done, write one to three ways to open and close an answer on this topic, a few multi-dimensional points each for positives, negatives and the way forward, and the keywords and frameworks you will break a question into. Fifteen to twenty points and a diagram are enough to reproduce in the exam. The strategy section of each topic on ghissu.com helps, but your own versions matter more.

APPSC Group-I Mains · Paper V

Model answers in handwriting style, 2008 to 2025, numbered as in the official papers · latest-syllabus arrangement

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at ghissu.com and join the waitlist (usually approved in 1 to 2 days).
Click a question to open its model answer. "Practice" opens the same question on ghissu.com.

No.	Question	Page	Practice online
2025	30 answers		
1(a)	Describe the research focus of the National Institute for Research on Commercial Agriculture (NIRCA) in Rajamahendravaram. How does it support India's agricultural...	22	Practice →
1(b)	Evaluate the impact of Internet of Things (IoT) technologies on the automation of household systems. Illustrate how everyday tasks have evolved due to IoT innovations.	25	Practice →
2(a)	Discuss the discovery of Raman effect by Dr. C.V. Raman. How did his work impact the global scientific community?	28	Practice →
2(b)	Which are the major scientific institutes located in Andhra Pradesh? Discuss the role of IISER Tirupati in the scientific and educational development of the state.	31	Practice →
3(a)	Discuss the evolution of e-governance in India, highlighting key initiatives. Analyze how these initiatives have transformed public service delivery and enhanced citizen...	34	Practice →
3(b)	Define cybercrime. Discuss the major types of cybercrimes prevalent in India today.	37	Practice →
4(a)	Discuss the key objectives and features of the National Policy on Information Technology, 2012. How does it aim to transform India into a digitally empowered...	40	Practice →
4(b)	Discuss the vision and mission of DRDO? Describe the role of DRDO in the contribution to national development.	43	Practice →
5(a)	Describe the role of the Indian Remote Sensing (IRS) satellite programme. How do these satellites assist in natural resource management?	46	Practice →
5(b)	Provide a comprehensive overview of the primary reasons for deforestation in Andhra Pradesh in the past two decades.	49	Practice →
6(a)	Investigate the relationship between mining activities and the occurrence of earthquakes. Explain how mining operations can be regulated to minimize seismic risks.	52	Practice →
6(b)	Define sustainable development and critically evaluate its relevance in the context of modern industrialization.	55	Practice →
7(a)	Describe the relationship between deforestation and the emergence of infectious diseases.	58	Practice →
7(b)	What is Environmental Impact Assessment (EIA)? Describe its key stages.	61	Practice →
8(a)	Explain the different types of natural resources in India. How are they being conserved through government initiatives?	64	Practice →
8(b)	What are the key principles of Integrated Solid Waste Management (ISWM)? Discuss the role of waste hierarchy in ISWM.	67	Practice →
9(a)	Analyze the potential of plant-derived biopharmaceuticals in promoting sustainable healthcare. Explain how the application of green pharmacy principles can help minimi...	70	Practice →
9(b)	Differentiate between conventional and non-conventional energy resources. Explain their relevance in India's energy sector.	73	Practice →

No.	Question	Page	Practice online
10(a)	Critically assess the effectiveness of major disaster management initiatives undertaken in India since the enactment of the Disaster Management Act, 2005.	76	Practice →
10(b)	Discuss the legislative framework for controlling air and water pollution in India. Highlight the key provisions and limitations of relevant acts.	79	Practice →
11(a)	Define global warming, and differentiate it clearly from climate change. Explain the key natural and human-induced causes of global warming and briefly discuss which factor...	82	Practice →
11(b)	Discuss the role of remote sensing in tracking the dispersion of environmental pollutants. Analyze its implications for human health in affected regions, highlighting...	85	Practice →
12(a)	Explain the regulatory framework governing genetically modified organisms in India and assess its adequacy in addressing modern challenges.	88	Practice →
12(b)	Trace the evolution of India's National Biotechnology Development Strategy (NBDS) and discuss its key objectives. Critically evaluate its impact on the growth of the...	91	Practice →
13(a)	Describe the components and functioning of a vaccine.	94	Practice →
13(b)	Define genetic engineering. Explain its applications in agriculture and medicine.	97	Practice →
14(a)	Describe the integrated strategies used in biotechnology to monitor and control outbreaks of bacterial diseases in poultry farming.	100	Practice →
14(b)	Examine how urban infrastructure influences the implementation of preventive measures during microbial disease outbreaks. Suggest technology-driven solutions.	103	Practice →
15(a)	Discuss the transmission dynamics of respiratory tract infections in densely populated urban settings. Propose effective public health strategies for their prevention.	106	Practice →
15(b)	Describe the importance of IPR in academic and industrial research collaborations.	109	Practice →
2023	30 answers		
1(a)	Do you think the problem in India is "not so much knowledge creation, as knowledge consumption" and Scientific Social Responsibility (SSR) will go a long way in remedyin...	113	Practice →
1(b)	Technology plays a key role in transforming healthcare sector and in addressing India's healthcare challenges — Discuss.	116	Practice →
2(a)	Digital platforms dramatically improve the quality and accountability of public service and facilitate citizen centric governance — Elaborate. Explain various Technical,...	119	Practice →
2(b)	Explain how 'Atmanirbhar Bharat' initiative is bringing a big change in India's IT sector. Suggest the steps to be taken to make India a Global technology hub.	122	Practice →
3(a)	What is Space Junk ? Discuss the challenges posed by increased space junk and the potential risk associated with it. What do you know about Project NETRA ?	125	Practice →
3(b)	Why is India's space industry looking for private sector investments even though ISRO is a successfully performing PSU ?	128	Practice →
4(a)	Though circumstances are strongly insisting for India to overturn decades of 'No first use' nuclear doctrine, still India is committed to No first use of nuclear weapons —...	131	Practice →
4(b)	What is Green Energy ? Discuss about the Green Energy corridors and their need in India.	134	Practice →
5(a)	Climate change is a global concern which needs urgent intervention to combat it — Explain why. Discuss how climate smart agriculture mitigates the devastating effects o...	137	Practice →
5(b)	What is meant by open cast mining ? Discuss the impacts of open cast mining on environment and solutions to overcome them. Add a note on the importance of post...	140	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
6(a)	What are various mitigation activities by the Government of India to lower CO2 emission for trading of carbon credits in the international market under Paris...	143	Practice →
6(b)	Explain various challenges associated with disposing of massive amounts of Municipal solid waste. Discuss action plan for municipal solid waste management in India for...	146	Practice →
7(a)	Biotechnology has unlocked a vast potential for improving human life, but the risks it poses are now a concern — Discuss throwing light on blessing and curse of...	149	Practice →
7(b)	Why is Ecotourism considered as a tool for conservation of Natural heritage ? Discuss challenges associated with Ecotourism.	152	Practice →
8(a)	Why is antibiotic resistance getting worse in India ? Discuss the strategies and action plans in India to deal with this.	155	Practice →
8(b)	Bio-pesticides are slower in action compared to conventional pesticides, with shorter persistence and susceptible to unfavourable environmental conditions, but still there i...	158	Practice →
9(a)	Why is the number of patents granted in India still a fraction compared to China and US, even though India's global position in innovations has been rising over the years ?	161	Practice →
9(b)	What are the tools required for the successful commercialization of Intellectual Property Rights in India ? Discuss the role of CIPAM in commercialization of Intellectu...	164	Practice →
10(a)	Inefficient farming and poor distribution are reasons for the world's food shortage and the genetically modified crops are a bad answer to the wrong problem — Comment.	167	Practice →
10(b)	Why is monitoring and weather forecasting crucial ? Discuss the role of ISRO in the weather prediction and capabilities in Disaster management. Add a note on Mission...	170	Practice →
11(a)	Discuss how Artificial Intelligence manipulated digital media can impact privacy, democracy and national security of a country. Suggest solutions to tackle this...	173	Practice →
11(b)	What is the current scenario of renewable power generation in India ? Discuss the factors responsible for the growing focus and shift towards Renewable energy.	176	Practice →
12(a)	What is coastal erosion ? Discuss factors causing coastal erosion. Why is erosion dominant in the Eastern coast than the Western coast in India ?	179	Practice →
12(b)	Discuss how technology translated into a medium for the destruction of environment through human exploitation.	182	Practice →
13(a)	The most important climate goal is limiting the Earth's warming to 1·5° Celsius. What happens to the world if we pass the crucial 1·5° Celsius climate threshold?	185	Practice →
13(b)	Block chain technology is the doorway to transforming the agriculture sector in India — Discuss.	188	Practice →
14(a)	What are the reasons behind the India's opposition to sign the Non-Proliferation of nuclear weapons (NPT) ? Discuss the impact of the civil nuclear agreement with the...	191	Practice →
14(b)	What are e-waste recycling practices in India ? Discuss the highlights of new e-waste management rules 2022 notified by the Government of India.	194	Practice →
15(a)	Discuss why India did not sign the COP26 pledge to stop deforestation and cut methane gas emission by 2030.	197	Practice →
15(b)	What are Gene Sanctuaries ? How do they protect the genetic diversity of plants in India ?	200	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
2020	30 answers		
1(a)	What is "life expectancy at birth"? Data shows that the global average life expectancy has more than doubled since 1900, and is now above 70 years. Describe three to five...	204	Practice →
1(b)	Telangana has several Central Government Institutions suitable for research funded by the Department of Science and Technology. These include BITS, CCMB, CDFD,...	207	Practice →
2(a)	What is the UIDAI? How has this government body harnessed IT to improve targeted delivery of government benefits (DBT)? How can the Aadhaar Card help in financial...	210	Practice →
2(b)	What is cybercrime? Describe with examples any two of the following and explain how the government helps victims of cybercrime: Cyber bullying, Cyber stalking, Phishing,...	213	Practice →
3(a)	Write a brief account of the history of ISRO. Name any three ISRO centers and their location. Add a note about the Mangalyaan or Chandrayaan missions of ISRO.	216	Practice →
3(b)	To date, ISRO has launched over 100 satellites. Explain what are GEO, LEO, MEO, and SSO. Use diagrams to illustrate your answer. Add a note on GSLV and PLSV launches.	219	Practice →
4(a)	What is nuclear energy? At the end of 2019, seven nuclear reactors were under construction in India, with a combined capacity of 5.4 GW. What are the advantages...	222	Practice →
4(b)	What is "Clean Energy"? List the sources of clean energy and explain how they work. Which of these sources can be used or are already widely used in India?	225	Practice →
5(a)	All member states of the UN adopted 17 goals in the 2030 Agenda for Sustainable Development. Name any four of these goals and explain how India can reach them.	228	Practice →
5(b)	What causes earthquakes? Can we prevent or predict where and when an earthquake will happen? What should relief agencies do after an earthquake occurs in an inland...	231	Practice →
6(a)	Classify solid waste in terms of source and types of solid wastes. Explain the steps to manage this waste. Add a note on advantages and disadvantages of landfills.	234	Practice →
6(b)	What do you know about the agents, stages, and effects of soil erosion? What are the factors affecting soil erosion and measures to conserve soil?	237	Practice →
7(a)	Is there any difference between gene therapy, transgenic organisms, and gene enhancement/genome editing? With example, explain the ethical concerns in each of...	240	Practice →
7(b)	In October 2020, the Nobel Prize for Chemistry was awarded to the CRISPR technology. Explain how CRISPR functions in nature and how it has been adapted for...	243	Practice →
8(a)	Write short notes on the causes, treatment, and methods of prevention of any 3 of the following diseases: 1. Cholera; 2. Malaria, 3. Encephalitis, 4. Tuberculosis, and 5. Avian...	246	Practice →
8(b)	What is recombinant DNA? Draw a schematic to show the process of creating rDNA and explain the roles of restriction enzymes and vectors in this process. Give an...	249	Practice →
9(a)	What is Intellectual Property? Explain with examples the differences between copyrights, patents, and trademarks. What are the rights an inventor has on his...	252	Practice →
9(b)	What is a patent? Give examples of patents. What is the difference between process patents and product patents? How are GI tags related to patents?	255	Practice →
10(a)	What is the status of utilization of Solar energy in India and the World? What are some factors hindering utilization of this green energy source?	258	Practice →
10(b)	Traditional methods of in-person education have been disrupted by the Covid-19 pandemic. How can India's Space Program help in restoring and expanding access to...	261	Practice →
11(a)	What do you know about the NeGP (National e-Governance Action Plan) operating under the aegis of Digital India? Explain about the role of any three of the following...	264	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
11(b)	What are biofuels and why are they important to India's economy? What do you know about the Indian National Policy on Biofuels?	267	Practice →
12(a)	Please interpret the following data in terms of contribution by countries to Science and Engineering publications. Identify any one factor for the state of Indian Science and...	270	Practice →
12(b)	Is there a difference between "forest cover" and "forest area"? What does the following graph show? (Graph data provided in the original PDF)	273	Practice →
13(a)	What is Climate Justice? How is this matter covered by news media and influential publications like Time magazine? Suggest at least 2 measures that are aligned to this...	276	Practice →
13(b)	Explain what you see in the following graphic. (a) How does human activity affect the substance shown in the graphic? (b) How does this substance affect human health?...	279	Practice →
14(a)	What are the most common and hazardous pollutants in outdoor and indoor air? Mention the health hazards from each of the pollutants. What is AQI (Air Quality Index)...	282	Practice →
14(b)	What is Covid-19 and how is it spread? How are immuno-diagnostic methods used in context of the Covid19 pandemic?	285	Practice →
15(a)	Explain the mechanism of action of vaccines. Give two examples where vaccines have eradicated diseases in India. What do you know about the Global Vaccine Action Plan...	288	Practice →
15(b)	The 2020 Nobel Prize for Physiology and Medicine was awarded to three scientists for their work on HCV. What do you know about this virus and the disease it causes? Is...	291	Practice →
2017	30 answers		
P-IV 1(a) old P-IV	What do you understand by "Big Data" ? Discuss India's initiatives in this regard; name the challenges and constraints.	295	Practice →
P-IV 1(b) old P-IV	The Internet of Things (IoT) is expected to be the biggest technological revolution. Examine its likely impact on India.	298	Practice →
P-IV 2(a) old P-IV	Explain the background and salient features of "Pradhan Mantri Gramin Digital Saksharta Abhiyan".	301	Practice →
P-IV 2(b) old P-IV	Cutting-edge developments in nanotechnology offer new ways of preventing and treating lifestyle diseases, including cancer. Elaborate.	304	Practice →
P-IV 3(a) old P-IV	What is GPS ? Explain the recent efforts made by India to create its own GPS system.	307	Practice →
P-IV 3(b) old P-IV	What is the difference between GSLV and PSLV launch vehicles ? Examine the advantages of GSLV over PSLV.	310	Practice →
P-IV 4(a) old P-IV	Discuss India's present composition of energy sources and future energy security scenario in the context of growing energy demand.	313	Practice →
P-IV 4(b) old P-IV	What are the benefits of "Smart Grid" ? Examine the challenges and strategies for "Smart Grid" operations in India.	316	Practice →
P-IV 5(a) old P-IV	Why are urban floods a recurrent feature in India ? Discuss the main causes of urban floods and measures taken by the Government(s) for urban flood control.	319	Practice →
P-IV 5(b) old P-IV	Critically examine how illegal mining of river sand in various parts of India is affecting the ecology of rivers and lives of people in the vicinity of these rivers.	322	Practice →
P-IV 6(a) old P-IV	Write a brief account of Bentham and Hooker's classification and discuss its demerits.	325	Practice →
P-IV 6(b) old P-IV	Which hormone regulates blood glucose levels in the body and how is it synthesized artificially ?	328	Practice →
P-IV 7(a) old P-IV	Describe the medicinal qualities of mushrooms.	331	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
P-IV 7(b)	Give an account on chemicals and drugs produced from animals.	334	Practice →
old P-IV 8(a)	Describe the major types of fungus that cause Mycoses in humans.	337	Practice →
old P-IV 8(b)	Describe the most common protozoan diseases, causative agents and preventive measures.	340	Practice →
P-IV 9(a)	Describe the methods of gene transfer in developing Genetically Modified crops.	343	Practice →
old P-IV 9(b)	What is a Biopesticide ? How are fungal parasites of insects useful in Biological control ?	346	Practice →
P-IV 10(a)	What is Major Histocompatibility Complex (MHC) ? Describe the role of MHC classes.	349	Practice →
old P-IV 10(b)	What are monoclonal antibodies ? How are they useful in detecting parasitic diseases ?	352	Practice →
old P-IV 11(a)	Invasive alien species are the second largest threat to biodiversity after habitat destruction. Discuss the causes with examples.	355	Practice →
P-IV 11(b)	Describe the carbon and nitrogen biogeochemical cycles. Cite examples in the Indian context.	358	Practice →
old P-IV 12(a)	What are the advantages and disadvantages of solar power over other renewable energy resources ? Why is the cost of solar energy decreasing ? What is its effect on...	361	Practice →
old P-IV 12(b)	"India's traditional water management methods are key in the present context." Examine the statement with special focus on water harvesting and disaster...	364	Practice →
old P-IV 13(a)	What effects does PM2.5 fine particulate matter have on the health of individuals in urban regions of India and across the world ? Discuss.	367	Practice →
old P-IV 13(b)	Critically discuss the various challenges and issues arising out of waste management in Indian towns and cities.	370	Practice →
old P-IV 14(a)	Recent observations have shown that the ozone layer over Antarctic shows signs of healing. Critically examine.	373	Practice →
P-IV 14(b)	Describe the role of Information Technology in environmental conservation.	376	Practice →
old P-IV 15(a)	Briefly describe the Sendai Framework for Disaster Risk Reduction and its adoption in the disaster management policy of India.	379	Practice →
old P-IV 15(b)	Examine the role of the National Green Tribunal in providing speedy justice in environmental matters by quoting its recent decisions.	382	Practice →
2016	60 answers		
P-IV 1(d)	Write a short note on the following: (i) Post Harvest Technology (PHT) (ii) Oilseeds Production Programme (OPP)	386	Practice →
old P-IV 1(a)	Briefly state the salient features of the National Data Sharing and Accessibility Policy, 2012 and its benefits?	389	Practice →
old P-IV 1(b)	Write a brief note on the following: (i) Indian Science Congress Association (ISCA) (ii) Council of Scientific and Industrial Research (CSIR)	392	Practice →
old P-IV 1(c)	Mention the main objectives of Science, Technology and Innovation (STI) Policy, 2013 of India.	395	Practice →
old P-IV 2(d)	Write a short note on: (i) Mobile Satellite Services (MSS) (ii) Environmental monitoring	398	Practice →
old P-IV 2(a)	Describe the MOM (Mars Orbiter Mission – Mangalyaan) programme of India? What are its distinguishing features?	401	Practice →

No.	Question	Page	Practice online
P-IV 2(b) old P-IV	Write a short note on the following: (i) Broadband digital communication (ii) Fixed-Satellite Services (FSS)	404	Practice →
P-IV 2(c) old P-IV	Explain the applications of Remote Sensing technology in agriculture.	407	Practice →
P-IV 3(d) old P-IV	Write a note on Local Area Network (LAN).	410	Practice →
P-IV 3(a) old P-IV	What is cloud computing? Explain briefly about AP Cloud Initiative?	413	Practice →
P-IV 3(b) old P-IV	Describe Fiber Grid Project of Andhra Pradesh.	416	Practice →
P-IV 3(c) old P-IV	What is ERP (Enterprise Resource Planning)? Narrate the critical issues in execution of ERP Projects.	419	Practice →
P-IV 4(d) old P-IV	Give a detailed list of barriers to renewable energy?	422	Practice →
P-IV 4(a) old P-IV	Write a short note on Shale Gas Revolution.	425	Practice →
P-IV 4(b) old P-IV	Narrate Nuclear Power growth in India. Add a note on Nuclear Fuel reserves in the country.	428	Practice →
P-IV 4(c) old P-IV	Discuss the scope of renewable energy in the scenario of energy crisis?	431	Practice →
P-IV 5(d) old P-IV	Write the main points of the agreement reached at United Nations Climate Change Conference (COP 21) held at Paris in December, 2015. What was India's stand at the...	434	Practice →
P-IV 5(a) old P-IV	Distinguish between a Hazard and a Disaster? Narrate various types of natural disasters, the coastal seashore areas are prone to?	437	Practice →
P-IV 5(b) old P-IV	Explain the details of Tsunami Warning System of India.	440	Practice →
P-IV 5(c) old P-IV	Discuss the common challenges in any disaster management plan.	443	Practice →
P-IV 6(d) old P-IV	What is Lactic acid fermentation? Explain how different products can be made by fermentation of milk?	446	Practice →
P-IV 6(a) old P-IV	What are Vitamins? Give a detailed classification of vitamins mentioning their functions and sources of availability?	449	Practice →
P-IV 6(b) old P-IV	What is yeast fermentation? Explain industrial use of yeast in making different products?	452	Practice →
P-IV 6(c) old P-IV	What is a phytohormone? Mention different groups of phytohormones with examples? Discuss the present scenario on commercial use of hormones in artificial fruit ripening?	455	Practice →
P-IV 7(d) old P-IV	Write a short note on the following: (i) Homeostatis (ii) Animal response to Osmotic conditions of the medium	458	Practice →
P-IV 7(a) old P-IV	What are essential oils? Enumerate the important uses of essential oils?	461	Practice →
P-IV 7(b) old P-IV	Give an account of the following: (i) Autotrophic Vs. Heterotrophic plants (ii) Symbiosis	464	Practice →
P-IV 7(c) old P-IV	(i) What are Lichens? Write their economic importance. (ii) Explain how insects are useful to human beings.	467	Practice →
P-IV 8(d) old P-IV	What is DNA Finger Printing? Explain how it is done. Write its forensic applications.	470	Practice →
P-IV 8(a) old P-IV	What is plant tissue culture? Write applications of tissue culture in agriculture?	473	Practice →
P-IV 8(b) old P-IV	What is 'Golden Rice'? Describe its advantages?	476	Practice →
P-IV 8(c) old P-IV	What are Bio fertilizers? Explain different types of bio fertilizers? Write advantages of bio fertilizers?	479	Practice →
P-IV 9(d) old P-IV	Explain the preventive measures to be taken to avoid common infections.	482	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
P-IV 9(a) old P-IV	Explain different patterns of disease outbreak with examples? Differentiate between epidemic and endemic diseases?	485	Practice →
P-IV 9(b) old P-IV	What is the cause of Chikungunya? What are its preventive measures and treatment? Give a comparative note on symptomology of Chikungunya and Dengue?	488	Practice →
P-IV 9(c) old P-IV	Write a brief note on aetiology, mode of spread, symptoms, treatment and preventive measures of malarial parasite?	491	Practice →
P-IV 10(d) old P-IV	Biotechnological intervention has transformed the vaccination technology' – Discuss.	494	Practice →
P-IV 10(a) old P-IV	What is immunisation? Briefly write the history of development of vaccines?	497	Practice →
P-IV 10(b) old P-IV	Write a short note on: (i) Edible vaccines (ii) ELISA test (iii) Autoimmune diseases	500	Practice →
P-IV 10(c) old P-IV	Give a detailed note on 'Triple Vaccination'.	503	Practice →
P-IV 11(d) old P-IV	List-out the projects/activities for which public consultation is not required for issue of prior Environmental Clearance.	506	Practice →
P-IV 11(a) old P-IV	Narrate the functions of State Pollution Control Board.	509	Practice →
P-IV 11(b) old P-IV	Describe the procedure for obtaining Environmental Clearance under Environment Impact Assessment Notification, 2006.	512	Practice →
P-IV 11(c) old P-IV	What is Environmentalism? Briefly state the problems does our planet face and suggest the solutions.	515	Practice →
P-IV 12(d) old P-IV	What is meant by mineral resources? How they form? Write about different types of mineral resources.	518	Practice →
P-IV 12(a) old P-IV	Classify dams based on structure and design.	521	Practice →
P-IV 12(b) old P-IV	Write a note on Andhra Pradesh Sand Mining Policy, 2016.	524	Practice →
P-IV 12(c) old P-IV	Discuss the basis and justification required for suitability of a crop and cropping pattern.	527	Practice →
P-IV 13(d) old P-IV	Explain the following: (i) National Bureau of Plant Generic Resources (NBPGR) (ii) Wild-life sanctuaries.	530	Practice →
P-IV 13(a) old P-IV	What is eco system? Explain the food chain system in eco systems?	533	Practice →
P-IV 13(b) old P-IV	Narrate the impact of urbanisation on bio-diversity.	536	Practice →
P-IV 13(c) old P-IV	Describe the major threats to bio-diversity. Explain different types of bio-diversities.	539	Practice →
P-IV 14(d) old P-IV	Describe noise pollution and explain the various control and preventive measures to be taken to control noise pollution.	542	Practice →
P-IV 14(a) old P-IV	Narrate various types and the characteristics of Solid Waste.	545	Practice →
P-IV 14(b) old P-IV	Write a short note on the following: (i) Pyrolysis (ii) Incineration	548	Practice →
P-IV 14(c) old P-IV	What is Soil Pollution and how it is caused?	551	Practice →
P-IV 15(d) old P-IV	Give a brief note on the following: (i) Sustainable agriculture (ii) Acid Rain	554	Practice →
P-IV 15(a) old P-IV	What is El Nino? What are the causes of El Nino? Discuss the scenario of impact of El Nino on global agriculture.	557	Practice →
P-IV 15(b) old P-IV	Describe Impact of Ozone layer depletion on different living organisms?	560	Practice →
P-IV 15(c) old P-IV	Describe impact of human beings on water resources.	563	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
2012	52 answers		
P-IV 1(d) old P-IV	Write a short note on the recent development taking place in the most popular National Solar Mission.	567	Practice →
P-IV 1(a) old P-IV	What are the important technological missions of India? Add a note on their objectives.	570	Practice →
P-IV 1(b) old P-IV	Write an essay on National Literacy Mission.	573	Practice →
P-IV 1(c) old P-IV	Enumerate the policy objectives in the development of Science and Technology in India.	576	Practice →
P-IV 2(d) old P-IV	What is the resolution of CARTOSAT-2 and for what purpose are its images useful? What is the main difference between CARTOSAT-2 and IKONOS?	579	Practice →
P-IV 2(a) old P-IV	What is a Satellite? Explain the different types of Satellites in the Space programme. Add a note on GSAT-5.	582	Practice →
P-IV 2(b) old P-IV	How many satellites are tracked at a time for a GPS? Explain how and for what purpose the GPS is used.	585	Practice →
P-IV 2(c) old P-IV	Define band width. What are the different band widths used in INSAT series? Mention the significance of thermal band in Agriculture.	588	Practice →
P-IV 3(d) old P-IV	Write an essay on the modem, wireless, and types of networking systems.	591	Practice →
P-IV 3(a) old P-IV	Discuss briefly the role of IT in rural development in India with a note on effective implementation.	594	Practice →
P-IV 3(b) old P-IV	Define Telemedicine. Write a note on the different types of telemedicine.	597	Practice →
P-IV 3(c) old P-IV	Which are the five major IT city hubs in India? What are the main factors required for the growth of software industry?	600	Practice →
P-IV 4(d) old P-IV	Define nuclear energy. Discuss briefly on the mitigation of nuclear hazards and nuclear waste.	603	Practice →
P-IV 4(a) old P-IV	What is the difference between renewable energy and non-renewable energy? What are the different types of non-conventional energy? Add a note on their resources.	606	Practice →
P-IV 4(b) old P-IV	What are the factors for the increase in the worldwide energy demand? Discuss briefly to mitigate such demands.	609	Practice →
P-IV 4(c) old P-IV	What are the new emerging renewable energy technologies?	612	Practice →
P-IV 5(d) old P-IV	Explain briefly on the role and responsibilities of National Disaster Management Authority (NDMA) in achieving its goal.	615	Practice →
P-IV 5(a) old P-IV	Explain various causes for the changes in climate and its effect on both the living and non-living bodies on the Earth.	618	Practice →
P-IV 5(b) old P-IV	What are the different types of indicators of climatic changes? Write a short note on cryogenic indicators.	621	Practice →
P-IV 5(c) old P-IV	What are the parameters considered to declare as a disaster? List out the various disasters which considered for declaration as national disasters.	624	Practice →
P-IV 6(d) old P-IV	Differentiate between virus and bacteria. What are the harmful and useful effects caused by both in the humans and animals?	627	Practice →
P-IV 6(a) old P-IV	Differentiate between old and modern Biotechnology. Describe the discovery of Louis Pasteur on fermentation.	630	Practice →
P-IV 6(b) old P-IV	What are antibiotics? Write advantages and disadvantages of antibiotics use.	633	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
P-IV 6(c) old P-IV	What is the difference between Penicillin and Insulin? Describe the Industrial production of Penicillin.	636	Practice →
P-IV 7(d)	Write an essay on stem cells research and the significance of cloning in various fields.	639	Practice →
old P-IV P-IV 7(a)	Define bioinsecticides, What are the different types of bioinsecticides?	642	Practice →
old P-IV P-IV 8(d)	What is a transgenic animal? List out the benefits from transgenic animals.	645	Practice →
old P-IV P-IV 8(a)	Write briefly on the types and uses of biopesticides.	648	Practice →
old P-IV P-IV 8(b)	Mention briefly the arguments in favour of and against GM crops.	651	Practice →
old P-IV P-IV 8(c) old P-IV	What are the different types of sterilization methods adopted for different types of materials?	654	Practice →
P-IV 9(d)	Discuss briefly on the morphology and biological characters of measles virus.	657	Practice →
old P-IV P-IV 9(a) old P-IV	What is meant by infection? Explain the terms Acute, chronic, fulminating, localized, generalized, primary and secondary used for infections with examples.	660	Practice →
P-IV 9(b)	Write a short note on Pathogenesis.	663	Practice →
old P-IV P-IV 9(c) old P-IV	What is hydrophobia? how is it caused? What remedial or precautionary measures are to be taken?	666	Practice →
P-IV 10(d)	Write a short note on the adverse consequences of vaccination.	669	Practice →
old P-IV P-IV 10(a) old P-IV	Enumerate the mechanisms of humoral immunity system and the usefulness of the antibodies.	672	Practice →
P-IV 10(b)	Give an account of the adverse effects of passive immunization.	675	Practice →
old P-IV P-IV 10(c) old P-IV	What are the organs of immune system? Give a schedule of immunization for newborn children.	678	Practice →
P-IV 11(b)	What are the salient features of Forest Act of 1980?	681	Practice →
old P-IV P-IV 11(c)	What are the measures to be taken in prevention and control of air pollution?	684	Practice →
old P-IV P-IV 13(d) old P-IV	In the Project Tiger, what are the administrative measures taken by the Ministry of Environment and Forests.	687	Practice →
P-IV 13(a)	What are the Principles of Ecology?	690	Practice →
old P-IV P-IV 13(b)	Define biodiversity. Discuss briefly on the consumptive use of biodiversity.	693	Practice →
old P-IV P-IV 13(c)	Discuss briefly various reasons leading to the loss of bio-diversity.	696	Practice →
old P-IV P-IV 14(d)	Write a short note on Eutrophication.	699	Practice →
old P-IV P-IV 14(a)	What are the major Air Pollutants? Add a note on their effects.	702	Practice →
old P-IV P-IV 14(b)	What is meant by Electrostatic precipitator? Explain its working principle.	705	Practice →
old P-IV P-IV 14(c)	Differentiate between BOD and COD and discuss their importance.	708	Practice →
old P-IV P-IV 15(d)	What is Brundtland Report? Enumerate.	711	Practice →
old P-IV P-IV 15(a) old P-IV	What are the salient benefits of the application of Information Technology in the field of medicine and medical health?	714	Practice →
P-IV 15(b) old P-IV	What is meant by watershed? How can watershed management be done effectively with modern Information Technology?	717	Practice →
P-IV 15(c) old P-IV	What are the effects of climate change on nature, sea level fluctuations, plants and animal migration, changes in insect pests?	720	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
2011	30 answers		
P-IV 1(a) old P-IV	(i) Illustrate the National Policy on Science and Technology. (ii) Discuss the role of technology missions in India with a suitable example.	724	Practice →
P-IV 1(b) old P-IV	Discuss on (i) Intellectual Property Rights. (ii) National Policy on Technology Missions.	727	Practice →
P-IV 2(a) old P-IV	(i) Comment on "Chandrayan" Programme. (ii) Discuss two salient features of IRS-5 Satellite.	730	Practice →
P-IV 2(b) old P-IV	Illustrate the role of Indian Space Programmes on (i) Rural India (ii) Natural Resources	733	Practice →
P-IV 3(a) old P-IV	Comment on the impact of the growth of Computer industry on the economic growth of India.	736	Practice →
P-IV 3(b) old P-IV	Write on (i) Communications and remote sensing (ii) Influence of Information Technology on Indian Rural Sector.	739	Practice →
P-IV 4(a) old P-IV	(i) Of late there is lot of public resistance towards the establishment of coal based power plants. Comment. (ii) Discuss the reasons for the sparse use of renewable...	742	Practice →
P-IV 4(b) old P-IV	Deliberate on (i) Wind power in India (ii) Biomass management in India (iii) Ocean energy	745	Practice →
P-IV 5(a) old P-IV	Write on (i) Disaster Management Programmes in India (ii) Mitigation methods to address Global warming.	748	Practice →
P-IV 5(b) old P-IV	Discuss the problems associated with unabated urbanization and suggest the ways and means to arrest this.	751	Practice →
P-IV 6(a) old P-IV	Discuss the Agricultural research in India mentioning its linkages. Describe the research and educational activities in the field of agriculture, animal husbandry and...	754	Practice →
P-IV 6(b) old P-IV	Elucidate a low volume high value product giving the example of insulin. Mention the role of Indian companies for its production.	757	Practice →
P-IV 7(a) old P-IV	Discuss the applications of plants in different areas with examples.	760	Practice →
P-IV 7(b) old P-IV	(i) Differentiate between wild and domestic animals. (ii) Give an overview of how animals are exploited in medical research.	763	Practice →
P-IV 8(a) old P-IV	Give a brief description of GM crops and explain the regulations and concerns associated with the release of a Genetically Modified crop.	766	Practice →
P-IV 8(b) old P-IV	Define biofertilizers. Give an account of technology involved for the mass production of different biofertilizers, with their application.	769	Practice →
P-IV 9(a) old P-IV	Discuss in detail about HIV virus; also mention the symptoms associated with the infection, preventive measures and diagnostic test for disease confirmation.	772	Practice →
P-IV 9(b) old P-IV	Describe the causative agent for malaria, emphasize its life cycle in different hosts and discuss the preventive measures for the disease.	775	Practice →
P-IV 10(a) old P-IV	Define immunity. Discuss the concept of vaccination and explain how the vaccine for OPT is produced conventionally.	778	Practice →
P-IV 10(b) old P-IV	What do you understand by a vaccine? Describe the journey of development from conventional to modern day (Recombinant vaccines).	781	Practice →
P-IV 11(a) old P-IV	In your own words explain the steps to bring awareness and in educating towards Environmental Protection in India	784	Practice →
P-IV 11(b) old P-IV	Write the salient features of Water Act and Water pollution Cess Act.	787	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
P-IV 12(a) old P-IV	Give an account on mineral resources of India. Explain the impacts of mining activity on the environment.	790	Practice →
P-IV 12(b) old P-IV	Explain Integrated Water Resource Management. Give the importance of Water user Associations.	793	Practice →
P-IV 13(a) old P-IV	What are the natural and human induces threats to biodiversity? Give an example of each.	796	Practice →
P-IV 13(b) old P-IV	What is meant by the term natural Selection? Draw a food chain that has you as the top consumer.	799	Practice →
P-IV 14(a) old P-IV	Explain Urban Governance and role of community in Municipal Solid Waste Management.	802	Practice →
P-IV 14(b) old P-IV	Discuss the sources, impacts and control measures of Marine pollution.	805	Practice →
P-IV 15(a) old P-IV	Explain National Action Plan on Climate Change (NAPCC), objectives and recommendations.	808	Practice →
P-IV 15(b) old P-IV	Discuss the strategies for wasteland reclamation.	811	Practice →
2008	30 answers		
P-IV 1(a) old P-IV	What are the strategies of development, transfer and diffusion of technology in our country ?	815	Practice →
P-IV 1(b) old P-IV	Discuss our national policy on generation and management of intellectual property.	818	Practice →
P-IV 2(a) old P-IV	What are the salient features of IRS-P5 ?	821	Practice →
P-IV 2(b) old P-IV	What is the role of Indian Space Program in agricultural and rural developmental activities ?	824	Practice →
P-IV 3(a) old P-IV	Discuss the role of software development in the economic growth of our country.	827	Practice →
P-IV 3(b) old P-IV	Discuss the various information technology applications with special reference to rural India.	830	Practice →
P-IV 4(a) old P-IV	What are the various renewable energy resources ? What is their potential in our country ?	833	Practice →
P-IV 4(b) old P-IV	What are the challenges faced by energy sector in our country ?	836	Practice →
P-IV 5(a) old P-IV	What are the potential impacts of climate change in India ?	839	Practice →
P-IV 5(b) old P-IV	What are the outcomes of crop science research in India ?	842	Practice →
P-IV 6(a) old P-IV	Define fermentation. Discuss the general procedure for the production of alcohol in the industry.	845	Practice →
P-IV 6(b) old P-IV	(i) What is a low volume, high value product in biotechnology ? (ii) Write about interferons, their applications and a general scheme for the industrial production of...	848	Practice →
P-IV 7(a) old P-IV	Write the general characteristics of terrestrial plants. Write about the uses of plants to mankind.	851	Practice →
P-IV 7(b) old P-IV	Write about the definition, criteria, history and process of domestication of some common animals by humans.	854	Practice →
P-IV 8(a) old P-IV	What is meant by genetically modified crops ? Write their role in agriculture with a note on the legitimate concerns about transgenic agriculture.	857	Practice →
P-IV 8(b) old P-IV	Write about the role of biotechnology in environmental clean-up processes.	860	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

No.	Question	Page	Practice online
P-IV 9(a) old P-IV	Write about the causative agent for diarrhoea, its symptoms, modes of infection and the required preventive measures by the government during the outbreak.	863	Practice →
P-IV 9(b) old P-IV	Write about different types of tuberculosis, modes of infection, symptoms, common diagnostic tests and causes for its prevalence in India.	866	Practice →
P-IV 10(a) old P-IV	What are vaccines ? Write the fundamental concept of vaccination and list some of the compulsory vaccines for children.	869	Practice →
P-IV 10(b) old P-IV	Write about the different types of modern vaccines available today. How are the modern vaccines better than the conventional vaccines ?	872	Practice →
P-IV 11(a) old P-IV	Write in detail about various segments of environment.	875	Practice →
P-IV 11(b) old P-IV	(i) Write in detail about the Water (Prevention and Control of Pollution) Act. (ii) What are the issues involved in the enforcement of environmental legislation and law ?	878	Practice →
P-IV 12(a) old P-IV	(i) Explain about the types of forests present in India. (ii) How are forests helping the human society ? Explain.	881	Practice →
P-IV 12(b) old P-IV	Explain about the mineral resources present in India.	884	Practice →
P-IV 13(a) old P-IV	(i) What do you understand by the term ecosystem ? (ii) Write the structure of ecosystem. (iii) List out different types of aquatic ecosystems existing in India. (iv) Wha...	887	Practice →
P-IV 13(b) old P-IV	(i) What are the threats to Indian biodiversity ? (ii) How can Indian biodiversity be conserved ?	890	Practice →
P-IV 14(a) old P-IV	As an individual what is your role in the prevention of environmental pollution ? Explain.	893	Practice →
P-IV 14(b) old P-IV	(i) Explain the effects of air pollution. (ii) Suggest various measures to control air pollution.	896	Practice →
P-IV 15(a) old P-IV	Write about the following : (i) Climate change (ii) Acid rain	899	Practice →
P-IV 15(b) old P-IV	(i) What is the need for the development of wastelands ? (ii) Write about reclamation of wastelands.	902	Practice →

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

PAPER V AT A GLANCE

Trend analysis

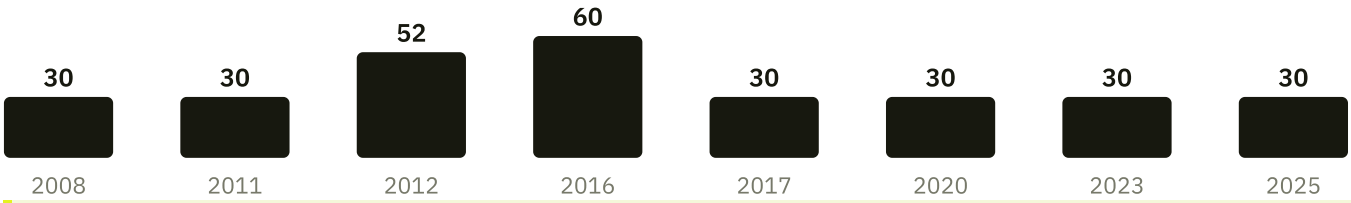
292
APPSC QUESTIONS

8
SITTINGS, 2008 TO 2025

9
CHAPTERS

59
SUB-TOPICS ASKED

QUESTIONS BY SITTING



Darker cells mark the sub-topics the commission returns to most. Older sittings (2008 to 2017) are counted under today's syllabus.
To practice any sub-topic with all its past questions together, sign in at ghissu.com and join the waitlist (usually approved in 1 to 2 days).

CHAPTER AND SUB-TOPIC (LATEST SYLLABUS)	2008	2011	2012	2016	2017	2020	2023	2025	TOTAL
1. Science, Technology & Innovation: policies, institutions, achievements	**1**	**3**	**3**	**4**	**·**	**3**	**2**	**4**	**20**
1.1 Integration of Science, Technology and Innovation for better human life; Scien...	·	·	·	·	·	1	1	1	3
1.2 National Policies on proliferation of Science, Technology and Innovation	·	2	3	3	·	·	·	·	8
1.3 India's contribution in the field of Science and Technology	·	·	·	·	·	1	·	·	1
1.4 Concerns and challenges in the proliferation and use of science and technology	·	·	·	·	·	·	1	·	1
1.5 Role and Scope of Science and Technology in nation building	1	·	·	·	·	·	·	·	1
1.6 Major Scientific institutes for Science and Technology in AP and India. Major S...	·	1	·	1	·	1	·	2	5
1.7 Achievements of Indian Scientist in the field of Science and Technology- Indig...	·	·	·	·	·	·	·	1	1
2. Information and Communication Technology (ICT)	**2**	**2**	**3**	**4**	**3**	**3**	**4**	**3**	**24**
2.1 Information and Communication Technology (ICT) - its importance, advantag...	1	1	2	2	2	·	1	·	9
2.2 E-governance and India	·	·	·	·	1	2	1	1	5
2.3 Cyber Crime and policies to address security concerns	·	·	·	·	·	1	1	1	3
2.4 Government of India Policy on Information Technology (IT)	·	·	·	·	·	·	·	1	1
2.5 IT Development in AP and India	1	1	1	2	·	·	1	·	6
3. Indian Space Programme, ISRO, Satellites and DRDO	**2**	**2**	**5**	**4**	**2**	**3**	**3**	**2**	**23**
3.1 Indian Space Programme - Past, Present and Future	1	1	·	·	·	·	2	·	4
3.2 Indian Space Research Organization (ISRO): it's activities and achievements	·	1	·	1	1	2	·	·	5
3.3 Satellite Programmes of India and Use of Satellites in different fields like Heal...	1	·	5	3	1	1	1	1	13
3.4 Defence Research and Development Organization (DRDO)	·	·	·	·	·	·	·	1	1
4. Energy: needs, resources, policy; Nuclear programme and policy	**2**	**2**	**5**	**4**	**3**	**3**	**4**	**1**	**24**
4.1 India's energy needs, efficiency and resources; Energy demands, Indian Ener...	1	·	1	·	2	·	·	·	4
4.2 Conventional and Non-Conventional energy resources; Clean energy resource...	1	2	2	3	1	2	1	1	13
4.3 Energy policy of India - Government Policies and Programmes; energy policie...	·	·	1	·	·	·	1	·	2
4.4 Salient features of Nuclear Policy of India; Development of Nuclear programm...	·	·	1	1	·	1	·	·	3
4.5 Nuclear Policies at the International level and India's stand on them	·	·	·	·	·	·	2	·	2
5. Development vs Environment: natural resources, climate change, EIA, natural dis...	**7**	**6**	**5**	**12**	**3**	**5**	**4**	**6**	**48**
5.1 Development Vs. Nature / Environment	·	1	·	1	·	·	1	·	3
5.2 Depletion of Natural Resources- Metals, Minerals: Conservation Policy. Types ...	1	1	·	1	·	·	·	1	4
5.3 Mining in AP and India	·	·	·	1	1	·	1	1	4

PAPER V TREND ANALYSIS, CONTINUED

CHAPTER AND SUB-TOPIC (LATEST SYLLABUS)	2008	2011	2012	2016	2017	2020	2023	2025	TOTAL
5.4 Environmental Pollution Natural and Anthropogenic and Environmental degra...	2	1	·	·	·	1	·	·	**4**
5.5 Sustainable Development: possibilities and challenges (and Sustainable Devel...	·	·	1	1	·	1	·	1	**4**
5.6 Climate Change and Its effect on the world; Climate justice: a global phenome...	2	1	3	·	·	1	2	·	**9**
5.7 Environment Impact Assessment	·	·	·	2	·	·	·	1	**3**
5.8 Natural Disasters: Cyclones, Earth Quakes, Landslides & Tsunamis: Prediction...	·	·	·	3	1	1	·	·	**5**
5.9 Correlation between Health & Environment	·	·	·	·	·	·	·	1	**1**
5.10 Forest resources; Social Forestry, Afforestation and deforestation	1	·	·	·	·	1	·	1	**3**
5.12 Water resources: Types, Water shed management	·	1	1	2	1	·	·	·	**5**
5.13 Land resources: types of soils and soil reclamation	1	1	·	1	·	·	·	·	**3**
6. Pollution & solid waste; global issues; environmental legislation; disaster manag...	**3**	**4**	**9**	**8**	**6**	**3**	**5**	**5**	**43**
6.1 Sources, impacts and control of air pollution	1	·	3	·	1	1	·	·	**6**
6.2 Sources, impacts and control of water pollution	·	1	2	·	·	·	·	·	**3**
6.3 Sources, impacts and control of soil pollution; Noise pollution	·	·	·	2	·	·	·	·	**2**
6.4 Solid waste management - Types of solid waste, impacts of solid waste, recycl...	·	1	·	2	1	1	2	1	**8**
6.5 Remedial measurers for Soil erosion and Costal erosion	·	·	·	·	·	1	1	·	**2**
6.6 Global Environmental issues and role of information Technology in Environme...	·	·	1	1	2	·	·	2	**6**
6.7 Environmental legislation: International Law, Montreal protocol, Kyoto protoc...	·	·	·	1	·	·	2	·	**3**
6.8 The Environment (Protection) act 1986, Forest conservation Act, Wildlife prot...	1	1	1	1	1	·	·	1	**6**
6.9 National Disaster Management Policy, 2016 of India and Disaster manageme...	·	1	2	1	1	·	·	1	**6**
6.10 White Revolution, Green Revolution, and Green Pharmacy	1	·	·	·	·	·	·	·	**1**
7. Biotechnology & Nanotechnology: scope, ethics, genetic engineering, biodiversity...	**7**	**5**	**7**	**13**	**9**	**3**	**3**	**3**	**50**
7.1 Nature, Scope and Applications of Biotechnology in India; Government policies	1	·	·	·	·	·	·	2	**3**
7.2 Nature, Scope and Applications of Nanotechnology in India	·	·	·	·	1	·	·	·	**1**
7.3 Ethical, Social and Legal concerns, Government policies	·	1	·	·	·	·	1	1	**3**
7.4 Genetic engineering, issues related to it and its impact on human life	·	·	1	·	·	2	·	·	**3**
7.5 Bio – diversity	2	2	**4**	**4**	2	·	2	·	**16**
7.6 fermentation	2	1	2	2	1	·	·	·	**8**
7.7 Immuno – diagnosis techniques	·	·	·	1	2	1	·	·	**4**
7.8 Legacy (pre-2018 syllabus): general botany, zoology, nutrition and physiology:...	2	1	·	**6**	3	·	·	·	**12**
8. Human diseases; genetic engineering basics; biotechnology in agriculture; vaccin...	**5**	**6**	**15**	**11**	**4**	**5**	**3**	**5**	**54**
8.1 Human diseases-microbial infections. Common infections and preventive me...	·	·	**5**	2	2	·	1	2	**12**
8.2 Basic knowledge of infections-diarrhoea, dysentery, cholera, tuberculosis, m...	2	2	2	2	·	2	·	·	**10**
8.3 Introduction to Genetic Engineering and Biotechnology. Basic concepts of gen...	·	·	·	1	1	1	·	1	**4**
8.4 Tissue culture methods and applications	·	·	·	1	·	·	·	·	**1**
8.5 Biotechnology in agriculture- Bio-pesticides, Bio-fertilizers, Bio-fuels, Genetic...	1	2	3	2	1	1	2	·	**12**
8.6 Animal husbandry- transgenic animals	·	·	1	·	·	·	·	1	**2**
8.7 Vaccines: Introduction to immunity, Fundamental concepts in vaccination, Pro...	2	2	**4**	3	·	1	·	1	**13**
9. Intellectual Property Rights; Promotion of Science in AP and India	**1**	**·**	**·**	**·**	**·**	**2**	**2**	**1**	**6**
9.1 Issues related to Intellectual Property Rights in the field of Science and Techn...	1	·	·	·	·	2	2	1	**6**
Paper V total	**30**	**30**	**52**	**60**	**30**	**30**	**30**	**30**	**292**

1 question 2 3 or more · not asked that year

To practice any sub-topic with all its past questions together,
sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

APPSC GROUP-I MAINS · PAPER V

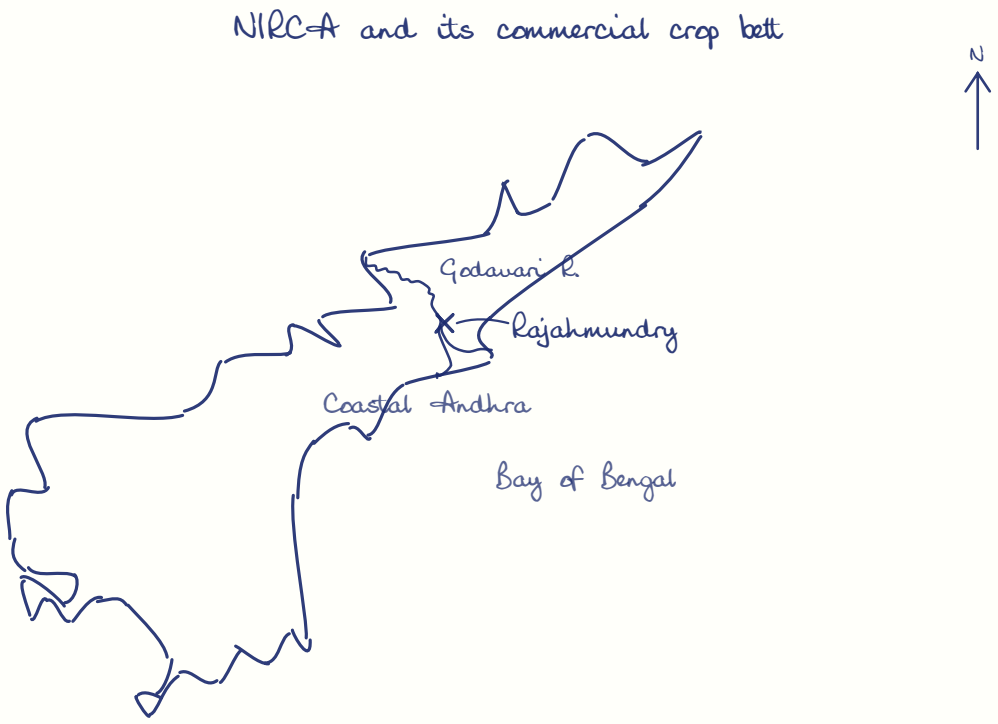
2025

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

1(a) Describe the research focus of the National Institute for Research on Commercial Agriculture (NIRCA) in Rajamahendravaram. How does it support India's agricultural sector?

10

Introduction



NIRCA at Rajamahendravaram on the Godavari

The National Institute for Research on Commercial Agriculture (NIRCA) in Rajamahendravaram researches commercial crops tailored to market demand. As commercial agriculture grows in importance for farm incomes and export potential, its work supports national priorities of productivity, resource efficiency and climate resilience.

Research Focus of NIRCA

1. Commercial Crop Improvement: Developing high-yielding, disease-resistant and climate-resilient varieties of commercial crops such as spices and horticultural produce for domestic and export markets.
2. Location-Specific Technology: Agro-climatic zone-based research tailored to regions like Rajamahendravaram, ensuring higher adoption and impact among local farmers.
3. Sustainable Resource Management: Optimising water use, soil health and nutrients, supporting the Soil Health Card and Per Drop More Crop to raise input efficiency.
4. Value Chains and Markets: Post-harvest technology, processing and supply chain research to reduce losses, raise profitability and integrate farmers into value chains.

Support to India's Agricultural Sector

1. Productivity and Profitability: Improved varieties and agronomic practices raise returns, aiding diversification under Rashtriya Krishi Vikas Yojana.
2. Climate Resilience: Research on abiotic stress tolerance supports the National Mission for Sustainable Agriculture and helps vulnerable regions like coastal Andhra Pradesh.

3. Technology Transfer: Collaboration with Krishi Vigyan Kendras and extension agencies takes innovations to farmers and raises adoption.

4. Livelihoods: Market-driven research and value addition help small and marginal farmers shift to commercial agriculture, raising rural incomes and employment.

Way Forward

1. Public-Private Partnerships: Collaborate with agribusinesses and startups to scale technologies.

2. Climate-Smart Research: Varieties suited to Andhra Pradesh's agro-ecological zones.

3. Farmer-Centric Extension: Use digital platforms and KVK networks for real-time dissemination.

4. Value Chain Integration: Processing and marketing infrastructure for better returns.

Conclusion

NIRCA's research makes commercial crops more productive, sustainable and market-oriented, improving farmer incomes in coastal Andhra Pradesh. Integrating it with national missions will advance zero hunger and inclusive growth.

Stuck at "Critically examine"?
See the thinking before you write.
ghissu.com →

1(b) Evaluate the impact of Internet of Things (IoT) technologies on the automation of household systems. Illustrate how everyday tasks have evolved due to IoT innovations.

10

Introduction

The rapid spread of Internet of Things (IoT) technologies has made household automation a key area of change. In Andhra Pradesh, IoT in homes aligns with Digital India and smart city projects. Connected devices now automate routine tasks, promising convenience, efficiency and energy savings as urban lifestyles evolve.

Impact on Household Automation

1. Convenience: Smart thermostats, lighting and security cameras are controlled remotely via smartphones in cities like Visakhapatnam.
2. Resource optimisation: Systems monitor electricity and water use in real time, cutting bills, as seen in Amaravati residential complexes.
3. Voice assistants: Smart speakers enable hands-free chores, helping elderly and differently-abled members.
4. Safety alerts: Sensors detect gas leaks or fire and alert homeowners and emergency services.

Practice 1(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

Household tasks before and with IoT

Before IoT	With IoT
Manual cleaning	Robotic vacuums
Switching devices by hand	Remote smartphone control
Guessing food stock	Smart fridge inventory
Late leak detection	Instant gas, fire alerts

Evolution of Everyday Tasks

1. Cleaning: Robotic vacuums and smart washing machines reduce the time spent on chores.
2. Cooking: Smart refrigerators manage inventory and suggest recipes, cutting food wastage.
3. Health at home: Wearables track vital signs, aiding elderly and chronic care.
4. Leisure: Smart TVs offer personalised content.

Challenges

1. Security: Weak protocols expose homes to hacking.
2. Privacy: Continuous data collection is poorly regulated in India.
3. Digital divide: Rural and poorer households lack access.

4. Dependence: Malfunctions trouble older users.

Way Forward

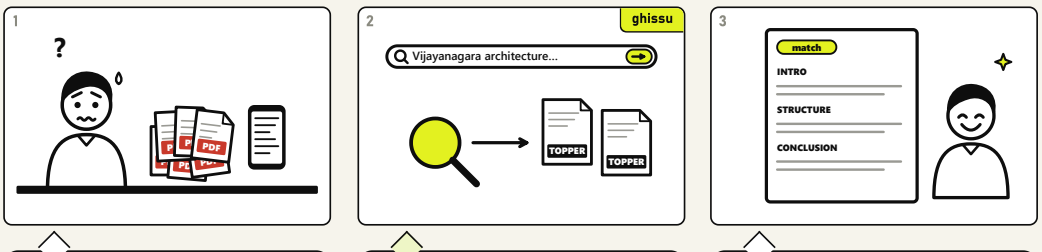
1. Cybersecurity: Enforce cybersecurity standards for IoT devices with awareness campaigns.
2. Access: Expand rural internet penetration and affordability.
3. Education: Promote digital literacy across all age groups.

Conclusion

IoT has transformed household tasks, but security and digital inequality must be tackled for inclusive benefits.

A STORY YOU MAY KNOW TOPPERS' ANSWERS, PER QUESTION

The topper copy hunt



Telegram, YouTube, forty PDFs... nothing on this question.

Type the question into ghissu.

Five topper answers to this exact one!

— Borrow the topper's approach, not the topper's sentences.

Try it on your question →

ghissu.com

14 days free · no card

SCAN

2(a) Discuss the discovery of Raman effect by Dr. C.V. Raman. How did his work impact the global scientific community?

10

Introduction

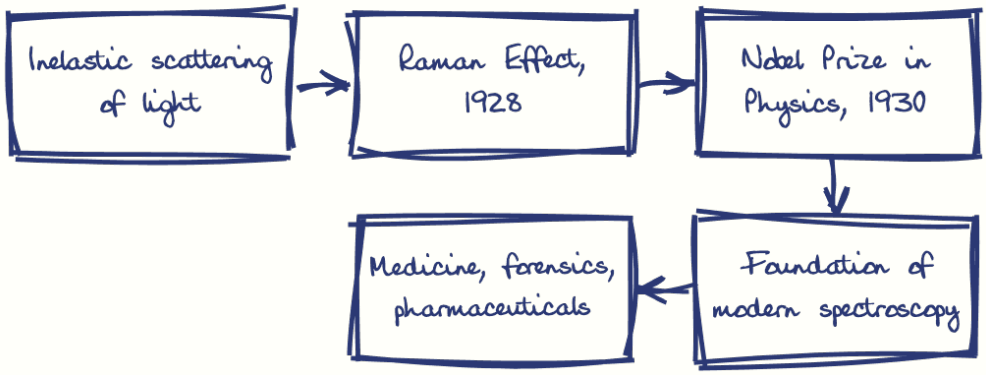
Dr. C.V. Raman's discovery of the Raman Effect in 1928 was a watershed in Indian and global science. At a time when India was under colonial rule, it showed that the country could contribute original and fundamental knowledge to the world, and it still inspires research across India.

The Discovery and its Recognition

1. **Indian Research Environment:** Raman conducted his experiments at the Indian Association for the Cultivation of Science, Kolkata, proving that world-class discoveries could emerge from Indian institutions despite limited resources.
2. **The Effect:** He discovered the inelastic scattering of light, now known as the Raman Effect.
3. **Nobel Prize:** It earned him the Nobel Prize in Physics in 1930, making him the first Asian and Indian to win a Nobel in the sciences.
4. **Rigour over Resources:** Despite limited colonial-era infrastructure and funding, his meticulous experiments showed that intellectual rigour and

creativity could overcome material shortages, inspiring later Indian scientists.

From discovery to global impact



Impact on the Global Scientific Community

1. **Foundation of Modern Spectroscopy:** It revolutionised the understanding of molecular vibrations and material properties, becoming a cornerstone technique in laboratories worldwide and driving nanotechnology and quantum materials research.
2. **Interdisciplinary Research:** Bridging physics and chemistry, it enabled breakthroughs in medical diagnostics and forensic science.
3. **Applied Sciences:** It is used in chemistry, material science and biology, with applications in pharmaceuticals and environmental monitoring.
4. **Scientific Exchange:** His recognition encouraged exchanges with global centres, including efforts to

bring Max Born to India.

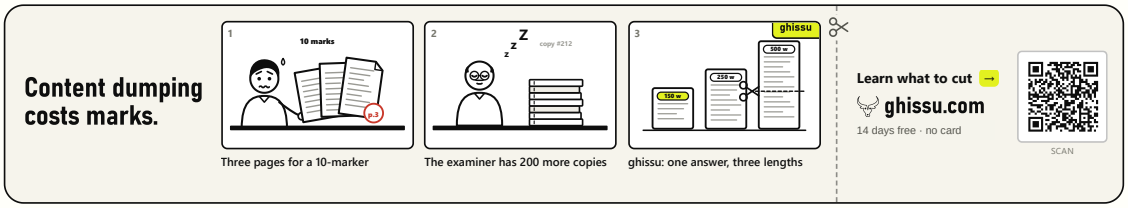
5. Inspiration for Developing Countries: It showed that breakthroughs can emerge outside Western hubs, democratising global science.

Legacy in India

1. Investment in Science: His work encouraged government and academia to create research bodies and funding for indigenous innovation, such as the Raman Research Institute in Bengaluru.
2. National Science Day: Celebrated on 28 February, it promotes public engagement with science and inspires students.
3. Scientific Autonomy: It underlines indigenous capability, in line with Atmanirbhar Bharat.

Conclusion

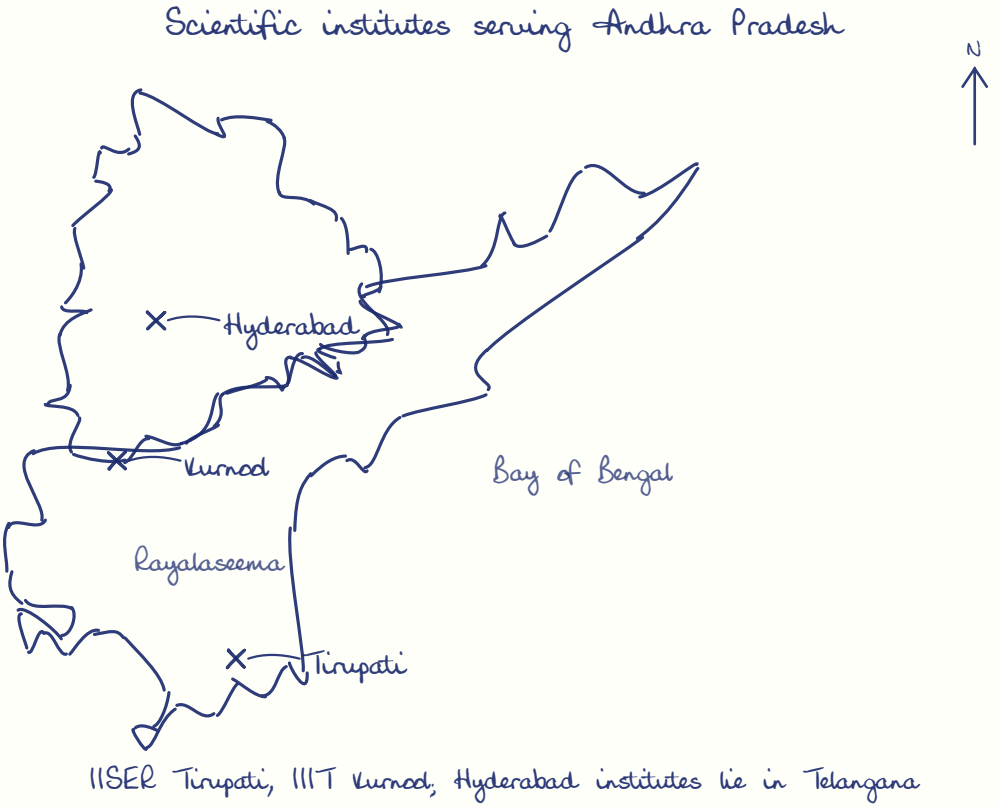
Raman's discovery placed India on the world's scientific map and gave the world a lasting tool, inspiring generations to pursue high-impact research.



2(b) Which are the major scientific institutes located in Andhra Pradesh? Discuss the role of IISER Tirupati in the scientific and educational development of the state.

10

Introduction



Science and technology are pivotal to Andhra Pradesh's vision of sustainable, innovation-led growth. Premier institutes like IISER Tirupati reflect the state's ambition to nurture advanced research and skilled human capital, supporting its higher-than-national-average GSDP growth and per capita income.

Major Scientific Institutes

1. **IISER Tirupati:** One of the seven IISERs in India, integrating teaching with cutting-edge research and offering autonomous Masters and Doctoral degrees in a creative atmosphere.
2. **IIIT Kurnool:** A centrally funded institute for IT education and research, meeting the growing demand for IT professionals.
3. **Hyderabad Institutes Serving AP:** CRIDA, NAARM, IIT Hyderabad and ICRISAT are located in Telangana, but their work aids Rayalaseema's dryland farming, AP's semi-arid zones and engineering research collaborations.

Role of IISER Tirupati

1. **Research Infrastructure:** State-of-the-art laboratories attract talented students and faculty and promote interdisciplinary research, bridging the gap in advanced science education.
2. **Human Capital:** It develops skilled scientists and researchers who contribute to the state's knowledge economy, supporting AP's higher enrolment in PhD and engineering courses than many states.
3. **Innovation Ecosystem:** AP has limited private R&D units and low state-funded R&D. IISER's research output and collaborations with industry

and government aid technology transfer and fill these gaps.

Linking the Two

IISER's infrastructure and programmes produce a skilled scientific workforce, which in turn boosts the state's innovation and economic growth.

Way Forward

1. Expand Funding: More state and central funding for IISER Tirupati and other institutes.
2. Industry Linkages: Partnerships with local industry for innovation-driven entrepreneurship.
3. Science Outreach: Popularisation programmes to draw rural youth into STEM fields.

Conclusion

Led by IISER Tirupati, the state's institutes are building research excellence and skilled manpower. Strengthening them serves Swarna Andhra Vision 2047 and positions Andhra Pradesh as a hub of scientific knowledge and technological advancement.

Correct, but average? One line from your answer

YOUR LINE
~~The 1857 revolt failed.~~
× true · forgettable

→

WITH A VALUE ADDITION
The 1857 revolt failed for want of a centre: no common plan, no modern arms, and the loyalty of the Sikhs, the Gurkhas and most princes to the Company.
✓ cause · evidence · specifics

ILLUSTRATIVE

Add value to your next answer →
ghissu.com
14 days free · no card
SCAN

Write your own answer to 2(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

3(a) Discuss the evolution of e-governance in India, highlighting key initiatives. Analyze how these initiatives have transformed public service delivery and enhanced citizen engagement.

10

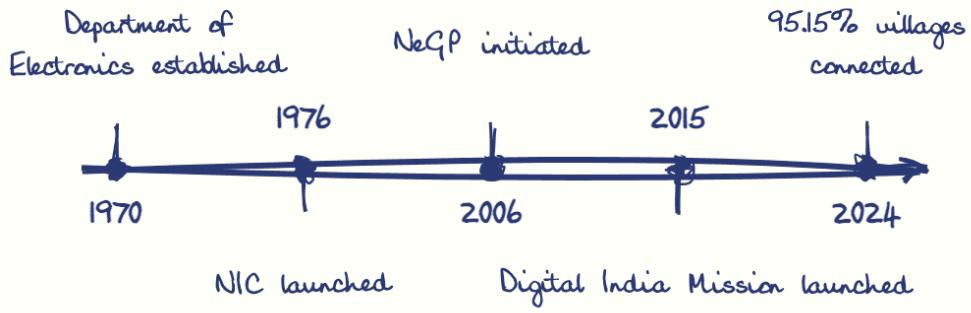
Introduction

E-governance has become central to efficient, citizen-centric governance in the digital age. With over (95%) of villages connected to the internet by April (2024) and Digital India bridging the urban-rural divide, it is vital for transparency, accountability and inclusion in the world's largest democracy.

Evolution of E-Governance

1. **Early Foundations:** The Department of Electronics (1970) and the National Informatics Centre (1976) began by computerising district offices.
2. **Expansion:** The NICNET satellite network (1987) linked government offices, and the National e-Governance Plan (2006) aimed at accessible, transparent services.
3. **Digital India:** The (2015) Mission accelerated infrastructure, services and literacy through DigiLocker and Mobile Seva.

Evolution of E-Governance in India



Transforming Public Service Delivery

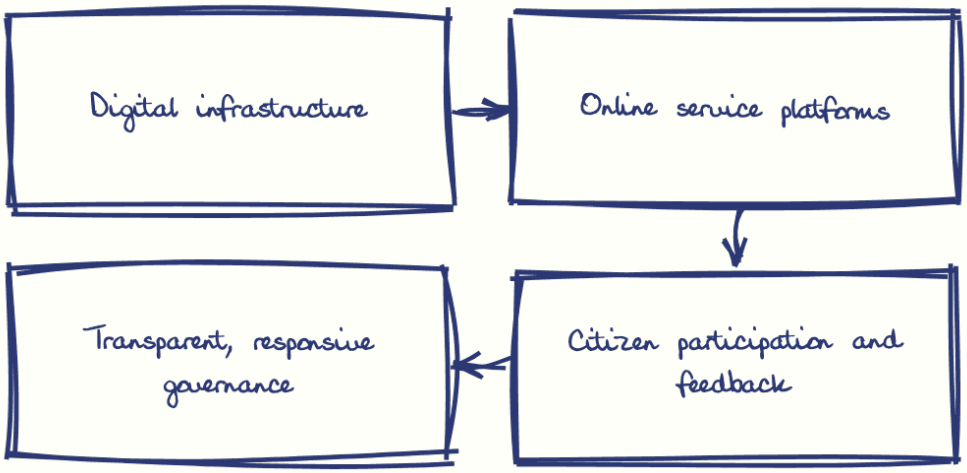
1. **JAM Trinity:** Jan Dhan, Aadhaar and Mobile enable direct benefit transfers, reducing leakages in subsidies.
2. **DigiLocker and e-District:** Digital certificates and electronic land records and revenue services.
3. **AP's e-Seva:** G2C and B2C services that cut processing times.
4. **Rural Governance:** 2.63 crore SVAMITVA property cards and 2.54 lakh panchayat plans on eGramSwaraj.

Enhancing Citizen Engagement

1. **MyGov:** Direct engagement with policymakers.
2. **Grievance Redressal:** RailMadad lets citizens track complaints.
3. **SabhaSaar:** An AI tool launched in August

2025 by the Ministry of Panchayati Raj summarises Gram Panchayat meetings for transparency.

E-governance impact chain



Way Forward

1. Bridge the digital divide: Extend affordable connectivity and digital literacy to women, the elderly and remote villages.
2. Protect data: Implement the Digital Personal Data Protection Act, 2023 effectively to build citizen trust.

Conclusion

E-governance has moved India from computerised offices to citizen-centric governance, cutting corruption and enabling real-time monitoring; digital literacy and vernacular interfaces will deepen it.

Write your own answer to 3(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

3(b) Define cybercrime. Discuss the major types of cybercrimes prevalent in India today.

10

Introduction

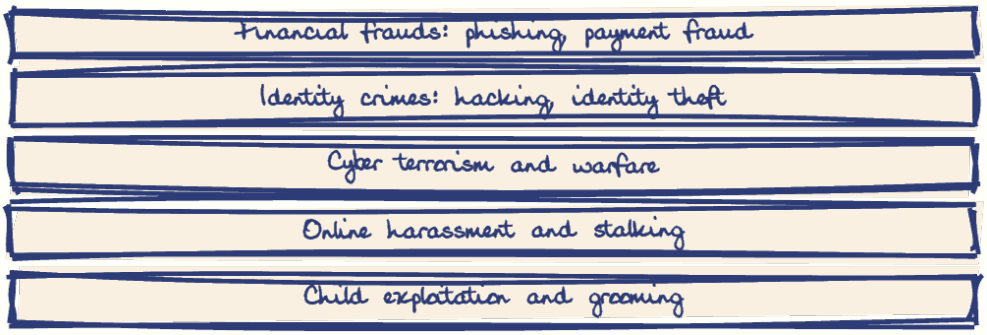
With over 969 million internet users as of March 2025, India's rapid digital transformation brings immense opportunity and serious risk. Cybercrime exploits weaknesses in systems and human behaviour, threatening individuals, businesses and critical infrastructure, and its rising incidence demands urgent attention.

Defining Cybercrime

1. **Meaning:** Cybercrime refers to illegal activities using computers, networks or digital devices that target data, systems or individuals for financial gain, disruption or harm, such as hacking, identity theft and online harassment.
2. **Legal Basis:** The Information Technology Act, 2000 and its amendments define and punish cybercrimes, and the Digital Personal Data Protection Act, 2023 protects data privacy.
3. **Electronic Evidence:** Investigations depend on digital evidence, the National Cyber Crime Reporting Portal and I4C aid reporting and coordination.

4. Rights at Stake: When data is stolen or harassment targets the vulnerable, it threatens privacy, free expression and dignity.

Major Cybercrime Types in India



Major Types in India Today

1. Hacking and Identity Theft: Unauthorised access steals or alters data, and identity theft fuels financial fraud and privacy violations, straining police capacity.
2. Phishing and Financial Fraud: Social engineering tricks users into revealing sensitive details, and the threat has grown with UPI; over 9.42 lakh SIM cards linked to frauds were blocked.
3. Cyber Terrorism: State-sponsored attacks on critical infrastructure and government networks, like the 2020 Mumbai power grid attack, disrupt services and spread misinformation.

4. Cyberstalking: Gendered trolling and harassment of women and activists cause trauma, yet underreporting is high, as cases from Visakhapatnam show.

5. Child Exploitation: 2,188 child pornography cases in 2021, with grooming and cyberbullying rising, prompting CCPWC and awareness campaigns in Andhra Pradesh schools.

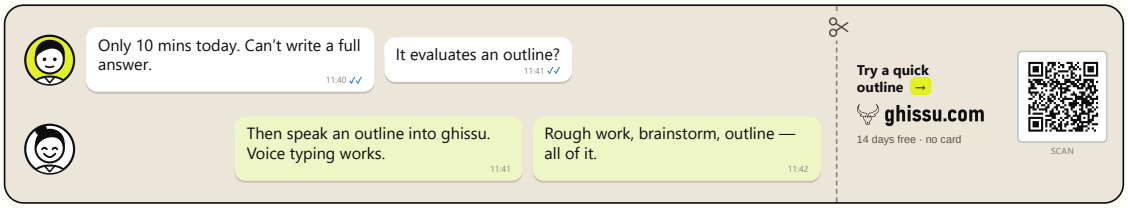
Way Forward

1. Capable cyber units: Train and equip cybercrime units in Andhra Pradesh, using the I4C framework.

2. Digital literacy: Teach women, students and senior citizens safe payments and reporting through the 1930 helpline.

Conclusion

Better-trained cyber units, digital literacy and stronger laws, as the Standing Committee on Home Affairs urged, are needed to keep India's cyberspace safe.



4(a) Discuss the key objectives and features of the National Policy on Information Technology, 2012. How does it aim to transform India into a digitally empowered society?

10

Introduction

The National Policy on Information Technology, 2012 was introduced as India emerged as a global IT hub but still lagged in digital inclusion and infrastructure. It aimed to harness ICT for better social services and to revive electronics manufacturing, laying the foundation for Digital India and the National e-Governance Plan.

Key Objectives

1. ICT for Social Services: Use existing and evolving ICT to improve education, health, skills and welfare for all sections.
2. Electronics Manufacturing: Revive the ESDM sector to reduce imports and build self-reliance.
3. E-Governance: Turn service delivery citizen-centric, transparent and efficient.
4. Digital Inclusion: Universal digital literacy and access, especially in rural areas.

Main Features

1. E-Governance Framework: Support for NeGP's (31)

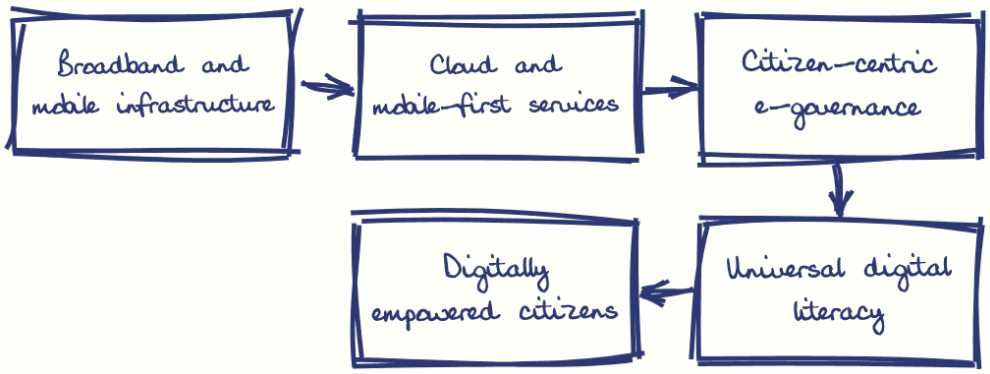
Practice 4(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

Mission Mode Projects across central and state departments.

2. Infrastructure: Broadband highways and mobile connectivity, even in remote regions.
3. Cloud and Mobile-First: Government applications hosted on cloud and designed for mobile.
4. Localisation and Open Standards: Services in many Indian languages, with open-source software and open APIs for interoperability.
5. Cybersecurity: Stronger security infrastructure and capacity.

Path to a digitally empowered society



Transforming India into a Digitally Empowered Society

1. Bridging the Divide: Its connectivity vision was carried forward by BharatNet and the National Broadband Mission, letting rural and remote

citizens join the digital economy.

2. Service Delivery: Platforms like DigiLocker and UMANG simplify access and reduce corruption.
3. Digital Literacy: Campaigns like PMGDISHA and similar state initiatives in Andhra Pradesh empower rural youth and women.
4. Growth and Innovation: ESDM revival and start-ups create jobs and lift India's place in the digital economy.

Way Forward

1. Rural Infrastructure: Faster high-speed internet in underserved areas through funding and public-private partnerships.
2. Digital Literacy: Scale up targeted literacy programmes for marginalised groups.

Conclusion

The 2012 policy linked objectives, features and inclusion into one path from infrastructure to better governance and economic empowerment, the base on which Digital India was built.

STUDY TIP · BEFORE YOU OPEN A CHAPTER
List what was asked from it in the last ten years. Read that first; the rest second.

ghissu
ghissu has already done this for every topic — with the trend drawn out.

See the trend for your chapter →
ghissu.com
14 days free · no card
SCAN

4(b) Discuss the vision and mission of DRDO? Describe the role of DRDO in the contribution to national development.

10

Introduction

In an era where technological superiority shapes security and economic progress, the Defence Research and Development Organisation (DRDO) is pivotal to India's strategic autonomy and innovation. With rising geopolitical challenges and the push for Atmanirbhar Bharat, its contribution now extends well beyond the military.

Vision of DRDO

1. Self-Reliance: To develop state-of-the-art weapons and systems that reduce dependence on imports. The IGMDP, which produced Agni and Prithvi missiles, reflects this vision.
2. Technology Leadership: To stay ahead of evolving threats through indigenous capability.

Mission of DRDO

1. Continuous R&D: Research, design and development across missiles, aeronautics, electronics and life sciences through its network of laboratories.
2. Collaboration: Partnerships with Indian industry and academia to train scientists and develop AI and

Practice 4(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

robotics.

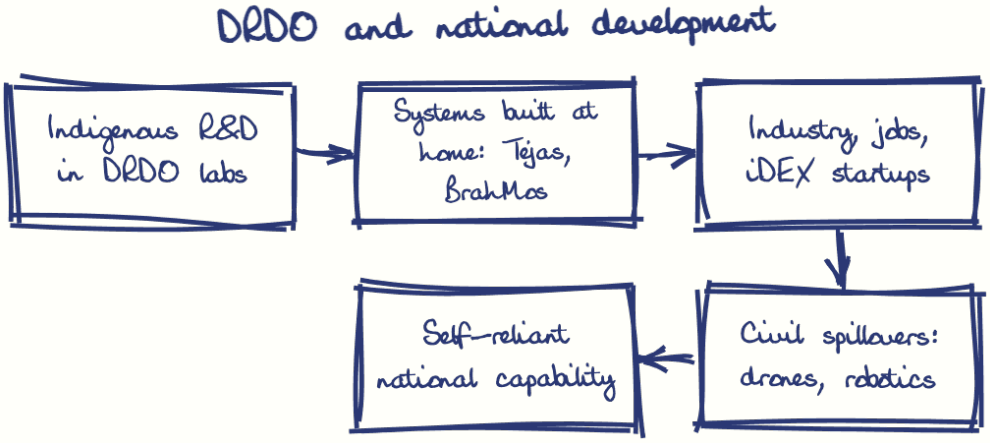
3. Execution Gaps: Complex reviews and limited strategic guidance cause delays, and some critical platforms are still imported, though 80–90% indigenisation is projected within a decade.

Role in National Development

1. Defence Modernisation: LCA Tejas, BrahMos and anti-drone technologies strengthen military capability and deterrence.
2. Industrial Base: Indigenous production creates jobs and innovation, while the Technology Development Fund and iDEX draw private firms and startups into defence manufacturing.
3. Civil Spillovers: Dual-use work on drones and robotics aids healthcare, agriculture, disaster management and the environment.
4. Regional Development: Proximity to research hubs, the Sriharikota launch centre and industrial corridors in Andhra Pradesh supports technology-driven jobs and skills.

Way Forward

1. Strategic Guidance: Clearer civilian direction to prioritise projects.



2. Emerging Technologies: AI, robotics and lithium-ion batteries as mission-mode projects.

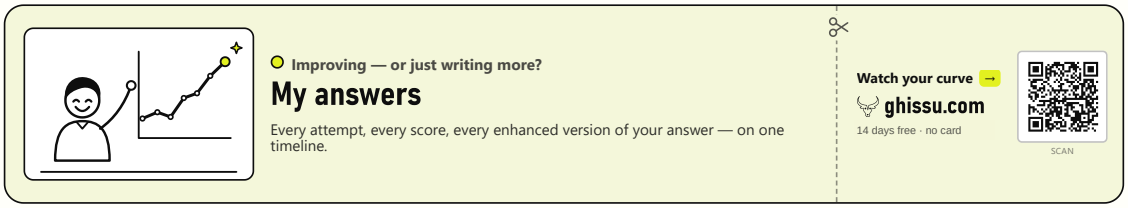
3. Human Capital: Academic partnerships for a skilled workforce.

Significance

Every system built at home instead of imported keeps money, jobs and know-how in India, so defence research also builds national capability.

Conclusion

DRDO embodies India's quest for defence self-reliance, strengthening security while catalysing industrial and national progress.



5(a) Describe the role of the Indian Remote Sensing (IRS) satellite programme. How do these satellites assist in natural resource management?

10

Introduction

The Indian Remote Sensing (IRS) programme, begun in the 1980s, is a cornerstone of India's space-based capability, especially for natural resource management. Amid climate variability and resource scarcity, IRS satellites provide geospatial data for precise monitoring and planning. ISRO's Sriharikota launch facility in Andhra Pradesh has been pivotal in deploying them.

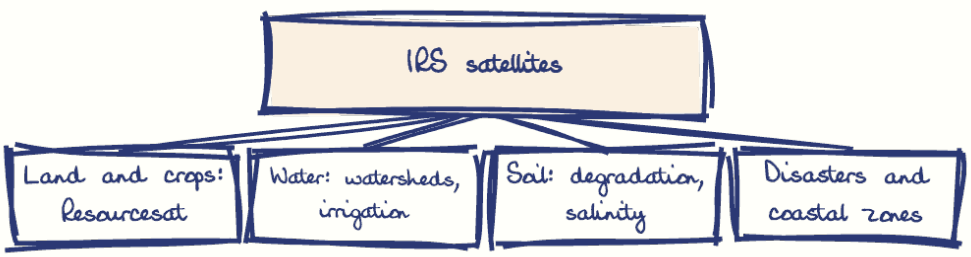
Role of the IRS Programme

1. **ISRO and NRSC:** ISRO operates the programme, with the NRSC as nodal agency for data acquisition, processing and distribution under the Remote Sensing Data Policy (2011).
2. **Policy and Licensing:** The policy licenses private remote sensing satellites and allows open access to imagery up to one-metre resolution while screening finer data, balancing security and development.
3. **National Integration:** The NNRMS, set up by the Planning Commission, coordinates ministries to use satellite data for resource inventories.

How IRS Assists Natural Resource Management

1. **Land and Crop Monitoring:** Resourcesat-1 and 2A give multispectral data at scales from 1:10,000 to 1:250,000 for land use mapping and for identifying kharif, rabi and zaid crop areas and multiple cropping zones, crucial for Andhra Pradesh's horticulture and chilli belts.
2. **Water Resources:** Surface and groundwater inventory, watershed management and irrigation planning support the Jal Shakti Abhiyan and drought mitigation.
3. **Soil and Environment:** Detection of land degradation, erosion, salinity and waterlogging aids sustainable land management in drought-affected Rayalaseema.
4. **Disasters and Coasts:** Early warning, hazard zonation and damage assessment, plus coastal zone management along Andhra Pradesh's 974 km coastline for fisheries and marine ecosystems.

IRS and natural resources



Way Forward

1. Better Data: Higher-resolution sensors with faster turnaround for localised use in diverse agro-climatic zones.
2. State Capacity: Training and infrastructure so state agencies and local bodies in rural and coastal districts can interpret IRS data for planning.
3. Ground Integration: Combining data with field surveys and schemes like Soil Health Card and PMFBY.
4. Partnerships: Private participation in technology and data services to expand applications.

Conclusion

The IRS programme turns space technology into evidence-based planning for land, water, soil and coasts, helping India use its resources sustainably amid climate variability.

STUDY TIP · WHERE TO WRITE A NOTE

Write the note against the question that taught it. That is the only place revision will look for it.

ghissu

On ghissu every question has a notes tab — and your notes gather into chapter notes by themselves.

Open the notes tab →

ghissu.com

14 days free · no card

SCAN

5(b) Provide a comprehensive overview of the primary reasons for deforestation in Andhra Pradesh in the past two decades.

10

Introduction

Deforestation in Andhra Pradesh over the past two decades has become a critical concern for biodiversity, climate resilience and local livelihoods. Rapid economic growth and population pressure have intensified land-use change, making it vital to understand the drivers of forest loss.

Hotspots of forest loss in Andhra Pradesh

Primary Reasons for Deforestation

1. Agricultural Expansion: Forest land has been

converted for cash crops like chillies and horticulture, especially in Rayalaseema and other drought-prone districts, causing fragmentation and soil degradation.

2. Industry and Infrastructure: The Visakhapatnam-Chennai Industrial Corridor and port expansion along the 974 km coastline, along with the Polavaram irrigation project, have diverted large forest tracts.

3. Illegal Logging and Encroachment: Weak enforcement and local livelihood dependence let logging persist, while encroachment for settlements and small farms degrades forests.

4. Mining: Mining of barytes and road metal has expanded into forested zones; the CAG audit report (2021-22) flagged compliance gaps and revenue shortfalls.

5. Urbanisation and Population Pressure: Urban sprawl around Visakhapatnam and Amaravati encroaches on forests, while rural growth raises demand for fuelwood and grazing.

How These Causes Deepen Forest Loss

1. Permanent Diversion: Project land is diverted under the Forest Conservation Act, and compensatory afforestation under APCAMPA rarely matches the

ecological value lost.

2. Governance Gaps: Weak accountability, noted in state audits, lets illegal logging and unsustainable mining continue.
3. Over-Extraction: Demand for fuelwood and non-timber produce degrades what remains, while urban waste contaminates land.

Way Forward

1. Enforcement: Stronger monitoring with community participation.
2. Sustainable Farming: Agroforestry and AP Community Managed Natural Farming to ease pressure on forests.
3. Strict EIA: Rigorous compliance for industry, mining and infrastructure, with transparent consultation.
4. Green Urban Planning: Green belts that protect existing forest patches.

Conclusion

Addressing these drivers together can reconcile Andhra Pradesh's growth with ecological balance.

Wrote the answer. Now what?
Upload it — every line marked, in minutes.
ghissu.com →

6(a) Investigate the relationship between mining activities and the occurrence of earthquakes. Explain how mining operations can be regulated to minimize seismic risks.

10

Introduction

India recorded (159) earthquakes from November (2024) to February (2025), which makes anthropogenic triggers such as mining an urgent concern. Understanding this link is vital to protect communities and infrastructure while pursuing development.

Relationship between Mining and Earthquakes

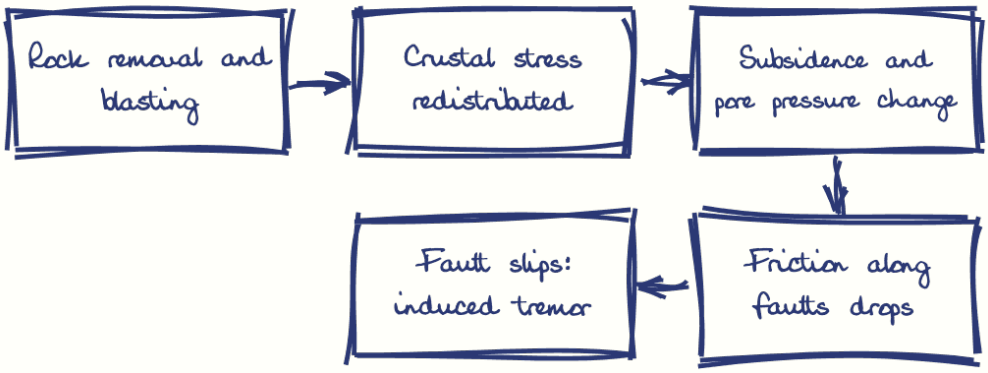
1. Evidence: Deep underground extraction and blasting have been correlated with more seismic events because they disturb geological stress fields, as in intensive coal mining regions.
2. Induced seismicity: The (1967) Koyna earthquake in Maharashtra, triggered by reservoir filling, shows how human interventions can induce major quakes.
3. Mining clusters: Many leases operating together cumulatively alter subsurface stress, so their impacts must be assessed collectively, not in isolation.
4. Andhra Pradesh: Mining in the Rayalaseema belt

has raised concerns of localised tremors, calling for state-specific monitoring.

Mechanisms

1. **Stress redistribution:** Removing large volumes of rock redistributes crustal stress and can trigger fault slippage.
2. **Blasting:** Shock waves from controlled explosions can destabilise nearby faults and induce quakes of varying magnitude.
3. **Subsidence:** Collapse of mined-out areas can reactivate dormant faults.
4. **Water table change:** Mining alters groundwater levels, and changed pore pressure reduces friction along faults.

How mining can trigger earthquakes



Regulating Mining to Minimise Seismic Risk

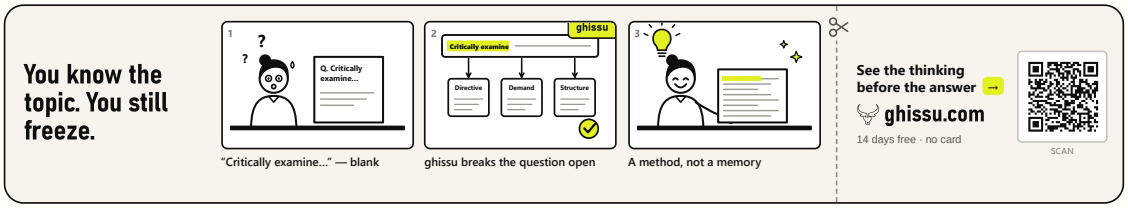
1. **Cluster-based EIA:** The EIA Amendment (2016)

assesses mining clusters together and lets regulators cap leases in sensitive zones.

2. Licensing controls: The MMDR Act limits lease size and number; AP's Directorate of Mines and Geology enforces this.
3. Curbing illegal mining: Simpler, cheaper legal procedures discourage illegal mining that ignores safety norms.
4. Capacity building: The DGMS-SIMTARS MoU promotes risk-based safety management.
5. Monitoring and alternatives: Seismic observatories in mining zones and use of M-sand to reduce extraction.

Conclusion

Mining can induce earthquakes by redistributing crustal stress, blasting and altering groundwater. Cluster-based assessment, real-time monitoring and strict enforcement can minimise this risk, especially in seismically sensitive regions like Andhra Pradesh.



6(b) Define sustainable development and critically evaluate its relevance in the context of modern industrialization.

10

Introduction

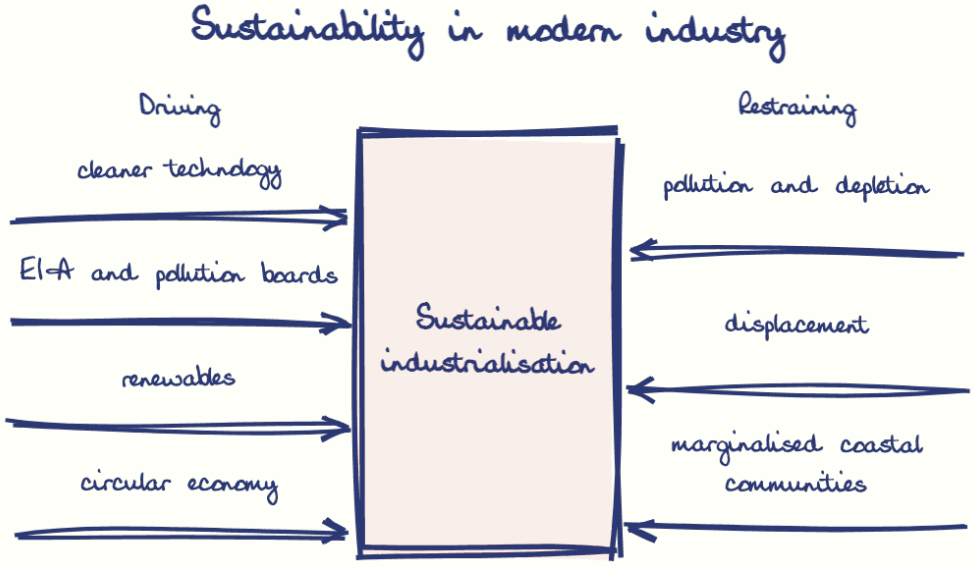
With climate change and environmental degradation threatening economic and social progress, sustainable development has become a critical framework for balancing growth with ecology. Rapid industrialisation, as in Andhra Pradesh's port-led industrial corridors, makes this urgent so that present gains do not cost future generations.

Defining Sustainable Development

1. Brundtland definition: Development that meets present needs without compromising future generations' ability to meet theirs, stressing intergenerational equity (1987).
2. Three pillars: It harmonises economic growth, social equity and environmental protection.
3. Needs and limits: It prioritises the essential needs of the poor and respects the environment's capacity.
4. Quality of life: It covers living standards, health, education and cultural well-being, so development is inclusive and ecologically viable.

Relevance to Modern Industrialisation

1. Environmental costs: Visakhapatnam's industrial hubs show pollution, resource depletion and biodiversity loss that need mitigation.
2. Social equity: Farming and fishing communities in coastal AP risk marginalisation.
3. Economic resilience: Avoiding overexploitation sustains production and jobs in a state that relies on agriculture and aquaculture alongside industry.
4. National goals: The SDGs, Panchamrit pledges and net zero by 2070 rest on this framework.



Critical Evaluation

1. Tension: Industry drives prosperity but strains ecosystems; cleaner technology resolves part of this.
2. Governance: Pollution control boards and EIA must

embed sustainability in industrial planning.

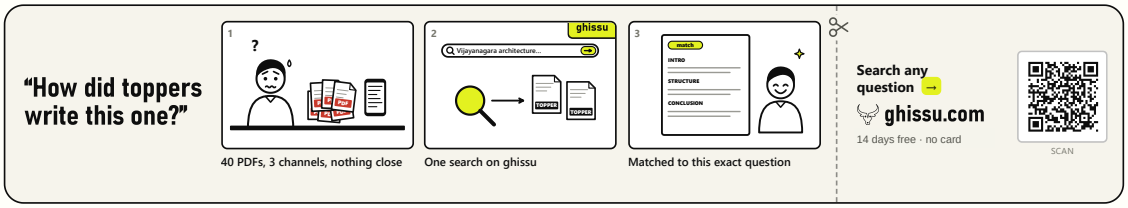
3. Green industry: Renewables and the circular economy show growth and sustainability can align.
4. Scale and equity: Displacement around Polavaram shows inequality deepens when sustainability is ignored.

Way Forward

1. Governance: Sustainability criteria in industrial approvals in AP's corridors.
2. Green technology: Subsidies and public–private partnerships for renewables and pollution control.
3. Inclusive planning: Social impact assessment and community participation in regions like Rayalaseema.

Conclusion

Sustainable development remains deeply relevant to modern industrialisation. It is not anti-growth; it offers a path to balance industrial expansion with environmental preservation and social equity, provided governance and green technology give it substance.



7(a) Describe the relationship between deforestation and the emergence of infectious diseases.

10

Introduction

Accelerating forest loss has raised concern about its effect on human health. Deforestation disrupts ecosystems, increases human–wildlife interaction and enables zoonotic spillover, which lies behind most recent pandemics. India, including biodiverse Andhra Pradesh with its forest-fringe communities, faces this growing risk.

How Deforestation Contributes to Disease

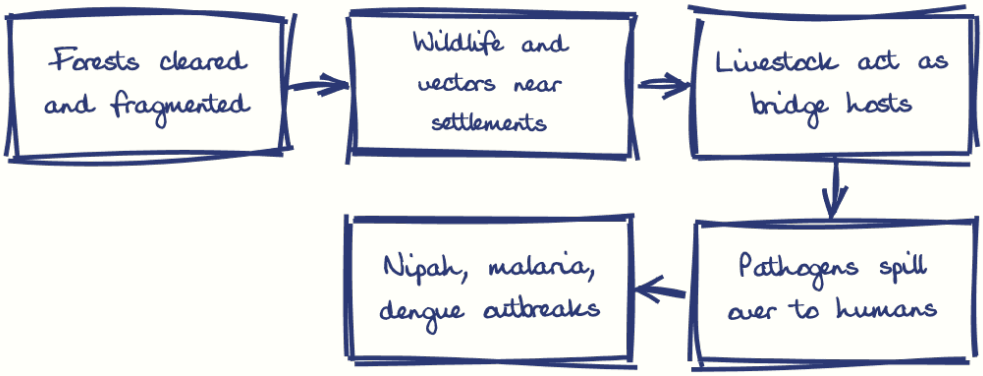
1. **Habitat Disruption:** Fragmented habitats push animals near settlements, as with encroachment in Andhra Pradesh's Eastern Ghats.
2. **Biodiversity Loss:** Diverse ecosystems regulate disease through balanced predator–prey relations; when deforestation destroys them, rodents, common pathogen reservoirs, multiply unchecked and raise transmission risks.
3. **Agriculture at Forest Edges:** Clearing for crops and livestock creates interfaces where pathogens jump from wildlife to domestic animals and then to humans, as land-use change in Rayalaseema shows.

Practice 7(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

4. Microbial Change: Forest destruction disrupts the microbial diversity essential for ecosystem health, letting pathogens thrive and spread to humans.

From forest loss to outbreaks



Mechanisms of Outbreaks

1. Vector Expansion: Altered habitats let mosquitoes and ticks expand their range into new areas, raising malaria and dengue, as seen in forest-fringe districts of Andhra Pradesh.
2. Defaunation: Loss of large animals leaves ecological voids filled by small mammals like rodents, efficient reservoirs that aid hantavirus and leptospirosis.
3. Encroachment and Hunting: Bushmeat hunting increases direct contact with wildlife.
4. Livestock Bridges: Domestic animals near degraded forests act as intermediate hosts, significant in rural Andhra Pradesh where

mixed farming is common.

Examples

1. Nipah in Malaysia: Deforestation pushed fruit bats near human habitats, spreading Nipah virus to livestock and then to people.
2. Ebola and HIV: Ebola in West Africa and HIV in Central Africa have been traced to human intrusion into remote forests.
3. Andhra Pradesh: Dengue and malaria have risen in forest-bordering districts, and leptospirosis and scrub typhus in degraded parts of Rayalaseema.

Way Forward

1. Forest Protection: Enforce forest laws with community forest resource management.
2. Disease Surveillance: Monitor health at forest interfaces to catch outbreaks early.

Conclusion

Deforestation turns forests from buffers into sources of disease. Sustainable land-use planning in forest-fringe zones is equally needed, since forest health and human health are linked.

Write your own answer to 7(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

7(b) What is Environmental Impact Assessment (EIA)? Describe its key stages.

10

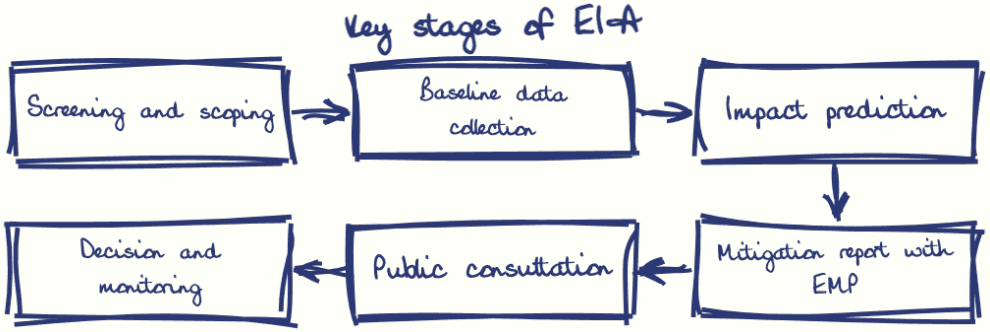
Introduction

Environmental Impact Assessment (EIA) is a tool that foresees and mitigates the potential environmental consequences of a project before it begins, ensuring sustainable development. In Andhra Pradesh, where Polavaram and port-led industrial corridors are underway, EIA helps harmonise development with ecology.

Legal and Institutional Basis

1. **Legal Foundation:** EIA is mandated under the Environment (Protection) Act, 1986, with the first EIA Notification issued in 1994.
2. **Authorities:** MoEF&CC oversees EIA nationally, while State Environmental Impact Assessment Authorities implement and monitor compliance at the regional level.

Key Stages of EIA



1. Screening: Decides whether a project needs EIA based on its size, type and location.
2. Scoping: Identifies key issues and sets the terms of reference for the study.
3. Baseline Data Collection: Records air, water, soil, biodiversity and socio-economic conditions as the reference point.
4. Impact Prediction: Forecasts direct, indirect, reversible and irreversible effects.
5. Mitigation and Reporting: The report sets out ways to avoid or compensate harm, with an Environment Management Plan (EMP).
6. Public Consultation: Hearings bring in local knowledge, as in Visakhapatnam port expansion.
7. Decision Making: The authority approves, rejects or seeks modifications.
8. Post-Clearance Monitoring: Ensures compliance, critical for projects like Polavaram.

Challenges

1. Weak Participation: Language barriers and short notice limit local inputs in Andhra Pradesh.
2. Conflict of Interest: Reliance on private consultants leads to biased reports, as in corridors near

Visakhapatnam.

3. Weak Monitoring: Enforcement gaps let violations persist, as in some aquaculture projects along AP's coastline.

4. Exemptions and Dilutions: The draft EIA Notification 2020 cut public hearing periods and allowed post-facto clearance, raising fears that safeguards are being compromised.

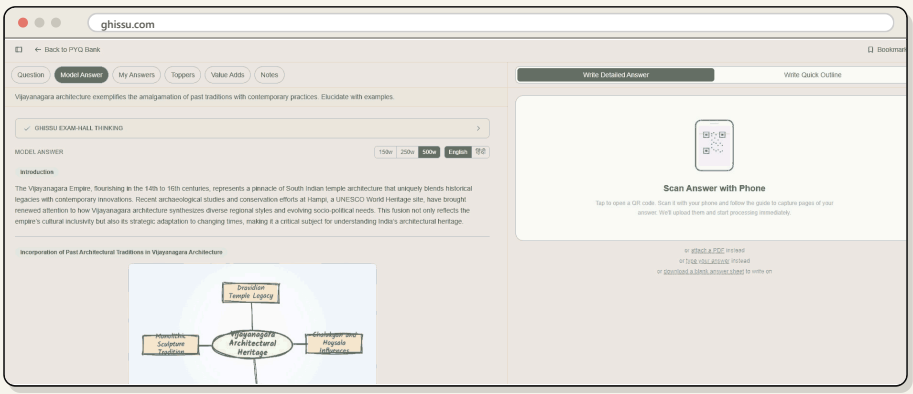
Conclusion

A robust EIA with genuine public participation, credible studies and strict monitoring balances growth with environmental sustainability.

THE REAL PRODUCT · NOT A MOCK-UP

MODEL ANSWERS IN 150 · 250 · 500

Writing 400 words for a 150-word question?



1 One question, three lengths: 150 · 250 · 500 words.

2 Two languages at a tap: English and हिंदी.

3 Keywords, structure and diagrams the examiner rewards.

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

8(a) Explain the different types of natural resources in India. How are they being conserved through government initiatives?

10

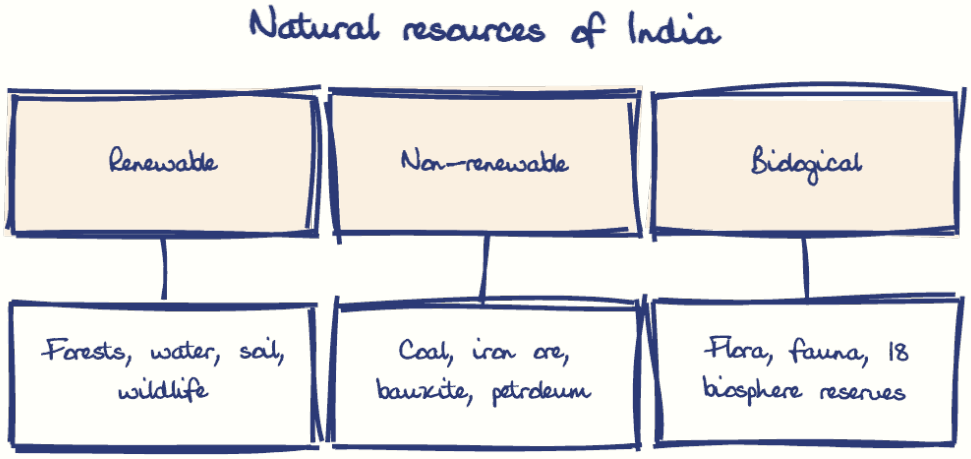
Introduction

India's vast and diverse natural resources form the backbone of its economy and ecological stability. Rising population pressure, industrialisation and climate change have made their sustainable conservation urgent, both for future generations and for socio-economic development.

Types of Natural Resources in India

1. Renewable resources: Forests, water, soil and wildlife; forests cover about 24% of the area, and social forestry is improving green cover in districts like Visakhapatnam.
2. Non-renewable resources: Coal, iron ore, bauxite and petroleum are finite; Andhra Pradesh holds 86.11% of India's barytes and 18.49% of its road metal reserves.
3. Water resources: Average rainfall is about 1100 mm, but little is stored; groundwater meets nearly 62% of irrigation needs and 85% of rural water use.
4. Biological resources: Rich flora and fauna are protected in national parks, sanctuaries and 18

biosphere reserves.



Conservation through Government Initiatives

1. **Forests and biodiversity:** Forest policy aims at one-third forest cover; the Green India Mission under the NAPCC drives afforestation, and AP runs Joint Forest Management.
2. **Water:** Jal Shakti Abhiyan, Atal Bhujal Yojana and Mission Amrit Sarovar promote rainwater harvesting and recharge; Polavaram develops river water but faces rehabilitation challenges.
3. **Minerals:** The MMDR Act regulates extraction, though CAG reports show compliance gaps in AP.
4. **Communities and law:** The Forest Rights Act empowers tribals, and Articles 48-A and 51-A(g) oblige state and citizens to protect

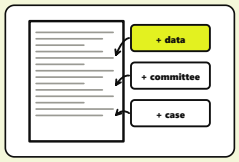
nature.

Way Forward

1. Community conservation: Build capacity of panchayats to manage forests and water.
2. Monitoring: Use satellite remote sensing for real-time tracking of extraction and environmental impact.
3. Resource efficiency: Build efficient practices into agriculture, industry and urban planning.

Conclusion

Tailored, community-led conservation backed by strong monitoring can secure India's resources for future generations.



No data, no committee, no case?

Value additions

For every question: the number, the report, the example — from standard and government sources, linked.

Make your answer memorable →

ghissu.com

14 days free · no card

SCAN

8(b) What are the key principles of Integrated Solid Waste Management (ISWM)? Discuss the role of waste hierarchy in ISWM.

10

Introduction

Rapid urbanisation has sharply raised municipal solid waste in India. Andhra Pradesh's 123 ULBs generated about 6,890 tons per day in 2022, with legacy waste reaching 85.89 lakh metric tons. Integrated Solid Waste Management (ISWM) combines technical, institutional and social strategies, with the waste hierarchy as its guiding principle.

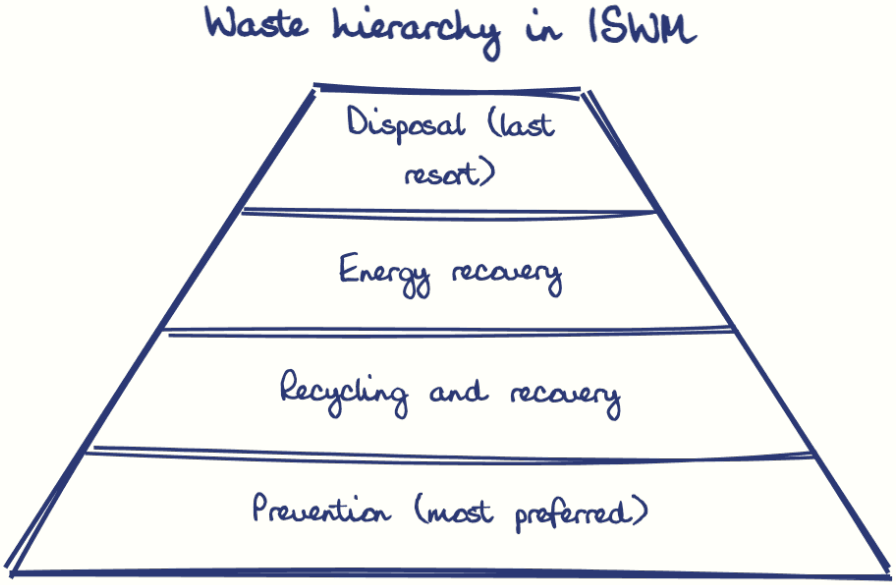
Key Principles of ISWM

1. **Multi-tier governance:** Central, state and local bodies must coordinate, with ULBs pivotal; yet only 5 of 17 test-checked ULBs had framed bye-laws by 2022.
2. **Policy and regulation:** The SWM Rules, 2016 and Swachh Bharat Mission (Urban) mandate segregation and promote Extended Producer Responsibility to involve producers.
3. **Capacity and inclusion:** Training staff and integrating informal waste pickers, who collect over 10,000 tonnes of reusable waste daily, are central.

4. Data and monitoring: Actual segregation was near (75%) against a claimed (98.17%), showing the need for accountability.

5. Segregation and infrastructure: Source segregation is critical, but (82) of (123) ULBs lacked wet waste plants as of (2022).

Role of the Waste Hierarchy



1. Prevention at the top: Waste minimisation and reuse through conscious consumption and product design cut impact and cost.

2. Recycling and recovery: Recovering materials reduces the need for virgin resources, aided by the informal sector.

3. Energy recovery: Waste-to-energy is a

transitional step that reduces landfill volume and yields renewable energy, if pollution is managed.

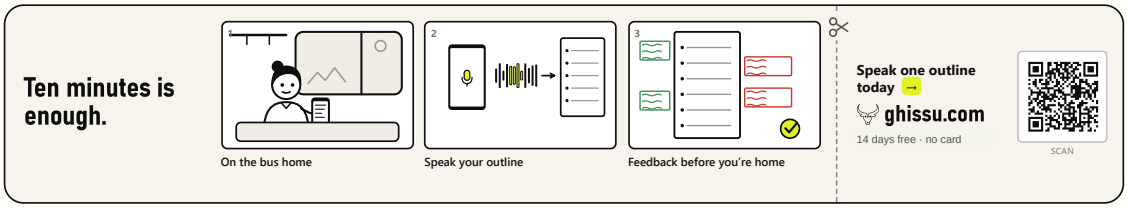
4. Disposal as last resort: Only inert residue goes to sanitary landfills, with just (4) ULBs having working sites by 2022, open dumps persist.

Challenges

1. Financial and technical gaps: Limited funds and expertise hinder advanced processing, though private sector participation is encouraged to bring efficiency and innovation.
2. Behavioural barriers: Public apathy and the NIMBY syndrome obstruct facility siting and community participation, needing behaviour change campaigns.

Conclusion

Empowering ULBs with technical, financial and legal tools, sustained awareness and incentives for segregation will let the waste hierarchy drive ISWM in practice.



Write your own answer to 8(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

Scan to practice

9(a) Analyze the potential of plant-derived biopharmaceuticals in promoting sustainable healthcare. Explain how the application of green pharmacy principles can help minimize environmental harm at each stage of the pharmaceutical lifecycle.

10

Introduction

Growing demand for healthcare in India, along with environmental concerns, calls for sustainable solutions. Plant-derived biopharmaceuticals can cut the ecological footprint of medicine without losing efficacy, while green pharmacy principles limit harm across the pharmaceutical lifecycle.

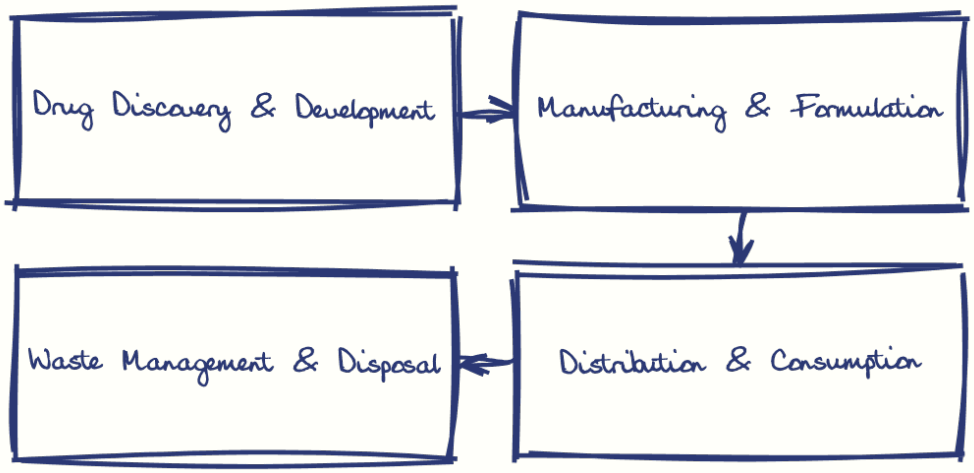
Potential of Plant-Derived Biopharmaceuticals

1. Cost-Effective Production: Molecular farming uses plants as natural bioreactors for therapeutic proteins and vaccines, reducing reliance on costly fermentation and aiding affordable care in states like Andhra Pradesh.
2. Reduced Footprint: It needs less energy and water and creates fewer hazardous wastes than traditional manufacture.
3. Biodiversity and Livelihoods: India's roughly 7,000 to 8,000 medicinal plant species offer a vast base, sustainable cultivation in AP supports local economies and habitats.
4. Innovation: Biopharma SHAKTI and the National

BioPharma Mission aim to raise India's global market share.

Green Pharmacy Across the Lifecycle

Green Pharmacy in Pharmaceutical Lifecycle



1. Discovery and Development: Renewable plant resources and biotechnological methods reduce chemical synthesis and toxic solvents from the outset.
2. Manufacturing: Energy-efficient processes, water recycling and waste reduction lower emissions; the Bulk Drug Parks scheme in AP promotes cleaner production.
3. Distribution and Use: Minimal packaging and rational drug use reduce contamination, and telemedicine in AP cuts patient travel emissions.
4. Disposal: Segregation and treatment under the Biomedical Waste Management Rules, 2016 prevent

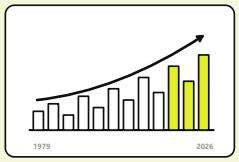
soil and water pollution.

Way Forward

1. Medicinal Plant Cultivation: Promote Good Agricultural Practices and sustainable harvesting of medicinal plants in AP.
2. Green Manufacturing: Add renewable energy and waste minimisation to Bulk Drug Parks.
3. Standards and Training: Train providers in green pharmacy principles.

Conclusion

Plant-derived biopharmaceuticals and green pharmacy together make healthcare affordable and eco-friendly, so growth does not come at the planet's expense, advancing SDG (3) on good health.



Which sub-topic will they ask next?

Trend analysis

Every syllabus topic: the direction of travel, the most tested sub-topic, the probable next theme.

Prepare the pattern →

ghissu.com

14 days free · no card

SCAN

9(b) Differentiate between conventional and non-conventional energy resources. Explain their relevance in India's energy sector.

10

Introduction

India's energy sector is in transition, balancing growing demand with sustainability and energy security. Per capita emissions are low but total emissions are rising, so the shift from conventional to non-conventional energy is vital for climate commitments and growth.

Conventional vs Non-Conventional

Conventional vs non-conventional energy

Conventional	Non-conventional
Coal, oil, gas, large hydro	Solar, wind, biomass, geothermal
Finite and depletable	Renewable and abundant
High emissions	Low or zero emissions
Centralised supply	Decentralised, reaches remote areas

1. Conventional: Coal, petroleum, natural gas and large hydropower, which have historically dominated India's mix.
2. Non-Conventional: Solar, wind, biomass, small

hydro, geothermal and fuel cells with minimal footprint.

3. Sustainability: Conventional sources are finite and depletable and emit heavily; non-conventional ones are renewable and clean.

4. Supply: Non-conventional energy can be decentralised, reaching remote and rural areas.

Relevance in India's Energy Sector

1. Conventional Backbone: Coal gives about 69% of commercial electricity; petroleum drives transport and industry. Andhra Pradesh, too, relies heavily on coal-based thermal power plants.

2. Its Limits: Finite reserves, pollution and import dependence threaten energy security.

3. Rising Renewables: Solar rose from 3 GW in 2014 to over 140 GW by early 2026; wind reached 54.65 GW. The PM Surya Ghar mission links rural Andhra Pradesh homes to solar power.

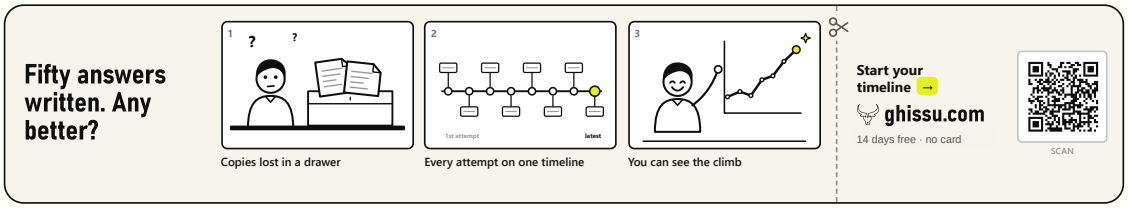
4. Policy and Economy: Renewables support Paris commitments, jobs and access in rural and tribal areas, backed by the PAT scheme.

Way Forward

1. Coastal and Rayalaseema Projects: Speed up solar and wind in Andhra Pradesh's best-endowed regions.
2. Decentralised Solutions: Off-grid systems for remote areas.
3. R&D: Indigenous storage and fuel cell technology.
4. Policy Alignment: Link state policy with the National Green Hydrogen Mission.

Conclusion

Conventional energy meets India's immediate needs, but non-conventional energy is essential for sustainable development, energy security and climate resilience.



10(a) Critically assess the effectiveness of major disaster management initiatives undertaken in India since the enactment of the Disaster Management Act, 2005.

10

Introduction

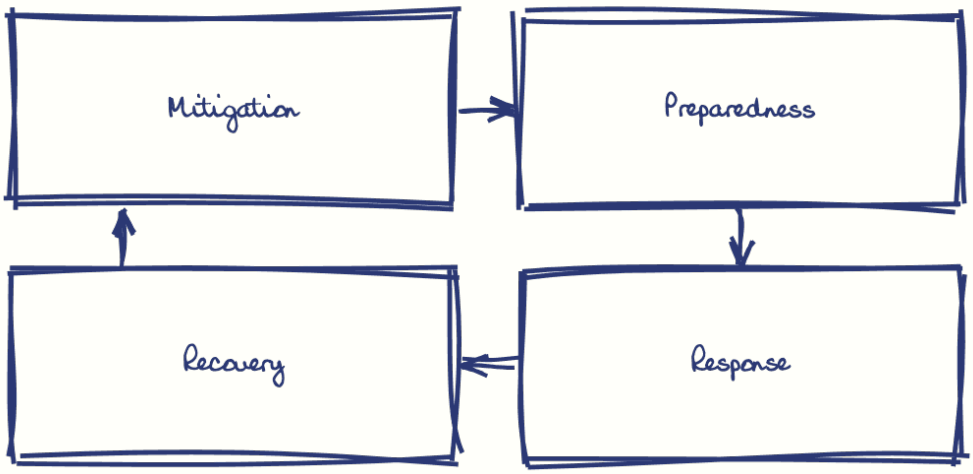
Frequent cyclones and floods made a robust framework urgent. Since the Disaster Management Act, 2005, India has sought to move from reactive relief to proactive risk reduction, critical for states like Andhra Pradesh with cyclone and flood risks along its coast.

Major Initiatives Since 2005

1. **Institutional Framework:** The Act created NDMA, SDMAs and DDMAs to coordinate preparedness and response at every level.
2. **National Policy (2009):** It gives a roadmap for mainstreaming resilience into development and community participation.
3. **Response and Technology:** The NDRF grew from 8 to 16 battalions, and seismic observatories rose from 80 in 2014 to 168 by 2025.
4. **Financial Mechanism:** The State Disaster Response Fund provides timely relief money to states, including Andhra Pradesh.

Assessment of Effectiveness

Disaster management cycle since 2005



1. Preparedness: NDMA's Earthquake Risk Indexing and digital warnings improved readiness, timely cyclone warnings cut casualties in AP's coastal districts.
2. Mitigation: Risk reduction is mandated in projects, but the Polavaram audit revealed delays and rehabilitation problems.
3. Response: NDMA, NDRF and states run better rescue operations, and the Act enabled COVID-19 lockdowns and resource mobilisation, though overlapping roles hurt coordination.
4. Recovery: SDRF helps, but AP audits show administrative delays in welfare and rehabilitation.

Strengths and Weaknesses

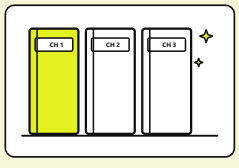
1. Strengths: Clear roles, better inter-agency coordination and a culture of preparedness.
2. Weaknesses: Centralisation limits state autonomy and jurisdictions overlap.
3. Equity Concerns: Vulnerable communities stay under-protected, and weak community engagement limits scalability.

Way Forward

1. Decentralise: Give state and local bodies, like those in AP, more autonomy and resources.
2. Clarify Roles: Reduce overlap between NDMA, NDRF and states.
3. Climate Integration: Mainstream adaptation, especially for AP's coastal and drought-prone areas.

Conclusion

Since 2005, India has built strong institutions and response capacity, but mitigation and recovery lag. Closing these gaps will complete the shift from relief to resilience.



Notes in five notebooks, none at hand?

Notes that live with the question

Note from toppers' answers, exam-hall thinking and model answers — your notes gather into topic and chapter notes on their own.

Build notes that find you →

ghissu.com

14 days free · no card

SCAN

10(b) Discuss the legislative framework for controlling air and water pollution in India. Highlight the key provisions and limitations of relevant acts.

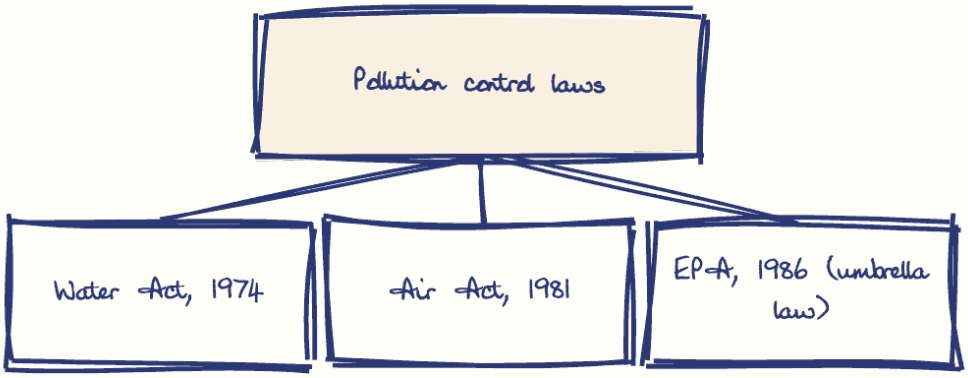
10

Introduction

India faces a severe crisis of air and water pollution, with over 80% of the population exposed to unsafe air and many rivers critically polluted. Laws enacted since the 1970s have built a framework, but implementation gaps persist, especially in industrialising regions like Visakhapatnam.

Legislative Framework and Key Provisions

Pollution control laws in India



1. **Pollution Control Boards:** The Water Act, 1974 and the Air Act, 1981 established the CPCB and SPCBs, which set standards, grant consents to industries and enforce compliance.
2. **Two-Stage Consent:** Industries and local bodies

must obtain consent to establish and consent to operate from SPCBs, so that controls exist before operations begin.

3. Environment Protection Act, 1986: Enacted after the Bhopal disaster, this umbrella legislation empowers the Centre to set standards and coordinate action, filling gaps in the Water and Air Acts.

4. Recent Amendments: The Water Amendment Act, 2024 and changes under the Jan Vishwas Act, 2023 replaced many criminal sanctions with monetary penalties and gave adjudication powers to designated officers.

Limitations

1. Capacity Constraints: SPCBs suffer understaffing, lack of technical expertise and funds; in Andhra Pradesh this shows in poor compliance by urban local bodies.

2. Compliance Gaps: Many industries operate without consent or exceed limits; CPCB's 2023 data shows persistent non-compliance among Grossly Polluting Industries.

3. Reduced Deterrence: Replacing imprisonment with fines may embolden violations where monitoring is weak.

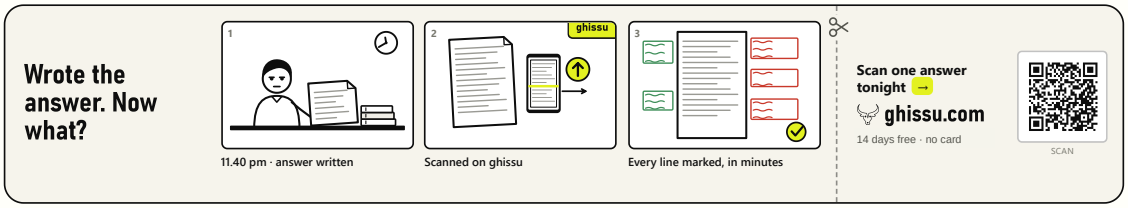
4. Fragmented Jurisdiction: States' competence over water leads to uneven adoption of amendments, and inter-state basins like the Krishna and Godavari, which affect Andhra Pradesh, lack coordinated action.

Way Forward

1. Strengthen SPCBs: Invest in training, staff and finances, focusing on industrial centres.
2. Transparency: Promote real-time pollution data and community engagement.
3. Harmonise Laws: Encourage uniform adoption of amendments and inter-state river basin mechanisms.
4. Balanced Reform: Keep strong penalties so ease of business does not dilute standards.

Conclusion

The framework has built vital institutions but weak enforcement undermines it. Stronger capacity and public accountability will secure the right to a healthy environment under Article 21.



11(a) Define global warming, and differentiate it clearly from climate change. Explain the key natural and human-induced causes of global warming and briefly discuss which factors are currently dominant in driving global temperature rise.

10

Introduction

Rising temperatures and more frequent extreme weather make it vital to understand global warming and climate change. In Andhra Pradesh, worsening droughts in Rayalaseema and rising sea levels along its 974 km coastline hurt agriculture and coastal communities.

Defining Global Warming

1. Meaning: Global warming is the long-term rise in Earth's average surface temperature, mainly due to greenhouse gases like CO_2, methane and nitrous oxide that trap heat.
2. Evidence: CO_2 has risen from about 280 ppm before the Industrial Revolution to over 400 ppm, as Mauna Loa Observatory data show, alongside warming of about 1°C since 1900.

Global Warming vs Climate Change

Climate change covers erratic monsoons and more cyclones, as Andhra Pradesh sees, which follow from warming but go beyond temperature.

Practice 11(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

Global Warming vs Climate Change

Global Warming	Climate Change
Temperature increase	Long-term shifts in weather patterns
Primarily caused by GHGs	Natural and anthropogenic causes
Focus on atmospheric warming	Rainfall, storms and sea levels
A subset of climate change	Includes warming and broader effects

Causes of Global Warming

1. Natural causes: Volcanic eruptions, solar variations and ocean cycles like El Niño cause short-term swings but not the sustained warming since the 19th century.
2. Human-induced causes: Fossil fuel combustion, industry and deforestation emit large amounts of GHGs; coal power and vehicles in Visakhapatnam are local examples.
3. Land use changes: Deforestation in forested districts reduces carbon sequestration and worsens warming, while afforestation and sustainable farming can cut emissions.

Dominant Factors Today

The overwhelming driver is human-induced

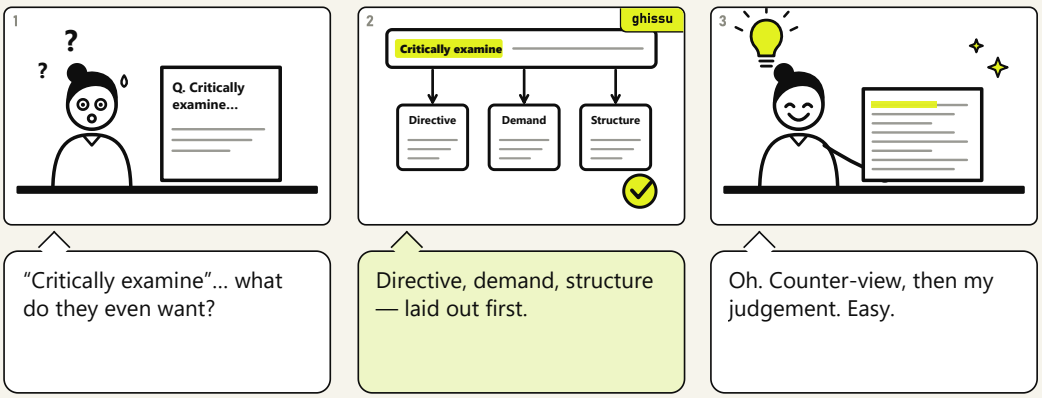
GHG emissions, chiefly CO_2 from energy and land-use change, as IPCC assessments confirm; natural causes play a minor role.

Conclusion

Clear definitions help target climate action. Expanding solar and wind under the National Action Plan on Climate Change, afforestation, climate-resilient agriculture for Rayalaseema and early warning for heatwaves and cyclones can address the dominant human drivers in Andhra Pradesh.

A STORY YOU MAY KNOW EXAM-HALL THINKING

Three minutes staring at one word



— Walk into the hall with a method, not a memory.

Walk in with a method →
ghissu.com
14 days free · no card
SCAN

11(b) Discuss the role of remote sensing in tracking the dispersion of environmental pollutants. Analyze its implications for human health in affected regions, highlighting limitations and future improvements.

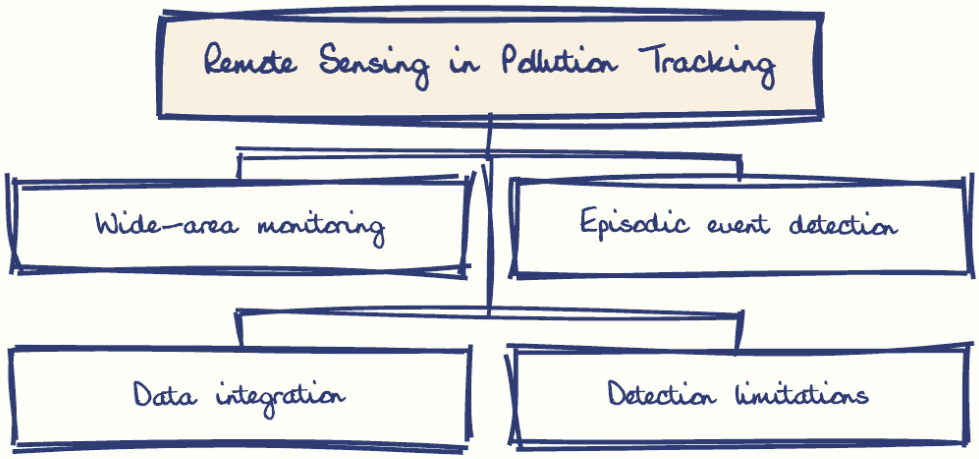
10

Introduction

Rising pollution from industrial growth and farm practices makes tracking pollutant dispersion critical for public health. Remote sensing tracks pollutants over large areas and enables timely action, which matters for Andhra Pradesh's industrial corridors and agrarian regions.

Role in Tracking Pollutant Dispersion

1. Wide-area monitoring: Satellite and aerial sensors detect particulate matter and gases across regions where ground stations are sparse, from the coast to drought-prone Rayalaseema.
2. Episodic events: Satellites tracked wheat stubble burning in the National Capital Region and guided policy; this can aid paddy residue management in Andhra Pradesh.
3. Ground integration: The National Clean Air Programme (NCAP) combines satellite data with low-cost sensors and regulatory monitors for precise dispersion maps.



Implications for Human Health

1. Respiratory and heart disease: Dispersed PM2.5 and toxic gases raise chronic respiratory disease, heart conditions and mortality.
2. Vulnerable groups: Children, the elderly and the sick suffer most, as near Visakhapatnam's industrial corridor with air and water pollution.
3. Multi-pollutant exposure: Mixtures such as carcinogenic diesel exhaust harm more than single pollutants.
4. Policy gap: Air quality policies often ignore health sector realities.

Limitations and Future Improvements

1. Detection limits: Cloud cover, detection windows

and mixed sources reduce accuracy, fires outside windows go uncounted, and ground-level and indoor pollution are missed.

2. Improvements: Better sensors and algorithms, a regional and source-specific focus, and indigenous satellite products under NCAP.

3. Transparency: Public access to the data builds accountability.

Conclusion

Combined with ground monitoring and health policy, remote sensing can guide timely action and protect vulnerable people.

FIVE-MINUTE LESSON TOPPERS' ANSWERS, PER QUESTION

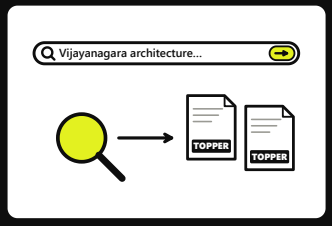
Read a topper copy like an examiner

Don't read a topper's answer for its facts. Read it for its moves — they transfer to any question.

1. **First two lines** — How does the intro frame the demand?
2. **Headings** — How is the body split — by dimension, by actor, by time?
3. **Proof** — Which fact, report or case holds up each paragraph?
4. **Last two lines** — What way forward, and how specific is it?

ON GHISSU, THIS IS DONE FOR YOU — ON EVERY QUESTION.

1.5 lakh+ topper answers are matched to each question — with their intro, structure and conclusion laid bare.



Search any question →
ghissu.com
14 days free · no card
SCAN

12(a) Explain the regulatory framework governing genetically modified organisms in India and assess its adequacy in addressing modern challenges.

10

Introduction

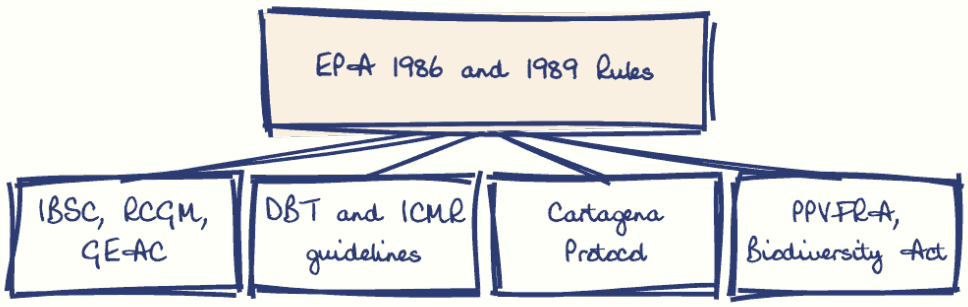
Biotechnology has made genetically modified organisms central to India's food security and sustainable development. With growing trade and environmental concerns, regulation must balance innovation with biosafety, public health and ecology, especially across diverse agro-ecological zones like Andhra Pradesh.

Regulatory Framework

1. **Legal Instruments:** GMOs are regulated under the Environment Protection Act, 1986 and the 1989 Rules on hazardous microorganisms and genetically engineered organisms, in force from 1993.
2. **Multi-Tier Institutions:** Six committees, including IBSC, RCGM and the GEAC, the apex body for environmental release, work under the MoEFCC as nodal agency.
3. **DBT and ICMR Guidelines:** DBT guides research, confined field trials, toxicity and allergenicity tests; ICMR advises on health risk.
4. **International and Allied Laws:** India is party

to the Cartagena Protocol, and the PPVFR Act, (2001) and Biodiversity Act, (2002) cover IP and biodiversity.

GMO regulation in India



Assessment of Adequacy

1. Strengths: Case-by-case risk assessment and long-term monitoring address gene flow, allergenicity and superweeds; the (1998) guidelines require proof of safety and economic viability.
2. Conflict of Interest: The Standing Committee on Agriculture found the same scientists developing and approving GM technology eroding trust and delaying approvals, as in the Bt brinjal controversy.
3. Weak State Role: Though agriculture is a state subject, state consultation on field trials is not mandatory, causing resistance and litigation.
4. Innovation Burden: Costly compliance deters public

and small innovators, as with the delayed GM mustard (DMH-11), and rules lag gene editing like SDN-1 and SDN-2.

Way Forward

1. Biosafety Act: Give regulators statutory autonomy and conflict-free decisions.
2. State Coordination: Make state permission mandatory for field trials, crucial for states like Andhra Pradesh with diverse agro-climatic zones.

Conclusion

India's GMO framework is science-based but slow and short of independence. A Biosafety Act and clear rules for gene editing will make it fit for modern challenges.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 12(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

Scan to practice

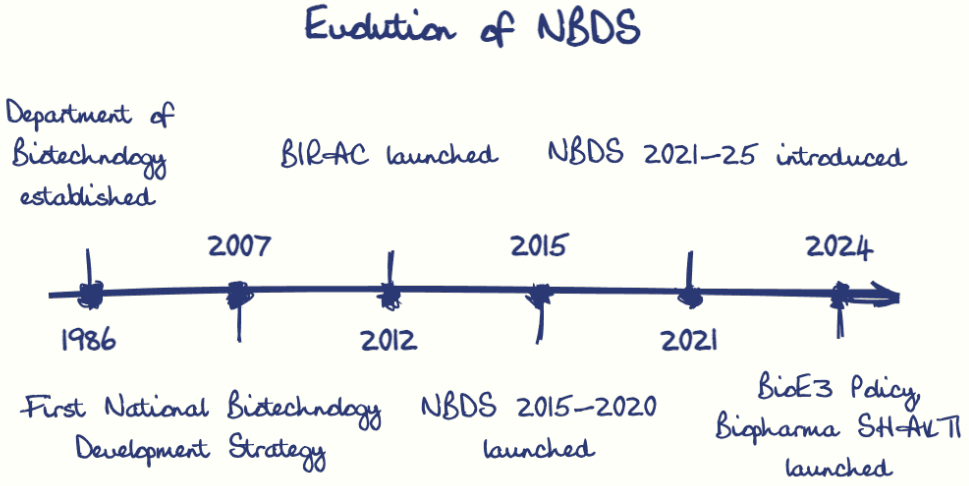
12(b) Trace the evolution of India's National Biotechnology Development Strategy (NBDS) and discuss its key objectives. Critically evaluate its impact on the growth of the biotechnology sector in India.

10

Introduction

The National Biotechnology Development Strategy is a framework to harness biotechnology for growth, healthcare, agriculture and sustainability. With India's bioeconomy above $165 billion, 4.2% of GDP, it serves Atmanirbhar Bharat, backed by government initiatives and innovation ecosystems.

Evolution of NBDS



This shows a move from foundational research to strategic policy and industry support, with BIRAC backing startups.

Key Objectives and Positive Impact

1. **Research Expansion:** Bio-RIDE and Emerging Frontiers support over 150 projects across 21 states, including AP's research institutions, strengthening genomics and bio-manufacturing.
2. **Economic Growth:** It aims to grow the bioeconomy to $300 billion by 2030. Biopharma SHAKTI (Rs 10,000 crore) backs startups and manufacturing, and India leads in vaccines.
3. **Sectoral Applications:** Drought-tolerant chickpea in farming, genome sequencing in health, and biofuels under the National Biofuel Policy.
4. **Institutions:** The Open Access Policy aids collaboration and BIRAC moves technology from lab to market, though dedicated legislation is still pending after the Biotechnology Regulatory Authority Bill lapsed.

Critical Evaluation: Limitations

1. **Regulatory Delays:** Overlapping rules slow approvals for GM crops like Bt brinjal, and gene-editing ethics raise resistance.
2. **Funding Gaps:** R&D spend is only about 0.65% of GDP, with limited advanced infrastructure such as BSL-4 labs, which slows translational research.

3. Talent Shortage: Skill mismatch and brain drain limit innovation.

4. Access and Environment: Patents may restrict access for small farmers, and GM crops raise biodiversity concerns.

Way Forward

Strengthen State S&T Councils, including AP's, through BioE3 Cells, streamline regulation, raise R&D funding and retain skilled people.

Conclusion

NBDS has catalysed significant growth in India's biotech sector. Removing regulatory, funding and talent bottlenecks will turn it into a truly inclusive bioeconomy.

REVIEWER'S NOTE · VALUE ADDITION

Examiners read hundreds of answers a day. The one with a number, a committee and a case is the one they remember.

ghissu

ghissu gives you the data point, the report and the example for every question — each linked to its source.

Get the value adds for any question →

ghissu.com

14 days free · no card

SCAN

13(a) Describe the components and functioning of a vaccine.

10

Introduction

Vaccines are a cornerstone of public health, as the COVID-19 pandemic and India's immunisation drive showed. They prime the immune system to recognise and fight pathogens, preventing disease spread and severity, including under Andhra Pradesh's Universal Immunisation Programme.

Components of a Vaccine

1. Live attenuated and inactivated: Weakened pathogens, as in BCG, or dead ones, as in Covaxin made with Vero Cell technology.
2. Sub-unit vaccines: Only parts of the pathogen are used, as in the HPV vaccine.
3. Genetic vaccines: DNA (Zydus Cadila) and mRNA vaccines (Pfizer, Moderna) carry instructions for body cells to make pathogen proteins.
4. Viral vectored vaccines: A carrier virus delivers the antigen gene, as in AstraZeneca.

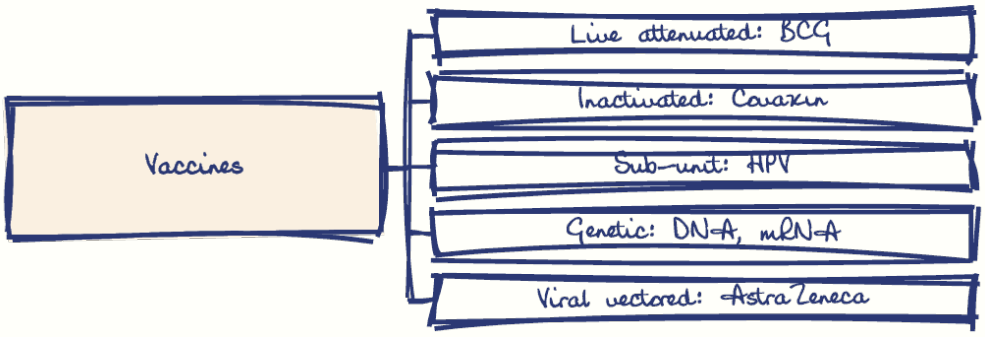
Functioning of a Vaccine

1. Immune activation: Vaccines target the adaptive

Practice 13(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

Types of vaccines



immune system by simulating infection without disease.

2. Memory formation: The body produces memory B and T cells that respond rapidly on real exposure, contributing to herd immunity.
3. Delivery and dosage: Correct schedules, often intramuscular injections, are essential. UIP gives free access to children and pregnant women.
4. Variable efficacy: Genetic, environmental and nutritional factors affect results, while hesitancy and logistics affect uptake.

Implementation

1. Regulation: The CDSCO approves vaccines, including emergency use authorisations that sped up COVID-19 vaccines.
2. Coverage: India's immunisation coverage rose from 35% in the early 1990s to over 76%,

helped by Andhra Pradesh's targeted outreach, though disparities remain.

3. Cold chain: Potency needs robust cold chains, with platforms like CoWIN improving delivery.

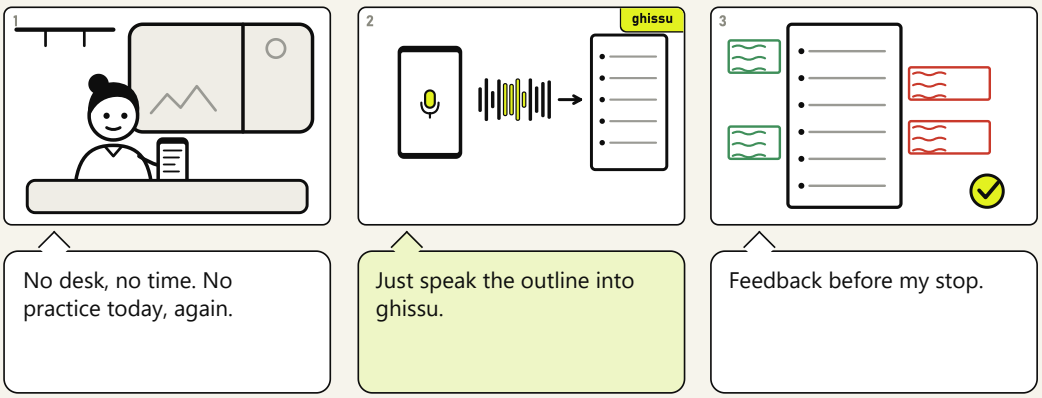
4. Hesitancy: Misinformation and fear in rural and low-literacy areas are tackled through community sensitisation.

Conclusion

A vaccine combines a well-chosen antigen with the body's own memory to give protection in advance. Strong regulation, cold chains and trust keep this working.

A STORY YOU MAY KNOW QUICK OUTLINE — TYPE, SPEAK OR SCAN

Practice on the bus



— An outline a day beats a full answer a week.

Try a quick outline →
ghissu.com
14 days free · no card

SCAN

13(b) Define genetic engineering. Explain its applications in agriculture and medicine.

10

Introduction

Genetic engineering has emerged as a transformative technology of the 21st century, driving innovation in agriculture and medicine. Backed by the National Biotechnology Development Strategy (2021–25) and climate-smart gene-edited crops, it is crucial for food security and healthcare.

Defining Genetic Engineering

1. **Meaning:** Genetic engineering is the deliberate manipulation of an organism's DNA using recombinant DNA technology, gene cloning and CRISPR-Cas9 gene editing to introduce, remove or alter specific genes for desired traits.
2. **GMOs:** The modified organisms, called GMOs, can express traits such as pest resistance or enhanced nutrition that conventional breeding cannot achieve.

Applications in Agriculture

1. **Climate-Resilient Crops:** Drought-tolerant varieties like the SAATVIK (NC9) chickpea help farmers in drought-prone Rayalaseema adapt to climate stress.

Applications of genetic engineering

Agriculture	Medicine
Bt cotton	Insulin, growth hormone
SAATVIK drought-tolerant chickpea	Gene therapy, sickle-cell anaemia
Golden rice	Hepatitis B vaccines
Gene-edited MTU-1010 rice	Monoclonal antibodies

2. Pest Resistance: Bt cotton resists pests, reducing pesticide use and raising yields, benefiting Andhra Pradesh farmers.
3. Higher Yield: ICAR has developed CRISPR gene-edited rice lines in the MTU-1010 background with higher yield and larger grain spikes.
4. Nutrition: Golden rice, enriched with vitamin A precursors, addresses micronutrient deficiencies.
5. Aquaculture: In Andhra Pradesh's aquaculture sector, feed formulations such as fermented soybean meal in shrimp feed enhance growth.

Applications in Medicine

1. Therapeutic Proteins: Insulin and human growth hormone are mass-produced by genetically modified

bacteria.

2. Gene Therapy: Correcting defective genes treats hereditary diseases like sickle-cell anaemia, moving towards personalised medicine.
3. Vaccines: Engineered hepatitis B vaccines and monoclonal antibodies have transformed prevention and treatment, with the Serum Institute playing a global role.
4. Diagnostics and Research: Model organisms and diagnostic kits help understand genetic disorders and develop targeted therapies.

Way Forward

1. Research and Outreach: Andhra Pradesh should fund research through ICAR and biotech hubs and expand Biotech-KISAN hubs to link scientists with farmers.

Conclusion

Genetic engineering offers precise, efficient and sustainable solutions, as seen in gene-edited crops and engineered medicines. Robust biosafety regulation and training of farmers and health professionals will ensure its benefits reach farmers and patients.

Which sub-topic comes next?
Trend analysis for every syllabus topic.
ghissu.com →

Write your own answer to 13(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
Scan to practice

14(a) Describe the integrated strategies used in biotechnology to monitor and control outbreaks of bacterial diseases in poultry farming.

10

Introduction

The poultry sector is crucial for food security and rural livelihoods in India and Andhra Pradesh, but bacterial outbreaks threaten flock health, productivity and incomes. With antibiotic resistance rising, biotechnology offers integrated strategies for early detection, monitoring and control.

Monitoring and Early Detection

1. Molecular Diagnostics: PCR and next-generation sequencing increasingly used in Andhra Pradesh, rapidly identify bacterial pathogens and track their spread, enabling timely action.
2. Environmental Monitoring: Biosensors and environmental DNA sampling detect bacterial contamination in poultry environments.
3. Data and Predictive Modelling: AI and machine learning integrate epidemiological data from farms in Visakhapatnam and other districts to predict outbreaks, though infrastructure gaps limit use.

Control Strategies

1. Vaccine Platforms: Biotechnology enables rapid vaccines

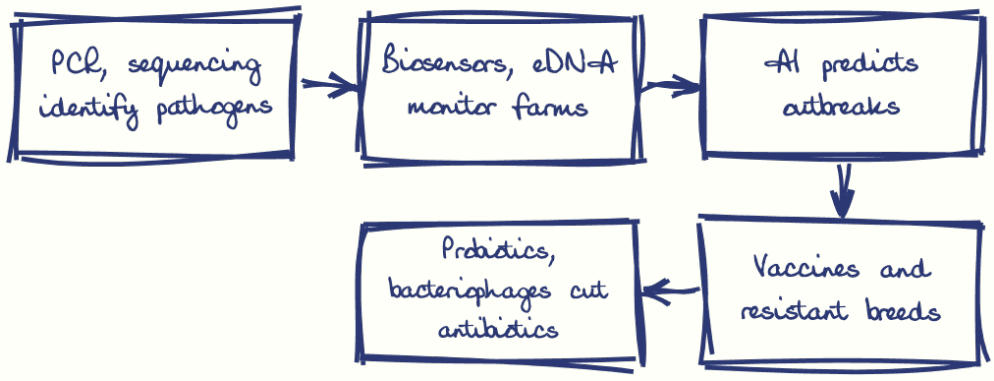
against bacterial pathogens, some field-tested in Andhra Pradesh's poultry clusters against Salmonellosis and Mycoplasmosis.

2. Disease-Resistant Breeds: Imported grandparent stocks combined with indigenous breeding in Andhra Pradesh improve resistance and productivity.

3. Alternatives to Antibiotics: Probiotics, bacteriophages and genome-edited birds are being piloted in Andhra Pradesh's commercial farms to reduce antibiotic dependence.

4. One Health Approach: Given zoonotic risk, strategies link animal, human and environmental health surveillance.

Biotech chain against poultry bacteria



Institutional Support

ICAR and the Central Avian Research Institute work with Sri Venkateswara Veterinary University on resistant breeds and diagnostics, while the Innovative Poultry

Productivity Project under the National Livestock Mission backs biosecure stocks and vaccines suited to Indian conditions, including Andhra Pradesh.

Challenges

Biosecurity is hard to enforce on small and backyard farms in Rayalaseema and coastal Andhra, where hygiene and feed quality vary.

Way Forward

1. Diagnostic Infrastructure: Expand diagnostic and genomic sequencing facilities in Andhra Pradesh's veterinary universities for rapid outbreak detection.
2. Farmer Training: Extension programmes in Rayalaseema and coastal Andhra on biosecurity, vaccination and reduced antibiotic use.
3. Data Platforms: Integrate poultry health data from across Andhra Pradesh for predictive modelling and real-time monitoring.

Conclusion

Integrated biotechnology, from diagnostics to vaccines and antibiotic alternatives, can make poultry farming safer and more sustainable.

Are you actually improving?
Every attempt scored. Watch the curve.
ghissu.com →

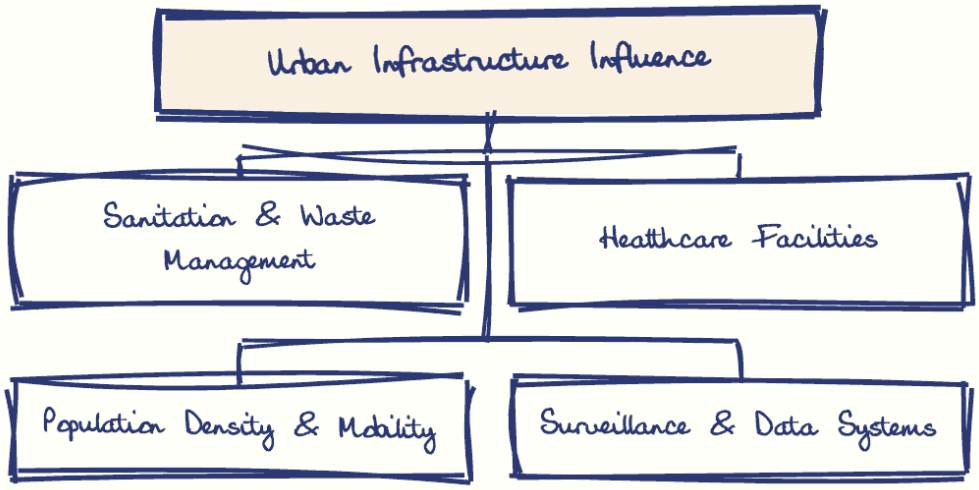
14(b) Examine how urban infrastructure influences the implementation of preventive measures during microbial disease outbreaks. Suggest technology-driven solutions.

10

Introduction

The COVID-19 pandemic underscored the critical role of urban infrastructure in managing microbial disease outbreaks in dense cities. Rapid urbanisation and climate change intensify health risks. In Andhra Pradesh, where urban local bodies (ULBs) face gaps in waste management and healthcare, resilient systems backed by technology are urgent.

Influence of Urban Infrastructure on Preventive Measures



1. **Sanitation and waste:** Andhra Pradesh's 123 ULBs generated about 6,890 tons of solid waste daily in 2022, with only 4 having operational

sanitary landfills. Poor waste handling breeds vectors and weakens environmental hygiene.

2. Healthcare gaps: In Andhra Pradesh, 743 specialist posts were vacant in secondary healthcare in 2022, and only 175 CHCs existed against a need of 412, limiting isolation, treatment and surveillance.

3. Density and mobility: High density and inter-city movement make contact tracing and quarantine hard, and sprawl in Visakhapatnam complicates containment.

4. Data integration: Fragmented health data and weak public-private links, with no unified portal in India including Andhra Pradesh, delay detection.

Technology-Driven Solutions

1. AI early warning: AI models for real-time outbreak prediction, like those of the UK Health Security Agency, can map spread and forecast hotspots.

2. Telemedicine: Applying Health Ministry telemedicine guidelines reduces hospital burden and exposure risk.

3. Smart waste management: IoT-enabled segregation

and monitoring can close gaps in wet waste processing plants found by Andhra Pradesh's audit.

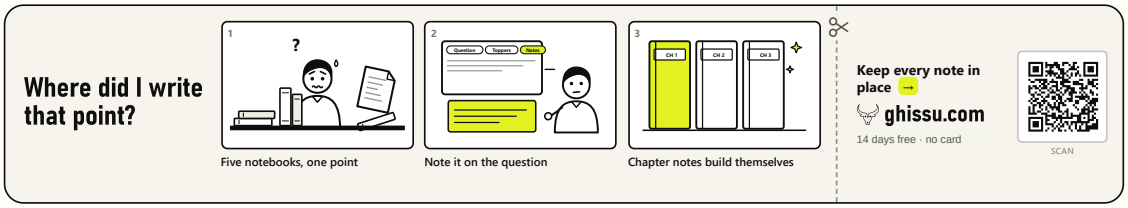
4. Unified data portals: Combining clinical, genomic and epidemiological data with AI analytics strengthens contact tracing.

Way Forward

1. Build capacity: Expand sanitary landfills and wet waste plants in Andhra Pradesh's ULBs, fill medical vacancies and add CHCs.

Conclusion

Urban infrastructure shapes outbreak prevention through sanitation, healthcare, population management and data. AI surveillance, telemedicine and smart waste systems can build health-secure cities, in line with SDG (9.)



15(a) Discuss the transmission dynamics of respiratory tract infections in densely populated urban settings. Propose effective public health strategies for their prevention.

10

Introduction

Rapid urbanisation in India, with the urban population exceeding 55 percent, has intensified the transmission of respiratory tract infections (RTIs) in cities like Visakhapatnam. COVID-19 showed how urban density, social habits and infrastructure gaps accelerate spread, making prevention critical for health and economic stability.

Transmission Dynamics in Dense Cities

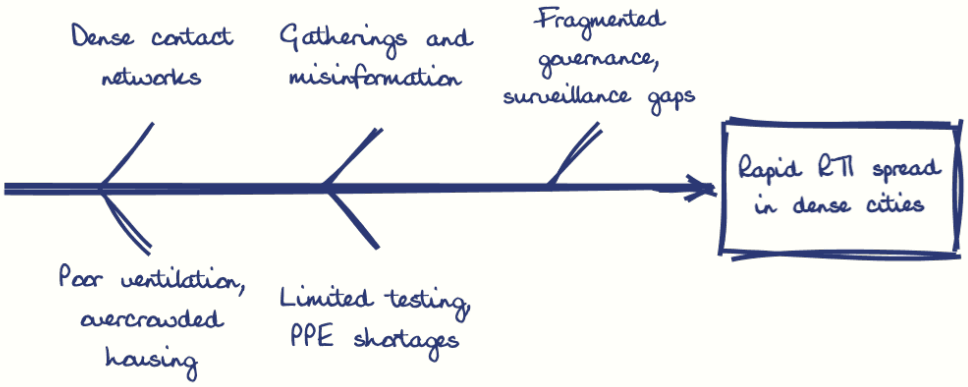
1. **Density and contact networks:** Dense settlements like Dharavi in Mumbai create extensive contact networks that enable rapid airborne and droplet transmission, overwhelming testing and isolation capacity.
2. **Environmental factors:** Poor ventilation, overcrowded housing and inadequate sanitation in informal settlements raise risk, while crowded markets and public transport amplify spread.
3. **Behaviour:** Communal gatherings increase exposure, and misinformation or low risk perception lowers adherence to masks and hygiene.

4. Resource constraints: Limited testing and PPE shortages delay case finding and contact tracing.

Institutional Weaknesses

1. Fragmented governance: In India, including Andhra Pradesh, split roles of municipal bodies and state health departments hamper surveillance and response.
2. Surveillance gaps: No integrated multi-disciplinary surveillance exists to catch outbreaks early in slums and high-density localities.
3. Weak integration: Private providers are poorly linked to public emergency response, and community participation is overlooked.

Why RTIs spread fast in cities



Public Health Strategies for Prevention

1. Integrated surveillance: Real-time systems combining community health workers, hospitals and private clinics can detect outbreaks early in urban hotspots.

2. Better urban infrastructure: Affordable housing, sanitation and ventilation in Andhra Pradesh's rapidly urbanising areas address the environmental determinants of RTIs.

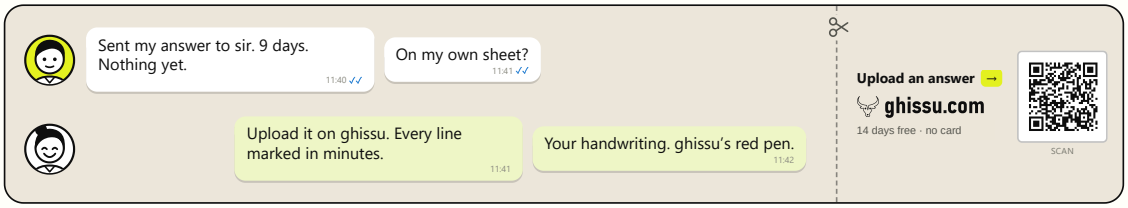
3. Community health education: Campaigns through local leaders and digital platforms can sustain mask use and hand hygiene.

4. Testing and care: Scaling up rapid antigen and antibody tests with dedicated RTI care centres ensures timely treatment.

5. Governance and technology: A unified urban health governance model in Andhra Pradesh and AI-based contact tracing can map transmission networks.

Conclusion

RTI spread in cities reflects social, environmental and institutional factors. Surveillance, infrastructure, community engagement and access to care can break transmission chains, advancing SDG (3.)



15(b) Describe the importance of IPR in academic and industrial research collaborations.

10

Introduction

In an era of rapid innovation, intellectual property rights (IPR) are crucial for collaboration between academia and industry. Initiatives like the National IPR Policy 2016 ensure that research is commercialised and rewarded, so IPR bridges academic research and industrial application, driving growth and self-reliance.

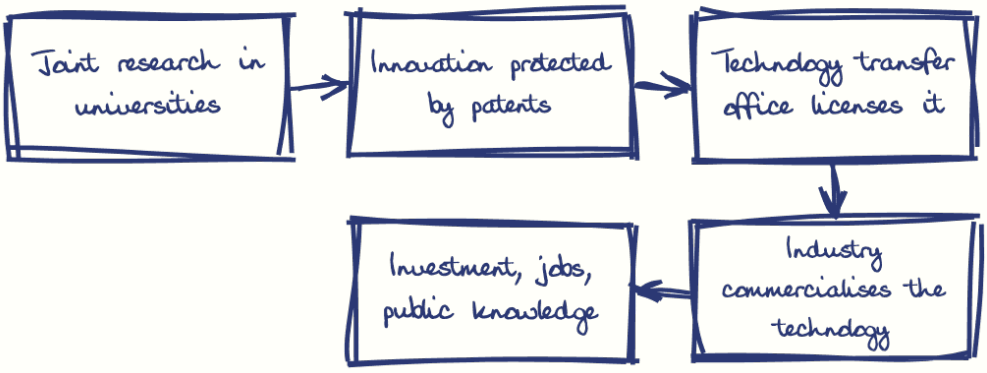
Importance of IPR: Institutional Framework

1. Legal foundation: India's IPR regime, aligned with TRIPS and the Paris Convention, safeguards innovations from joint research; the 2016 policy promotes their generation and commercialisation.
2. Institutional IP policies: Universities such as Andhra University and Sri Venkateswara University adopt IP policies and use technology transfer offices for licensing and joint ventures.
3. Capacity building: Training in patent filing, licensing and benefit-sharing helps researchers and administrators manage partnerships smoothly.
4. Innovation ecosystems: Industrial parks in the Visakhapatnam–Chennai industrial corridor host start-ups and R&D units linked to academia.

Practice 15(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2025 · Paper V

Scan to practice

IPR in academia–industry collaboration



Challenges and Outcomes

1. Balancing access and incentives: Exclusive rights reward innovators, while disclosure after the term keeps outputs socially accessible, as in Andhra Pradesh's horticulture research.

2. Complex licensing: Royalty sharing and commercialisation rights are hard to negotiate; NARIs must manage patented biotechnologies without infringing existing IP.

3. Cultural shift: Researchers focused on pure science may resist IP protocols, needing leadership and training.

4. Economic outcomes: Effective IPR management has raised patent filings and technology transfers, attracting investment and jobs in Andhra Pradesh's growing biotech and pharma sectors.

Way Forward

1. IP policies in academia: Every university in Andhra Pradesh should adopt a comprehensive IP policy aligned with national standards to streamline collaboration with industry.
2. Training: Expand capacity building on IPR management, licensing and commercialisation to build a culture of innovation and legal compliance.
3. Collaborative platforms: State-level innovation hubs and digital platforms can bring academia, industry and government together for IP sharing, dispute resolution and technology transfer.
4. Infrastructure: Expand industrial parks and incubation centres in Andhra Pradesh's key innovation corridors to support IP-driven collaborations.

Conclusion

IPR turns academic research into industrial innovation, rewarding creators while society gains new technology. Stronger IP policies, training and innovation hubs will advance Atmanirbhar Bharat and SDG (9.)

REVIEWER'S NOTE · CRITICALLY EXAMINE

"Critically examine" pays for the counter-view. An answer that stops at the first view has stopped at half the marks.

ghissu

ghissu exam-hall thinking decodes every directive — Discuss, Examine, Elucidate — before you write a line.

Decode your next question →

ghissu.com

14 days free · no card

SCAN

APPSC GROUP-I MAINS · PAPER V

2023

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

1(a) Do you think the problem in India is "not so much knowledge creation, as knowledge consumption" and Scientific Social Responsibility (SSR) will go a long way in remedying this situation? Justify your stand. Discuss the challenges associated with the implementation of Scientific Social Responsibility.

10

Introduction

India's vast talent pool and scientific legacy demand a shift from being mainly a consumer of knowledge to a producer and disseminator of it. Scientific Social Responsibility (SSR) asks scientists and institutions to engage actively with society, so that knowledge is created and also consumed and applied, vital for climate change, public health and digital transformation.

Stand: Consumption is the Bigger Gap, and SSR Helps

1. Socio-economic barriers: Disparities in education and digital literacy, especially in drought-prone Rayalaseema, keep available research from reaching people.
2. Public trust: COVID-19 showed the role of SSR in timely vaccine communication, but also revealed misinformation and resistance.
3. Mandate for engagement: The constitutional duty to nurture scientific temper and Vigyan Prasar's

work across media show how SSR can build an informed citizenry. Andhra Pradesh has used local languages for this.

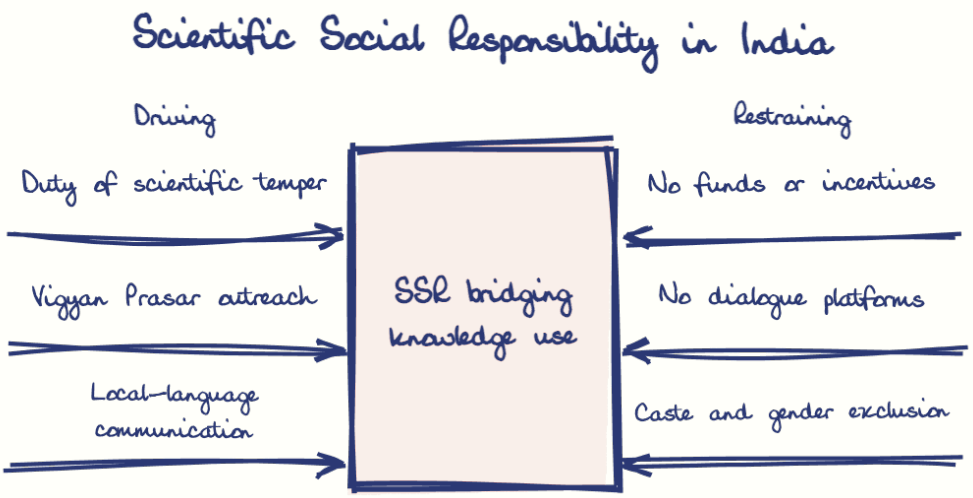
4. Creation still matters: R&D at about 0.69% of GDP, behind the USA and South Korea, constrains both creation and dissemination, so SSR must go with more investment.

Challenges in Implementing SSR

1. Resources and incentives: Funding for popularisation is limited and scientists get no reward for public engagement.
2. No dialogue platforms: Dedicated forums between researchers and communities are missing.
3. Fragmented governance: Short tenures of administrators and sidelined social sciences impede sustained effort.
4. Social stratification: Caste and gender exclusions hinder inclusive knowledge networks.
5. Impact assessment: No robust mechanism measures SSR outcomes.

Way Forward

1. Institutional linkages: Connect scientific institutions, social scientists and community



groups.

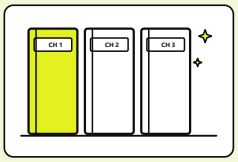
2. Incentives: Dedicate funds for science communication and embed SSR in academic career frameworks.

3. Literacy: Use vernacular content and digital tools in regions like Rayalaseema.

4. Governance: Create multi-sectoral bodies to oversee SSR.

Conclusion

India's challenge is less creation than consumption, and SSR can significantly bridge it, building an informed society in line with ~~Atmanirbhar~~ Bharat.



Notes in five notebooks, none at hand?

Notes that live with the question

Note from toppers' answers, exam-hall thinking and model answers — your notes gather into topic and chapter notes on their own.

Build notes that find you →

ghissu.com

14 days free · no card

SCAN

1(b) Technology plays a key role in transforming healthcare sector and in addressing India's healthcare challenges — Discuss.

10

Introduction

Technology has emerged as a critical enabler in transforming India's healthcare, especially after COVID-19 accelerated digital adoption. Through the Ayushman Bharat Digital Mission (ABDM) and telemedicine, it is improving access, affordability and quality. Given India's vast population and infrastructure gaps, technology is essential for efficient and equitable care.

How Technology Transforms Healthcare Delivery

1. **Digital health records:** EHRs under the National Digital Health Blueprint allow data sharing across providers. ABDM has created over 78 crore ABHA accounts, linking more than 55 crore records.
2. **Telemedicine:** eSanjeevani has enabled over 282 million teleconsultations, letting remote patients consult specialists without travel.
3. **AI in diagnostics:** AI gives second opinions and risk stratification; the National TB Elimination Programme saw a 27% fall in adverse outcomes after AI deployment.

4. Wearables and IoT: Devices monitor heart conditions, glucose and epilepsy attacks in real time, aiding self-care and reducing hospital visits.

Health challenges and technology answers

Challenge	Technology response
Doctors far from villages	eSanjeevani telemedicine
Scattered patient records	ABHA digital records
Late diagnosis	AI-supported screening
Staff skill gaps	Online training, simulation

Addressing India's Healthcare Challenges

1. Access gaps: Andhra Pradesh's CAG (2024) audit found (743) vacant specialist posts and only (175) CHCs against (412) needed. Telemedicine connects underserved people to expert care.

2. Affordability and quality: Early diagnosis cuts costs, and the Digital Health Incentive Scheme encourages adoption, vital where health gets (3.11%) of the state budget against an (8%) target.

3. Disease management: AI-supported screening raised TB and diabetic retinopathy detection by

12–16%, tackling the dual disease burden.

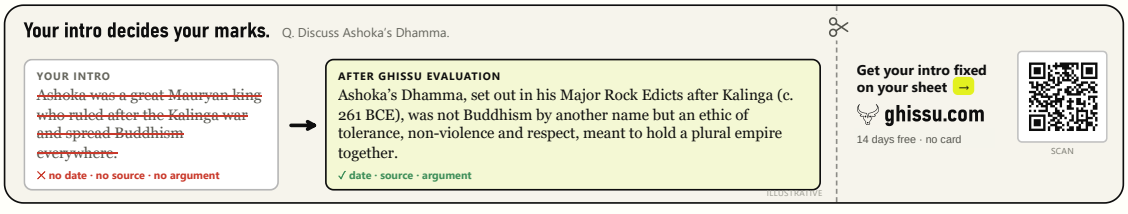
4. Workforce training: Online training and simulation, including haptic feedback, build skills, vital given vacancies in medical colleges and health posts.

Way Forward

1. Rural digital infrastructure: Improve connectivity and digital literacy.
2. Partnerships: Scale innovative health technologies through public–private partnerships, optimising scarce resources.
3. Regulation: Build data privacy and ethical AI frameworks to protect patients.
4. Local R&D: Promote indigenous health technology and continuous training of health workers.

Conclusion

Technology is pivotal for overcoming India's challenges of access, affordability and quality. By integrating records, telemedicine, AI and wearables, India can move towards universal health coverage in line with SDG (3).



2(a) Digital platforms dramatically improve the quality and accountability of public service and facilitate citizen centric governance — Elaborate. Explain various Technical, Economic and Social bottlenecks in the implementation of e-governance programmes in India.

10

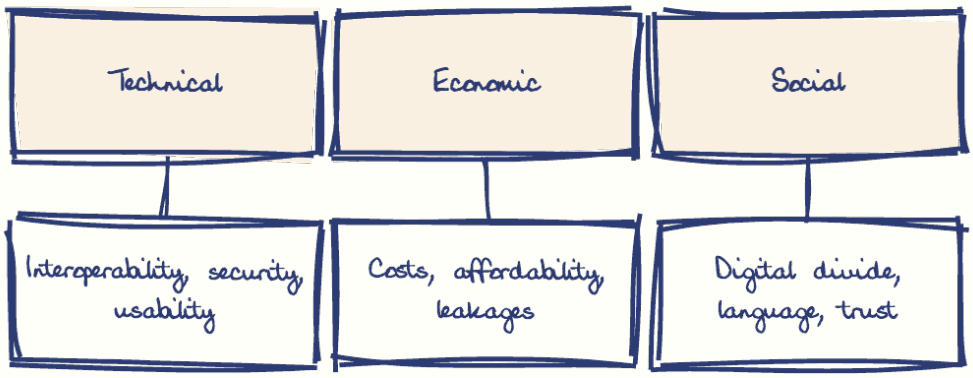
Introduction

Digital infrastructure has made Indian governance more accessible, transparent and efficient. Through Digital India and platforms such as Aadhaar, UPI and Direct Benefit Transfer, digital platforms are now central to citizen-centric services. Yet technical, economic and social bottlenecks limit e-governance.

How Platforms Improve Quality and Accountability

1. **Real-Time Transparency:** Live data on service delivery allows continuous monitoring. e-Taal tracks electronic transactions, curbing corruption.
2. **DBT Efficiency:** Over Rs 43.95 lakh crore has been transferred directly to bank accounts since 2013, minimising leakages and middlemen.
3. **Grievance Redressal:** RailMadad lets citizens lodge and track complaints, and MyGov brings their ideas to policymakers.
4. **Accessibility:** DigiLocker gives digital documents, and AP's e-District issues digitally signed certificates without office visits.

Bottlenecks in e-governance



Technical Bottlenecks

1. Infrastructure: Over 2.15 lakh Gram Panchayats have optical fibre, yet remote gaps persist.
2. Interoperability: Fragmented central, state and local systems cause duplication.
3. Security: Weak data protection and cybersecurity risk breaches and erode trust.
4. Usability: Low digital literacy and few vernacular interfaces exclude non-English speakers.

Economic Bottlenecks

1. Investment: Building and maintaining infrastructure strains state budgets, as AP's post-bifurcation fiscal constraints show.
2. Affordability: Costly smartphones and data exclude the poor.

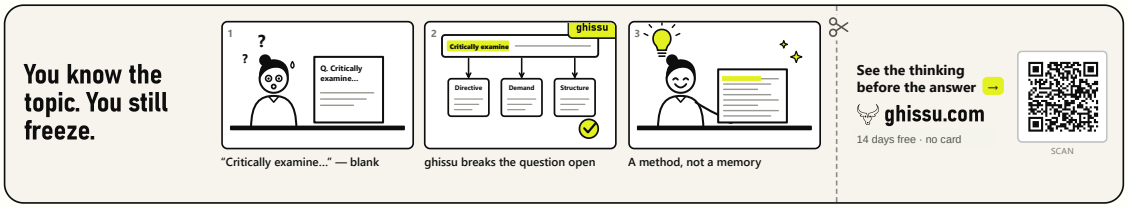
3. Leakages: A CAG audit of AP's CFMS found Rs 968 crore in excess payments due to system lapses.
4. Sustainability: Weak funding and little private collaboration stall scaling.

Social Bottlenecks

1. Digital Divide: Rural people, women and the elderly lack devices and skills.
2. Resistance to Change: Citizens and officials cling to paper processes.
3. Language: English-dominated content excludes many in linguistically diverse states like Andhra Pradesh.
4. Trust Deficit: Low awareness and privacy fears reduce uptake.

Conclusion

Digital platforms have raised the quality and accountability of public services. Faster rural broadband, vernacular interfaces and strong data protection will make them truly citizen-centric.



2(b) Explain how 'Atmanirbhar Bharat' initiative is bringing a big change in India's IT sector. Suggest the steps to be taken to make India a Global technology hub.

10

Introduction

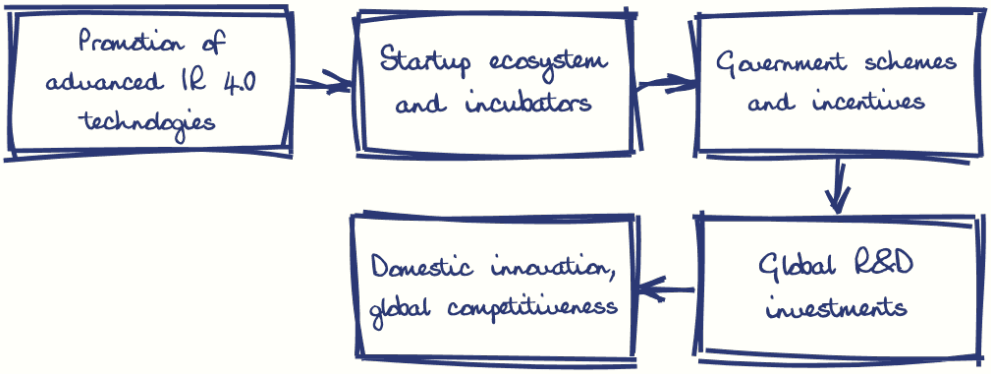
The Atmanirbhar Bharat initiative is transforming India's IT sector by emphasising self-reliance, indigenous capability and a robust startup ecosystem, crucial for growth and technological sovereignty. Andhra Pradesh reflects this, with industry GVA growing 6.58% in 2024-25.

Impact on the IT Sector

1. Indigenous Innovation: Promotion of IR 4.0 technologies like AI, blockchain, IoT and biotech has pushed firms to build products for domestic and global markets, reducing technology imports.
2. Startup Ecosystem: Over 30,000 startups and 250 incubators make India the third-largest tech startup hub, with Tier-II and Tier-III cities contributing.
3. Government Schemes: Digital India, the PLI Scheme and the India Semiconductor Mission give incentives for scale-up and R&D.
4. Global R&D Hub: MNCs are setting up large R&D centres, as seen in Visakhapatnam's IT

parks.

Atmanirbhar Bharat's Impact on IT Sector



Steps to Make India a Global Technology Hub

1. R&D Infrastructure: Higher investment and innovation clusters linking academia, industry and government.
2. Skills and Talent: Upskilling through FutureSkills PRIME and the Visvesvaraya PhD Scheme, and retaining talent to stop brain drain.
3. Ease of Doing Business: Simpler regulation, with PM Gati Shakti and the National Logistics Policy cutting costs.
4. Public–Private Partnerships: Collaboration among government, industry and startups to speed commercialisation, expanding models like the Atal Innovation Mission.
5. Data Governance: Joining global

standard-setting bodies with clear cybersecurity rules.

Way Forward

1. Innovation ecosystems: More Atal Tinkering Labs and incubation centres in Andhra Pradesh to nurture grassroots innovation.
2. Public R&D spending: Higher budgets for semiconductors and AI under the India Semiconductor Mission 2.0.
3. Inclusive reskilling: Large-scale programmes for emerging technologies in urban and rural areas.

Conclusion

By pairing self-reliance with global integration, and pursuing these steps with stable, predictable policy, India can become a global technology hub.

1.5 lakh+

Borrow the approach, not the answer.

Topper answers matched by meaning to your question — with how each one opened, structured and concluded.

See how toppers wrote it → ghissu.com

14 days free · no card

SCAN

3(a) What is Space Junk ? Discuss the challenges posed by increased space junk and the potential risk associated with it. What do you know about Project NETRA ?

10

Introduction

Space junk refers to debris such as defunct satellites and fragments orbiting the Earth. With India's growing satellite fleet and launches from ISRO's Sriharikota centre, it threatens the safety and life of assets that provide communication, navigation and earth observation services millions depend on.

Challenges Posed by Space Junk

1. **Collision Hazards:** Over 34,000 trackable pieces larger than 10 cm and nearly a million smaller fragments travel at 29,000 km/h, risking catastrophic collisions in Low Earth Orbit.
2. **Orbital Congestion:** Debris near the ISS orbit can remain for decades, and in higher orbits for millennia, making key slots unusable.
3. **Tracking Constraints:** Small debris needs costly advanced radars and telescopes.
4. **Weak Global Regime:** No binding agreement exists, creating a tragedy of the commons.

Potential Risks

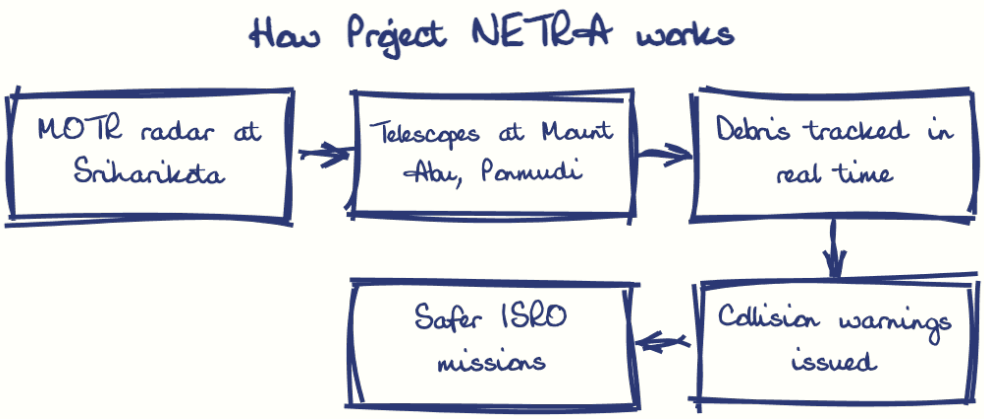
1. Economic Disruption: Damage to satellites disrupts finance, health and governance.
2. Strategic Risk: National security suffers as India raises its launch rate and constellation size.
3. ASAT Debris: Destructive anti-satellite tests create large debris clouds, which India opposes.

Project NETRA

1. Indigenous SSA System: Project NETRA is India's own Space Situational Awareness system, giving real-time tracking and collision warnings and reducing dependence on foreign data.
2. Infrastructure: The Multi Object Tracking Radar at Sriharikota and optical telescopes at Mount Abu and Ponmudi track debris as small as (30) cm.
3. Impact: Collision alerts improve mission assurance for ISRO satellites and position India as a responsible space actor.

Way Forward

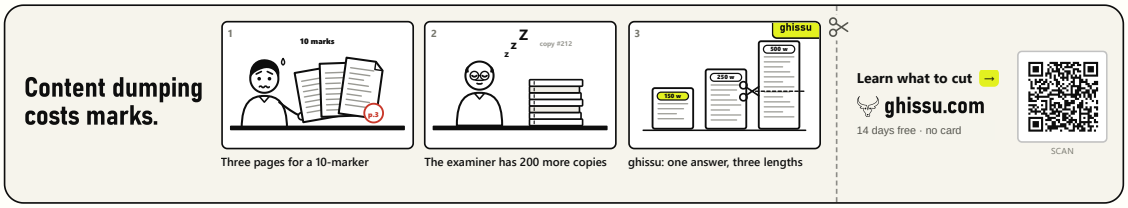
1. Expand NETRA: More advanced radars and telescopes to track smaller debris.



2. Global Norms: Lead on binding debris rules in the IADC and UNCOPUOS.
3. Indigenous Technology: Support startups and research institutions in debris removal and traffic management.
4. Capacity Building: Use ISRO's facilities and academic links to build skilled manpower.

Conclusion

Expanding NETRA and shaping global norms will protect India's assets and ensure space sustainability.



3(b) Why is India's space industry looking for private sector investments even though ISRO is a successfully performing PSU ?

10

Introduction

India's space sector has achieved remarkable feats through ISRO, including the historic Chandrayaan-3 landing. Yet despite ISRO's success as a public sector unit, the government has encouraged private participation since 2020 to accelerate innovation, meet growing demand and enhance global competitiveness.

Why the Industry Seeks Private Investment

1. **Expanding Demand:** Space services now serve urban planning, agriculture, defence and disaster management, needing more agile providers.
2. **Import Dependency:** Space technology imports of 2114 crore in FY 2021-22 far exceeded exports, so indigenous manufacturing is needed.
3. **Employment and Innovation:** Over 229 startups by 2024, like Skyroot Aerospace and Agnikul Cosmos, created around 2,500 jobs.
4. **Capacity Constraints:** ISRO's limited launch pads and manufacturing capacity restrict mission frequency and scale.

ISRO's Strengths Remain

1. Cost-Effective Excellence: PSLV and Chandrayaan-3, which made India the first to land near the lunar south pole, show ISRO's leadership.
2. Industry Catalyst: Godrej & Boyce (Vikas engines) and Larsen & Toubro (launch infrastructure at Sriharikota) have long supported ISRO.
3. Focus on Research: ISRO can now concentrate on complex interplanetary missions and cutting-edge research while routine manufacturing and launch services open to private players.

ISRO and private players

ISRO	Private sector
Complex interplanetary missions	Routine manufacturing, launches
Limited launch pads	Added launch capacity
Proven trust, know-how	Agile startups like Skyroot
Research focus	Jobs and indigenous manufacturing

Why Both Are Needed

1. Complementarity: ISRO's success built trust and know-how that private firms can build on, in a

symbiotic relationship.

2. Scaling Ambitions: Growing missions demand capacity beyond ISRO's resources.
3. New Institutions: IN-SPACe as single-window regulator and NSIL as commercial arm enable technology transfer.
4. Avoiding Overdependence: Sole reliance on ISRO risks bottlenecks in launch frequency and technology diffusion; private entry diversifies capabilities and investment.

Way Forward

1. Joint Ventures: ISRO and private firms in launch vehicle manufacturing and satellite production, combining their strengths.
2. Regulation: A comprehensive Space Activities Bill on liability and intellectual property.

Conclusion

A clear legal framework, joint ventures and skill development will help India move from a state-centric model to a collaborative ecosystem where private investment complements ISRO.

Correct, but average?
A number, a committee, a case — for every question.
ghissu.com →

4(a) Though circumstances are strongly insisting for India to overturn decades of 'No first use' nuclear doctrine, still India is committed to No first use of nuclear weapons — Comment.

10

Introduction

India's No First Use (NFU) doctrine, declared in 1999 and formalised in 2003, commits India to use nuclear weapons only in retaliation to a nuclear attack. The growing conventional and nuclear power of China and Pakistan challenges it. Yet India's adherence reflects a balance between restraint and deterrence.

Circumstances Pressing India to Overturn NFU

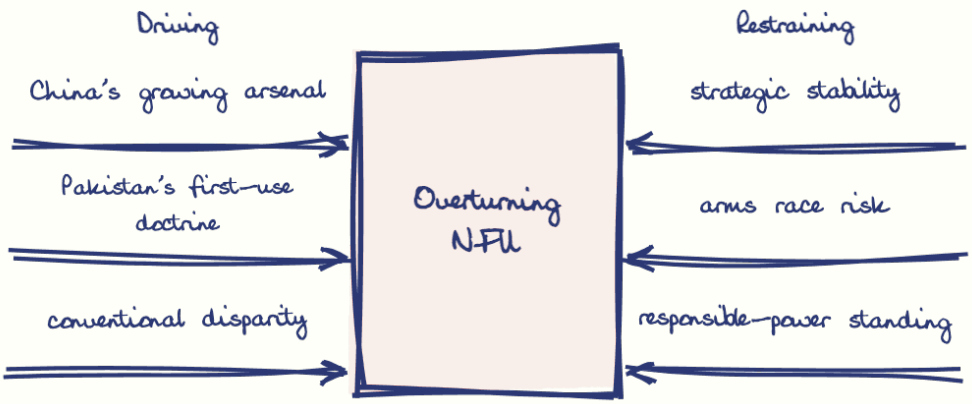
1. **Two-Front Threat:** China's arsenal is projected to reach 1,000 warheads by 2030, and Pakistan's tactical weapons lower deterrence thresholds.
2. **Conventional Disparity:** China's superior forces and aggression along the LAC in Ladakh strain purely conventional deterrence.
3. **Pakistan's First-Use Doctrine:** Its declared policy and the Hatf IX missile create a coercive setting where NFU may seem a disadvantage.
4. **Strategic Debate:** Senior officials and analysts

have hinted at ambiguity or counterforce options to pre-empt threats.

Why India Remains Committed

1. Moral Foundation: Nuclear weapons are political tools for deterrence, not battlefield weapons, in line with India's support for disarmament.
2. Strategic Stability: NFU reduces the risk of accidental or pre-emptive strikes and keeps a credible retaliatory deterrent, with platforms like INS Arihant.
3. Avoiding an Arms Race: Dropping NFU needs costly command and control changes and could trigger a South Asian arms race.
4. International Standing: It shows India as a responsible power, enabling deals like the India-Australia uranium deal.

Should India drop No First Use?

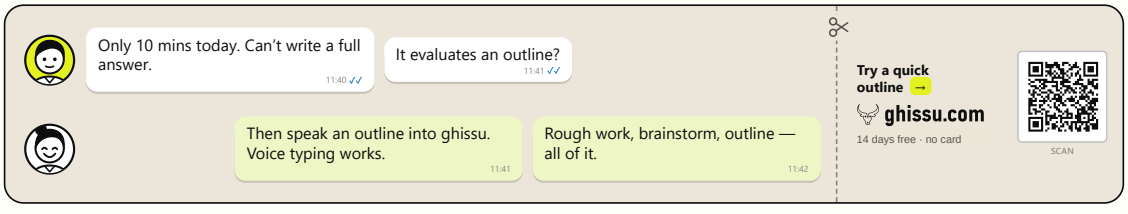


Implications

1. **Maintaining NFU:** It reinforces deterrence by punishment and India's standing as global arms control weakens, with New START expiring in 2026.
2. **Overturning Risks Escalation:** A first-use posture could provoke adversaries, raise miscalculation and destabilise South Asia.
3. **Fiscal Burden:** New command structures and larger arsenals would divert resources from development.
4. **Middle Path:** Calibrated ambiguity may aid deterrence but erodes India's moral high ground.

Conclusion

Despite real pressure from China and Pakistan, NFU continues to serve India's security, regional stability and strategic autonomy. A careful posture review that strengthens deterrence, not abandonment, is the prudent course.



4(b) What is Green Energy ? Discuss about the Green Energy corridors and their need in India.

10

Introduction

India's growth and rising demand make the shift to sustainable energy urgent. With net-zero by 2070 and over 500 GW of renewables planned, green energy and its efficient transmission are critical. Green energy corridors integrate renewable power into the grid reliably.

Understanding Green Energy

1. Definition: Power from renewable, non-polluting sources like solar, wind, biomass and small hydro; capacity excluding large hydro crossed 103 GW in 2021.
2. Benefits: Fewer emissions, cleaner air and jobs; the renewable workforce grew nearly fivefold in 2014-19.
3. Challenges: Variable solar and wind need advanced grid management and storage, and high upfront costs complicate financing.

Green Energy Corridors

1. Concept: Dedicated transmission networks of high-capacity lines and substations carrying

Practice 4(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2023 · Paper V

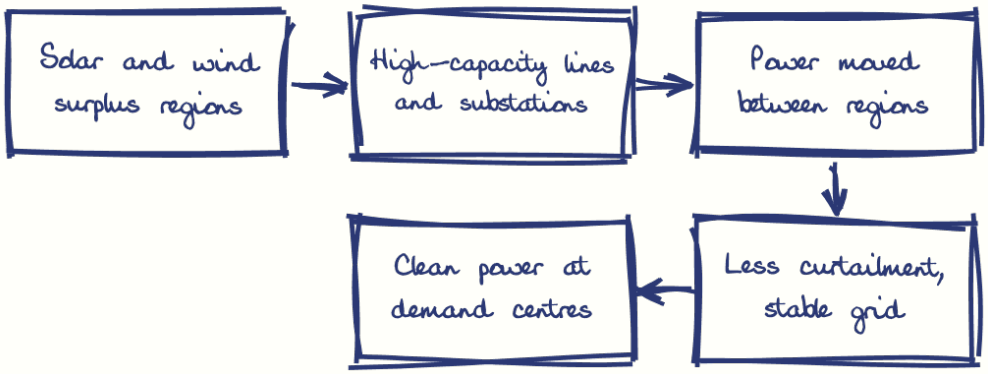
Scan to practice

renewable power to demand centres.

2. Progress: Inter-state corridors of 3,200 ckm and 17,000 MVA were completed by 2020; intra-state corridors target 9,700 ckm and 22,600 MVA.

3. Technology and Finance: Moving power between regions with complementary profiles cuts reliance on costly batteries; the National Clean Energy Fund and green bonds fund them.

How a green energy corridor works



Need in India

1. Grid Integration: Renewables give about 27% of generation, but variability destabilises the grid; corridors move surplus power to deficit regions, cutting curtailment.

2. Energy Security: Fossil imports meet 40% of primary energy needs; corridors support the National Green Hydrogen Mission.

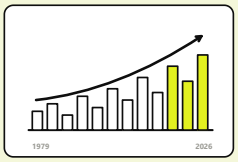
3. Regional Development: Andhra Pradesh's coastal solar potential and the Visakhapatnam–Chennai corridor benefit, along with clean energy access in rural areas.
4. Economic Impact: Efficient transmission lowers power costs, attracts investment and creates green jobs in rural and semi-urban areas, though land acquisition remains a hurdle.

Way Forward

1. Faster Completion: Smart grids and real-time monitoring for flexibility.
2. Finance: Investment-grade green bonds and PPPs.
3. Renewable Hubs: Energy parks in Andhra Pradesh linked to corridors.

Conclusion

Green energy corridors are the backbone of India's clean energy transition, turning renewable capacity into reliable power for every region.



Which sub-topic will they ask next?

Trend analysis

Every syllabus topic: the direction of travel, the most tested sub-topic, the probable next theme.

Prepare the pattern →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 4(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

Scan to practice

5(a) Climate change is a global concern which needs urgent intervention to combat it — Explain why. Discuss how climate smart agriculture mitigates the devastating effects of climate change.

10

Introduction

Climate change has moved from a distant threat to an immediate global crisis of extreme weather, rising temperatures and disrupted ecosystems. Its impacts are critical for India, where nearly 55% of the workforce depends on agriculture. The threat to food, water and socio-economic stability demands urgent, innovative solutions such as climate-smart agriculture.

Why Climate Change Needs Urgent Intervention

1. **Agricultural vulnerability:** Yields of staples are already falling, with projected losses of up to 35% in rice and 60% in maize by 2039.
2. **Water scarcity:** Agriculture uses 80–90% of India's water; climate variability deepens stress and conflict in regions like Rayalaseema.
3. **Socio-economic disparity:** Small and marginal farmers, with holdings averaging 1.08 hectares, have little adaptive capacity.
4. **Feedback loops:** Agriculture emits about 14% of national greenhouse gases through methane and

Practice 5(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2023 · Paper V

Scan to practice

nitrous oxide, and drives deforestation, accelerating warming.

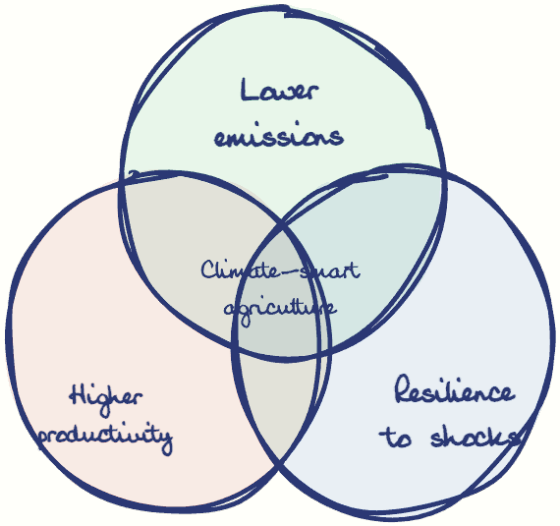
How Climate Smart Agriculture Mitigates Its Effects

1. Sustainable productivity: Conservation tillage and integrated soil fertility management raise yields and soil health, as with AP's soil health cards and organic farming.
2. Resilience to shocks: Drought-resistant varieties and drip irrigation stabilise incomes in drought-hit Rayalaseema.
3. Emission reduction: Agroforestry and reduced tillage build soil organic carbon; CAMPA afforestation in AP sequesters carbon.
4. Institutional support: CSA needs enabling policy, credit and extension suited to local conditions; the NMSA in AP links climate risk management with farmer welfare.

Way Forward

1. Scale CSA: Targeted subsidies and farmer training in vulnerable districts of AP.
2. Water management: Water-efficient technology and watershed development to ease stress in

Three wins of climate-smart agriculture



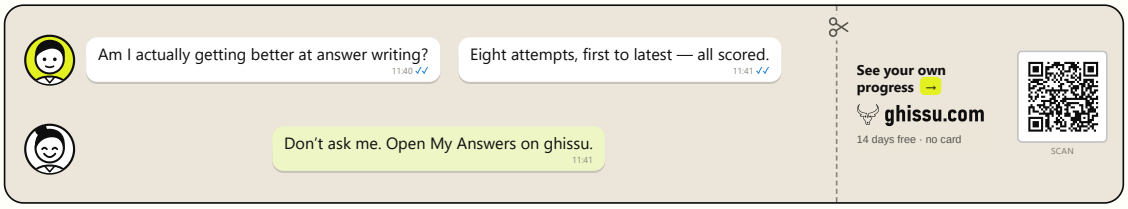
Rayalaseema.

3. Data: Use soil health card data to tailor interventions.

4. Community participation: Farmer collectives should co-create adaptation strategies.

Conclusion

Climate change is an urgent, multifaceted threat that needs immediate global and local action. Climate-smart agriculture raises productivity, builds resilience and cuts emissions together, supporting India's commitment to net zero by 2070.



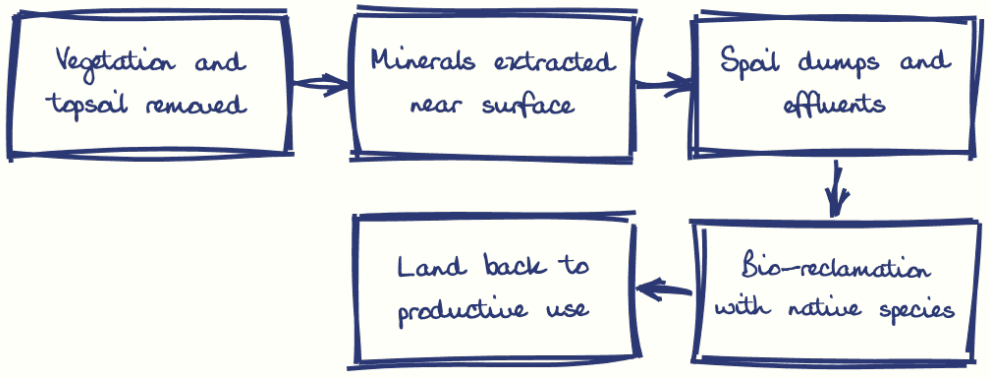
5(b) What is meant by open cast mining ? Discuss the impacts of open cast mining on environment and solutions to overcome them. Add a note on the importance of post mining reclamation.

10

Introduction

Open cast mining, also called surface mining, extracts minerals by removing large areas of soil and rock to reach deposits near the surface. Rising demand for coal and other minerals has made it prominent, including in Andhra Pradesh, but it raises serious environmental and social concerns.

Open cast mining and reclamation



Environmental Impacts

1. Land degradation and deforestation: Removal of vegetation and topsoil causes forest loss and habitat fragmentation, as in Rayalaseema's mineral-rich districts.

2. Water pollution and depletion: Wastes and

effluents contaminate water, while large-scale pumping lowers groundwater tables for irrigation and drinking.

3. Air pollution: Dust and emissions cause respiratory diseases among workers and nearby residents.

4. Socio-economic disruption: Displacement of local people and loss of agricultural land threaten livelihoods, calling for alternative employment and social security.

Solutions to Overcome the Impacts

1. Legal safeguards: The MCDR, 1988 mandate an Environment Management Plan before mining begins, covering every stage including restoration.

2. Rigorous EIA: Strong impact assessment, with EMPs, minimises ecological damage and protects wildlife corridors and water bodies in AP.

3. Surveillance: The iMSS and Mine Mitra portal improve transparency and curb illegal mining.

4. Waste management: Dump sites away from forests and water, and treatment of mine effluents.

5. Independent regulators: Autonomous authorities at centre and state to oversee clearances and mine closure, since states like AP grant leases and

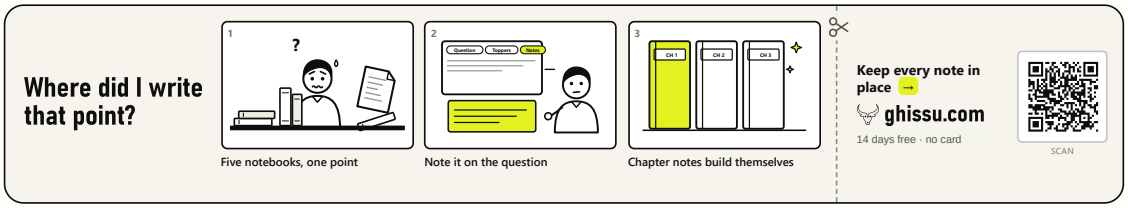
enforce safeguards.

Importance of Post-Mining Reclamation

1. Eco-restoration: Bio-reclamation through gabion check-dams, contour trenches and native species stabilises mine spoils and restores biodiversity, as shown by the Central Soil and Water Conservation Research Institute.
2. Sustainable land use: Reclaimed land returns to productive use, and proper dumping and effluent treatment keep it fit for reuse.
3. Livelihoods: Involving locals in reclamation and skill development provides alternative employment.
4. Intergenerational equity: Restoration safeguards the livelihoods of future generations.

Conclusion

Open cast mining is vital but costly. Strict oversight, post-mining reclamation and community participation can align it with SDG (13) and SDG (15) and protect future generations.



Write your own answer to 5(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2023 · Paper V

Scan to practice

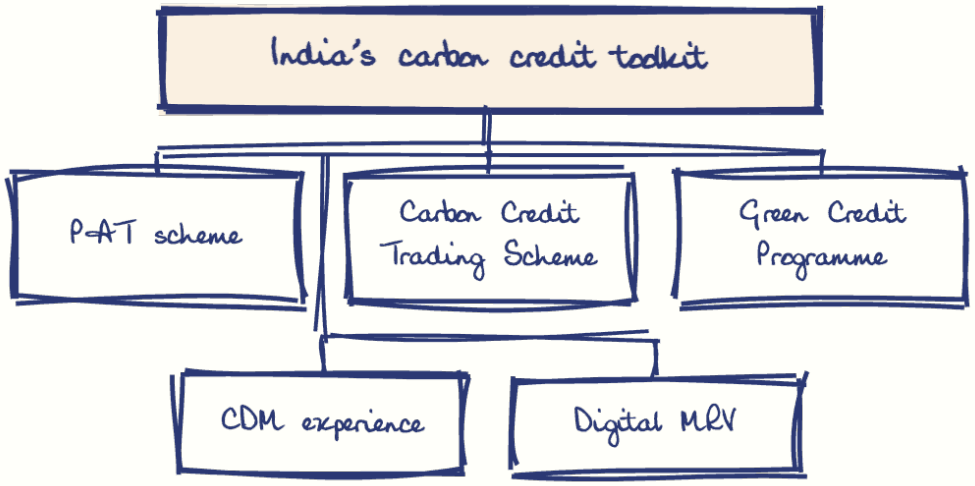
6(a) What are various mitigation activities by the Government of India to lower CO2 emission for trading of carbon credits in the international market under Paris Agreement ?

10

Introduction

India's Paris commitments have driven efforts to cut CO2 emissions through market mechanisms that allow carbon credits to be traded internationally. India is aligning with Article 6 of the Paris Agreement, which promotes voluntary cooperation and emission trading to meet NDCs, supporting its goal of net zero by 2070.

Institutional Measures



1. **Legal foundation:** The Energy Conservation (Amendment) Act, 2022 empowers the central

government to establish a carbon credit trading scheme.

2. National Steering Committee: The NSCICM coordinates policy design and implementation and ensures alignment with international standards.
3. PAT scheme: The Bureau of Energy Efficiency administers Perform, Achieve and Trade, which rewards energy-intensive industries with tradable energy saving certificates.
4. Digital MRV: Blockchain-based monitoring, reporting and verification of greenhouse gases ensures transparency and prevents double counting.

Market and Mitigation Activities

1. Carbon Credit Trading Scheme: Launched in 2024, the CCTS covers nine energy-intensive sectors, letting firms trade allowances by abatement cost.
2. CDM experience: Over 1,500 registered CDM projects lay the groundwork for linking with Article 6 mechanisms.
3. Green Credit Programme: It awards green credits for environment-positive actions by individuals, communities and industries.
4. State action: Andhra Pradesh's industrial hubs

like Visakhapatnam and its port-led corridors have adopted PAT efficiency measures and renewable energy obligations, showing regional commitment to national policy.

Challenges

1. Market readiness: CCTS participants, especially in Andhra Pradesh's emerging industrial clusters, need continuous capacity building to trade allowances well.
2. International rules: Ongoing Article (6) negotiations complicate seamless integration.
3. Equity: Scaling across sectors and ensuring equitable access to carbon markets remain hard.

Way Forward

Train regulators and firms, expand blockchain MRV, scale up Mission LiFE for community credits, and engage actively to finalise Article (6) rules.

Conclusion

Anchored in strong institutions and innovative market tools, these activities cut CO_2 emissions, help India trade credible credits abroad and build a low-carbon economy.

Write your own answer to 6(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2023 · Paper V

Scan to practice

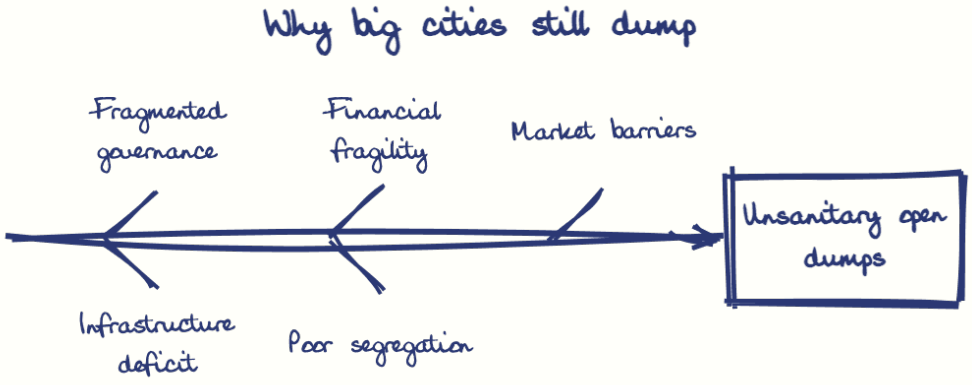
6(b) Explain various challenges associated with disposing of massive amounts of Municipal solid waste. Discuss action plan for municipal solid waste management in India for cities generating waste > 500 tonnes per day.

10

Introduction

Rapid urbanisation and economic growth have caused an unprecedented rise in municipal solid waste (MSW) in India. Cities generating over 500 tonnes a day face acute difficulty in disposing of it safely, and improper disposal drives pollution, disease and climate impacts, undermining urban livability.

Challenges in Disposing of MSW



1. Fragmented governance: ULBs are constitutionally responsible but lack clear roles and accountability, and only (5) of (17) test-checked Andhra Pradesh ULBs had bye-laws by (2022).
2. Infrastructure deficit: The state's (123) ULBs

generated about 6,890 tonnes daily in 2022, yet only 4 had operational sanitary landfills and 82 lacked wet waste plants.

3. Financial fragility: Heavy dependence on government grants and tight budgets limit investment in modern technology and long-term planning.

4. Poor segregation: About 75–80% of MSW is collected nationally, but audits show 75% segregation against a claimed 98%, leaving mixed waste.

5. Unsanitary dumps: Open dumps without liners cause leachate contamination of groundwater and methane emissions.

6. Health and market barriers: Rats and flies spread cholera and dysentery near dumpsites, and compost plants sit underused due to low demand.

Action Plan for Cities above 500 TPD

1. Integrated systems: Every such city needs its own MSW plan with source segregation, decentralised processing units, biomethanation and waste-to-energy with environmental safeguards, ending in scientific landfill sites.

2. Institutional capacity: Empower ULBs with

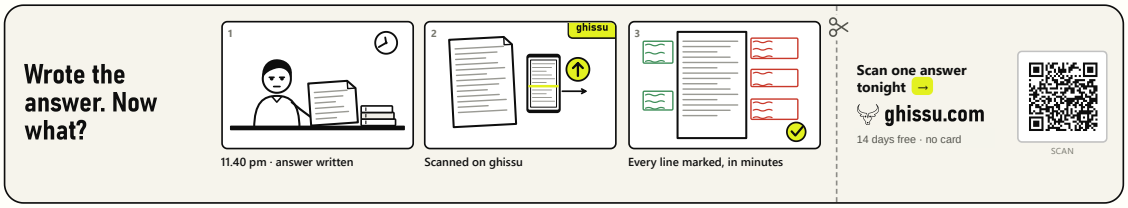
capacity building, clear regulatory frameworks and financial support, including public–private partnerships for efficiency and innovation.

3. Legacy waste: Remediate old dumps under the Dumpsite Remediation Accelerator Programme (DRAP), reclaiming land.
4. Community engagement: Run awareness campaigns and incentivise segregation at source, involving citizens, informal waste workers and NGOs in an inclusive system.

Together these steps break the cycle: segregation feeds processing, and processing shrinks what reaches landfills.

Conclusion

Disposal in large cities needs a holistic mix of institutional reform, technology and community participation. Building on Swachh Bharat Mission, sustained effort can protect health, cut pollution and support sustainable urban development.



7(a) Biotechnology has unlocked a vast potential for improving human life, but the risks it poses are now a concern — Discuss throwing light on blessing and curse of Biotechnology.

10

Introduction

India's bioeconomy has crossed USD 165 billion, driven by the BioE3 Policy and Bio-RIDE scheme. Yet concerns over biosafety, ethics and misuse make it both a blessing and a curse.

Blessings of Biotechnology

Blessing and curse of biotechnology

Blessings	Curses
Healthcare	Biosafety risks
Agriculture	Ethical dilemmas
Environment	Regulatory and IP issues
Economy and jobs	Dual-use threat

1. Healthcare: It enabled personalised medicine, vaccines and diagnostics; India produces 60% of global vaccines, and the Serum Institute was pivotal in COVID-19.
2. Agriculture: Bt cotton covers 95% of India's

cotton area, raising yields and incomes, while drought-tolerant SAATVIK chickpea and genome-edited rice aid climate resilience.

3. Environment: Bio-based chemicals and carbon capture under BioE3 promote a circular bioeconomy.
4. Economy and Jobs: The sector is projected to reach USD 300 billion by 2030, and Bio-RIDE fosters startups and skilled jobs.

Curses of Biotechnology

1. Biosafety Risks: Released GMOs may cause pesticide resistance, harm non-target species and reduce biodiversity.
2. Ethical Dilemmas: Gene editing such as CRISPR faces resistance over "playing God", complicating policy.
3. Regulatory and IP Issues: Overlapping rules delay approvals, as in the long Bt brinjal moratorium, and patent disputes hinder farmers' access.
4. Dual-Use Threat: Synthetic biology could be misused for biowarfare.

Balancing Promise and Peril

1. Risk-Benefit Analysis: Compare products with alternatives, such as Bt crops against chemical

pesticides.

2. Public Engagement: Transparent communication counters misinformation and builds trust.
3. Unified Governance: A single Biotechnology Authority can streamline approvals and speed safe innovation, including in AP's biotech hubs.
4. Biosecurity: Ethical guidelines, robust biosafety protocols and international cooperation prevent misuse.

Way Forward

1. State-Level Innovation: Expand biotech research infrastructure and skills in Andhra Pradesh.
2. Awareness Campaigns: Targeted outreach can build trust and address ethical concerns.

Conclusion

Biotechnology can greatly improve health, food security and sustainability, but its risks are real. Sound regulation, public trust and strict biosecurity will ensure its blessings outweigh its curses.

REVIEWER'S NOTE · CRITICALLY EXAMINE

"Critically examine" pays for the counter-view. An answer that stops at the first view has stopped at half the marks.

ghissu

ghissu exam-hall thinking decodes every directive — Discuss, Examine, Elucidate — before you write a line.

Decode your next question →

ghissu.com

14 days free · no card

SCAN

7(b) Why is Ecotourism considered as a tool for conservation of Natural heritage ? Discuss challenges associated with Ecotourism.

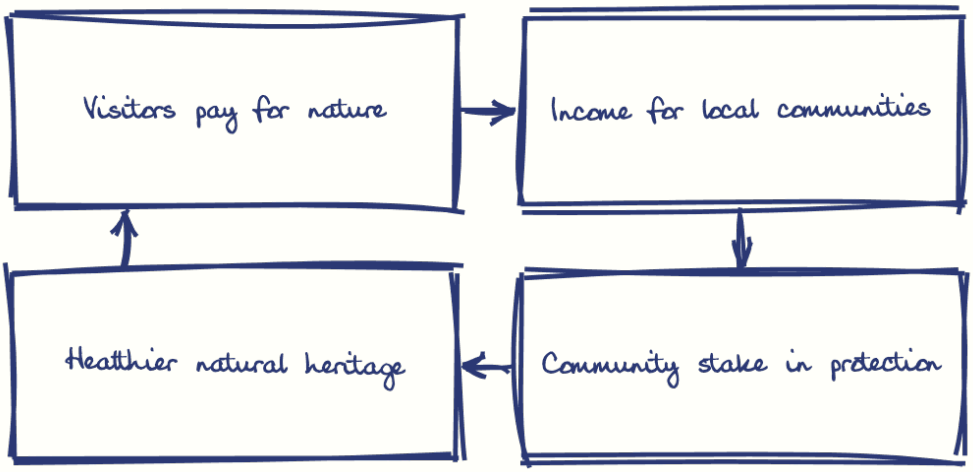
10

Introduction

Ecotourism has gained prominence as a sustainable alternative to conventional tourism in ecologically sensitive regions. Amid environmental degradation and biodiversity loss, it promotes responsible travel that supports ecosystems and local livelihoods. Andhra Pradesh, with rich biodiversity and coastal ecosystems, can use it to balance conservation and development.

Why Ecotourism Conserves Natural Heritage

Ecotourism as a conservation loop



1. Policy integration: Forest and tourism departments jointly promote protected areas like the Coringa Wildlife Sanctuary, tying tourism income to conservation of mangroves and coastal ecosystems.

2. Community participation: Tribal and rural people manage and benefit from natural resources, so they gain a stake in protecting them rather than exploiting them.
3. Regulation and standards: Waste management protocols and visitor limits in sensitive zones minimise ecological footprints.
4. Capacity building: Government and NGO training in hospitality, conservation education and sustainable resource use fosters stewardship of natural heritage among local stakeholders.

Challenges of Ecotourism

1. Environmental degradation: Poor management causes habitat disturbance, pollution and resource depletion, as in over-visited hill stations where waste and infrastructure strain fragile ecosystems.
2. Economy versus ecology: Benefits to communities must not come at the cost of nature, which needs continuous monitoring and adaptive management to keep ecosystems healthy.
3. Infrastructure gaps: Remote sites in Andhra Pradesh lack visitor management and emergency response, undermining both conservation and the tourist experience.

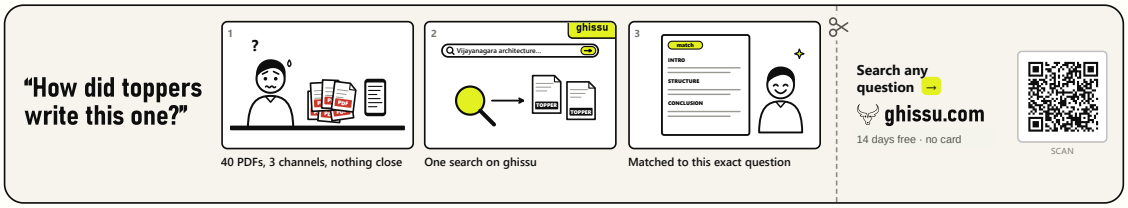
4. Cultural impact: Commodification of traditions risks social alienation and loss of cultural heritage.

Way Forward

1. Community-led governance: Give communities in Andhra Pradesh decision-making power and revenue-sharing models.
2. Environmental monitoring: Use technology and scientific methods to track ecological impacts and act before degradation sets in.
3. Sustainable infrastructure: Build eco-friendly facilities and connectivity.
4. Education: Run visitor awareness programmes on the value of natural heritage and local cultures to foster responsible behaviour.

Conclusion

Ecotourism conserves natural heritage by linking environmental protection with community welfare. Well managed, it safeguards biodiversity and cultural landscapes while providing livelihoods, in line with SDG (15.)



8(a) Why is antibiotic resistance getting worse in India ? Discuss the strategies and action plans in India to deal with this.

10

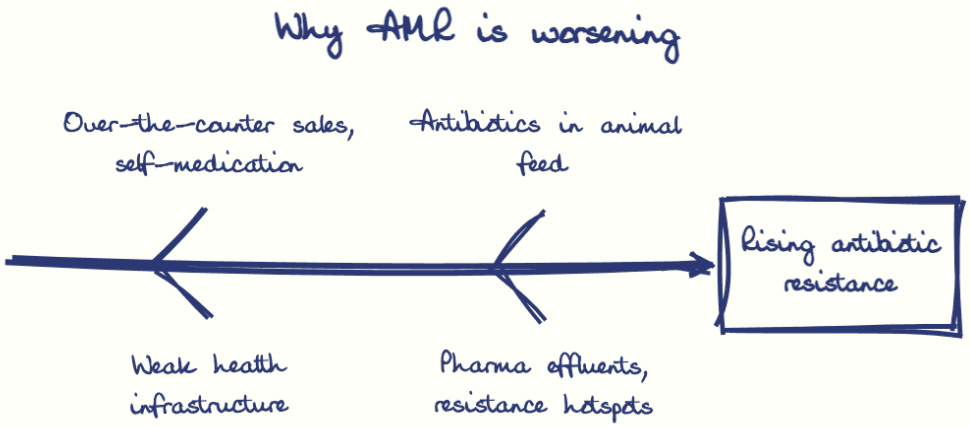
Introduction

Antibiotic resistance has become a critical public health challenge in India, threatening to undermine decades of medical progress. Misuse and overuse of antibiotics, weak healthcare infrastructure and regulatory gaps have accelerated it. Tackling it is urgent, especially in states like Andhra Pradesh where healthcare delivery is stretched.

Why Resistance Is Getting Worse

1. **Unregulated sales:** Easy over-the-counter availability without prescription leads to rampant self-medication, worsened by informal providers and pharmacies acting as primary care in rural areas.
2. **Weak infrastructure:** Andhra Pradesh had (743) specialist vacancies and only (175) Community Health Centres against a need of (412) (CAG 2024). Poor diagnosis encourages empirical and often wrong antibiotic use.
3. **Animal and veterinary use:** Antibiotics added to animal feed promote resistance that can transfer to humans.

4. Environmental contamination: Pharmaceutical effluents in industrial corridors like Visakhapatnam release residues into water, creating resistance hotspots.



Strategies and Action Plans

1. NAP-AMR 2017: The National Action Plan on AMR, aligned with WHO's Global Action Plan, covers awareness, surveillance, rational use, infection control and research through a One Health approach.
2. Surveillance: ICMR collects AMR data from (25) hospitals, though wider coverage is needed, including in Andhra Pradesh.
3. Regulation: Schedule H1 restricts over-the-counter sales, and national treatment guidelines standardise use, though enforcement is uneven in rural areas.

4. State action: Kerala's State Action Plan on AMR is a model, but few states, including Andhra Pradesh, have formal plans.

Way Forward

1. Infrastructure: Fill specialist vacancies and expand CHCs in Andhra Pradesh so accurate diagnosis replaces empirical antibiotic use.
2. Surveillance: Widen ICMR's AMR network to more public and private facilities in Andhra Pradesh for real-time monitoring.
3. Awareness: Sustained community campaigns in Andhra Pradesh on the dangers of self-medication and the need to complete prescribed courses.
4. Enforcement: Enforce Schedule H1 strictly and regulate pharmaceutical effluents in industrial hubs like Visakhapatnam.

Conclusion

AMR needs coordinated action across sectors and states. These measures, in line with N-AP-AMR, advance SDG ③ and keep antibiotics working for future generations.

150 · 250 · 500 words.
Same question, three model answers.
ghissu.com →

8(b) Bio-pesticides are slower in action compared to conventional pesticides, with shorter persistence and susceptible to unfavourable environmental conditions, but still there is a need to shift towards Bio-pesticides — Explain why.

10

Introduction

Environmental and health concerns over chemical pesticides have intensified the focus on sustainable farming. Pesticide residues threaten food safety and ecosystems in India, so the shift to bio-pesticides is gaining momentum despite their slower action, in line with the BioE3 Policy and Mission LiFE.

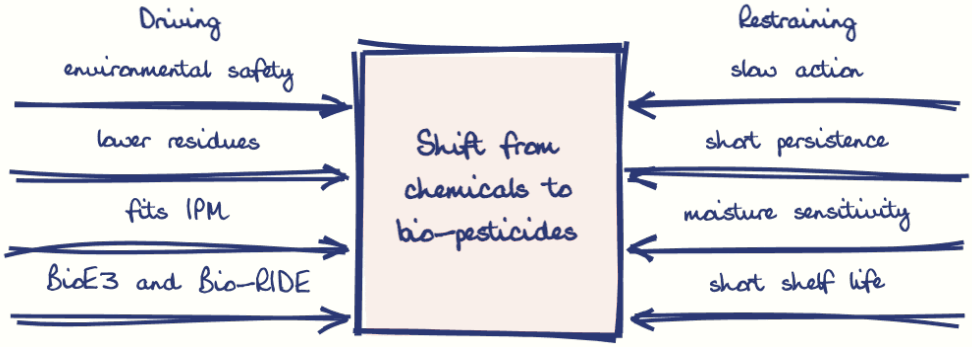
Limitations of Bio-pesticides

1. Slower Action: They work through infection or metabolic disruption; Bt must be ingested and disrupt the pest gut, unlike the immediate knockdown of chemicals.
2. Short Persistence: Ultraviolet radiation, temperature swings and rainfall degrade them, so repeated sprays raise costs.
3. Environmental Sensitivity: Entomopathogenic fungi need specific moisture, making them unpredictable in dry Rayalaseema.
4. Shelf Life: Living agents often need cold storage, unlike chemicals that last up to two years.

Why the Shift Is Still Necessary

1. **Environmental Safety:** Being biodegradable and target-specific, they spare beneficial insects and soil microbiota and prevent pest resistance and secondary pest outbreaks common with chemicals.
2. **Health and Residues:** Chemicals leave harmful residues in food and water; bio-pesticides lower them, addressing food safety concerns raised by the CIBRC.
3. **IPM Compatibility:** They work with cultural and mechanical controls under integrated pest management, as in chillies and mangoes in Andhra Pradesh's horticulture, improving quality without chemical overuse.
4. **Policy Alignment:** BioE3 and the Bio-RIDE scheme promote bio-based farm solutions, encouraging innovation and commercialisation, while Sri Venkateswara University in Andhra Pradesh runs research and field trials.

Shift to bio-pesticides



Overcoming the Limitations

1. Robust Formulations: Research on formulations with better environmental tolerance and longer shelf life, through programmes like Emerging Frontiers in Biotechnology.
2. Farmer Training: Extension services in Andhra Pradesh's Rayalaseema and coastal regions on use, waiting periods and integration with IPM.
3. Cold Chains and Incentives: Proper storage, subsidies and market linkages to keep products viable and affordable.

Conclusion

The weaknesses of bio-pesticides are technical and can be fixed, while the damage from chemicals is lasting. Short-term speed matters less than long-term soil, food and ecosystem health, which is why the shift is necessary.

Correct, but average? One line from your answer

YOUR LINE
~~The 1857 revolt failed.~~
✕ true · forgettable

→

WITH A VALUE ADDITION
The 1857 revolt failed for want of a centre: no common plan, no modern arms, and the loyalty of the Sikhs, the Gurkhas and most princes to the Company.
✓ cause · evidence · specifics

ILLUSTRATIVE

Add value to your next answer →
ghissu.com
14 days free · no card
SCAN

9(a) Why is the number of patents granted in India still a fraction compared to China and US, even though India's global position in innovations has been rising over the years ?

10

Introduction

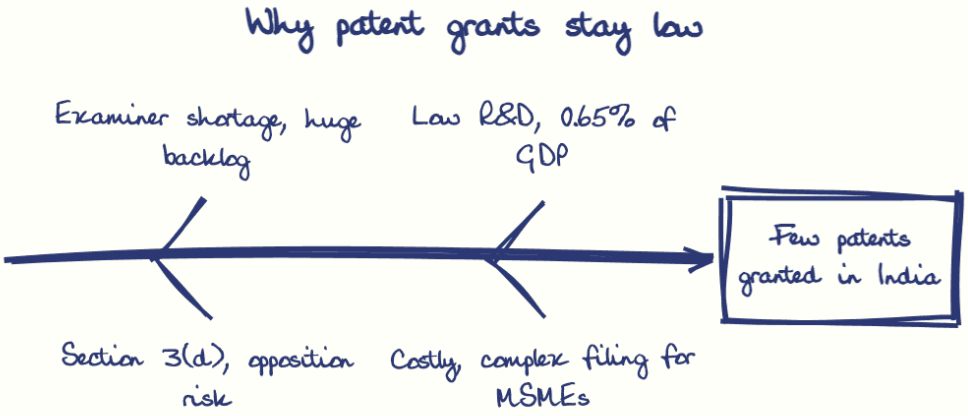
India climbed to 39th in the Global Innovation Index (2024), driven by rising patent filings, a vibrant start-up ecosystem and initiatives like the National IPR Policy and the Bharat 6G Alliance. Yet patents granted in India remain far fewer than in China and the US, due to structural and systemic factors.

Factors Behind Low Patent Grants

1. **Examiner shortage:** Only (132) examiners handled nearly (2 lakh) pending applications in 2016-17, with delays often over five years that discourage innovators.
2. **Stringent criteria:** Section 3(d) restricts incremental innovations, and pre- and post-grant oppositions raise litigation risk; these protect access to medicines, as in Novartis AG v. Union of India (2013).
3. **Low R&D investment:** GERD is about (0.65%) of GDP against China's (2.68%) and government dominates R&D with little private

participation.

4. Complex, costly filing: Cost and lack of expertise deter start-ups and MSMEs in tier-2 and tier-3 cities; filings rose 50% in five years, but conversion to grants stays low.



Why Innovation Rank and Patents Diverge

1. Broader metrics: The ranking reflects ICT services, market scale and venture capital; India ranks first in ICT services exports and third in late-stage venture capital deals.
2. Service-based innovation: Much innovation is in software, business models and services, which are less patent-intensive than manufacturing or pharmaceuticals, so many innovations never seek patents.
3. Policy contrast: China's state-led R&D, streamlined processes and MIC 2025 speed

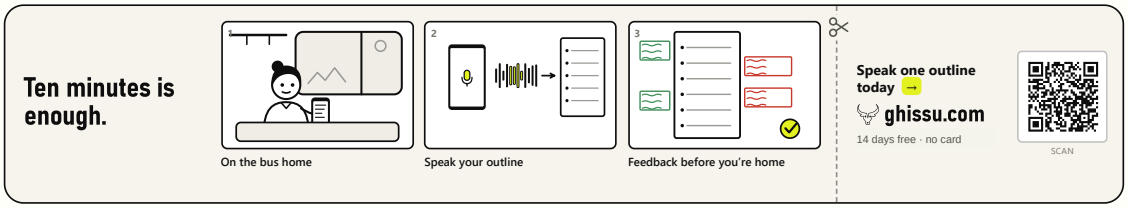
grants, while India balances innovation with public interest.

Implications

1. Weaker incentives: Delays and uncertainty reduce investment in R&D and commercialisation, hurting start-ups and SMEs that need patents to attract investors.
2. Strategic autonomy: Fewer grants limit India's ability to assert technological sovereignty and compete in high-value sectors like semiconductors and 6G technology, affecting the Bharat 6G Alliance.

Conclusion

Hiring over 450 examiners and expedited filing for Indian residents are steps forward. Higher private R&D and simpler, cheaper processes for start-ups and MSMEs can align patent grants with India's innovation rise.



Write your own answer to 9(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

Scan to practice

9(b) What are the tools required for the successful commercialization of Intellectual Property Rights in India ? Discuss the role of CIPAM in commercialization of Intellectual Property Rights.

10

Introduction

With rising patent filings and innovation indices, India needs tools and institutions to turn intellectual assets into marketable products and services. Commercialisation of Intellectual Property Rights is thus vital for growth and competitiveness, and the Cell for IPR Promotion and Management (CIPAM) bridges policy, awareness and facilitation.

Tools Required for Commercialisation

1. Robust legal framework: TRIPS adherence and the National IPR Policy 2016 strengthened patent laws, trademark protection and enforcement, so innovators can secure and monetise inventions.
2. Simplified procedures: Shorter examination timelines and fee concessions for start-ups and MSMEs lower entry barriers; the Startup Intellectual Property Protection Scheme speeds market access.
3. Innovation support centres: Technology and Innovation Support Centres give access to patent information and technical expertise, linking research to industry.
4. Awareness and capacity: Public campaigns and

IPR education in curricula build a culture of innovation and respect for IP.

Role of CIP-AM

1. Policy coordination: Under DPIIT, CIP-AM operationalises the National IPR Policy, coordinating ministries, states and stakeholders for a unified IP ecosystem.
2. Awareness and outreach: Workshops and digital campaigns reach rural innovators and industry, raising IP literacy, especially in states like Andhra Pradesh where innovation hubs are emerging.
3. Supporting start-ups and MSMEs: It manages the start-up protection scheme, providing financial aid and fast-track examination that cut costs and timelines for start-ups in Visakhapatnam and other Andhra Pradesh regions.
4. Digitalisation: Tools like the IP Saarthi Chatbot and virtual hearings improve transparency, though access remains uneven in rural Andhra Pradesh.

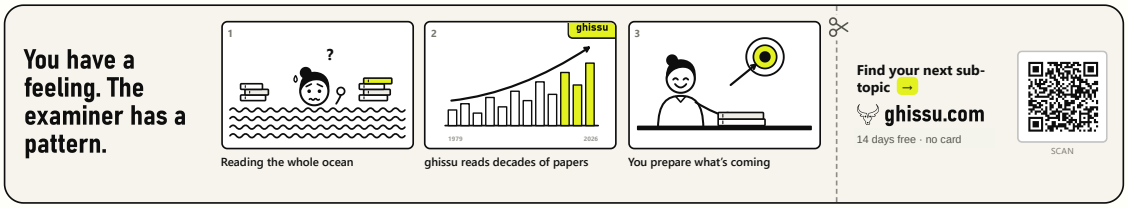
From invention to market

Way Forward

1. State-level facilitation: Stronger IP facilitation centres in Andhra Pradesh can give localised support and link research institutions with industry.
2. Financial incentives: More funding and subsidies for patent filing and commercialisation, especially for start-ups and MSMEs in corridors like Visakhapatnam–Chennai.
3. Industry-academia linkages: University–industry partnerships, coordinated by CIP-AM, can turn research into commercial products.
4. Digital inclusion: Better digital literacy and infrastructure in rural areas can widen access to CIP-AM's digital IP services.

Conclusion

Commercialisation rests on strong laws, simple processes, innovation support and awareness. CIP-AM orchestrates these elements, driving growth and technological self-reliance.



10(a) Inefficient farming and poor distribution are reasons for the world's food shortage and the genetically modified crops are a bad answer to the wrong problem — Comment.

10

Introduction

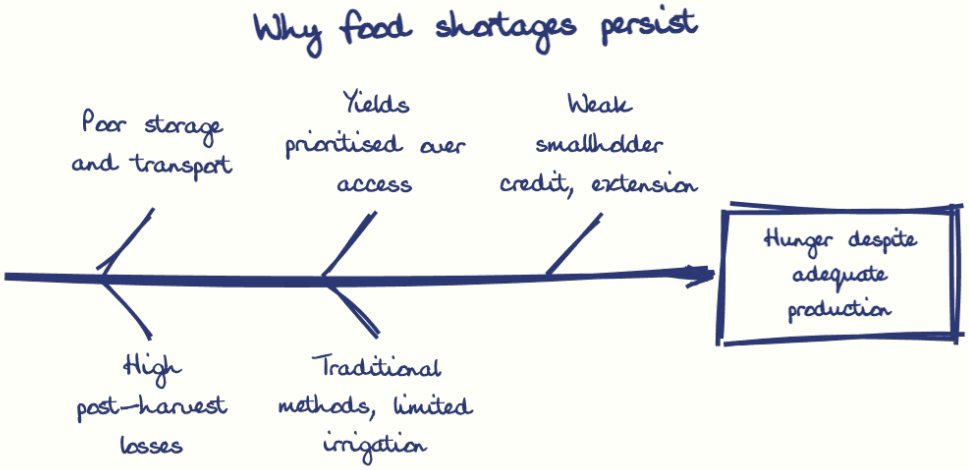
Food insecurity persists despite major gains in agricultural productivity. Over 2.6 billion people still lack access to nutritious and sufficient food, showing the problem is less about production and more about distribution and inefficiency. GM crops, promoted as a technological fix, therefore appear misplaced.

Food Shortage Is a Problem of Farming and Distribution

1. Fragmented Distribution: Inadequate storage and transport waste large quantities of food; Andhra Pradesh's supply chains suffer high post-harvest losses despite e-NAM.
2. Production over Access: Policy in India and Andhra Pradesh has prioritised yields through high-input farming rather than equitable access, neglecting rural poverty and market access.
3. Inefficient Farming: Despite the Green Revolution, many farmers in Andhra Pradesh still use traditional methods with limited irrigation, so

productivity stagnates and climate risk rises.

4. Weak Smallholder Support: Insufficient credit and extension in drought-prone Rayalaseema prevent small farmers from adopting technology or diversifying, worsening food insecurity despite production gains.
5. Lost Harvests: Poor cold storage and transport mean much food in Andhra Pradesh and across India is lost after harvest, causing shortages despite adequate production.



Why GM Crops Are a Bad Answer

1. Economic Risk: Expensive Bt cotton seeds and inputs trapped Andhra Pradesh farmers in debt cycles.
2. Environmental Harm: Chemical-intensive GM packages caused soil degradation and pest

resistance, even in the fertile delta regions of Andhra Pradesh, threatening long-term food security.

3. Regulatory Caution: India's approval of only Bt cotton reflects concerns over biosafety, seed dependency and ecological risk.

A Balanced View

Yields still matter for climate-hit farmers, so GM crops need not be rejected outright. But they cannot fix storage, roads, credit or poverty, and adopting them without robust biosafety frameworks and farmer-centric policies could deepen socio-economic inequality and environmental risk.

Conclusion

The statement is largely valid: hunger is fundamentally a problem of inefficient farming systems and poor distribution, not insufficient production. Cold storage, rural roads, agroecological practices and credit and insurance for smallholders, with prudent GM regulation, will better achieve SDG 2.

Improving — or just writing more?

My answers

Every attempt, every score, every enhanced version of your answer — on one timeline.

Watch your curve →

ghissu.com

14 days free · no card

SCAN

10(b) Why is monitoring and weather forecasting crucial ? Discuss the role of ISRO in the weather prediction and capabilities in Disaster management. Add a note on Mission NISAR.

10

Introduction

Accurate weather monitoring and forecasting have become indispensable as climate variability and extreme weather events rise. For India's agrarian economy and vulnerable coasts, timely information saves lives and protects livelihoods, and ISRO's satellites have greatly improved preparedness.

Why Monitoring and Forecasting Are Crucial

Weather risks and space assets of Andhra Pradesh

Cyclone-prone coast, drought-prone interior

1. Protecting Lives: Early warnings allow evacuation from cyclones and floods in coastal regions.
2. Agricultural Productivity: Weather-based advisories guide irrigation and cropping, as for farmers in drought-prone Rayalaseema and coastal Andhra.
3. Water Management: Satellite and ground data aid planning in the Krishna and Godavari basins.

ISRO's Role in Weather Prediction

1. Satellite Infrastructure: Cartosat and Oceansat-2 provide continuous earth observation for weather and ocean state forecasting, launched from Sriharikota.
2. Partnership: ISRO supplies data to IMD and NCMRWF for numerical weather prediction models, improving reliability and lead time.
3. Better Accuracy: IMD's accuracy has improved by (40%) in a decade; the Bharat Forecasting System gives (6 km) resolution forecasts, doubling spatial detail.

Capabilities in Disaster Management

1. Early Warning: GIS-based MHEW-DSS uses satellite data for impact-based, location-specific alerts covering nearly (80%) of the population.

2. Outcomes: Cyclone deaths have fallen to near zero, with faster response in coastal districts.

Note on Mission NISAR

The joint NASA-ISRO Synthetic Aperture Radar mission will provide high-resolution, all-weather imaging of Earth's surface. It will monitor floods, landslides and earthquakes, enabling near real-time damage assessment and strengthening disaster risk reduction.

Way Forward

1. Integration: Link satellite data with ground observation networks for hyper-local forecasts and early warnings.
2. Operationalise NISAR: Use its data to track floods, landslides and coastal erosion.

Conclusion

ISRO's satellites, better models and NISAR have transformed disaster management. Hyper-local forecasting, AI models and trained local authorities will further build climate resilience.

STUDY TIP · WHERE TO WRITE A NOTE

Write the note against the question that taught it. That is the only place revision will look for it.

ghissu

On ghissu every question has a notes tab — and your notes gather into chapter notes by themselves.

Open the notes tab →

ghissu.com

14 days free · no card

SCAN

11(a) Discuss how Artificial Intelligence manipulated digital media can impact privacy, democracy and national security of a country. Suggest solutions to tackle this infodemic.

10

Introduction

AI's integration into digital media has transformed how information spreads. As the IndiaAI Mission and India Semiconductor Mission speed AI adoption, manipulated content poses urgent challenges to privacy, democracy and national security in a diverse democracy like India.

Impact on Privacy

1. Data Exploitation: Manipulated media relies on vast personal data, risking unauthorised harvesting and breaches, as the Aadhaar data leak affecting millions, including residents of Andhra Pradesh, showed.

2. Deepfake Violations: Hyper-realistic deepfakes enable non-consensual image morphing and identity theft, hurting women most, as AI-generated sexualised images on 'X' showed.

3. Legal Gaps: IT Act sections 66C, 66E and (67) cover identity theft and privacy, but no AI-specific law exists.

4. Surveillance: AI facial recognition risks privacy and

freedoms, needing a balance with constitutional rights.

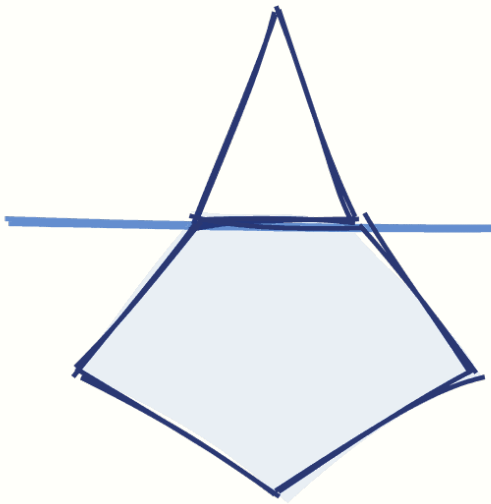
Impact on Democracy

1. Electoral Integrity: Deepfakes and disinformation can distort political narratives, mislead voters and potentially sway election outcomes.
2. Polarisation: Filter bubbles and misinformation, as with WhatsApp groups in communal tensions, erode trust in institutions.
3. Liar's Dividend: Spreaders of falsehood evade accountability by casting doubt on genuine evidence.

Impact on National Security

1. Cyber Threats: AI-powered misinformation can trigger panic or destabilise financial markets, a concern for Visakhapatnam port security.
2. Weaponisation: Autonomous weapons and AI surveillance create new vulnerabilities.
3. Foreign Interference: Cross-border disinformation can inflame internal tensions.
4. Institutional Gaps: The IT Act and voluntary codes are insufficient; a national AI strategy spanning defence, intelligence and civil governance is needed.

The AI infodemic



Visible
Deepfake videos
Morphed images
Viral fake news

Beneath the surface
Personal data harvesting
Liar's dividend
Filter bubbles
Foreign disinformation

Solutions to the Infodemic

1. AI-Specific Legislation: Address deepfakes, privacy and misinformation with effective enforcement.
2. Digital Literacy: Public campaigns and university courses on disinformation.
3. Technological Countermeasures: AI tools to detect and flag deepfakes, through public-private partnerships.
4. Multi-Stakeholder Governance: Government, industry, civil society and academia building transparent, accountable AI rules.

Conclusion

AI-manipulated media threatens the foundations of a free society. Combining law, technology and an informed public can contain this infodemic while keeping AI's democratic benefits.

Write your own answer to 11(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2023 · Paper V

Scan to practice

11(b) What is the current scenario of renewable power generation in India ? Discuss the factors responsible for the growing focus and shift towards Renewable energy.

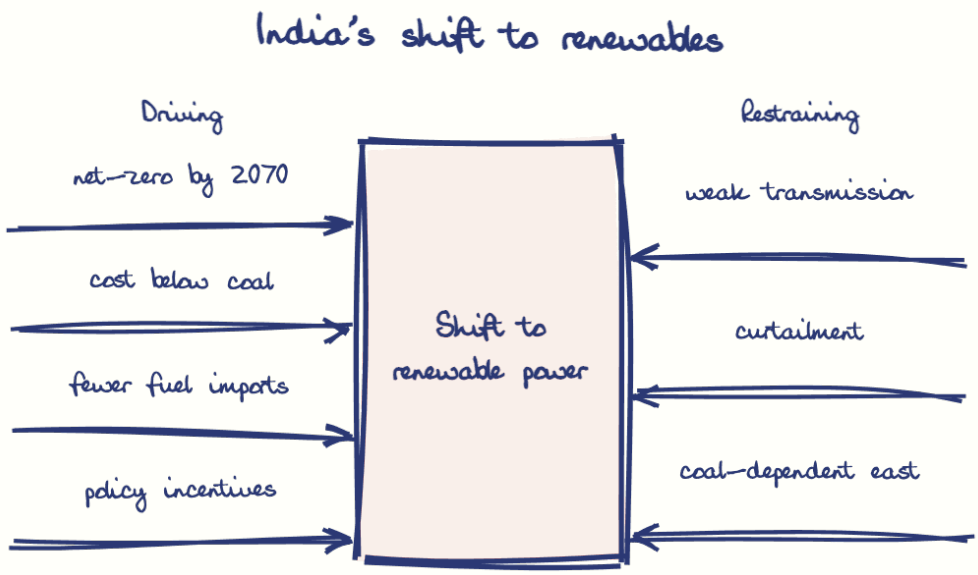
10

Introduction

India's renewable energy sector has transformed in recent years, driven by climate change, energy security and sustainable development. The shift from fossil fuels is vital to meet growing demand and international climate commitments.

Current Scenario of Renewable Power

1. **Rapid Expansion:** Renewables reached 156 GW by October 2024, led by solar and wind; Andhra Pradesh is among the top contributors, using its coastal and climatic advantages.
2. **Diverse Mix:** Solar reached 119.02 GW by mid-2025 and wind about 54.65 GW, alongside biomass, small hydro and green hydrogen.
3. **Regional Disparity:** Capacity is concentrated in western and southern states, with large solar parks and wind farms in Andhra Pradesh, while eastern India remains coal-dependent.
4. **Grid Challenges:** Weak transmission and limited storage cause curtailment.



Factors Driving the Shift

1. **Climate Commitments:** The net-zero by 2070 goal favours low-carbon alternatives to coal.
2. **Economic Competitiveness:** Solar and wind costs now match or undercut coal, drawing private investment.
3. **Energy Security:** Renewables cut fuel imports, a big part of the trade deficit.
4. **Technology:** Better solar PV, wind turbines and storage have improved efficiency and reliability.
5. **Global Leadership:** The International Solar Alliance raises India's standing and attracts FDI in clean technology.
6. **Policy Support:** The National Solar Mission, PM Kusum Yojana and CPSU Scheme Phase II

give incentives.

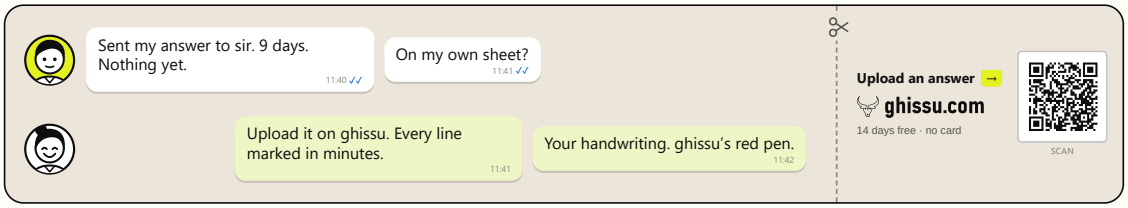
7. Coal's Limits: Supply constraints, price volatility and pollution make fossil power less attractive.

Way Forward

1. Grid and Storage: Modernise transmission to reduce curtailment.
2. Regional Balance: Targeted incentives for eastern India.
3. Local Manufacturing: Domestic solar modules and wind components.

Conclusion

Renewables are now central to India's power sector. Stronger grids, regional balance and stable policy will sustain the shift and secure a clean energy future.



12(a) What is coastal erosion ? Discuss factors causing coastal erosion. Why is erosion dominant in the Eastern coast than the Western coast in India ?

10

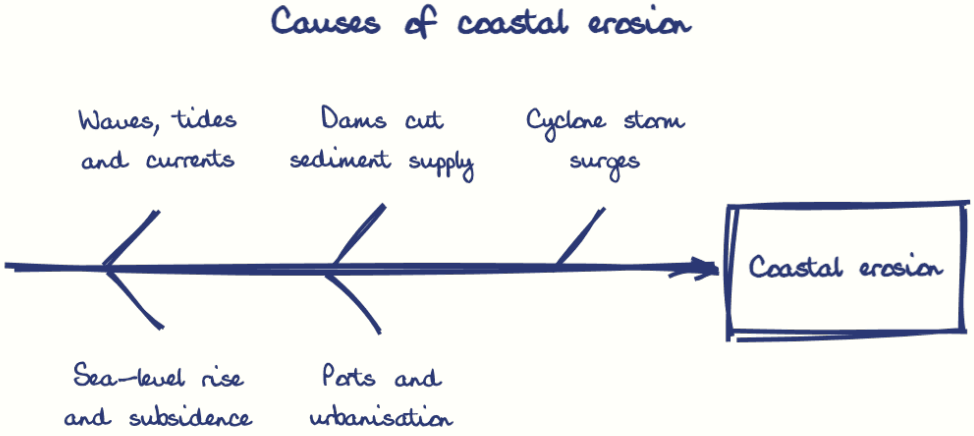
Introduction

Coastal erosion is a pressing concern in India. About 20% of India's population lives along the coast, so land loss threatens livelihoods, infrastructure and ecology. Studies show erosion is more pronounced on the Eastern coast, including Andhra Pradesh's long shoreline.

What is Coastal Erosion

1. Process: It is the wearing away and redistribution of sand and sediment by waves, tides and currents, causing the landward retreat of the coastline, rapidly in storms or gradually with sea-level rise.

Factors Causing Coastal Erosion



1. Natural forces: Waves, tides and near-shore currents remove sediment, while storm surges and cyclones cause sudden, severe events.
2. Sea-level rise and subsidence: Climate change raises sea levels, and sinking land raises relative sea level, especially on fine-sediment coasts.
3. Human activities: Ports, seawalls and groynes interrupt sediment flow, causing sediment starvation down-drift; removing mangroves and dunes exposes shores.
4. Sediment supply: Dams and river regulation reduce the sediment that replenishes beaches.

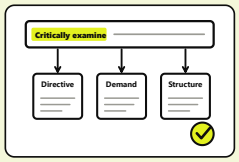
Why Erosion Dominates the Eastern Coast

Erosion-prone east coast of Andhra Pradesh

1. Cyclone frequency: More frequent and intense Bay of Bengal cyclones bring storm surges and strong waves.
2. Sediment deficit: Upstream dams and diversions on the Godavari and Krishna limit beach nourishment.
3. Geomorphology: Low-lying, gently sloping shores with finer sediment erode more easily than the rocky, steep Western coast.
4. Human interventions: Port development around Visakhapatnam and Kakinada disrupts sediment transport.

Conclusion

Integrated Coastal Zone Management, mangrove restoration, permeable groynes, monitoring and community engagement can protect the vulnerable Eastern coast.



Stuck at the directive?

Exam-hall thinking

The directive, the demand, the structure — before your first line. Full for beginners, Crux for veterans.

Walk in with a method →

ghissu.com

14 days free · no card

SCAN

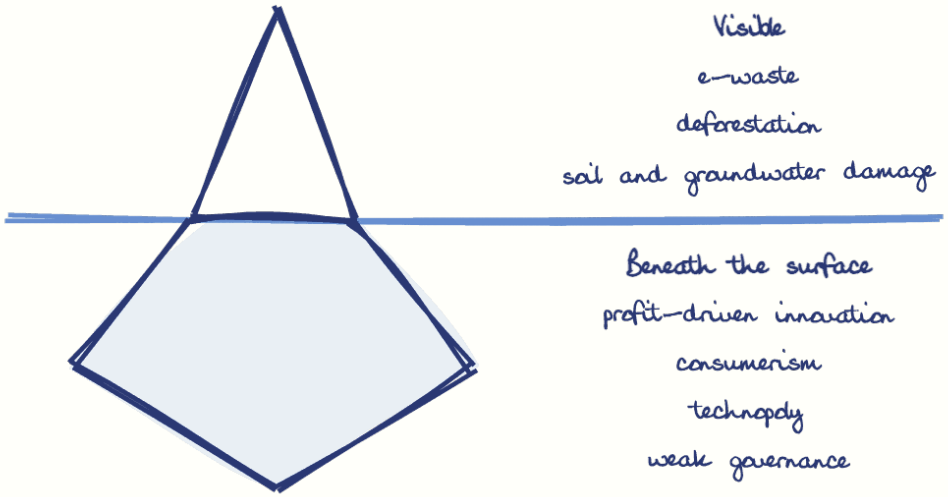
12(b) Discuss how technology translated into a medium for the destruction of environment through human exploitation.

10

Introduction

Rising heatwaves, erratic monsoons and biodiversity loss demand scrutiny of technology's role. While technology has propelled human progress, it has also become a conduit for the intensified exploitation of natural resources. This duality matters in India, and in Andhra Pradesh with its rich biodiversity and agrarian communities.

Technology and environmental destruction



Technology's Role in Environmental Destruction

1. Industrial pollution: Advances in machinery and energy production have raised air, water and soil contamination; India generates about 20 lakh tons

of e-waste a year, much of it improperly disposed.

2. Deforestation and habitat loss: Mechanised agriculture and infrastructure clear forests; the Polavaram dam caused significant forest loss, hurting wildlife and tribal communities.

3. Resource depletion: Overuse of chemical fertilisers and pesticides in drought-prone Rayalaseema degrades soil health and long-term productivity.

4. Technological fixes: Irrigation technology raises water efficiency but encourages unsustainable cropping and groundwater depletion.

Technology as a Medium of Human Exploitation

1. Large-scale extraction: Machinery and digital monitoring aid exploitation of forests, minerals and water; aquaculture along AP's 974 km coastline harms fisherfolk and coastal ecology.

2. Market-driven expansion: Innovation for profit ignores environmental cost, echoing Hardin's Tragedy of the Commons.

3. Displacement: Large projects such as Amaravati raise land acquisition and sustainability concerns.

4. Unequal benefits: A digital divide keeps the gains of sustainable technology away from marginalised groups, though Digital India has

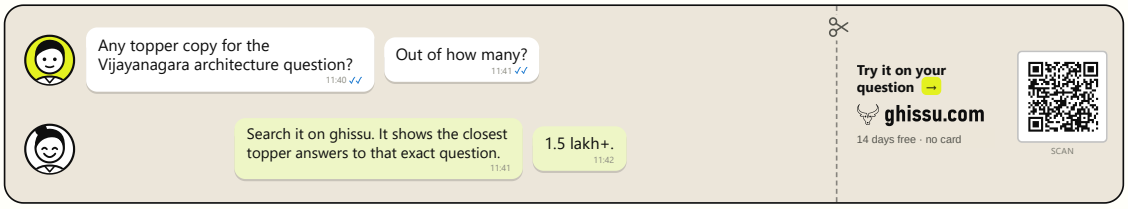
improved connectivity in rural AP.

Technology and Degradation: The Link

1. Technopoly: Neil Postman warned that technology's logic can override social and environmental values.
2. Consumerism: Technology fuels resource-intensive lifestyles, and the rapid turnover of electronic devices worsens e-waste.
3. Potential for good: ISRO's satellite technology supports environmental monitoring and disaster management in Andhra Pradesh.
4. Governance gaps: Weak enforcement of the Forest Conservation Act and Green India Mission limits technology's potential to curb degradation.

Conclusion

Technology has become a medium of destruction because profit and consumption, not ecology, have steered it. Promoting green technologies under effective governance can turn it from an instrument of exploitation into one of sustainability.



13(a) The most important climate goal is limiting the Earth's warming to 1·5° Celsius. What happens to the world if we pass the crucial 1·5° Celsius climate threshold?

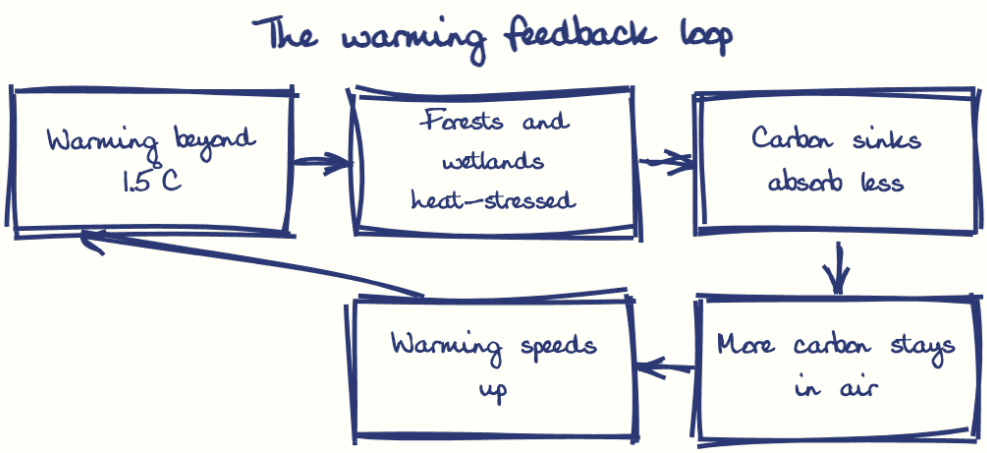
10

Introduction

2024 was the hottest year on record, reaching 1.55°C above pre-industrial levels. The Paris Agreement's goal of limiting warming to 1.5°C is critical to avoid catastrophic environmental and socio-economic consequences, especially for vulnerable regions such as Andhra Pradesh.

Environmental Impacts beyond 1.5°C

1. **Accelerated biodiversity loss:** Habitat degradation and extinctions disrupt pollination and nutrient cycling. AP's 974 km coastline faces sea-level rise and salinisation.
2. **Extreme weather:** Heatwaves, droughts and intense cyclones grow more frequent; Rayalaseema's water scarcity and soil degradation worsen.
3. **Marine damage:** Ocean warming and acidification cause coral bleaching and hurt fisheries in places like Visakhapatnam.
4. **Disrupted carbon sinks:** Heat-stressed forests and wetlands absorb less carbon, creating a feedback loop that speeds warming.



Social and Economic Consequences

1. Food security: Lower yields and more pest outbreaks threaten agrarian economies like AP's, where horticulture, chilli farming and aquaculture support millions.
2. Health and inequality: Heat stress, vector-borne disease and malnutrition rise, hitting the marginalised in drought-prone areas hardest and straining public health infrastructure.
3. Economic setbacks: Disaster relief diverts development funds, extreme weather disrupts Visakhapatnam's port-led corridors.
4. Migration: Degradation and livelihood loss drive internal migration, straining housing, sanitation and jobs in cities like Amaravati.

Why the Threshold Matters

1. Irreversible damage: The IPCC warns that passing

1.5°C risks crossing tipping points with cascading, irreversible effects.

2. Carbon budget: The remaining global carbon budget for 1.5°C is depleting fast, demanding immediate emission cuts.

3. Climate justice: Rural communities in climate-sensitive sectors contributed least to emissions but face the greatest risks.

4. Local action: AP's use of the National Adaptation Fund for Climate Change shows local action within the global framework.

Conclusion

Passing 1.5°C would turn manageable risks into irreversible ones, from collapsing ecosystems to mass displacement. Scaling up renewables, with AP contributing to India's 450 GW target through its solar and wind potential, and strong local adaptation are the surest way to stay within the limit.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 13(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

Scan to practice

13(b) Block chain technology is the doorway to transforming the agriculture sector in India — Discuss.

10

Introduction

With the National Blockchain Framework launched in September 2024 and deployed across NIC data centres including Hyderabad, blockchain has become a tool for transparent and efficient governance. Agriculture, which employs a large share of Indians and suffers from fragmented holdings, poor traceability and limited credit, can gain immensely from its secure and decentralised features.

Applications and Opportunities

1. Traceability and Certification: Immutable records of crop lifecycles enable farm-to-market tracing and can integrate organic certification under PGS and third-party systems, cutting costs and widening market access.

2. Digital Identity and Credit: Blockchain identities help small farmers build verifiable credit histories for microloans, as BanQu has shown.

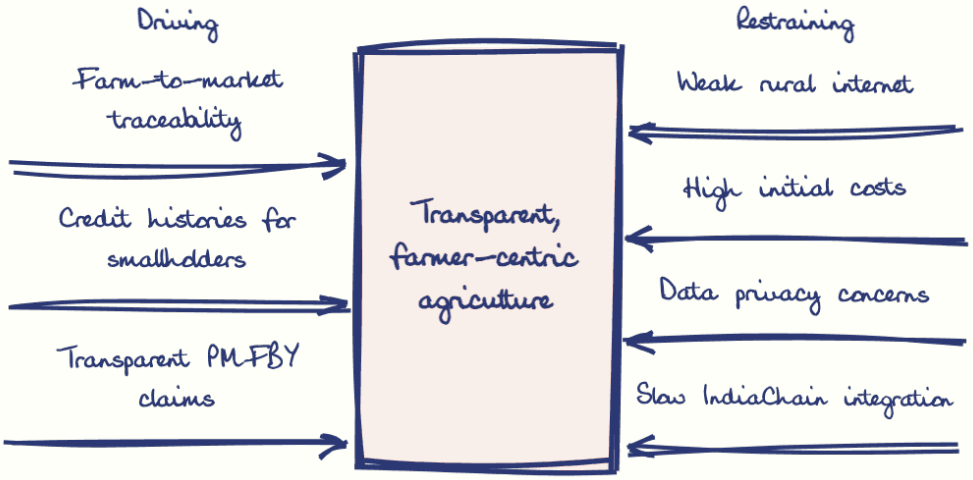
3. Supply Chains and Payments: With IoT tracking, it reduces intermediaries, lowers transaction costs and ensures timely payments, cutting post-harvest losses in AP's horticulture and aquaculture.

4. Scheme Delivery: Transparent PM-FBY claim processing reduces fraud, the Framework has already verified over 34 crore documents.

Challenges and Risks

1. Infrastructure and Literacy: Blockchain needs reliable internet and digital literacy, both uneven in rural areas, so the most vulnerable farmers risk exclusion.
2. High Initial Costs: Technology, training and devices are costly for smallholders, so government subsidy for devices and training is crucial.
3. Data Privacy: Immutability raises questions of data ownership, needing clear regulation.
4. Integration: Slow progress of IndiaChain shows the difficulty of aligning with existing databases.

Blockchain in Indian agriculture



Why it is a Doorway

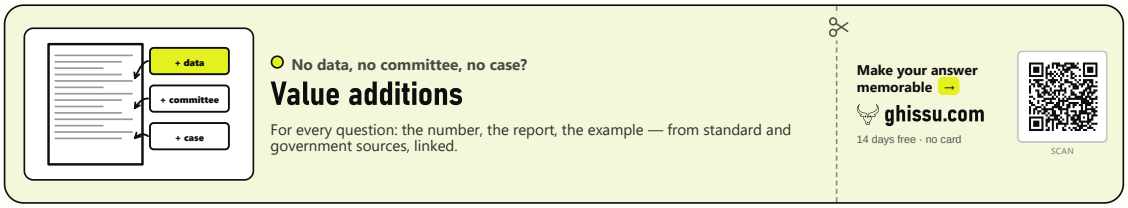
1. Transparency and Trust: Farmers can verify land records, ownership and transactions securely, reducing disputes and improving tenure security.
2. Markets: Proven authenticity and quality help farmers reach premium domestic and international markets, including growing organic exports.
3. Sustainability: Tracking farm waste for conversion into energy or compost supports GOBARdhan.
4. Digital Ecosystem: With Bharat VISTAAR and the Digital Agriculture Mission, it builds a digital agriculture ecosystem.

Way Forward

1. Rural Digital Infrastructure: Broadband and affordable smartphones in rural areas.

Conclusion

If connectivity, cost and regulatory gaps are bridged, blockchain can indeed be the doorway to a transparent, farmer-centric agriculture.



14(a) What are the reasons behind the India's opposition to sign the Non-Proliferation of nuclear weapons (NPT) ? Discuss the impact of the civil nuclear agreement with the Nuclear Supplier Group (NSG) on nuclear power generation in India.

10

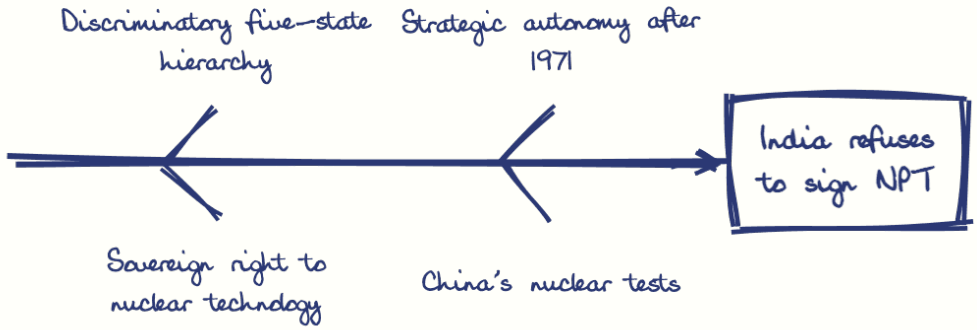
Introduction

India's nuclear policy has been shaped by its quest for strategic autonomy and equitable recognition. It opposes the NPT because it legitimises nuclear weapons for only five states. Meanwhile, civil nuclear deals after the NSG waiver have opened new paths for nuclear power, vital for growing demand and climate goals.

Reasons for India's Opposition to the NPT

1. Discriminatory Framework: The NPT gives exclusive rights to five nuclear weapon states and restricts others, undermining India's sovereign right to nuclear technology.
2. Strategic Autonomy: After the 1971 conflict and China's nuclear tests, India rejected any treaty limiting its deterrent.
3. Independent Path: Staying out brought isolation and sanctions but strengthened India's resolve, leading to the 1974 peaceful explosion and the 1998 tests.

Why India stays out of the NPT



Impact of the NSG Agreement on Nuclear Power

1. Access to Fuel and Technology: The 2008 NSG waiver recognised India's non-proliferation record and allowed civil nuclear imports under IAEA safeguards.
2. Capacity Expansion: Capacity is about 8.7 GW, planned to reach 22.48 GW by 2031–32 and 100 GW by 2047.
3. Implementation Hurdles: Liability laws and regulatory ambiguity slow investment and technology transfer.
4. Strategic Benefits: The US deal strengthened partnerships and opened scope for exporting nuclear equipment.

Way Forward

1. Liability Reform: Clarity for foreign partners.

2. NSG Membership: Sustained diplomacy for full membership.

3. Indigenous Technology: R&D in Small Modular Reactors to cut import dependence.

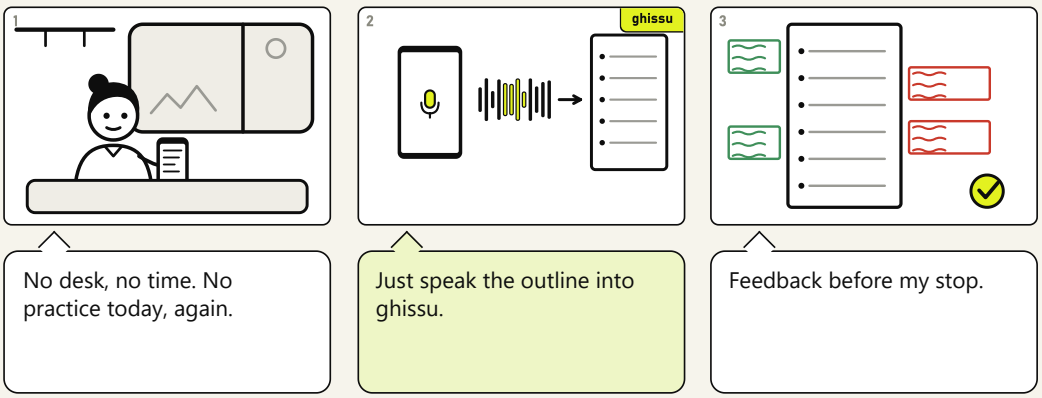
4. Regulation: A stronger Atomic Energy Regulatory Board to build global confidence.

Conclusion

India's opposition to the NPT reflects its insistence on autonomy and equitable nuclear governance, while the NSG waiver has boosted nuclear power through access to technology and fuel, strengthening energy security.

A STORY YOU MAY KNOW QUICK OUTLINE — TYPE, SPEAK OR SCAN

Practice on the bus



— An outline a day beats a full answer a week.

Try a quick outline →
ghissu.com
14 days free · no card
SCAN

14(b) What are e-waste recycling practices in India ? Discuss the highlights of new e-waste management rules 2022 notified by the Government of India.

10

Introduction

The rapid growth of electronic device use has caused a surge in e-waste in India, posing environmental and health risks. India produced about 3.2 million tonnes of e-waste in 2024, but about 90% of it was undocumented. This calls for strong institutions and updated rules for sound recycling and disposal.

E-waste Recycling Practices in India

E-waste recycling in India

Informal sector	Formal sector
Large share of recycling	Inadequate capacity
Lacks safety standards	CPCB standards, monitoring
Worker exposure, pollution	EPR and take-back

1. **Informal sector dominance:** A large share of recycling is handled by informal workers who lack safety and environmental standards, leading to hazardous exposure and pollution.

2. Regulated formal sector: MoEFCC and CPCB set standards and monitor compliance across states, but formal capacity is inadequate, with few environmentally sound facilities and no model installations in many states, including Andhra Pradesh.

3. Producer responsibility: Extended Producer Responsibility, introduced in the 2011 rules, makes producers collect and recycle e-waste and encourages eco-friendly design.

4. Collection and take-back: Consumer awareness and participation are low, so municipal corporations and social entrepreneurs are needed to expand take-back programmes.

5. Data gaps: Unreliable data hampers enforcement and infrastructure planning.

Highlights of the E-Waste Management Rules, 2022

1. Expanded scope: EPR now covers 106 categories of electrical and electronic equipment.

2. Mandatory registration: All manufacturers, producers, refurbishers, dismantlers and recyclers must register on a CPCB online portal, replacing authorisations and ensuring traceability.

3. Annual EPR targets: Producers get specific yearly

collection and recycling targets.

4. Ban on landfill and incineration: Only sound recycling and refurbishment are allowed.
5. Informal sector integration: Training and partnerships formalise informal recyclers and cut health hazards.

Way Forward

1. Local capacity: Set up dedicated e-waste aggregation and processing centres with modern technology in municipal bodies of Andhra Pradesh and other states.
2. Partnerships: Government, private enterprises and social entrepreneurs should build model recycling facilities and effective take-back systems.
3. Data and awareness: Digital tracking for accurate data, plus campaigns and incentives for consumers and producers.

Conclusion

Backed by the 2022 rules, sound e-waste management can reduce pollution and health risks and advance a circular economy.

STUDY TIP · BEFORE YOU OPEN A CHAPTER
List what was asked from it in the last ten years. Read that first; the rest second.

ghissu
ghissu has already done this for every topic — with the trend drawn out.

See the trend for your chapter →
ghissu.com
14 days free · no card
SCAN

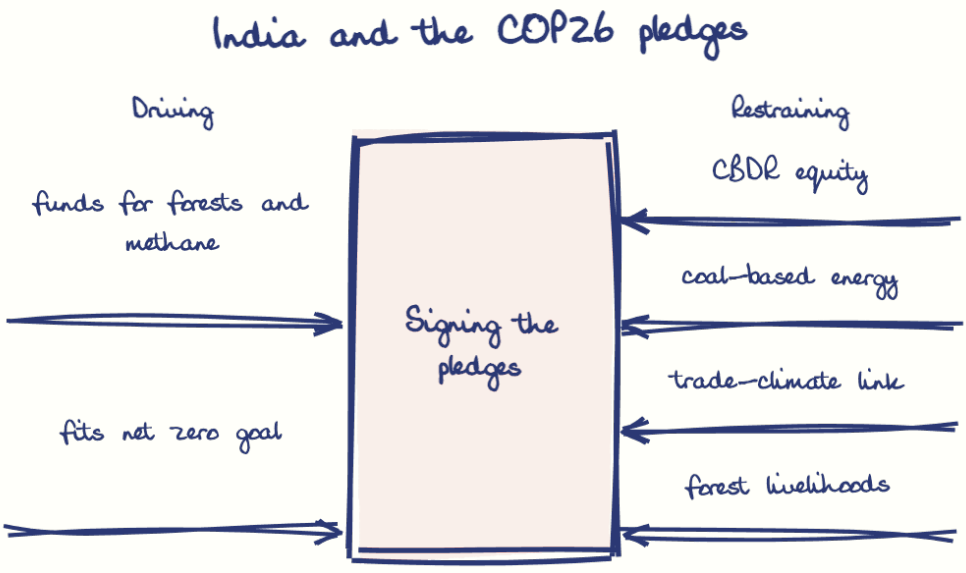
15(a) Discuss why India did not sign the COP26 pledge to stop deforestation and cut methane gas emission by 2030.

10

Introduction

India's decision not to sign the COP26 pledges to stop deforestation and cut methane emissions by 2030 reflects a balancing act between environmental commitments and development priorities. India has progressed on renewables and emission intensity but is wary of agreements that may constrain growth or sovereignty especially given vulnerable people such as those in Andhra Pradesh's Rayalaseema drought belt.

Reasons for Not Signing



1. Equity and historical responsibility: India's cumulative emissions are only about 4%

since 1850 though it holds 17.5% of the world's population, so under CBDR developed countries should bear more of the burden and finance.

2. Development and energy security: With 75% of energy coal-based, a rapid methane cut or deforestation halt would risk energy access, growth and forest-based livelihoods; India prefers a phased approach.

3. Trade and sovereignty: The pledge linked trade with climate, which India sees as the WTO's domain. It asked for trade language to be removed, fearing unfair export restrictions.

4. Existing commitments: The National ~~Afforestation~~ Programme and Green India Mission aim at a carbon sink of 2.5 to 3 billion tonnes CO_2 equivalent, and India launched Mission LiFE at COP26.

Implications of the Pledge for India

1. Growth constraints: Binding targets could limit farming and industry in rural and forest-dependent areas of Rayalaseema and the Eastern Ghats.

2. Forest communities: Strict controls could hurt tribal people who rely on forests, needing care

as in joint forest management.

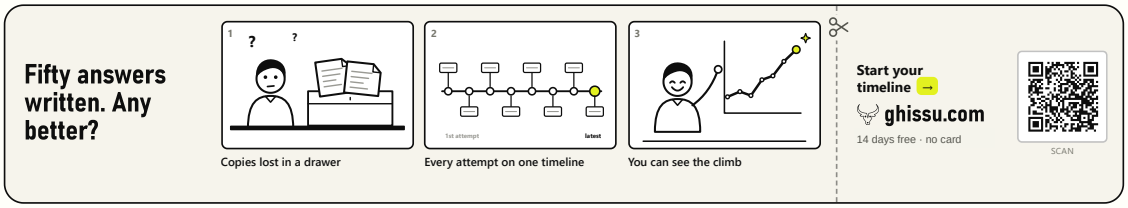
3. Feasibility: Tight timelines and international monitoring could force transitions not feasible without finance and technology.

National Interest

1. Poverty and climate: India seeks lower emissions intensity alongside energy access and poverty reduction, especially in Andhra Pradesh's Rayalaseema drought belt and coastal aquaculture zones.
2. Leadership on its terms: Through the International Solar Alliance and Mission LiFE, India leads without compromising sovereignty or growth.
3. Support needed: Methane and deforestation targets need advanced technology and finance, which India expects from developed nations.

Conclusion

India's refusal was not a retreat from climate action but a defence of equity and development, consistent with its net zero by 2070 goal.



15(b) What are Gene Sanctuaries ? How do they protect the genetic diversity of plants in India ?

10

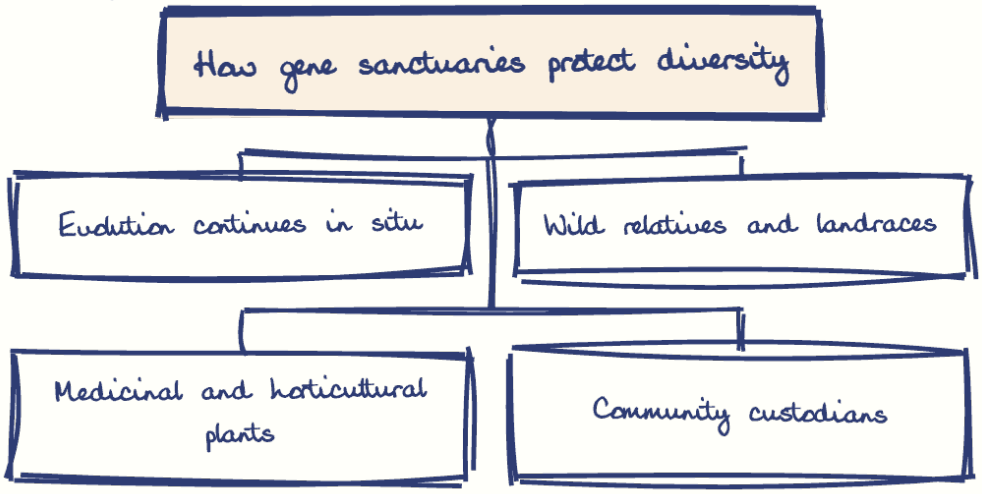
Introduction

Conservation of plant genetic diversity is critical for sustainable agricutture, food security and ecological balance. With habitat loss and climate change rising, gene sanctuaries have emerged as a vital in situ strategy that preserves plant diversity in natural habitats, complementing ex situ gene banks.

What Gene Sanctuaries Are

1. Meaning: Natural habitats protected to conserve wild plants and their relatives where they grow, letting evolution continue.
2. Institutional support: The Biological Diversity Act, 2002, the Convention on Biological Diversity, the Ministry of Environment and ICAR's NBPGR guide their creation; the citrus sanctuary in the Garo Hills arose under the Man and Biosphere programme.
3. Link with protected areas: They lie within or beside biosphere reserves and parks; the Nokrek Biosphere Reserve in Meghalaya conserves wild citrus relatives.

How They Protect Genetic Diversity



1. Evolutionary processes: Plants in natural habitats undergo natural selection, maintaining adaptive traits that ex situ collections may lose.
2. Wild relatives and landraces: These are preserved for crop breeding and resilience, sanctuaries exist for banana, sugarcane, rice and manga.
3. Medicinal and horticuttural plants: Success in Meghalaya and Sikkim for orchids and rhododendrons shows their ecological and cuttural value.
4. Community custodians: Women seed-guardians in Meghalaya preserve heirloom varieties, linking conservation with livelihoods.

5. Research and monitoring: NBPGR and ICAR centres keep germplasm and assess genetic erosion risks.

Limitations

1. Partial coverage: They cannot conserve a species' entire variability, so seed banks and field gene banks are needed too.
2. Management pressures: Agricultural expansion in Andhra Pradesh's Rayalaseema drought belt and scarce funds hamper management.

Way Forward

1. Institutional coordination: Link central agencies, state governments and local communities, especially in Andhra Pradesh, with adequate funds and technical support.
2. Combined approach: Integrate in situ and ex situ conservation through gene banks and field gene banks across agro-ecological zones.

Conclusion

Gene sanctuaries keep India's plant genetic diversity alive and evolving, and with gene banks and communities they secure future food security.

Where did you write that point?
Notes that live with the question.
ghissu.com →

Write your own answer to 15(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2023 · Paper V

APPSC GROUP-I MAINS · PAPER V

2020

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

1(a) What is "life expectancy at birth"? Data shows that the global average life expectancy has more than doubled since 1900, and is now above 70 years. Describe three to five factors which have contributed to improvement in life expectancy.

10

Introduction

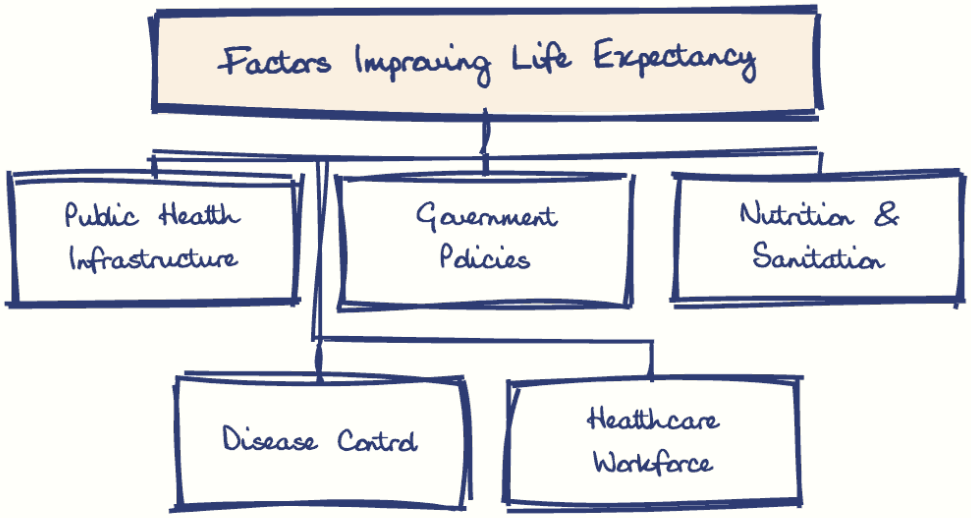
Life expectancy at birth is a key demographic indicator: the average number of years a newborn is expected to live under current mortality conditions. Since 1900 the global average has more than doubled to above 70 years, reflecting advances in health and living standards, including targeted gains in Andhra Pradesh.

Factors Behind the Improvement

1. Public health infrastructure: Immunisation and maternal-child health services drastically cut infant deaths; the National Health Mission improved access in Andhra Pradesh.
2. Decline in infant mortality: IMR fell from 30 per 1,000 live births in 2019 to 24 in 2024 nationally, so more children survive past infancy.
3. Government schemes: Janani Suraksha Yojana raised institutional deliveries in Andhra Pradesh and reduced neonatal deaths.
4. Nutrition and sanitation: Nutrition programmes and the Swachh Bharat Mission cut

malnutrition and waterborne diseases.

5. Contrd of infectious diseases: Vaccination and better treatment curbed tuberculosis and malaria, historically major killers, as Andhra Pradesh's active TB contrd programme shows.
6. NCD management: Awareness and care for hypertension and diabetes improved adult survival
7. Workforce and surveillance: More trained health workers reach underserved regions, and ICMR surveillance enables evidence-based policy against high-mortality conditions.



Way Forward

1. Rural healthcare: Expand PHCs and mobile units

in remote areas.

2. Nutrition and sanitation: Intensify state nutrition missions and sanitation drives.

3. Prevention: Run community awareness campaigns on lifestyle diseases for early diagnosis.

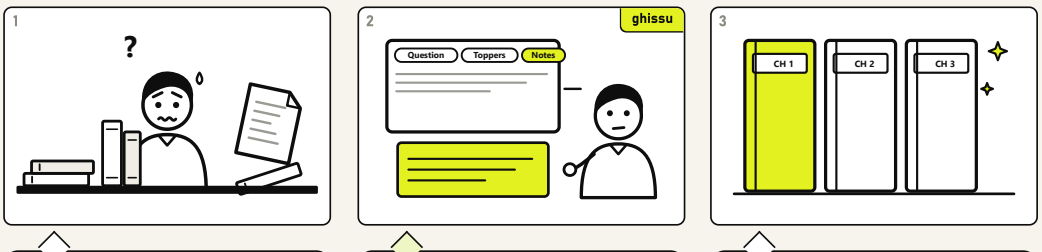
4. Health data: Upgrade information systems for real-time monitoring.

Conclusion

Life expectancy reflects the combined impact of health, social and economic factors. Equitable healthcare and prevention can make longer lives healthier too.

A STORY YOU MAY KNOW NOTES THAT LIVE WITH THE QUESTION

Where did I write that?



I KNOW I noted this. Which notebook...?

Note it right on the question.

And chapter notes build themselves.

— Note it where you learnt it.

Keep every note in place →

ghissu.com

14 days free · no card

SCAN

1(b) Telangana has several Central Government Institutions suitable for research funded by the Department of Science and Technology. These include BITS, CCMB, CDFD, CRIDA, DRDO, ICRISAT, IICT, NALSAR, NGRAU, and NIN. (i) How should these institutions contribute towards education of B.A./B.Com./B.Sc. students in TS? (ii) Write short notes about the mission and contributions of any two of these Institutions in Science/Technology in India.

10

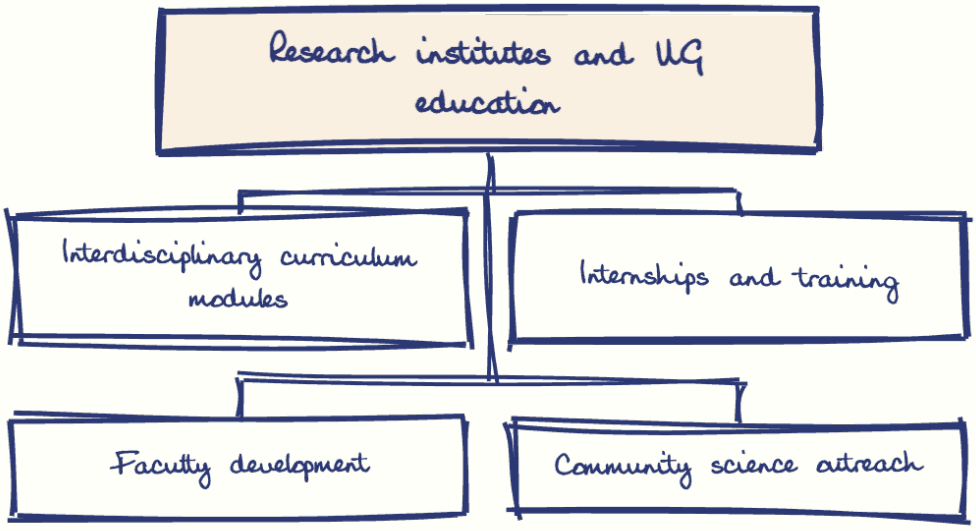
Introduction

Telangana combines an expanding higher education system with a strong presence of premier Central research institutions. Integrating them into B.A., B.Com. and B.Sc. learning can bridge the gap between theory and practical research, building a research culture and skills aligned with the state's socio-economic growth.

(i) Contribution to Undergraduate Education

1. **Curriculum Enrichment:** CCMB and ICRISAT can co-develop interdisciplinary modules with universities, exposing students to cutting-edge research and its applications.
2. **Internships and Training:** Structured internships in biotechnology, agriculture and materials science raise employability and research aptitude.
3. **Faculty Development:** Joint workshops and faculty exchanges can upgrade teaching methods in colleges short of qualified teachers.

4. Community Outreach: NIN and DRDO can involve students in community projects and science popularisation, building awareness of science in public health and national security.



Challenges

1. Weak Research Culture: Low doctoral enrolment and weak research environments in universities.
2. Resources: Many colleges lack infrastructure and qualified faculty.
3. Coordination: Clear policy links between DST, state education bodies and institutions are needed.
4. Equity: High privatisation makes access for poorer students a concern.

(ii) Mission and Contributions of Two Institutions

1. **CCMB, Hyderabad:** Its mission is frontier research in cellular and molecular biology, contributing to genomics and molecular medicine. It has pioneered work on genetic disorders, infectious diseases and bioinformatics, trains scientists and partners industry in biotechnology.
2. **ICRISAT, Hyderabad:** It works on agriculture in the dryland tropics for food security and livelihoods, developing drought-tolerant varieties and integrated natural resource management vital to Telangana's farms.

Way Forward

1. **Formal MoUs:** Institutionalise internships, joint projects and curriculum work.
2. **Project-Based Learning:** Add research components to undergraduate programmes.
3. **Inclusive Access:** Scholarships for underrepresented groups.

Conclusion

Linking these institutions with undergraduate education can transform Telangana's colleges, supporting Atmanirbhar Bharat and SDG (9) on industry, innovation and infrastructure.

Write your own answer to 1(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
Scan to practice

2(a) What is the UIDAI? How has this government body harnessed IT to improve targeted delivery of government benefits (DBT)? How can the Aadhaar Card help in financial transactions?

10

Introduction

A reliable identity system is crucial for inclusive governance and efficient service delivery. The Unique Identification Authority of India (UIDAI) built a robust digital identity infrastructure, with over 140 crore Aadhaar numbers issued, it has transformed how benefits reach citizens and eased financial transactions.

What is UIDAI

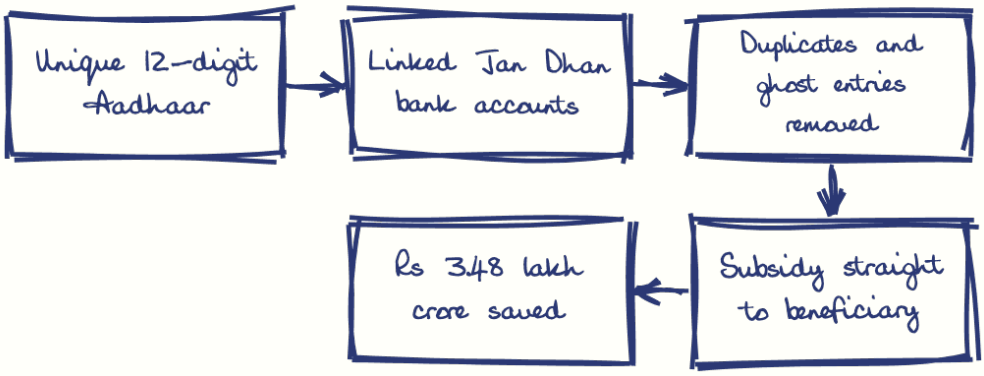
1. Statutory Authority: Set up in 2009 as an executive body and made statutory under the Aadhaar Act, 2016, it issues a unique 12-digit number to all residents to eliminate fake and duplicate identities.
2. Data Collected: Minimal demographic data (name, date of birth, gender, address) plus fingerprints and iris scans for reliable verification.
3. Scale: The world's largest biometric ID system, built on "one resident, one Aadhaar".
4. Concerns: Privacy, exclusion of those with worn

fingerprints, and surveillance fears need legal safeguards.

Harnessing IT for Targeted Delivery

1. DBT Backbone: Direct Benefit Transfer, launched in 2013, sends subsidies straight to Aadhaar-linked bank accounts, reducing leakages.
2. Savings: Cleansing databases of duplicates and ghost entries in LPG subsidy and MGNREGA saved over Rs 3.48 lakh crore by April 2025.
3. State Experience: AP uses Aadhaar-based DBT against fraud, though a 2024 CAG report found excess payments in its CFMS.
4. Limits: Literacy gaps, rural connectivity and data security still restrict benefits.

How Aadhaar powers DBT



Aadhaar in Financial Transactions

1. AePS: The Aadhaar Enabled Payment System allows cash withdrawals and transfers through biometric authentication, without cards.
2. Financial Inclusion: Linkage with over 57 crore Jan Dhan accounts brings millions to credit, insurance and subsidies.
3. Digital Payments: Integration with UPI enables secure, real-time payments.
4. Safeguards: Breach risks call for consent frameworks like DEPA.

Way Forward

1. Privacy: Comprehensive data protection and stronger UIDAI compliance.
2. State IT Governance: Real-time validation and duplicate checks in systems like CFMS.

Conclusion

UIDAI has made identity the foundation of targeted welfare and digital finance. Strong privacy laws and digital literacy will keep it inclusive and trusted.

Your intro decides your marks. Q. Discuss Ashoka's Dhamma.

YOUR INTRO
~~Ashoka was a great Mauryan king who ruled after the Kalinga war and spread Buddhism everywhere.~~
✕ no date · no source · no argument

→

AFTER GHISSU EVALUATION
Ashoka's Dhamma, set out in his Major Rock Edicts after Kalinga (c. 261 BCE), was not Buddhism by another name but an ethic of tolerance, non-violence and respect, meant to hold a plural empire together.
✓ date · source · argument

ILLUSTRATIVE

Get your intro fixed on your sheet →
ghissu.com
14 days free · no card
SCAN

2(b) What is cybercrime? Describe with examples any two of the following and explain how the government helps victims of cybercrime: Cyber bullying, Cyber stalking, Phishing, Ransomware.

10

Introduction

With over 969 million internet users as of March 2025, India's digital transformation has also raised its exposure to cybercrime. Rising threats erode trust in digital platforms and endanger people's privacy, safety and economic security, so protecting victims is essential.

What is Cybercrime

1. **Meaning:** Cybercrime is any unlawful act committed using computers, communication devices or networks to harm individuals, organisations or the state, such as hacking, identity theft, data breaches and online harassment.

2. **Legal Basis:** The IT Act, 2000 and IPC provisions such as Section 67 on obscene material and Section 354D on stalking enable prosecution.

3. **Impact:** It threatens privacy and free expression, causes economic loss and trauma, and hurts women and children most, cybercrimes against

women rose 239% between 2017 and 2022.

4. Underreporting: Fear, stigma and low awareness mean few cases are reported or end in conviction.

Two Examples

1. Cyber Bullying: Using digital platforms to harass or intimidate through rude messages, false information or non-consensual images. Gendered trolling and image-based abuse push women offline and can cause depression.
2. Ransomware: Malware that encrypts data and demands cryptocurrency. Attacks rose 69% between 2022 and 2023; the 2022 attack on AIIMS New Delhi crippled operations and exposed patient data. Ransomware-as-a-Service now lets unskilled criminals attack.

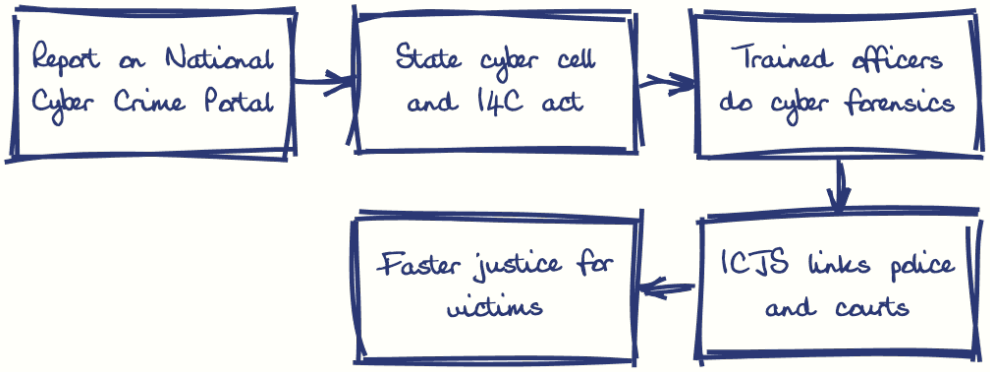
Government Support for Victims

1. Reporting: The National Cyber Crime Reporting Portal lets victims report anonymously and links to state cyber cells, including in Andhra Pradesh.
2. Coordination: I4C coordinates agencies and has trained over 25,000 officers in cyber forensics.

3. Awareness: Cyber Swachhta Kendra and CyTrain promote cyber hygiene, and the CCPWC scheme protects women and children.

4. Justice: The Inter-operable Criminal Justice System links courts, police and prisons to speed cases.

How the government helps victims



Conclusion

Both bullying and ransomware exploit technical gaps and human fear. Wider digital literacy in vulnerable communities and better-equipped state cyber cells will help victims get faster and fuller justice.

REVIEWER'S NOTE · CRITICALLY EXAMINE

"Critically examine" pays for the counter-view. An answer that stops at the first view has stopped at half the marks.

ghissu

ghissu exam-hall thinking decodes every directive — Discuss, Examine, Elucidate — before you write a line.

Decode your next question →

ghissu.com

14 days free · no card

SCAN

3(a) Write a brief account of the history of ISRO. Name any three ISRO centers and their location. Add a note about the Mangalyaan or Chandrayaan missions of ISRO.

10

Introduction

The Indian Space Research Organisation (ISRO) has been a cornerstone of India's scientific progress since 1969. It grew from modest beginnings into a global space player, and missions like Chandrayaan and Mangalyaan have enhanced India's strategic autonomy and international stature.

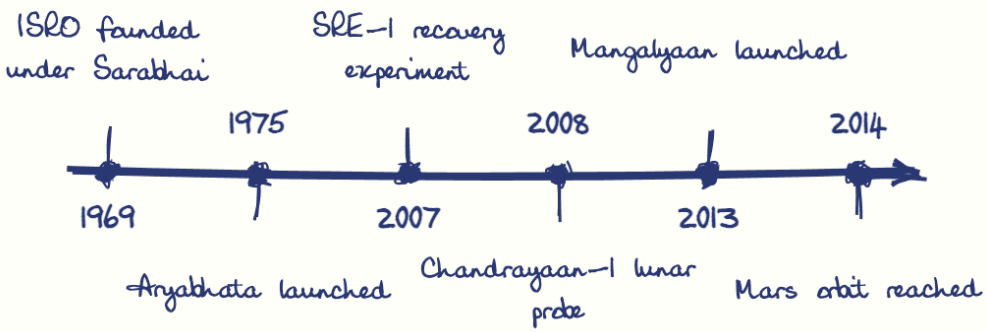
Brief History of ISRO

1. Founding and Vision: ISRO was established in 1969 under Dr. Vikram Sarabhai, who envisioned space technology as a tool for national development and scientific progress.
2. First Satellite: India entered the space age with Aryabhata in 1975.
3. Launch Vehicles: ISRO built indigenous vehicles, from the SLV to the PSLV, which became the workhorse for placing satellites in orbit.
4. Space Science: The Space Capsule Recovery Experiment (SRE-1) in 2007 demonstrated technologies for reusable launch vehicles and human spaceflight.

Practice 3(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2020 · Paper V

Scan to practice

Milestones of ISRO



Three ISRO Centres and Their Locations

1. Vikram Sarabhai Space Centre, Thiruvananthapuram: Develops launch vehicles and related technologies.
2. U R Rao Satellite Centre, Bengaluru: Designs, develops and integrates satellites.
3. Satish Dhawan Space Centre, Sriharikota (Andhra Pradesh): The primary launch centre for ISRO's satellite missions, including PSLV and GSLV launches.

Note on Chandrayaan and Mangalyaan

1. Chandrayaan Missions: Chandrayaan-1, launched in (2008) from Sriharikota, was India's first lunar probe and discovered water molecules on the Moon's surface. Chandrayaan-2 (2019) carried an orbiter with eight instruments to study the Moon's evolution and mineralogy.

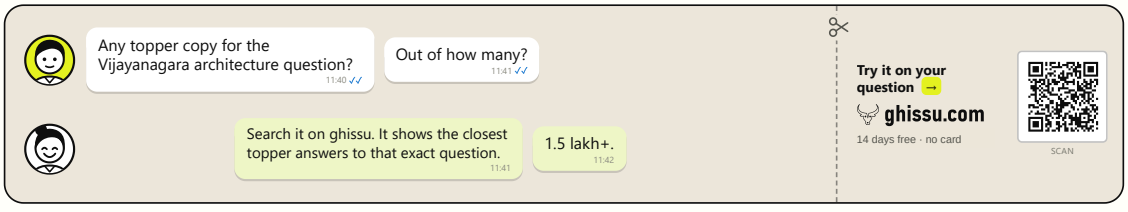
2. Mangalyaan: Launched in 2013, the Mars Orbiter Mission was India's first interplanetary mission and entered Mars orbit in 2014. It showed India could build, launch and navigate a spacecraft to another planet at a fraction of the cost of other countries.

Way Forward

Stronger R&D in indigenous launch vehicle and satellite technology, more planetary missions, wider use of Sriharikota for commercial launches and public–private partnerships will build on this record.

Conclusion

ISRO's history reflects India's commitment to using space for national development. Chandrayaan and Mangalyaan have raised its global status and strengthened technological self-reliance, and continued innovation will keep India at the forefront of space exploration.



3(b) To date, ISRO has launched over 100 satellites. Explain what are GEO, LEO, MEO, and SSO. Use diagrams to illustrate your answer. Add a note on GSLV and PLSV launches.

10

Introduction

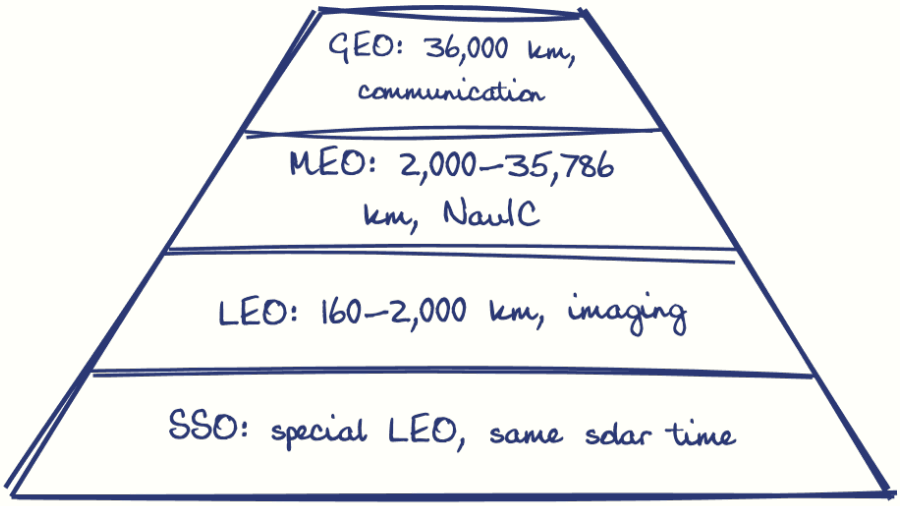
ISRO has launched over 100 satellites, a milestone in India's technological and strategic progress. They operate in different orbits, GEO, LEO, MEO and SSO, each serving distinct purposes in communication, navigation, earth observation and research.

Satellite Orbits and Their Characteristics

1. **Geostationary Orbit (GEO):** About 36,000 km above the equator, matching Earth's rotation period so the satellite appears stationary. It suits communication, weather forecasting and broadcasting with continuous coverage.
2. **Low Earth Orbit (LEO):** Between 160 and 2,000 km, satellites move fast relative to the surface, allowing high-resolution imaging, but continuous coverage needs a constellation.
3. **Medium Earth Orbit (MEO):** Between 2,000 and 35,786 km, it balances coverage and signal delay and is used for navigation systems like NavIC.

4. Sun-Synchronous Orbit (SSO): A special LEO that passes over the same area at roughly the same local solar time, giving consistent lighting for imaging, used for agricultural and environmental monitoring in Andhra Pradesh.

Satellite orbits by attitude



Note on PSLV and GSLV

1. PSLV: ISRO's reliable workhorse for LEO and SSO, with over (46) successful missions and multiple satellites in a single launch.
2. GSLV: Designed for heavier payloads to GEO and GTO, it uses indigenously mastered cryogenic technology and launches GSAT communication satellites. The LVM3 carries up to (4) tons and launched CMS-03 for naval communication.

3. Strategic Importance: Both have made India self-reliant in launch technology, reducing dependence on foreign services, and earn revenue by launching foreign satellites.

4. Launch Site: New launch pads and integration facilities at Sriharikota allow multiple launches.

Conclusion

From the Satish Dhawan Space Centre at Sriharikota, PSLV and GSLV place satellites in every orbit India needs, supporting agriculture, connectivity and India's place in the global space market.

A STORY YOU MAY KNOW MODEL ANSWERS IN 150 · 250 · 500

Three pages for ten marks

More words, more marks. Right?

...copy number 212. Page three. Skimming.

Same answer at 150 words. Now I see what to cut.

— Learn what to cut before the examiner does.

Learn what to cut →
ghissu.com
14 days free · no card

SCAN

4(a) What is nuclear energy? At the end of 2019, seven nuclear reactors were under construction in India, with a combined capacity of 5.4 GW. What are the advantages and disadvantages of using nuclear energy in India? What is the NPT and how does this impact India which is not a member of the NPT?

10

Introduction

Nuclear energy is the power released from the atom in reactors, a reliable low-carbon source. With seven reactors totalling 5.4 GW under construction by the end of 2019, it is central to balancing India's rising demand with its Paris commitment to cut carbon intensity, while ensuring energy security and technological autonomy.

Advantages and Disadvantages

Nuclear energy in India

Advantages	Disadvantages
Low carbon, reliable	Accident risk
Indigenous fuel cycle	High cost, long gestation
Largest thorium reserves	Radioactive waste, public fear

1. Reliable Clean Power: Large-scale, low-carbon supply cuts fossil dependence, states like Andhra Pradesh need it for industrial

corridors.

2. Expansion: 22 reactors (about 8.7 GW) today, 22.48 GW by 2031–32 and 100 GW by 2047 under the Nuclear Energy Mission.
3. Indigenous Technology: PHWRs and the Prototype Fast Breeder Reactor at Kalpakkam (2024) support Aatmanirbhar Bharat.
4. High Costs: Heavy upfront investment and long construction strain public finances and delay capacity.
5. Safety and Waste: Accident risk and radioactive waste remain, and public fear after Fukushima hurts acceptance.
6. Fuel: Uranium is scarce, but India has the world's largest thorium reserves, though thorium technology is still in R&D.

NPT and Its Impact on India

1. What Is the NPT: A treaty to prevent the spread of nuclear weapons, promote disarmament and enable peaceful cooperation.
2. India's Stand: India never signed it, seeing it as discriminatory since it recognises only five nuclear-weapon states.
3. Impact: It caused trade embargoes and

technology denial, eased by the 2008 NSG waiver and deals with the US, Russia, France and Australia.

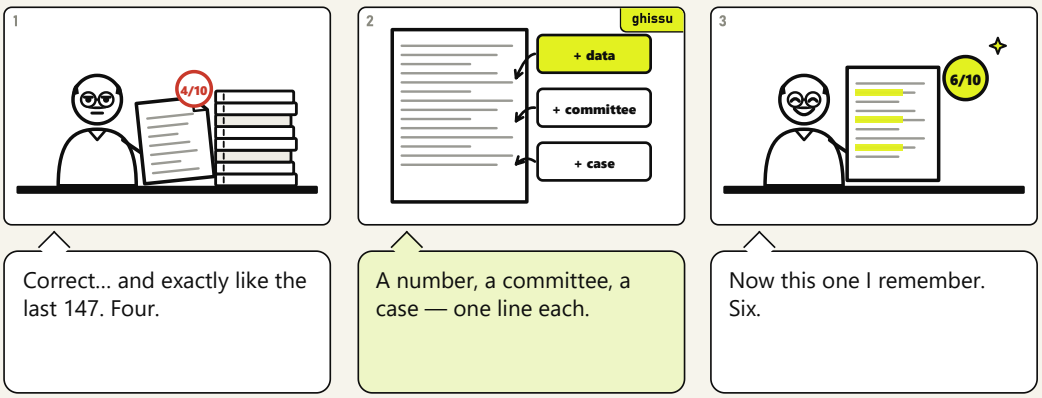
4. Balance: Staying out keeps India's sovereign right to develop nuclear technology while it engages in civilian nuclear commerce.

Conclusion

Nuclear power can secure India's clean energy needs if cost, safety and fuel limits are managed, while India preserves its strategic autonomy outside the NPT.

A STORY YOU MAY KNOW VALUE ADDITIONS

Copy number 148



— One line richer per paragraph: that is the gap between 4 and 6.

Make your answer memorable →
ghissu.com
14 days free · no card
SCAN

4(b) What is "Clean Energy"? List the sources of clean energy and explain how they work. Which of these sources can be used or are already widely used in India?

10

Introduction

Climate change and sustainable development have driven a global shift to clean energy, energy produced with little or no greenhouse gas emission or pollution. With rising demand, India targets 50% non-fossil installed capacity by 2030 and net-zero by 2070.

Sources of Clean Energy and How They Work

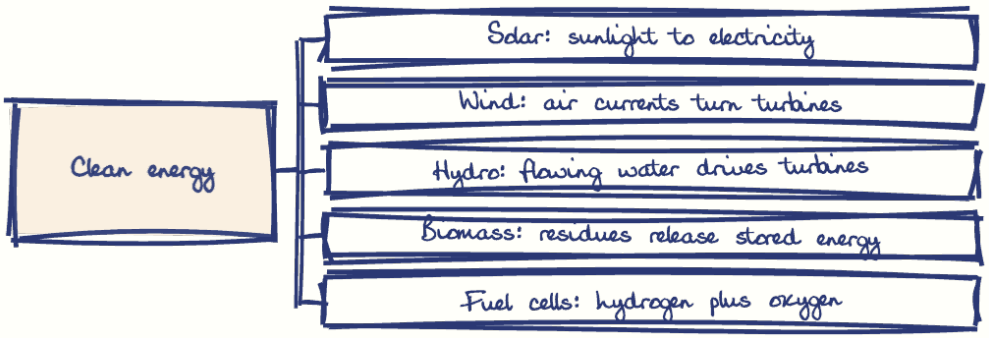
1. Solar: Photovoltaic cells or concentrating solar power systems convert sunlight into electricity.
2. Wind: Turbines convert the kinetic energy of air currents into electrical energy.
3. Hydropower: The potential energy of flowing water drives turbines.
4. Biomass: Burning agricultural residues and wood releases stored solar energy as heat or electricity.
5. Geothermal: Heat from the earth's interior is tapped for power.
6. Fuel Cells: Hydrogen reacts with oxygen to produce electricity, with only water as

Practice 4(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2020 · Paper V

Scan to practice

byproduct.

Clean energy sources and how they work



Use in India

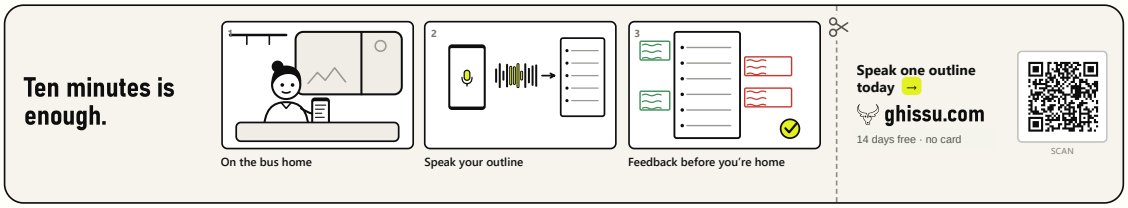
1. Sdar (Widely Used): Capacity rose from 3 GW in 2014 to 140 GW by January 2026, driven by falling costs; Andhra Pradesh's sdar parks along its 974 km coastline add to it.
2. Wind (Widely Used): 54.65 GW by January 2026, including Rayalaseema and coastal Andhra Pradesh.
3. Hydropower (Widely Used): Large projects on the Krishna and Godavari.
4. Biomass (Widely Used): Over 70% of people depend on it for daily energy; about 750 million tonnes are available yearly.
5. Geothermal and Fuel Cells (Potential): Emerging, with limited deployment so far.

Enablers and Challenges

1. Policy: The National Solar Mission aims for 500 GW by 2030, executed by MNRE and BEE.
2. Incentives: Generation Based Incentives and accelerated depreciation draw private investment, while the PAT scheme promotes efficiency.
3. Grid: Intermittency needs storage and smart grids to reduce curtailment.
4. Benefits: Lower emissions, air pollution and health risks.

Conclusion

Clean energy is vital for India's energy security and climate goals. Solar, wind, hydro and biomass are already widely used, and R&D in green hydrogen and storage can widen the rest.



5(a) All member states of the UN adopted 17 goals in the 2030 Agenda for Sustainable Development. Name any four of these goals and explain how India can reach them.

10

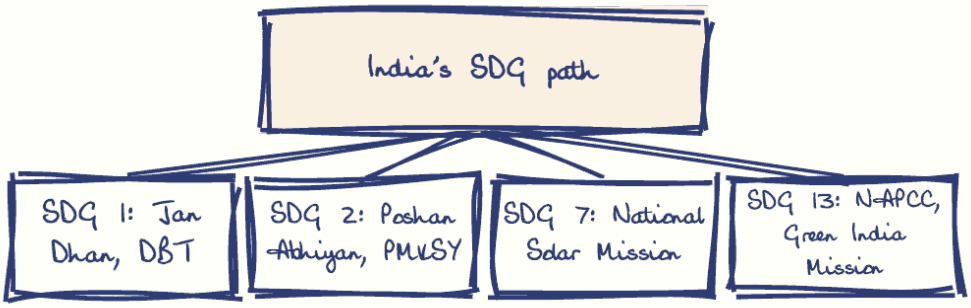
Introduction

Amid climate challenges and socio-economic disparities, the 2030 Agenda adopted by all UN member states integrates social, economic and environmental goals for inclusive, sustainable growth. For India, with vast development needs and ecological vulnerability, achieving these goals is essential for a resilient future.

Four Sustainable Development Goals

1. **No Poverty (SDG 1):** Eradicate extreme poverty, measured as living on less than $1.90 a day, and build social protection.
2. **Zero Hunger (SDG 2):** End hunger, achieve food security and nutrition, and promote sustainable agriculture.
3. **Affordable and Clean Energy (SDG 7):** Ensure universal access to reliable, sustainable, modern energy.
4. **Climate Action (SDG 13):** Combat climate change through mitigation, adaptation and resilience.

Four SDGs and India's route



How India Can Reach Them

1. Poverty: Jan Dhan Yojana and direct benefit transfers target subsidies; AP's DBT-aligned welfare reaches the marginalised, and the SDG India Index 2023-24 shows progress.
2. Hunger: Poshan Abhiyan improves nutrition for women and children in AP districts; horticulture and aquaculture support food production, and PM Krishi Sinchayee Yojana raises irrigation efficiency in drought-prone Rayalaseema.
3. Clean energy: The National Solar Mission expands renewable capacity, with AP using its coastal geography and solar potential, while the Smart Grid Mission and FAME India promote efficiency and electric mobility.
4. Climate action: The N-APCC and state action plans guide resilience; afforestation under the Green India Mission and Green Highways

Practice 5(a) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2020 · Paper V

Scan to practice

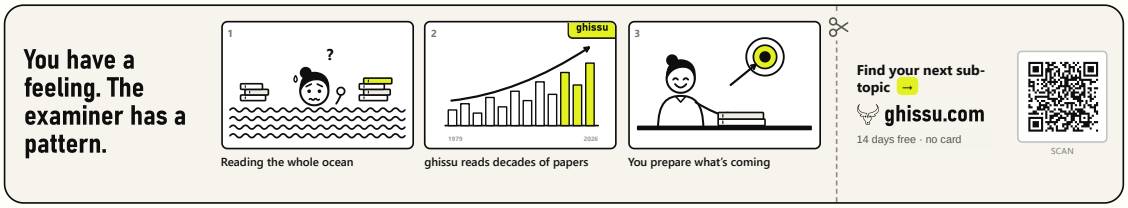
sequester carbon, and AP manages water through the Polavaram project and protects its 974 km coast.

Way Forward

1. Monitoring: Better data collection and real-time state-level tracking to tailor interventions.
2. Partnerships: Industry and civil society collaboration in AP to scale renewable energy projects and sustainable agriculture.
3. Infrastructure: Climate-resilient urban and rural assets, with PM Gati Shakti for logistics and disaster preparedness.
4. Community engagement: Education and capacity building for grassroots participation in sustainable practices.

Conclusion

Through targeted schemes and state initiatives, and guided by Panchamrit and net zero by 2070, India can reach these goals and leave no one behind.



5(b) What causes earthquakes? Can we prevent or predict where and when an earthquake will happen? What should relief agencies do after an earthquake occurs in an inland and a coastal region?

10

Introduction

About 59% of India is vulnerable to seismic activity, and India recorded 159 earthquakes between November 2024 and February 2025. Understanding their causes, the limits of prediction and effective relief is vital for states like Andhra Pradesh, with its long coastline and varied inland terrain.

Causes of Earthquakes

1. **Tectonic Plate Movements:** Most earthquakes occur along plate boundaries, where accumulated stress is suddenly released as seismic waves.
2. **Volcanism and Faulting:** Eruptions and the folding and faulting of rocks, as along the San Andreas Fault.
3. **Anthropogenic Causes:** Reservoir-induced seismicity, as in Koyna (1967), plus underground mining and nuclear testing.
4. **Local Geology:** Soil, rock type and topography shape how waves travel and how much damage occurs.

Can We Prevent or Predict Them?

1. **Prevention Impossible:** Their origin lies in deep Earth processes, so efforts focus on mitigation and preparedness.
2. **Prediction Elusive:** Exact time and place cannot be forecast, observatories rose from (80) in (2014) to (168) by (2025), improving monitoring, not prediction.
3. **Risk Zoning and Early Warning:** Hazard maps identify high-risk zones, while early warning systems give seconds to minutes of notice.

What Relief Agencies Should Do

1. **Rescue and Medical Aid:** Search and rescue, first aid and temporary shelters, led by the NDRF, now (16) battalions.
2. **Restore Services:** Quickly repair water, power and communication to prevent fires and epidemics, a priority in inland towns.
3. **Coastal Measures:** Add tsunami warning dissemination, evacuation, and rehabilitation of fisheries and aquaculture, vital to Andhra Pradesh's coastal economy.
4. **Community Participation:** Involve locals in reconstruction, as Gujarat's (2001) earthquake

showed.

Relief after an earthquake

Inland region	Coastal region
NDRF rescue and medical aid	Rescue plus tsunami warning
Restore water, power, communication	Evacuate coastal settlements
Prevent fires and epidemics	Rehabilitate fisheries, aquaculture

Way Forward

1. Seismic Monitoring: Modernise observatories and add GPS-based tracking for real-time data.
2. Building Codes: Enforce the National Building Code in inland and coastal Andhra Pradesh.
3. Awareness: Regular preparedness drills in inland and coastal areas, so communities respond well in the first hours.

Conclusion

Since earthquakes cannot be prevented or precisely predicted, resilient construction, stronger monitoring and community training are the best defence.

Are you actually improving?
Every attempt scored. Watch the curve.
ghissu.com →

6(a) Classify solid waste in terms of source and types of solid wastes. Explain the steps to manage this waste. Add a note on advantages and disadvantages of landfills.

10

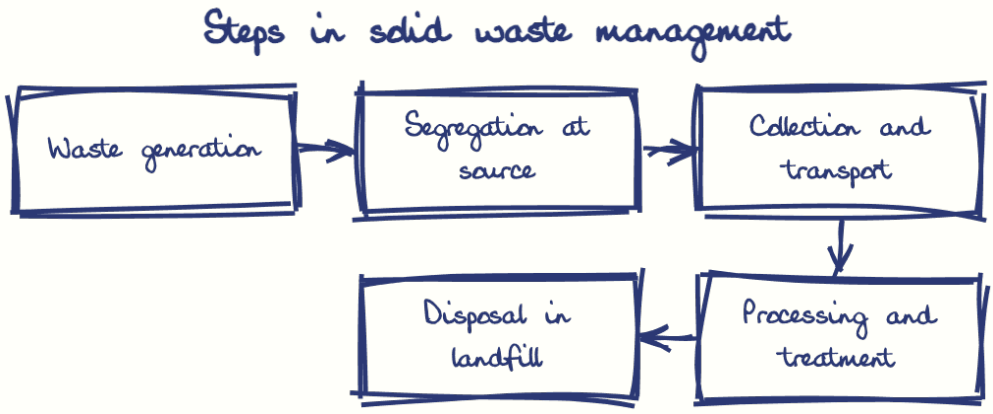
Introduction

Rapid urbanisation and rising living standards have sharply increased solid waste in India. Andhra Pradesh's 123 ULBs generated about 6,890 tons per day in 2022. Classifying waste and managing it systematically prevents pollution, disease and resource depletion.

Classification of Solid Waste

1. **By source:** Municipal waste from households, markets and shops; industrial waste from manufacturing; agricultural crop residues; biomedical waste from healthcare; and e-waste from discarded electronics.
2. **By type:** Biodegradable (food and garden waste), non-biodegradable (plastics, metals, glass), recyclable (paper, some plastics), hazardous (chemicals, batteries) and inert (construction debris).
3. **State context:** Much waste is biodegradable due to the agrarian economy, but Visakhapatnam produces growing non-biodegradable and hazardous waste.

Steps in Management



1. Segregation at source: Separating waste at generation cuts contamination; a 2022 audit found only about 75% actual segregation against claims of over 98%.
2. Collection and transport: Green, white and black bins carry waste to processing or disposal sites; efficient collection reduces littering.
3. Processing: Composting or biomethanation yields manure or biogas; recyclables go to units; hazardous waste is neutralised.
4. Disposal: Residual waste goes to landfills or incineration; proper disposal prevents pollution and health hazards.

Landfills: Advantages and Disadvantages

1. Advantages: Simple and economical; daily soil cover reduces odour and pests; closed sites can

become parks or parking lots.

2. Disadvantages: Mixed waste forms leachate that contaminates soil and groundwater, and organics release methane, a potent greenhouse gas. Only (4) of (123) ULBs had operational sanitary landfills.

3. Sanitary landfills: Lined with clay and plastic to stop leakage, they are costlier but more sustainable than open dumps.

Conclusion

Classification enables targeted strategies from segregation to sound disposal, and better sanitary landfills and processing in line with the SWM Rules, (2016) and Swachh Bharat Mission (Urban) (2.0,) will make the state's waste management sustainable.

STUDY TIP · WHERE TO WRITE A NOTE

Write the note against the question that taught it. That is the only place revision will look for it.

ghissu

On ghissu every question has a notes tab — and your notes gather into chapter notes by themselves.

Open the notes tab →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 6(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

Scan to practice

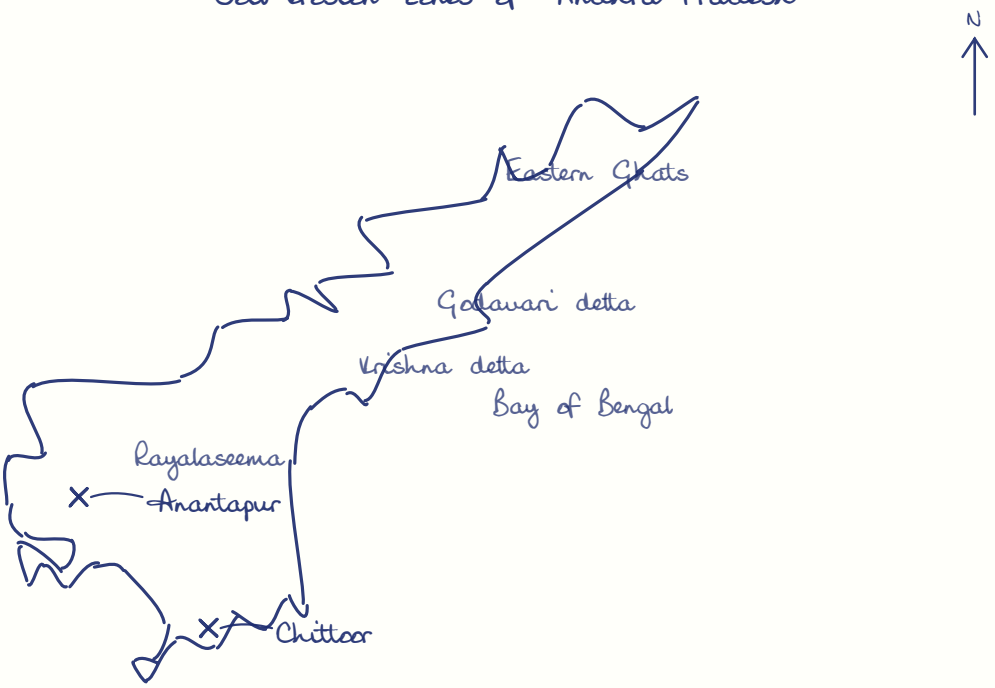
6(b) What do you know about the agents, stages, and effects of soil erosion? What are the factors affecting soil erosion and measures to conserve soil?

10

Introduction

Soil erosion threatens farming and ecological balance in Andhra Pradesh. Its terrain, from the drought-prone Rayalaseema belt to the fertile Krishna and Godavari deltas, sees varying soil loss that hurts crops and water resources.

Soil erosion zones of Andhra Pradesh



water erosion in deltas, wind erosion in Rayalaseema

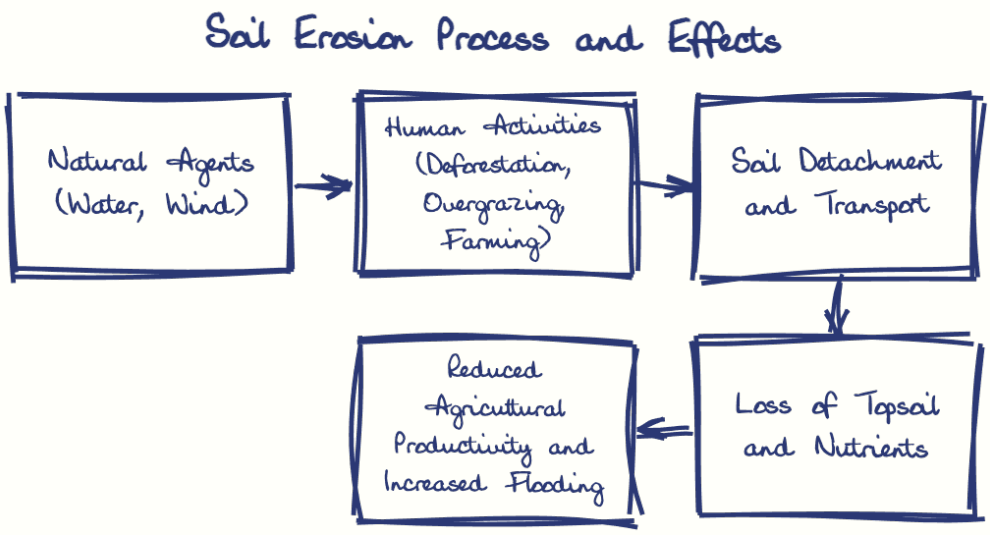
Agents of Soil Erosion

1. Water: Water erosion dominates the delta regions due to heavy monsoon rainfall and river

flooding and the undulating Eastern Ghats are also susceptible to it.

2. Wind: Wind erosion is significant in the dry Rayalaseema belt.

Stages and Effects



1. Stages: Agents detach soil particles and transport them away, stripping topsoil and nutrients.

2. Effects: Loss of topsoil reduces agricultural productivity, harms water resources and increases flooding.

Factors Affecting Soil Erosion

1. Climate and topography: High-intensity monsoon rain in Coastal Andhra and steep Eastern Ghats slopes increase runoff velocity.

2. Soil and vegetation: Highly erodible soils like the red soils of Chittoor are vulnerable once cover is lost.

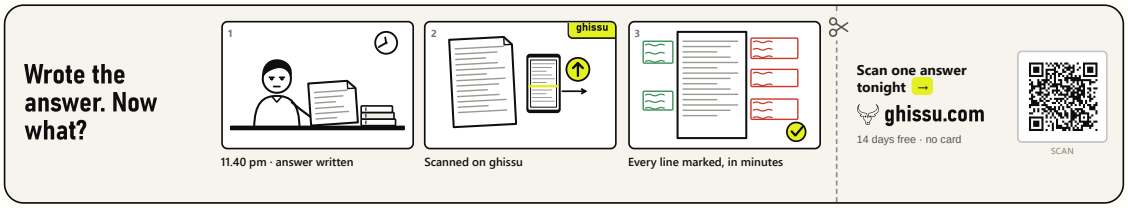
3. Land use: Overgrazing, deforestation and slope farming without terraces, as in parts of Visakhapatnam district, accelerate loss.

Measures to Conserve Soil

1. Field techniques: Contour bunding, terracing, afforestation and cover cropping, with shelter belts cutting wind velocity in Rayalaseema.
2. Institutions: The Central Soil Conservation Board and the state Soil and Water Conservation Department run region-specific plans, aided by the Soil Health Card Scheme.

Conclusion

Contour bunding and check dams have cut soil loss in pilot watersheds of Anantapur, but fragmented landholdings limit wider adoption. Sustained community engagement, alongside government and research bodies, is therefore the key to conserving soil.



7(a) Is there any difference between gene therapy, transgenic organisms, and gene enhancement/genome editing? With example, explain the ethical concerns in each of these approaches? When can genome editing be considered ethical?

10

Introduction

Genetic engineering has transformed medicine, agriculture and biotechnology, making it possible to treat disease, improve crops and even enhance human traits. With tools like CRISPR-Cas9, the ethics of manipulating genes have become urgent, including in agrarian states like Andhra Pradesh.

Differences Between the Approaches

Genetic Modification Approaches

Gene Therapy	Transgenic Organisms
Somatic cells only	Foreign gene insertion
Treats genetic diseases	New cross-species traits
Not heritable	Heritable if germline targeted
Vector gene delivery	Recombinant DNA technology

1. **Gene Therapy:** It corrects or replaces defective genes in an individual's somatic cells using

viral or non-viral vectors, so changes are not inherited.

2. Transgenic Organisms: Foreign genes from one species are stably integrated into another, creating GMOs with new traits for farming and research.

3. Genome Editing: Tools like CRISPR directly modify specific DNA, often without foreign DNA, in somatic or germline cells, allowing heritable changes and enhancement.

Examples and Ethical Concerns

1. Gene Therapy: The first success in 1990 treated ADA deficiency, and AP institutions now explore it for inherited disorders. Concerns include unintended mutations, cancer, exploitation of desperate patients and unequal access.

2. Transgenic Organisms: Bt cotton, approved in India including AP, resists pests, but raises gene flow to wild species, biodiversity loss and corporate control over small farmers.

3. Genome Editing: DRR Dhan 100 rice is gaining approval, in humans, it raises fears of eugenics, inequality and "designer babies".

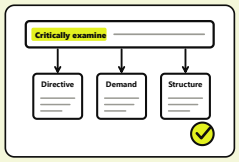
4. Common Themes: All raise worries about "playing God", lost genetic diversity and genetic discrimination in insurance and jobs.

When Genome Editing Is Ethical

1. Therapeutic Purpose and Safety: It must prevent or cure serious disease, with robust evidence of safety and few off-target effects.
2. Consent and Equity: Informed consent and equitable access are essential.
3. Oversight and Engagement: Strong regulation and public dialogue must align it with social values.

Conclusion

The three approaches differ in method and heritability, but all carry ethical risks. Genome editing is ethical when it heals serious disease safely, with consent, fairness and oversight, not when it seeks enhancement.



Stuck at the directive?

Exam-hall thinking

The directive, the demand, the structure — before your first line. Full for beginners, Crux for veterans.

Walk in with a method →

ghissu.com

14 days free · no card

SCAN

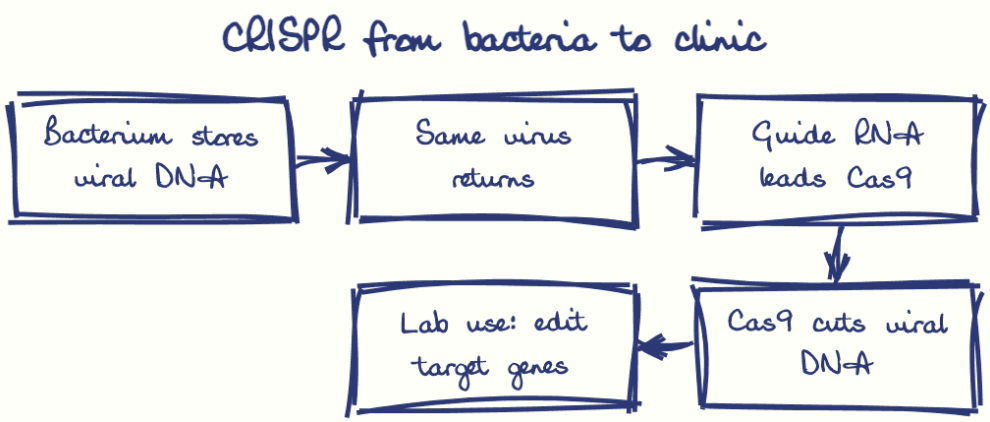
7(b) In October 2020, the Nobel Prize for Chemistry was awarded to the CRISPR technology. Explain how CRISPR functions in nature and how it has been adapted for research and clinical applications.

10

Introduction

The 2020 Nobel Prize in Chemistry for CRISPR marked a revolutionary milestone in genetics. Its ability to edit DNA rapidly, cheaply and precisely offers solutions to genetic disease, agricultural problems and environmental issues.

How CRISPR Works in Nature



1. Bacterial Adaptive Immunity: Bacteria capture snippets of viral DNA and store them in their genome as a genetic memory.
2. Guide RNA and Cas9: On meeting the same virus again, the stored sequences produce guide RNA that directs the Cas9 enzyme, the molecular scissors, to cut the matching viral

DNA.

3. Repair and Immunity: The cut triggers DNA repair that removes or alters the viral DNA, stopping replication.
4. Discovery: Jennifer Daudna and Emmanuelle Charpentier explained this in 2012, leading to its use as a gene-editing tool.

Adaptation for Research and Clinical Use

1. Precise Gene Editing: Synthetic guide RNAs target specific DNA in human, animal or plant cells, allowing deletions, insertions or replacements faster and cheaper than earlier methods.
2. Therapy and Diagnostics: Gene therapy corrects mutations behind sickle cell anaemia and muscular dystrophy, and edits tumour-linked genes. Tata Memorial's RAPID-CRISPR test detects Acute Promyelocytic Leukemia.
3. Agriculture: Drought-resistant rice with higher proline content is crucial for Rayalaseema's drought-prone areas.
4. Biotechnology and Environment: Algae with doubled fat content improve biofuels, and gene drives can control malaria-carrying mosquitoes.

Challenges

1. **Ethics:** The (2018) gene-edited babies case in China shows regulation lagging behind.
2. **Safety:** Off-target effects need rigorous trials.
3. **Access and Funding:** High costs may exclude small farmers in Andhra Pradesh, and R&D at about (0.65%) of GDP limits scale.

Way Forward

1. **Ethical Guidelines:** Frame comprehensive rules for research and clinical use, drawing on international best practice.
2. **Public R&D:** Invest in biotechnology institutions in Andhra Pradesh to build indigenous capability.

Conclusion

CRISPR turned a bacterial defence into a powerful tool for medicine and farming. Clear ethical rules, more public R&D and inclusive access will help India use it responsibly.

1.5 lakh+

Borrow the approach, not the answer.

Topper answers matched by meaning to your question — with how each one opened, structured and concluded.

See how toppers wrote it → ghissu.com

14 days free · no card

SCAN

8(a) Write short notes on the causes, treatment, and methods of prevention of any 3 of the following diseases: 1. Cholera; 2. Malaria, 3. Encephalitis, 4. Tuberculosis, and 5. Avian flu.

10

Introduction

Communicable diseases like chdera, malaria and tuberculosis continue to pose significant public heatth challenges in India, including Andhra Pradesh, despite advances in medical science. Their persistence is linked to environmental, social and infrastructural factors, so contrd needs a clear grasp of causes, treatment and prevention.

(1.) Chdera

1. Cause: Chdera is caused by the bacterium Vibrio chderae. Outbreaks are linked to contaminated water sources and poor sanitation.
2. Treatment: Laboratory identification of the pathogen allows early detection and timely treatment, though rural areas of Andhra Pradesh struggle to access advanced diagnostics.
3. Prevention: Community education on hygiene and safe drinking water, with better sanitation, waste disposal and water storage practices.

Practice 8(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2020 · Paper V

Scan to practice

(2.) Malaria

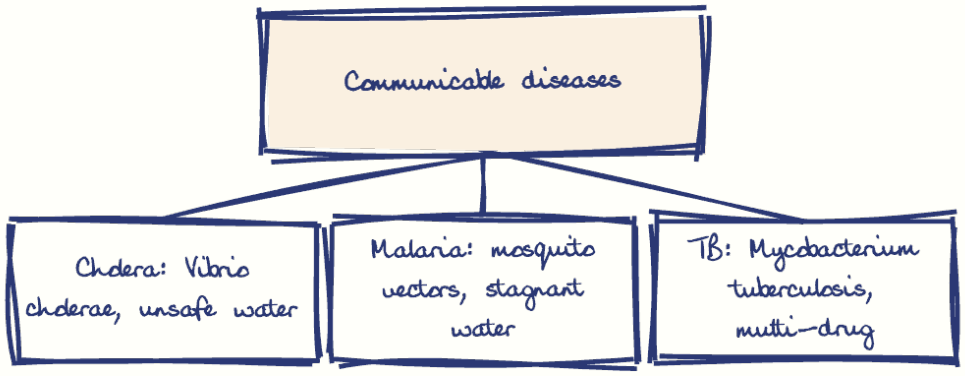
1. **Cause:** Malaria transmission depends on mosquito vectors breeding in stagnant water, common in parts of Andhra Pradesh where waste management and water storage are poor.
2. **Treatment:** Early diagnosis and treatment are pursued under the Intensified Malaria Elimination Project (IMEP)–3, in which Andhra Pradesh actively participates.
3. **Prevention:** Vector control through insecticide-treated nets and elimination of breeding sites has seen partial success in Andhra Pradesh but needs scaling up.

(3.) Tuberculosis

1. **Cause:** TB is caused by the bacterium Mycobacterium tuberculosis, identified through laboratory diagnosis.
2. **Treatment:** It needs prolonged multi-drug regimens under the National Tuberculosis Elimination Programme (NTEP), often complicated by drug resistance and non-compliance in Andhra Pradesh's rural and urban populations.
3. **Prevention and Support:** Early detection, culturally sensitive campaigns to reduce stigma, and nutritional support through Nikshay Poshan Yojana

improve outcomes.

Three diseases at a glance



Common Challenge

The 2024 CAG audit found large vacancies in medical staff and a shortage of Community Health Centres in Andhra Pradesh, which hamper timely diagnosis and treatment of all three diseases.

Conclusion

All three persist because environmental, social and institutional factors are intertwined. Stronger health infrastructure, better surveillance and community participation are essential to control and eventually eliminate these diseases, in line with SDG 3.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths →
ghissu.com
14 days free · no card
SCAN

8(b) What is recombinant DNA? Draw a schematic to show the process of creating rDNA and explain the roles of restriction enzymes and vectors in this process. Give an example of rDNA technology.

10

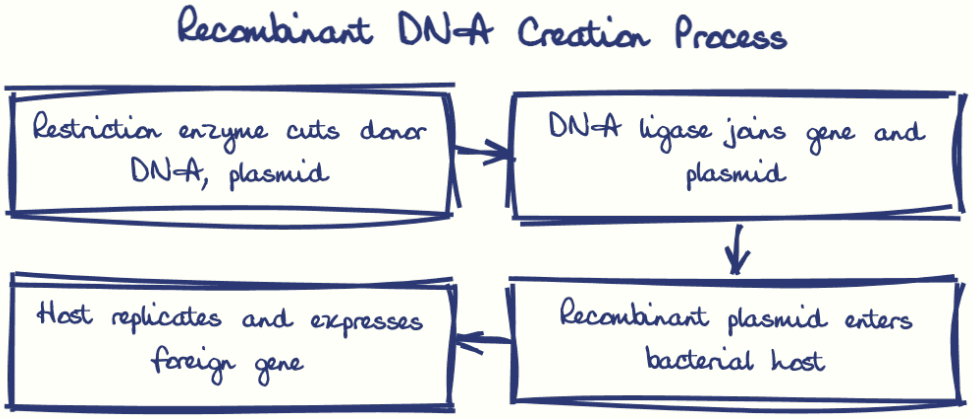
Introduction

Recombinant DNA (rDNA) technology is a cornerstone of modern biotechnology, enabling precise genetic changes that drive advances in medicine, agriculture and industry. Its safe use in India is ensured by the Department of Biotechnology's regulatory framework.

What Is Recombinant DNA

rDNA is artificially created DNA formed by joining DNA fragments from different species. It exploits the universal structure and function of DNA across organisms to transfer genetic information.

Process of Creating rDNA



1. Role of Restriction Enzymes: They act as molecular scissors that recognise specific DNA sequences and cut at these sites, producing fragments with complementary sticky or blunt ends. This isolates the gene of interest, such as the human insulin gene.
2. Role of Vectors: Plasmid vectors, often from bacteria like Escherichia coli, carry the foreign DNA. Cut with the same enzyme, they have matching ends, and they replicate independently within host cells.
3. Ligation and Expression: DNA ligase seals the fragments to form rDNA, which is introduced into a host bacterium that copies the plasmid and expresses the gene.

Example and Applications

1. Human Insulin: Inserting the human insulin gene into E. coli enables large-scale, cost-effective insulin for diabetes treatment, a practice adopted by Indian biotech firms and research centres in Andhra Pradesh.
2. Agriculture: rDNA techniques develop GM crops with pest resistance and drought tolerance, which can help regions like Rayalaseema that face climatic challenges.

3. Research and Diagnostics: Gene cloning and functional analysis aid the understanding of genetic diseases prevalent in India, with Andhra Pradesh's research institutions contributing.

Regulation

The Recombinant DNA Advisory Committee and the Review Committee on Genetic Manipulation frame safety guidelines, while Institutional Biosafety Committees approve projects in research institutions, including those in Andhra Pradesh.

Conclusion

rDNA technology turns living cells into factories for useful products. It must be backed by advanced infrastructure, skilled manpower and biosafety training to realise its promise safely and ethically.

REVIEWER'S NOTE · VALUE ADDITION

Examiners read hundreds of answers a day. The one with a number, a committee and a case is the one they remember.

ghissu

ghissu gives you the data point, the report and the example for every question — each linked to its source.

Get the value adds for any question →

ghissu.com

14 days free · no card

SCAN

9(a) What is Intellectual Property? Explain with examples the differences between copyrights, patents, and trademarks. What are the rights an inventor has on his invention, and how are these rights gained and enforced globally?

10

Introduction

In an era of rapid innovation, protecting creations of the mind is crucial for economic growth and competitiveness. Intellectual Property rights give inventors and creators exclusive contrd over their work, encouraging research and development.

What is Intellectual Property

1. Concept: Intellectual Property covers creations of the mind such as inventions, literary and artistic works, designs, symbols, names and images used in commerce, protected by law as intangible assets.
2. Limited, territorial rights: IPRs exclude others for a limited period and apply only where registered, though treaties allow cross-border recognition.
3. State support: The Andhra Pradesh Innovation Society helps local innovators with IP registration and awareness.

Copyrights, Patents and Trademarks

Copyright vs Patent

Copyright	Patent
Protects expression of ideas	Protects inventions and processes
Books, music, software	Pharmaceutical drugs, machinery
Life of author plus 60 years	20 years from filing
Reproduce, distribute, display	Make, use, sell

1. Copyright: Protects expression in literary, artistic, musical and software works, as with a novel or a film.
2. Patent: Protects novel, non-obvious and industrially applicable inventions, such as a new agricultural biotechnology from Acharya N.G. Ranga Agricultural University.
3. Trademark: Protects names or logos like Amul that distinguish goods, renewable indefinitely while in use.

Rights of an Inventor and Their Enforcement

1. Exclusive rights: The inventor can stop others making, using, selling or importing the invention during the patent term, and can

license it.

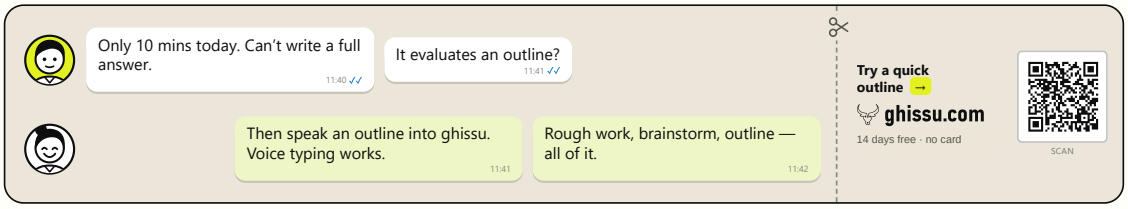
2. How rights are gained: By filing with a national office like the Indian Patent Office, which examines novelty, inventive step and industrial applicability.

3. Global enforcement: The Paris Convention and TRIPS Agreement under the WTO set minimum standards and dispute resolution; courts and customs curb counterfeits.

4. Balancing access: TRIPS led India to amend its patent laws, and Novartis AG v. Union of India (2013) balanced patents with public health.

Conclusion

Balanced IP rights reward creators, attract investment and keep knowledge flowing to the public after expiry.



9(b) What is a patent? Give examples of patents. What is the difference between process patents and product patents? How are GI tags related to patents?

10

Introduction

Protecting intellectual property is crucial for creativity and economic growth. Patents give inventors exclusive rights and so encourage research, which matters for India and Andhra Pradesh under Make in India and Startup India.

What is a Patent

1. Definition: A patent is a government-granted exclusive right that lets the inventor stop others making, using, selling or importing the invention, usually for (20) years, in return for disclosure.
2. Criteria: The invention must be novel, non-obvious and industrially applicable; the Patents Act, (1970) excludes methods of agriculture and biological processes.

Examples of Patents

1. Pharmaceuticals: Post-TRIPS, product patents are recognised, and Andhra Pradesh's pharmaceutical firms increasingly patent new drug formulations.

Practice 9(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2020 · Paper V

Scan to practice

2. Biotechnology: Diamond v. Chakrabarty (1980) allowed a patent on an oil-dissolving microbe.

3. Utility and design patents: Utility patents cover machines or chemical compounds, while design patents protect the ornamental look of products, used by start-ups in Visakhapatnam.

Process vs Product Patents

Process Patents vs Product Patents

Process Patents	Product Patents
Protect method or procedure	Protect the actual product
Innovation in manufacturing	Innovation in new products
New drug synthesis method	New pharmaceutical molecule
Duration: 20 years	Duration: 20 years

1. Process patents: Others may make the same product by a different method, which let India's early regime build a generic drug industry.

2. Product patents: Recognised for drugs after TRIPS, they protect the product however it is made.

GI Tags and Patents

1. **Distinct protection:** GI tags protect regional products like Kalamkari textiles and traditional knowledge, not inventions.
2. **Complementary roles:** Both add market value and prevent unauthorised use.
3. **Overlap:** A patented method to improve a GI product can coexist, but needs legal clarity.

Conclusion

Patents reward new inventions while GI tags guard collective heritage; protecting both supports innovation and inclusive growth in states like Andhra Pradesh.

FIVE-MINUTE LESSON TREND ANALYSIS, PER SUB-TOPIC

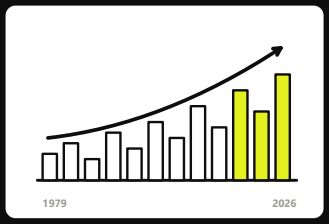
Read a chapter's past before its pages

Twenty minutes of this before a new chapter tells you where to spend the next twenty hours.

1. **List** — Every question asked from the chapter in the last ten years.
2. **Circle** — Sub-topics asked twice or more.
3. **Notice the shift** — Facts to analysis? National to State? Old to current?
4. **Read** — The circled sub-topics first; the rest second.

ON GHISSU, THIS IS DONE FOR YOU — ON EVERY QUESTION.

Every syllabus topic already has its trend analysis: direction of travel, most tested sub-topic, probable next theme.



Prepare the pattern →
ghissu.com
14 days free · no card
SCAN

10(a) What is the status of utilization of Solar energy in India and the World? What are some factors hindering utilization of this green energy source?

10

Introduction

Solar energy has become pivotal to the global clean energy transition, driven by climate goals and energy security. India, with abundant solar radiation, now ranks third worldwide in installed solar power, and its Paris commitment of 500 GW of renewables by 2030 makes solar a cornerstone.

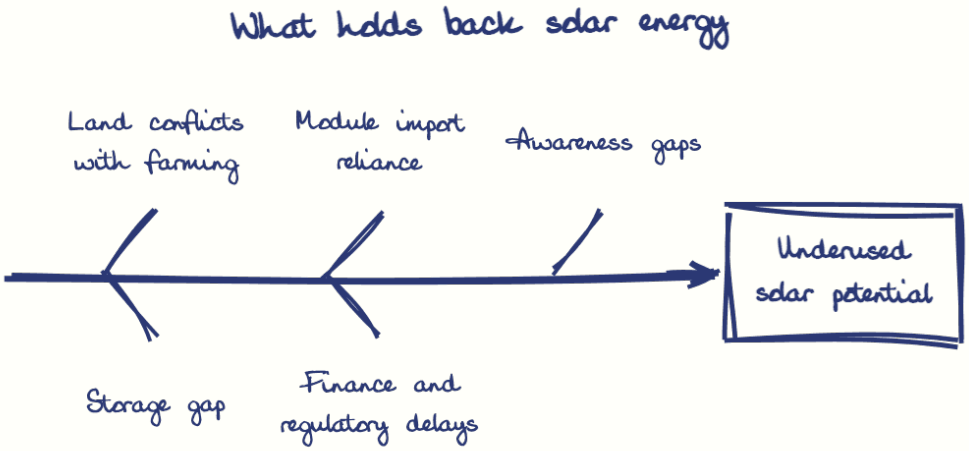
Status of Solar Utilisation

1. **World:** Solar is central to the global transition; India's leadership of the International Solar Alliance attracts funding and technology sharing.
2. **India's Policy Push:** The National Solar Mission, PM Surya Ghar and PM-KUSUM give subsidies and incentives; Andhra Pradesh saw over 6.3 lakh rooftop installations in 2024.
3. **Manufacturing:** PLI raised module capacity from 38 GW in 2024 to 74 GW in 2025.
4. **Grid and Rural Use:** The Green Energy Corridor builds transmission, and solar

irrigation in Andhra Pradesh supports farmers.

Factors Hindering Utilisation

1. **Land Conflicts:** Solar farms need large land, clashing with agriculture and delaying projects.
2. **Storage Gap:** About 5.5 GW of storage against 243 GW of renewables, far below the 61 GW needed by 2030.
3. **Import Reliance:** $1.1 billion of modules imported in FY24, half from China.
4. **Finance and Regulation:** High upfront costs, inconsistent state policies and delays slow rooftop solar.
5. **Awareness Gaps:** Digital divides and finance gaps limit rural adoption.



Candidates must not write on this margin

Candidates must not write on this margin

Way Forward

1. Storage: ~~Affordable~~ large-scale batteries.
2. Land: Fast-track rules and agrivoltaics.
3. Raw Materials: Domestic polysilicon and wafers.
4. Finance and Awareness: More subsidies, affordable loans and outreach in rural and semi-urban Andhra Pradesh to speed rooftop solar.

Conclusion

India's solar use has grown impressively, making it a global leader. Fixing storage, land, finance and supply chain gaps will make this growth lasting.

Improving — or just writing more?

My answers

Every attempt, every score, every enhanced version of your answer — on one timeline.

Watch your curve → ghissu.com

14 days free · no card

SCAN

10(b) Traditional methods of in-person education have been disrupted by the Covid-19 pandemic. How can India's Space Program help in restoring and expanding access to education in India?

10

Introduction

The Covid-19 pandemic kept Indian schools closed for over 73 weeks by late 2021, one of the longest closures globally. This magnified inequality, causing learning loss and higher dropouts in rural and disadvantaged areas. India's Space Program, led by ISRO, offers satellite connectivity and space science outreach to restore and expand access.

Disruption to In-Person Education

1. Prolonged Closures: Over a year of closure broke the classroom-based model.
2. Digital Divide: Rural and poorer students lacked devices and internet, leading to dropouts and learning regression, as ASER 2020 highlighted.
3. Shift to Public Schools: Financial distress moved children from private to government schools, straining rural public education.
4. Gendered Impact: Rural girls faced higher dropout risks, as UNICEF and NFHS data note.

Restoring Access through the Space Program

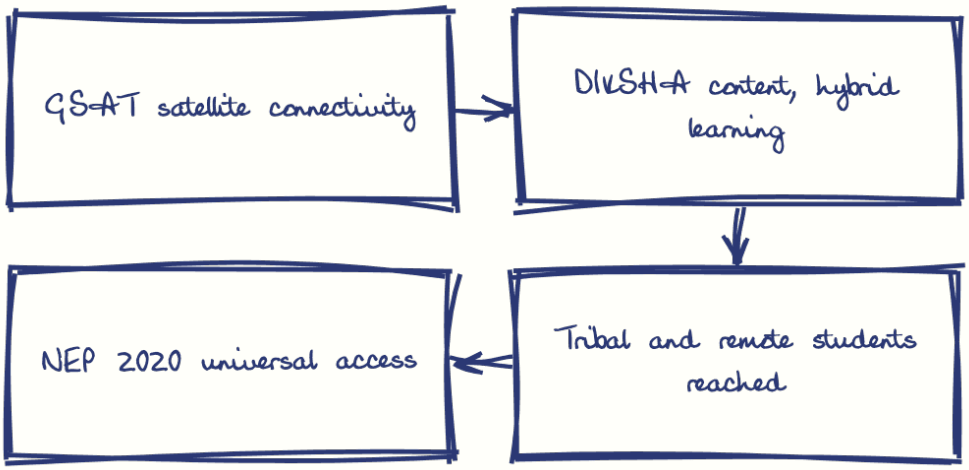
1. **Satellite Connectivity:** ISRO's GSAT series brings internet to remote areas without terrestrial networks, such as the Rayalaseema drought belt.
2. **Digital Platforms:** Satellite communication supports DIKSHA, which delivered curriculum content during closures.
3. **Space Education:** ISRO's YUVIKA programme brings space science to students and teachers, building scientific temper.
4. **Local Collaboration:** The Sriharikota launch centre enables regional educational collaborations and awareness programmes.

Expanding Access beyond Pre-Pandemic Levels

1. **Hybrid Learning:** Space-enabled infrastructure supports blended learning, raising enrolment and retention.
2. **Inclusive Education:** Satellite links reach tribal and remote communities, matching the NEP 2020 goal of universal access.
3. **Experiential Learning:** Like NASA and JAXA, ISRO can promote AR tools using satellite data.
4. **Scheme Support:** Data-driven monitoring and remote learning complement PM-SHRI and Samagra

Shiksha Abhiyan.

Space links for learning



Way Forward

1. Satellite Broadband: Faster rollout in rural and remote areas for equitable access.
2. Curriculum: Space science modules with state education departments.
3. Partnerships: ISRO and EdTech firms for affordable devices and local-language content.
4. Teacher Training: Equip rural educators to use these tools.

Conclusion

India's Space Program can restore disrupted learning and expand educational access beyond pre-pandemic levels, especially for marginalised students.

Write your own answer to 10(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2020 · Paper V

Scan to practice

11(a) What do you know about the NeGP (National e-Governance Action Plan) operating under the aegis of Digital India? Explain about the role of any three of the following projects in this plan: 1. Jan-Dhan Yojana, 2. Digital Locker, 3. National Digital Literacy Mission, 4. E-Bhasha, and 5. BharatNeT (National Optical Fibre Network).

10

Introduction

Digital transformation is a cornerstone of inclusive growth in India. Under Digital India, launched in 2015, the National e-Governance Plan (NeGP) has become the key framework to digitise government services and make them accessible to all citizens, especially in rural and underserved areas, improving transparency and reducing corruption.

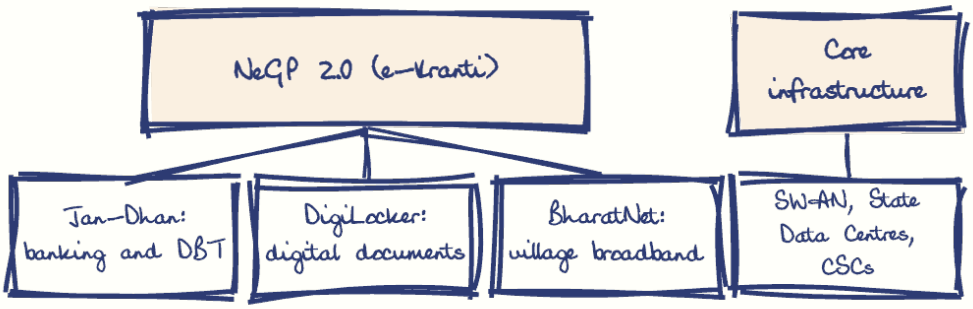
About NeGP

1. Comprehensive Framework: Launched in 2006 and revitalised as NeGP 2.0 or e-Kranti, it aims to make all services electronically accessible through common outlets, integrating 31 Mission Mode Projects across central and state governments.
2. Outcome-Oriented Approach: Cloud, mobile and geospatial technologies create an interoperable, agile system focused on real outcomes.
3. Decentralised Implementation: The Department of Information Technology assists while ministries and states own projects, encouraging partnerships, as

with e-Procurement in Andhra Pradesh.

4. Core Infrastructure: State Wide Area Networks, State Data Centres and Common Service Centres extend services to rural areas, as AP's extensive use of CSCs shows.

NeGP and its key projects



Role of Three Projects

1. Jan-Dhan Yojana: Aadhaar-based identity gives universal banking access and enables direct benefit transfers, so subsidies reach beneficiaries without leakage; in Andhra Pradesh it channels welfare payments digitally.
2. DigiLocker: A secure cloud platform to store and share official documents; with over 62 crore users, it simplifies verification and supports paperless governance.
3. BharatNet: The National Optical Fibre Network brings high-speed broadband to all gram

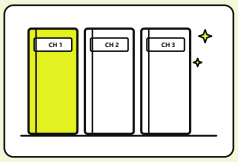
panchayats, forming the backbone of NeGP services. Higher rural internet penetration enables e-governance, telemedicine and digital education.

Way Forward

1. Digital Literacy: Tailor the National Digital Literacy Mission to local languages and contexts, especially in rural areas.
2. Interoperability: Unified protocols across Mission Mode Projects.
3. Partnerships: Private technology firms to scale infrastructure and services.
4. Cybersecurity: Robust frameworks to protect citizen data and build trust.

Conclusion

NeGP under Digital India has made services accessible, transparent and efficient; Jan-Dhan, DigiLocker and BharatNet show how technology bridges gaps in inclusion.



Notes in five notebooks, none at hand?

Notes that live with the question

Note from toppers' answers, exam-hall thinking and model answers — your notes gather into topic and chapter notes on their own.

Build notes that find you →

ghissu.com

14 days free · no card

SCAN

11(b) What are biofuels and why are they important to India's economy? What do you know about the Indian National Policy on Biofuels?

10

Introduction

India's energy demand is rising with economic growth, urbanisation and more vehicles. With over 85% of crude oil imported, biofuels offer a strategic way to reduce import dependence, enhance energy security and promote sustainable development.

What Are Biofuels

Biofuels are fuels such as ethand and biodiesel produced from biomass, including mdasses, agricuttural and forest residues, surplus food grains, non-edible oils, municipal sdid waste and cow dung. Second-generation (2G) ethand uses lignocellulosic biomass.

Importance to India's Economy

1. Energy Security: Replacing part of imported crude saves foreign exchange and cushions the economy.
2. Rural Prosperity: Biofuels generate rural employment and better farmer remuneration, supporting Make in India and doubling farmers' income.
3. Waste Management: Turning residues and waste into fuel supports Swachh Bharat Abhiyan and

reduces pollution.

4. Regional Potential: Andhra Pradesh's agricultural residue and biomass, its coastline easing bio-refinery logistics and the Visakhapatnam-Chennai industrial corridor make it a key player.

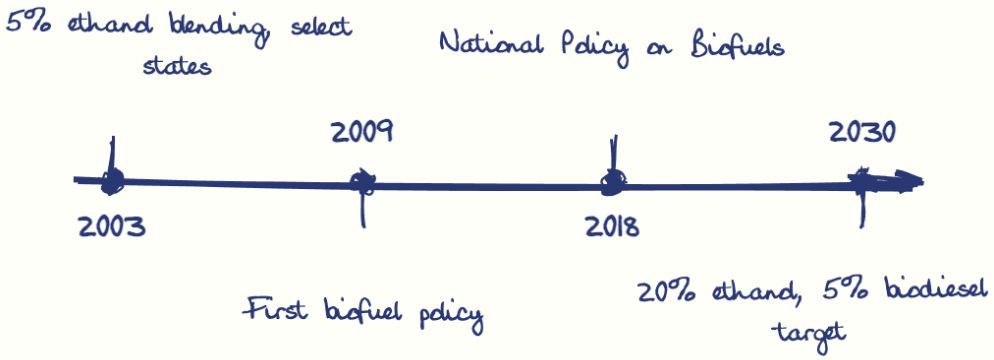
National Policy on Biofuels, 2018

1. Evolution: From 5% ethanol blending in select states in 2003, through the 2009 policy, to the 2018 policy.
2. Targets: 20% ethanol blending in petrol and 5% biodiesel by 2030.
3. Feedstock Diversification: Raw materials expanded beyond molasses to agro-waste, surplus grains and non-edible oils, while restricting grains to damaged ones to protect food security.
4. Advanced Biofuels: Pradhan Mantri JI-VAN Yojana supports 2G ethanol projects.
5. Institutions: The National Biofuel Coordination Committee under the Ministry of Petroleum and Natural Gas oversees implementation.

Challenges

Cyclical sugarcane output, high feedstock costs and

Evolution of biofuel policy



food-crop competition mean only about 50% of petrol is blended at E10.

Way Forward

1. Feedstock Supply Chains: Build reliable chains for agro-waste and non-edible oils in Andhra Pradesh to keep feedstock steady and cheap.
2. 2G Bio-refineries: Promote investment in second-generation bio-refineries in Andhra Pradesh to reduce reliance on food-based feedstock.

Conclusion

Biofuels link energy security with farm incomes and clean air. With R&D under BioE3 and Bio-RIDE, they can become a pillar of a cleaner, self-reliant economy.

Write your own answer to 11(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2020 · Paper V

Scan to practice

12(a) Please interpret the following data in terms of contribution by countries to Science and Engineering publications. Identify any one factor for the state of Indian Science and recommend a relevant policy measure to address the situation.

10

S&E articles in all fields, for 15 largest producing regions, countries, or economies: 2008 and 2018

(Number)

Rank	Region, country, or economy	2008	2018	Average annual growth rate 2008–18 (%)	2018 world total (%)	2018 cumulative total (%)
-	World	1,755,850	2,555,959	3.83	-	-
1	China	249,049	528,263	7.81	20.67	20.67
2	United States	393,979	422,808	0.71	16.54	37.21
3	India	48,998	135,788	10.73	5.31	42.52
4	Germany	91,904	104,396	1.28	4.08	46.61
5	Japan	108,241	98,793	-0.91	3.87	50.47

Introduction

The table compares Science and Engineering (S&E) articles of the five largest producers in 2008 and 2018. World output grew from 17.6 lakh to 25.6 lakh articles, at 3.83% a year, and these five countries produced just over half of it.

Interpreting the Data

1. **China leads:** Output more than doubled to 5.28 lakh articles, giving the largest world share of 20.67%.
2. **US stagnates:** Output rose only 0.71% a year, and its share is now 16.54%.
3. **India rises fastest:** Articles nearly tripled

from 48,998 to 1,35,788. India's growth of 10.73% a year is the highest in the table, taking it to third place with a 5.31% share.

4. Mature systems slow: Germany grew 1.28% a year, while Japan's output fell by 0.91% a year.

5. Gap remains: Despite its pace, India produces about one-fourth of China's output.

India and China in S&E articles

India	China
48,998 in 2008	2,49,049 in 2008
1,35,788 in 2018	5,28,263 in 2018
10.73% annual growth	7.81% annual growth
5.31% world share	20.67% world share

One Factor: Low Investment in Research

India's gross expenditure on R&D was only about 0.64% of GDP in 2020–21, far below that of China and the US. Funding is concentrated in a few government agencies, and private investment and university research

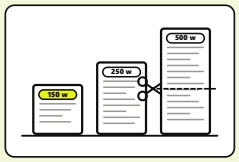
remain weak. This limits both the volume and the impact of Indian science.

Policy Measure

1. Anusandhan National Research Foundation: Fully operationalise the ANRF created by the 2023 Act to fund competitive, peer-reviewed research in universities.
2. Industry participation: Use ANRF and the Rs 1 lakh crore Research, Development and Innovation fund (2025) to draw private investment into research.
3. Research capacity: Expand fellowships and laboratories in state universities, where most students study.

Conclusion

India is the fastest-growing producer in the table, but its share remains modest. Higher, steadier research funding through ANRF can turn this growth into quality and impact.



Writing 400 words for a 150-word question?

Model answers in three lengths

150, 250 and 500 words, in English and हिंदी. Learn what to cut before the examiner does.

Read one question three ways →

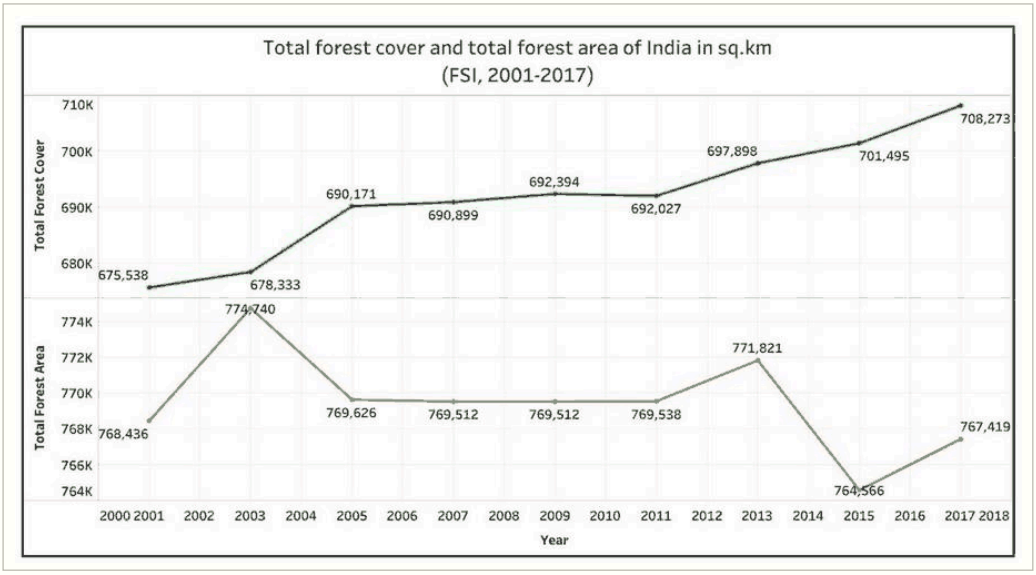
ghissu.com

14 days free · no card

SCAN

12(b) Is there a difference between "forest cover" and "forest area"? What does the following graph show?

10



Introduction

Yes. The Forest Survey of India (FSI) uses two separate measures. Forest cover is what satellites see on the ground, while recorded forest area is what government records classify as forest. The graph compares the two from 2001 to 2017.

Forest Cover and Forest Area

1. Forest cover: All land over one hectare with tree canopy density above 10%, irrespective of ownership or legal status. It includes orchards, bamboo and palm.
2. Recorded forest area: Land recorded as Reserved, Protected or Unclassed forest,

whether or not it carries trees.

3. Key difference: One is a physical measure from satellite imagery; the other is a legal and administrative status.

What the Graph Shows

1. Rising cover: Forest cover rose from 6,75,538 sq km in 2001 to 7,08,273 sq km in 2017, or 21.54% of India's area, with only a slight dip around 2011.

Forest cover vs forest area

Forest cover	Forest area
Seen by satellite	Recorded in government land records
Canopy above 10%	Legal status as forest
Any ownership	Reserved, Protected, Unclassed
7,08,273 sq km in 2017	7,67,419 sq km in 2017

2. Flat area: Recorded forest area stayed near 7.7 lakh sq km, peaking in 2003 and dipping in 2015.

3. Narrowing gap: The gap between the two lines shrinks from about 93,000 to 59,000 sq km, so tree cover is growing faster

Candidates must not write on this margin

Candidates must not write on this margin

than the legal forest estate.

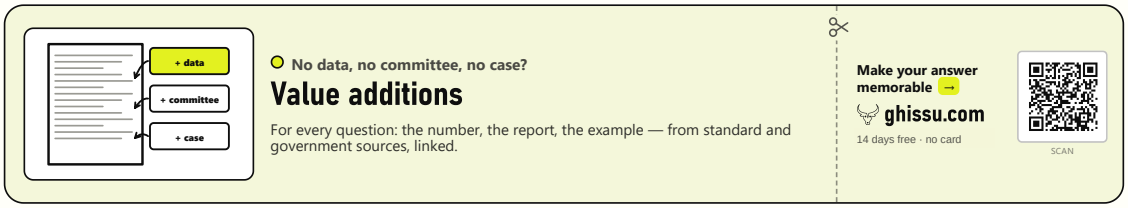
4. Hidden story: Much of the gain is likely from plantations and agroforestry outside recorded forests, while some recorded forests carry little or degraded tree cover.
5. Methods matter: Better satellite resolution and changing methods also affect year-to-year figures.

Significance

1. Policy goal: The National Forest Policy, 1988 aims for one-third of the country under forest or tree cover, so cover is still well short.
2. Better monitoring: Reporting canopy density classes alongside area helps separate dense forests from scattered trees.

Conclusion

Forest cover and forest area answer different questions. Rising cover is welcome, but forest quality and the health of recorded forests must be tracked too.



How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 12(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2020 · Paper V

Scan to practice

13(a) What is Climate Justice? How is this matter covered by news media and influential publications like Time magazine? Suggest at least 2 measures that are aligned to this goal.

10

Introduction

With extreme weather growing more frequent and awareness rising, climate justice has gained urgency in India and globally. Vulnerable communities, as in coastal Andhra Pradesh with its long coastline and agrarian distress, bear a disproportionate burden. Media coverage shapes how the public and policymakers understand this.

Understanding Climate Justice

1. Equity in responsibility: Climate change is an ethical issue, those least responsible suffer most, so obligations must reflect historical emissions and capacity.
2. Structural inequality: It exposes climate colonialism, where developed nations control agendas through finance and technology. AP's rich clean energy resources stay underused for lack of infrastructure and finance.
3. Inclusive governance: Women, persons with disabilities and low-income groups must share in policy and adaptation benefits.

4. Systemic change: Economic systems must change so climate policy reduces, not deepens, poverty and disparity.

Coverage by News Media and Publications

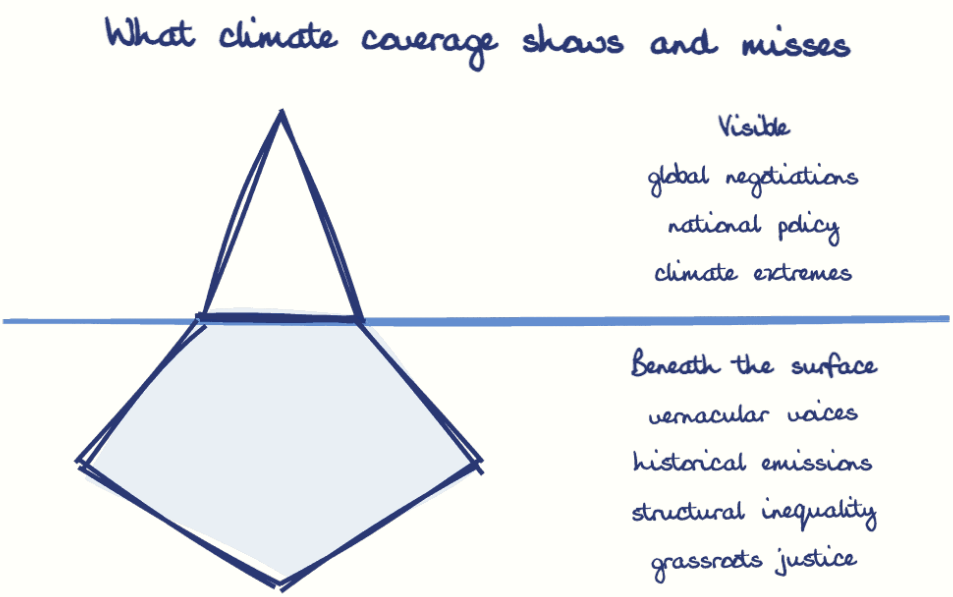
1. Growing but uneven: Down to Earth and The Third Pole report on climate, but coverage is concentrated in elite English media, missing vernacular voices from AP's rural and coastal communities.
2. Global over local: Influential publications like Time magazine highlight international negotiations and national policy, often at the cost of grassroots justice issues.
3. Data and misinformation: Journalists struggle for disaggregated data, and audiences worry about false climate news.
4. Framing: Terms like "climate extremes" replace "climate change", widening reach but diluting ethical urgency.

Measures Aligned with Climate Justice

1. Local climate journalism: Fund vernacular media in regions like AP, create climate beats and train reporters in data and ethics.
2. Inclusive policy: Gender-sensitive disaster

Candidates must not write on this margin

Candidates must not write on this margin



management that includes voices from Rayalaseema and coastal areas.

3. Finance and technology: Equitable access to climate finance for AP's renewables.

4. Transparency: Public access to impact data under the RTI Act.

Conclusion

Climate justice demands a shift from uniform responsibility to an equitable framework that recognises historical emissions and social vulnerability. Local media, inclusive policy and open data, guided by the Panchamrit vision, can build a just, low-carbon future.

How did toppers write this one?
1.5 lakh+ topper answers, matched to each question.
ghissu.com →

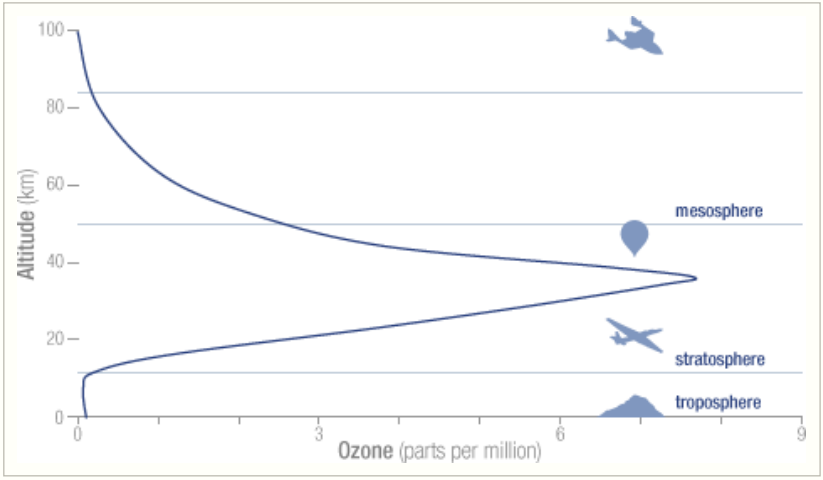
How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 13(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
Scan to practice

13(b) Explain what you see in the following graphic. (a) How does human activity affect the substance shown in the graphic? (b) How does this substance affect human health?

10



Introduction

The graphic shows the vertical distribution of ozone in the atmosphere. Ozone concentration is low near the ground, rises sharply in the stratosphere to a peak of nearly (8) parts per million at about (35 km), and falls again towards the mesosphere. This band is the ozone layer.

(a) How Human Activity Affects Ozone

1. Depletion above: Chlorofluorocarbons, halons and other ozone-depleting substances reach the stratosphere, where chlorine and bromine atoms destroy ozone molecules in chain reactions.
2. Ozone hole: This produces the seasonal ozone hole over Antarctica each spring.

3. Excess below: Vehicles, industries and biomass burning emit nitrogen oxides and volatile organic compounds that form ground-level ozone in sunlight.
4. Global response: Under the Montreal Protocol (1987), ozone-depleting substances are being phased out, and the layer is expected to recover in the coming decades.
5. India's part: India phased out CFCs in 2008, ahead of the 2010 deadline, is phasing out HCFCs by 2030, and includes ozone in its National Ambient Air Quality Standards.

Good ozone and bad ozone

Stratospheric ozone	Ground-level ozone
Shields Earth from UV-B	Forms from vehicle and industry emissions
Destroyed by CFCs and halons	Increased by NOx and VOCs
Depletion harms health	Its presence harms health

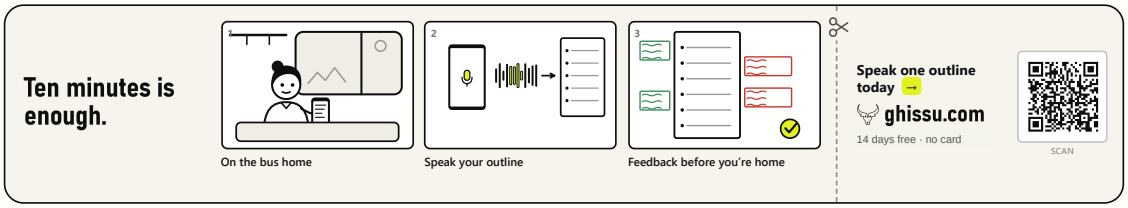
(b) Effects on Human Health

1. More UV-B: A thinner ozone layer lets more UV-B radiation reach the ground, raising the risk of skin cancer.

2. Eyes and immunity: Greater UV exposure causes cataracts and weakens the immune system.
3. Lungs: Ground-level ozone irritates the airways, triggering asthma attacks, coughing and reduced lung function.
4. Vulnerable groups: Children, the elderly and outdoor workers face the highest exposure.
5. Wider harm: Ground-level ozone also damages crops, which affects food security.

Conclusion

Ozone protects life high above but harms it near the ground. Continued compliance with the Montreal Protocol and tighter control of vehicle and industrial emissions are needed to keep both in balance. Guarding the ozone layer above and cleaning the air we breathe below are two sides of the same public health goal.



14(a) What are the most common and hazardous pollutants in outdoor and indoor air? Mention the health hazards from each of the pollutants. What is AQI (Air Quality Index)? Who monitors AQI in India? What is the contribution of PM to AQI?

10

Introduction

Air pollution threatens public health and economic development in India, causing millions of premature deaths each year. Both outdoor and indoor pollution drive respiratory and cardiovascular disease. The Air Quality Index (AQI) communicates air quality to the public and guides policy.

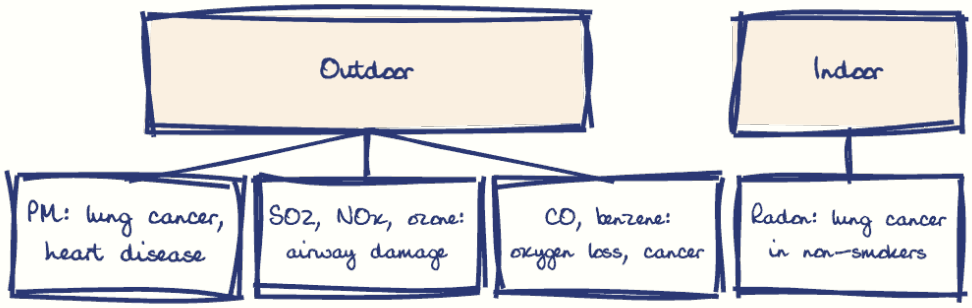
Common Pollutants and Their Health Hazards

1. PM2.5 and PM10: Fine particles reach deep into lungs and blood, causing infections, lung cancer, heart disease and premature death; PM2.5 also carries toxic substances.
2. Sulphur Dioxide: From industry and power plants; irritates airways, aggravates asthma and can cause chronic bronchitis.
3. Nitrogen Oxides: From vehicles and thermal plants; inflame airways, reduce lung function and form ozone and smog.
4. Ozone: A secondary pollutant from NOx and VOCs; causes chest pain and coughing and

worsens asthma and emphysema.

5. Carbon Monoxide: From vehicles and indoor cooking, reduces oxygen delivery causing headaches, dizziness and even death.
6. Benzene and VOCs: From industry and fuel vapours, carcinogenic, damaging blood and immunity.
7. Radon (Indoor): From building materials and soil, a leading cause of lung cancer in non-smokers.

Hazardous air pollutants



What Is AQI and Who Monitors It?

1. AQI: Launched in 2015 under the Swachh Bharat Mission, it gives real-time data in six categories from Good to Severe, based on eight pollutants: PM10, PM2.5, NO2, SO2, CO, O3, NH3 and Pb.
2. Monitoring Bodies: The Central Pollution

Contrd Board, under the Air Act, 1981, sets NAAQS and oversees AQI; State Pollution Contrd Boards and local bodies monitor locally, and IIT Kanpur helped develop the method.

Contribution of PM to AQI

PM2.5 and PM10 are the largest contributors to poor air in Indian cities, often exceeding safe limits and pushing AQI into Poor to Severe. Yet many cities, including in Andhra Pradesh, still lack full real-time data, and indoor radon and biomass smoke remain under-monitored, burdening rural and low-income homes.

Conclusion

AQI has made air quality visible and shows that PM drives India's bad air days. Expanding monitoring with advanced sensors and covering indoor pollutants would make it a fuller guide to action.

Correct, but average? One line from your answer

YOUR LINE
~~The 1857 revolt failed.~~
✕ true · forgettable

→

WITH A VALUE ADDITION
The 1857 revolt failed for want of a centre: no common plan, no modern arms, and the loyalty of the Sikhs, the Gurkhas and most princes to the Company.
✓ cause · evidence · specifics

ILLUSTRATIVE

Add value to your next answer →
ghissu.com
14 days free · no card
SCAN

14(b) What is Covid-19 and how is it spread? How are immuno-diagnostic methods used in context of the Covid19 pandemic?

10

Introduction

COVID-19 is caused by the novel coronavirus SARS-CoV-2, which emerged in late 2019 and spread worldwide. WHO declared it a pandemic in March 2020. Its high transmissibility and unpredictable outcomes strained health systems, including in densely populated Andhra Pradesh.

How COVID-19 Spreads

1. Respiratory droplets: The virus spreads mainly through droplets generated by coughing, sneezing, talking or breathing, which settle quickly on surfaces.
2. Contact transmission: People touch contaminated surfaces and then their face.
3. Timing and silent spread: It is most contagious in the first three days after symptom onset, and asymptomatic transmission and pre-symptomatic spread complicate containment.

Immuno-Diagnostic Methods in the Pandemic

1. Antibody detection: Tests detect IgM and IgG antibodies; IgM indicates recent infection, while

Antibody tests vs RT-PCR

Antibody tests	RT-PCR
IgM and IgG antibodies	Viral RNA
Faster and cheaper	Specialised labs, costlier
Seroprevalence surveys	Gold standard, active infection

IgG develops later and suggests past exposure.

2. Population surveillance: Being faster, cheaper and suited to point-of-care use, they assess seroprevalence and population exposure rather than diagnose individuals.

3. Complement to RT-PCR: RT-PCR, standardised by ICMR and NIV, remains the gold standard for active infection but needs specialised labs and costs more.

4. Limitations: Antibody tests are qualitative and cannot show antibody levels or immunity duration.

5. Validation: FIND evaluated many rapid tests, and ICMR protocols require rigorous multi-site validation.

Institutional Support for Diagnosis

1. **Lab expansion:** Over 2,400 laboratories were testing by March 2021, including in Andhra Pradesh, where medical colleges became testing centres through AIIMS and PGIMER training.
2. **Surveillance:** The IDSP kept real-time data from district to centre, and the IHIP added GIS-based outbreak mapping for targeting hotspots in Andhra Pradesh and beyond.
3. **Data systems:** The Special Surveillance System linked Aarogya Setu contact tracing and CoWIN.

Way Forward

1. **Regional labs:** Upgrade facilities in Andhra Pradesh under the PM-Ayushman Bharat Health Infrastructure initiative.
2. **Quality:** Validate immuno-diagnostic kits strictly as per ICMR guidelines.

Conclusion

Immuno-diagnostics complemented molecular testing in mapping the pandemic. Regional labs, validated kits and integrated data systems will strengthen future response.

Only ten minutes today?
Speak an outline. It gets evaluated too.
ghissu.com →

15(a) Explain the mechanism of action of vaccines. Give two examples where vaccines have eradicated diseases in India. What do you know about the Global Vaccine Action Plan 2011-2020 and Mission Indradhanush?

10

Introduction

Vaccination is a cornerstone of public health in a populous country like India, where infectious diseases weigh heavily on child mortality. Initiatives such as Mission Indradhanush have accelerated immunisation coverage and reduced vaccine-preventable diseases.

Mechanism of Action of Vaccines

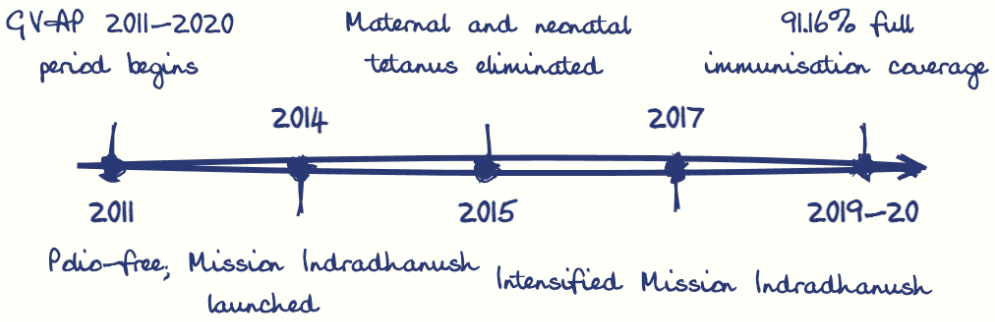
1. **Immune priming:** A harmless form or component of a pathogen, such as an inactivated virus or protein subunit, stimulates immunity without causing disease.
2. **Antibody production:** The body makes specific antibodies that neutralise the pathogen and reduce disease severity.
3. **Memory cells:** Long-lived memory B and T cells mount a faster, stronger response on real exposure.
4. **Herd immunity:** Wide coverage breaks the chain of infection, protecting even the unvaccinated.

Diseases Eradicated in India

1. **Polio:** India was declared polio-free in 2014 after decades of the Pulse Polio Programme, mop-up rounds and surveillance.

2. **Maternal and neonatal tetanus:** Eliminated in 2015 through high tetanus toxoid vaccination of pregnant women and better hygiene during childbirth, significantly reducing neonatal deaths. Andhra Pradesh implemented these drives, improving child health despite infrastructure gaps flagged by the 2024 CAG audit.

Immunisation milestones in India



GV-AP 2011–2020 and Mission Indradhanush

1. **GV-AP:** A WHO plan to prevent millions of deaths by 2020 through universal access to vaccines, stronger systems and new vaccines, stressing country ownership, equity, integration with health systems and measurable targets.

2. **Mission Indradhanush:** Launched in 2014 for

full immunisation of children under two and pregnant women, it raised annual coverage growth from 1% to 6.7% and added rotavirus and pneumococcal vaccines.

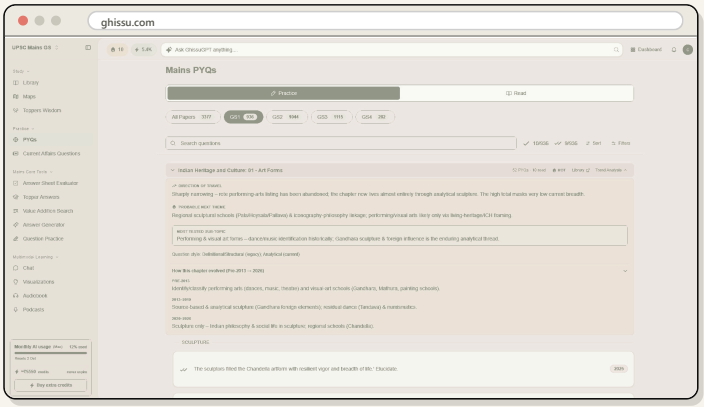
3. Intensified Mission Indradhanush: Launched in 2017 to aim for over 90% full immunisation, it reached 91.16% coverage by 2019–20. Andhra Pradesh took part in multiple phases through ASHAs and ANMs.

Conclusion

Vaccines have already ended polio and neonatal tetanus in India. Trained frontline workers and digital tracking like eVIN in Andhra Pradesh can sustain these gains.

THE REAL PRODUCT · NOT A MOCK-UP TREND ANALYSIS, PER SUB-TOPIC

Reading everything because you can't tell what's next?



1. Direction of travel: where the questions are heading.
2. The most tested sub-topic, named.
3. The probable next theme, from decades of papers.

See the trend for your chapter →

ghissu.com

14 days free · no card

SCAN

15(b) The 2020 Nobel Prize for Physiology and Medicine was awarded to three scientists for their work on HCV. What do you know about this virus and the disease it causes? Is there any vaccine available for this disease?

10

Introduction

The 2020 Nobel Prize in Physiology or Medicine was awarded to Harvey J. Alter, Michael Houghton and Charles M. Rice for discovering the Hepatitis C virus (HCV), a major cause of chronic liver disease worldwide. The discovery transformed the diagnosis, treatment and prevention of the disease.

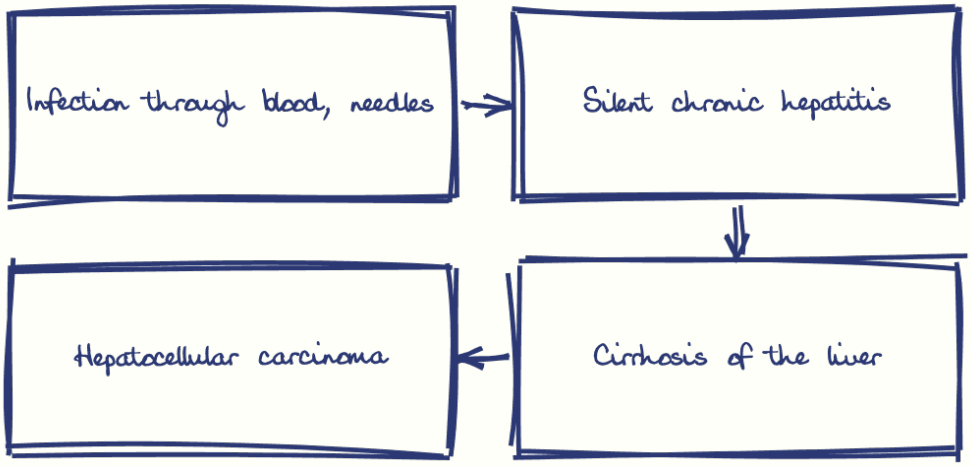
The Virus and the Disease It Causes

1. Chronic Liver Disease: HCV causes chronic hepatitis that progresses silently, leading to cirrhosis and hepatocellular carcinoma.
2. High-Risk Groups: Blood transfusion recipients and intravenous drug users are most at risk.
3. Treatment: Direct-acting antivirals (DAAs) have revolutionised therapy with high cure rates, and government hospitals and medical colleges in Andhra Pradesh offer HCV testing and these drugs.
4. Social Impact: In Andhra Pradesh, lack of awareness and stigma delay diagnosis, hurting

vulnerable groups most.

5. Access Barriers: Generic production has lowered DAA costs, but affordability and access remain hard in rural and tribal areas of Andhra Pradesh.

Course of Hepatitis C



Is a Vaccine Available?

1. No Approved Vaccine: Unlike hepatitis A and B, there is currently no approved vaccine for hepatitis C.

2. Prevention Without a Vaccine: Screening, safe blood transfusion practices and harm reduction become the core of control.

Response in India

1. NVHCP: The National Viral Hepatitis Control

Program targets hepatitis B and C, which Andhra Pradesh implements through district health systems, focusing on screening, awareness and linkage to care.

2. Research: ICMR leads hepatitis research, while PM-ABHIM strengthens health infrastructure in states including Andhra Pradesh.
3. Data Gaps: Underreporting and limited data on HCV prevalence in Andhra Pradesh impede targeted action.

Way Forward

1. Expand Screening: Scale up community screening of high-risk groups for early diagnosis.
2. Affordable Treatment: Ensure uninterrupted, subsidised DAAs through public facilities under Ayushman Bharat.

Conclusion

Though no vaccine exists yet, antiviral therapy offers real hope. Wider screening, subsidised DAAs and awareness in Andhra Pradesh will help achieve the SDG goal of ending viral hepatitis by 2030.

Are you actually improving?
Every attempt scored. Watch the curve.
ghissu.com →

APPSC GROUP-I MAINS · PAPER V

2017

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

P-IV
1(a)

What do you understand by "Big Data" ? Discuss India's initiatives in this regard; name the challenges and constraints.

10

Introduction

Big Data refers to vast volumes of structured and unstructured data generated continuously from diverse sources, which need advanced tools for storage, processing and analysis. It drives economic growth, governance and innovation, and India's digitalisation through Aadhaar and Digital India has positioned the country to harness it.

India's Initiatives

1. Open Data Ecosystem: data.gov.in offers over 0.16 million datasets across 33 sectors from 109 departments, aiding transparency and innovation.
2. Policy and Regulation: The Digital Personal Data Protection Act (2023) and the Draft National Data Governance Framework Policy balance privacy with data access for AI.
3. Institutional Coordination: A proposed India Data Office would unify data sharing among central and state agencies.
4. Service Platforms: UMANG and MyGov improve citizen engagement, while eSanjeevani and Kisan e-Mitra use data for healthcare and agriculture,

Practice 1(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

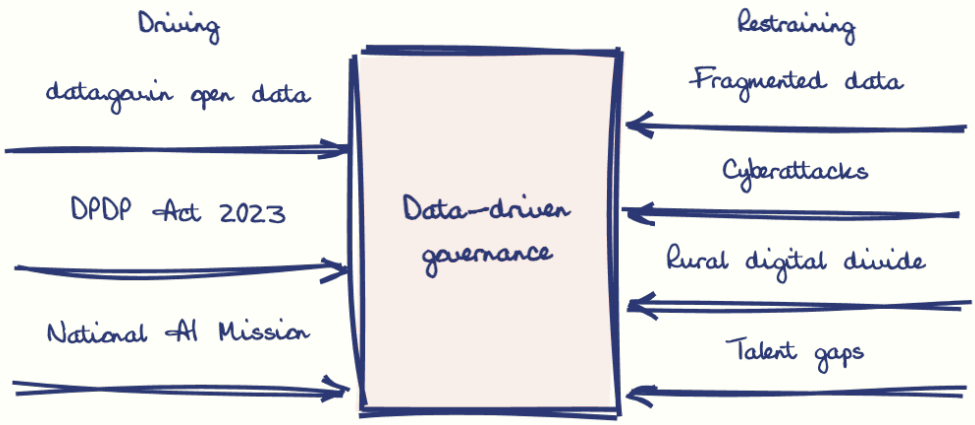
including in Andhra Pradesh.

5. Innovation Push: The National AI Mission and PLI schemes foster data-driven startups, while data localisation counters data colonisation.

Challenges and Constraints

1. Fragmentation and Quality: Ministries collect data separately with inconsistent standards, limiting integration and timely decisions.
2. Privacy and Security: Delayed data protection rules and rising cyberattacks threaten citizen trust.
3. Digital Divide: Despite 101.7 crore broadband subscribers, access is uneven in rural regions like Rayalaseema.
4. Talent and Datasets: A lack of annotated local datasets and skilled professionals hampers startups and indigenous innovation.

Big Data in India



Significance

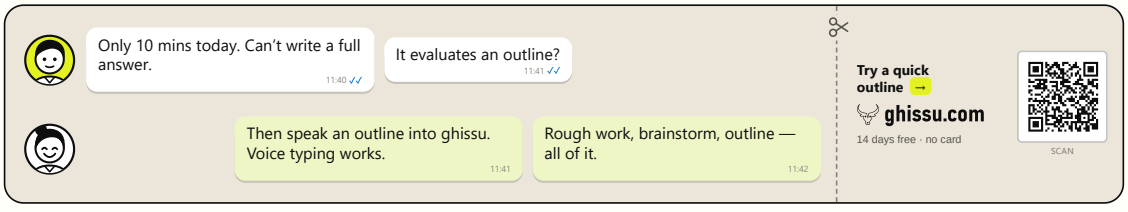
Big Data improves service efficiency, enables precision agriculture and personalised healthcare, and supports Atmanirbhar Bharat by building indigenous AI capacity and reducing dependence on foreign technology firms.

Way Forward

1. Data Governance: Finalise data protection and non-personal data policies.
2. Interoperability: Unified standards for sharing across central and state governments.
3. Skills and R&D: Capacity building in data science and patient capital for startups building indigenous datasets.
4. Digital Inclusion: Affordable broadband and digital literacy in underserved regions like Rayalaseema.

Conclusion

Big Data is a transformative asset for India, addressing privacy, quality and interoperability gaps will let it serve inclusive development.



P-IV
1(b)

The Internet of Things (IoT) is expected to be the biggest technological revolution. Examine its likely impact on India.

10

Introduction

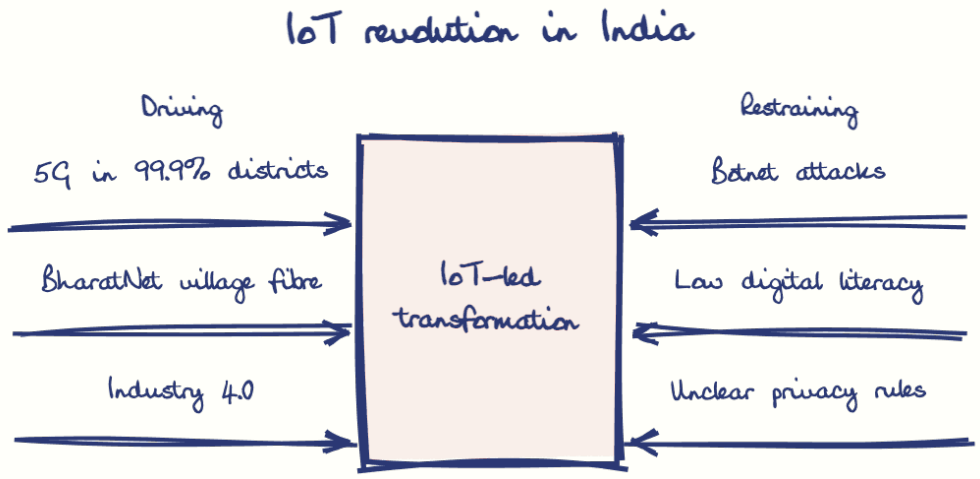
The Internet of Things (IoT) is emerging as a transformative technology for India. With the digital economy projected to cross US$1 trillion by 2025 and 5G covering 99.9% of districts, its integration into daily life and industry is increasingly feasible, and it is central to India's goals of inclusive growth and global competitiveness.

Likely Impact on India

1. Economic Productivity: Real-time data and automation raise output in manufacturing, logistics and agriculture; AP's aquaculture uses IoT sensors to monitor water quality, improving yield and cutting costs.
2. Service Delivery: IoT devices in telemedicine platforms like eSanjeevani widened healthcare access in rural Andhra Pradesh during COVID-19, supporting Digital India.
3. Resource Management: Soil moisture sensors in drought-prone Rayalaseema optimise irrigation and conserve water, in line with SDG 6.
4. Cybersecurity Risks: Unsecured devices invite

Practice 1(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V
1 / 3
Scan to practice

botnet attacks, a risk for smart city projects like Visakhapatnam.



Why it is a Revolution

1. Industry 4.0: IoT drives smart manufacturing, including mobile hubs on the Visakhapatnam–Chennai corridor, in a digital economy worth 11.74% of GDP.
2. Rural Connectivity: BharatNet, extended to over 2.15 lakh Gram Panchayats, lets IoT bridge the urban–rural divide.
3. Technological Sovereignty: Indigenous IoT supports Atmanirbhar Bharat, aided by the India AI Mission and FutureSkills Prime.
4. Hurdles: Low digital literacy, infrastructure gaps and unclear rules on data privacy remain.

Way Forward

1. **Security Standards:** Comprehensive IoT standards to protect infrastructure and citizen data.
2. **Digital Literacy:** Localised programmes in rural districts for equitable access.
3. **Indigenous R&D:** Support startups and research institutions building local IoT solutions.

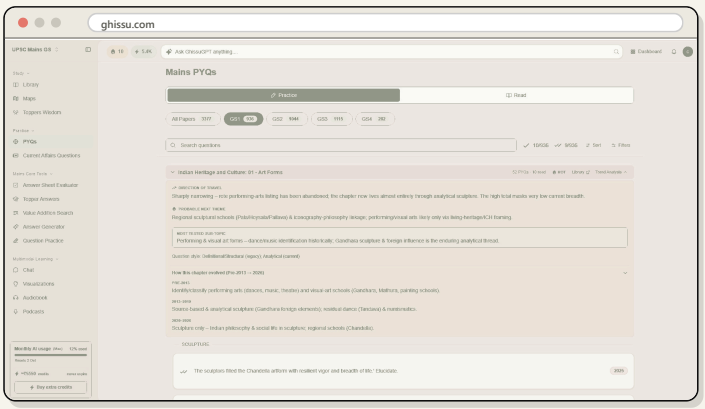
Conclusion

IoT can drive inclusive growth, social inclusion and sustainable development in India, if security and literacy gaps are closed.

THE REAL PRODUCT · NOT A MOCK-UP

TREND ANALYSIS, PER SUB-TOPIC

Reading everything because you can't tell what's next?



1. Direction of travel: where the questions are heading.
2. The most tested sub-topic, named.
3. The probable next theme, from decades of papers.

See the trend for your chapter →

ghissu.com

14 days free · no card

SCAN

P-IV
2(a)

Explain the background and salient features of "Pradhan Mantri Gramin Digital Saksharta Abhiyan".

10

Introduction

Digital literacy is a critical enabler of inclusive growth in rural India, where a large share of people remain digitally excluded. The Pradhan Mantri Gramin Digital Saksharta Abhiyan (PMGDISHA) was launched to equip rural citizens with essential digital skills so they can join the digital economy and access government services, in line with Digital India.

Background

1. Digital Exclusion: Rural internet coverage was about 31% against 67% in cities, and device costs and language barriers kept villagers out.
2. Enabling Infrastructure: BharatNet was bringing high-speed internet to villages, but people needed the skills to use it.
3. Approval: The Union Cabinet approved PMGDISHA in 2017.

Salient Features

1. Target: Make six crore rural individuals digitally literate by training one member per

Practice 2(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

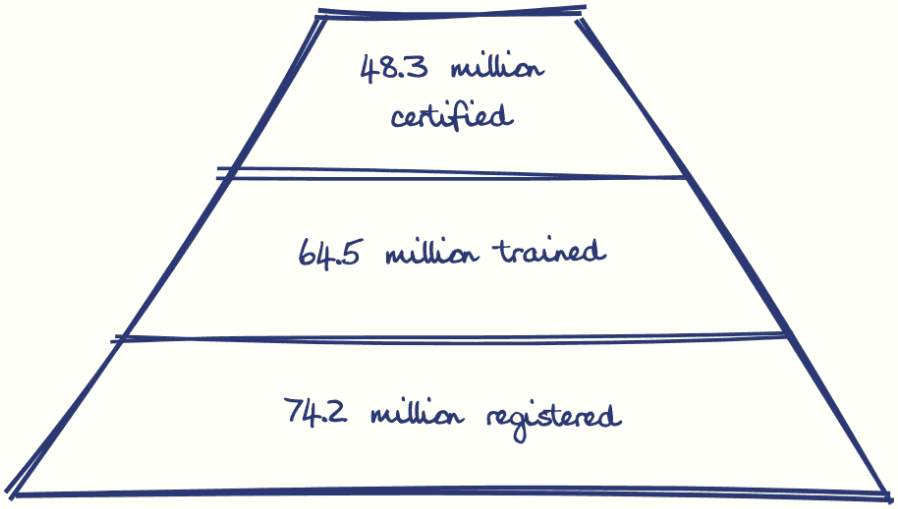
eligible household, covering about 40% of rural households.

2. Inclusive Focus: Priority to Scheduled Castes, Scheduled Tribes, women and the differently-abled.
3. Multi-Stakeholder Delivery: Government, NGOs and private partners train through Common Service Centres, rural schools and mobile digital vans, with MPs and MLAs leading awareness.
4. Certification and Monitoring: A certification process aims to certify one crore candidates a year, tracked on digital dashboards.
5. Synergy: It helps rural citizens use PM Jan Dhan Yojana and DigiLocker confidently.

Outcomes and Gaps

1. Scale: By 2024, 74.2 million registered, 64.5 million were trained and 48.3 million certified, among the largest programmes globally.
2. Accessibility: Costly devices and data and limited regional content still restrict participation of the remotest and poorest.
3. Training Quality: Variable training and grassroots infrastructure affect learning outcomes.

PMGDISHA reach by 2024



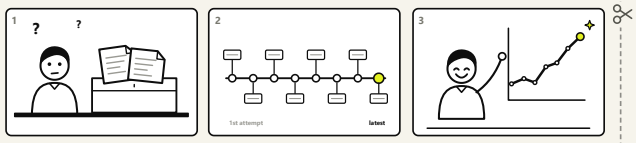
Way Forward

1. Local Content: Material in regional languages and dialects.
2. Trainer Capacity: Certify and upskill trainers for consistent quality.

Conclusion

PMGDISHA has laid a strong foundation for rural digital inclusion. Regional-language content and better trainers will help it bridge the rural-urban digital divide, in the spirit of Sabka Saath Sabka Vikas.

Fifty answers written. Any better?



Copies lost in a drawer · Every attempt on one timeline · You can see the climb

Start your timeline →
ghissu.com
14 days free · no card
SCAN

P-IV
2(b)

Cutting-edge developments in nanotechnology offer new ways of preventing and treating lifestyle diseases, including cancer. Elaborate.

10

Introduction

The rising burden of lifestyle diseases, including cancer, is a major public health challenge in India. Nanotechnology, which manipulates matter at the molecular and atomic levels, offers new avenues for early detection, targeted therapy and personalised medicine, in line with India's goal of technological self-reliance.

Key Advances in Nanotechnology

1. **Precision Nanomedicine:** Nanocarriers deliver drugs directly to affected cells, improving efficacy and reducing side effects.
2. **Nanosensors:** They monitor biomarkers of diabetes and cardiovascular disease in real time, enabling early diagnosis and continuous monitoring.
3. **Nano-Delivery Systems:** Liposomes and polymeric nanoparticles improve bioavailability and controlled release, aiding adherence.
4. **Digital Integration:** Nanotechnology combined with AI and data analytics supports predictive, personalised treatment.

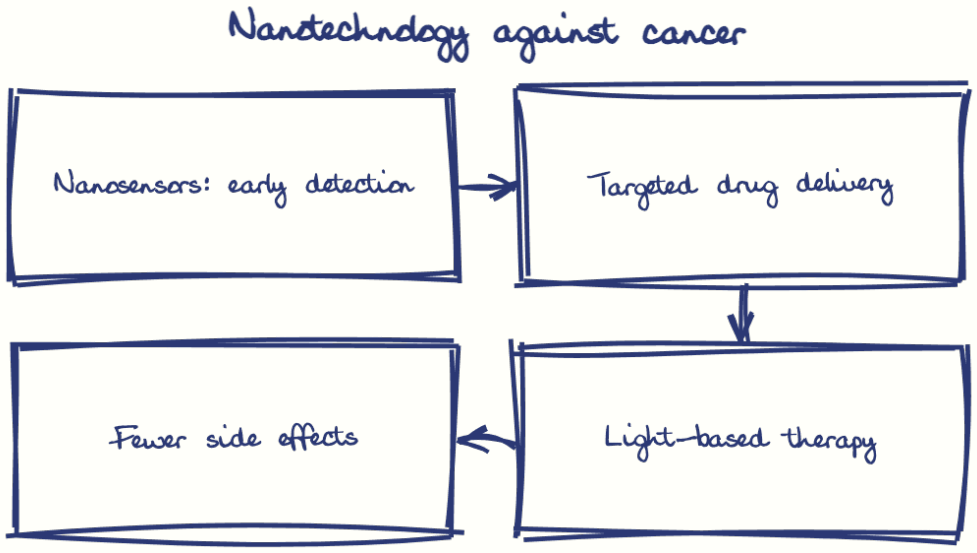
Practice 2(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Prevention and Treatment of Lifestyle Diseases

1. Nano-Nutraceuticals: Nano-encapsulated antioxidants and vitamins are absorbed better, helping prevent chronic inflammation linked to lifestyle diseases.
2. Better Diagnosis: Nanoparticles sharpen imaging and biosensors for earlier detection of diabetes and heart disorders.
3. Fewer Side Effects: Targeted delivery reduces the systemic toxicity of conventional therapy.
4. Smart Packaging: It keeps supplements stable and flags contamination.

Cancer Prevention and Therapy



1. Targeted Drug Delivery: Nanoparticles concentrate chemotherapy in tumours and spare normal tissue, crucial for tobacco-linked oral and lung cancers

common in India.

2. Early Detection: Nanobiosensors detect cancer biomarkers at very low concentrations, improving survival.

3. Light-Based Therapy: Photothermal and photodynamic therapies use nanoparticles to destroy cancer cells with fewer side effects.

4. AI Integration: The First Cancer Care initiative combines nanotech and AI for scalable cancer management, including in Andhra Pradesh.

Way Forward

1. Indigenous R&D: Fund research institutions in Andhra Pradesh to suit local disease patterns.

2. Regulation: Set clear guidelines on the safe use of nanomaterials.

3. Training: Build capacity of health professionals in Andhra Pradesh to use nano-diagnostics.

Conclusion

Nanotechnology can transform prevention and treatment of lifestyle diseases and cancer through precision care. Affordable translation through public-private partnerships will make its benefits reach all.

Where did you write that point?
Notes that live with the question.
ghissu.com →

Write your own answer to 2(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
3(a)

What is GPS ? Explain the recent efforts made by India to create its own GPS system.

10

Introduction

The Global Positioning System (GPS) is a satellite-based navigation technology that provides real-time location and timing information globally, underpinning sectors from defence to disaster management. India's dependence on foreign GPS was exposed in the Kargil War, when access was denied, showing the need for an indigenous system.

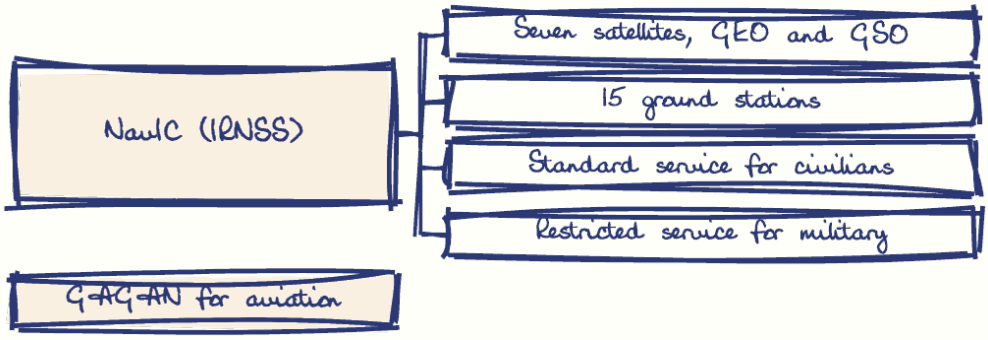
India's Efforts to Create Its Own System

1. **IRNSS or NAVIC:** India launched seven satellites in geostationary and geosynchronous orbits to give accurate positioning over India and a 1500 km radius.
2. **Two Service Tiers:** A Standard Positioning Service serves civilians, while a Restricted Service serves the military and authorised users.
3. **ISRO and Ground Infrastructure:** ISRO designs, launches and operates the satellites, supported by 15 ground stations for control and navigation parameters.
4. **GAGAN:** With the Airport Authority of India, ISRO built GAGAN to augment GPS

and improve aviation safety over Indian airspace.

5. Strategic Policy: The Ministry of Defence and Department of Space prioritise reducing dependence on GPS, GLONASS and Beidou, especially for the military.

India's own navigation system



Outcomes and Challenges

1. Precision: NAVIC gives accuracy better than (20) metres in all weather, vital for defence, disaster management and maritime navigation.
2. Dual Use: It aids Andhra Pradesh's coastal fisheries, early disaster warnings and mariners' safety.
3. Interoperability: Receivers work with GPS and GLONASS, though compatibility and interference issues remain.
4. Adoption: Full operational status and use in commercial devices and government services are

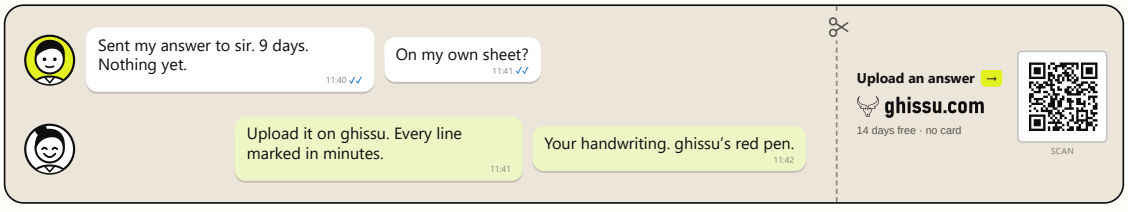
still in progress.

Way Forward

1. **More Satellites:** Expand the constellation and upgrade ground infrastructure.
2. **Indigenous Receivers:** Local manufacturing under Make in India.
3. **Civil–Military Coordination:** Optimise dual-use potential in coastal security and disaster response.
4. **International Collaboration:** Interoperability and data-sharing agreements with other navigation providers.

Conclusion

NAVIC marks a significant stride towards technological sovereignty, reducing vulnerability to foreign denial and advancing Atmanirbhar Bharat. Its use in defence, disaster management and maritime activity strengthens both security and development.



P-IV
3(b)

What is the difference between GSLV and PSLV launch vehicles ? Examine the advantages of GSLV over PSLV.

10

Introduction

India's space ambitions, shown in missions like Chandrayaan-3, depend on reliable launch vehicles. The PSLV and GSLV series are central to this, deploying satellites for communication, navigation and earth observation.

Differences between PSLV and GSLV

1. Design Purpose: PSLV launches satellites into polar and sun-synchronous orbits for earth observation, while GSLV places heavier payloads into geosynchronous transfer orbit (GTO) for communication satellites.

2. Technology: GSLV has a cryogenic upper stage using liquid hydrogen and liquid oxygen at very low temperatures, which PSLV lacks. ISRO mastered it with GSLV-D5 in (2014).

3. Payload Capacity: GSLV Mk III carries up to (4) tons to GTO and about (10) tons to LEO, well beyond PSLV.

4. Flight Record: PSLV has over (49) successful flights and a world record of (104) satellites in one mission; GSLV has six consecutive

successes since mastering cryogenics.

PSLV vs GSLV

PSLV	GSLV
Polar, sun-synchronous orbits	Geosynchronous transfer orbit
Solid and liquid stages	Cryogenic upper stage
Lighter payloads	Up to 4 tons to GTO
Over 49 successful flights	Fewer flights, costlier

Advantages of GSLV over PSLV

1. **Higher Payload:** It launches heavier communication and navigation satellites like CMS-03 from Sriharikota.
2. **Geostationary Access:** Satellites in GTO give continuous coverage for weather forecasting and broadband, supporting Digital India in rural areas.
3. **Strategic Autonomy:** Heavier communication and reconnaissance satellites strengthen national security.
4. **Commercial Opportunity:** It can launch heavier commercial satellites, though cryogenic complexity raises cost and limits launch cadence.

5. Technological Mastery: Handling solid, liquid and cryogenic stages places India among the few nations with such advanced technology.

Limitations

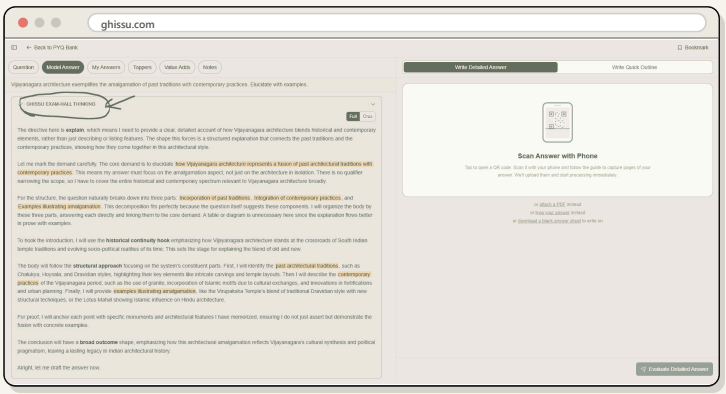
GSLV's costs are higher and its launch cadence harder to scale, while PSLV benefits from higher frequency, established supply chains and international customers.

Conclusion

PSLV remains the cost-effective workhorse for small and medium satellites, while GSLV gives India heavy-lift capability and greater strategic autonomy.

THE REAL PRODUCT · NOT A MOCK-UP EXAM-HALL THINKING

Know the topic, but freeze at the question?



Decode your next question →

ghissu.com

14 days free · no card

SCAN

1 The question broken open before a single line of the answer.

2 Directive, demand and structure highlighted in the text.

3 Full walks a beginner through it; Crux is a ten-second check.

P-IV
4(a)

Discuss India's present composition of energy sources and future energy security scenario in the context of growing energy demand.

10

Introduction

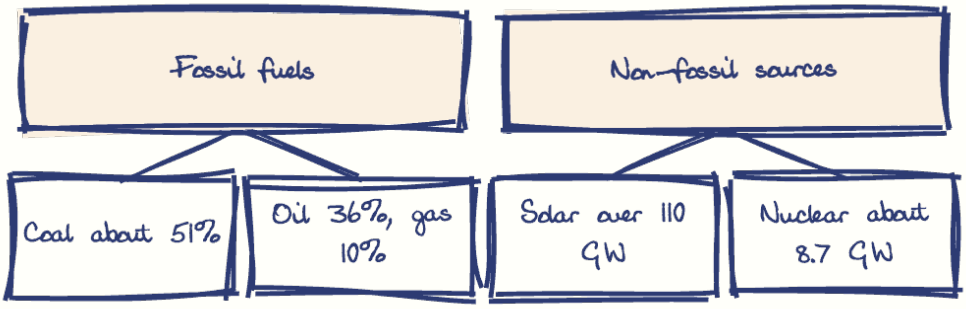
India's energy landscape is at a critical juncture as its economy and population grow. Energy demand is expected to rise two to two-and-a-half times by 2047. Ensuring energy security while honouring climate goals makes a diversified energy mix essential.

Present Composition of Energy Sources

1. **Dominance of Fossil Fuels:** Coal accounts for about 51%, oil 36% and natural gas around 10% of primary energy, with coal as the backbone of power generation.
2. **Rising Renewables:** Non-fossil sources form about 45.5% of installed power capacity (June 2024); solar grew from 2.82 GW in 2014 to over 110 GW in 2025 and wind more than doubled.
3. **Limited Nuclear:** About 1.8% of the power mix (8.7 GW), planned to exceed 22 GW by 2031–32 and possibly 100 GW by 2047.
4. **Andhra Pradesh:** Its long coastline and solar-friendly climate have driven solar and

wind growth, and the port-led Visakhapatnam-Chennai corridor aids energy logistics.

India's present energy mix



Future Energy Security Scenario

1. **Import Dependence:** Reliance on imported oil and gas exposes India to price volatility; LNG demand may reach (103) BCM by (2030).
2. **Rapid Demand Growth:** Urbanisation, industrialisation and rising living standards will at least double demand by (2047); in Andhra Pradesh, farming, industrial corridors and urban growth push demand up.
3. **Market Reforms:** Opening coal mining to private investment and a competitive wholesale power market can improve efficiency.
4. **Energy Trilemma:** Security, equity and sustainability must be balanced through grid

flexibility and storage.

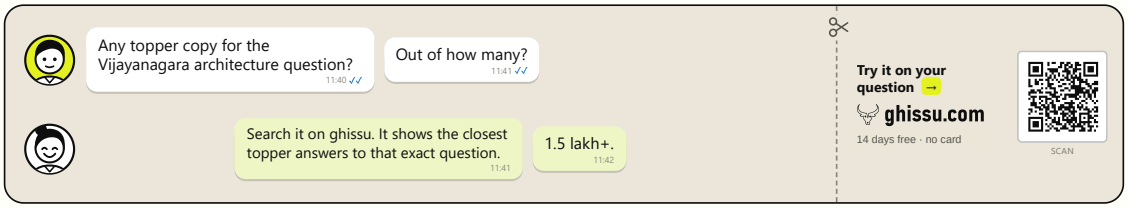
5. Health and Environment: The shift from biomass to LPG and electricity improves health but raises fossil fuel use unless renewables accelerate.

Way Forward

1. Accelerate Renewables: Scale up solar, wind and hydro, using Andhra Pradesh's geographic advantages, to cut fossil dependence.
2. New Technologies: Coal gasification and battery storage can reduce imports and emissions.
3. Efficiency: Strengthen the Perform Achieve and Trade scheme under the National Action Plan on Climate Change.

Conclusion

India's energy security depends on diversifying its mix, reforming markets and adopting clean technology so that rising demand is met reliably and sustainably.



P-IV
4(b)

What are the benefits of "Smart Grid" ? Examine the challenges and strategies for "Smart Grid" operations in India.

10

Introduction

With installed capacity above 520 GW in early 2026, India needs smarter grid management. The National Smart Grid Mission (NSGM) and Smart Meter National Programme (SMNP) drive adoption of technologies vital for renewables, loss reduction and consumer empowerment, including in fast-growing Andhra Pradesh.

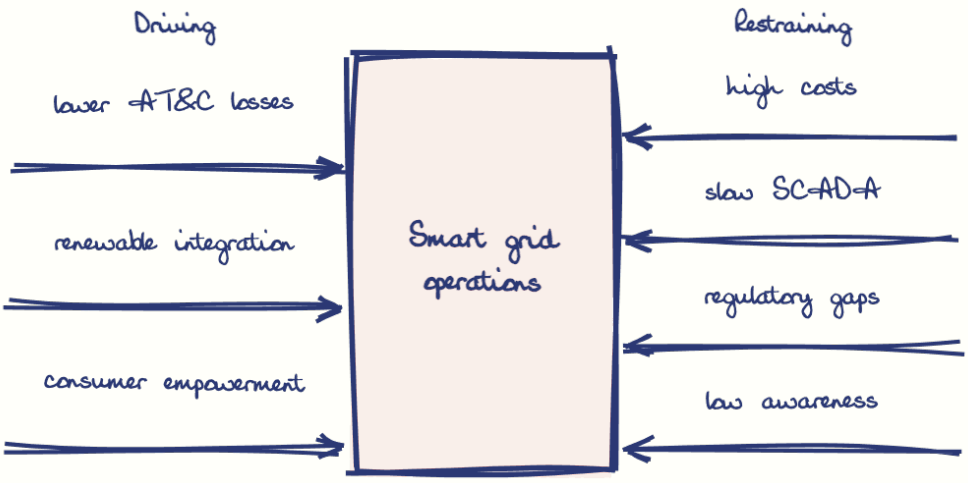
Benefits of Smart Grid

1. Efficiency: Real-time monitoring cuts ~~AT&C losses~~; NSGM pilots showed reductions of up to 17%, curbing theft.
2. Renewable Integration: Manages variability of solar and wind as distributed energy resources, keeping the grid stable as Andhra Pradesh expands renewables.
3. Consumer Empowerment: ~~Advanced Metering~~ Infrastructure lets consumers track use and respond to dynamic tariffs, saving cost and easing peak demand.
4. Reliability: Faster fault detection and restoration, vital for Visakhapatnam's industries.

Challenges

1. High Costs: Smart meters, automation and IT strain financially stressed discoms.
2. Technical Limits: SCADA updates every 5 seconds are too slow for advanced functions.
3. Regulatory Barriers: Weak inter-agency coordination, delayed tariff reform and aligning Andhra Pradesh's policy with NSGM.
4. Low Acceptance: Limited awareness and data privacy concerns, especially in rural and semi-urban areas.

Smart grid rollout in India



Strategies

1. Phased Pilots: Expand NSGM pilots before large rollouts.

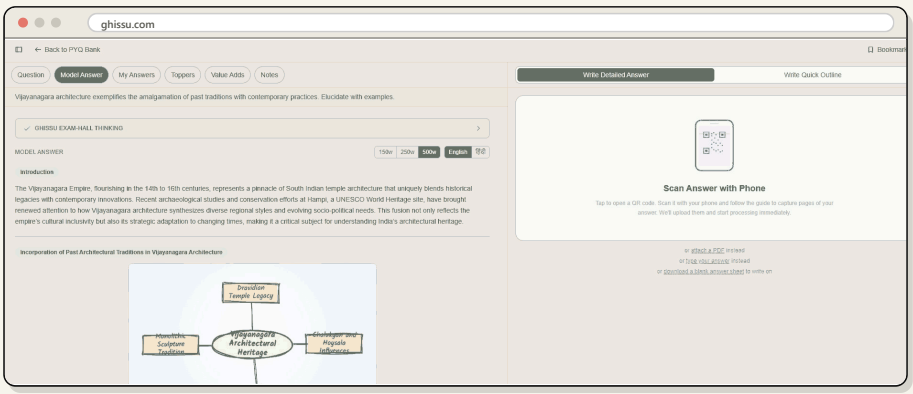
2. PPP Models: SMNP with EESL's BOOT model reduces upfront cost.
3. Capacity Building: Train utility staff and consumers.
4. Regulatory Reform: Stronger SERCs in tariff design and robust data privacy rules.
5. Microgrids: DER management suited to Andhra Pradesh's renewable potential.

Conclusion

Smart grids can transform India's power sector by cutting losses and absorbing renewables. Closing cost, technology and trust gaps through phased rollout will make it reliable and green.

THE REAL PRODUCT · NOT A MOCK-UP MODEL ANSWERS IN 150 · 250 · 500

Writing 400 words for a 150-word question?



1. One question, three lengths: 150 · 250 · 500 words.
2. Two languages at a tap: English and हिंदी.
3. Keywords, structure and diagrams the examiner rewards.

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

P-IV
5(a)

Why are urban floods a recurrent feature in India ? Discuss the main causes of urban floods and measures taken by the Government(s) for urban flood control.

10

Introduction

Urban floods have become increasingly frequent and severe in India, driven by climate change and rapid urbanisation. The IPCC notes that rising temperatures intensify monsoon precipitation, causing extreme rainfall events in cities like Mumbai, Hyderabad and Chennai, which disrupt economies, damage infrastructure and threaten lives.

Why Urban Floods Recur: Main Causes

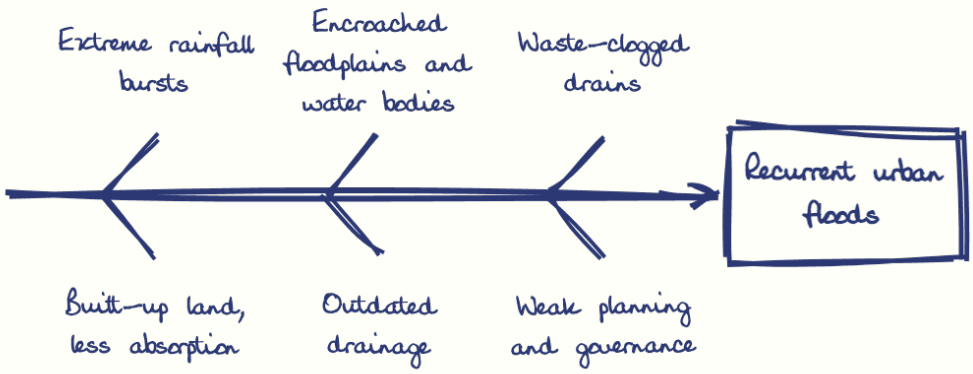
1. Climate Change: Heavier, more intense monsoon bursts now regularly exceed what city drains were built for.
2. Rapid Urbanisation: Built-up land has grown sharply in cities like Delhi, leaving less open ground to absorb rain.
3. Encroachment: Construction in low-lying areas, floodplains and water bodies blocks natural channels, as in Visakhapatnam and other Andhra Pradesh cities.
4. Inadequate Drainage: Outdated or insufficient networks cannot cope with intense rainfall, causing waterlogging.

5. Solid Waste: Of Andhra Pradesh's (123) ULBs, only (4) had operational sanitary landfills in (2022), so waste blocks drains.

6. Weak Planning and Governance: Many cities lack drainage master plans, and overlapping roles of municipal, state and central bodies delay response.

7. Little Green Infrastructure: Water-Sensitive Urban Design remains patchy, reducing natural stormwater capacity.

Why urban floods recur



Measures Taken by Governments

1. NDMA Guidelines: After the (2005) Mumbai floods, the (2008) National Guidelines for Urban Flood Management recognised urban floods as distinct and stressed preparedness.

2. UFRMP: The Urban Flood Risk Management

Programme supports cities in risk assessment and infrastructure upgrades, which Andhra Pradesh ULBs are gradually adopting.

3. Smart Cities Mission: Encourages urban watershed modelling and real-time flood monitoring.
4. Monsoon Action Plans: Cities are urged to prepare local plans and risk maps, though many Andhra Pradesh cities lack risk atlases.

Way Forward

1. Integrated Water Management: City drainage master plans with green infrastructure and natural retention.
2. Stronger ULBs: Resources for Andhra Pradesh ULBs to manage waste and maintain drains.
3. Real-Time Forecasting: Rain atlases with 15-minute interval data.

Conclusion

Urban floods recur because climatic stress meets unplanned growth and weak institutions. Coordinated governance, technology and sustainable planning can build resilient cities in line with SDG (11) and SDG (13).

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 5(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
5(b)

Critically examine how illegal mining of river sand in various parts of India is affecting the ecology of rivers and lives of people in the vicinity of these rivers.

10

Introduction

Illegal sand mining has become a critical environmental and social challenge, driven by the construction boom and weak enforcement. It disrupts river ecosystems, threatens biodiversity and undermines livelihoods, especially in AP, where the Godavari and Krishna sustain agriculture, drinking water and fisheries.

Impact on River Ecology

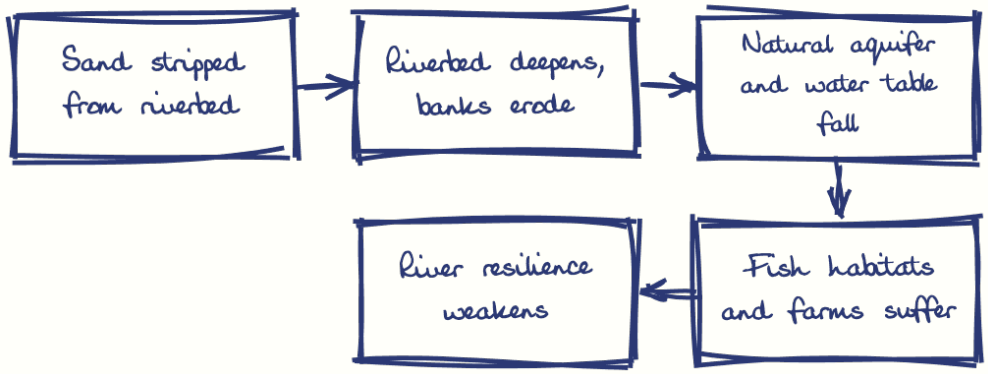
1. Riverbed degradation: Stripping sand deepens riverbeds, enlarges mouths and destabilises banks; rampant mining aggravated the Gwalior–Chambal floods of 2021.
2. Groundwater depletion: Riverbed sand is a natural aquifer; mining in the Papagni catchment lowered water tables in villages of AP and Karnataka.
3. Biodiversity loss: Micro-organism habitats critical for soil fertility and nutrient cycling are destroyed.
4. Erosion and coastal damage: Heavy machinery damages roads and bridges, while coasts face shoreline erosion and saline water ingress,

Practice 5(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

threatening mangroves.

How illegal sand mining harms rivers



Impact on People

1. Agriculture: Falling water tables cause water scarcity and lower yields for riverbank farmers in AP.
2. Fisheries: Destroyed breeding grounds cut fish populations and fishermen's income.
3. Workers' health: Manual miners suffer musculoskeletal pain and respiratory disease, as studies in Karnataka and UP show.
4. Revenue loss: Royalty and tax evasion, with audits finding compliance gaps.

Critical Assessment

1. River equilibrium: Mining upsets the sediment balance and flow regime.

2. Weak enforcement: The MMDR Act and the Sustainable Sand Mining Management Guidelines, 2016 exist, but compliance is poor.

3. Feedback loop: Mining, groundwater depletion and habitat loss together accelerate river decline and weaken resilience.

Way Forward

1. Surveillance: Use iMSS and Mine Mitra in AP.

2. Community committees: Participatory committees of riverbank communities should oversee sustainable sand mining.

3. Worker safety: Enforce occupational health and safety rules for sand miners.

Conclusion

Illegal sand mining trades long-term river health for short-term gain. Strict enforcement and community stewardship can protect both ecology and livelihoods.

REVIEWER'S NOTE · VALUE ADDITION

Examiners read hundreds of answers a day. The one with a number, a committee and a case is the one they remember.

ghissu

ghissu gives you the data point, the report and the example for every question — each linked to its source.

Get the value adds for any question →

ghissu.com

14 days free · no card

SCAN

P-IV
6(a)

Write a brief account of Bentham and Hooker's classification and discuss its demerits.

10

Introduction

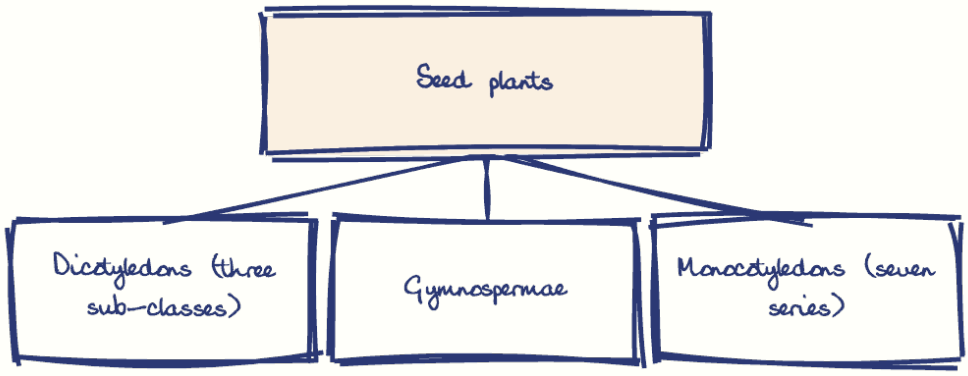
Bentham and Hooker's classification is the most widely used natural system of classifying seed plants. George Bentham and Joseph Dalton Hooker published it in Genera Plantarum (1862 to 1883) from specimens at Kew. Gamble's Flora of the Presidency of Madras, covering present-day Andhra Pradesh, follows it, so it shapes Indian herbaria and teaching.

Outline of the System

1. Basis: Plants are grouped on many similarities taken together, especially floral characters.
2. Scope: It covers only seed plants: about 202 families and 7,569 genera.
3. Three classes: Dicotyledons, Gymnospermae and Monocotyledons, with gymnosperms placed between the two angiosperm classes.
4. Dicotyledons: Sub-classes Polypetalae (Thalamiflorae, Disciflorae, Calyciflorae), Gamopetalae (Inferae, Heteromerae, Bicarpellatae) and Monochlamydeae.

5. Monocotyledons: Seven series, from Microspermae (orchids) to Glumaceae (grasses).

Bentham and Hooker's outline



Merits

1. Practical utility: Descriptions from real specimens make identification easy.
2. Natural grouping: Related families largely stay together, and Ranales first fits later views.
3. Wide adoption: Floras of the Eastern Ghats and Andhra Pradesh follow it, and it remains a teaching standard.

Demerits

1. Not phylogenetic: Though published after Darwin's Origin of Species, it does not show evolutionary relationships.
2. Position of gymnosperms: They sit between dicots

and monocots though more primitive than all angiosperms.

3. Artificial Monochlamydeae: Unrelated families are grouped only for absent petals, so Euphorbiaceae and Urticaceae lie far from their relatives.

4. Misplaced families: Advanced Orchidaceae opens the monocots, while primitive Liliaceae comes later.

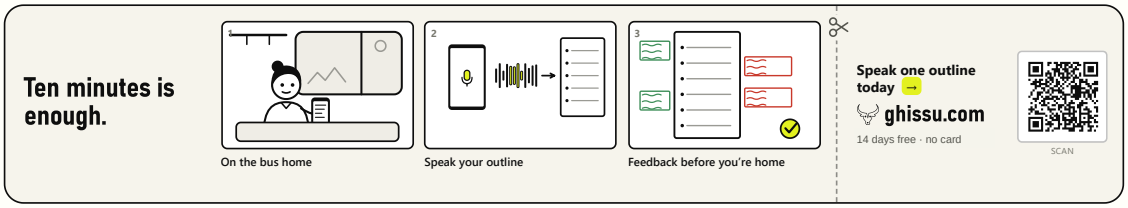
5. Gamopetalae over Polypetalae: Treating all gamopetalous families as more advanced does not always hold.

6. Limited characters: It ignores anatomy, embryology, cytology and chemistry, used by Engler, Hutchinson and the APG.

7. Seed plants only: Algae, fungi, bryophytes and pteridophytes are left out.

Conclusion

Its demerits, chiefly misplaced gymnosperms, the artificial Monochlamydeae and no evolutionary basis, led to phylogenetic systems, but its clarity keeps it valuable for teaching and identification.



P-IV
6(b)

Which hormone regulates blood glucose levels in the body and how is it synthesized artificially ?

10

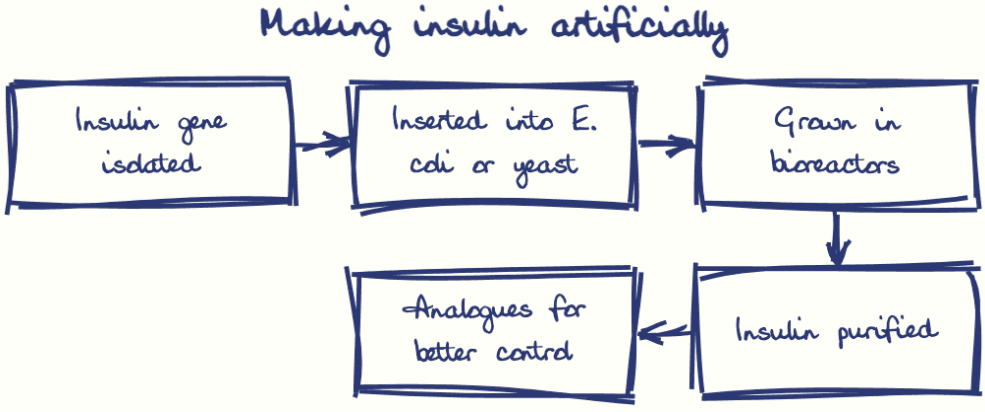
Introduction

Blood glucose regulation is vital for energy balance and metabolic health, especially with rising diabetes in India and Andhra Pradesh. Insulin is the key hormone that lowers blood glucose, and biotechnology now synthesises it artificially, transforming treatment.

Hormone Regulating Blood Glucose

1. Source: Insulin is secreted by the beta-cells of the Islets of Langerhans in the pancreas.
2. Action: It promotes cellular glucose uptake and glycogenesis, lowering blood sugar and maintaining glucose homeostasis.

Artificial Synthesis of Insulin



1. Recombinant DNA technology: The insulin gene is introduced into genetically engineered Escherichia coli or Saccharomyces cerevisiae.
2. Production in bioreactors: The engineered microbes are grown in sophisticated bioreactors under controlled conditions, where they multiply and produce insulin in bulk.
3. Purification: The insulin is separated and purified so that it is bioequivalent to human insulin and safe to inject.
4. Insulin analogues: The same technology makes rapid-acting and long-acting analogues that improve glycaemic control and quality of life, with research at Sri Venkateswara University and other AP institutions.

Because the human gene is used, the product matches natural insulin, which ended reliance on animal sources and made large-scale supply possible.

Regulation and Access

1. Oversight: CDSCO and NPPA regulate quality, pricing and availability.
2. Public schemes: National Health Mission programmes and Jan Aushadhi supply affordable insulin through government hospitals in Andhra

Pradesh's districts.

Challenges

1. Cost and cold chain: Production costs, cold chain needs and weak health infrastructure keep insulin costly for many in rural Andhra Pradesh, hurting adherence.
2. Quality risks: Counterfeits and variable bioactivity need stronger monitoring.
3. Digital health: Telemedicine helps timely dose adjustment and adherence tracking.
4. Scaling up: Indigenous production needs investment, skills and streamlined regulation.

Way Forward

Invest in recombinant research in biotech clusters, train primary health centre staff in insulin use, and link Jan Aushadhi with diabetes screening.

Conclusion

Insulin is the principal hormone regulating blood glucose. Recombinant synthesis has widened access, and stronger R&D, anti-counterfeit checks and public schemes can make it affordable for all.

Which sub-topic comes next?
Trend analysis for every syllabus topic.
ghissu.com →

Write your own answer to 6(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
7(a)

Describe the medicinal qualities of mushrooms.

10

Introduction

Mushrooms have gained attention for their dual role as nutritious food and potent medicinal agents. With rising health consciousness and demand for natural remedies, mushroom cultivation and research have expanded in India, including Andhra Pradesh, alongside the national push under AYUSH.

Key Medicinal Properties

1. **Immune modulation:** Polysaccharides enhance innate and adaptive immune responses, acting as bioresponse modifiers rather than direct cytocidal agents.
2. **Antitumor activity:** Species like Ganoderma lucidum and Lentinula edodes inhibit tumour growth, mainly through beta-glucans.
3. **Antiviral and antibacterial effects:** Extracts of Trametes versicolor and Inonotus obliquus help manage infections.
4. **Cardioprotective and anti-inflammatory:** Mushrooms help regulate blood pressure, reduce cholesterol and curb inflammation.

Therapeutic Applications

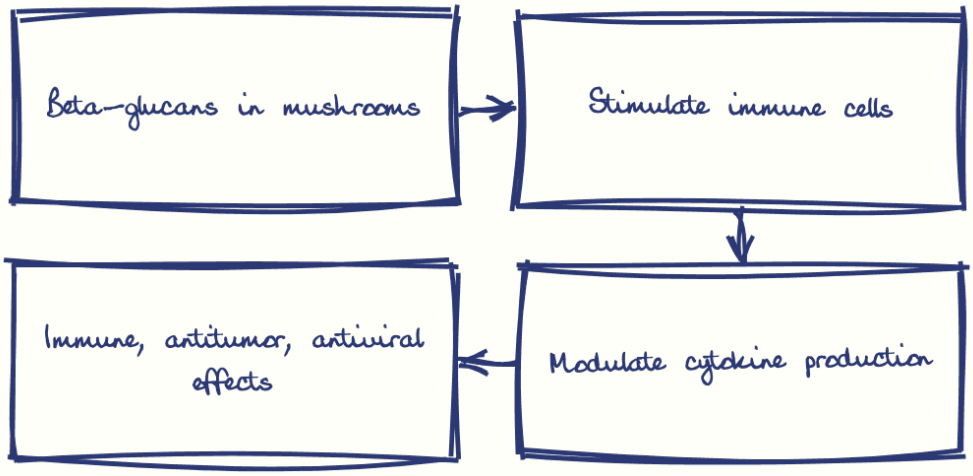
1. Cancer adjunct therapy: Polysaccharide-K and lentinan are used with chemotherapy and radiation in Japan and China, improving outcomes by boosting immunity.
2. Chronic disease management: Being low fat, cholesterol-free and high in fibre, mushrooms help diabetic and hypertensive patients control sugar and blood pressure.
3. Nutritional supplementation: Rich in essential amino acids, vitamin D and minerals, they address deficiencies common in vegetarian populations.

Bioactive Compounds Behind These Effects

1. Beta-glucans: The most studied compounds, they stimulate immune cells and modulate cytokine production.
2. Glycoproteins and proteoglycans: They aid immune modulation and reduce tumour growth and inflammation.
3. Ergosterol and fatty acids: Ergosterol, a vitamin D precursor, and polyunsaturated fatty acids support heart health and act as antioxidants.
4. Vitamins and minerals: Riboflavin, niacin, biotin, potassium and magnesium support metabolic and

physiological functions.

How beta-glucans act



Way Forward

1. Cultivation and research: Promote mushroom farming on local agro-waste in Andhra Pradesh and fund clinical validation of extracts within AYUSH.
2. Awareness: Campaigns on the medicinal benefits of mushrooms can raise consumption and acceptance as functional foods.

Conclusion

Mushrooms combine nutritional and medicinal qualities that can significantly improve public health. Harnessing them advances the Sustainable Development Goal of healthy lives and well-being for all.

Are you actually improving?
Every attempt scored. Watch the curve.
ghissu.com →

P-IV
7(b)

Give an account on chemicals and drugs produced from animals.

10

Introduction

Animal-derived chemicals and drugs have played a crucial role in medicine and industry for centuries, offering bioactive compounds that cannot easily be synthesised artificially. With India's growing pharmaceutical and animal husbandry sectors, they matter for healthcare, agriculture and the economy, especially in Andhra Pradesh where livestock and dairy support rural livelihoods.

Key Chemicals and Drugs from Animals

1. **Hormones:** Insulin, originally extracted from the pancreas of pigs and cows, remains vital for diabetes treatment, though synthetic versions are now common.

2. **Enzymes:** Rennet, derived from calf stomachs, is essential in cheese production, linking animal biochemistry to Andhra Pradesh's dairy sector.

3. **Antimicrobial agents:** Certain animal-derived peptides and enzymes have antimicrobial properties, used in veterinary medicine to control diseases in Andhra Pradesh's large livestock population.

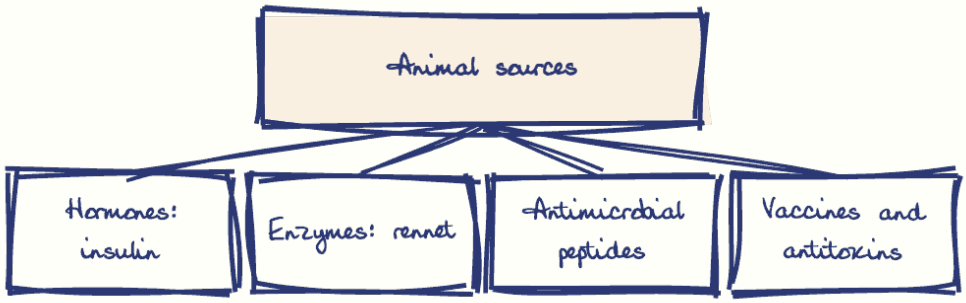
4. **Vaccines and antitoxins:** Animal tissues and blood components produce vaccines and antitoxins,

Practice 7(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

for example equine serum for rabies treatment.

Drugs and chemicals from animals



Significance

1. Medical therapeutics: Drugs like insulin are irreplaceable in managing chronic disease such as diabetes.
2. Livestock productivity: Animal-based antimicrobials and vaccines keep livestock healthy directly raising milk and meat output in a sector that contributes significantly to the state's agricultural GVA.
3. Industry and research: Animal chemicals feed pharmaceuticals and cosmetics in Visakhapatnam and other industrial corridors, and serve as models and raw material for drug development.

Challenges

1. Animal welfare: Extraction raises issues of pain

and exploitation. The CCSEA oversees ethical standards, but enforcement is weak in rural Andhra Pradesh.

2. Biodiversity: Overharvesting and illegal wildlife trade threaten species and risk zoonotic disease.
3. Quality control: Counterfeit and substandard products pose health risks.
4. Cost: Costly extraction and purification limit access, and R&D needs more investment to reduce import dependence.

Way Forward

1. Ethical oversight: Strengthen CCSEA and the Animal Welfare Board of India to ensure humane drug production.
2. Indigenous R&D: Raise investment in animal biotechnology to build affordable, home-grown animal-derived medicines.

Conclusion

Animal-derived chemicals and drugs remain indispensable for human and veterinary health. Ethical, regulated and research-led production can secure their benefits while protecting animals and biodiversity.

Where did you write that point?
Notes that live with the question.
ghissu.com →

Write your own answer to 7(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
8(a)

Describe the major types of fungus that cause Mycoses in humans.

10

Introduction

Mycoses, fungal infections of humans, have gained attention due to rising incidence among the immunocompromised and during the COVID-19 pandemic. The fungi causing them vary in biology and pathogenic mechanism, which makes diagnosis and treatment complex.

Major Types of Fungi Causing Mycoses

1. Phycomycetes (Zygomycetes): Primitive fungi with aseptate, coenocytic hyphae, reproducing by sporangiospores. Mucor and Rhizopus cause mucormycosis, a severe angioinvasive infection in diabetic or immunocompromised patients.

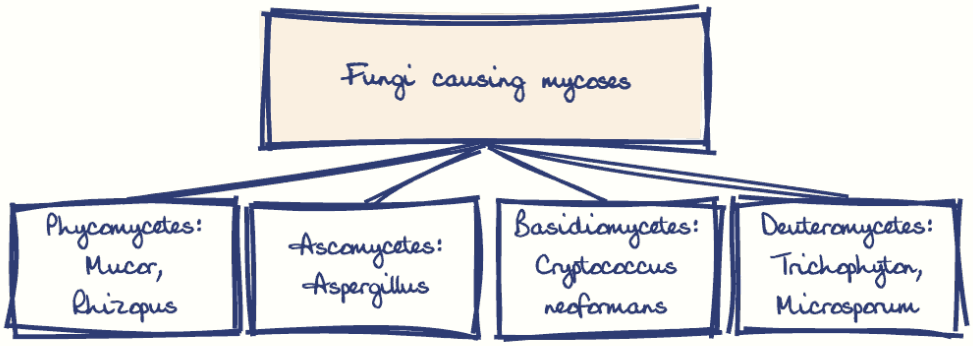
2. Ascomycetes (sac fungi): Septate hyphae with ascospores formed in asci. Aspergillus causes aspergillosis, from allergic reactions to invasive disease in lung patients.

3. Basidiomycetes: Advanced fungi bearing basidiospores on basidia, including mushrooms. Cryptococcus neoformans causes life-threatening cryptococcosis in HIV/AIDS patients.

4. Deuteromycetes (imperfect fungi): Known only by their asexual stage. Dermatophytes such as Trichophyton

and Microsporum cause ringworm and athlete's foot.

Major Fungal Groups Causing Mycoses



Clinical Context in Andhra Pradesh

1. Mucormycosis: Cases rose after COVID-19, as steroid use and diabetes increased susceptibility, affecting sinuses, lungs and brain.
2. Aspergillosis: Humid coastal regions like Visakhapatnam raise exposure to Aspergillus spores.
3. Cryptococcosis: Though less common, it is significant among HIV/AIDS patients, needing diagnostic capacity in tertiary hospitals.
4. Dermatophytoses: Tropical climate and socio-economic factors make skin, hair and nail infections widespread, mostly managed in primary care.

Challenges

1. Diagnosis: Culture, histopathology and molecular

assays are limited in rural Andhra Pradesh.

2. Treatment: Antifungal resistance and the toxicity of amphotericin B complicate management.

3. Law awareness: Law awareness among providers and patients in Andhra Pradesh leads to late presentation, which raises morbidity and mortality.

4. Resource settings: Drug toxicity and resistance hit hardest in resource-constrained settings, where monitoring of patients on amphotericin B is difficult.

Conclusion

The four fungal groups differ in reproductive structures and morphology, which guides laboratory identification and targeted therapy. Better diagnostics and awareness can cut the burden of mycoses.

Your intro decides your marks. Q. Discuss Ashoka's Dhamma.

YOUR INTRO
~~Ashoka was a great Mauryan king who ruled after the Kalinga war and spread Buddhism everywhere.~~
✕ no date · no source · no argument

→

AFTER GHISSU EVALUATION
Ashoka's Dhamma, set out in his Major Rock Edicts after Kalinga (c. 261 BCE), was not Buddhism by another name but an ethic of tolerance, non-violence and respect, meant to hold a plural empire together.
✓ date · source · argument

ILLUSTRATIVE

Get your intro fixed on your sheet →
ghissu.com
14 days free · no card

SCAN

Write your own answer to 8(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
8(b)

Describe the most common protozoan diseases, causative agents and preventive measures.

10

Introduction

Protozoan diseases remain a significant public health challenge in India, especially in tropical regions where vectors and poor sanitation facilitate spread. Malaria and amoebiasis cause substantial morbidity and mortality. Progress in malaria elimination in states like Andhra Pradesh shows the value of sustained institutional effort and community awareness.

Common Protozoan Diseases and Causative Agents

1. Malaria: Caused by a protozoan parasite whose life cycle involves human and mosquito hosts. It is transmitted by the Anopheles mosquito, which makes control complex.
2. Amoebiasis: Caused by Entamoeba histolytica, it spreads through contaminated water and food, especially in Andhra Pradesh communities where sanitation and hygiene are inadequate.
3. Risk factors: Seasonal monsoons and poverty in regions like Rayalaseema create conditions favourable to vectors and limit access to clean water.

Practice 8(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Preventive Measures

1. Personal hygiene and sanitation: Cleanliness, drinking boiled or treated water and proper disposal of human waste prevent amoebiasis and other protozoan infections, promoted by public health drives in Andhra Pradesh.

2. Vector control and protection: Mosquito nets, repellents, insecticide spraying and elimination of stagnant water are critical against malaria and are encouraged by Andhra Pradesh's state health programmes.

3. Health education: Community programmes on symptoms, transmission and prevention encourage early treatment-seeking.

4. Vaccines: Malaria vaccines RTS,S/AS01 and R21/Matrix-M are WHO-approved, though deployment in India is at an early stage.

Malaria vs Amoebiasis

Malaria	Amoebiasis
Protozoan parasite, two hosts	Entamoeba histolytica
Spread by Anopheles mosquito	Contaminated water and food
Nets, sprays, no stagnant water	Boiled water, sanitation, hygiene

Institutional Framework

1. **Malaria elimination:** The National Strategic Plan for Malaria Elimination (2023–2027) guides vector control, diagnosis and treatment, with Andhra Pradesh running IMEP-3 in endemic districts.
2. **Surveillance and care:** IDSP monitors outbreaks, while primary health centres and community health workers in Andhra Pradesh ensure early diagnosis.

Challenges

1. **Treatment gaps:** Delayed diagnosis and incomplete treatment adherence hinder control.
2. **Behavioural hurdles:** Vector management in rural Andhra Pradesh faces logistical and behavioural problems.

Way Forward

1. **Access and linkage:** Ensure antiprotozoal drugs in rural Andhra Pradesh and link control with sanitation drives like Swachh Andhra.

Conclusion

Malaria and amoebiasis still threaten vulnerable regions. Coordinated institutions, preventive habits and community participation are vital to control them.

Write your own answer to 8(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
9(a)

Describe the methods of gene transfer in developing Genetically Modified crops.

10

Introduction

Genetically modified (GM) crops have become pivotal to modern agricutture, addressing pest resistance, disease tolerance and abiotic stress. Advances in recombinant DNA technology allow precise gene transfer that introduces desirable traits without altering the plant's overall genetic architecture. In Andhra Pradesh, where agricutture is a key livelihood, this promises higher productivity.

Methods of Gene Transfer

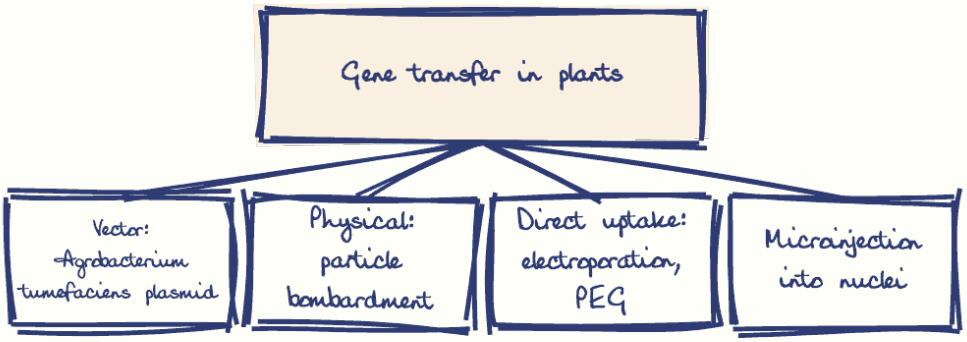
1. Agrobacterium-Mediated Transformation: This widely used plasmid method exploits the natural ability of Agrobacterium tumefaciens to transfer a gene construct into plant cells. It is effective in dicots and some monocots, giving stable integration with minimal disruption.

2. Particle Bombardment (Biolistics): This physical method propels DNA-coated particles into plant tissues, so it works even in species less amenable to Agrobacterium, such as rice and maize.

3. Direct DNA Uptake: Electroporation and polyethylene glycol (PEG)-mediated transformation help protoplasts, cells without walls, take up naked

DNA in vitro, after which whole plants are regenerated.

Methods of gene transfer



Supporting Framework

1. Regulation: The Genetic Engineering Appraisal Committee (GEAC) approves and monitors GM crops, while DBT guidelines cover transgenic development and biosafety.
2. Research Base: Acharya N.G. Ranga Agricuttural University in Andhra Pradesh offers tissue cutture and molecular bidogy facilities for transgenic crops suited to regional agro-climatic conditions.
3. Interdisciplinary Skills: Gene transfer combines plant molecular bidogy, tissue cutture, genetic engineering and biochemistry to build and deliver gene constructs.

Challenges

1. **Tissue Culture:** Each crop and cultivar needs tailored conditions, and elite germplasm often regenerates poorly, a bottleneck in Andhra Pradesh's diverse crop improvement programmes.
2. **Stable Expression:** Gene silencing or inactivation can switch off transgenes across generations, so long-term molecular evaluation is needed.
3. **Biosafety:** Gene flow to wild relatives and superweeds need case-by-case risk assessment.
4. **Farmer Concerns:** Bitech cotton has been adopted, including in regions of Andhra Pradesh, but seed costs, intellectual property rights and equitable access still limit wider acceptance.

Conclusion

Gene transfer methods are the engine of GM crop development. Locally adapted protocols, public-private partnerships between government institutions and biotech firms, and robust science-based monitoring will let gene transfer raise crop productivity and resilience in Andhra Pradesh.

REVIEWER'S NOTE · CRITICALLY EXAMINE

"Critically examine" pays for the counter-view. An answer that stops at the first view has stopped at half the marks.

ghissu

ghissu exam-hall thinking decodes every directive — Discuss, Examine, Elucidate — before you write a line.

Decode your next question →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 9(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
9(b)

What is a Biopesticide ? How are fungal parasites of insects useful in Biological control ?

10

Introduction

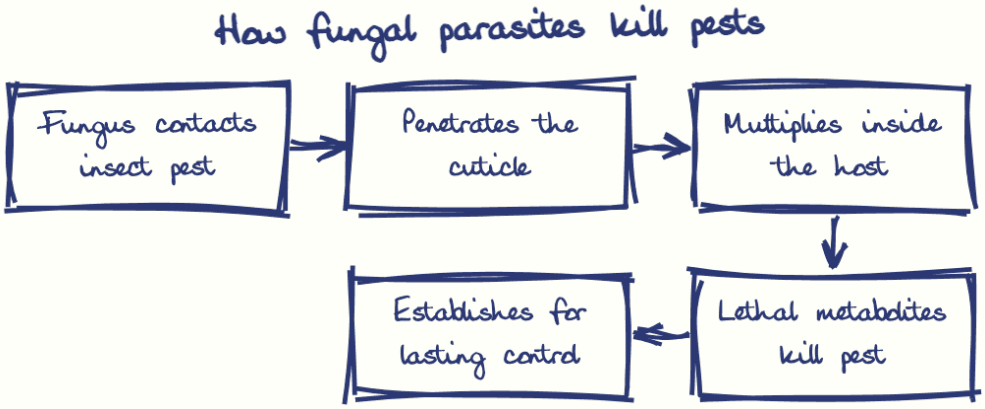
With rising concern over the environmental and health hazards of chemical pesticides, biopesticides have emerged as a sustainable alternative. The emphasis on integrated pest management (IPM) makes biological control, including fungal parasites, crucial for reducing chemical dependency in Andhra Pradesh.

What Is a Biopesticide

1. Concept: Biopesticides are formulations derived from natural organisms such as bacteria, fungi or viruses, or their metabolites, that control pests by targeting specific species. They are biodegradable and less toxic to non-target organisms.
2. Types: Microbial pesticides like Bacillus thuringiensis, biochemical pesticides like neem extracts, and plant-incorporated protectants.
3. Policy Support: The BioE3 Policy and Bio-RIDE scheme promote their development and commercialisation, supporting India's bioeconomy and sustainable agriculture.

Usefulness of Fungal Parasites in Biological Control

1. **Mechanism of Action:** Entomopathogenic fungi infect pests by penetrating the cuticle, multiplying inside the host and producing lethal metabolites. This specific action spares beneficial insects.
2. **Common Species:** Beauveria bassiana and Metarhizium anisopliae are formulated for field use against whiteflies, aphids and stem borers.
3. **Long-Term Suppression:** By establishing in the environment, they give lasting control and reduce pesticide resistance and contamination.
4. **Ecological Balance:** They keep pests below economic thresholds without disturbing predator-prey relationships.
5. **Economic and Health Gains:** Less chemical use lowers costs, cuts health risks and boosts residue-free exports.
6. **Integration:** They complement parasitoids and predators in a multi-pronged approach. In Andhra Pradesh they have been explored against pests of chilli and horticultural crops, reducing chemical load and improving soil health in regions like Rayalaseema.



Limitations

Their effectiveness depends on environmental conditions like humidity and temperature, and they act more slowly than chemical pesticides, so farmers need awareness and proper application techniques.

Conclusion

Fungal parasites turn the natural enemies of pests into a safe tool. Research under Bio-RIDE for local agro-climatic zones, local production units, subsidies and inclusion in state IPM programmes can make fungal biopesticides a pillar of sustainable farming in Andhra Pradesh.

1.5 lakh+

Borrow the approach, not the answer.

Topper answers matched by meaning to your question — with how each one opened, structured and concluded.

See how toppers wrote it → ghissu.com

14 days free · no card

SCAN

P-IV
10(a)

What is Major Histocompatibility Complex (MHC) ? Describe the role of MHC classes.

10

Introduction

The Major Histocompatibility Complex (MHC) is a genetic system that governs immune recognition and compatibility in vertebrates. Its importance has grown with organ transplantation, since matching donor and recipient MHC profiles is essential to prevent graft rejection.

What Is MHC

1. Location and polymorphism: The gene complex lies on the short arm of chromosome (6) and is so polymorphic that no two unrelated people share a profile, except identical twins.
2. HLA system: The human MHC is the HLA system, with loci A, B, C and D carrying many alleles; they encode cell surface markers of self and non-self.
3. Clinical use: The Sri Venkateswara Institute of Medical Sciences in Tirupati uses HLA typing in transplant protocols, improving graft survival through donor-recipient compatibility.

Practice 10(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Candidates must not write on this margin

Candidates must not write on this margin

Role of MHC Classes

MHC Class I vs Class II

Class I	Class II
Almost all nucleated cells	Antigen-presenting cells
Endogenous antigens	Exogenous antigens
To cytotoxic T cells	To helper T cells

1. Class I molecules: Found on almost all nucleated cells, they present endogenous antigens, such as viral proteins made inside infected cells, to cytotoxic T cells, which destroy infected or abnormal cells.
2. Class II molecules: Found mainly on antigen-presenting cells like macrophages and dendritic cells, they present exogenous antigens from engulfed pathogens to helper T cells, which coordinate responses including antibody production.
3. Immune surveillance: High variability lets the immune system recognise many pathogens and mutated cells, defending against infection and neoplasia.
4. Transplant relevance: The same variability

How to use this book with ghissu.com
see page 4

Index · page 6

Practice 10(a) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

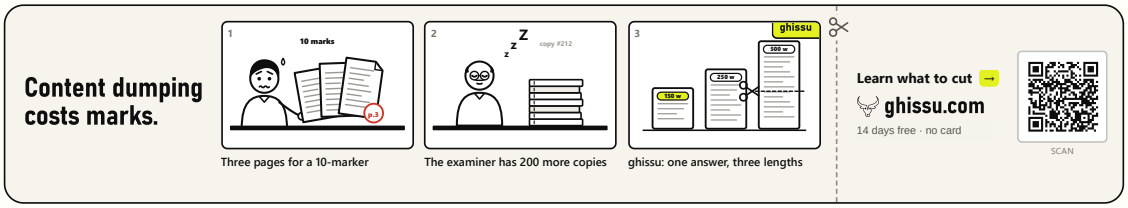
explains why unrelated grafts are rejected, so MHC typing is routine in Visakhapatnam and Vijayawada.

Challenges

1. Matching difficulty: Tribal and non-tribal groups in Andhra Pradesh show varied HLA frequencies.
2. Limited technology: DNA-based HLA typing is accurate but scarce in rural Andhra Pradesh.
3. Rejection risk: Minor mismatches still trigger rejection and need lifelong immunosuppressive therapy, raising costs.
4. Data gaps: Scarce HLA frequency data for many communities in the state hinders donor registries.

Conclusion

MHC is central to immunity and transplantation. More DNA-based HLA labs in Andhra Pradesh's tertiary centres and regional HLA databases will improve transplant outcomes.



P-IV
10(b)

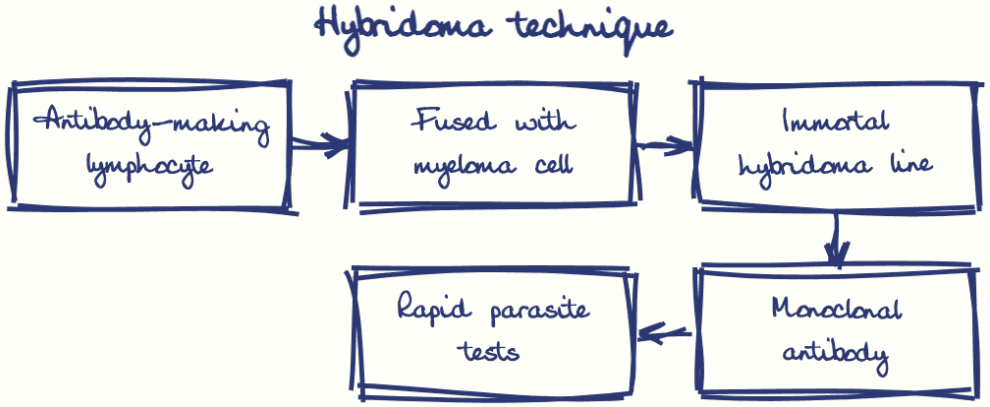
What are monoclonal antibodies ? How are they useful in detecting parasitic diseases ?

10

Introduction

Monoclonal antibodies (mAbs) mark a breakthrough in disease diagnosis and treatment. Their relevance has grown with the need for precise, rapid and cheap diagnostics, especially for parasitic diseases that remain a public health challenge in India and Andhra Pradesh.

What Are Monoclonal Antibodies



1. Meaning: Antibodies that are all identical and specific to a single antigen.
2. Hybridoma technique: Developed by Kohler and Milstein in 1975, it fuses an antibody-producing lymphocyte with a myeloma (cancer) cell, creating an immortal cell line that produces antibody continuously.

3. Indian capacity: ICMR and its collaborating centres in Andhra Pradesh have advanced mAb research and built it into programmes like the Intensified Malaria Elimination Project.

Use in Detecting Parasitic Diseases

1. Specificity: mAbs precisely identify parasites such as Plasmodium (malaria), Leishmania (kala-azar) and Wuchereria (elephantiasis), reducing false positives compared with microscopy or culture.
2. Rapid, cheap testing: They can be built into rapid diagnostic tests that give results in minutes at the point of care, crucial in rural Andhra Pradesh where labs are scarce.
3. Higher sensitivity with PCR: When parasite load is low, combining mAbs with polymerase chain reaction enables early diagnosis.
4. Surveillance: mAb diagnostics support the Test, Treat, Track strategy and India's progress towards malaria elimination.
5. Policy support: MedTech Mitra and the PM-Ayushman Bharat Health Infrastructure Mission aid mAb-based kits.

Challenge

Parasites alter surface antigens to evade detection, so antigenic variation demands continuous updating of m-Ab panels.

Way Forward

1. Indigenous R&D: Fund research in Andhra Pradesh institutions for region-specific kits suited to local parasite strains.
2. Public health integration: Use m-Ab diagnostics more widely in state surveillance and the National Leprosy Eradication Programme.
3. Training: Equip health workers in Andhra Pradesh to use and interpret m-Ab tests in the field.

Conclusion

Monoclonal antibodies offer specificity, speed and adaptability in detecting parasitic diseases, supporting SDG (3) and India's self-reliance in health technology.

Correct, but average? One line from your answer

YOUR LINE
~~The 1857 revolt failed.~~
✕ true · forgettable

→

WITH A VALUE ADDITION
The 1857 revolt failed for want of a centre: no common plan, no modern arms, and the loyalty of the Sikhs, the Gurkhas and most princes to the Company.
✓ cause · evidence · specifics

ILLUSTRATIVE

Add value to your next answer →
ghissu.com
14 days free · no card
SCAN

P-IV
11(a)

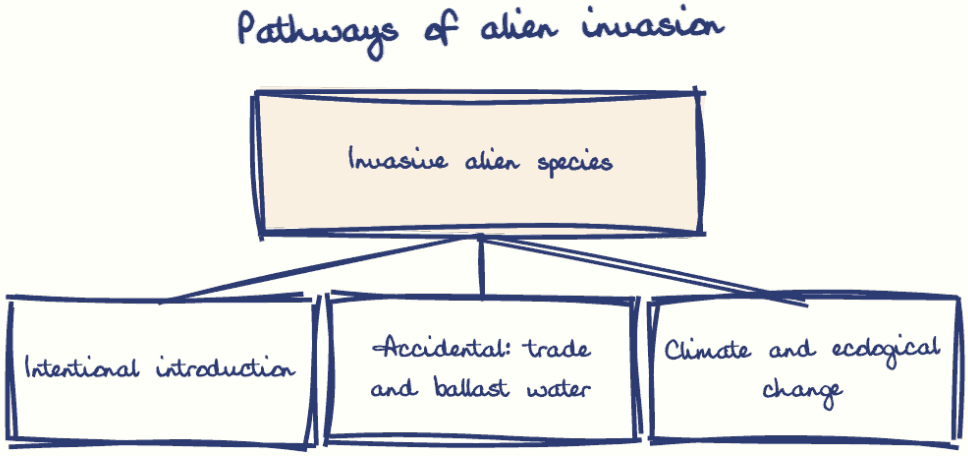
Invasive alien species are the second largest threat to biodiversity after habitat destruction. Discuss the causes with examples.

10

Introduction

Invasive alien species have emerged as the second biggest threat to biodiversity after habitat destruction. Globalisation, climate change and expanding human activity aid their spread and establishment, so understanding their causes is vital for conservation in India and Andhra Pradesh.

Causes of Invasive Alien Species



1. Intentional introduction by humans: Species are deliberately brought in for agriculture, horticulture or biological control without weighing ecological consequences. Example: water hyacinth, introduced for ornamental purposes, spread and choked aquatic

ecosystems.

2. Accidental transport through trade and travel: Ships' ballast water and cargo carry aquatic organisms, while insects and seeds hitchhike on goods across regions.
3. Ecological factors and climate change: Altered rainfall and temperature let invaders colonise habitats where natives are less competitive, helping species like the giant African snail establish themselves.

Examples Illustrating the Causes

1. Leptocybe invasa: This gall-forming wasp, brought in accidentally through trade, has spread in peninsular India, damaging eucalyptus plantations with leaf and stem galls, with economic and ecological costs.
2. Water hyacinth in Andhra Pradesh: It clogs rivers and reservoirs, reduces oxygen levels and threatens native aquatic life.
3. Giant African snail: Spread in Andhra Pradesh, it competes with native snails and damages crops, helped by ecological change and the absence of natural predators.
4. Ballast water invasions: Ships docking at ports

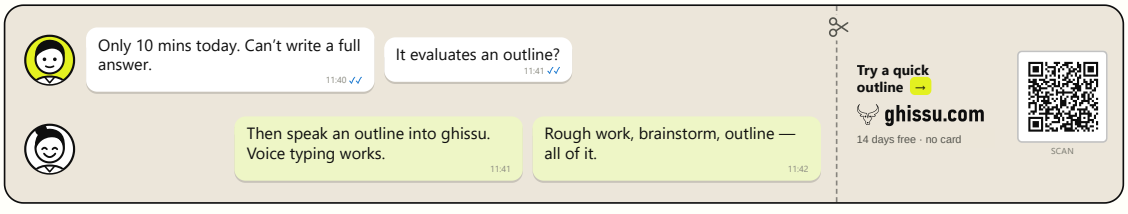
like Visakhapatnam can discharge invasive aquatic species, altering marine ecosystems.

Way Forward

1. Biosecurity: Rigorous inspection and quarantine protocols at ports and borders.
2. Early detection: Surveillance systems in vulnerable ecosystems of Andhra Pradesh.
3. Community awareness: Educate people in Rayalaseema and coastal Andhra to avoid introducing non-native species and to report sightings.
4. Climate adaptation: Build climate resilience into state biodiversity action plans.

Conclusion

Invasive alien species are a complex threat driven by human activity, global trade and ecological change. Biosecurity, monitoring and community engagement, aligned with the National Biodiversity Strategy and Action Plan, can protect native ecosystems and Andhra Pradesh's biological heritage.



P-IV
11(b)

Describe the carbon and nitrogen biogeochemical cycles. Cite examples in the Indian context.

10

Introduction

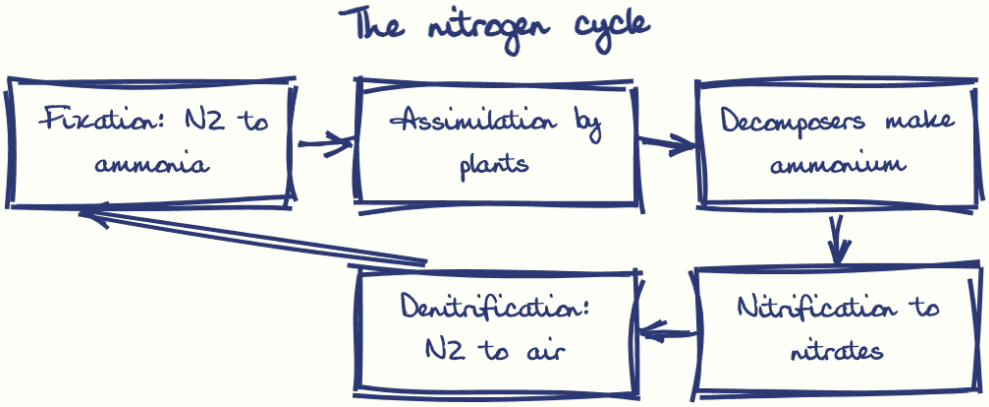
The carbon and nitrogen biogeochemical cycles sustain life by moving essential elements through natural processes. In India, with diverse ecosystems, intensive agriculture and rising pollution, managing them supports the NAPCC and sustainable development.

Carbon Cycle

1. Carbon Exchange: Carbon moves between atmosphere, plants, animals, soil and oceans through photosynthesis, respiration, decomposition and combustion; plants turn CO_2 into organic matter that enters the food chain.
2. Short and Long-Term Cycles: Most carbon cycles quickly, but some is stored in peat, sediments and fossil fuels; burning fuels releases it as CO_2, adding to greenhouse gases.
3. Decomposition and Respiration: Microbial decomposition of dead matter and respiration by plants and animals return CO_2 to the air, balancing the cycle.
4. Indian Context: The Green India Mission

promotes afforestation that helps neutralise about 11% of annual emissions, and AP's coastal mangroves are significant carbon sinks.

Nitrogen Cycle



1. Fixation: Inert atmospheric N_2 is converted into ammonia and nitrates by nitrogen-fixing bacteria in soil and legume root nodules.
2. Assimilation: Plants use nitrates and ammonium to make amino acids and proteins, which pass to animals.
3. Decomposition and Denitrification: Decomposers release ammonium, nitrifying bacteria form nitrates, and denitrifying bacteria return N_2 to the atmosphere.
4. Indian Context: Legumes like black gram and green gram enrich soil naturally, but excessive synthetic fertiliser in states including AP causes

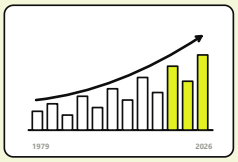
groundwater contamination and nitrous oxide emissions, the Soil Health Card scheme helps optimise use.

Way Forward

1. Agroforestry: Community-led carbon farming in AP raises sequestration and soil health.
2. Optimised Fertiliser: Expand Soil Health Cards and precision farming.
3. Monitoring: Support local universities to track carbon and nitrogen fluxes.
4. Climate Policy: Add nitrogen management to AP's State Action Plan on Climate Change, complementing carbon reduction under the N-APCC.

Conclusion

Balanced carbon and nitrogen cycles sustain ecology and farm productivity. Afforestation, wise fertiliser use and scientific monitoring support rural livelihoods and SDG (13) and SDG (15).



Which sub-topic will they ask next?

Trend analysis

Every syllabus topic: the direction of travel, the most tested sub-topic, the probable next theme.

Prepare the pattern →

ghissu.com

14 days free · no card

SCAN

P-IV
12(a)

What are the advantages and disadvantages of solar power over other renewable energy resources ? Why is the cost of solar energy decreasing ? What is its effect on climate change efforts ?

10

Introduction

Solar power is the defining feature of India's energy transition, driven by the 100 GW target and schemes like PM-KUSUM and PM Surya Ghar Yojana. Its scalability and declining cost make it vital for reducing carbon intensity under the Paris Agreement, especially in sunny Andhra Pradesh, where solar pumps are transforming farm livelihoods.

Advantages over Other Renewables

1. **Abundant and Predictable:** Sunshine for about 300 days a year is more predictable than wind or biomass.
2. **Scalable and Decentralised:** From parks to rooftops, it cuts transmission losses; PM Surya Ghar empowers Andhra Pradesh households.
3. **Low Maintenance:** Few moving parts, unlike wind turbines or hydro plants.
4. **Environment:** No operating emissions and a smaller footprint than large hydro or bioenergy.

Disadvantages

Solar vs other renewables

Solar power	Other renewables
Large land need	Hydro uses less land
Daytime, weather-dependent	Biomass is dispatchable
Needs storage	Hydro, biomass give baseload

1. Land: Solar farms compete with agriculture and habitats.
2. Intermittency: Output needs costly storage, unlike hydro or biomass.
3. Imports: About half of imports come from China.

Why Costs Are Falling

1. Technology and Scale: Better PV efficiency and (91) GW/year domestic capacity made solar competitive with coal since (2016.)
2. Policy: The National Solar Mission, PM-KUSUM and PLI scheme created economies of scale.
3. Global Supply: Subsidised Chinese production lowered prices.

4. Financing: Net metering and PM Surya Ghar cut upfront costs.

Effect on Climate Change Efforts

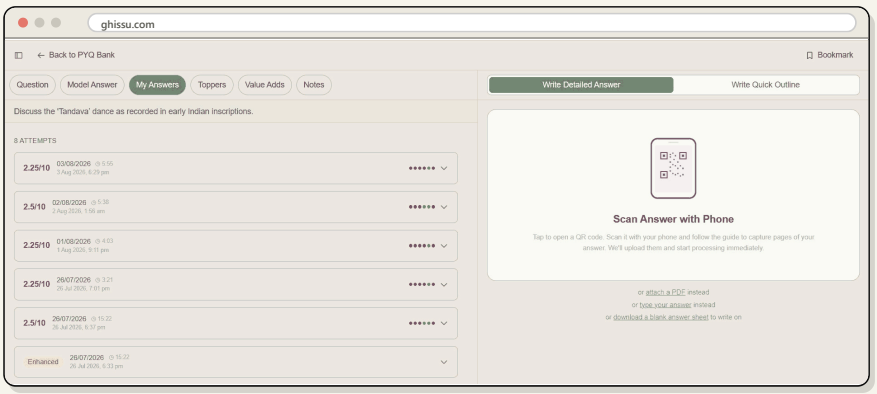
Solar displaces fossil-fuel electricity with zero operating emissions, reducing carbon intensity and helping meet Paris commitments. Falling costs also let India meet rising demand without adding emissions.

Conclusion

Cheaper, scalable solar is India's strongest climate tool if storage and manufacturing gaps close.

THE REAL PRODUCT · NOT A MOCK-UP MY ANSWERS — YOUR PROGRESS

Fifty answers written. Are you any better?



1 Eight attempts at one question, first to latest.

2 Every one scored — the curve is visible.

3 Enhanced versions kept beside your own.

Start your timeline →

ghissu.com

14 days free · no card

SCAN

P-IV
12(b)

"India's traditional water management methods are key in the present context." Examine the statement with special focus on water harvesting and disaster management.

10

Introduction

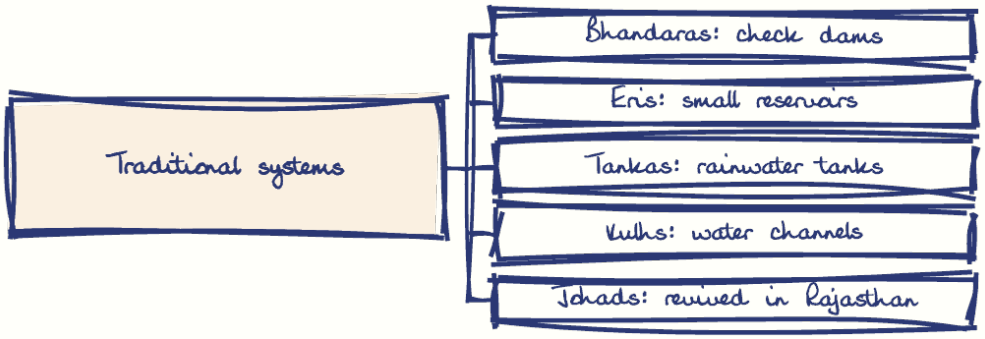
With rising climate variability, India faces more water-related disasters: floods and droughts have risen by over 100% and 29% since 2000. This makes India's traditional water management methods, especially water harvesting rooted in local knowledge and community participation, highly relevant and aligned with the SDGs and IPCC advice.

Traditional Water Harvesting Techniques

1. Diverse Systems: Bhandaras (check dams), eris (small reservoirs), tankas (rainwater tanks) and kulhs (water channels), adapted to local climate and terrain.
2. Community-Centric Management: Small and decentralised, built and maintained by communities, which ensures local ownership.
3. Agricultural Support: They recharge groundwater and give reliable irrigation in drought-prone areas like Rayalaseema.
4. Neglect: Rapid urbanisation and modern infrastructure have led to their neglect, with

traditional knowledge and technical skills being lost.

Traditional water harvesting systems



Contribution to Disaster Management

1. Flood and Drought Mitigation: They regulate flow, reduce runoff and recharge aquifers, easing both floods and droughts.
2. Ecological Buffers: They prevent soil erosion and preserve wetlands that buffer extreme weather.
3. Examples: Restored johads in Rajasthan revived water security, while small reservoirs and farm ponds in Andhra Pradesh helped communities cope with drought.
4. Limitations: Their capacity is insufficient against large-scale disasters, so they need modern forecasting and infrastructure.

Integrating Traditional and Modern Practices

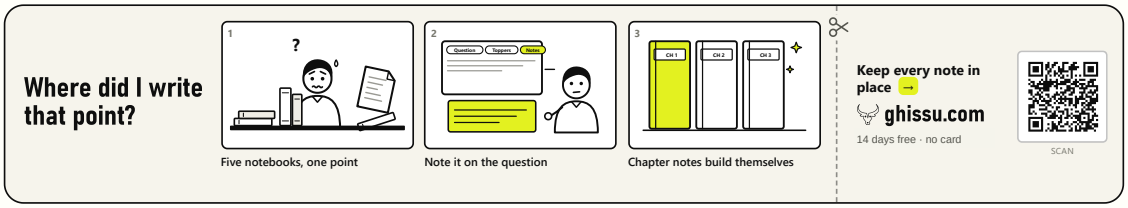
1. Technology: GIS, remote sensing and climate models can optimise design and placement.
2. Policy Support: The Jal Jeevan Mission and Atal Bhujal Yojana promote revival, with Andhra Pradesh taking part.
3. Barriers: Lack of technical knowledge, inadequate funding and institutional fragmentation hinder blending them with large projects like Polavaram.
4. Community Engagement: Capacity building and participatory governance ensure maintenance and equitable access.

Way Forward

Train local stakeholders to revive structures, use GIS to prioritise restoration, and spread traditional knowledge among youth.

Conclusion

The statement holds: traditional methods remain vital, but their full value comes when they are revived through community engagement and blended with modern science for climate resilience.



Write your own answer to 12(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

P-IV
13(a)

What effects does PM2.5 fine particulate matter have on the health of individuals in urban regions of India and across the world ? Discuss.

10

Introduction

Fine particulate matter, PM2.5, has become a critical public health challenge in urban India and worldwide. The Lancet estimated that air pollution caused 1.7 million premature deaths in India in 2019, nearly 18% of all deaths, with PM2.5 a major contributor. Dense industrial, transport and residential emissions make cities the worst affected.

Health Effects on Individuals in Urban Areas

1. **Respiratory and Cardiovascular Disease:** Particles smaller than 2.5 micrometres penetrate deep into lungs and the bloodstream, causing asthma, chronic bronchitis and reduced lung function, as well as hypertension and heart attacks. Delhi's Centre for Chest Surgery reports rising lung cancer among young non-smokers.
2. **Higher Mortality and Morbidity:** Long-term exposure is associated with higher deaths from lung cancer, heart disease and strokes.
3. **Vulnerable Groups:** Children, the elderly and the sick are most at risk; the urban poor live in dirtier areas with less healthcare.

4. Ultra-Fine Particles: PM1 penetrates even deeper, carrying toxic metals that may cause gene damage and cancer, yet standards for it are weak.

India Compared with Global Cities

1. Higher Exposure: Delhi, Mumbai and Visakhapatnam often exceed WHO guidelines, reaching severe AQI above 300, while many global cities are cleaner due to stricter norms.
2. Sources: India's PM2.5 comes from industry, residential combustion, transport, construction dust and seasonal stubble burning, abroad, mainly traffic and industry.
3. Policy Gaps: Weak enforcement and poor waste management, as in Andhra Pradesh's urban local bodies, contrast with robust systems abroad.
4. Health Systems: Developed countries manage pollution-related diseases better, while Andhra Pradesh struggles with vacancies in medical posts and too few community health centres.

Way Forward

1. Monitoring and Standards: Track PM2.5 and ultra-fine particles in all cities and enforce

PM2.5 exposure: India vs world

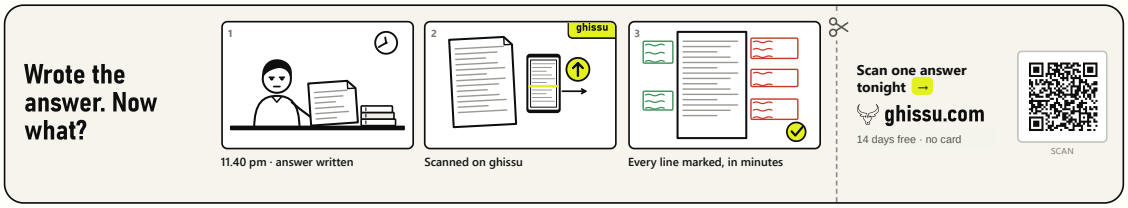
Indian cities	Many global cities
Often above WHO guidelines	Cleaner, stricter norms
Industry, stubble, dust, combustion	Mainly traffic and industry
Weak enforcement, care gaps	Robust systems, better care

stricter norms for industry and vehicles.

2. Waste and Construction: Better waste segregation and processing, acting on the 2022 CAG audit in Andhra Pradesh, to cut emissions from waste burning and dust.
3. Healthcare Access: Fill medical vacancies so pollution-related illness is treated in time.

Conclusion

PM2.5 harms urban people everywhere, but Indians face higher exposure and weaker care. Cleaner cities and stronger urban health infrastructure can reduce this burden.



P-IV
13(b)

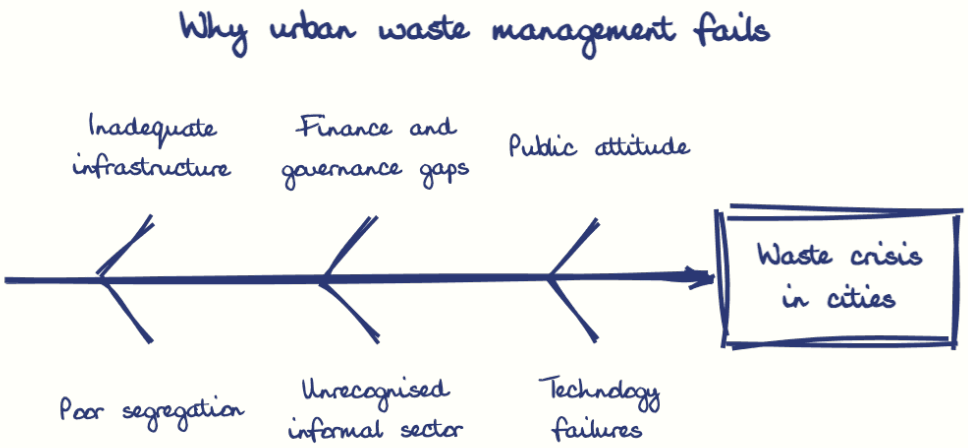
Critically discuss the various challenges and issues arising out of waste management in Indian towns and cities.

10

Introduction

Rapid urbanisation and rising consumption have sharply increased solid waste in India. Towns and cities generate about 62 million tons a year, projected to nearly triple by 2030. Despite the SWM Rules, 2016 and Swachh Bharat Mission (Urban), management remains inefficient and fragmented, hurting health and the environment, especially where ULBs face acute constraints, as in Andhra Pradesh.

Challenges in Urban Waste Management



1. **Inadequate infrastructure:** Most ULBs lack facilities for segregation, collection and processing, so cities rely on open dumping; only 4 of 123

Andhra Pradesh ULBs had sanitary landfills in March 2022.

2. Poor segregation: Rules mandate segregation into biodegradable, non-biodegradable and hazardous waste, but low awareness and weak enforcement keep compliance low.
3. Finance and governance: Fragile municipalities cannot invest in technology or land, and many uncoordinated agencies slow delivery.
4. Informal sector and public attitude: Waste pickers lack recognition and safety, and a callous public attitude leads to littering.

Issues Arising

1. Environmental degradation: Leachate contaminates soil and water, and unengineered dumps release methane, adding to climate change.
2. Public health risks: Unscientific handling exposes ragpickers and slum dwellers to disease and hazards; COVID-19 exposed gaps in safety protocols for waste workers.
3. Urban sustainability: Legacy waste of 85.89 lakh metric tons in Andhra Pradesh's dump yards locks up land and strains ecosystems.
4. Technology failures: Waste-to-Energy plants

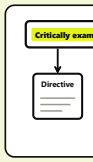
underperform on unsegregated, seasonally variable waste.

Way Forward

1. Infrastructure: Prioritise engineered sanitary landfills and upgraded wet waste processing plants, supported by targeted funding and technical assistance.
2. Segregation: IEC campaigns, schod education and strict enforcement of the rules.
3. Inclusion and governance: Formalise waste pickers with safety gear and social security, and use bankable recycling units and clear coordination.

Conclusion

Urban waste challenges are infrastructural, social and financial at once, so only a combined response can make Indian cities clean and move them towards a circular economy.



Stuck at the directive?

Exam-hall thinking

The directive, the demand, the structure — before your first line. Full for beginners, Crux for veterans.

Walk in with a method →

ghissu.com

14 days free · no card

SCAN

P-IV
14(a)

Recent observations have shown that the ozone layer over Antarctic shows signs of healing. Critically examine.

10

Introduction

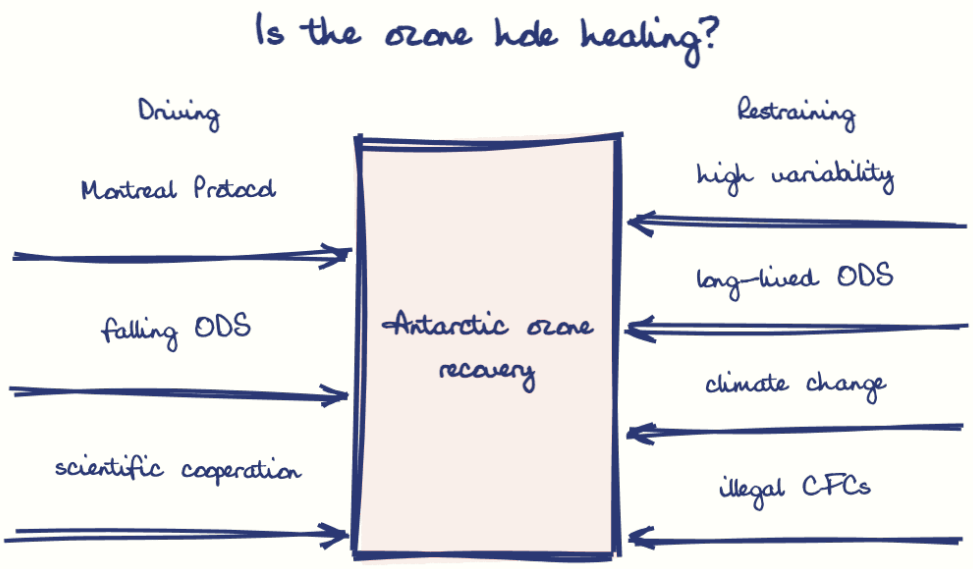
The Antarctic ozone layer, which shields Earth from harmful ultraviolet radiation, has been severely depleted since the late 20th century by human-made chemicals. Recent observations suggest signs of healing. Assessing how far this is real matters for global environmental governance and for UV-sensitive regions like coastal Andhra Pradesh.

Evidence of Healing

1. Recovery trends: Satellite and ground measurements since the early 2000s show a gradual rise in spring ozone and a shrinking ozone hole, with India's Antarctic stations Maitri and Bharati contributing data.
2. Declining ODS: CFC concentrations are falling after the Montreal Protocol and its amendments, so fewer halogen radicals destroy ozone.
3. Natural dynamics: The polar vortex stabilises the atmosphere, allowing gradual regeneration in spring as sunlight returns and radicals diminish.

Practice 14(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice



4. Policy and science: India ratified the Protocol in 1992, and its role in Antarctic research, including hosting the 46th Antarctic Treaty Consultative Meeting in Kochi in 2024, shows the value of cooperation in monitoring.

Critical Limitations

1. High variability: Volcanic eruptions and stratospheric temperature swings have produced larger holes than expected in some years.
2. Long-lived ODS: Banned substances persist for decades, so return to 1980 levels is projected only by mid-century.
3. Climate change: Greenhouse gases can cool the stratosphere, promoting polar stratospheric clouds and ozone loss.

4. Illegal production: Reports of unauthorised CFC production threaten recovery.

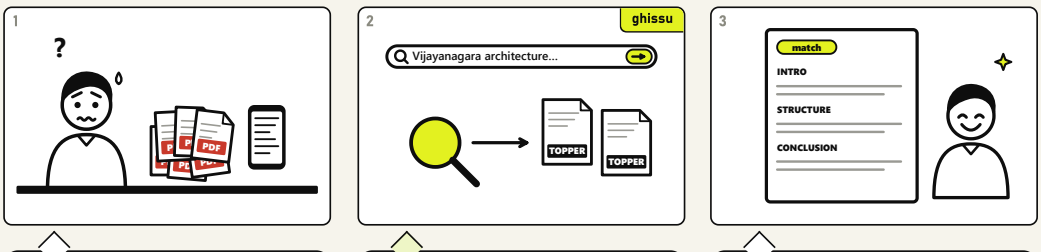
5. Regional risk: Coastal Andhra Pradesh faces UV risks to farming and public health from ozone fluctuations, yet lacks local UV monitoring and adaptation capacity.

Conclusion

The Antarctic ozone layer is healing, a success of global policy, but slowly and unevenly. Stronger enforcement against illegal ODS and integrated climate–ozone action are needed to secure it.

A STORY YOU MAY KNOW TOPPERS' ANSWERS, PER QUESTION

The topper copy hunt



Telegram, YouTube, forty PDFs... nothing on this question.

Type the question into ghissu.

Five topper answers to this exact one!

— Borrow the topper's approach, not the topper's sentences.

Try it on your question →

ghissu.com

14 days free · no card

SCAN

P-IV
14(b)

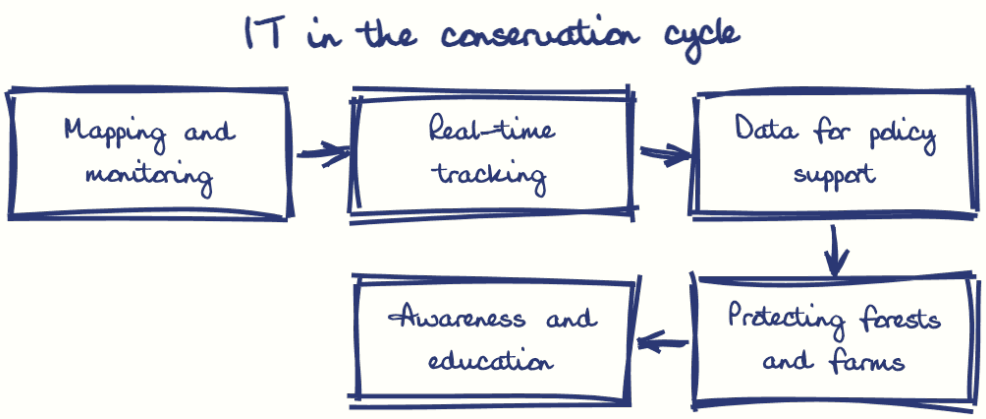
Describe the role of Information Technology in environmental conservation.

10

Introduction

The accelerating pace of environmental degradation and climate change demands innovative solutions. Information Technology (IT) has become a critical enabler, offering tools to monitor, manage and mitigate environmental problems. In Andhra Pradesh, with its rich biodiversity, long coastline and agrarian economy, IT is vital for sustainable resource management and disaster resilience.

Role of IT in Environmental Conservation



1. **Mapping and monitoring:** MoEFCC and the Andhra Pradesh Forest Department use GIS and remote sensing for biodiversity mapping and forest monitoring, improving decisions and enforcement.

2. **Real-time tracking:** Remote sensing enables near

Practice 14(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V
1 / 3
Scan to practice

real-time tracking of deforestation, land-use change and disaster impacts, improving preparedness in flood-prone coastal districts.

3. Awareness and education: The Centre for Environment Education uses digital platforms and social media to teach rural communities in Andhra Pradesh sustainable practices.

4. Sustainable agricutture: Digital Green uses mobile video to spread soil health and water conservation practices among farmers in Rayalaseema and Coastal Andhra.

5. Protecting forests: A CSR-funded IoT-based forest monitoring pilot by Hitachi India and the Institute of Wood Science and Technology protects valuable forest species in regions including Andhra Pradesh.

6. Policy support: The National Action Plan on Climate Change and digital monitoring of water and soil aid integrated resource management.

Challenges

1. Digital divide: Limited internet connectivity and digital literacy in many rural areas of Andhra Pradesh restrict the reach of IT-based initiatives.

2. Resources and coordination: Limited financial and human resources, and coordination gaps between state agencies, NGOs and local communities, slow the scaling up of programmes.

Way Forward

1. Infrastructure and skills: Expand broadband and power in rural and forest areas, and train forest officials and farmers in GIS and mobile tools.
2. Integrated data: Combine satellite data, ground surveys and community inputs on interoperable platforms for real-time decisions.
3. Partnerships: More CSR-driven innovation and academic collaboration can deploy IoT sensors and AI analytics for wildlife protection.

Conclusion

IT has transformed conservation by enabling precise monitoring, informed policy and community participation. Harnessed well, it supports sustainable resource management and disaster resilience in line with the Sendai Framework.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths → ghissu.com

14 days free · no card

SCAN

Write your own answer to 14(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2017 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
15(a)

Briefly describe the Sendai Framework for Disaster Risk Reduction and its adoption in the disaster management policy of India.

10

Introduction

Frequent cyclones on the eastern coast and floods in river basins demand a comprehensive risk reduction strategy. The Sendai Framework (2015–2030) is a global blueprint to reduce disaster losses and build resilience. India has adopted it as it shifts from relief to proactive prevention.

Key Features of the Sendai Framework

1. Four Priority Areas: Understanding disaster risk, strengthening risk governance, investing in risk reduction for resilience, and enhancing preparedness to build back better.
2. Focus on Risk Reduction: Unlike earlier frameworks, it targets underlying risks through resilient infrastructure and community awareness, not only post-disaster response.
3. Inclusive Approach: It involves local communities, women and vulnerable groups in planning and implementation.
4. Link with SDGs: Reducing disaster losses advances goals like poverty alleviation and climate action.

Adoption in India's Disaster Management Policy

Sendai priorities in Indian policy

Sendai Framework	India's adoption
Four priority areas	NDMP 2016 alignment
Focus on risk reduction	Legal, institutional base
Inclusive approach	Regional implementation
Link with SDGs	Capacity and early warning

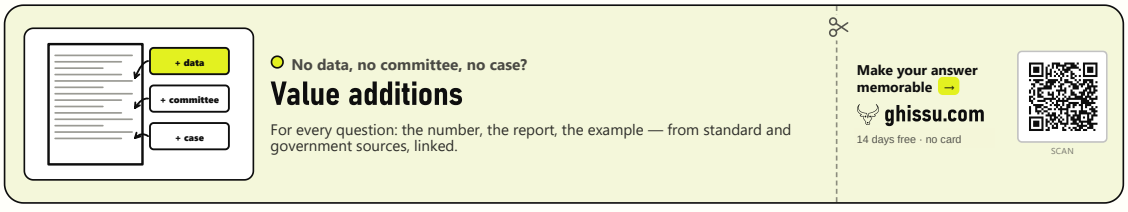
1. National Plan Alignment: India's first National Disaster Management Plan (2016) is explicitly based on Sendai's four priorities.
2. Legal and Institutional Base: The Disaster Management Act, 2005 created NDMA and SDMAs, including Andhra Pradesh's, which coordinate Sendai-aligned action.
3. Regional Implementation: Strategies suit local hazards, landslide-prone Eastern Ghats in AP use zonation mapping, reflecting context-specific risk understanding.
4. Capacity and Early Warning: Seismic observatories rose from (80) in (2014) to (168) by (2025), and the NDRF grew to (16) battalions.

Way Forward

1. Community Participation: Involve women and vulnerable groups in risk governance through capacity-building programmes and inclusive policy frameworks.
2. Resilient Infrastructure: Prioritise it in coastal and drought-prone areas like Visakhapatnam and Rayalaseema.
3. Technology for Early Warning: Use advanced sensing and data analytics for timely warnings at the grassroots level.
4. Ecosystem-Based Measures: Restore wetlands and forests in AP's river basins to reduce flood and landslide risk.

Conclusion

Sendai has driven a paradigm shift in India from reactive relief to risk reduction. Embedding its priorities in law, institutions and regional plans, including AP's tailored strategies, makes India better equipped to reduce disaster losses and protect vulnerable communities and supports India's broader sustainable development goals.



P-IV
15(b)

Examine the role of the National Green Tribunal in providing speedy justice in environmental matters by quoting its recent decisions.

10

Introduction

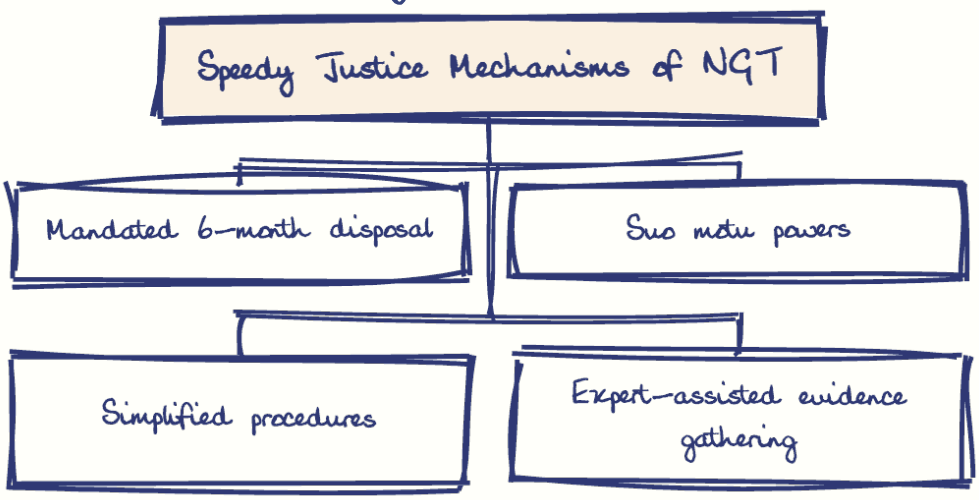
Pollution, deforestation and industrial hazards demand swift and specialised judicial intervention. The National Green Tribunal, established in 2010, was designed to provide expeditious justice in environmental matters and reduce the burden on traditional courts, which matters in fast-developing states like Andhra Pradesh with a long coastline and industrial corridors.

Role of the NGT

1. Specialised Adjudication: A quasi-judicial body under the NGT Act, 2010, it handles environment, forest and natural resource cases, guided by natural justice rather than the Code of Civil Procedure.
2. Expert Composition: Judicial members and expert members from scientific fields let it decide technical issues soundly.
3. Rights Protection: It enforces the right to a clean environment under Article 21 and gives marginalised communities a forum.

4. Compensation Powers: It can impose fines and order restoration, applying the polluter pays principle.

Mechanisms for Speedy Justice



1. Timely Disposal: It must dispose of cases within six months of filing.
2. Suo Motu Cognizance: It acts on media reports or letters without formal complaints.
3. Simplified Procedure: Freedom from strict evidentiary rules allows faster hearings.

Recent Decisions

1. Ludhiana Gas Leak: Its suo motu intervention showed rapid response to a deadly hazard.

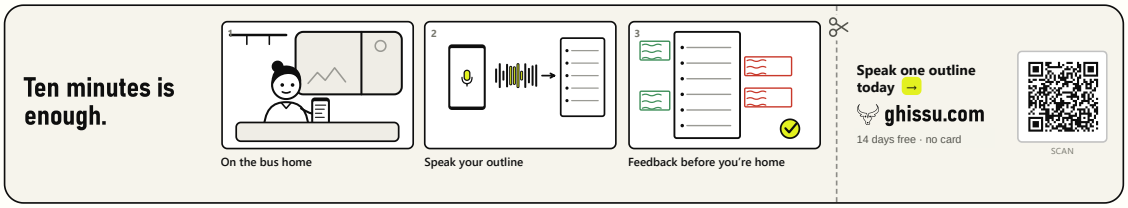
2. Vedanta Penalty: A 25 crore penalty on Vedanta's alumina refinery set a precedent for corporate accountability.
3. Afforestation Order: It directed the Border Roads Organisation to plant 10,000 trees for trees cut during road construction in Uttarakhand.
4. Coastal Pollution: It acted against illegal sand mining and industrial pollution, including on Visakhapatnam's coast.

Limitations

As of 2020, it had fewer than the mandated 10 judicial and 10 expert members, causing pendency and missed six-month targets.

Conclusion

The NGT has become a fast, expert environmental guardian. Filling vacancies will help it meet its mandate fully.



APPSC GROUP-I MAINS · PAPER V

2016

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

P-IV
1(d)

Write a short note on the following: (i) Post Harvest Technology (PHT) (ii) Oilseeds Production Programme (OPP)

10

Introduction

Post Harvest Technology (PHT) and the Oilseeds Production Programme (OPP) are critical to India's strategy for food security, farmer incomes and lower import dependence. India imports over half its edible oil and faces significant post-harvest losses, so these efforts matter under schemes like NMEO-Oilseeds and the Agriculture Infrastructure Fund, especially in Andhra Pradesh.

(i) Post Harvest Technology

1. Need: Perishables like fruits and vegetables lose 20-40% due to weak cold chains and processing, acute in horticulture-rich Andhra Pradesh.
2. Infrastructure support: PM Kisan SAMPADA Yojana and the Agriculture Infrastructure Fund finance cold storage, processing units and integrated post-harvest facilities.
3. Research and extension: CFTRI and Krishi Vigyan Kendras develop and disseminate post-harvest technologies, seed varieties and best practices to farmers.

Practice 1(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

4. Adoption barriers: Small farmers lack access and credit, and fragmented holdings hinder mechanisation and value addition.

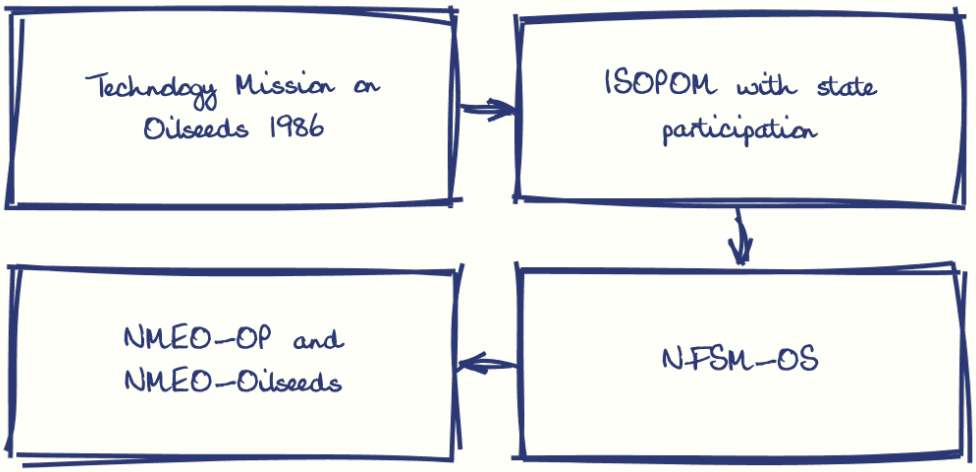
(ii) Oilseeds Production Programme

1. Origins: The Technology Mission on Oilseeds (1986) was a pioneering effort integrating research, input support and price stabilisation. It later evolved into ISOPOM with active participation of states including Andhra Pradesh.
2. Price support: The Integrated Policy on Oilseeds and Market Intervention Operations ensure minimum support prices and protect farmers from price volatility.
3. Outcomes: High-yielding, pest-resistant varieties raised production, supported by NFSM-OS and NMEO-OP.
4. Challenge: Acreage declined as rice and wheat gave better returns and cheaper imports depressed domestic prices.

Way Forward

1. Infrastructure: Build cold storage, pack houses and processing units in horticulture clusters using the Agriculture Infrastructure Fund and PMKSY.
2. Access: Use SATHI for seed quality and supply

Evolution of oilseeds programmes



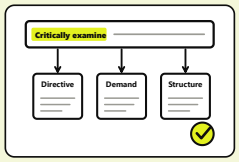
chain transparency, and extend Kisan Credit Card coverage to smallholders.

3. Procurement: Assure oilseed procurement and price support under PM-AASHA to raise acreage.

4. Partnerships: Link government, research institutions and private firms to scale value addition.

Conclusion

PHT and OPP are pivotal to transforming Indian agriculture. Together they can cut losses, raise incomes and build self-reliance in edible oils, supporting SDG (2.)



Stuck at the directive?

Exam-hall thinking

The directive, the demand, the structure — before your first line. Full for beginners, Crux for veterans.

Walk in with a method →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 1(d) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
1(a)

Briefly state the salient features of the National Data Sharing and Accessibility Policy, 2012 and its benefits?

10

Introduction

In the digital age, data is a critical asset for governance, innovation and development. The National Data Sharing and Accessibility Policy (NDSAP), notified in 2012, aims to unlock government-owned data by making it accessible proactively and in machine-readable form. It fosters transparency, accountability and public participation, relevant also to Andhra Pradesh's growing digital governance.

Salient Features

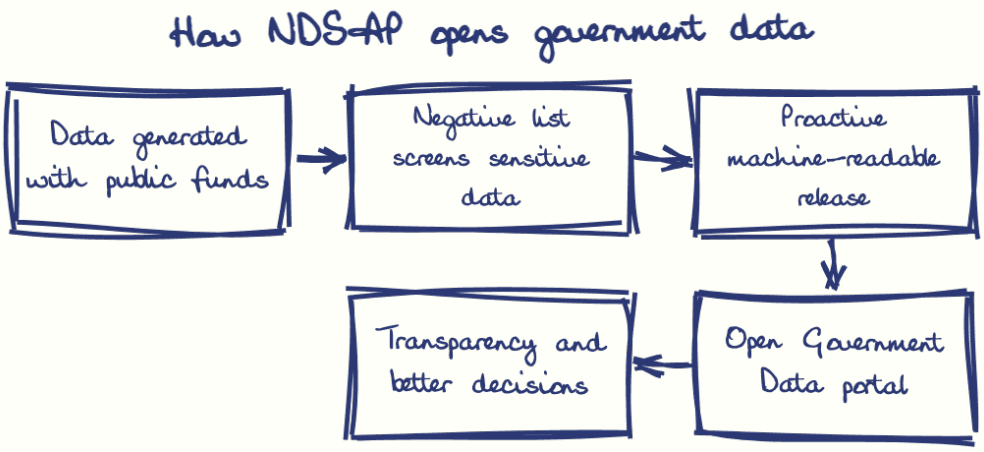
1. Sharing mandate: All non-sensitive data generated using public funds must be made accessible in both human-readable and machine-readable formats.
2. Proactive disclosure: Data is to be released proactively, not only on request, widening its use.
3. Negative list: Restricted datasets are placed on a negative list, balancing openness with privacy and security.
4. Implementation bodies: State Technical and Implementation Committees oversee data sharing at

Practice 1(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

the state level.

5. Open data platform: The policy led to the Open Government Data portal.



Benefits

1. Transparency and accountability: Citizens can see and scrutinise the data government collects with public money.
2. Public participation: Open data lets citizens, researchers and civil society engage with policy.
3. Better decisions: It enables data-driven decision-making in planning and service delivery.
4. Innovation: Open data has supported education and e-governance innovations like SWAYAM and SAMARTH.

Challenges

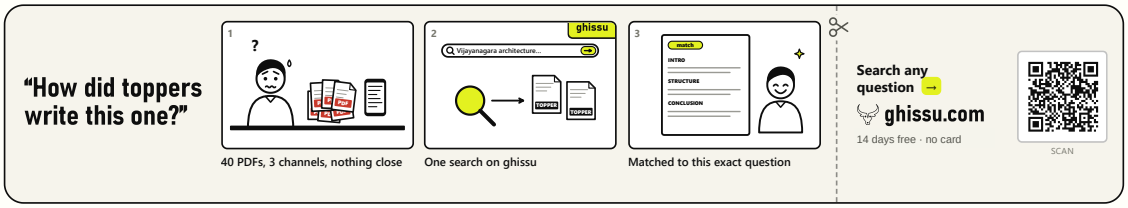
1. Data silos: Many agencies, including in Andhra Pradesh, struggle to share data internally.
2. Time lag: Delays between collection and release reduce real-time usefulness.
3. Privacy risks: The Cambridge Analytica scandal shows the need for safeguards.

Way Forward

1. Privacy law: Integrate data sharing with the Digital Personal Data Protection Act and DPDP Rules.
2. Real-time sharing: Invest in technology to cut release delays.
3. Partnerships: Use government data with academia, industry and civil society.

Conclusion

NDS-AP laid the foundation for transparent, data-driven governance. With privacy safeguards and institutional support, it can empower citizens in line with Digital India.



P-IV
1(b)

Write a brief note on the following: (i) Indian Science Congress Association (ISCA) (ii) Council of Scientific and Industrial Research (CSIR)

10

Introduction

Science and technology are pivotal to India's pursuit of strategic autonomy and innovation-led growth. The Indian Science Congress Association (ISCA) and the Council of Scientific and Industrial Research (CSIR) have historically shaped India's scientific landscape, fostering research, collaboration and industrial application, and remain crucial for national and regional challenges, including in Andhra Pradesh.

(i) Indian Science Congress Association

1. **Origin and mission:** Founded in 1914, ISCA is one of India's oldest scientific bodies, promoting and popularising science through annual congresses that bring together scientists, researchers and policymakers.
2. **Community building:** It fosters collaboration across disciplines and encourages regional universities and research centres, including from Andhra Pradesh, bridging central and state efforts.
3. **Science popularisation:** Public lectures, awards and outreach nurture scientific temper in society.
4. **Challenges:** Limited engagement with cutting-edge

research and young researchers, and a weak role in policy advocacy and technology transfer.

(ii) Council of Scientific and Industrial Research

1. Framework: Established in 1942, CSIR is a premier autonomous R&D organisation under the Ministry of Science and Technology, with a network of 37 national laboratories.

2. Contributions: It has developed indigenous crop varieties, pharmaceuticals and advanced materials, and recently battery recycling and green hydrogen catalysts.

3. Governance and funding: As a registered society with major government funding plus industry revenue, it is stable but needs reforms for agility.

4. Challenges and strategic role: It must modernise infrastructure, retain talent and draw private partners, while serving national security through defence collaborations.

Way Forward

1. Regional linkages: Connect ISCA and CSIR labs with Andhra Pradesh universities and industries.

2. Youth and inclusion: Expand CSIR's YUVA portal and ISCA outreach to women and young scientists.

ISCA and CSIR at a glance

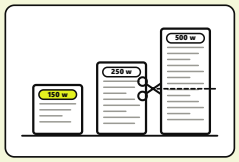
ISCA	CSIR
Founded 1914	Established 1942
Annual science congresses	37 national laboratories
Popularises science	Applied industrial R&D
Weak policy role	Needs reforms for agility

3. Infrastructure and funding: Upgrade facilities and adopt flexible funding to encourage PPPs and startup incubation.

4. Sustainable development: Align research with aquaculture, horticulture and environmental needs, supporting SDG (9).

Conclusion

ISCA and CSIR remain foundational pillars of Indian science, one fostering knowledge exchange and the other applied innovation. Regional integration and modernisation can make them drivers of a self-reliant India.



Writing 400 words for a 150-word question?

Model answers in three lengths

150, 250 and 500 words, in English and हिंदी. Learn what to cut before the examiner does.

Read one question three ways →

ghissu.com

14 days free · no card

SCAN

P-IV
1(c)

Mention the main objectives of Science, Technology and Innovation (STI) Policy, 2013 of India.

10

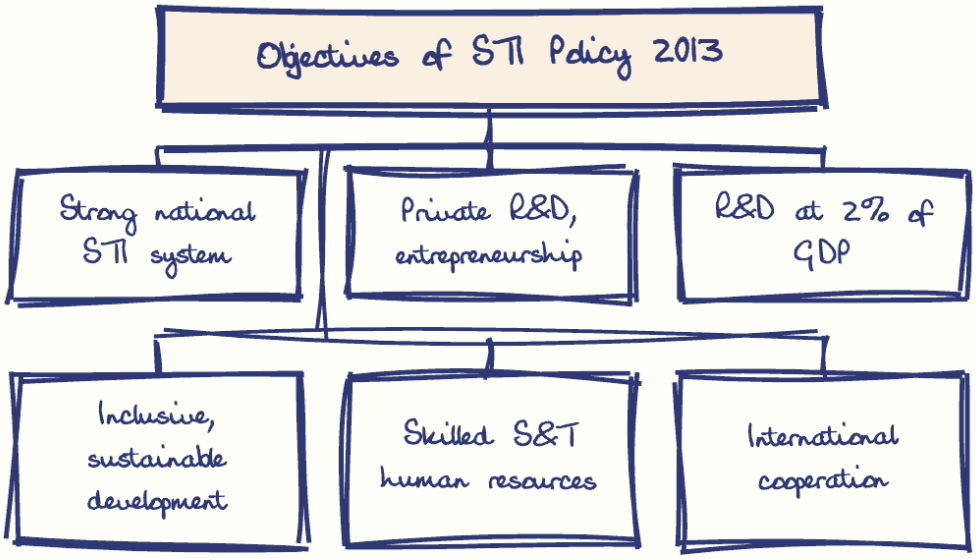
Introduction

The STI Policy, 2013 was formulated to harness science and technology for India's inclusive and sustainable development. With technology driving growth and competitiveness, it aims to strengthen the research ecosystem and innovation capacity, relevant to states like Andhra Pradesh that host institutions such as ISRO's Sriharikota centre.

Primary Objectives

1. National S&T capacity: Build a strong and viable science, technology and innovation system supporting high-technology-led growth.
2. Innovation and entrepreneurship: Encourage private sector R&D and youth entrepreneurship through DST skill programmes.
3. Research and development: Raise gross expenditure on R&D to at least 2% of GDP, with balanced public and private contributions.
4. Inclusive and sustainable development: Apply S&T for social inclusion and environmental sustainability, in line with initiatives like Digital India that widen equitable access to technology.

5. Human resources: Enhance S&T skills and nurture young researchers and women scientists.
6. International cooperation: Collaborate through the India–Japan Science Council and India–Russia Working Group on S&T.



Focus Areas and Goals

1. Knowledge economy: Integrate STI into economic development, as in start-up promotion under Start-up India.
2. Grassroots innovation: Support indigenous technologies and traditional knowledge for local needs.
3. Global competitiveness: Place five Indian research institutions among the world's top 100.
4. Digital platforms: Converge services through Common

Service Centres, aiding agricutture and fisheries.

5. Sustainable growth: Promote innovation for environmental sustainability and social mobility, aligned with the SDGs.
6. Public-private partnerships: Link government, academia and industry for faster commercialisation.

Way Forward

1. R&D investment: Raise funding through PPPs so states like Andhra Pradesh share in innovation budgets.
2. Skill development: Strengthen S&T education and entrepreneurship training at the grassroots.
3. Regional hubs: Build innovation clusters in Visakhapatnam and Sriharikota to spur local growth.
4. International collaboration: Deepen STI partnerships with Japan and Russia for technology transfer.

Conclusion

The STI Policy, 2013 lays the foundation for an innovation-driven knowledge-based economy that empowers all sections of society, in line with Atmanirbhar Bharat.

Write your own answer to 1(c) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
2(d)

Write a short note on: (i) Mobile Satellite Services (MSS) (ii) Environmental monitoring

10

Introduction

Mobile Satellite Services (MSS) and environmental monitoring have become critical to India's development amid rapid digital expansion and climate challenges. They support rural connectivity, disaster management and sustainable resource use, in line with inclusive growth and environmental resilience.

(i) Mobile Satellite Services

1. **Meaning and Role:** MSS enables real-time communication and data transmission via satellite, supporting timely decisions in defence, agriculture and disaster response, including coastal security.
2. **Institutional Capacity:** ISRO deploys MSS satellites such as CMS-03 for strategic communication in the Indian Ocean.
3. **Bridging the Divide:** With rural internet tele-density at 46.73%, satellites like GSAT-11 aim to give rural India high-bandwidth connectivity.
4. **Telemedicine:** With bodies like the National Health Resource Repository, it connects remote hospitals with urban super-speciality centres.

5. Constraints: High deployment cost, ground infrastructure and skilled staff limit scaling.

(ii) Environmental Monitoring

1. Disaster Management: Satellite flood early warning and high-resolution Bhuvan portal imagery improve preparedness and post-disaster assessment.
2. Agriculture: Weather-based crop advisories and soil health monitoring raise productivity in horticulture and aquaculture.
3. Conservation: Remote sensing tracks forest cover, water and biodiversity in sensitive zones like the Eastern Ghats.
4. Data Governance: NRSC regulates remote sensing data and licenses private operators, balancing security and commercial use.
5. Rural Planning: Andhra Pradesh uses GIS and satellite data in MGNREGA and PMGSY.

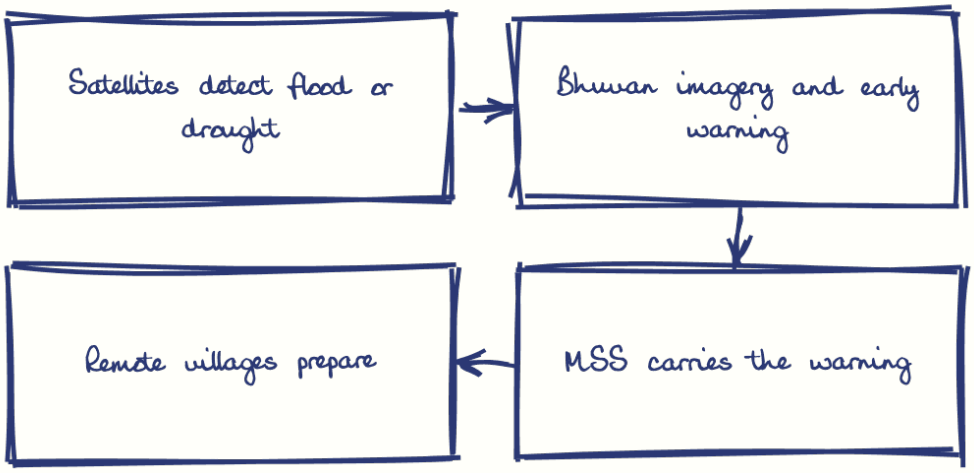
Way Forward

1. Indigenous Satellites: Cost-effective MSS satellites for rural and coastal needs.
2. Connectivity: Speed up BharatNet and GSAT-11 projects.
3. State Planning: Use satellite environmental data in

water management and farm schemes like APBP.

4. Skilled Workforce: Train staff to run ground infrastructure and interpret data.

Observation and MSS together



Why They Matter Together

Communication and observation reinforce each other: monitoring satellites detect a flood or drought, and MSS carries the warning to remote villages that lack terrestrial networks.

Conclusion

MSS and environmental monitoring together link remote citizens and protect resources, creating jobs in hubs like Visakhapatnam.

Correct, but average?
A number, a committee, a case — for every question.
ghissu.com →

P-IV
2(a)

Describe the MOM (Mars Orbiter Mission – Mangalyaan) programme of India? What are its distinguishing features?

10

Introduction

India's Mars Orbiter Mission (MOM), or Mangalyaan, was the country's first interplanetary mission, launched on 5 November 2013 from Sriharikota in Andhra Pradesh. It made India the first Asian nation to reach Mars orbit on its maiden attempt, showing growing technological prowess and strategic autonomy.

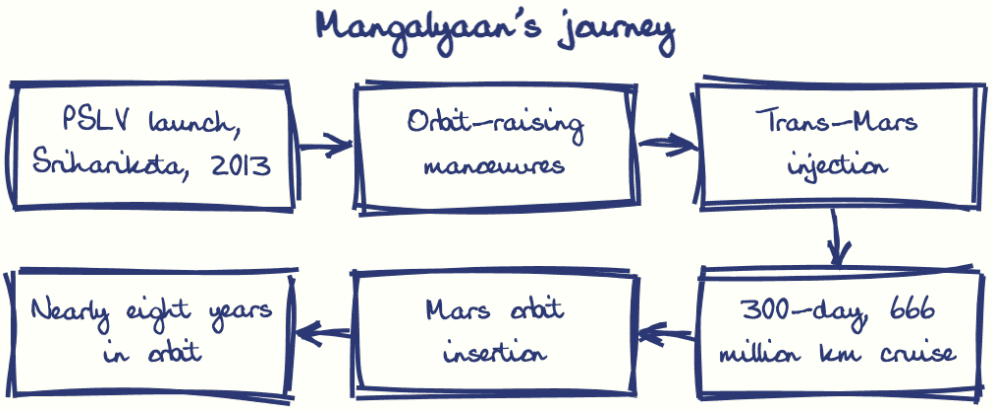
The MOM Programme

1. **Launch and Tracking:** ISRO launched MOM on the PSLV from the Satish Dhawan Space Centre. It was monitored from ISTRAC in Bengaluru with the Indian Deep Space Network antennae in Karnataka.
2. **Mission Design:** Complex phases included orbit-raising manoeuvres, trans-Mars injection, trajectory corrections and Mars orbit insertion, using ISRO's own force models and navigation algorithms.
3. **Journey:** The craft stayed healthy over a 300-day journey covering 666 million km.
4. **Scientific Payload:** Five instruments studied

Candidates must not write on this margin

Candidates must not write on this margin

Mars' surface features, morphology, mineralogy and atmosphere.



Distinguishing Features

1. Cost-Effectiveness: At about $73 million, it is one of the most cost-effective Mars missions globally.
2. Maiden-Attempt Success: India reached Mars orbit at the first try, a first for Asia.
3. Longevity: Planned for a short duration, it operated for nearly eight years until contact was lost in April 2022 during a long Martian eclipse.
4. Global Data Sharing: Over 7,200 registered users worldwide accessed its data through the ISRO Science Data Centre.
5. Scientific Value: Its data improved understanding of Mars' atmosphere and exosphere.

How to use this book with ghissu.com
see page 4

Index · page 6

Practice 2(a) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

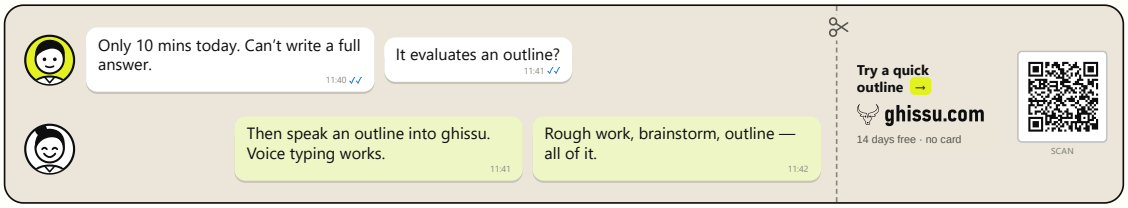
Scan to practice

Significance

1. **Strategic Autonomy:** Indigenous planning, launch and deep-space tracking showed India could run an interplanetary mission without foreign help.
2. **Role of Sriharikota:** The launch underlined Andhra Pradesh's place in India's space ambitions, and its infrastructure continues to support ISRO's expanding missions.
3. **Inspiration:** The success raised India's global status and inspired further lunar and interplanetary missions.

Conclusion

MOM marked India's transition from a space follower to a global innovator. Built on indigenous capability and assets like Sriharikota, it laid a strong foundation for future interplanetary missions in the spirit of Aatmanirbhar Bharat.



P-IV
2(b)

Write a short note on the following: (i) Broadband digital communication (ii) Fixed-Satellite Services (FSS)

10

Introduction

Digital communication has become crucial for India's socio-economic development, especially after the pandemic, as connectivity underpins remote work, education and governance. Broadband and Fixed-Satellite Services (FSS) are pivotal in bridging the digital divide and building a digitally empowered society.

(i) Broadband Digital Communication

1. **Meaning:** High-speed digital internet delivered mainly over optical fibre and wireless networks.
2. **Policy Support:** The National Broadband Mission and BharatNet have expanded fibre to over 2.15 lakh Gram Panchayats, including in Andhra Pradesh.
3. **Faster Rollout:** Amended Right of Way Rules, 2022 speed fibre, 5G and broadband deployment.
4. **Coverage Gaps:** Rural internet tele-density stands at 46.73%, and only 28% of rural women have internet access, due to last-mile and literacy gaps.
5. **Sectoral Impact:** It enables smart farming, telemedicine and digital service delivery through

Practice 2(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Common Services Centres.

(ii) Fixed-Satellite Services

1. Role: FSS provides broadband to islands, mountainous regions and maritime zones that terrestrial networks struggle to reach.
2. Regulatory Framework: TRAI and DoT allocate spectrum administratively under the Telecommunications Act, 2023, stressing affordability.
3. Private Investment: Up to 100% FDI in satellite operations draws private innovation into satellite broadband.
4. Charges: NGSO-based FSS pays 4% of Adjusted Gross Revenue, with rural subscriber fees exempted.
5. Technology: High Throughput Satellites integrated with 5G give high-speed, low-latency internet.
6. Strategic Value: Satellite links give resilient channels for defence and disaster management, and need less terrestrial infrastructure, suiting coastal and ecologically sensitive areas.
7. Launch Capacity: ISRO's Sriharikota facility plays a strategic role in deploying these satellites.

Broadband and FSS

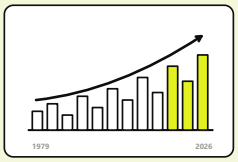
Terrestrial broadband	Fixed-Satellite Services
Optical fibre and wireless	Satellite links
Dense, connected areas	Islands, hills, sea
BharatNet, Right of Way Rules	Telecom Act 2023, 100% FDI
Last-mile gaps	Less ground infrastructure

Way Forward

1. Satellite R&D: Indigenous satellite manufacturing and launches, using Sriharikota's expertise to cut costs and improve reliability.
2. Digital Literacy: Focused programmes in rural and drought-affected areas to close the gender and rural-urban divide.

Conclusion

Broadband and FSS complement each other: fibre serves dense areas while satellites reach the remote. Indigenous satellites from Sriharikota and digital literacy will make digital inclusion real.



Which sub-topic will they ask next?

Trend analysis

Every syllabus topic: the direction of travel, the most tested sub-topic, the probable next theme.

Prepare the pattern →

ghissu.com

14 days free · no card

SCAN

P-IV
2(c)

Explain the applications of Remote Sensing technology in agriculture.

10

Introduction

Remote sensing has become indispensable in modern agriculture because it provides timely, accurate and large-scale data on crop health, soil conditions and water resources. Across India's diverse farm landscape, including Andhra Pradesh's varied agro-climatic zones and horticulture, it enables precision farming. Satellite imagery, drones and AI analytics are transforming traditional practices.

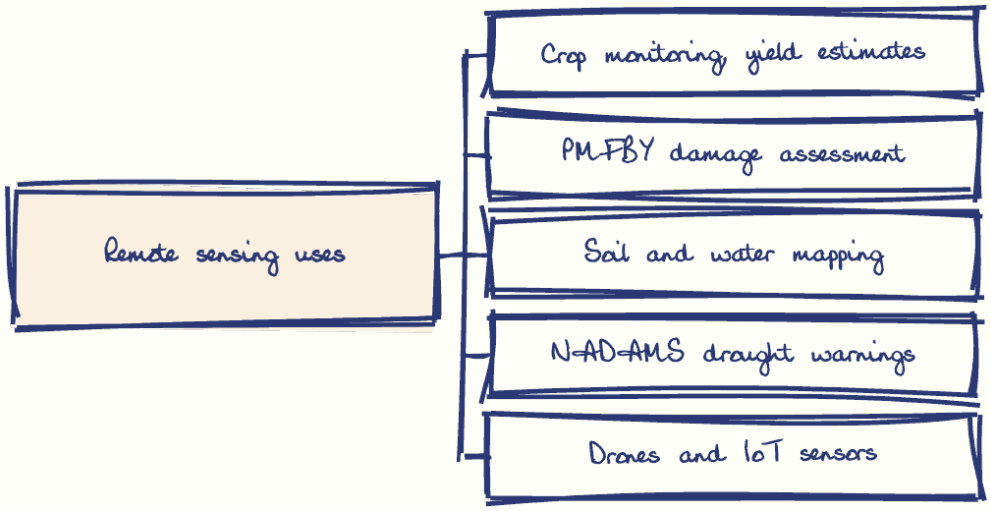
Institutional Support

1. **ISRO and NRSC:** IRS, Resourcesat and RISAT satellites, led by the National Remote Sensing Centre, Hyderabad, feed the Mahalanabis National Crop Forecast Centre.
2. **Data Policy:** The Remote Sensing Data Policy regulates access and promotes private participation, and its one window clearance streamlines data dissemination.
3. **Research:** Agricultural universities, including in Andhra Pradesh, work with ISRO on local applications.

Applications in Agriculture

1. **Crop Monitoring and Yield Estimation:** Continuous monitoring detects pests, nutrient deficiencies and water stress early, and supports accurate yield forecasts for procurement; in Andhra Pradesh it has improved chilli and horticulture management.
2. **Crop Insurance:** Imagery speeds damage assessment and claims under PMFBY, giving quicker compensation for drought and pest losses.
3. **Soil and Water Management:** Mapping soil type, moisture and fertility optimises fertiliser and irrigation use.
4. **Drought Assessment:** NADAMS issues drought warnings, helping Rayalaseema manage scarce water.
5. **Drones and IoT:** Drones capture detailed field images and IoT sensors automate irrigation, cutting input costs; the Namo Drone Didi scheme gives drones to women's self-help groups.
6. **Decision Support:** Combining satellite data with weather forecasts and AI gives precise yield predictions, risk assessment and real-time advisories.

Remote sensing on the farm



Way Forward

1. Farmer Training: Extension and digital literacy for small and marginal farmers.
2. Local Tailoring: State capacity and university research suited to local cropping patterns.
3. IoT and AI at Scale: Large pilots of precision farming to cope with climate variability.
4. Partnerships: Government, private firms and farmer organisations to scale solutions sustainably.

Conclusion

Remote sensing is making farming data-driven, precise and sustainable, raising productivity and resilience when linked to government schemes.

Are you actually improving?
Every attempt scored. Watch the curve.
ghissu.com →

P-IV
3(d)

Write a note on Local Area Network (LAN).

10

Introduction

A Local Area Network (LAN) connects multiple computers and devices within a limited area such as an office building or campus, enabling resource sharing and communication. With the rapid digitisation of government offices and industries in India, LANs have become the backbone of efficient and secure internal communication, making their components, uses and limits important to understand.

Components and Technologies

1. Key Components: Network interface cards in each device, switches or hubs, routers for external connectivity, and media such as Ethernet cables or optical fibre.

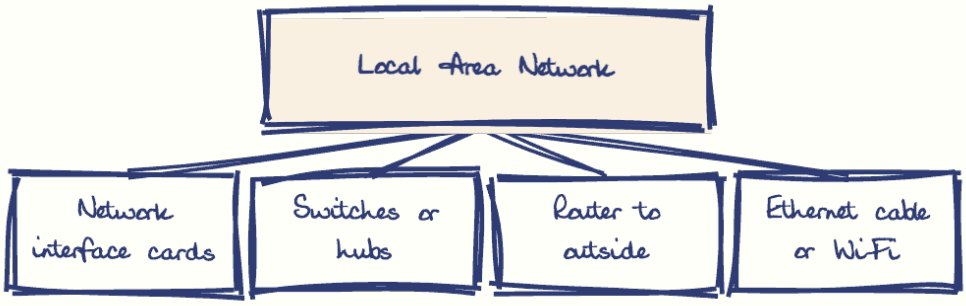
2. Wired and Wireless LANs: LANs use Ethernet cables or Wi-Fi. Wi-Fi access at the Planning Commission's Yojana Bhavan gave senior officers flexible, secure connectivity.

3. Protocols and Standards: TCP/IP, IEEE 802.3 (Ethernet) and IEEE 802.11 (Wi-Fi) ensure interoperability.

Practice 3(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Building blocks of a LAN



Applications and Benefits

1. Resource Sharing: Shared files, printers and applications improve collaboration, as with the web-based NetShare application.
2. E-Governance: The Desktop Executive Video Conferencing System over NICNET shows LANs aiding administration.
3. Cost Efficiency and Scalability: Centralised resources cut hardware costs, and fibre upgrades let LANs grow.
4. Security Management: Floor-wise VLANs segment the network to prevent spam and virus attacks.

Challenges and Limitations

1. Security Vulnerabilities: Wireless LANs remain open to unauthorised access and denial of service despite WEP.
2. Costs: Cabling, switches, routers and skilled staff

Practice 3(d) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

are expensive, and rural infrastructure is limited.

3. Limited Range: LANs must be integrated with WANs to connect dispersed locations.

4. Complexity: Compatibility with evolving standards needs continuous upgrades.

Way Forward

1. Cybersecurity: Encryption and authentication beyond WEP for sensitive data.

2. Optical Fibre: Expansion under BharatNet for urban and rural backbones.

3. Skills: Training IT staff in LAN management and security for public and private sectors.

Conclusion

LANs underpin efficient, collaborative workplaces and e-governance, securing and scaling them is essential for digital administration.

STUDY TIP · WHERE TO WRITE A NOTE

Write the note against the question that taught it. That is the only place revision will look for it.

ghissu

On ghissu every question has a notes tab — and your notes gather into chapter notes by themselves.

Open the notes tab →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 3(d) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

Scan to practice

P-IV
3(a)

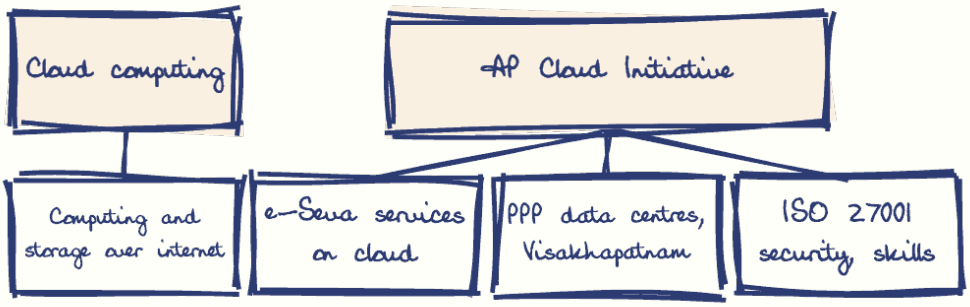
What is cloud computing? Explain briefly about AP Cloud Initiative?

10

Introduction

Cloud computing is the delivery of scalable, cost-effective and flexible IT services over the internet, so users draw on computing power and storage without owning it. Its importance has surged with data-intensive technologies like artificial intelligence and big data analytics. The AP Cloud Initiative harnesses it to boost the state's digital infrastructure, IT ecosystem and service delivery.

Cloud computing and the AP Cloud Initiative



Features of the AP Cloud Initiative

1. **Government-Led Infrastructure:** Platforms like e-Seva use the cloud to provide seamless G2C and B2C services, improving citizen access and administrative efficiency.
2. **Public-Private Partnerships:** The state works with

Practice 3(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

private cloud providers to build data centres, such as the Google AI data centre in Visakhapatnam, which received incentives worth over Rs 22,000 crore and 600 acres of land.

3. Security Frameworks: It follows national data security standards and certifications like ISO 27001, while addressing data sovereignty concerns.
4. Skill Development: Partnerships with educational institutions and industry build a workforce able to manage cloud technologies.

Outcomes

1. Governance: Numerous services digitised through e-Seva and UM-ANG give real-time delivery with more transparency and efficiency.
2. Industry: Scalable IT infrastructure supports port-led corridors like the Visakhapatnam-Chennai corridor, attracting investment and fostering innovation.

Challenges

1. Connectivity Gaps: Rural areas, especially drought-prone Rayalaseema, lack reliable broadband and power, which cloud adoption needs.
2. Data Privacy: Cross-border data flows must be balanced with localisation mandates and national

data protection laws.

Way Forward

1. Rural Connectivity: Accelerate broadband and power infrastructure in underserved areas of the state so that cloud access and digital inclusion are equitable.
2. Security Protocols: State-specific guidelines aligned with national frameworks on privacy and cybersecurity.
3. Cloud Skills: Specialised training and certifications for a talent pipeline in IT and startups.
4. Collaboration: Government, academia and industry partnerships for indigenous cloud solutions.

Conclusion

By expanding rural connectivity, strengthening security protocols and building a cloud skill ecosystem, Andhra Pradesh can use cloud technology for digital governance and inclusive growth, in line with SDG 9.

Your intro decides your marks. Q. Discuss Ashoka's Dhamma.

YOUR INTRO
~~Ashoka was a great Mauryan king who ruled after the Kalinga war and spread Buddhism everywhere.~~
✕ no date · no source · no argument

AFTER GHISSU EVALUATION
Ashoka's Dhamma, set out in his Major Rock Edicts after Kalinga (c. 261 BCE), was not Buddhism by another name but an ethic of tolerance, non-violence and respect, meant to hold a plural empire together.
✓ date · source · argument

ILLUSTRATIVE

Get your intro fixed on your sheet →
ghissu.com
14 days free · no card
SCAN

Write your own answer to 3(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

P-IV
3(b)

Describe Fiber Grid Project of Andhra Pradesh.

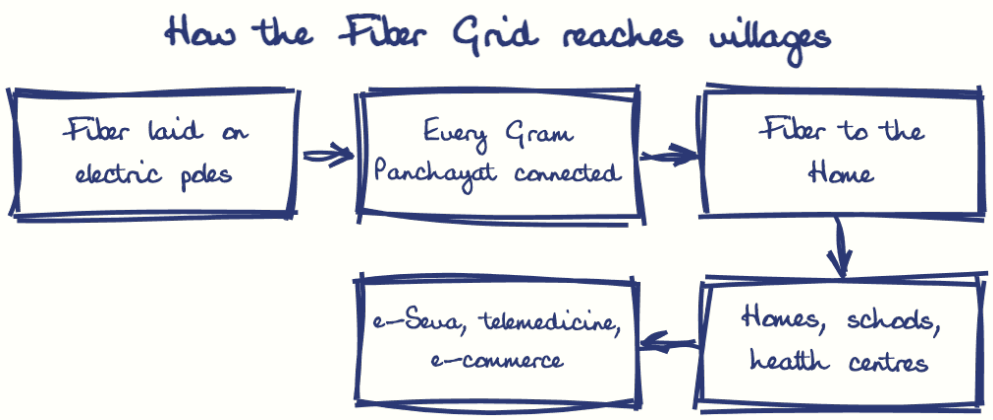
10

Introduction

Robust internet connectivity is essential for socio-economic development, especially in rural and remote areas. The Fiber Grid Project of Andhra Pradesh aims to bridge the digital divide by providing high-speed broadband to all Gram Panchayats and households, enabling e-governance, digital education, telemedicine and economic empowerment.

Components of the Project

1. Optical Fiber Network: Cables connect all Gram Panchayats, laid along existing electric poles to speed deployment and reduce cost.
2. Last-Mile Connectivity: Fiber to the Home connections reach rural homes, schools, health centres and government offices.
3. Public-Private Partnership: Private participation has accelerated network expansion and service delivery.
4. Satellite and Wireless Integration: Satellite media and Wi-Fi hotspots cover remote and tribal regions where fiber is hard to lay.



Economic and Social Impact

1. Rural Economy: Farmers and small entrepreneurs gain access to digital markets, e-commerce and financial services.
2. Governance: Platforms like e-Seva deliver welfare schemes efficiently, reducing corruption and delays.
3. Education and Health: It supports Samagra Shiksha ~~Abhiyan~~ and telemedicine in rural areas.
4. Social Inclusion: Marginalised communities can join the digital economy and access information, fostering social equity.

Challenges and Opportunities

1. Terrain and Cost: Remote, difficult areas raise logistical and financial burdens, needing solutions like electric poles and satellite links.

2. Digital Literacy: Rural users need skills and awareness to use services.

3. Maintenance: The extensive network needs continuous investment and technical expertise, which can strain state resources.

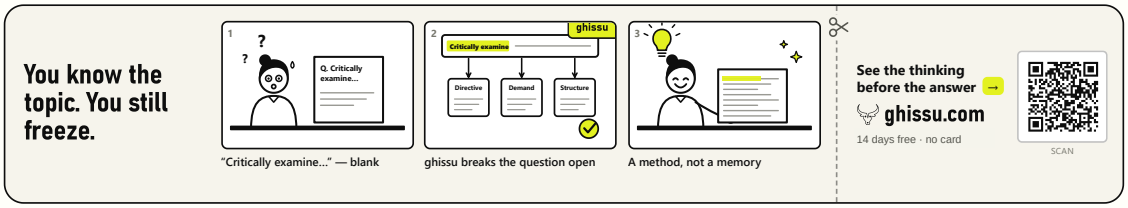
4. Innovation: Smart villages like Mori in East Godavari use fiber for cashless transactions and e-governance.

Way Forward

Digital literacy campaigns, stronger partnerships with the private sector and satellite links in tough terrain will maximise the benefits, while integration with state development schemes gives a holistic framework.

Conclusion

Aligned with Swarna Andhra Vision 2047 and SDG 9, the project catalyses growth, inclusion and better governance, laying the foundation for a knowledge-based economy.



P-IV
3(c)

What is ERP (Enterprise Resource Planning)? Narrate the critical issues in execution of ERP Projects.

10

Introduction

Enterprise Resource Planning (ERP) systems integrate an organisation's functions, such as finance, operations and content, into one unified digital platform. In India's growing digital governance and industrial ecosystem, ERP is critical for efficiency, transparency and scalability, as the customised ERP at TISS-SVE shows.

What ERP Involves

1. Customised Development: Software must suit the organisation; the TISS-SVE ERP integrates accounts, curricular content and examinations for a multi-partner delivery model.
2. Centralised Data: A single database backed by a data centre ensures seamless coordination across departments.
3. Role-Based Access: Defined administrative, supervisory and operational roles ensure accountability.
4. Benefits: Digital document management cuts paperwork and errors, streamlined fee collection improves accountability and real-time monitoring

Practice 3(c) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

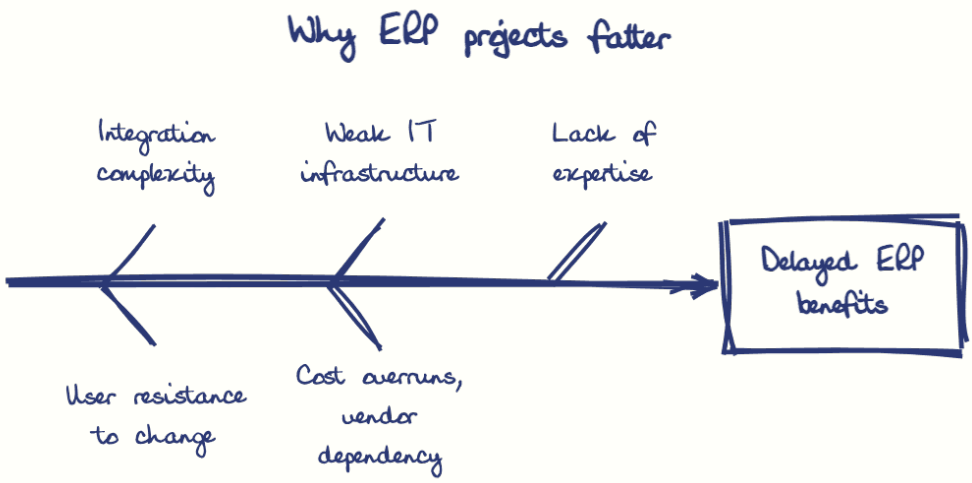
flags bottlenecks early.

Critical Issues in Executing ERP Projects

1. **Integration Complexity:** Linking finance, operations and content modules needs iterative customisation and close IT–user collaboration, as the phased TISS–SVE build showed.
2. **Change Management:** Users accustomed to legacy systems and manual processes resist the change, delaying benefits unless training and change strategies are thorough, a common problem in government rollouts.
3. **Infrastructure and Connectivity:** ERP needs reliable internet and IT infrastructure; gaps in rural and underdeveloped regions hamper consistent access and real-time data.
4. **Cost Overruns:** Unforeseen complexity, long timelines and vendor dependency push up costs and maintenance.
5. **Lack of Expertise:** Earlier public sector attempts failed without experts, prompting renewed efforts with professional vendors.

Way Forward

1. **IT Infrastructure:** Reliable connectivity and data centres for uninterrupted operation in urban and rural



areas.

2. Capacity Building: Specialised training for government and partner staff to build ERP literacy and ease the shift from legacy systems.
3. Knowledge Transfer: Vendor deals with source code handover and local maintenance.
4. Change Management: Institutional practices that address user resistance and build a culture of digital adoption.

Conclusion

ERP can transform governance and service delivery by integrating processes and enhancing transparency, if its execution issues are managed with change management and capacity building.

How did toppers write this one?
1.5 lakh+ topper answers, matched to each question.
ghissu.com →

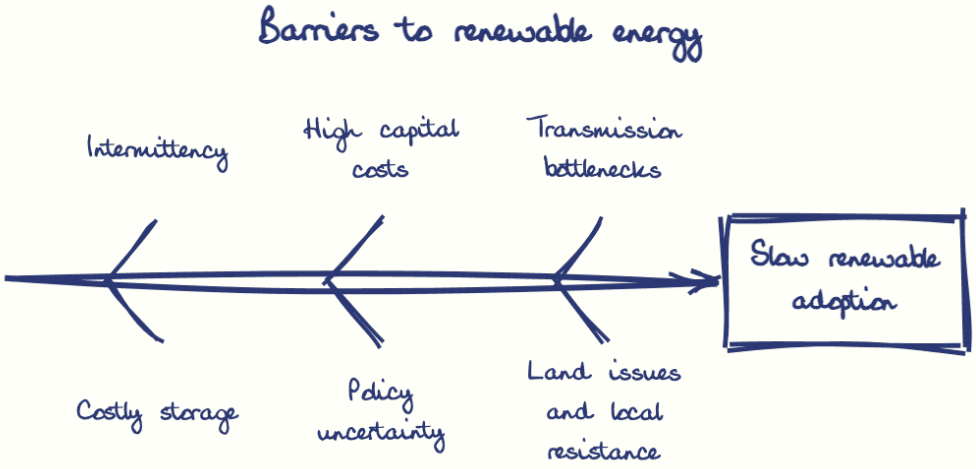
P-IV
4(d)

Give a detailed list of barriers to renewable energy?

10

Introduction

Rapid renewable expansion is critical for India's sustainable development and climate commitments as it aims for 450 GW of renewable capacity by 2030. Despite progress, including Andhra Pradesh's growing solar and wind projects, multiple barriers still hinder full adoption. Understanding them is essential to speed up the transition.



Technological and Economic Barriers

1. **Intermittency:** Intermittency makes solar and wind output fluctuate daily and seasonally with the weather, causing grid instability and needing advanced forecasting.

2. **Costly Storage:** Storage remains expensive and

underdeveloped; ReNew Power's battery installations show it is still nascent.

3. High Capital Costs: Large upfront investment and scarce affordable credit deter private and risk-averse investors.

4. Tariff Disputes: Legacy high-tariff contracts stress DISCOMs, and the tariff renegotiation dispute involving Andhra Pradesh DISCOMs hurt investor confidence.

Policy and Infrastructure Barriers

1. Policy Uncertainty: Frequent changes, misalignment with macroeconomic policy and weak state enforcement unsettle developers and financiers.

2. Payment Delays: The shift from feed-in tariffs to auctions added risk, and states like Maharashtra delayed payments to producers.

3. Transmission Gaps: Transmission bottlenecks keep power from reaching demand centres.

4. Land Issues: Poor land records and no land utilisation policy delay projects, as at the Bhadla solar park.

Social and Environmental Barriers

1. Local Resistance: Land conflicts and weak

community engagement provoke opposition.

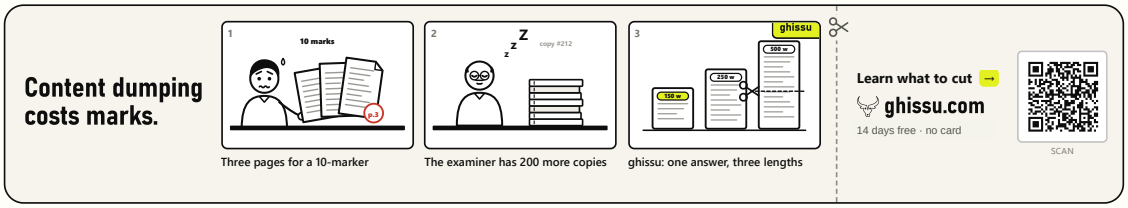
2. Law Awareness: Limited knowledge of rooftop solar and net metering restricts consumer participation.
3. Ecological Concerns: Large hydro causes habitat disruption, sedimentation and displacement, solar and wind farms affect local biodiversity.

Way Forward

1. Grid and Storage: Advanced storage and smart grids to manage intermittency.
2. Policy Stability: Consistent, transparent policies with stakeholder input and strict state enforcement.
3. Infrastructure: Robust transmission and clear land policies.
4. Public Participation: Awareness campaigns and community involvement.

Conclusion

Addressing these barriers together will let India, including Andhra Pradesh, accelerate its clean energy transition, cut fossil dependence and advance SDG 7.



P-IV
4(a)

Write a short note on Shale Gas Revolution.

10

Introduction

The shale gas revolution is a transformative shift in global energy, driven by technology that extracts natural gas from shale rock. It is relevant as India seeks energy security and lower import dependence, including in Andhra Pradesh with its sedimentary basins.

Understanding the Revolution

1. Technology: Hydraulic fracturing and horizontal drilling unlocked reserves once uneconomical.
2. Scale: US output rose from 3 billion cubic feet a day in 2000 to 70, making it a net energy exporter and reshaping global markets.
3. India's Potential: The USGS estimates about 6.1 trillion cubic feet in three basins, a significant untapped resource.
4. Policy: NELP and HELP simplify contracts and give pricing and marketing freedom.

Impacts

1. Energy Security: Less reliance on imported LNG, which meets a large share of consumption,

Practice 4(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

bringing self-reliance and steadier prices.

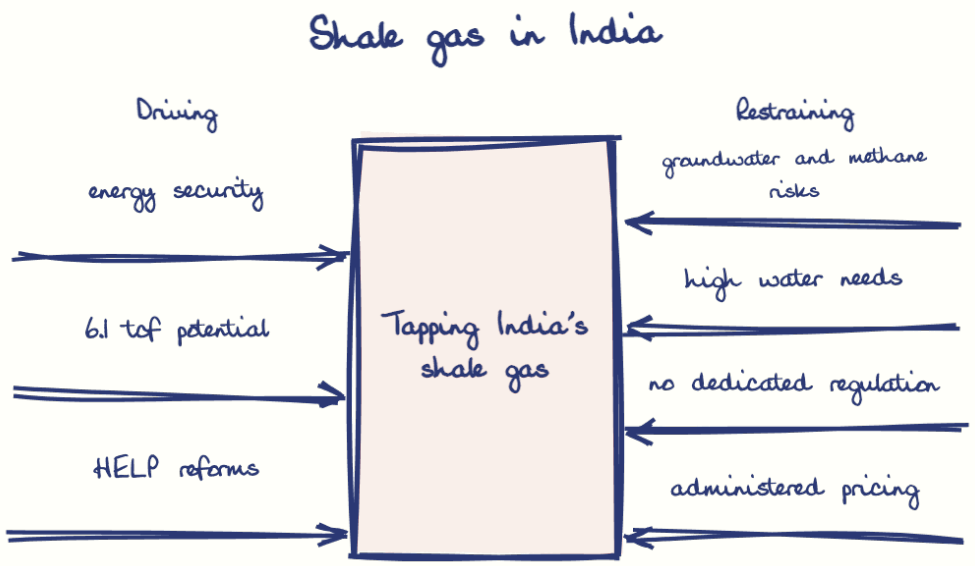
2. Growth and Jobs: The US boom added nearly $400 billion since 2016; similar gains could reach corridors like Visakhapatnam–Chennai.
3. Industry: More gas for fertilisers, power and city gas distribution.
4. Pricing: Consumption-weighted basket pricing follows international trends but not local production costs, hurting viability.

Challenges

1. Environment: Groundwater contamination, methane leakage and induced seismicity.
2. Water and Infrastructure: High water needs strain areas like Rayalaseema.
3. Regulation: No dedicated regime and public apprehension delay approvals.
4. Viability: High costs and administered pricing deter private investment.

Way Forward

1. Incentives: Contracts that reward exploration risk and allow cost recovery in India's sedimentary basins.



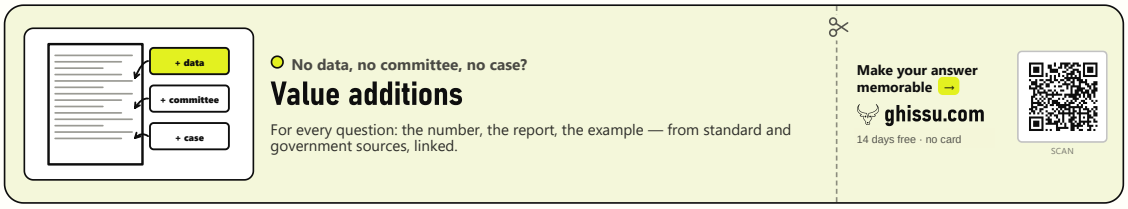
2. Safeguards: Monitor groundwater, methane and seismic activity.

3. Technology and Water: Indigenous drilling technology and water management in drought-prone areas.

4. Pricing Reform: Reflect production costs to draw private investment.

Conclusion

Shale gas can transform India's energy security and economy, including in Andhra Pradesh's basins, but only if immediate needs are balanced with environmental and regulatory safeguards.



P-IV
4(b)

Narrate Nuclear Power growth in India. Add a note on Nuclear Fuel reserves in the country.

10

Introduction

Nuclear power is emerging as a critical part of India's clean energy transition as it works to cut carbon intensity and raise low-carbon capacity to 40% by 2030. With demand rising, it offers reliable, emission-free power to complement renewables, and the target of 100 GW by 2047 shows its strategic importance.

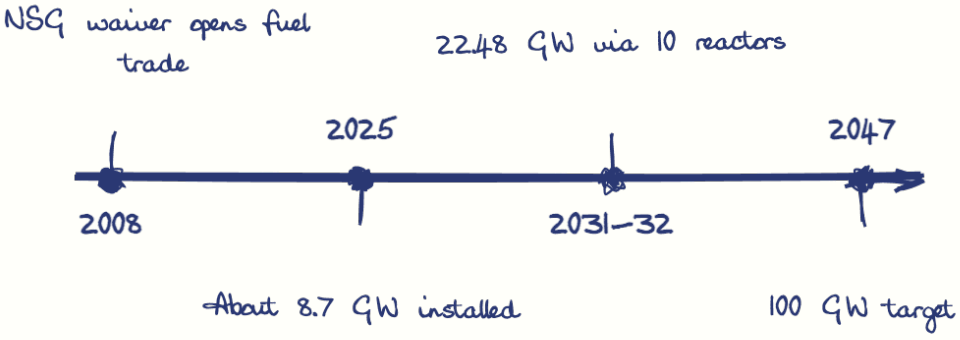
Growth of Nuclear Power in India

1. Institutions: The DAE and NPCIL oversee reactor construction, operation and the fuel cycle.
2. Indigenous Technology: PHWRs and the Prototype Fast Breeder Reactor at Kalpakkam mark progress in the three-stage programme.
3. International Cooperation: After the 2008 NSG waiver, deals with the US, Russia, France and Kazakhstan gave access to advanced reactors and fuel.
4. Capacity Growth: About 8.7 GW in early 2025, planned to reach 22.48 GW by 2031-32 through 10 new reactors, both

collaborative and indigenous.

5. Challenges: Liability legislation, stronger safety protocols and public trust are critical for new projects.

Growth of nuclear power in India



Nuclear Fuel Reserves

1. Uranium: Domestic reserves are limited, so India imports mainly from Russia, agreements and reprocessing reduce the risk.
2. Thorium: India has the world's largest thorium reserves, concentrated in coastal states including parts of Andhra Pradesh, ideal for future thorium reactors.

Outcomes

1. Energy Security: Steady large-scale power cuts fossil imports and emissions under the NDC.

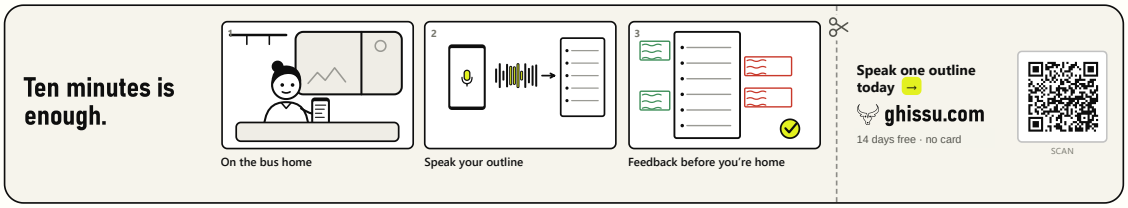
2. Growth and Jobs: High-technology manufacturing and skilled jobs, with industrial corridors like Andhra Pradesh's gaining power.
3. Strategic Autonomy: Indigenous fuel cycle skills reduce exposure to geopolitical pressure.

Way Forward

1. Thorium R&D: Pilot thorium-based reactors for long-term fuel sustainability.
2. Fuel Resilience: Diversify uranium imports and expand domestic exploration and mining, including in Andhra Pradesh's coastal areas.
3. Public Engagement: Transparent communication and robust safety rules.

Conclusion

Nuclear power is pivotal for a low-carbon, secure energy future, and thorium can make India a leader in advanced nuclear technology.



P-IV
4(c)

Discuss the scope of renewable energy in the scenario of energy crisis?

10

Introduction

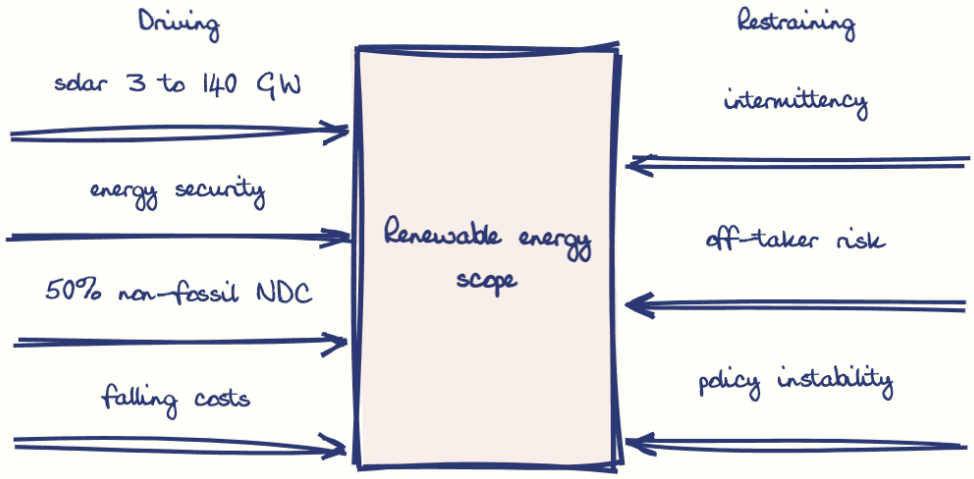
The global energy crisis, worsened by geopolitical tensions and finite fossil fuels, has made sustainable alternatives urgent. India, the world's third-largest energy consumer, must meet rising demand while cutting emissions. The rapid growth of solar and wind offers a path to energy security and sustainability.

Scope of Renewable Energy in India

1. **Expanding Capacity:** Solar rose from (3) GW in (2014) to (140) GW by January (2026) and wind reached (54.65) GW, backed by (55) solar parks in (13) states and the National Green Hydrogen Mission (5 MMT by 2030).
2. **Energy Security:** Renewables now exceed half of installed capacity, cutting dependence on volatile fuel imports. Andhra Pradesh's RPO and REC mechanisms have built local demand and investment.
3. **Climate Goals:** Renewables are inherently low-carbon, and India's updated NDC targets (50%) non-fossil capacity by (2030).
4. **Economic Feasibility:** Falling costs and zero

import duty on solar components make renewables competitive, though off-taker risk and policy instability worry investors.

Renewables amid the energy crisis



Role Amid the Energy Crisis

1. Buffer Against Shocks: The Russia–Ukraine conflict exposed fossil supply chains to price shocks and shortages; renewables are domestic and inexhaustible, so they make supply more resilient.
2. Grid Stability: Europe's 2025 grid failures show intermittency needs storage and smart grids; Andhra Pradesh is pairing solar and wind with battery storage.
3. Decarbonisation: Solar and wind may form 90–98% of installed capacity by 2070, cutting

pollution-related health risks.

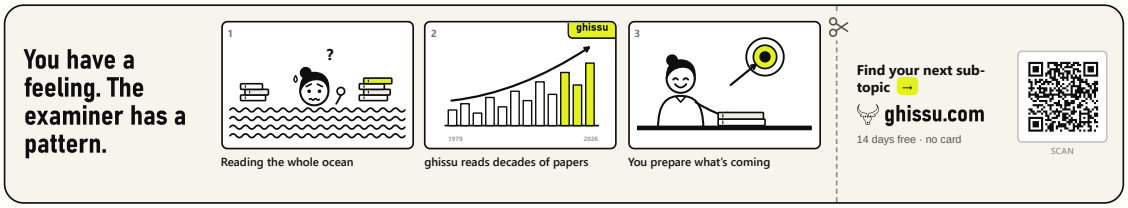
4. Rural Development: Decentralised power for areas without grid access under PM-KUSUM in Andhra Pradesh improved electricity and farm productivity.

Way Forward

1. Grid and Storage: Smart grids and large-scale batteries to manage intermittency.
2. Policy Stability: Consistent rules to cut investor risk and attract long-term finance.
3. Decentralised Solutions: Off-grid and mini-grid solar in rural areas.
4. Indigenous Manufacturing: Domestic R&D for Atmanirbhar Bharat.

Conclusion

Renewables have vast scope to strengthen India's energy security, sustainability and growth. In line with climate commitments and the International Solar Alliance, they can turn the crisis into an opportunity for a resilient, low-carbon future.



P-IV
5(d)

Write the main points of the agreement reached at United Nations Climate Change Conference (COP 21) held at Paris in December, 2015. What was India's stand at the conference?

10

Introduction

The UN Climate Change Conference (COP 21) in Paris in December 2015 marked a historic effort to fight climate change by uniting nearly all countries under a common framework. Rising temperatures and extreme weather threaten vulnerable regions such as coastal Andhra Pradesh with its 974 km coastline, so the Agreement sought to balance sustainability with development.

Main Points of the Paris Agreement

1. Temperature goal: Hold the global average rise well below 2°C above pre-industrial levels while pursuing 1.5°C, to cut risks for regions like drought-prone Rayalaseema.
2. NDCs: Each country submits and updates its Nationally Determined Contributions every five years, a bottom-up approach unlike the top-down Kyoto Protocol.
3. Climate finance: Developed countries reaffirmed at least USD 100 billion a year until 2025

for developing countries, reflecting CBDR-RC.

4. Comprehensive coverage: It spans mitigation, adaptation, loss and damage, technology development and transfer, capacity building and transparency, ensuring a balanced approach.

5. Global stocktake: A global stocktake every five years reviews collective progress.

India's Stand at COP (21)

Paris deal and India's stand

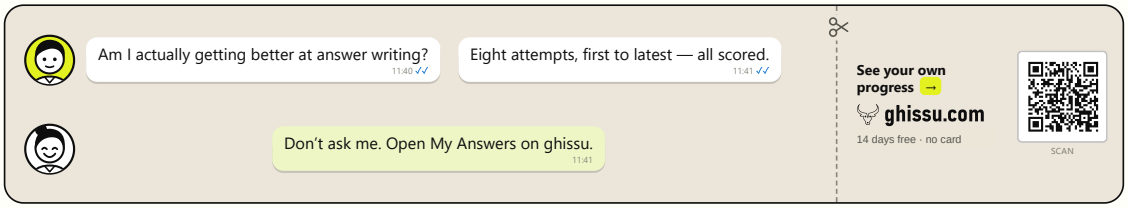
Paris Agreement	India's stand
Well below 2°C goal	Right to development
NDCs every five years	Climate justice, CBDR-RC
Climate finance	Developed nations bear more
Global stocktake	International Solar Alliance

1. Right to development: India pledged to reduce the emissions intensity of its GDP by 33-35% by (2030) from (2005) levels and to raise non-fossil power capacity to (40%) by (2030); it ratified the Agreement on October (2, 2016.)

2. Climate justice: It argued developed countries must bear more responsibility for cuts and finance, as India's per capita emissions are only about 1.6 tonnes.
3. Renewable leadership: India launched the International Solar Alliance at COP 21 and set a target of 175 GW of renewables by 2022, with solar and wind projects growing in Andhra Pradesh.
4. Poverty and energy access: It stressed that poverty reduction and energy access for vulnerable groups, such as Andhra Pradesh's agrarian and coastal communities, must be balanced with emission cuts.
5. Multilateralism: Its Environment Minister called the Agreement a "victory of multilateralism", reflecting India's active engagement.

Conclusion

The Paris Agreement is a landmark consensus that respects development needs, and India's stand helped turn it from a perceived naysayer into a proactive leader in climate diplomacy.



P-IV
5(a)

Distinguish between a Hazard and a Disaster? Narrate various types of natural disasters, the coastal seashore areas are prone to?

10

Introduction

With rising climate variability and encroachment on fragile ecosystems, the distinction between hazards and disasters is crucial for risk management. Coastal areas like Andhra Pradesh, with its 974 km coastline, face disaster risks that threaten livelihoods, infrastructure and ecology.

Distinguishing Hazard and Disaster

1. Hazard: A natural or anthropogenic event with the potential to cause harm, such as a cyclone, which causes no damage unless it meets vulnerable elements.
2. Disaster: The actual occurrence of a hazard that causes loss of life and property and overwhelms a community's capacity to cope without outside help.
3. Vulnerability and Exposure: A cyclone is a hazard; when it devastates Visakhapatnam it becomes a disaster, which is why dense coastal cities are more disaster-prone.

Practice 5(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

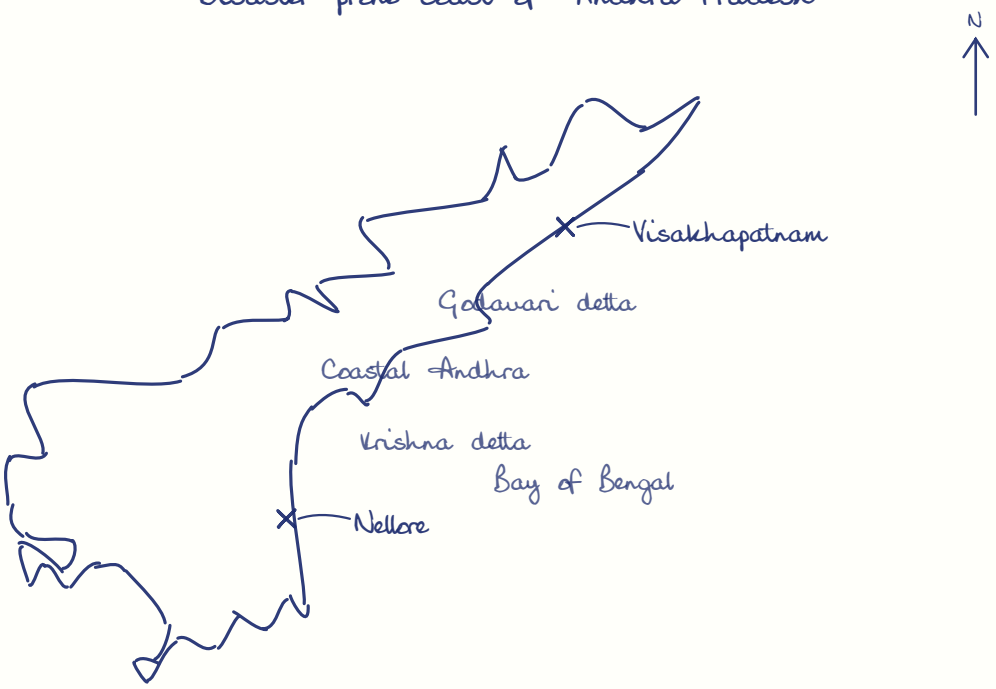
Candidates must not write on this margin

Candidates must not write on this margin

Hazard vs disaster

Hazard	Disaster
Potential event	Actual event
May cause no damage	Loss of life and property
Community copes	Overwhelms coping capacity

Disaster-prone coast of Andhra Pradesh



Cyclones, surges, floods, tsunami, erosion

Natural Disasters Coastal Areas Are Prone To

1. Cyclones and Storm Surges: Visakhapatnam and Nellore face destructive winds, surges and flooding that threaten fisheries and ports.

How to use this book with ghissu.com
see page 4

Index · page 6

Practice 5(a) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

2. Floods: Low-lying Krishna and Godavari deltas face recurrent flooding from monsoon rains, worsened by sea-level rise due to climate change.

3. Tsunamis: Rare but catastrophic waves from undersea earthquakes, the 2004 Indian Ocean tsunami severely hit the eastern coast, including Andhra Pradesh.

4. Coastal Erosion and Salinisation: Gradually degrade farmland and habitats.

5. Earthquakes and Landslides: Ground shaking causes structural collapse, while landslides triggered by heavy rain disrupt transport and settlements in hilly coastal areas.

These disasters are interlinked: cyclones bring floods and surges, while erosion and salinisation leave the coast weaker against the next event.

Conclusion

Since population density and ecosystem loss amplify impacts, early warning systems and community preparedness are key to keeping coastal hazards from becoming disasters.

Where did you write that point?
Notes that live with the question.
ghissu.com

Write your own answer to 5(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
5(b)

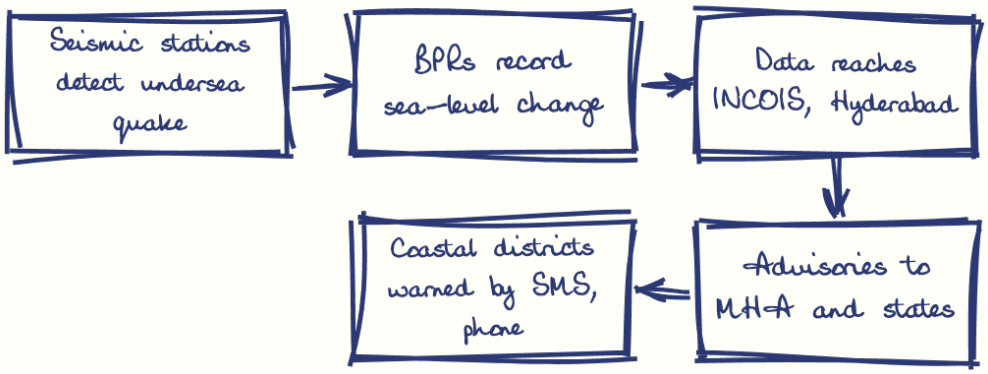
Explain the details of Tsunami Warning System of India.

10

Introduction

The 2004 Indian Ocean tsunami exposed India's vulnerability to sudden oceanic disasters. With a 7,516 km coastline, including Andhra Pradesh's 974 km, India built its own Tsunami Early Warning System for early detection and timely warnings.

How India's tsunami warning works



Components and Functioning

1. **Seismic and Ocean Monitoring:** Seismic stations and Bottom Pressure Recorders (BPRs), four in the Bay of Bengal and two in the Arabian Sea, detect undersea earthquakes above magnitude 6 and sea-level changes.
2. **Data Transmission and Alerts:** Sensor data reaches INCOIS, Hyderabad, which processes it

Practice 5(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

and generates advisories, relayed swiftly by satellite links such as Iridium transceivers.

3. Dissemination: INCOIS alerts the Ministry of Home Affairs and state emergency centres, which warn coastal districts like Visakhapatnam by telephone, SMS, fax and email. This multi-tier chain gives vulnerable people time to evacuate.

Technological and Institutional Framework

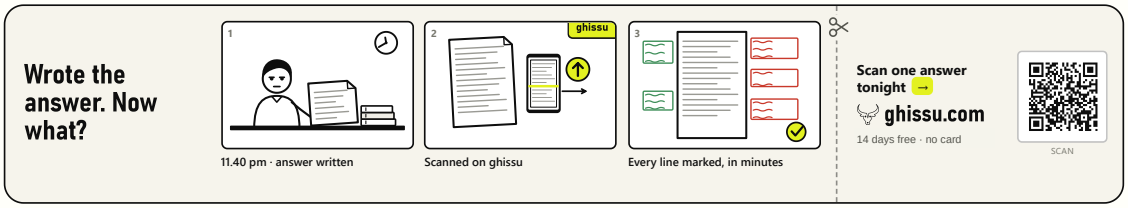
1. Sensor Network and Models: Seismic sensors, tide gauges and BPRs feed computational models at INCOIS that analyse seismic data and predict wave propagation, enabling real-time decisions.
2. Ministry of Earth Sciences: Provides policy guidance and funding; INCOIS, the nodal agency with a 24x7 centre, coordinates with IMD's seismic network and international oceanographic bodies.
3. State Integration: Alerts link to Andhra Pradesh's State Emergency Operations Centre for local evacuation, while NDMA's Tsunami Risk Management Guidelines fix roles across agencies.
4. MHA and States: They coordinate evacuation and relief; Andhra Pradesh has built tsunami alerts into its coastal disaster plans.

Effectiveness and Gaps

1. **INCOIS as Hub:** Its 24x7 centre uses Indian Ocean seismic data and international collaboration to improve accuracy and lead time.
2. **Proven Readiness:** It detected the 2007 Indonesia earthquake tsunami threat, validating its operational readiness.
3. **Community Measures:** Public education, drills, mangrove restoration and coastal bio-shields complement technology, in line with NDMA guidelines, so that warnings lead to action.
4. **Gaps:** Audits show fewer active buoys than recommended and funding shortfalls, which limit ocean data coverage and real-time accuracy.

Conclusion

Since 2004, the system has greatly reduced potential loss of life and property. Expanding BPRs and coastal observatories, assuring funding and building community preparedness will keep India's warning system a reliable shield for its coasts.



P-IV
5(c)

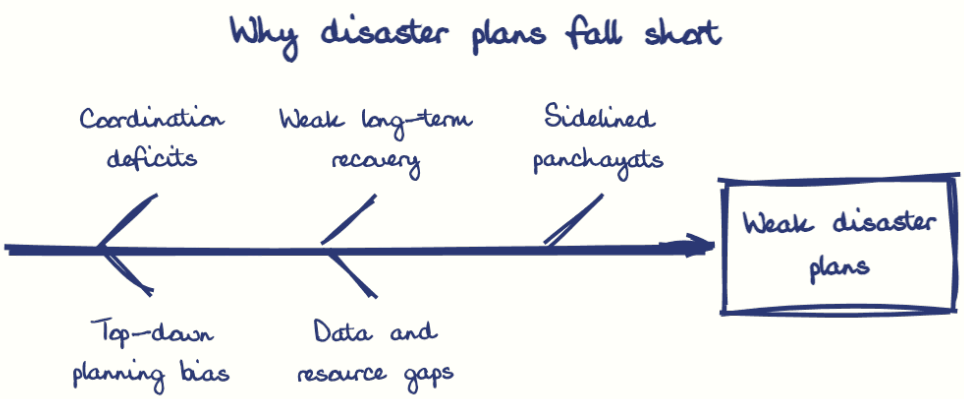
Discuss the common challenges in any disaster management plan.

10

Introduction

As climate change intensifies, India faces more frequent cyclones, floods and droughts. Andhra Pradesh, with its long coastline and experience of Cyclone Hudhud (2014), shows the need for robust plans. Despite better early warning, common challenges persist across all phases.

Common Challenges in Disaster Management Plans



1. **Coordination Deficits:** Poor coordination among central, state and local agencies fragments efforts and delays response, as seen in the bureaucratic limits faced by DDMAs in AP.
2. **Top-Down Planning Bias:** Centralised plans neglect community participation and local knowledge, weakening preparedness and mitigation in drought-prone rural Rayalaseema.

3. Weak Long-Term Recovery: Rehabilitation and livelihood restoration are poorly planned, as in slow post-cyclone recovery in Visakhapatnam.
4. Data and Resource Gaps: Outdated data and little village-level information hamper targeted interventions and resource allocation in vulnerable regions of AP.

Challenges Across Phases

1. Prevention and Mitigation: Embankments in the Godavari delta fail through poor maintenance, while zoning and ecosystem-based risk reduction are underused.
2. Preparedness and Response: Despite IMD cyclone warnings, dissemination gaps in remote Srikakulam villages lead to weak evacuation.
3. Recovery: Build Back Better is applied inconsistently, so reconstruction ignores climate resilience and social inclusion, perpetuating vulnerability and delaying sustainable recovery.
4. Institutional Capacity: Panchayats and elected representatives are sidelined in district disaster authorities.

Way Forward

1. Decentralised Governance: Give Gram Panchayats in AP funds and technical resources to build risk reduction into development.
2. Data and Early Warning: Create village-level databases and warnings in local languages.
3. Resilient Infrastructure: Apply Build Back Better to housing and infrastructure in districts like East Godavari.
4. Inter-Agency Coordination: Set clear protocols and joint training for state departments, disaster response forces and civil society, so that coordination is seamless in every phase.

Conclusion

Overcoming these challenges needs inclusive, data-driven and climate-resilient planning that empowers local institutions. In line with the Sendai Framework, this will help AP protect vulnerable people and infrastructure against the growing threat of climate-induced disasters.

REVIEWER'S NOTE · CRITICALLY EXAMINE

"Critically examine" pays for the counter-view. An answer that stops at the first view has stopped at half the marks.

ghissu

ghissu exam-hall thinking decodes every directive — Discuss, Examine, Elucidate — before you write a line.

Decode your next question →

ghissu.com

14 days free · no card

SCAN

P-IV
6(d)

What is Lactic acid fermentation? Explain how different products can be made by fermentation of milk?

10

Introduction

Lactic acid fermentation has renewed importance for food preservation, nutrition and sustainable food systems. In India, where dairy farming is a major livelihood and milk output has surged, it transforms perishable milk into shelf-stable products enriched with probiotics.

What Is Lactic Acid Fermentation

1. Mechanism: Lactic acid bacteria convert lactose in milk into lactic acid through homofermentation.
2. Effect: The lower pH creates an acidic environment that preserves milk, develops characteristic flavour, inhibits spoilage microbes and improves digestibility.
3. Why it matters: It extends shelf life and enriches milk with probiotics and bioactive compounds, in line with traditional diets and modern nutrition science.

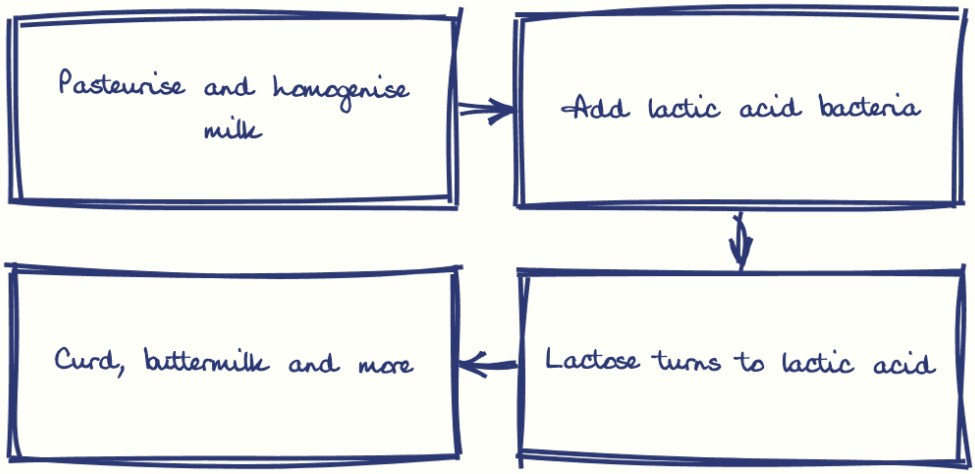
How Milk Products Are Made

1. Milk preparation: Processing units pasteurise and homogenise milk to keep bacterial counts low

Practice 6(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

From milk to curd



before fermentation.

2. Culture and conditions: Different microbial cultures, fermentation durations and temperatures produce different products, reflecting regional preferences.

3. Curd (dahi): Lactic acid bacteria acidify the milk, giving curd, the commonest fermented milk product.

4. Buttermilk: A fermented drink produced widely through dairy cooperatives, including in Andhra Pradesh.

5. Quality control: Standards and R&D on bacterial strains ensure microbial safety and consistency.

Benefits

1. Nutrition: Products supply probiotics, calcium and

vitamins, improving micronutrient bioavailability and gut health, and helping tackle malnutrition.

2. Livelihoods: Women, about 70% of the dairy workforce, earn beyond raw milk sales through cooperatives and self-help groups, especially in rural Andhra Pradesh.

3. Less waste: Surplus milk becomes value-added products in Andhra Pradesh, where the cold chain is limited.

Challenges

1. Process gaps: Inadequate cold storage, variable cultures and non-standard village processes limit scale and market reach for small producers in Andhra Pradesh.

2. Opportunity: Growing demand for traditional and probiotic dairy products opens new markets and room for innovation.

Conclusion

Lactic acid fermentation preserves milk, enriches nutrition and adds value for dairy farmers. Stronger cooperatives with quality starter cultures, better processing and digital tools can scale it for nutrition security and rural livelihoods.

Write your own answer to 6(d) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
6(a)

What are Vitamins? Give a detailed classification of vitamins mentioning their functions and sources of availability?

10

Introduction

Vitamins are essential micronutrients that play a crucial role in maintaining health and metabolic functions. With rising nutritional awareness in India, including Andhra Pradesh, understanding their classification, functions and sources is the foundation for tackling malnutrition.

What Are Vitamins

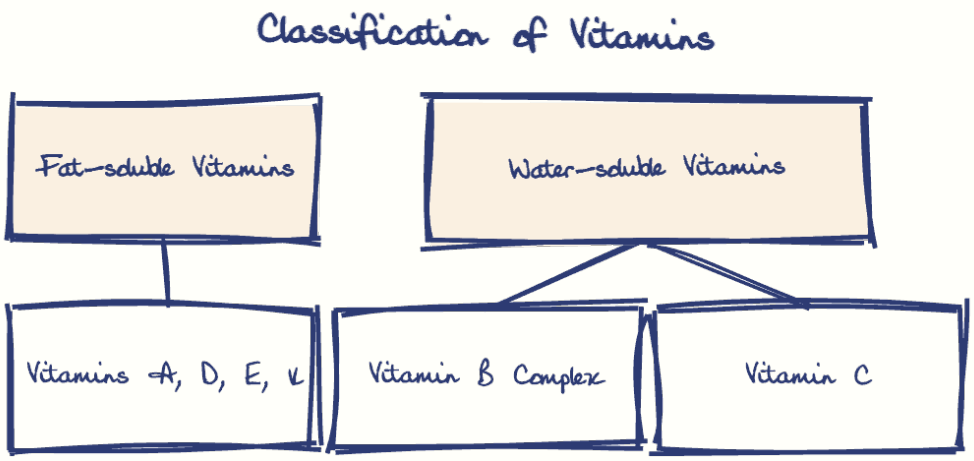
1. Essential nutrients: Organic compounds required in small amounts, acting mainly as enzyme cofactors to regulate metabolism and growth.
2. Micronutrient role: They provide no energy but help convert fats, carbohydrates and proteins into ATP.
3. Health impact: Deficiency causes specific diseases, such as night blindness from lack of vitamin A.

Classification of Vitamins

1. Fat-soluble vitamins: They dissolve in fats and are stored in the liver and fatty tissue. A supports vision and epithelial health, D

Practice 6(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice



aids calcium absorption and bone formation, E protects cell membranes as an antioxidant, and K is essential for blood clotting. Sources: liver, egg yolk, butter, green leafy vegetables, and sunlight for vitamin D.

2. Water-soluble vitamins: They are not stored much, so regular intake is needed. The B complex (B1 thiamine, B2 riboflavin, B3 niacin, B5, B6, B9 folic acid, B12 cobalamin) acts as coenzymes in energy metabolism and red blood cell formation, while C supports collagen and immunity. Sources: grains, legumes, guava and amla, and animal products for B12.

3. Deficiency diseases: B12 deficiency causes pernicious anaemia and lack of C causes scurvy, as ICMR and NIN guidelines stress.

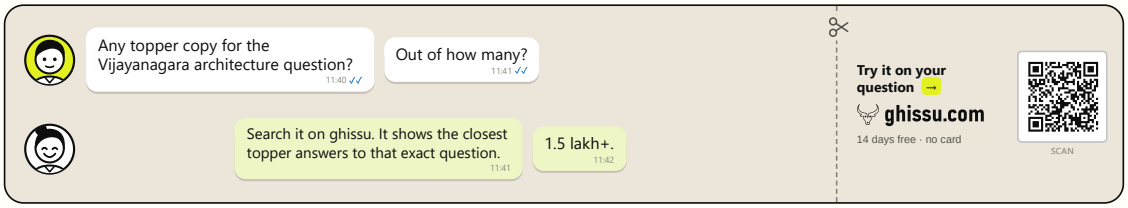
4. Regional sources: In Andhra Pradesh mango, papaya, spinach, drumstick leaves and aquaculture products add vitamins, though gaps persist in Rayalaseema.

Way Forward

1. Supplementation and fortification: Expand vitamin A and iron-folic acid supplementation, support biofortified crops through research in Andhra Pradesh and link efforts with ICDS in the state.

Conclusion

Vitamins are indispensable for health. Informed diets and targeted interventions can prevent deficiencies and improve public health outcomes.



P-IV
6(b)

What is yeast fermentation? Explain industrial use of yeast in making different products?

10

Introduction

Yeast fermentation is a cornerstone of biotechnology with ancient origins, now pivotal to modern industry and agriculture. Its importance has grown as India, including Andhra Pradesh, seeks sustainable bio-based solutions for energy, food and pharmaceuticals, so its mechanism and uses matter for the state's agro-industrial landscape.

What Is Yeast Fermentation

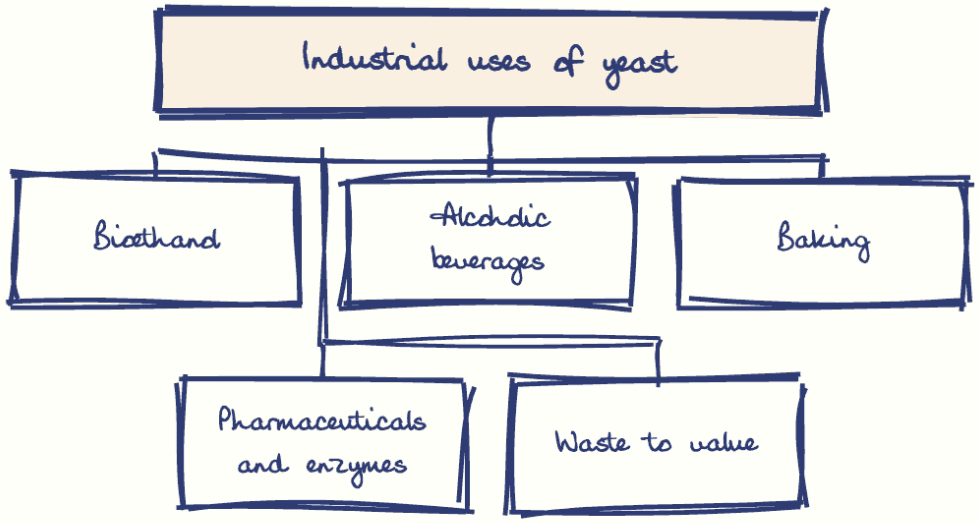
1. **Process:** Yeast, chiefly Saccharomyces cerevisiae, ferments sugars from high-sugar substrates like molasses or fruit must into ethanol.
2. **Conditions:** Precise control of temperature, about 30°C, and pH maximises ethanol yield, as pilot plants near Tirupati show.

Industrial Uses of Yeast

1. **Bioethanol:** Yeast ferments sugars from sugarcane and fruit residues into ethanol, blended with petrol under national mandates, implemented in Andhra Pradesh, to cut fossil fuel dependence.
2. **Alcoholic beverages:** Wine, beer and spirits depend

Practice 6(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice



on yeast, local wineries in Nellore district use selected strains for quality and aroma.

3. Baking industry: Baker's yeast is essential for bread and bakery products, supporting food processing in Vijayawada.
4. Pharmaceuticals and enzymes: Biotech firms in Visakhapatnam use yeast strains for industrial synthesis of antibiotics and enzymes.
5. Waste to value: Using agro-waste as substrate reduces pollution and adds value to by-products, supporting a circular bioeconomy.

Challenges

1. Substrate supply: Consistent high-sugar substrate is scarce in drought-prone Rayalaseema.

2. Scale-up cost: Moving from pilot to industrial scale is capital-intensive.
3. Quality standards: The Andhra Pradesh State Food Safety and Drug Administration makes fermented products meet benchmarks for consumer trust and exports.

Way Forward

1. Strain improvement: Molecular research in AP's universities, such as Sri Venkateswara University and biotech parks on robust strains tolerant to local substrates and stress.
2. Agro-residue units: Fermentation units near farm hubs in Rayalaseema and Coastal Andhra.
3. Skills: Train rural youth in fermentation technology.
4. Industry links: Partnerships in Visakhapatnam's industrial corridor for technology transfer.

Conclusion

Yeast fermentation is a vital biotechnological process that underpins diverse products from bioethanol, beverages and bread to pharmaceuticals, and it and offers sustainable economic opportunities for Andhra Pradesh.

150 · 250 · 500 words.
Same question, three model answers.
ghissu.com →

P-IV
6(c)

What is a phytohormone? Mention different groups of phytohormones with examples? Discuss the present scenario on commercial use of hormones in artificial fruit ripening?

10

Introduction

Phytohormones, or plant hormones, are naturally occurring organic substances that regulate plant growth and development at very low concentrations. In Andhra Pradesh, where horticulture including mango and banana is vital, they are crucial for fruit yield and postharvest management, in line with the Mission for Integrated Development of Horticulture (MIDH).

Groups of Phytohormones with Examples

1. Auxins: For example indole-3-acetic acid, produced in growing stem tips.
2. Cytokinins: Adenine derivatives synthesised mainly in roots.
3. Gibberellins: Formed through the mevalonate pathway.
4. Abscisic acid and ethylene: Abscisic acid (ABA) is produced in leaves and roots, while ethylene is a gaseous hormone synthesised in ripening fruits.
5. Interactions: Cytokinins promote stomatal opening while ABA induces closure, auxins inhibit

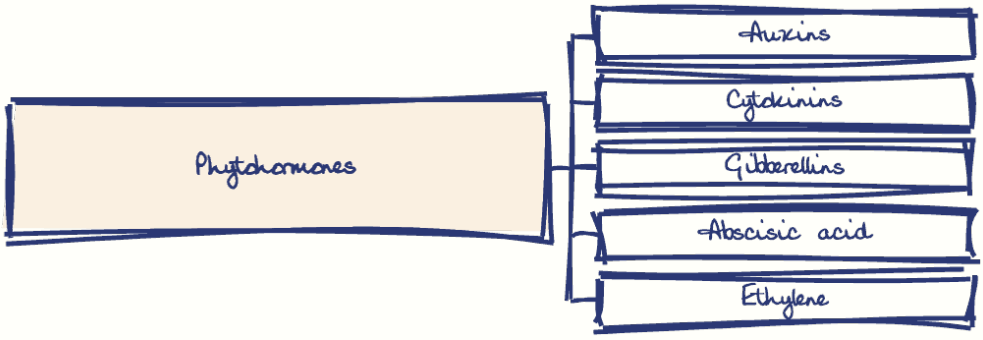
Practice 6(c) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

abscission while ethylene promotes it.

6. Other hormones: Brassinosteroids, jasmonates, salicylic acid and polyamines fine-tune ripening and defence.

Five primary phytohormones



Present Scenario of Commercial Artificial Ripening

1. Ethylene as the key hormone: It is the most commercially exploited hormone, triggering uniform post-harvest ripening in climacteric fruits like mangoes and bananas.

2. Current practices: Andhra Pradesh markets use ethylene gas or ethephon in sealed chambers, safer than hazardous calcium carbide, though illegal carbide use persists.

3. Health concerns: Risks of carbide-ripened fruit have prompted stricter enforcement and awareness campaigns in Andhra Pradesh promoting ethylene ripening.

4. Economic impact: Ripening extends shelf life, supporting small fruit merchants and farmers and horticuttural exports under the National Horticutture Mission.

5. Technological gaps: Weak ripening chamber infrastructure and limited farmer access to safe technology hold back compliance with food safety standards.

Way Forward

1. Enforcement and training: Agricutture and health departments should jointly penalise illegal ripening chemicals, while state agencies scale up training on safe ethylene use.

2. Research: Acharya N.G. Ranga Agricuttural University (ANGRAU) should extend its protocols for ripening mango and banana, so that safe ethylene methods protect consumer health and market standards.

Conclusion

Phytohormones are central to plant growth and to the fruit trade. Regulated, ethylene-based ripening can protect consumers while raising the incomes of Andhra Pradesh's fruit growers.

Correct, but average?
A number, a committee, a case — for every question.
ghissu.com →

P-IV
7(d)

Write a short note on the following: (i) Homeostatis (ii) Animal response to Osmotic conditions of the medium

10

Introduction

Homeostasis explains how organisms maintain a stable internal environment despite external fluctuations in temperature and water availability. This ability is crucial for survival and optimal functioning. In Andhra Pradesh, with diverse climatic zones and aquatic ecosystems, it is vital for managing biodiversity and the livelihoods that depend on it.

(i) Homeostasis

1. Meaning: Homeostasis is the dynamic constancy of an organism's internal environment, such as temperature and osmotic concentration. The concept, refined by Walter Cannon, underpins modern physiology.
2. Mechanisms: Physiological processes like sweating or shivering in mammals restore the internal balance when outside conditions change.
3. Energetic cost: Thermoregulation and osmoregulation are energy-intensive, so small animals and some aquatic species cannot afford them.
4. Research: The Andhra Pradesh Biodiversity Board and Sri Venkateswara University study such

Practice 7(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

adaptations in local fauna, informing conservation and fisheries policy.

(ii) Animal Response to Osmotic Conditions of the Medium

1. Regulators: Regulators actively maintain a stable internal osmotic concentration whatever the medium.
2. Conformers: Conformers let their internal conditions fluctuate with the environment, which limits their distribution and survival in extreme conditions.
3. Salinity stress: Salinity intrusion and temperature swings in coastal and estuarine zones disrupt the osmotic balance of aquatic animals, reducing fitness and harming fisheries and aquaculture.
4. Limited adaptive capacity: Many invertebrates lack mechanisms to regulate internal osmotic conditions, leaving them vulnerable to habitat change.
5. Human pressures: Urbanisation around Visakhapatnam and coastal industry alter water quality, while pollution and habitat fragmentation cut resilience to osmotic stress.

Way Forward

1. Research and adaptation: Study osmotic regulation in native species of the Pdavaram reservoir and

Response to osmotic conditions

Regulators	Conformers
Keep internal osmolarity stable	Internal state follows the medium
Active regulation, energy cost	Limited range of survival

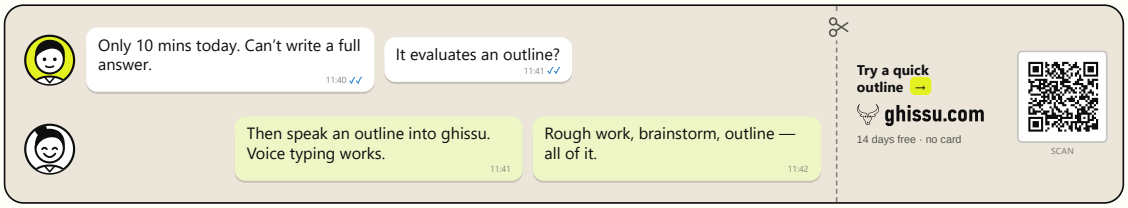
coastal wetlands, and use it to manage salinity intrusion in the Krishna delta.

2. Pollution control: Expand water quality monitoring and pollution abatement to keep osmotic conditions stable for aquatic life.

3. Awareness: Educate fishers and farmers on maintaining ecological balance for sustainable livelihoods.

Conclusion

Homeostasis and osmotic responses are central to species survival in Andhra Pradesh's diverse environments. Strengthening research and community participation supports SDG (14) and SDG (15).



P-IV
7(a)

What are essential oils? Enumerate the important uses of essential oils?

10

Introduction

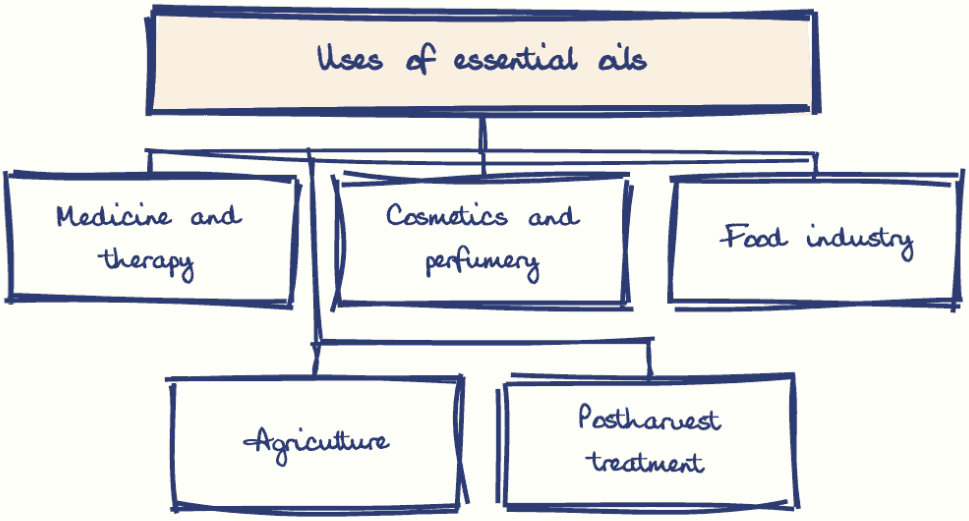
Essential oils have gained importance for their diverse uses across health, agriculture and industry. India ranks third in global production and second in the value of aromatic oil exports, and rising demand for natural products has raised their relevance further.

What Are Essential Oils

1. Nature: They are odoriferous, volatile compounds extracted from aromatic plants, mainly complex mixtures of acyclic and monoterpenoids that evaporate at ambient temperature.
2. Extraction methods: Water or steam distillation, enfleurage (cold fat), maceration (hot fat), solvent extraction, expression and supercritical CO_2 extraction.
3. India's position: India holds a 16–17% share of world production, backed by the National Aromatic and Medicinal Plant Board and CIM-AP, Lucknow.

Important Uses of Essential Oils

1. Pharmaceutical and therapeutic: Antibacterial,



antifungal, antiviral, insecticidal and antioxidant properties make them valuable in medicines, aromatherapy and natural remedies.

2. Cosmetics and perfumery: Used in perfumes, colognes, deodorants, toothpastes and skincare for fragrance and therapeutic benefit.
3. Food industry: With oleoresins, they serve as natural flavourings, preservatives and aroma enhancers.
4. Agricutture: Some act as natural insecticides and nematicides, offering eco-friendly pest control.
5. Postharvest treatment: Citrus peel extracts and other oils reduce microbial contamination and extend the shelf life of fruits and vegetables.

Challenges

1. Raw material: Lack of genuine planting material and dependence on wild collection for 90% of medicinal plants.
2. Quality: Consistent quality of oils and oleoresins needs expertise in raw material selection, handling and processing, as in spice oil production.

Way Forward

1. Sustainable cultivation: Good Agricultural Practices for aromatic plants.
2. R&D: Research on cultivation, extraction and quality control, as at the National Research Centre for Medicinal and Aromatic Plants, Anand.
3. Skills and markets: Train farmers and processors and promote exports and value addition.

Conclusion

Essential oils are vital natural products for health, cosmetics, food and farming, offering India, including Andhra Pradesh's horticulture, a chance to grow exports and rural livelihoods.

Which sub-topic comes next?
Trend analysis for every syllabus topic.
ghissu.com →

Write your own answer to 7(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
7(b)

Give an account of the following: (i) Autotrophic Vs. Heterotrophic plants (ii) Symbiosis

10

Introduction

Understanding how plants obtain nutrition and interact with other organisms is fundamental to ecosystem dynamics and farm productivity. In Andhra Pradesh's diverse agro-ecological zones, this knowledge supports climate-resilient agriculture and biodiversity conservation.

(i) Autotrophic versus Heterotrophic Plants

1. **Autotrophic plants:** Green plants and algae synthesise their own food by photosynthesis using chlorophyll, carbon dioxide and sunlight, forming the base of food chains.
2. **Heterotrophic plants:** Parasitic and insectivorous species depend on other organisms for nutrition, which complicates their ecological role and management.
3. **Key contrast:** Autotrophs capture solar energy and act as primary producers, while heterotrophs rely on organic compounds made by others.

(ii) Symbiosis

1. **Mutual benefit:** Symbiosis is an interaction that

Practice 7(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Autotrophic vs Heterotrophic Plants

Autotrophic Plants	Heterotrophic Plants
Food Source	Depend on photosynthesis
Examples	Green plants, algae
Role in Ecosystem	Primary producers
Energy Source	Solar energy

benefits both partners and the ecosystem.

2. Nitrogen fixation: Bacteria in legume root nodules convert atmospheric nitrogen into usable forms, enriching soil and reducing synthetic fertiliser use.

3. Mycorrhizae: Fungi associated with plant roots enhance nutrient uptake.

4. Soil conservation: It adds organic matter and microbial diversity, countering erosion in Rayalaseema and coastal Andhra.

5. Farm incomes: Lower fertiliser costs and better yields raise incomes, vital in drought-prone Rayalaseema.

6. Barriers: Low awareness, poor access to quality biofertilizers and rural infrastructure gaps in

Andhra Pradesh slow adoption, and efficiency varies with soil and climate.

Institutional Support

1. Research: ANGRAU studies plant nutrition and symbiosis, focusing on legumes that enrich soil nitrogen.
2. Extension: Extension services promote symbiotic nitrogen fixation alongside the Soil Health Card scheme, and M-KISAN spreads this knowledge to small farmers.

Conclusion

Autotrophs sustain food chains, while symbiosis sustains soil fertility. Local research by ANGRAU and wider extension can spread these practices for sustainable farming.

Improving — or just writing more?

My answers

Every attempt, every score, every enhanced version of your answer — on one timeline.

Watch your curve →

ghissu.com

14 days free · no card

SCAN

P-IV
7(c)

(i) What are Lichens? Write their economic importance. (ii) Explain how insects are useful to human beings.

10

Introduction

With growing environmental awareness, symbiotic organisms like lichens and insects have become crucial to sustainable resource use. Lichens act as bioindicators and resource contributors, while insects serve as pollinators and biological control agents. Their value is especially relevant for Andhra Pradesh, where agriculture and forestry are key livelihoods.

(i) Lichens and Their Economic Importance

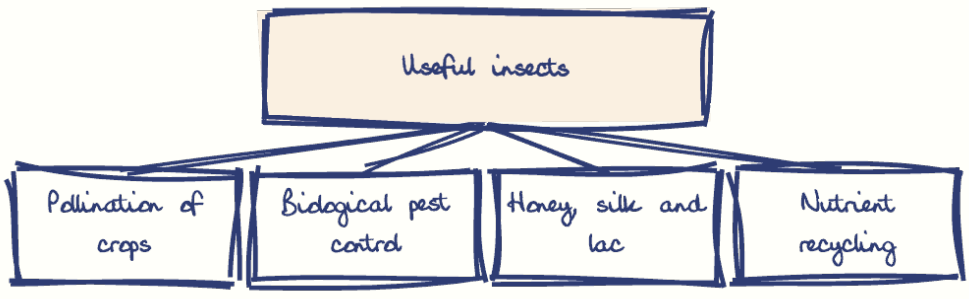
1. **Definition:** Lichens are composite organisms formed by a mutualistic association between a fungus and a photosynthetic partner, usually an alga or cyanobacterium.
2. **Soil formation:** This partnership lets them colonise harsh habitats, aiding soil formation and nutrient cycling, as in the Eastern Ghats.
3. **Commercial products:** They are harvested for dyes, perfumes and traditional medicines, providing income to local communities.
4. **Bioindicators:** Their sensitivity to air pollution makes them valuable bioindicators for environmental monitoring in industrial corridors.

5. Soil fertility: Cyanobacterial lichens fix nitrogen, reducing the need for chemical fertilisers, in line with the Andhra Pradesh State Organic Farming Policy.

(ii) How Insects Are Useful to Humans

1. Pollination services: Bees, butterflies and wasps pollinate crops such as mangoes and chillies in Rayalaseema and Coastal Andhra, supporting food security and farm incomes.
2. Biological pest control: Predators like ladybird beetles and parasitic wasps regulate pests, reducing reliance on chemical pesticides.
3. Economic products: Insects give honey, silk and lac. Sericulture in Anantapur district provides rural employment and export revenue.
4. Nutrient recycling: Detritivorous insects break down organic matter, maintaining soil health in drought-prone Rayalaseema.

How insects are useful

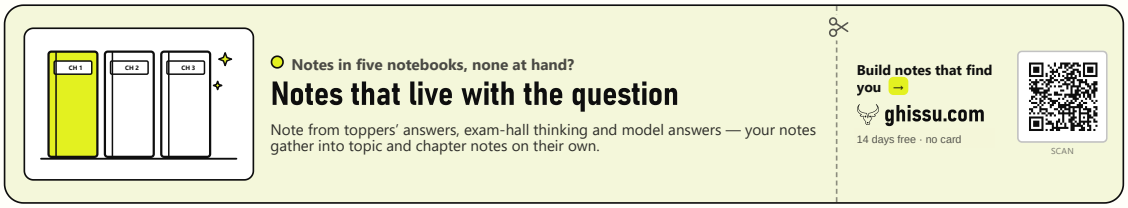


Challenges and Way Forward

1. Threats: Pollution and habitat loss endanger lichens and beneficial insects in industrial and agricultural zones.
2. Low awareness: Limited farmer awareness of biological pest control and lichen-based products restricts their wider economic potential.
3. Research gaps: Lichen-based biofertilisers and insect-based pest management tailored to local agro-climatic conditions need focused R&D.
4. Remedies: Promote integrated pest management training, community-based lichen conservation in forest fringes, incentives for sericulture and apiculture, and research through Acharya N.G. Ranga Agricultural University.

Conclusion

Lichens and insects are indispensable to ecological balance and economic sustainability. Conserving them and using their services wisely can support Andhra Pradesh's farmers and environment.



P-IV
8(d)

What is DNA Finger Printing? Explain how it is done. Write its forensic applications.

10

Introduction

DNA fingerprinting has emerged as a revolutionary tool in forensic science, enabling precise identification of individuals from their genetic material. As crime grows more complex, it has become crucial in criminal investigations, paternity disputes and disaster victim identification.

What Is DNA Fingerprinting

It is the technique of producing a unique pattern from regions of DNA that vary between individuals, so that two samples can be matched to the same person.

How It Is Done

1. **DNA Extraction:** DNA is extracted from biological samples such as blood, hair roots, saliva or skin cells from crime scenes or individuals, ensuring the material is uncontaminated and sufficient.
2. **Restriction Enzyme Digestion:** Restriction endonucleases cut DNA into fragments at known sequences, targeting variable regions like

Practice 8(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Variable Number Tandem Repeats (VNTRs) and Short Tandem Repeats (STRs).

3. Gel Electrophoresis: The fragments are separated by size, producing a unique banding pattern that can be visualised and compared.

4. Comparison and Statistics: Patterns from the crime scene are compared with suspects or reference samples, and probabilities are calculated; current methods reduce coincidental matches to near zero.

Steps in DNA fingerprinting

Forensic Applications

1. Criminal Investigations: Used extensively in Andhra Pradesh, it identifies perpetrators of rape, murder and other violent crimes, leading to convictions or exonerations.

2. Disaster Victim Identification: In mass casualty events, it identifies victims when conventional

methods fail, giving families in Andhra Pradesh closure and legal documents.

3. Paternity Disputes: Comparing a child's DNA with alleged parents resolves disputed parentage, a service increasingly sought in Andhra Pradesh's legal and social welfare systems.
4. Missing Persons: It traces missing individuals and identifies unidentified bodies, important in regions like Rayalaseema with migration and displacement.

Safeguards

Samples are collected in the presence of witnesses, including police and community representatives, to maintain the chain of custody and prevent tampering, as Andhra Pradesh's forensic procedures require. The state's labs work with national institutions like the Central Forensic Science Laboratory in Hyderabad.

Conclusion

More DNA labs and mobile units, trained forensic staff and interoperable platforms like ICJS will make DNA evidence faster and more reliable in Andhra Pradesh.

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 8(d) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
8(a)

What is plant tissue culture? Write applications of tissue culture in agriculture?

10

Introduction

Plant tissue culture has emerged as a pivotal biotechnology in modern agriculture, enabling rapid and disease-free propagation. With pest infestations, slow multiplication of certain crops and demand for quality planting material, its relevance is underscored by initiatives like the Clean Plant Programme.

What Is Plant Tissue Culture

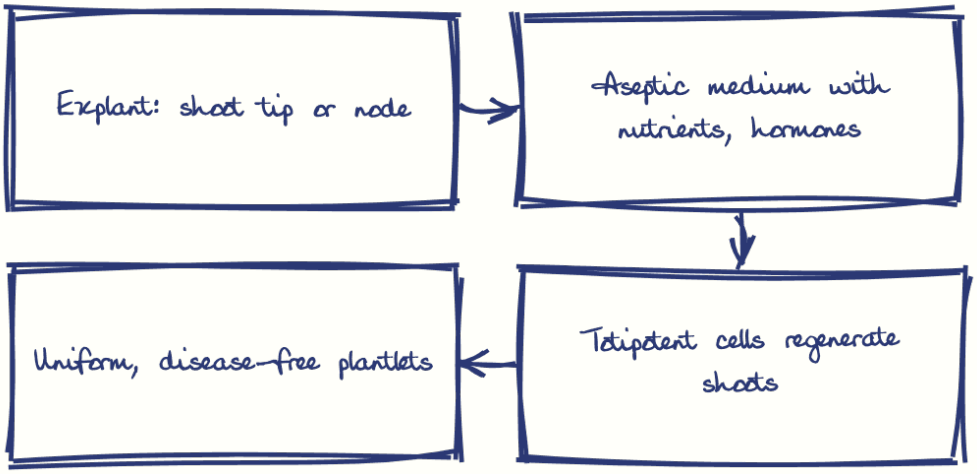
1. Definition: It is the aseptic cultivation of plant cells, tissues or organs in a controlled medium of nutrients, hormones and sterility, allowing growth and differentiation outside the parent plant.
2. Principle: It exploits totipotency, the ability of a single plant cell to regenerate into a whole plant under suitable conditions.
3. Micropropagation: Plants are multiplied from small explants such as leaves or nodal cuttings, speeding clonal propagation.

Applications in Agriculture

1. Disease-Free Planting Material: Meristem culture

Practice 8(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V
1 / 3
Scan to practice

Micropropagation in tissue culture



isolates tiny pathogen-free shoot tips to give virus-free plants, vital for sugarcane, potato and banana where clonal propagation builds up disease.

2. Rapid Multiplication of Elite Varieties: Micropropagation multiplies superior cultivars quickly with genetic uniformity, helping Andhra Pradesh's horticulture sector meet demand.

3. Germplasm Conservation and Breeding: It preserves genetic resources and enables doubled haploid production and genetic transformation for climate-resilient varieties such as drought-tolerant chickpea.

4. Pharmaceutical Compounds: Beyond propagation, it produces secondary metabolites of pharmaceutical value, diversifying farm output

and adding value.

Limitations

Tissue culture plants cost more than conventional ones, and protocols for crops like mango and coconut are not yet standardised, confining use to high-value crops. Infrastructure gaps in regions like Andhra Pradesh add further hurdles.

Way Forward

1. Clean Plant Centres: Set up more centres in Andhra Pradesh to supply affordable, disease-free planting material.
2. Partnerships and Training: Public-private partnerships to cut costs, standard protocols for Andhra Pradesh crops like mango and coconut, and extension programmes in the state to train farmers.

Conclusion

Plant tissue culture offers solutions for disease management, rapid propagation and genetic improvement. Backed by the Clean Plant Programme and regional research, it can become a driver of productivity and of SDG (2): Zero Hunger.

Wrote the answer. Now what?
Upload it — every line marked, in minutes.
ghissu.com →

P-IV
8(b)

What is 'Golden Rice'? Describe its advantages?

10

Introduction

Vitamin A deficiency remains a major public health problem in many developing regions, including parts of India, causing blindness and higher mortality among children. Golden Rice is a breakthrough in biofortification that tackles this through a staple crop.

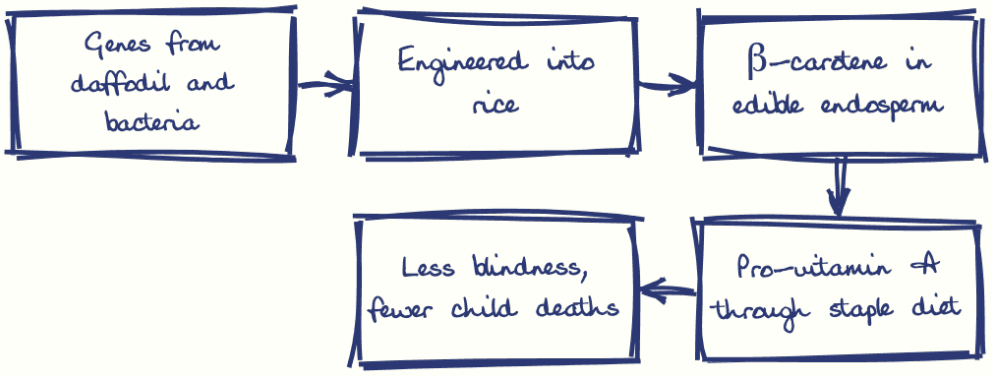
What Is Golden Rice

1. **Definition:** It is a genetically modified rice engineered to produce pro-vitamin A (β-carotene) in its edible endosperm.
2. **Development:** Professor Ingo Potrykus of the Swiss Federal Institute of Technology and Professor Peter Beyer of the University of Freiburg combined genes from daffodil and bacteria, a first in metabolic engineering of staple crops.
3. **Oversight:** Funded by the Rockefeller Foundation, the Swiss government and the EU, it was tested by IRRI under a Humanitarian Board.

Practice 8(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

How Golden Rice works



Advantages of Golden Rice

1. Tackles Vitamin A Deficiency: It provides bioavailable pro-vitamin A directly through a widely eaten staple.
2. Prevents Blindness and Deaths: It improves child health outcomes and reduces healthcare burdens in vulnerable populations.
3. Suits Rice-Dependent Diets: It helps where rice is the staple and dietary diversity is limited, such as the agrarian belts of Andhra Pradesh.
4. Complements Existing Programmes: It adds to supplementation and food-based approaches rather than replacing them.
5. Humanitarian Access: Over (70) patents held by about (40) organisations were cleared through free licences under a freedom to operate clause.

Challenges

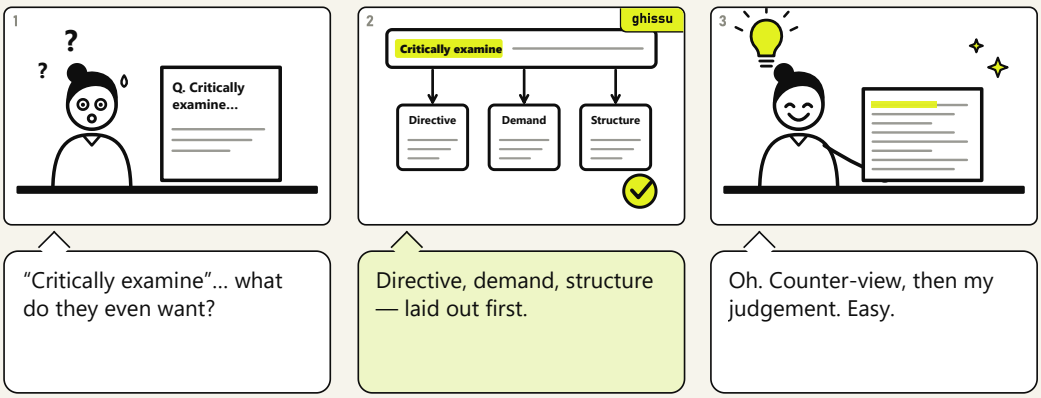
The first generation's 1.6–2.0 micrograms of β-carotene per gram was below children's daily needs, prompting improved varieties. Stringent biosafety rules, public scepticism about GMOs and complex IPR have delayed its reach to farmers and consumers in India, including Andhra Pradesh.

Conclusion

Streamlined approvals, awareness campaigns among Andhra Pradesh's farmers and consumers, and R&D for higher β-carotene can make Golden Rice an effective biofortification tool for nutritional security.

A STORY YOU MAY KNOW EXAM-HALL THINKING

Three minutes staring at one word



— Walk into the hall with a method, not a memory.

Walk in with a method →
ghissu.com
14 days free · no card
SCAN

P-IV
8(c)

What are Bio fertilizers? Explain different types of bio fertilizers? Write advantages of bio fertilizers?

10

Introduction

With rising costs and environmental harm from chemical fertilizers, biofertilizers have gained prominence as a sustainable alternative. The push for organic farming and soil health, including the Soil Health Management Scheme, makes them highly relevant in Andhra Pradesh, where agriculture is the backbone of the economy.

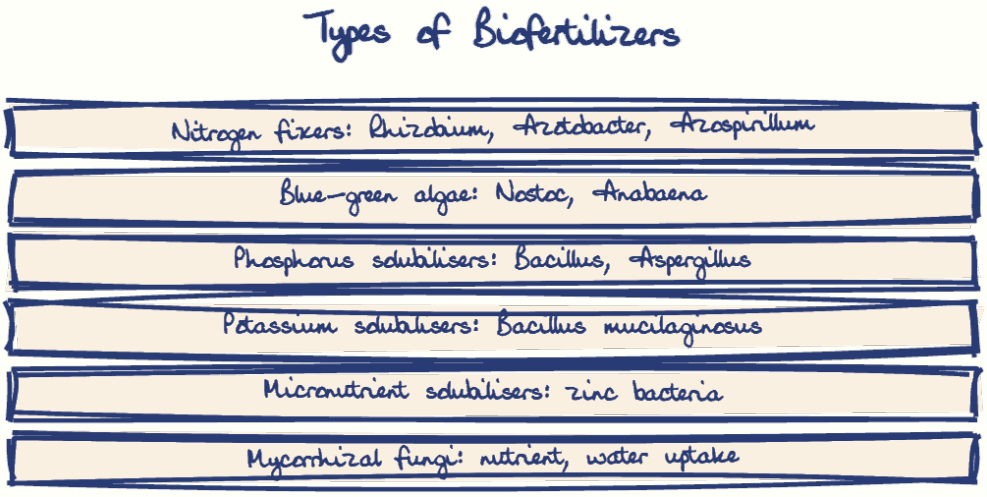
What Are Biofertilizers

1. Definition: They are living microbial inoculants of beneficial bacteria, fungi or algae that enrich soil by fixing atmospheric nitrogen, solubilising phosphorus or mobilising other nutrients.
2. Mechanism: Applied to seeds or soil, the microbes multiply around roots and convert unavailable nutrients into usable forms.
3. Not Organic Fertilizers: Organic fertilizers are decomposed matter; biofertilizers act through live microbes.
4. State Relevance: Andhra Pradesh's agricultural research institutions promote them to raise rice and pulse yields.

Practice 8(c) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Types of Biofertilizers



1. Nitrogen Fixers: Rhizobium forms root nodules in legumes, while Azotobacter and Azospirillum fix nitrogen non-symbiotically for cereals.
2. Phosphorus Solubilisers: They convert insoluble phosphorus into soluble forms for root growth.
3. Potassium and Micronutrient Solubilisers: Bacteria free locked potassium and zinc.
4. Mycorrhizal Fungi: These symbiotic fungi enhance nutrient uptake, especially phosphorus, and improve resistance to environmental stresses.

Advantages

1. Environmental Sustainability: They reduce chemical use, soil and water pollution and greenhouse gas emissions.
2. Soil Health: They improve microbial diversity

Candidates must not write on this margin

Candidates must not write on this margin

structure and moisture retention, vital for drought-prone Rayalaseema.

3. Cost-Effective: Local production needs lower capital than chemical fertilizers, cutting input costs and raising profits for small and marginal farmers.

4. Better Yields: Improved nutrient uptake raises yield and produce quality, as in Andhra Pradesh's pulses and rice.

Conclusion

Biofertilizers combine ecology with economics. Wider adoption, backed by soil health schemes and farmer awareness, offers an eco-friendly path to soil fertility, lower chemical dependence and farm profitability in Andhra Pradesh.

1.5 lakh+

Borrow the approach, not the answer.

Topper answers matched by meaning to your question — with how each one opened, structured and concluded.

See how toppers wrote it → ghissu.com

14 days free · no card

SCAN

P-IV
9(d)

Explain the preventive measures to be taken to avoid common infections.

10

Introduction

Preventing common infections is crucial in densely populated regions like Andhra Pradesh, where environmental and social factors accelerate spread. COVID-19 underscored proactive measures such as vaccination and hygiene. Prevention protects individuals, curbs community transmission and preserves economic productivity.

Individual and Community Measures

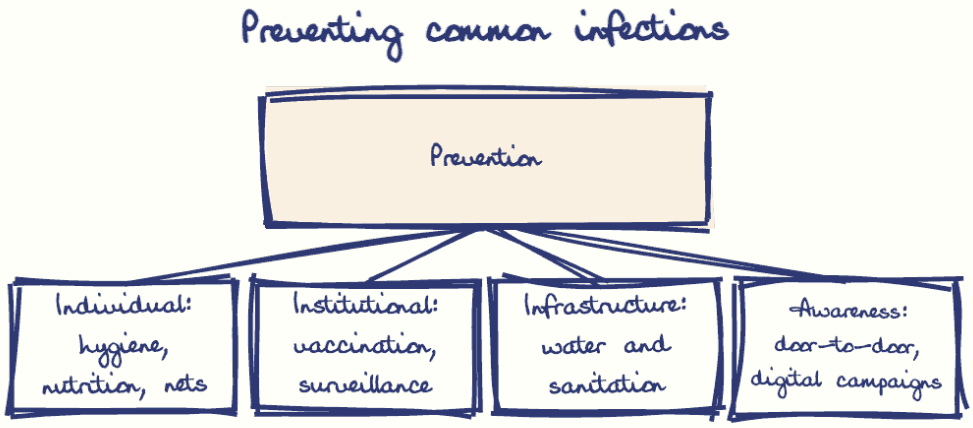
1. Personal hygiene: Regular handwashing, use of masks and clean surroundings reduce exposure to airborne and contact-transmitted pathogens.
2. Nutrition and immunity: Adequate nutrition strengthens immunity. Andhra Pradesh's welfare schemes aim to improve food security for the vulnerable.
3. Vector avoidance: Mosquito nets, window screens and elimination of breeding sites prevent dengue and malaria in coastal and rural Andhra Pradesh.
4. Early care: Consulting early for cough or fever enables timely diagnosis and reduces spread.

Practice 9(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Institutional Measures

1. Vaccination: Mission Intensified Indradhanush (4.0), implemented in Andhra Pradesh, has raised coverage against polio, diphtheria and tetanus.
2. Surveillance: The Integrated Disease Surveillance Programme enables early detection and timely isolation and containment.
3. Water and sanitation: Investment in safe drinking water, sanitation and waste management reduces contamination and mosquito breeding.
4. Health education: Digital media and door-to-door outreach promote hygiene, masks and social distancing.



Challenges

1. Rural gaps: Remote parts of Rayalaseema and other regions of Andhra Pradesh lack clean water and

sanitation, which sustains transmission cycles.

2. Vaccine hesitancy: Cultural mistrust and misinformation, as seen with HPV vaccines nationally, also reduce uptake in Andhra Pradesh and need tailored communication.

3. Vectors and capacity: Stagnant water in the Krishna and Godavari basins sustains mosquitoes, and COVID-19 exposed staff and facility gaps in Visakhapatnam and other districts.

Way Forward

1. Rural sanitation: Accelerate water purification and sanitation projects in underserved Andhra Pradesh districts.

2. Vaccine outreach: Run culturally sensitive campaigns to tackle vaccine hesitancy.

3. Vectors and workforce: Join community action with government spraying along river basins, and train health workers in preventive care.

Conclusion

Prevention combines institutional readiness, community engagement and individual responsibility to break transmission chains, advancing SDG (3) on good health and well-being.

Write your own answer to 9(d) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
9(a)

Explain different patterns of disease outbreak with examples? Differentiate between epidemic and endemic diseases?

10

Introduction

Understanding patterns of disease outbreak is crucial for public health planning in Andhra Pradesh, with its diverse ecological and demographic profile. Outbreaks vary in occurrence, spread and persistence, and distinguishing epidemic from endemic disease helps tailor interventions.

Patterns of Disease Outbreak

1. Epidemic pattern: A sudden increase in cases above the expected baseline in an area, rising and falling over weeks or months, as in the Nipah outbreak in Kerala.
2. Pandemic pattern: The COVID-19 pandemic of 2020-21 showed exponential spread of a novel pathogen with high reproductive number (R0).
3. Endemic pattern: Disease stays at a baseline level, as in the seasonal endemicity of malaria and dengue in Andhra Pradesh. Endemics can flare into epidemics.
4. Temporal and spatial patterns: Epidemics show sudden spikes and clustering; endemics show stable, persistent patterns, like vector-borne

Practice 9(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

disease in coastal and Rayalaseema regions linked to monsoon cycles.

Epidemic versus Endemic

Epidemic vs Endemic Diseases

Epidemic Diseases	Endemic Diseases
Sudden surge in cases	Constant baseline presence
Short duration, outbreak phase	Long-term persistence
Often novel or re-emerging pathogens	Established pathogens
Requires emergency response	Requires ongoing control measures
Spatial clustering common	Geographically widespread or localized

1. Flare-ups: Endemic diseases maintain a steady presence but can flare into epidemics when conditions are conducive, so the two patterns are linked.
2. Transmission: Epidemics involve new pathogens or strains with high R0, while endemics have stable transmission, though drug resistance can upset it, as in tuberculosis.

Management in Andhra Pradesh

1. Surveillance: The Integrated Disease Surveillance Programme and IHIP give real-time data, and

in Andhra Pradesh they were strengthened after COVID-19 for early detection of outbreaks such as dengue.

2. Legal basis: The Epidemic Diseases Act, 1897 and the Disaster Management Act, 2005 provide the legal basis, which Andhra Pradesh's health department implements to suit district-level needs.

3. Gaps: Weak data from private providers and rural areas delays detection, and during the COVID-19 waves Andhra Pradesh struggled to balance PPE, isolation beds and vaccines between epidemic hotspots and endemic disease control.

Conclusion

Epidemics demand rapid mobilisation, while endemics need sustained control. Stronger surveillance and trained rapid response teams can stop endemic disease from turning epidemic.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 9(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

P-IV
9(b)

What is the cause of Chikungunya? What are its preventive measures and treatment? Give a comparative note on symptomology of Chikungunya and Dengue?

10

Introduction

Vector-borne diseases like Chikungunya and Dengue pose major public health challenges in India, especially in the monsoon when mosquito breeding intensifies. Their rise in tropical states such as Andhra Pradesh makes understanding their cause, prevention and treatment urgent.

Cause of Chikungunya

1. **Viral Origin:** It is caused by the Chikungunya virus, an alphavirus transmitted by Aedes aegypti and Aedes albopictus mosquitoes, the same vectors that spread Dengue, which breed in stagnant water in urban and semi-urban Andhra Pradesh in the rainy season.
2. **Climatic Factors:** Rising temperatures and erratic rainfall extend the breeding season, while dense cities like Visakhapatnam aid spread.
3. **Outbreak Trends:** Andhra Pradesh reports periodic outbreaks, often alongside Dengue, tracked under IDSP. Insecticide resistance in mosquitoes and low public awareness hinder

contrd of the transmission cycle in the state.

Prevention and Treatment

1. Vector Contrd: Eliminate breeding sites and run community awareness campaigns in Andhra Pradesh's urban localities.
2. Personal Protection: Long-sleeved clothing, repellents and insecticide-treated nets reduce bites, as state public health drives stress in peak seasons.
3. Symptomatic Treatment: There is no specific antiviral, analgesics and antipyretics relieve fever and joint pain, with hydration and rest.
4. Diagnosis: In Andhra Pradesh's health system, overlap with Dengue demands better diagnostic facilities and trained health workers.

Comparative Symptomology

Chikungunya vs Dengue

Chikungunya	Dengue
Sudden high fever, 2–7 days	Biphasic fever pattern
Severe, lasting polyarthritis	Mild to moderate joint pain
Bleeding rare, platelets near normal	Marked thrombocytopenia, haemorrhage risk
Lasts weeks to months	Resolves in 1–2 weeks

Candidates must not write on this margin

Candidates must not write on this margin

1. Fever: Chikungunya brings sudden high fever for 2–7 days; Dengue fever often follows a biphasic pattern.
2. Joint Pain: Chikungunya causes severe polyarthritis that persists for weeks to months; Dengue causes mild to moderate joint and muscle pain.
3. Rash: A maculopapular rash is common in Chikungunya; Dengue's rash resembles measles, sometimes with petechiae.
4. Bleeding and Platelets: Haemorrhage is rare and platelets near normal in Chikungunya, while Dengue shows marked thrombocytopenia and can turn into hemorrhagic fever and shock.
5. Duration: Chikungunya symptoms can last weeks to months because of joint pain, whereas Dengue usually resolves in 1–2 weeks.

Conclusion

Since fever and rash overlap, laboratory confirmation is vital, though it is hard in resource-limited settings. Integrated vector control, surveillance and awareness can curb both diseases in Andhra Pradesh.

Correct, but average?
A number, a committee, a case — for every question.
ghissu.com →

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 9(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
9(c)

Write a brief note on aetiology, mode of spread, symptoms, treatment and preventive measures of malarial parasite?

10

Introduction

Malaria remains a critical public health challenge in tropical regions like Andhra Pradesh, where climatic conditions favour mosquito breeding. Despite a global rise in 2023, India has made remarkable progress, significantly reducing cases and deaths between 2017 and 2023.

Aetiology

1. Parasite: Malaria is caused by Plasmodium parasites, mainly P. falciparum and P. vivax in India.
2. Life cycle: The cycle involves the human liver and red blood cells and stages in the mosquito, offering several points for intervention.

Mode of Spread

1. Vector: The female Anopheles mosquito transmits the parasite through its bite.
2. Environmental drivers: Climate variability, deforestation and water stagnation in coastal Andhra Pradesh increase breeding sites, while poverty and poor access sustain transmission.

Symptoms

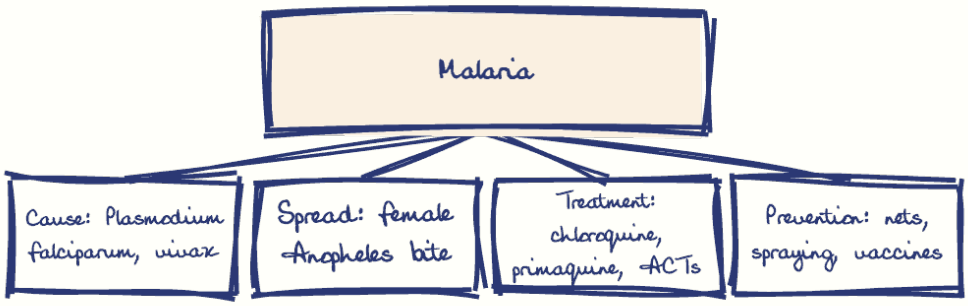
1. Clinical signs: Fever, chills, headache and fatigue. They overlap with other febrile illnesses, which delays diagnosis.

Treatment

1. Standard drugs: Chloroquine and primaquine remain standard.

2. Resistance: Drug-resistant strains, especially of P. falciparum, need artemisinin-based combination therapies (ACTs) and continuous monitoring.

Malaria at a glance



Preventive Measures

1. Vector control: Insecticide-treated nets, indoor residual spraying and environmental management, with integrated vector management against insecticide resistance.

2. Vaccines: WHO-recommended RTS,S/AS01 and

R21/Matrix-M can cut incidence in high-risk groups.

3. Surveillance: The Test, Treat, Track strategy and IMEP-3 strengthen diagnosis and treatment, including in Andhra Pradesh.

4. Community education: Awareness of symptoms and treatment adherence reduces transmission.

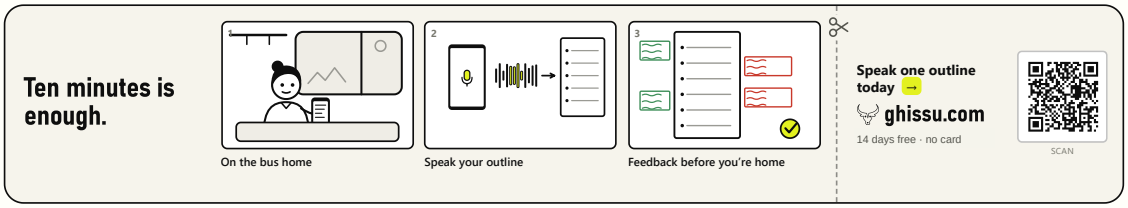
Way Forward

1. Digital surveillance: Real-time data tools can improve outbreak prediction in malaria-endemic zones of Andhra Pradesh.

2. Research: Develop new antimalarial drugs and insecticides and monitor resistance.

Conclusion

Malaria's aetiology, spread, treatment and prevention are interlinked. With the National Strategic Plan (2023-27) and India's exit from the WHO High Burden to High Impact group, elimination by 2030 is within reach.



P-IV
10(d)

Biotechnological intervention has transformed the vaccination technology' – Discuss.

10

Introduction

Biotechnological interventions have revolutionised vaccination technology as the race for vaccines during COVID-19 showed. India's role as the pharmacy of the world, producing over 60% of global vaccines, shows how biotechnology widens access and strengthens public health.

Nature of the Transformation

1. Novel vaccine platforms: DNA vaccines (ZyCoV-D), protein subunit vaccines (CORBEVAX) and mRNA vaccines (GEMCOVAC-19) elicit immunity more precisely and safely, a paradigm shift from live-attenuated or inactivated vaccines.
2. Clinical trial networks: 54 trial sites across India, including institutions in Andhra Pradesh, with GCP compliance and electronic data capture allow rapid, robust testing.
3. Genomic surveillance: Bioinformatics enables pathogen and host sequencing, and the One Day One Genome project of the Department of Biotechnology maps genetic diversity, helping design vaccines for diverse populations, including those in

Andhra Pradesh.

4. Policy frameworks: New biotechnology policies like Bio-E3 and Bio-RIDE promote translational research and commercialisation of vaccine technologies.

How biotechnology changed vaccines

Conventional	Biotech platforms
Live-attenuated or inactivated pathogen	DNA, subunit, mRNA vaccines
Development took years	Development in months
Injected, cold-chain dependent	Thermostable, intranasal options

Impact on Development and Delivery

1. Efficacy and safety: Biotechnology has improved immunogenicity and safety, as Bharat Biotech's Covaxin showed by passing rigorous trials and winning WHO emergency use listing, which matters for India's large and genetically diverse population.
2. Faster timelines: Recombinant DNA and mRNA platforms cut development from years to months, enabling indigenous COVID-19 vaccines.
3. Easier delivery: Thermostable and intranasal vaccines like iNCOVACC simplify storage in rural and

remote areas of Andhra Pradesh, complementing the Universal Immunisation Programme that reaches 2.9 crore pregnant women and 2.54 crore newborns a year.

Remaining Gaps

1. Regulation and supply: Regulatory delays, fragmented supply chains and limited animal testing facilities slowed the pandemic rollout.
2. Weak PPPs: Links between research and bulk manufacturing remain unstructured, and data-sharing agreements are missing.

Way Forward

1. PPP framework: A dedicated framework can connect translational research with manufacturing and technology transfer, using Andhra Pradesh's biotech hubs.
2. Regulatory agility: Harmonised emergency use protocols can speed approvals without compromising safety.

Conclusion

Biotechnology has made vaccines faster, safer and more accessible. Policies like Bio-E3 and Bio-RIDE, stronger partnerships and agile regulation can complete this transformation.

Write your own answer to 10(d) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
10(a)

What is immunisation? Briefly write the history of development of vaccines?

10

Introduction

Immunisation is a cornerstone of public health, as the global fight against COVID-19 showed. Vaccines have saved millions of lives, and in India, including Andhra Pradesh, the Universal Immunisation Programme has reduced the disease burden.

What is Immunisation

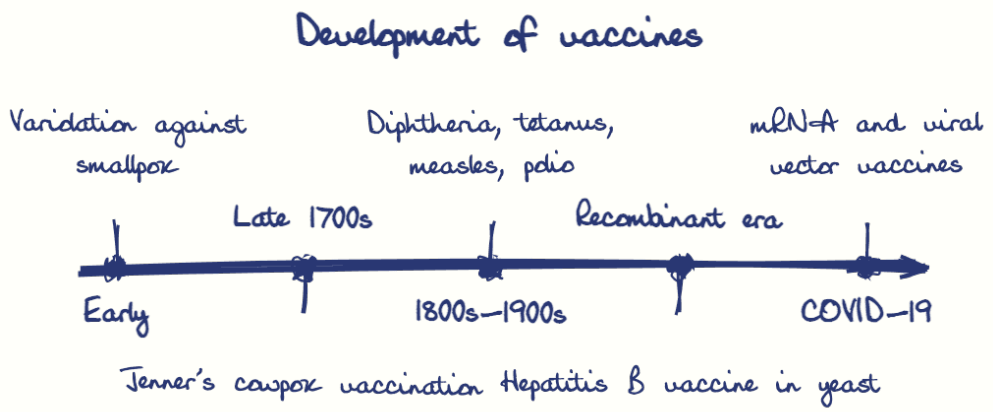
1. Definition: Immunisation makes a person immune to an infectious disease, usually by a vaccine that stimulates the adaptive immune system to fight the pathogen on future exposure.
2. Immune memory: The immune system remembers a pathogen and mounts a faster, stronger response later, the scientific basis of immunisation.
3. Active and passive immunity: Vaccines induce active immunity using weakened or inactivated pathogens; passive immunity, through colostrum or placenta, is immediate but temporary.
4. Herd immunity: It also reduces community transmission; in Andhra Pradesh, UIP has raised coverage, protecting children and pregnant women from polio, measles and tetanus.

History of Development of Vaccines

1. Variolation: Traditional practice deliberately exposed people to smallpox material to induce immunity, one of the earliest practices.
2. Jenner's breakthrough: Edward Jenner found in the late 18th century that cowpox protected against smallpox, 'vaccination' comes from 'vaccinia'. This marked the birth of modern vaccination.
3. Expanding range: In the 19th and 20th centuries, vaccines for diphtheria, tetanus, whooping cough, measles and polio shaped childhood schedules.
4. Recombinant DNA: The hepatitis B vaccine made in yeast cells allowed safer large-scale production.
5. Platform technologies: mRNA and viral vector vaccines enabled the rapid creation of COVID-19 vaccines.
6. India's initiatives: UIP and the free HPV vaccination against cervical cancer widen access, and Andhra Pradesh has actively implemented them.

Way Forward

1. Local R&D: Support institutions in Andhra



Pradesh to develop indigenous vaccines and reduce import dependence.

2. Outreach: Better infrastructure and awareness in rural Andhra Pradesh can tackle hesitancy and reach universal coverage.
3. Surveillance: Robust adverse event monitoring keeps vaccines safe and effective.

Conclusion

Immunisation, rooted in centuries of scientific progress, remains vital for disease prevention. Expanding outreach and surveillance will reduce the infectious disease burden and advance the Sustainable Development Goals.

STUDY TIP · BEFORE YOU OPEN A CHAPTER
List what was asked from it in the last ten years. Read that first; the rest second.

ghissu
ghissu has already done this for every topic — with the trend drawn out.

See the trend for your chapter →
ghissu.com
14 days free · no card
SCAN

P-IV
10(b)

Write a short note on: (i) Edible vaccines (ii) ELISA test (iii) Autoimmune diseases

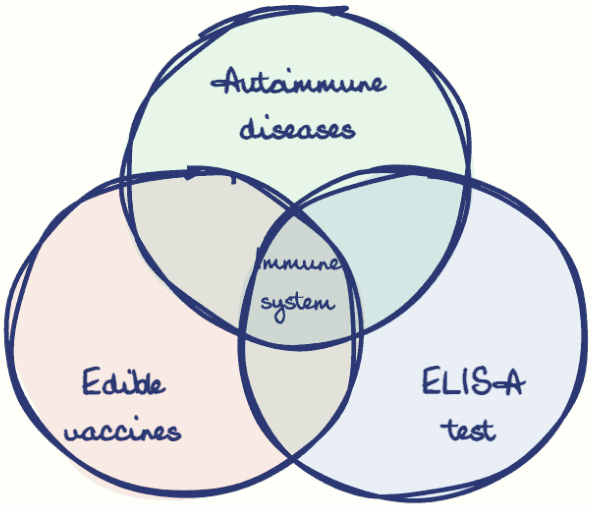
10

Introduction

Advances in biotechnology and immunology have transformed healthcare. Edible vaccines, ELISA testing and autoimmune diseases are increasingly relevant in India and Andhra Pradesh, where public health and surveillance remain priorities under the Universal Immunisation Programme.

(i) Edible Vaccines

Three tools of immunology



1. Concept: Genetically modified plants produce antigenic proteins, offering a cost-effective, easily distributed alternative to traditional vaccines. Department of Biotechnology-linked institutions in Andhra Pradesh are exploring it.

2. Merits: Easier storage and administration, grown in staple crops, they suit local food habits and could raise vaccine uptake in rural areas.

3. Concerns: Consistent dosage, regulatory approval, public acceptance and ethical worries about GMOs.

(ii) ELISA Test

1. Concept: The Enzyme-Linked Immunosorbent Assay detects antibodies or antigens in blood samples.

2. Uses: It diagnoses infections and autoimmune disorders and tracks disease prevalence and vaccine efficacy; antibody tests guided immunity assessment during COVID-19, and ICMR's regional centres in Andhra Pradesh support such testing.

3. Strengths and limits: It is cheap and sensitive but needs laboratories and trained staff; district hospitals in Andhra Pradesh now offer it, though rural reach is limited.

(iii) Autoimmune Diseases

1. Meaning: The immune system attacks the body's own tissues, so specialised healthcare services are required.

2. Diagnosis gap: Complex symptoms delay diagnosis, worsened in Andhra Pradesh by limited

awareness and few specialists.

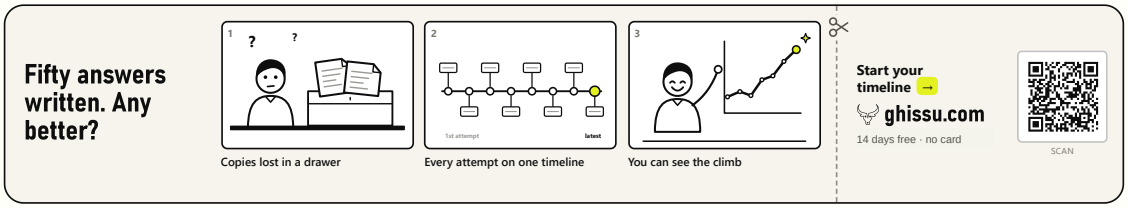
3. Social impact: Chronic disability reduces quality of life and productivity.
4. Response: National Health Mission screening and treatment protocols improve outcomes, while awareness campaigns and support groups reduce stigma.

Way Forward

1. R&D: Fund edible vaccines for local disease burdens.
2. Testing: Expand laboratory networks and train health workers to take ELISA to rural and underserved areas.
3. Integrated care: Bring autoimmune screening and management into existing public health programmes for early diagnosis.

Conclusion

These are critical frontiers of healthcare. Indigenous research, wider diagnostics, ethical guidelines and public awareness will build a resilient health system in line with SDG (3).



P-IV
10(c)

Give a detailed note on 'Triple Vaccination'.

10

Introduction

Triple vaccination refers to the triple antigen (DPT) vaccine, a single combined vaccine protecting children against diphtheria, pertussis and tetanus. It has been a pillar of India's Universal Immunisation Programme since 1985, and DPT3 coverage is the main indicator of how well immunisation reaches children, including in Andhra Pradesh.

Diseases Covered

1. Diphtheria: The toxin of Corynebacterium diphtheriae forms a grey throat membrane that can block breathing and damage the heart.
2. Pertussis: Bordetella pertussis causes violent whooping cough, fatal in infants through pneumonia.
3. Tetanus: Clostridium tetani toxin from wounds or an unclean cord causes spasms and lockjaw.

Composition

1. Toxoids: Diphtheria and tetanus toxins inactivated with formaldehyde raise antibodies

Practice 10(c) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

without toxicity.

2. Pertussis component: Killed whole-cell bacteria (DTwP, used in India) or acellular antigens (DTaP).

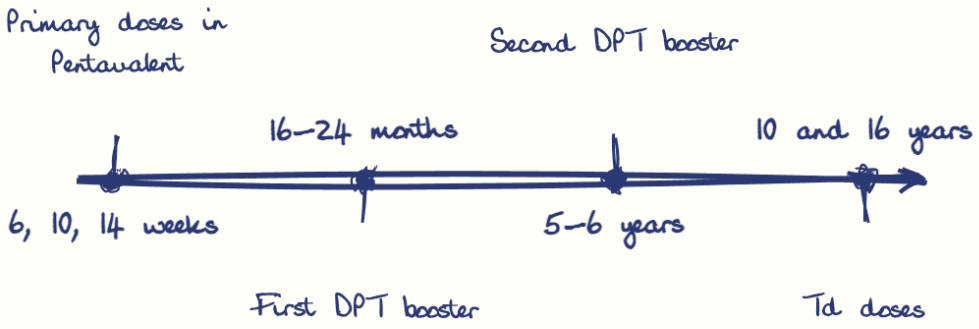
3. Adjuvant: An aluminium salt strengthens the response.

Schedule in India

1. Primary doses: Given in the Pentavalent vaccine at (6), (10) and (14) weeks.

2. Boosters: At (16) to (24) months and (5) to (6) years, then Td at (10) and (16) years; Td also protects pregnant women.

DPT schedule in India



Significance

1. Child survival: Deaths from diphtheria and whooping cough fell sharply.

2. Tetanus elimination: India eliminated maternal and neonatal tetanus in 2015.
3. Cost-effective: One injection protects against three diseases.

Challenges

1. Side effects: Fever and swelling from whole-cell pertussis can feed hesitancy.
2. Dropouts: Missed boosters leave pockets where diphtheria recurs.
3. Cold chain: DPT is damaged by freezing.

Way Forward

1. Track every child: Digital records and frontline workers in Andhra Pradesh can trace dropouts.
2. Catch-up drives: Mission Indradhanush rounds reach tribal and remote areas.

Conclusion

Triple vaccination is one of the most successful public health tools, and sustaining coverage will keep these diseases under control in Andhra Pradesh and India.

Where did you write that point?
Notes that live with the question.
ghissu.com →

Write your own answer to 10(c) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

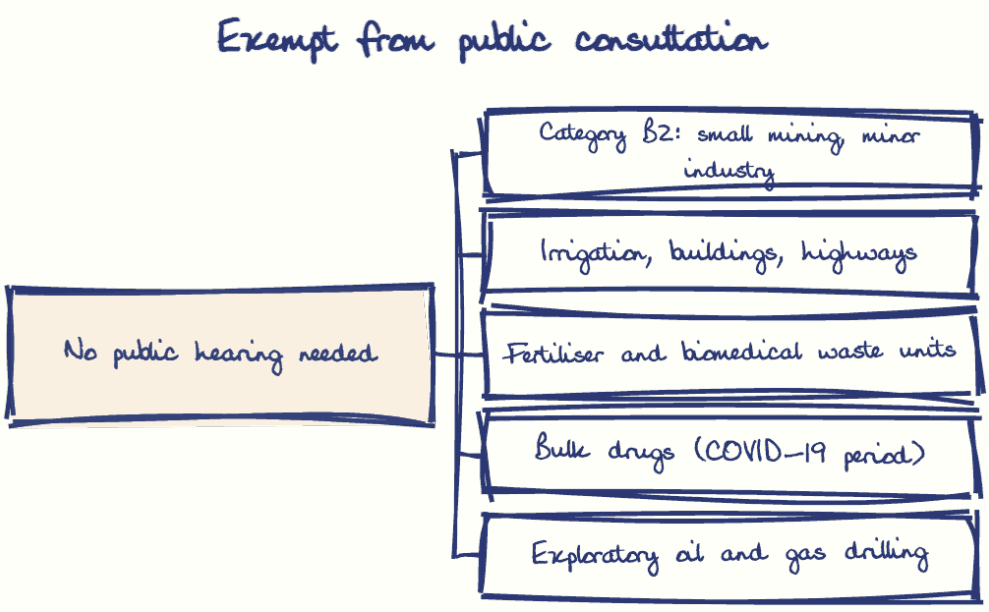
P-IV
11(d)

List-out the projects/activities for which public consultation is not required for issue of prior Environmental Clearance.

10

Introduction

Prior Environmental Clearance (EC) balances growth with ecological sustainability, and public consultation within it ensures local stakeholders' concerns are addressed. However, certain projects are exempted from consultation to streamline approvals and reduce delays, especially where large infrastructure and industrial projects in Andhra Pradesh demand timely clearance.



Projects and Activities Exempt from Public Consultation

1. Category B2 Projects: Small-scale mining leases and minor industrial activities need neither EIA nor

consultation, as their impact is considered minimal.

2. Irrigation Projects: Certain canal construction and minor water resource works, to expedite agriculture and water management.
3. Halogen and Chemical Fertiliser Units: Exempted under the amended notification to avoid bottlenecks in essential chemical manufacturing.
4. Biomedical Waste Treatment Facilities: Exempted for timely waste management.
5. Building and Infrastructure: Building construction, elevated roads, highways and expressways, including projects in the Visakhapatnam–Chennai industrial corridor.
6. Bulk Drug Projects: Re-categorised from Category A to B2 during the COVID-19 crisis.
7. Oil and Gas Exploratory Drilling: The 2019 amendment exempted onshore and offshore exploratory drilling.

Legal and Regulatory Basis

1. EIA Notification 2006: MoEFCC's notification sorts projects into A, B1 and B2, with B2 exempt due to limited impact.
2. Rule 5: Defines public consultation as ascertaining concerns of local affected persons; exemptions rest on

negligible impacts.

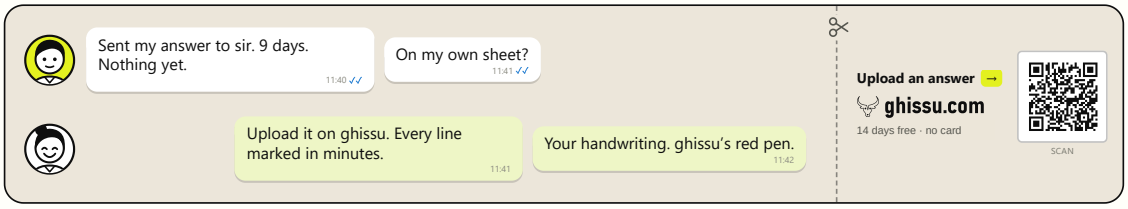
3. Draft EIA Notification 2020: This expired draft proposed exempting nearly (40) projects, such as sand extraction and solar thermal power plants.

Way Forward

1. Sharper Criteria: Exempt only genuinely low-impact projects, so that real environmental harm is not waved through.
2. Transparency: Publish clear, accessible lists of exemptions with reasons to maintain public trust and accountability.
3. Local Inputs: Engage communities in sensitive zones like the Rayalaseema drought belt, even for exempted projects.
4. Virtual Consultations: Use digital platforms where physical hearings are impractical.

Conclusion

Applied judiciously and aligned with local ecological realities, exemptions speed infrastructure and industrial growth while keeping environmental safeguards and community welfare intact.



P-IV
11(a)

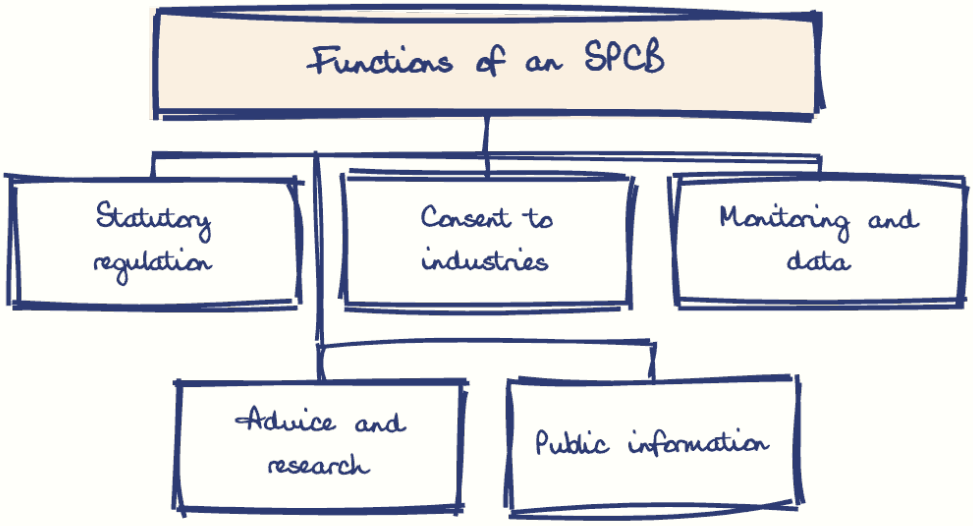
Narrate the functions of State Pollution Control Board.

10

Introduction

Rising air and water pollution has made State Pollution Control Boards (SPCBs) crucial for public health and ecological balance. With rapid industrialisation and urbanisation, as in Andhra Pradesh with its industrial corridors and coastal activities, SPCBs act as frontline regulators that monitor, control and prevent pollution at the state level.

Functions of SPCBs



1. **Statutory regulation:** SPCBs are constituted under the Water Act, 1974 and the Air Act, 1981, with expanded roles under the Environment (Protection) Act, 1986, to regulate water, air,

Practice 11(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V
1 / 3
Scan to practice

noise and waste pollution.

2. Consent and compliance: They grant mandatory consent to establish and operate, ensuring control measures are in place; units in the Visakhapatnam–Chennai corridor need SPCB clearance.

3. Monitoring and data: They run air and water quality networks that feed the National Air Quality Monitoring Programme and National Water Quality Monitoring Programme, tracking trends and compliance.

4. Emission oversight: They enforce Continuous Emission Monitoring Systems in industries.

5. Advisory and research: They advise the state government, coordinate with central agencies and sponsor region-specific abatement technology, such as for river flows in Andhra Pradesh's basins.

6. Public information: They are expected to disclose pollution data and engage communities.

Challenges

1. Capacity gaps: Limited resources, staff and technical skill weaken enforcement and real-time monitoring, including in Andhra Pradesh, where only a few ULB sanitary landfills were

operational in 2022.

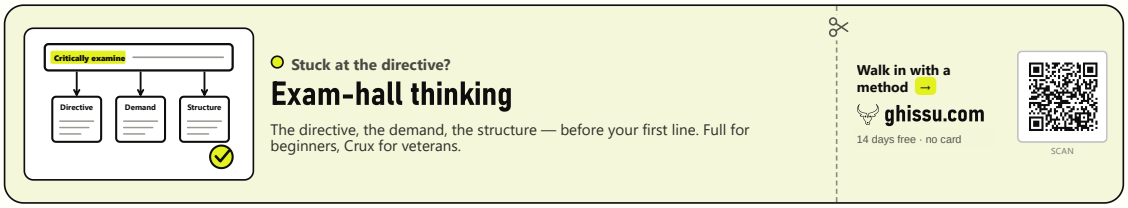
2. Data and coordination: Poorly maintained devices compromise data, and inter-agency coordination is weak.
3. Transparency: Limited disclosure reduces public pressure for compliance.

Way Forward

1. Institutional capacity: Invest in technical infrastructure, skilled personnel and digital tools for accurate monitoring.
2. Industry accountability: Regular audits, incentives for cleaner technology and penalties for CEMS data manipulation.
3. Coordination and disclosure: Shared data platforms with municipal and central agencies, and proactive disclosure with community grievance redressal.

Conclusion

Empowered SPCBs can ensure cleaner air and water and support sustainable industry, in line with SDG 6 and SDG 11.



P-IV
11(b)

Describe the procedure for obtaining Environmental Clearance under Environment Impact Assessment Notification, 2006.

10

Introduction

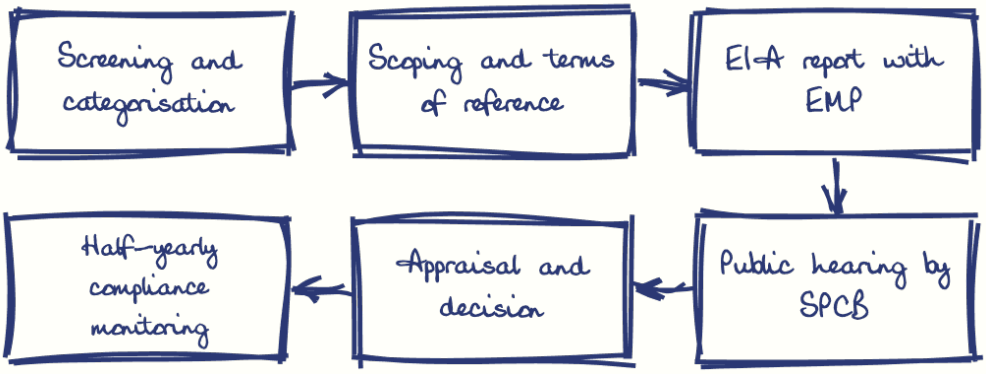
The EIA Notification, 2006, issued by MoEFCC, mandates prior Environmental Clearance (EC) for projects likely to have significant environmental impacts. This ensures adverse effects are assessed and mitigated before implementation. In Andhra Pradesh, with its long coastline and the Visakhapatnam–Chennai corridor, the process manages risks amid rapid development.

Institutional Framework

1. Category A: Large mining leases or national highways are cleared by the Central Government through the Expert Appraisal Committee (EAC).
2. Category B: Appraised at state level by the State Environment Impact Assessment Authority (SEIAA) and SEAC, as for state highway expansions and smaller mining projects in Andhra Pradesh.
3. Expert Committees: Ecologists, social scientists and technical experts evaluate EIA reports and EMPs.

Procedure and Stages of Clearance

Environmental Clearance under EIA 2006



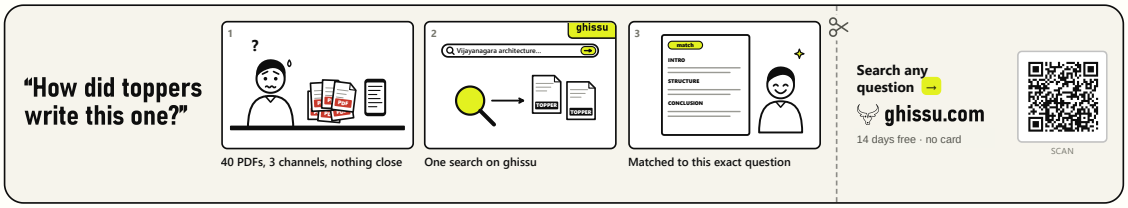
1. **Screening and Scoping:** Category B projects are screened for the need of a full EIA, while Category A goes straight to scoping, which fixes the terms of reference.
2. **EIA Report:** The proponent assesses impacts, mitigation and an EMP; since 2018 a Corporate Environment Responsibility plan is also required.
3. **Public Hearing:** The State Pollution Control Board lets affected communities voice concerns, as needed for Polavaram.
4. **Appraisal and Decision:** EAC or SEAC recommends clearance with conditions or rejection; MoEFCC or SEIAA then grants EC.
5. **Monitoring:** Proponents file half-yearly compliance reports, and violations can lead to withdrawal of clearance.

Challenges

1. Quality Issues: Single-season studies and proponent funding weaken reports.
2. Participation Gaps: In Andhra Pradesh, hearings sometimes exclude key stakeholders or come after decisions, and documents rarely appear on time in local languages.
3. Monitoring Deficiencies: Enforcement remains weak, as the delays and environmental concerns around the Polavaram Project show.

Conclusion

Credible reports, timely disclosure and firm monitoring would make the clearance process a true safeguard for ecological and social interests.



P-IV
11(c)

What is Environmentalism? Briefly state the problems does our planet face and suggest the solutions.

10

Introduction

Environmentalism is a broad social and moral movement to protect and improve the natural environment through sustainable management and conservation. It is urgent as climate change, resource depletion and biodiversity loss threaten human well-being. In Andhra Pradesh, with its long coastline, biodiversity and farm economy, these issues directly affect livelihoods.

Framework for Environmental Protection

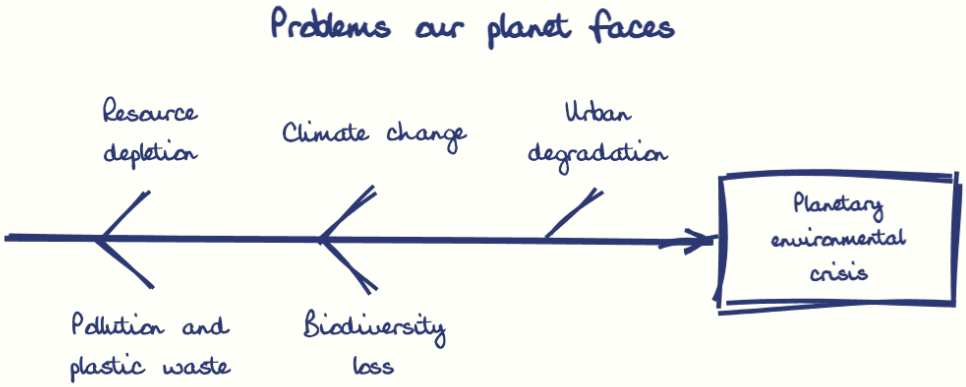
1. **Laws:** The Environment (Protection) Act, 1986, the Air and Water Acts and the Wildlife Protection Act, 1972 regulate pollution and conserve biodiversity. Andhra Pradesh's State Pollution Control Board enforces them in the Krishna and Godavari basins.

2. **Institutions:** MoEFCC, the Central Pollution Control Board and the National Green Tribunal oversee enforcement.

3. **Enforcement Gaps:** Limited capacity, corruption and competing development priorities weaken implementation, as urban growth around Visakhapatnam shows.

Problems Our Planet Faces

1. **Resource Depletion and Pollution:** Population growth and overconsumption cause shortages of water, fuel and food, hurting Andhra Pradesh's aquaculture and horticulture; plastic waste and industrial effluents contaminate soil and water.
2. **Climate Change:** Greenhouse gases cause warming, ocean acidification and extreme weather, seen in Rayalaseema's droughts and coastal cyclones.
3. **Biodiversity Loss:** Deforestation and habitat destruction drive extinctions and shrink carbon sinks.
4. **Urban Degradation:** Sprawl and corridors like Visakhapatnam–Chennai outpace infrastructure, worsening waste and air quality.
5. **Health and Equity:** Poor air and water cause respiratory and waterborne diseases, hurting the marginalised most.



Candidates must not write on this margin

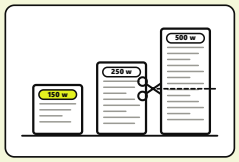
Candidates must not write on this margin

Solutions

1. **Stronger Institutions:** Better funding, training and technology for pollution control boards and environmental agencies.
2. **Renewable Energy:** Solar and wind projects in line with Panchamrit targets reduce fossil dependence, as Andhra Pradesh's solar parks and irrigation modernisation show.
3. **Community Engagement:** Public hearings, joint forest management and the LiFE movement in schools and communities.
4. **Integrated Planning:** Mainstream environment in development, with circular economy principles.

Conclusion

Environmentalism must guide development so that growth and conservation go together. Strong institutions, clean energy and active communities can protect the planet for future generations.



Writing 400 words for a 150-word question?

Model answers in three lengths

150, 250 and 500 words, in English and हिंदी. Learn what to cut before the examiner does.

Read one question three ways →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 11(c) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
12(d)

What is meant by mineral resources? How they form? Write about different types of mineral resources.

10

Introduction

Mineral resources are naturally occurring substances that are fundamental to modern industry, supplying raw materials for manufacturing, energy and infrastructure. With rising demand, understanding what they are, how they form and their types is crucial. In Andhra Pradesh, deposits of barytes and road metal contribute significantly to the economy.

Meaning and Governance

1. **National asset:** The MMDR Act, 1957 treats mineral wealth as a national asset to be exploited in a balanced way for present and future needs.
2. **Andhra Pradesh:** The state holds 86.11% of India's barytes and 18.49% of its road metal reserves, regulating leases under central guidelines.
3. **Exploration bodies:** The Geological Survey of India and the National Mineral Development Corporation coordinate exploration and mining.
4. **Audit gaps:** A CAG audit of Andhra

Practice 12(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Pradesh's mineral sector (2017–22) found revenue shortfalls of 2% to 52% against budget estimates.

Formation of Minerals

1. Geological processes: Minerals form naturally through heat, pressure and chemical reactions over millions of years in igneous, metamorphic and sedimentary rocks.
2. Veins and lodes: In igneous and metamorphic rocks, metallic minerals like copper and tin fill cracks and fissures.
3. Beds and layers: In sedimentary rocks, minerals like coal and gypsum accumulate in layers.
4. Residual deposits: Weathering removes soluble parts of rocks and leaves residual deposits behind.

Types of Mineral Resources

1. Metallic minerals: Ferrous ones contain iron, like iron ore and manganese; non-ferrous ones include copper and bauxite. They are crucial for metal production.
2. Non-metallic minerals: Organic mineral fuels such as coal and petroleum, and inorganic minerals like mica and limestone, serve

Candidates must not write on this margin

Candidates must not write on this margin

construction, agriculture and chemicals.

3. Local significance: AP's limestone, though small nationally, supports cement industries.

4. Balancing act: Andhra Pradesh must weigh extraction against its environmental and social costs in forest and tribal areas.

Types of mineral resources

Conclusion

Minerals form slowly and are finite. Institutions like the District Mineral Foundation and the National Critical Mineral Mission (2024–31) must ensure sustainable mining and community welfare.

REVIEWER'S NOTE · VALUE ADDITION

Examiners read hundreds of answers a day. The one with a number, a committee and a case is the one they remember.

ghissu

ghissu gives you the data point, the report and the example for every question — each linked to its source.

Get the value adds for any question →

ghissu.com

14 days free · no card

SCAN

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 12(d) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
12(a)

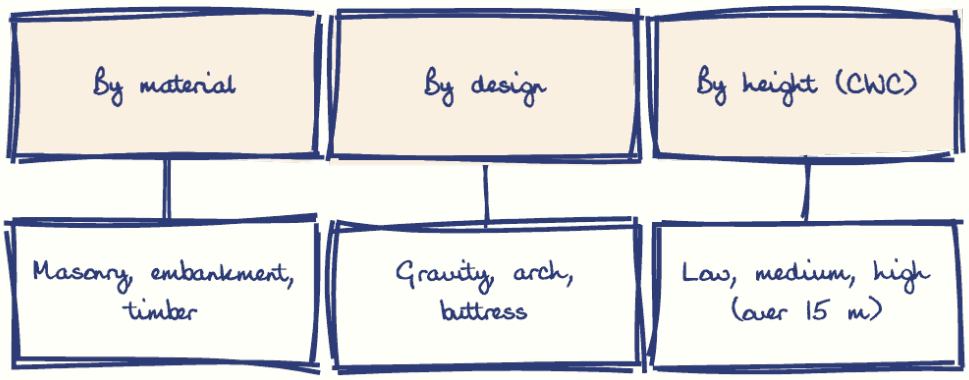
Classify dams based on structure and design.

10

Introduction

Dams are critical for water management, flood control, irrigation and hydropower, as in Andhra Pradesh's Polavaram multipurpose project. Their structure and design vary with geographic, geological and functional needs, so classification is essential for planning, safety and sustainability.

Classification of dams



Classification by Structure (Material)

1. **Masonry Dams:** Built of concrete or stone.
2. **Embankment Dams:** Built of earth or rock-fill; they often face sedimentation that reduces reservoir capacity, an issue for Polavaram.
3. **Timber Dams:** Less common today.

Classification by Design

1. **Gravity Dams:** Rely on their own weight to hold back water.
2. **Arch Dams:** Transfer water pressure to the abutments.
3. **Buttress Dams:** Use supports to resist water pressure.

Each design suits specific topography and seismic conditions.

Classification by Height

The Central Water Commission classes dams as low, medium or high, with large dams typically exceeding (15) metres.

Design Challenges and Adaptations

1. **Geology and Seismicity:** Seismic zones like the Himalaya need special designs, and Andhra Pradesh's varied geology needs tailored embankment or gravity designs.
2. **Multipurpose Complexity:** Spillways and flood cushions balance irrigation, hydropower and flood control.
3. **Ageing Dams:** Many dams are over (25) years old; DRIP funds rehabilitation and DHARMA

Practice 12(a) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

gives real-time health data.

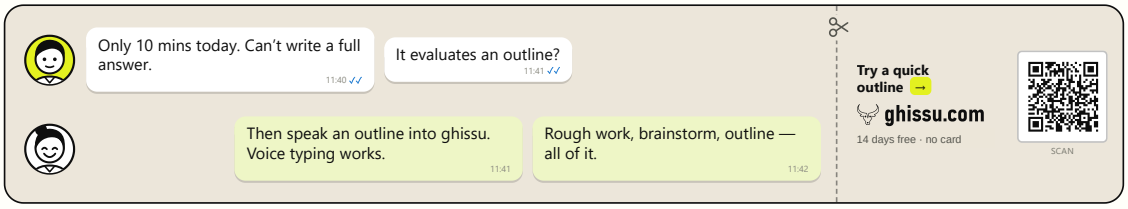
4. Sustainable Design: Fish passages and sediment management reduce ecological harm.
5. Flood Management: Adjustable spillways and reservoir schedules cut flood risk in basins like the Godavari.
6. Seismic Resilience: Flexible materials and reinforced foundations protect dams in quake-prone areas.

Regulation

The National Dam Safety Authority, set up under the Dam Safety Act 2021, oversees structural safety, including Polavaram.

Conclusion

Classifying dams by material, design and height guides the right choice for each site. Matching structure to local conditions, with climate-resilient standards and community engagement, keeps dams safe engines of development.



P-IV
12(b)

Write a note on Andhra Pradesh Sand Mining Policy, 2016.

10

Introduction

Sand mining is vital for construction, but unregulated extraction causes environmental damage and social conflict. Given AP's rivers like the Godavari and Krishna and its long coastline, the AP Sand Mining Policy, 2016 was introduced to regulate extraction, prevent illegal mining and protect riverine ecosystems while supporting livelihoods.

Rivers under the AP Sand Mining Policy, 2016

Key Features of the Policy

1. **Manual mining only:** Mechanised equipment such

as machinery is banned, only manual and animal-driven methods are allowed, protecting rivers like the Godavari and Vamsadhara.

2. Decentralised allocation: Sand blocks are allotted by open bidding through committees led by district joint collectors.
3. Price control: A maximum retail price tag curbs black marketing, with penalties for violations.
4. Environmental safeguard: Mining below three metres in riverbeds is banned to prevent erosion and protect groundwater recharge.

Challenges in Implementation

1. Illegal mining: Weak enforcement and corruption let illegal mining persist along the Godavari and Nagavali, hurting revenue and the environment.
2. Revenue shortfalls: A CAG audit found gaps of 2% to 52% in AP's mineral revenue realisation in 2017–22, showing weak monitoring.
3. Livelihood pressure: The manual labour rule supports traditional livelihoods, but mechanised illegal mining displaces local workers and fuels conflict with sand mafias.
4. Clearance delays: Delays and unclear EIA guidelines hamper sustainable operations and

compliance with Supreme Court directives on illegal mining.

Strengthening the Policy

1. Technology: Satellite monitoring and GIS mapping to identify sand blocks and track mining in real time.
2. Local bodies: Empower Panchayats and Gram Sabhas to grant permissions and monitor mining, reducing illegal extraction.
3. Capacity building: Train district officials and mining committees in regulations and environmental safeguards.
4. DMF funds: Use the District Mineral Foundation for restoration of mining-affected areas.

Conclusion

The (2016) policy is a crucial step towards sustainable resource management, balancing economic needs with environmental protection and social equity. Effective implementation, better monitoring and community engagement can curb illegal mining, safeguard rivers and advance SDG (15) on life on land.

Write your own answer to 12(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
12(c)

Discuss the basis and justification required for suitability of a crop and cropping pattern.

10

Introduction

With climate variability and population pressure, sustainable agriculture is a critical concern, especially in Andhra Pradesh with its diverse agro-climatic zones. The shift from the Green Revolution's high-input model to ecologically sound practice demands careful assessment of crop suitability and cropping patterns to ensure food security and resource efficiency.

Crop zones of Andhra Pradesh

Basis for Determining Crop Suitability

1. **Agro-Climatic Conditions:** Soil type, depth, drainage, rainfall, temperature and attitude decide crops; low, erratic rain in Rayalaseema favours drought-resistant millets.
2. **Soil and Irrigability:** Salinity and water-holding capacity matter; FAO land classification guides paddy in well-irrigated coastal and deltaic zones.
3. **Water Availability:** Water-scarce regions need low-water or rainfed crops, while Polavaram's improved supply shifts choices towards rice.
4. **Economic Viability:** Profitability and market access count; chilli in Guntur reflects both suitability and demand.

Justification for Selecting a Cropping Pattern

1. **Land Use Efficiency:** Multiple cropping and crop rotation raise productivity without degrading soil, as with horticulture combined with food crops in irrigated zones.
2. **Risk Mitigation:** Diversification reduces losses from pests, disease and climate shocks; millets insure Rayalaseema against monsoon failure.
3. **Infrastructure and Technology:** Irrigation, storage, transport, better seeds and schemes like M-KISAN

and the Soil Health Card support suitable sequences.

Other Influencing Factors

1. Geography: Soil fertility, topography and microclimate matter; Krishna and Godavari deltaic soils support paddy while uplands favour pulses and oilseeds.
2. Markets: Price swings, input costs and demand for horticulture near the Visakhapatnam industrial corridor have altered traditional patterns.
3. Policy: PMKSY irrigation projects have expanded irrigated area, enabling diversification and intensification.

Way Forward

Promote climate-resilient, less water-intensive crops in vulnerable regions through research and extension services.

Conclusion

A crop must suit the soil, climate and water, and a cropping pattern must justify itself through efficiency, risk reduction and farmer welfare. Such planning makes farming both sustainable and profitable.

Which sub-topic comes next?
Trend analysis for every syllabus topic.
ghissu.com →

Write your own answer to 12(c) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
13(d)

Explain the following: (i) National Bureau of Plant Generic Resources (NBPGR) (ii) Wild-life sanctuaries.

10

Introduction

Conservation of biodiversity and genetic resources is critical for agriculture, ecological balance and wildlife heritage. Institutions like NBPGR and protected areas like wildlife sanctuaries preserve genetic diversity and species, a role that has grown amid habitat loss, including in Andhra Pradesh, where conservation supports agrarian livelihoods.

(i) National Bureau of Plant Genetic Resources

1. Premier ICAR institute: It collects, conserves, evaluates and documents plant genetic resources in India.
2. National repository: It safeguards indigenous crop varieties and wild relatives that are critical for food security and crop improvement.
3. Coordination with states: It works with State Biodiversity Boards and Biodiversity Management Committees on in situ and ex situ conservation, documenting region-specific resources such as those in Andhra Pradesh's agro-ecological zones.
4. Challenges: Changing land use and climate threaten in situ conservation, and ex situ collections risk

gene pool stagnation, so dynamic strategies are needed.

(ii) Wildlife Sanctuaries

Sanctuary vs national park

Wildlife sanctuary	National park
Declared by state government	Stricter protection
Regulated grazing allowed	No grazing
Private land possible	No private land

1. Definition: Protected areas declared by state governments under the Wildlife (Protection) Act, 1972 to conserve specific species and their habitats.
2. Difference from national parks: Sanctuaries allow limited human activity like regulated grazing and private land ownership, provided wildlife is not harmed.
3. Administration: The state notifies boundaries and manages them through forest and wildlife departments; hunting is generally prohibited.
4. Importance: They are critical habitats for endangered and endemic species, such as the

grizzled giant squirrel in Andhra Pradesh, and support eco-tourism and local livelihoods.

5. Challenges: Fragmentation, conflict, poaching and invasive species, along with funding and manpower gaps.
6. Outcomes: Sanctuaries have aided the recovery of several species and preserved critical habitats; nationally, tiger numbers have risen due to protected area networks.

Way Forward

1. In situ conservation: Expand protected areas and community-led management, especially in biodiversity-rich parts of Andhra Pradesh.
2. Capacity and technology: Use GIS-based monitoring and research.
3. Public participation: Run awareness campaigns and involve communities to reduce conflict and encourage sustainable use of genetic resources.

Conclusion

Both function within India's NBSAP 2024-30, which stresses sustainable use, equitable benefit sharing and ecosystem resilience.

Are you actually improving?
Every attempt scored. Watch the curve.
ghissu.com →

P-IV
13(a)

What is eco system? Explain the food chain system in eco systems?

10

Introduction

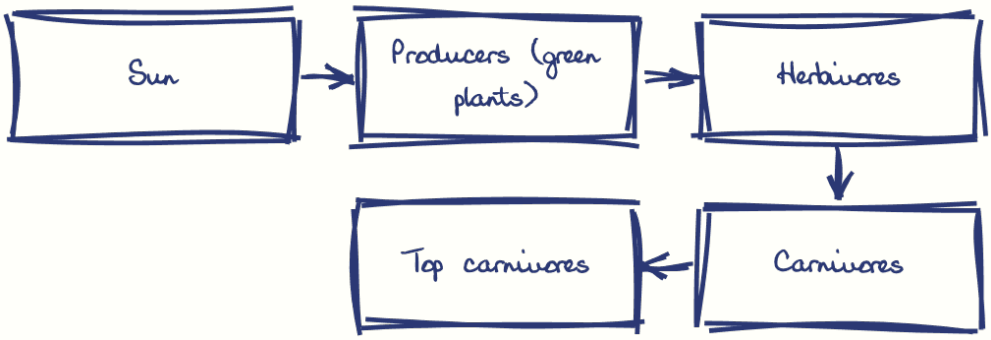
Ecosystems are fundamental to life, representing the complex interactions between living organisms and their environment. Amid pollution and biodiversity loss, understanding ecosystems and their food chains is crucial, especially in Andhra Pradesh, whose agriculture, aquaculture and forests depend on them.

What Is an Ecosystem

1. Definition: An interacting, interdependent complex of living communities and their physical environment in which energy flows and materials cycle. Arthur Tansley coined the term in 1935.
2. Components: Biotic producers, consumers and decomposers interact with abiotic soil, water and climate.
3. Energy Flow and Cycling: Producers capture solar energy by photosynthesis; energy flows one way through trophic levels while matter cycles.
4. Examples in AP: The 974 km coast, the Krishna and Godavari basins and Eastern Ghats forests.

The Food Chain System

A grazing food chain



1. Concept: A linear sequence in which each member feeds on the one before it, transferring energy from producers to top carnivores.
2. Types: The grazing food chain begins with green plants eaten by herbivores; the detritus food chain begins with dead matter broken down by microbes and detritivores. Both recycle nutrients.
3. Trophic Levels: Producers, primary consumers (herbivores), secondary consumers (carnivores) and tertiary consumers; energy transfer efficiency falls at higher levels.
4. Food Web: Many interlinked chains form a food web, raising stability and resilience.

Threats to Food Chains

1. Pollution: Heavy pesticide use in Rayalaseema

causes bioaccumulation and biomagnification.

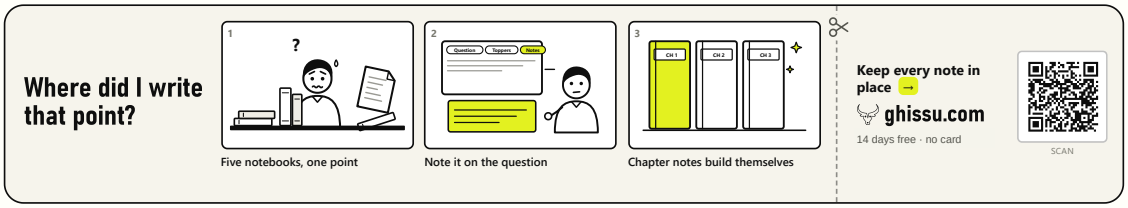
2. Habitat Loss: Habitat degradation and overexploitation in AP's forests and aquatic ecosystems break predator–prey links and weaken ecosystem services.
3. Climate Variability: El Nino alters monsoons and primary productivity.
4. Unsustainable Farming: Monocropping and chemical dependency degrade soil and water, weakening the base of food chains.

Way Forward

Promote zero budget natural farming, monitor pollutants in the Krishna delta, and restore forests and wetlands.

Conclusion

Ecosystems and their food chains sustain both life and livelihoods. Protecting them through ecological farming, monitoring and habitat restoration is key to sustainable development in Andhra Pradesh and beyond.



P-IV
13(b)

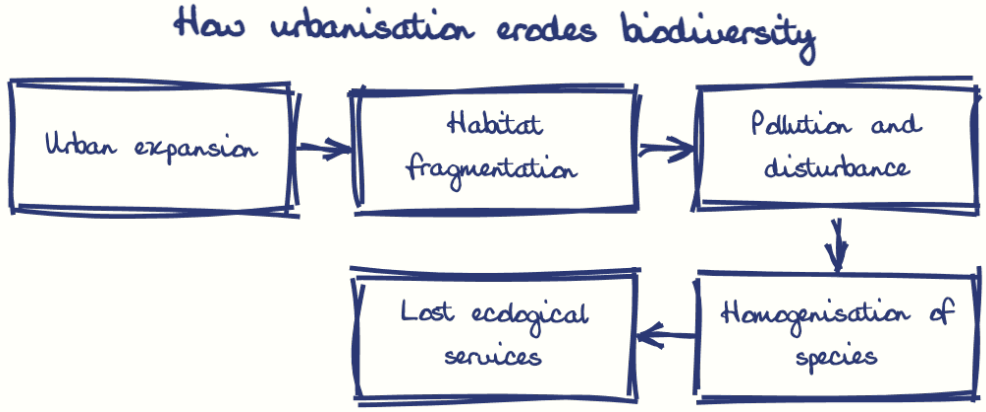
Narrate the impact of urbanisation on bio-diversity.

10

Introduction

Urbanisation in India is accelerating, replacing natural habitats with concrete infrastructure and altering the balance of flora and fauna. This matters now, as the State of India's Birds 2023 and the WWF Living Planet Report 2024 highlight an alarming decline in species diversity amid urban expansion.

How Urbanisation ~~Affects~~ Biodiversity



1. **Habitat destruction and fragmentation:** Built environments replace large ecosystems and isolate green patches, causing decline of specialist species such as forest specialists and grassland birds in Indian cities.

2. **Homogenisation of species:** Cities favour generalists like the house crow, feral rock pigeon

and Asian koel, as they adapt well to human-altered habitats, while native and specialist birds decline, reducing overall diversity and resilience.

3. Pollution and disturbance: Air, noise and light pollution disrupt wildlife; urban noise drives birds from habitats and glass facades cause fatal collisions.

4. Predation by domestic animals: Free-ranging cats and dogs prey on birds and small wildlife, creating a landscape of fear.

Biodiversity Loss and Change

1. Link with species decline: The most urbanised areas have the lowest diversity of birds, including rare and insectivorous species.

2. Altered ecological functions: Loss of birds weakens seed dispersal, pollination and pest control, vital for ecosystem health and human well-being.

3. Weak compensation: Afforestation often turns grasslands into monocutture plantations that do not support the original biodiversity.

4. Climate synergy: Urbanisation worsens climate change through greenhouse gas emissions and

heat island effects, further stressing species and accelerating their decline.

Way Forward

1. Native vegetation: Use native flora in city landscaping to support local wildlife and maintain ecological balance.
2. Monitoring: Run systematic urban biodiversity studies in cities like Visakhapatnam.
3. Green infrastructure: Build interconnected green corridors and parks for species movement.
4. Pet management: Control free-ranging pets through awareness and policy.

Conclusion

Urbanisation alters habitats and ecological dynamics, but with conscious planning cities can become havens for diverse species. Biodiversity-friendly development fulfils SDG (15) and the Fundamental Duty to protect the environment.

Your intro decides your marks. Q. Discuss Ashoka's Dhamma.

YOUR INTRO
~~Ashoka was a great Mauryan king who ruled after the Kalinga war and spread Buddhism everywhere.~~
✕ no date · no source · no argument

→

AFTER GHISSU EVALUATION
Ashoka's Dhamma, set out in his Major Rock Edicts after Kalinga (c. 261 BCE), was not Buddhism by another name but an ethic of tolerance, non-violence and respect, meant to hold a plural empire together.
✓ date · source · argument

ILLUSTRATIVE

Get your intro fixed on your sheet →
ghissu.com
14 days free · no card
SCAN

Write your own answer to 13(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
13(c)

Describe the major threats to bio-diversity. Explain different types of bio-diversities.

10

Introduction

Biodiversity, the variety of life at genetic, species and ecosystem levels, is fundamental to ecological balance and human well-being. India, one of the (12) mega-diverse countries, harbours a vast array of species across ten biogeographic zones. Rapid human activity and environmental change have escalated threats, making its types and challenges a national priority.

Types of Biodiversity

1. Genetic diversity: Variation of genes within species lets populations adapt to change, high diversity in India's agro-biodiversity is crucial for food security and resilience against pests.
2. Species diversity: It measures the number and abundance of species, India hosts over (96,000) animal and (47,000) plant species, with endemics in the Western Ghats and Himalayas.
3. Ecosystem diversity: India's ten biogeographic zones, from deserts to coasts, sustain ecosystems that provide nutrient cycling, climate regulation and water purification.

4. Ecological subtypes: Alpha, beta and gamma diversity refer to variety within a habitat, between habitats and across landscapes, adding layers of resilience.

Major Threats to Biodiversity

1. Habitat loss and fragmentation: Urbanisation, deforestation and projects like the Great Nicobar Project (Rs 80,000 crore) cause species decline, in Andhra Pradesh, fragmentation in the Eastern Ghats hurts endemic species and forest connectivity.
2. Overexploitation and poaching: Unsustainable harvesting and illegal wildlife trade endanger tigers and elephants, while human-wildlife conflict persists in Srikakulam and Visakhapatnam districts.
3. Invasive species and pollution: Exotic species disrupt native ecosystems, while pesticides and heavy metals degrade habitats and complicate restoration.

Institutional Response

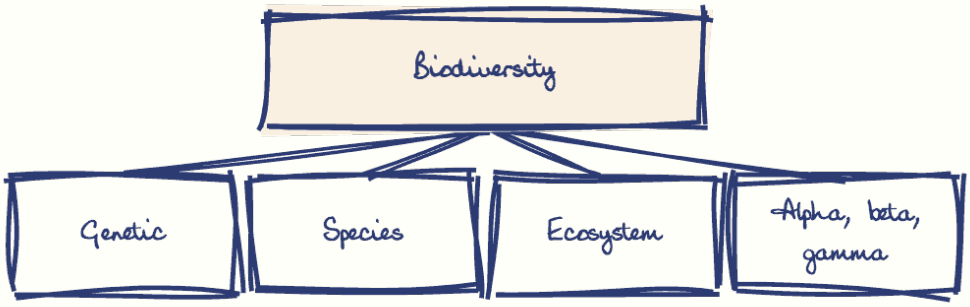
1. Legal framework: The Biological Diversity Act, 2002 and Rules of 2004 create a three-tier system of NBA, State Biodiversity Boards

and BMCs, with over 2,76,653 BMCs set up.

2. Protected areas: These grew from 745 in 2014 to 1,134 in 2025, covering over 1.87 lakh sq km and safeguarding tigers and elephants.

3. Scientific advice: The National Board for Wildlife, the apex scientific body, advises on policy and reviews progress.

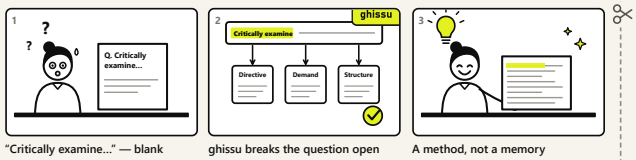
Types of biodiversity



Conclusion

Biodiversity in all its types faces mounting pressure. Strengthening this three-tier governance, scientific advice through the NBWL and community participation will help India safeguard its genetic, species and ecosystem wealth.

You know the topic. You still freeze.



"Critically examine..." — blank | ghissu breaks the question open | A method, not a memory

See the thinking before the answer →
ghissu.com
14 days free · no card
SCAN

P-IV
14(d)

Describe noise pollution and explain the various control and preventive measures to be taken to control noise pollution.

10

Introduction

Noise pollution is excessive, unwanted sound that has become a major environmental and public health concern in India due to rapid urbanisation, industrialisation and vehicular traffic. It causes hearing loss, stress and cardiovascular problems, undermining the right to a healthy environment, especially in Visakhapatnam and other urban centres of Andhra Pradesh.

Nature and Impact

1. Sources: Factories, construction and heavy traffic raise ambient noise, worsened by inadequate zoning and poorly maintained machinery.
2. Health and social effects: Prolonged exposure causes hearing impairment, hypertension, mental stress and reduced productivity, hurting children and workers in industrial clusters.
3. Weak enforcement: Limited infrastructure, manpower and low priority, plus low public awareness, blunt the laws in cities like Vijayawada.

Legal Framework

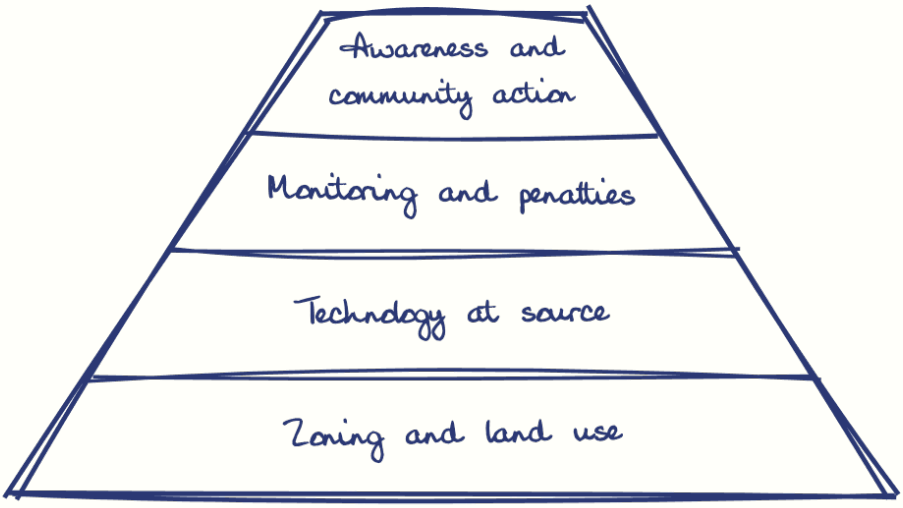
1. Noise Rules: The Noise Pollution (Regulation and Control) Rules, 2000 set permissible limits for industrial, commercial, residential and silence zones.

2. Constitutional backing: The Supreme Court has upheld a noise-free environment under Article 21, and courts direct stricter enforcement during festivals.

3. Pollution Control Boards: The APPCB monitors noise and acts against unauthorised loudspeakers and vehicular noise.

Control and Preventive Measures

Hierarchy of noise control



1. Zoning and land use: Separate noisy industries and transport corridors from residential areas, with

silence zones around hospitals and schools.

2. Technology at source: Use soundproofing, noise-absorbing materials and quieter machinery; enforce night horn bans and promote electric vehicles in cities like Amaravati.

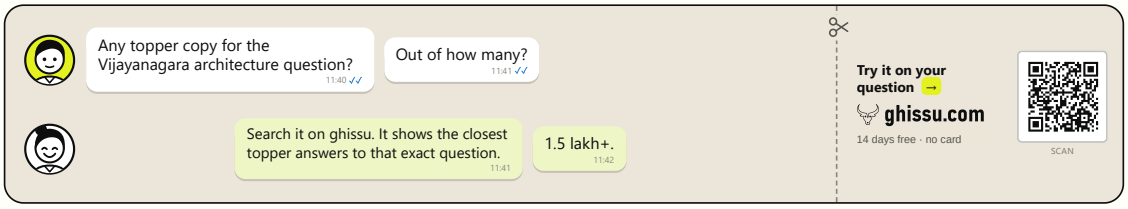
3. Monitoring and penalties: Empower local authorities to seize violating equipment and impose fines, backed by real-time monitoring data.

4. Awareness: Campaigns in schools and communities before festivals build voluntary compliance.

5. Planning and community action: Build noise mitigation into city master plans, map hotspots with sensors and GIS, and involve Resident Welfare Associations.

Conclusion

Noise control needs a coordinated mix of robust laws, technology and active public participation. Implementing the rules firmly in Andhra Pradesh's urban and industrial areas will protect public health.



P-IV
14(a)

Narrate various types and the characteristics of Solid Waste.

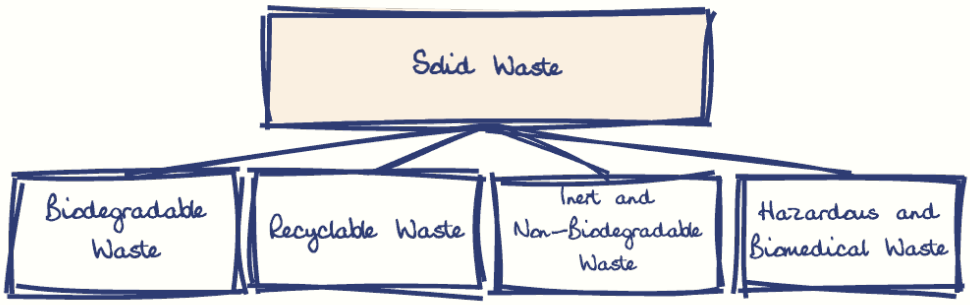
10

Introduction

Rapid urbanisation and industrial growth in India, including Andhra Pradesh's expanding urban local bodies, have sharply raised solid waste generation; the state's 123 ULBs generated about 6,890 tons per day in 2022. Understanding the types and characteristics of waste is essential for designing collection, segregation and disposal systems.

Types and Characteristics

Types of Solid Waste



1. **Biodegradable waste:** Food scraps, kitchen waste and garden trimmings that decompose naturally. It is the largest share, about 52% nationally and 55–60% in many cities, and suits composting and bio-methanation once segregated at source.

2. Recyclable waste: Plastics, paper, glass, metals and textiles form about (17%) of waste. Plastic, about (9.4 million) tonnes a year in India, is non-biodegradable and releases toxic by-products when burned.

3. Inert and non-biodegradable waste: Construction and demolition debris and dirt, about (32%), do not decompose and need landfilling or reuse in construction; many landfills lack liners and leachate management.

4. Hazardous and biomedical waste: From hospitals, industries and agriculture, it is toxic and infectious, needs careful handling and is often neglected in municipal systems, posing health risks.

Management Gaps in Andhra Pradesh

1. Weak rules adoption: Only (5) of (17) test-checked ULBs had framed bye-laws under the SWM Rules by (2022).

2. Capacity constraints: Many ULBs lack technical and financial capacity; only (4) had operational sanitary landfills and (82) lacked wet waste processing plants as of (2022).

3. Low segregation: Audits found only about (75%) segregation despite claims of higher rates, and

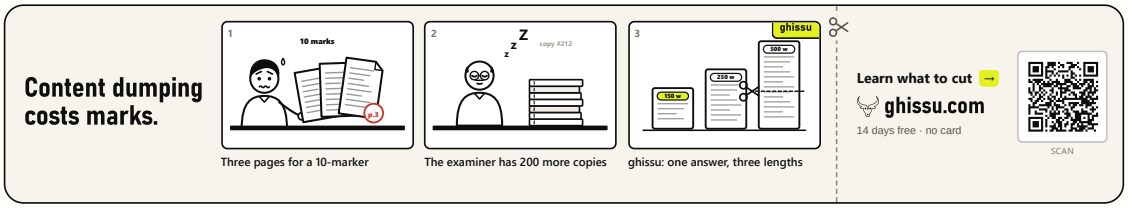
a lack of composting and recycling units leads to dumping and open burning, which pollute and spread disease.

Way Forward

1. Type-specific processing: Composting and bio-methanation for biodegradable waste, material recovery for recyclables and C&D plants for inert debris, under Swachh Bharat Mission (Urban) 2.0.

Conclusion

This classification shows that each type needs a differentiated strategy suited to its physical and chemical nature, built on segregation at source and appropriate treatment technology.



P-IV
14(b)

Write a short note on the following: (i) Pyrolysis (ii) Incineration

10

Introduction

With rising urbanisation and industrialisation, waste management is a critical challenge in Andhra Pradesh. Pyrolysis and incineration are thermal technologies that reduce waste volume and recover energy, though their environmental and economic impacts remain debated.

(i) Pyrolysis

1. **Process:** Thermal decomposition of waste in an oxygen-limited environment, producing charcoal, tar and fuel gases.
2. **Merits:** It is considered cleaner than incineration and is relatively inexpensive for diverse feedstocks, including agricultural residues common in Rayalaseema.
3. **Limits:** Residues like fine carbon particles and pyrolysis oil need proper handling, and it requires investment and skilled operation.

(ii) Incineration

1. **Process:** Waste is burnt at high temperatures, sharply reducing its volume.

Pyrolysis vs Incineration

Pyrolysis	Incineration
Heat with little oxygen	Burn at high temperature
Charcoal, tar, fuel gases	Volume cut, some energy
Cleaner and cheaper	Dioxins, furans, toxic ash
Oil and carbon residues	Costly, public opposition

2. Pollution: It generates toxic ash and air pollutants such as dioxins and furans, so in India it is used mainly for infectious biomedical waste, with concerns in cities like Visakhapatnam.

3. Cost: Plants are capital-intensive and often face public opposition over pollution fears.

Regulatory Framework and Challenges

1. Oversight: MoEFCC and CPCB guidelines control emissions, enforced locally by the Andhra Pradesh Pollution Control Board.

2. Rules: The SWM Rules, 2016 mandate segregation and promote waste-to-energy in cities like Visakhapatnam and Vijayawada, and some projects use PPP models.

Practice 14(b) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

3. Constraints: ULBs often lack technical expertise and funds, leading to poor performance, and mixed municipal waste with high organic content limits efficiency unless segregation is rigorous.

Way Forward

1. Cleaner technology: Modern pyrolysis units with emission controls can turn agricultural and municipal waste into fuel, supporting circular economy goals.
2. Segregation and capacity: Better source segregation improves feedstock, while training and funds help ULBs run plants.
3. Awareness and monitoring: IEC campaigns can educate communities on the benefits and risks of thermal treatment and build acceptance, while stronger APPCB monitoring keeps plants within prescribed pollution limits.

Conclusion

Under strict monitoring, pyrolysis and incineration can work as complementary technologies within an integrated waste strategy, in line with the national Waste-to-Energy guidelines.

Correct, but average?
A number, a committee, a case — for every question.
ghissu.com →

P-IV
14(c)

What is Soil Pollution and how it is caused?

10

Introduction

Soil pollution has emerged as a critical challenge with intensifying industrialisation and agricutture. In Andhra Pradesh, where agricutture and aquacutture form the economic backbone, it directly affects livelihoods, heatth and ecological balance.

What is Soil Pollution

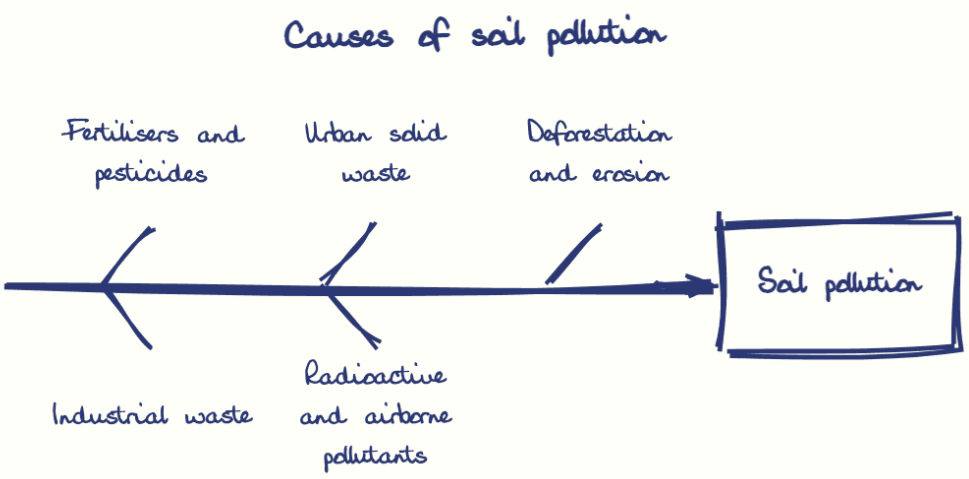
1. Definition: It is the addition of harmful substances to soil that adversely affect its physical, chemical and biological properties, reducing fertility and productivity.
2. Pollutants: These include persistent toxic compounds, chemicals, salts, radioactive materials and disease-causing agents that disrupt the natural balance of soil organisms and degrade soil quality.
3. Heatth impact: Contaminated soil hampers plant growth and enters the food chain, causing bioaccumulation and biomagnification of toxins in humans and animals.

How it is Caused

1. Fertilisers and pesticides: Indiscriminate use leaves

Practice 14(c) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V

Scan to practice



harmful residues, reduces beneficial microbes and raises salinity, as seen with endosulfan in cashewnut plantations.

2. Industrial waste: Effluents with heavy metals such as mercury, lead and cadmium are dumped on land, polluting soil and groundwater near the Visakhapatnam corridor.

3. Urban solid waste: Rapid urbanisation in Vijayawada and Guntur produces waste and sludge that, dumped carelessly, cause foul odours and pathogenic outbreaks.

4. Radioactive and airborne pollutants: Wastes from mining and nuclear activities, along with acid rain and airborne pollutants, deposit toxic substances on the land surface.

5. Deforestation and erosion: In drought-prone Rayalaseema, loss of vegetation cover leaves soil

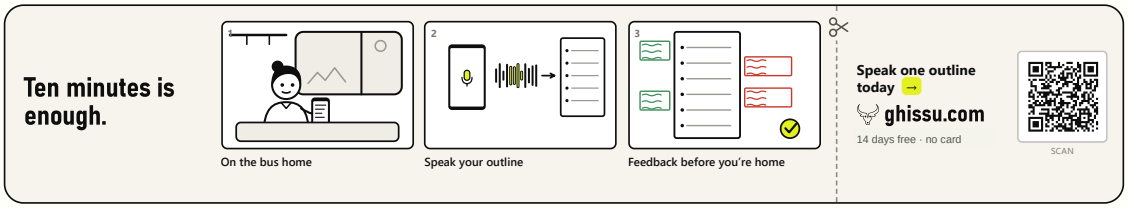
more vulnerable.

Way Forward

1. Organic farming: Promote organic manures and test-based nutrients under the Soil Health Card scheme.
2. Industrial regulation: Enforce zero liquid discharge norms and effluent monitoring in industrial clusters.
3. Waste and soil conservation: Use scientific waste disposal, with contour bunding, afforestation and watershed management in Rayalaseema.

Conclusion

Soil pollution undermines agricultural sustainability, public health and ecological balance. Integrated action on its causes, through organic farming, industrial regulation and urban waste management, supports land degradation neutrality and will restore soil health and ensure food security.



P-IV
15(d)

Give a brief note on the following: (i) Sustainable agriculture (ii) Acid Rain

5

Introduction

Sustainable agriculture and acid rain are critical issues for India's ecological balance and agricultural productivity. With climate variability intensifying, sustainable farming is essential for food security and environmental health. Acid rain, caused by industrial emissions, threatens ecosystems, infrastructure and human health, especially in industrially active regions and metropolitan cities.

(i) Sustainable Agriculture

1. **Meaning:** Farming that meets food needs while protecting soil, water and ecosystems, so productivity lasts.
2. **NMSA:** The National Mission for Sustainable Agriculture promotes climate-resilient farming through Rainfed Area Development and Per Drop More Crop, benefiting millions and improving water-use efficiency, including in Andhra Pradesh.
3. **Soil Health Card Scheme:** Over 25 crore soil health cards guide balanced fertiliser use to prevent soil degradation.
4. **State action:** AP links national schemes with

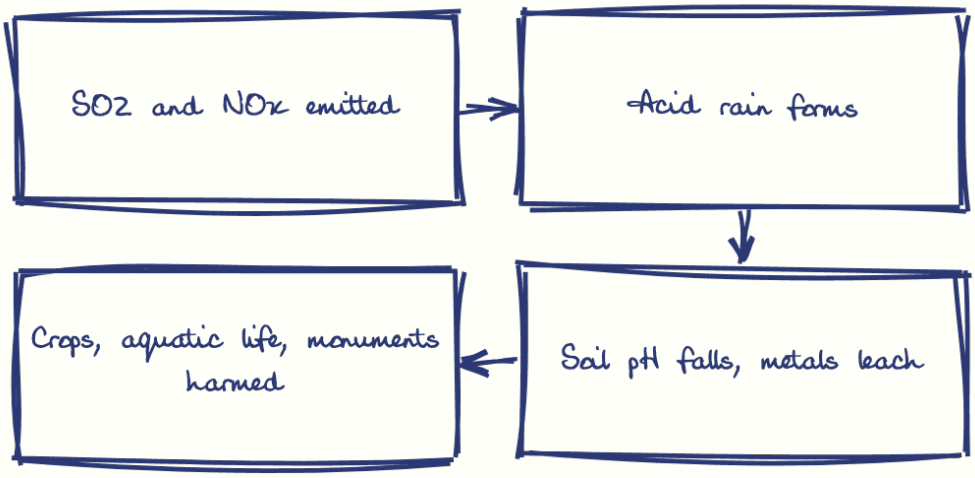
its agricuttural extension services to spread sustainable practices and monitor acid rain effects on crops and water bodies.

5. Challenge: Indiscriminate fertiliser and pesticide use still pollutes soil and water, especially in drought-prone Rayalaseema.

(ii) Acid Rain

1. Causes: Sulphur dioxide and nitrogen oxides from industries and vehicles are the primary precursors.
2. Ecological impact: It lowers soil pH, damages crops and aquatic life, and leaches heavy metals into drinking water.
3. Damage to heritage: It corrodes buildings, infrastructure and monuments like the Taj Mahal, causing lasting cuttural loss.
4. Contrd: India enforces BS-VI emission norms for industries and vehicles, and real-time air quality monitoring helps manage these gases.
5. Gaps: Weak enforcement and low awareness leave northern industrial betts and urban centres most at risk.

Acid rain and its damage



Common Thread

Changing rainfall patterns and temperature extremes worsen both problems: they stress farms already weakened by soil degradation and intensify acid rain's effects. Integrated climate-resilient strategies are therefore needed rather than separate fixes.

Conclusion

Sustainable agriculture and acid rain control are vital for India's environmental security. Stronger implementation of the NMSA and strict pollution control can safeguard ecosystems and livelihoods, in line with SDG (13) and SDG (15).

Which sub-topic will they ask next?

Trend analysis

Every syllabus topic: the direction of travel, the most tested sub-topic, the probable next theme.

Prepare the pattern →

ghissu.com

14 days free · no card

SCAN

P-IV
15(a)

What is El Nino? What are the causes of El Nino? Discuss the scenario of impact of El Nino on global agriculture.

5

Introduction

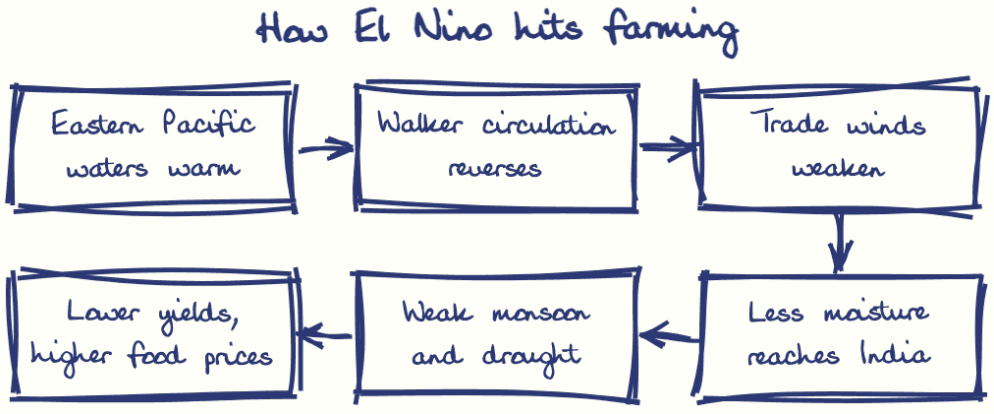
With growing climate variability, El Nino has become crucial to managing agricultural risk. El Nino is a periodic warming of surface waters in the central and eastern tropical Pacific that disrupts normal ocean–atmosphere interactions, altering rainfall and temperature worldwide and affecting crops and food prices.

Understanding El Nino

1. ENSO System: It is the oceanic warm phase of the El Nino–Southern Oscillation (ENSO), with atmospheric pressure changes forming the Southern Oscillation.
2. Global Effects: It typically weakens the Indian monsoon, bringing drought to India and Southeast Asia, while causing heavy rain and floods in the Americas.
3. Duration: Events last about 9–12 months and are often followed by La Nina, which can enhance India's monsoon.

Causes of El Nino

1. **Ocean Warming:** Abnormally warm eastern Pacific waters replace the cold, nutrient-rich Humboldt Current near Peru, disrupting marine ecosystems.
2. **Walker Circulation Reversal:** The warming reverses the Walker circulation, weakening trade winds and shifting atmospheric pressure patterns.
3. **Reduced Moisture to India:** This shift cuts moisture transport to the subcontinent, causing deficient southwest monsoon rain and agricultural drought.
4. **Other Oscillations:** The Indian Ocean Dipole and Southern Oscillation modulate its intensity, complicating prediction.



Impact on Global Agriculture

1. **Production Decline:** Droughts in India and Southeast Asia cut kharif and rabi yields,

causing grain deficits, as in Andhra Pradesh's drought-prone Rayalaseema.

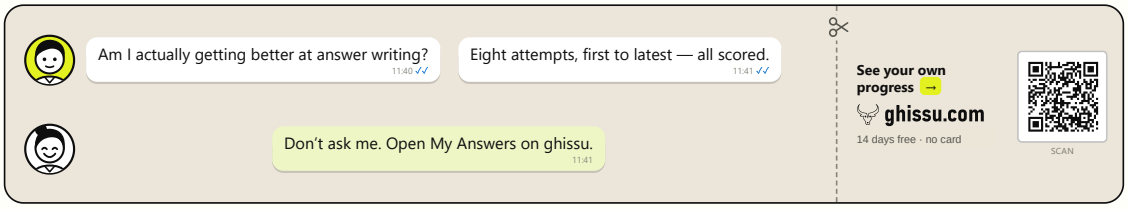
2. Uneven Global Effects: Argentina may see higher maize output, while heavy rains and floods hit the Americas.

3. Economic Consequences: Lower output raises global food prices and inflation, hurting GDP and rural livelihoods in agrarian economies like India.

4. Adaptive Measures: Ecological monitoring, water conservation and the Soil Health Card in Andhra Pradesh promote resilient farming.

Conclusion

El Nino's disruption of global weather threatens farm output and food security, especially in monsoon-dependent regions like Andhra Pradesh. Climate-resilient agriculture, aligned with SDG (2) and SDG (13), can safeguard livelihoods amid climatic uncertainty.



P-IV
15(b)

Describe Impact of Ozone layer depletion on different living organisms?

5

Introduction

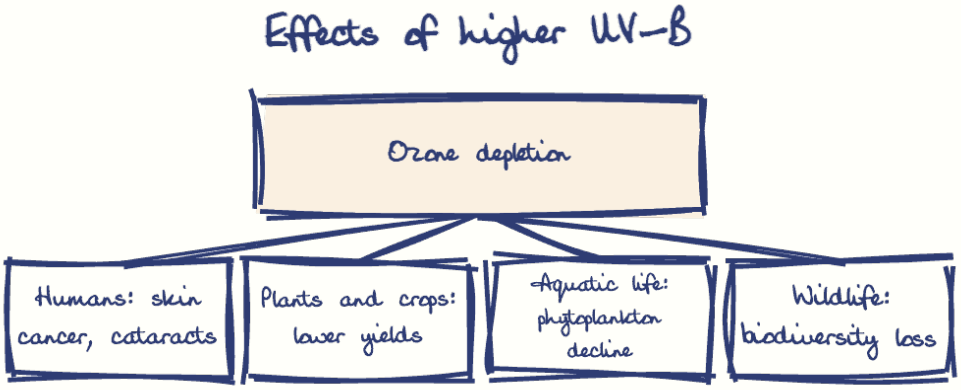
Ozone layer depletion allows more ultraviolet radiation to reach the Earth's surface, posing serious risks to living organisms. The stratospheric ozone layer absorbs 97–99% of harmful UV-B rays, shielding life from DNA damage. Despite the Montreal Protocol, thinning, most severe over the poles, still affects ecosystems and health, especially in Andhra Pradesh's coastal areas where agriculture and aquaculture are vital.

Impact on Living Organisms

1. Human health: Higher UV-B exposure raises skin cancers, including melanoma, and cataracts that impair vision, and suppresses immunity against infection, a serious risk in the Rayalaseema drought belt with limited healthcare.
2. Plants and crops: UV damages plant DNA and photosynthesis, reducing growth; chilli and horticulture lose yield and nutrient content, threatening food security.
3. Aquatic ecosystems: Phytoplankton, the base of aquatic food webs, are highly sensitive to UV-B; their decline hurts fish, marine

biodiversity and fisheries along the 974 km coastline.

4. Wildlife and biodiversity: UV causes genetic mutations and reproductive failures in animals, threatening biodiversity in the Eastern Ghats.



Human health effects are the most immediate, crop and aquatic losses threaten food security, and biodiversity loss erodes long-term ecosystem resilience.

Challenges

1. Enforcement gaps: The ODS (Regulation and Control) Rules, 2000 phase out harmful chemicals, but compliance lags in informal sectors.
2. Awareness deficit: Limited awareness in rural Andhra Pradesh restricts community participation in mitigation.
3. Monitoring deficits: Inadequate local monitoring of UV radiation in coastal and drought-prone

regions limits data-driven policy.

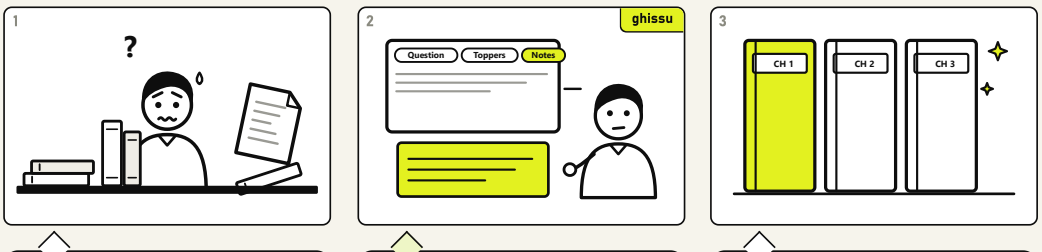
4. Climate interaction: Ozone depletion worsens climate change effects, complicating disaster management where erratic rainfall and temperature shifts already strain farming and water.

Conclusion

Ozone depletion harms human health, agriculture, aquatic life and biodiversity. Stronger ODS enforcement, local research and community awareness can mitigate these impacts, in line with SDG (3) and SDG (15).

A STORY YOU MAY KNOW NOTES THAT LIVE WITH THE QUESTION

Where did I write that?



I KNOW I noted this. Which notebook...?

Note it right on the question.

And chapter notes build themselves.

— Note it where you learnt it.

Keep every note in place →
ghissu.com
14 days free · no card

SCAN

P-IV
15(c)

Describe impact of human beings on water resources.

5

Introduction

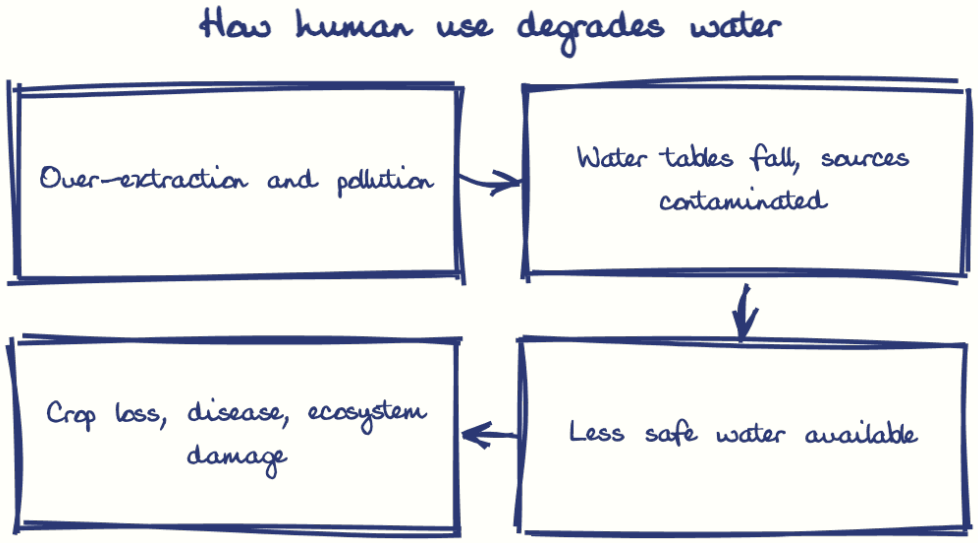
Water resources sustain life, agriculture and industry, yet human activity increasingly strains them. In India, including Andhra Pradesh, population growth, urbanisation and economic development have intensified demand, leading to over-extraction and pollution that threaten both availability and quality.

Human Activities Affecting Water Resources

1. **Groundwater Over-Extraction:** Withdrawal beyond natural recharge for irrigation and urban use has lowered water tables, as in the Rayalaseema drought belt.
2. **Pollution:** Untreated industrial effluents and farm runoff carrying pesticides and fertilisers contaminate surface and groundwater and harm aquatic life.
3. **Large Water Infrastructure:** Projects like the Polavaram Project alter river flows and sediment transport and submerge habitats.
4. **Urbanisation:** Rapid urban growth in cities like Visakhapatnam raises domestic and industrial consumption, often outpacing supply and stressing existing infrastructure.

Candidates must not write on this margin

Candidates must not write on this margin



Consequences for Availability and Quality

1. **Reduced Availability:** Unsustainable extraction and diversion of river flows reduce accessible freshwater, causing scarcity in rural and urban Andhra Pradesh.
2. **Poorer Quality:** Agricultural and industrial pollutants make water unsafe for drinking and irrigation, raising health risks and lowering productivity.
3. **Ecosystem Disruption:** Altered flows and habitat loss from dams and embankments impair natural purification, flood regulation and biodiversity support.

Socio-Economic and Environmental Implications

1. **Agricultural Stress:** Scarcity and poor quality cut crop yields in Rayalaseema and other drought-prone districts, driving rural distress and migration.

Practice 15(c) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2016 · printed in old-syllabus Paper IV, now Paper V
Scan to practice

2. Health Impacts: Waterborne diseases hit women and children in rural areas hardest.

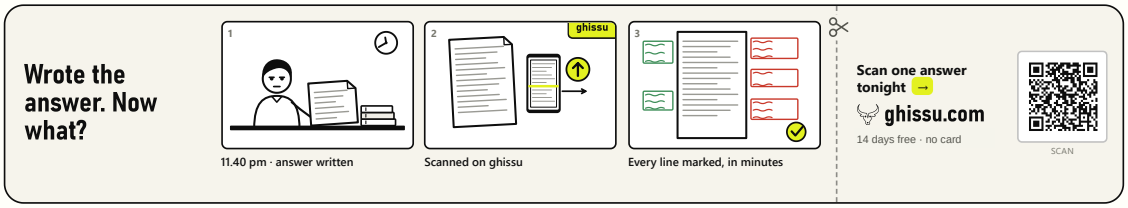
3. Displacement and Conflict: Polavaram displaces over 1 lakh families, causing social upheaval and straining rehabilitation.

4. Environmental Degradation: Loss of wetlands and aquatic biodiversity through altered hydrology and pollution weakens long-term sustainability.

These impacts form a chain: over-extraction and pollution lower water tables and contaminate sources, which leaves less safe water for farms, homes and ecosystems.

Conclusion

Human activity has deeply altered water systems, threatening both availability and quality with wide social and environmental costs. Integrated water resources management and sustainable practices, as under the National Water Mission being implemented in Andhra Pradesh, are essential to secure water and achieve SDG (6.)



APPSC GROUP-I MAINS · PAPER V

2012

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

P-IV
1(d)

Write a short note on the recent development taking place in the most popular National Solar Mission.

10

Introduction

The National Solar Mission, a flagship of the National Action Plan on Climate Change, has gained fresh momentum. With solar capacity at 94.17 GW by November 2024, and new schemes widening rural adoption, it is pivotal for energy security and climate commitments.

Recent Developments

1. **Phased Design:** Three phases (2010–13, 2013–17, 2017–22) with periodic reviews to adjust targets.
2. **New Schemes:** PM Surya Ghar: Muft Bijli Yojana installed over 6.3 lakh rooftop units in nine months; PM-KUSUM spreads solar to farmers, including in Andhra Pradesh.
3. **Manufacturing Push:** The PLI scheme and Make in India boost domestic panels, building technological sovereignty and cutting imports.
4. **Centre–State Collaboration:** MNRE works with states like Andhra Pradesh, which uses its coastal geography for large projects.

Practice 1(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

5. Falling Tariffs: Tariffs fell 65% in a decade, attracting private investment.

Where the National Sdar Mission stands

On track	Lagging
Ground-mounted 64,415 MW	Rooftop 11,870 of 40,000 MW
Tariffs down 65%	Sdar parks only 28.60% built
PM Surya Ghar 6.3 lakh roofs	Storage 5.5 GW, 61 GW needed

Outcomes and Gaps

1. Ground-Mounted Success: 64,415 MW by March 2024 exceeded targets.
2. Rooftop Lag: Only 11,870 MW against a 40,000 MW target, showing uneven progress.
3. Sdar Park Delays: Only 28.60% of 40,000 MW installed, and only 40% of 8,100 crore released, reflecting weak appraisal and monitoring.
4. Storage Gap: 5.5 GW of storage against 243 GW of renewables, 61 GW needed by 2030.

Implications

1. Climate: Supports the NDC to cut emission

intensity by 45% from 2005 levels by 2030.

2. Jobs: New markets and jobs in manufacturing and installation, as in Visakhapatnam's port-led corridor.

3. Environment: Lower fossil use protects drought-prone Rayalaseema.

Way Forward

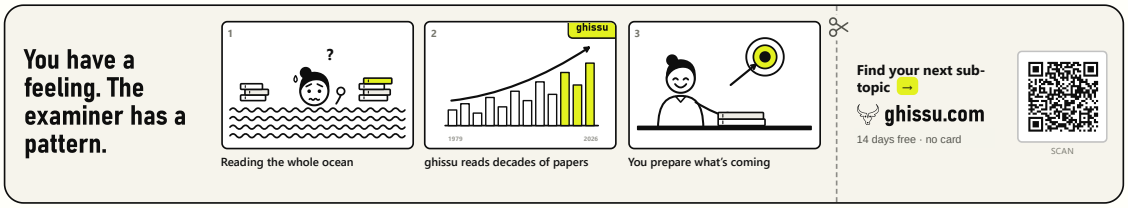
1. Fund Release: Faster disbursement, monitoring and accountability.

2. Rooftop Push: Targeted subsidies and awareness, especially in urban and peri-urban Andhra Pradesh.

3. Storage: Advanced storage and grids to manage intermittency.

Conclusion

The mission has transformed India's solar landscape through ground-mounted success and falling tariffs. Fixing rooftop, solar park and storage gaps will sustain it.



P-IV
1(a)

What are the important technological missions of India? Add a note on their objectives.

10

Introduction

India's technological missions have become pivotal to economic growth, strategic autonomy and social development. With achievements like Chandrayaan-3 and the launch of the India-AI Mission, technology is a decisive lever for India's ambition to be a global innovation hub. The missions address national challenges and aim for leadership in critical fields.

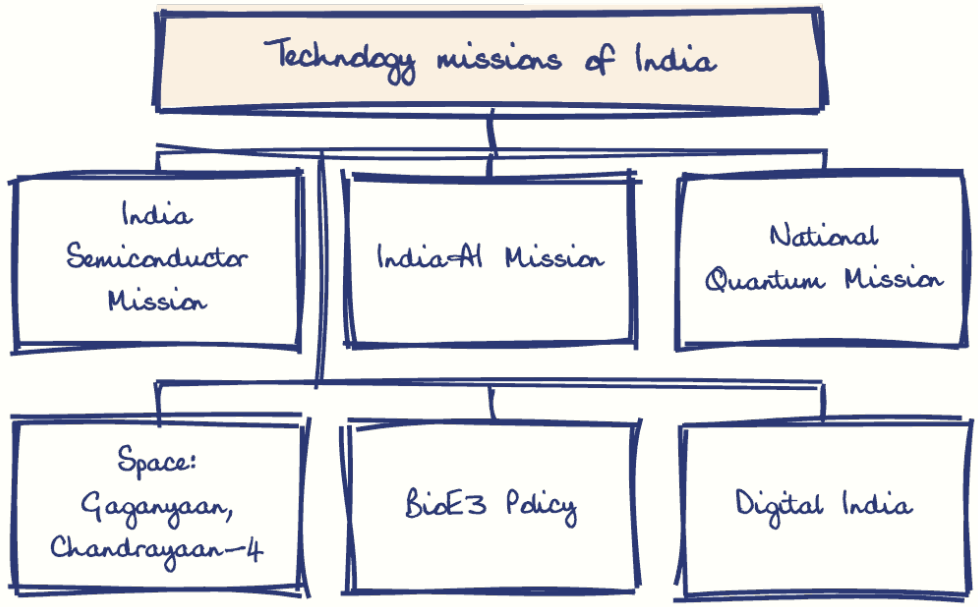
Important Missions and Their Objectives

1. **India Semiconductor Mission:** Develop indigenous chip manufacturing to fix supply chain vulnerabilities, training 85,000 engineers, including from Andhra Pradesh.
2. **India-AI Mission:** Boost AI research and applications; it has deployed over 38,000 GPUs and supports start-ups.
3. **National Quantum Mission:** Accelerate innovation in quantum technology and reduce import dependence.
4. **Space missions:** ISRO at Sriharikota plans Chandrayaan-4, Gaganyaan and the Bharatiya Antariksha Station, with private participation targeting a 15% share of the global space economy.

Practice 1(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

5. BioE3 Policy: Promote bio-based chemicals and precision biotherapeutics.
6. Digital India: DigiLocker and PRAGATI improve service delivery and transparency, including in rural Andhra Pradesh.
7. Defence technology: Indigenous missile systems and electronic warfare reduce reliance on imports.



Common Objectives

Across missions the shared aims are self-reliance, import substitution, national security and inclusive service delivery.

Institutional Support

1. ANRF: The ANRF Act 2023 aligns research

with national priorities and builds capacity in central and state universities.

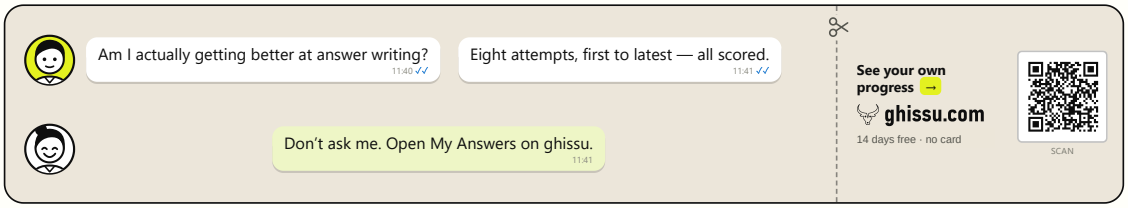
2. Digital health: The Ayushman Bharat Digital Health Mission improves healthcare access.

Way Forward

1. R&D funding: Strengthen research centres and academia–industry links.
2. Skills: Train talent in AI, quantum and semiconductors to reduce brain drain.
3. Responsible innovation: Build regulatory frameworks for the ethical, privacy and security concerns of emerging technologies.
4. Private participation: Encourage more private players in mission programmes to speed commercialisation.

Conclusion

These missions move India from technology adoption to creation, fostering self-reliance and strategic sovereignty. By joining innovation with inclusive development, they build resilient infrastructure in line with SDG (9.)



P-IV
1(b)

Write an essay on National Literacy Mission.

10

Introduction

The National Literacy Mission (NLM), launched in 1988 by Prime Minister Rajiv Gandhi, was a pivotal effort against adult illiteracy, targeting functional literacy for 80 million people aged 15–35. Illiteracy was then a major barrier to socio-economic development and empowerment. As India pursues inclusive growth and digital empowerment, the NLM's legacy remains relevant, including in Andhra Pradesh.

Institutional Framework

1. Comprehensive setup: Central and state governments, voluntary agencies, universities and youth centres collaborated to reach diverse groups, including Scheduled Castes and Scheduled Tribes.

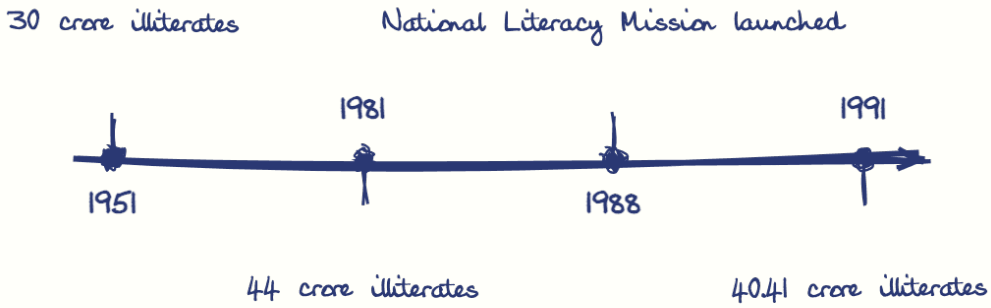
2. Voluntary agencies: Over 500 voluntary organisations took part, with several active in Andhra Pradesh, reflecting the mission's stress on community participation.

3. Resource centres: 16 state resource centres and Shramik Vidyapeeths provided technical support and worker education.

4. Continuing education: Post-literacy work through

libraries and reading rooms, backed by media and academic evaluation, linked literacy to lifelong learning.

Adult illiteracy and the NLM



Challenges and Outcomes

1. Scale of the problem: Rapid population growth raised illiterates from 30 crore in 1951 to 44 crore in 1981, the number fell to 40.41 crore by 1991, showing slow progress.
2. Gender barriers: Social mindsets held back female literacy; Beti Bachao Beti Padhao and separate toilets for girls under the Swachh Vidyalaya Initiative have since helped.
3. Declining volunteerism: The voluntary spirit waned, reducing effectiveness.
4. Social impact: Awareness of health, rights and family planning improved, though unevenly, and

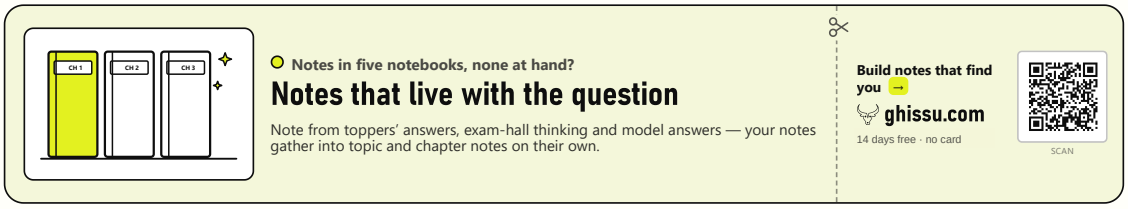
was reinforced by Saakshar Bharat and Padhna Likhna Abhiyan.

Way Forward

1. Community engagement: Revive volunteerism and local participation in rural and tribal areas of Andhra Pradesh.
2. Technology: Use Digital India and BharatNet to bring adult learning to remote and marginalised groups.
3. Female literacy: Tackle socio-cultural barriers through safe schools and incentives.
4. Skills: Link literacy to livelihoods under Deendayal Antyodaya Yojana to motivate adult learners.

Conclusion

The NLM laid the foundation of India's adult education. Sustained and linked to technology, gender equity and skills, literacy efforts can advance SDG 4 and informed citizenship.



P-IV
1(c)

Enumerate the policy objectives in the development of Science and Technology in India.

10

Introduction

Science and technology are pivotal to India's economic growth, strategic autonomy and social development, given global technological competition and the digital revolution. With the India Semiconductor Mission and ISRO's Sriharikota facility, India aims to use innovation for inclusive progress. Its policy objectives reflect a vision of becoming a global innovation hub.

Key Policy Objectives

1. Strategic autonomy and self-reliance: Reduce dependence on foreign technology, as stressed in the Technology Policy Statement of (1983) and Atmanirbhar Bharat, by developing semiconductors and space technology at home.

2. Innovation and research: Through the Atal Innovation Mission and SERB funding, reach the top (50) of the Global Innovation Index and place five institutions in the world's top (100).

3. Economic and industrial growth: Boost technology-driven manufacturing through Make in India, corridors like Visakhapatnam-Chennai use

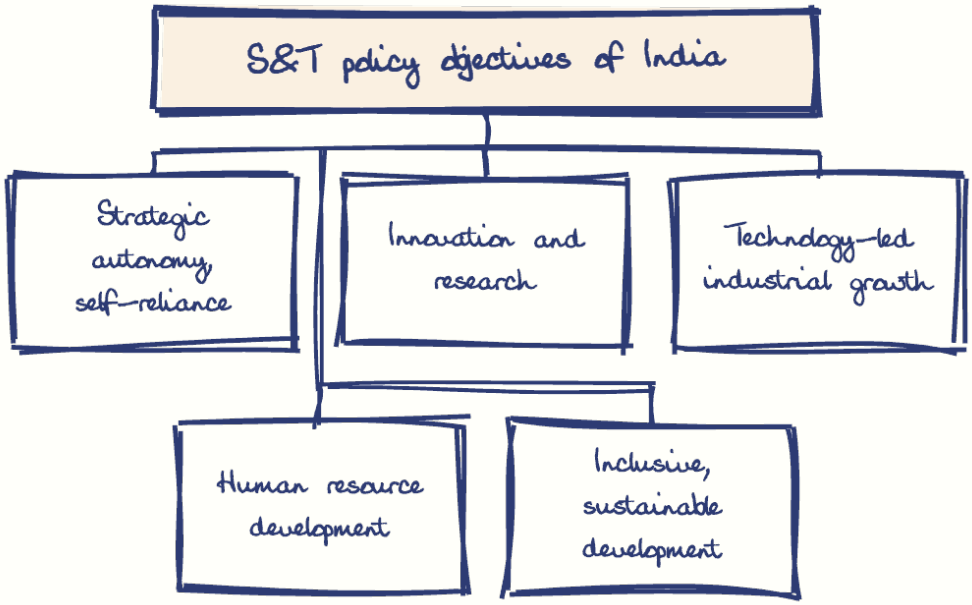
Practice 1(c) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

technology for competitiveness.

4. Human resource development: Train and retain talent, including women scientists, as technical manpower grows about 9% a year through IITs and IISc.

5. Inclusive and sustainable development: Digital India improves governance for all, while renewable energy balances growth with ecology.



How the Objectives Shape Strategy

1. Innovation ecosystem: Technology parks, research clusters and PPPs link academia, industry and government for translational research.

2. Investment target: Spending 2% of GDP on R&D, equally public and private.

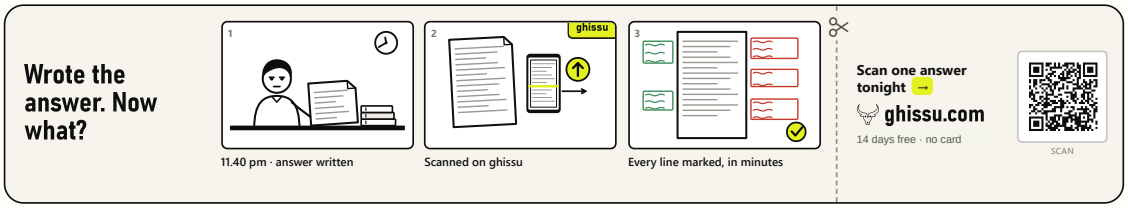
3. Strategic sectors: Space, defence, electronics and biotechnology get priority.
4. State strengths: Andhra Pradesh's aquaculture technology and Vizag pharma capacity feed national goals.

Way Forward

1. R&D funding: Increase and diversify funding for frontier technologies and research infrastructure.
2. Skills: Expand specialised training in AI and quantum computing and retain talent through incentives.
3. Industry–academia links: Speed up adoption and commercialisation of research.
4. Sustainable technologies: Prioritise green and inclusive digital solutions.

Conclusion

These objectives aim to build a self-reliant, innovative and inclusive technological ecosystem, making India a global leader in science and technology.



P-IV
2(d)

What is the resolution of CARTOSAT-2 and for what purpose are its images useful? What is the main difference between CARTOSAT-2 and IKONOS?

10

Introduction

The Cartosat series, developed by ISRO, marks a major advance in India's remote sensing, crucial for land management, urban planning and national security. As demand for precise geospatial data grows in agriculture, infrastructure and defence, Cartosat-2's high-resolution imagery has become indispensable.

Resolution of Cartosat-2

Cartosat-2 offers a spatial resolution of about 0.8 metres, enabling detailed observation of terrestrial features and accurate mapping of land boundaries and urban infrastructure.

Purposes for Which Its Images Are Useful

1. Cartography and Planning: Cartographic mapping, urban and rural planning and coastal land use regulation, useful for Andhra Pradesh's long coastline.

2. Utility Management: Monitoring road networks and water distribution.

3. Land Records: Precise geo-referenced data supports DILRMP, which has digitised about 95% of

land records, and the SVAMITVA Scheme.

4. National Security: Continuous surveillance of sensitive areas and tactical battlefield monitoring.

Difference between Cartosat-2 and IKONOS

1. Resolution: IKONOS offers about (1) metre, slightly coarser than Cartosat-2's (0.8) metres.
2. Operational Focus: IKONOS, a US satellite, was one of the first commercial high-resolution satellites giving global imagery. Cartosat-2 is tailored to India's strategic and developmental needs.
3. Origin and Cost: Cartosat-2 is indigenous, while IKONOS means dependence on foreign data at higher cost.

Cartosat-2 vs IKONOS

Cartosat-2	IKONOS
About 0.8 metre resolution	About 1 metre resolution
Indigenous, ISRO	US commercial satellite
Land governance, security	Global commercial imagery
No foreign data cost	Costlier foreign data

Limitations

Optical sensors cannot see through cloud or at night, which RISAT radar covers, and a four-day revisit delays real-time monitoring.

Way Forward

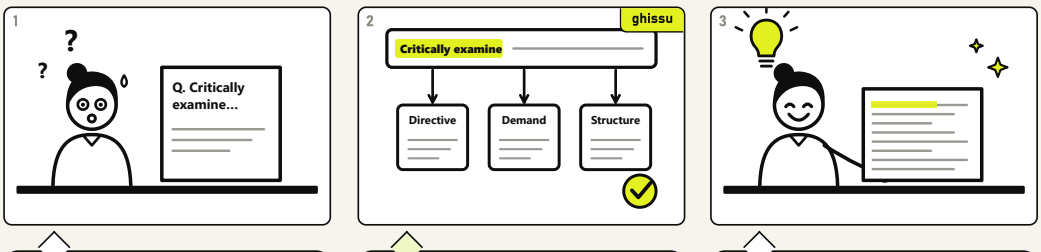
Small satellite constellations with sub-metre imagery and higher revisit rates, combined with radar data, can fill these gaps for disaster management and precision agriculture.

Conclusion

Cartosat-2 reflects technological self-reliance, serving land governance and security alike.

A STORY YOU MAY KNOW · EXAM-HALL THINKING

Three minutes staring at one word



"Critically examine"... what do they even want?

Directive, demand, structure — laid out first.

Oh. Counter-view, then my judgement. Easy.

— Walk into the hall with a method, not a memory.

Walk in with a method →
ghissu.com
14 days free · no card

SCAN

P-IV
2(a)

What is a Satellite? Explain the different types of Satellites in the Space programme. Add a note on GSAT-5.

10

Introduction

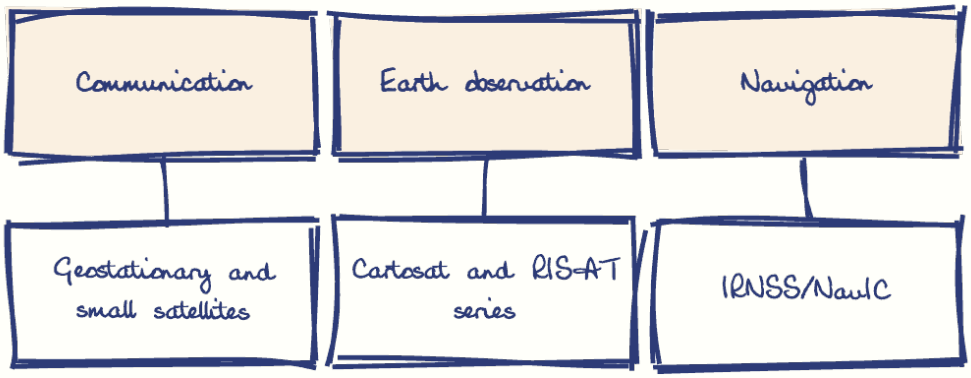
A satellite is an artificial object placed in orbit around Earth or another celestial body to collect data, relay communication or perform scientific observations. With ISRO's launches from Sriharikota, satellites have become critical infrastructure for telecommunication, navigation, earth observation and defence.

Types of Satellites in the Space Programme

1. **Geostationary Satellites:** Orbiting about 36,000 km above the equator, they stay fixed relative to Earth, giving wide coverage for communication and missile tracking, but suffer high latency.
2. **Small Satellites:** Under 500 kg and in LEO between 500 and 1,500 km, they offer lower latency and real-time data, and are cheaper and replaceable with the SSLV.
3. **Earth Observation Satellites:** The Cartosat and RISAT series provide imaging and Synthetic Aperture Radar data.
4. **Navigation Satellites:** IRNSS or NavIC provides

precise positioning services.

Types of Satellites in India's Space Programme



Performance and Challenges

1. Launch Capacity: India launches 5–7 satellites a year with high success; CMS-03 is the heaviest satellite launched from Indian soil.
2. Latency Trade-offs: GEOSATs cover wide areas but lag, while LEO SmSats cut latency but need constellations.
3. Bandwidth Gap: GSAT satellites serve DTH and broadband, yet India still leases much bandwidth.
4. Defence: GSAT-7 and RISAT strengthen C4ISR capability.

Note on GSAT-5

GSAT-5 is part of the GSAT series of

geostationary communication satellites, designed for telecommunication, broadcasting and broadband services. It is intended for satellite telephony, television and internet, helping bridge the digital divide in remote areas, with transponders in multiple frequency bands for civilian and strategic users.

Conclusion

India balances large, high-capacity satellites with small, agile ones. High-throughput satellites like GSAT-11 and GSAT-29 can reduce dependence on leased bandwidth.

FIVE-MINUTE LESSON TOPPERS' ANSWERS, PER QUESTION

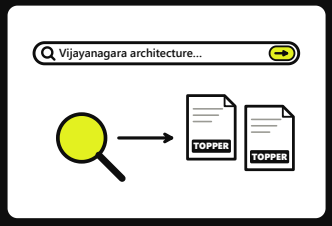
Read a topper copy like an examiner

Don't read a topper's answer for its facts. Read it for its moves — they transfer to any question.

1. **First two lines** — How does the intro frame the demand?
2. **Headings** — How is the body split — by dimension, by actor, by time?
3. **Proof** — Which fact, report or case holds up each paragraph?
4. **Last two lines** — What way forward, and how specific is it?

ON GHISSU, THIS IS DONE FOR YOU — ON EVERY QUESTION.

1.5 lakh+ topper answers are matched to each question — with their intro, structure and conclusion laid bare.



Search any question →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 2(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

P-IV
2(b)

How many satellites are tracked at a time for a GPS? Explain how and for what purpose the GPS is used.

10

Introduction

The Global Positioning System (GPS) has become indispensable, underpinning navigation, communication and strategic operations worldwide. India's indigenous NavIC complements it to ensure resilience and autonomy, with uses from defence to governance.

How Many Satellites Are Tracked

The GPS constellation consists of (24) operational satellites, but a receiver typically tracks signals from (4) to (12) satellites simultaneously. At least four are needed for an accurate position, and tracking more adds redundancy and precision in urban or mountainous areas.

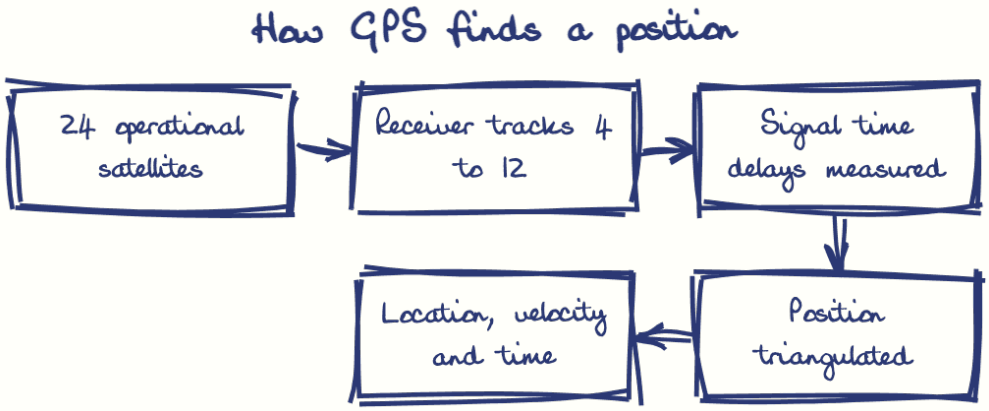
How GPS Works

1. Signal Timing: The receiver uses the time delay of signals from several satellites to triangulate exact location, velocity and time.
2. Real-Time Accuracy: Tracking many satellites at once improves reliability for real-time navigation and timing.
3. Interoperability: Devices combine GPS with other

Practice 2(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

GNSS such as GLONASS, Beidou and NavIC to improve coverage.



Purposes for Which GPS Is Used

1. **Defence:** Guiding precision munitions and troop movements; the military also uses NavIC to avoid dependence in crises.
2. **Civil Aviation:** GAGAN, developed by ISRO and the Airports Authority of India, augments GPS for safer air traffic management.
3. **Disaster Management and Governance:** Search and rescue, resource mapping, drought management in Rayalaseema and coastal monitoring along the 974 km shoreline.
4. **Commercial Use:** Smartphone navigation, fleet management and location-based services.

Candidates must not write on this margin

Candidates must not write on this margin

Challenges

1. Signal Vulnerability: Weak signals are prone to jamming in dense areas.
2. Strategic Risk: Dependence on foreign GNSS is risky in geopolitical tensions, so India builds the seven-satellite NavIC constellation.

Way Forward

1. Expand NavIC: More satellites and ground infrastructure for wider coverage.
2. Augmentation: Strengthen GAGAN for local needs.
3. Cybersecurity: Anti-jamming and anti-spoofing protection.

Conclusion

Expanding NavIC, strengthening GAGAN and adopting anti-jamming measures will secure navigation for defence and development.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths →
ghissu.com
14 days free · no card
SCAN

P-IV
2(c)

Define band width. What are the different band widths used in INSAT series? Mention the significance of thermal band in Agriculture.

10

Introduction

Satellite communication and remote sensing have become crucial for India's development, especially in agriculture and disaster management. The INSAT series provides communication, broadcasting and meteorological services across the country, and its bands and thermal sensing have direct uses for farmers.

Definition of Bandwidth

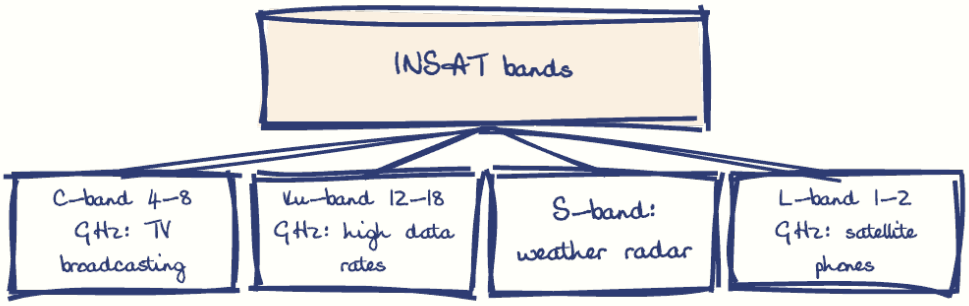
Bandwidth is the range of frequencies within a given band that a satellite communication channel can use to transmit data. It determines the data transmission capacity.

Bands Used in the INSAT Series

1. C-Band (4–8 GHz): Widely used for television broadcasting and satellite feeds, less affected by rain fade, so reliable.
2. Ku-Band (12–18 GHz): Supports higher data rates, used in GSAT-7A for military and civilian links between radar stations and airbases.

3. S-Band: Used for weather radar, some communication satellites and ship radar functions.
4. L-Band (1-2 GHz): Used for satellite phones and GPS carriers at sea, air and land, with low data rates but little rain fade.

Bands used in the INSAT series



Significance of Thermal Band in Agriculture

1. Water Stress Detection: Thermal bands capture infrared radiation from crops; canopy temperature reveals crop water stress and guides irrigation.
2. Disease Monitoring: Temperature variations flag diseases early, preventing crop loss.
3. Crop Health Assessment: Non-invasive monitoring of transpiration and stomatal conductance supports high-throughput phenotyping in breeding.
4. Yield Prediction: Data on drought response helps select stress-tolerant varieties and improve yield

forecasts.

Application in Andhra Pradesh

1. **Crop Insurance:** Thermal data improves damage assessment and claim accuracy under PMFBY.
2. **Extension:** Krishi Vigyan Kendras help farmers interpret thermal imagery for water and disease management.
3. **Cost Barrier:** Costly equipment and expertise limit small and marginal farmers, needing subsidies and support.

Way Forward

KVK awareness drives, user-friendly platforms linking INSAT thermal data to local advisories, and subsidies for smallholders will widen use.

Conclusion

INSAT uses C, Ku, S and L bands for varied needs, while the thermal band enables precision farming. Wider farmer access can raise productivity and food security.

REVIEWER'S NOTE · VALUE ADDITION

Examiners read hundreds of answers a day. The one with a number, a committee and a case is the one they remember.

ghissu

ghissu gives you the data point, the report and the example for every question — each linked to its source.

Get the value adds for any question →

ghissu.com

14 days free · no card

SCAN

P-IV
3(d)

Write an essay on the modem, wireless, and types of networking systems.

10

Introduction

India's digital connectivity has expanded rapidly under the National Broadband Mission and Digital India. With over 1.3 billion telecom subscribers and broadband crossing one billion by late 2025, modems and wireless systems form the backbone of this revolution and help bridge the urban–rural digital divide.

Modem, Wireless and Types of Networks

1. **Modem:** A modulator–demodulator converts digital data into analog or carrier signals for telephone lines, cable, fibre or radio, and back again; DSL, cable, fibre and 4G/5G modems are common types.

2. **Wireless Systems:** Data travels by radio waves, microwaves or infrared through Wi-Fi, Bluetooth, cellular and satellite links, using bands from kilohertz to gigahertz, with spectrum allocated on TRAI advice.

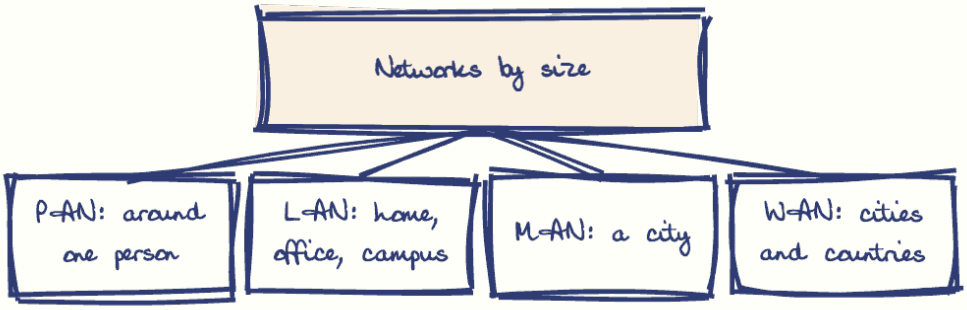
3. **Types of Networks:** A PAN links devices around one person, a LAN covers a home, office or campus, a MAN spans a city and a WAN such as the internet connects countries;

Practice 3(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

WL-AN, WW-AN and WP-AN are their wireless forms.

Types of networks by size



Institutional Framework

1. National Broadband Mission: Launched in 2019, with over 2.15 lakh Gram Panchayats connected under BharatNet.
2. Private Sector: Over 92% of subscribers, driving modem and wireless deployment.
3. Standards: IEEE developed security protocols like WEP.

Challenges

1. Coverage Gaps: Rural tele-density in Rayalaseema lags urban areas.
2. Security Risks: Eavesdropping and denial of service attacks exploit open-air transmission.
3. Digital Divide: Data fell from Rs 269 per GB

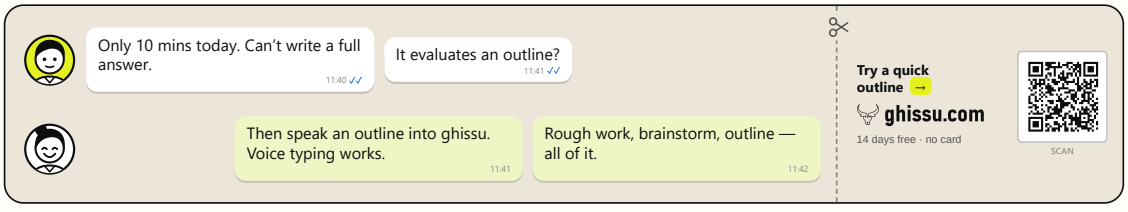
in 2014 to Rs 8–10 in 2025, yet only 28% of rural women have internet.

Applications

1. Agriculture: Wireless sensors aid precision farming and aquaculture.
2. Governance: Common Service Centres give digital access to welfare schemes and legal aid through tele-law.
3. Health and Education: Telemedicine and online learning reach underserved people, a need highlighted by COVID-19.

Conclusion

Wider fibre, stronger end-to-end encryption and digital literacy will make networking inclusive.



P-IV
3(a)

Discuss briefly the role of IT in rural development in India with a note on effective implementation.

10

Introduction

Information technology has become a cornerstone of rural development, especially in the post-pandemic era when connectivity and digital services became vital. With the Digital India Programme and BharatNet bridging the rural-urban divide, IT is transforming agriculture, governance, education and healthcare for the majority of Indians who live in villages.

Role of IT in Rural Development

1. **Digital Infrastructure:** BharatNet has connected over 2.15 lakh Gram Panchayats with optical fibre, enabling service delivery at the grassroots, including in remote Andhra Pradesh.

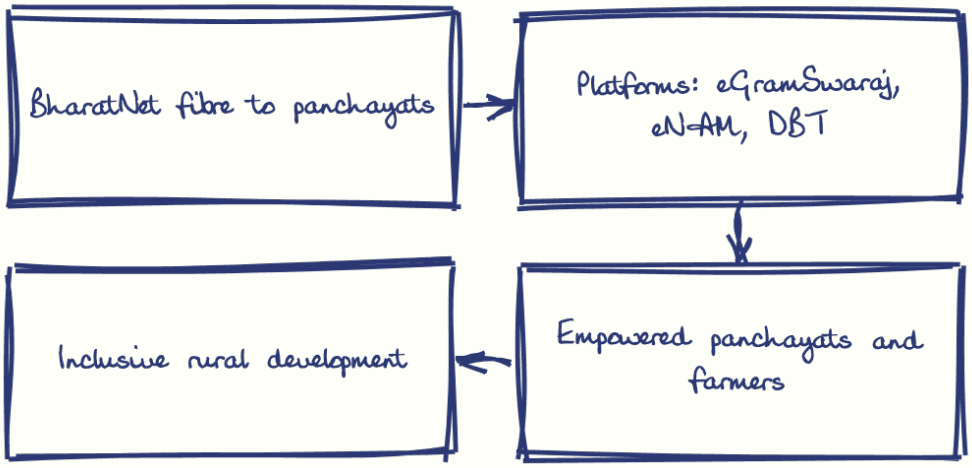
2. **Local Governance Platforms:** eGramSwaraj has onboarded over 2.53 lakh panchayats for transparent governance and fund management, while AIKosh and BHASHINI support local languages and decisions.

3. **Institutional Capacity:** MeitY's India AI Governance Guidelines and Andhra Pradesh's land records modernisation under DILRMP build IT-enabled rural administration.

4. Agriculture: eNAM and the Kisaan Knowledge Management System bring market access and advisory services to farmers, helping them sell better and farm better.

5. Service Delivery: Direct Benefit Transfers have reduced leakages in LPG subsidy and MGNREGA payments.

IT chain in rural development



Implementation Challenges

1. Literacy Gaps: Despite infrastructure, low digital literacy among farming communities, as in Andhra Pradesh, limits the use of eNAM and welfare platforms.

2. Connectivity and Gender: Slow, unreliable internet hampers e-governance and telemedicine; only 28% of rural women have internet access.

3. Usability: Complex systems like GST filing burden small rural entrepreneurs.

4. Uneven Outcomes: Digital welfare platforms in Andhra Pradesh have improved monitoring, but audits still reveal gaps in execution and fund utilisation.

Ensuring Effective Implementation

1. Simplify Platforms: Intuitive, vernacular interfaces for rural users.

2. Digital Literacy: Scale up PMGDISHA with a focus on women and marginalised groups.

3. Connectivity: Accelerate BharatNet Phase-II and PM-WANI for affordable high-speed internet.

4. Participatory Governance: Use MyGov and grievance redressal systems so that initiatives meet local needs and stay accountable.

Conclusion

IT can transform rural governance, farming, education and health, the chain running from infrastructure to platforms to empowered panchayats. Effective implementation that tackles literacy, connectivity and usability supports Atmanirbhar Bharat and SDG (9.)

Which sub-topic comes next?
Trend analysis for every syllabus topic.
ghissu.com →

Write your own answer to 3(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
3(b)

Define Telemedicine. Write a note on the different types of telemedicine.

10

Introduction

The COVID-19 pandemic accelerated telemedicine in India as distancing limited access to facilities. The Ministry of Health and NITI Aayog issued Telemedicine Practice Guidelines in 2020, enabling millions to consult doctors remotely and helping bridge the urban-rural divide.

Definition of Telemedicine

1. **Meaning:** Telemedicine is the remote delivery of healthcare through electronic communication, allowing diagnosis, treatment and management without in-person visits.
2. **Formal Definition:** It is the use of ICT by healthcare professionals to exchange valid information for diagnosis, treatment, prevention, research and continuing education, where distance is a critical factor.
3. **Versus Telehealth:** Telehealth covers health education and information; telemedicine involves clinical care.

Types of Telemedicine

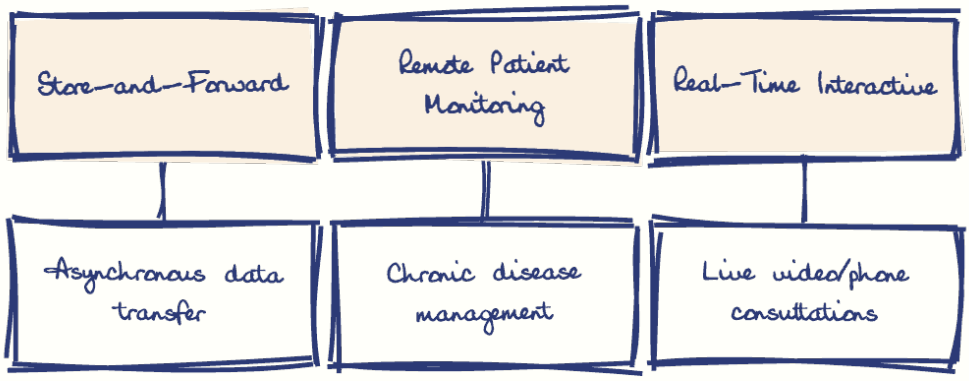
1. **Store-and-Forward:** Asynchronous, medical

history, lab reports and images are sent securely to specialists for later analysis. It is cost-effective where connectivity is weak, but delays urgent care. Primary health centres forward data to hospitals in Visakhapatnam and Vijayawada.

2. Remote Patient Monitoring: Wearables and sensors collect health data outside hospitals, enabling chronic disease management and early detection of complications. Infrastructure, privacy and patient literacy are hurdles; Andhra Pradesh's mobile health units in tribal areas show its potential.

3. Real-Time Interactive: Live video or phone consultations allow immediate diagnosis and build patient trust, but need robust internet and raise licensure and security concerns. eSanjeevani OPD, widely used in Andhra Pradesh, connected rural patients with urban specialists during the pandemic.

Types of Telemedicine



Practice 3(b) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Way Forward

1. Digital Infrastructure: Expand high-speed internet and telemedicine facilities in rural areas.
2. Data Privacy: Protection laws and consent mechanisms to build trust.
3. Scheme Integration: Use eSanjeevani within state health missions to complement physical care.

Conclusion

With better rural internet, data privacy frameworks and uniform licensure, telemedicine can democratise healthcare access.

FIVE-MINUTE LESSON MY ANSWERS — YOUR PROGRESS

Measure progress, don't guess it

Writing more is not the same as writing better. This is how to know which one you're doing.

1. **Score everything** — Even a rough attempt gets a number.
2. **Rewrite** — The same question, one week later.
3. **Compare** — Intro, dimensions, conclusion — side by side.
4. **One fix at a time** — Track a single weakness until it's gone.

ON GHISSU, THIS IS DONE FOR YOU — ON EVERY QUESTION.

My Answers keeps every attempt, every score and every enhanced version on one timeline.

Watch your curve →

ghissu.com

14 days free · no card

SCAN

P-IV
3(c)

Which are the five major IT city hubs in India? What are the main factors required for the growth of software industry?

10

Introduction

India's IT sector is a key driver of growth, innovation and global competitiveness, contributing about 8% of GDP and employing over 5 million people. Its concentration in a few cities reflects policy support, skilled human capital and infrastructure, which explains both the hubs and the factors behind them.

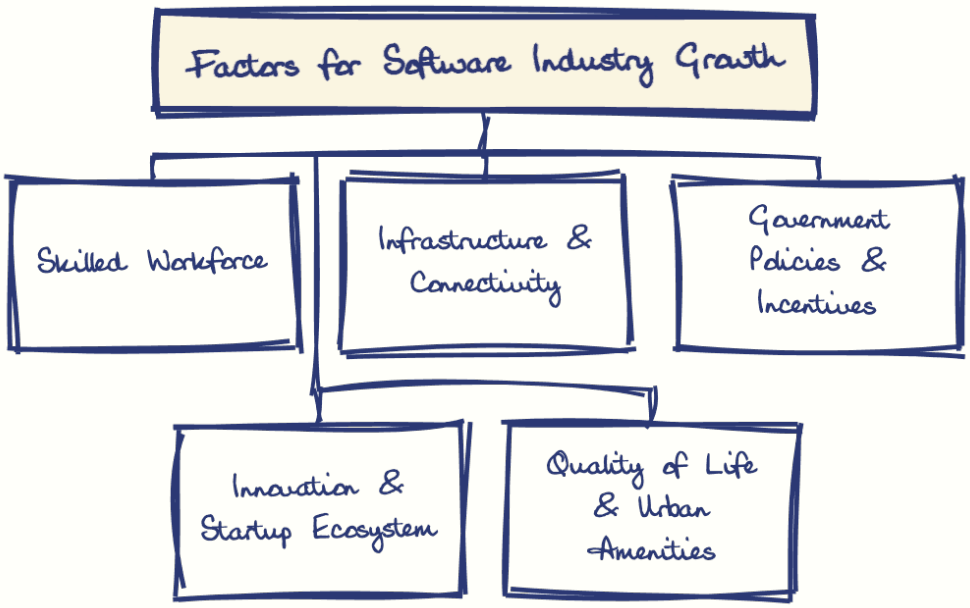
Five Major IT Hubs

1. Bengaluru: The Silicon Valley of India, led by Karnataka's early IT policy, institutions like IISc and IIIT-B, early MNCs such as IBM and Microsoft, and IT corridors with metro links.
2. Hyderabad: Rapid growth through HITEC City and Genome Valley, strong digital infrastructure, low disaster risk and a growing startup culture.
3. Pune: Anchored in strong educational institutions and nearby manufacturing, with a moderate climate and incentives.
4. Chennai: A strong industrial base and port

connectivity aid software exports, backed by a skilled workforce.

5. Delhi NCR: Gurgaon and Noida gain from nearness to the capital, connectivity, IT parks and SEZs.

Factors for Software Industry Growth



1. Skilled Workforce: Premier institutions and continuous upskilling.
2. Infrastructure: Reliable power, water and digital connectivity.
3. Policy Support: Tax incentives, SEZs and the IT Acts of 2000 and 2008 attract investment.
4. Startup Ecosystem: Fosters innovation and

entrepreneurship.

5. Quality of Life: Climate, housing and transport attract and retain talent.

Challenges

Congestion and water scarcity strain big hubs, and about 70% of IT exports come from seven cities, causing regional imbalance.

Conclusion

Sustainable planning and STPI parks in emerging cities like Visakhapatnam and Amaravati can spread the gains of the software industry more evenly.

FIVE-MINUTE LESSON NOTES THAT LIVE WITH THE QUESTION

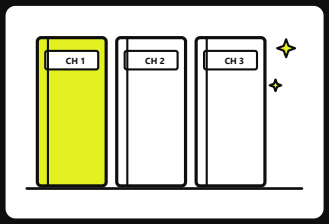
Notes that come back when you need them

Most notes are written once and never found again. Four habits fix that.

1. **Where** — On the question that taught you the point.
2. **How long** — One line. If it needs a paragraph, it's two notes.
3. **Tag** — With its syllabus sub-topic.
4. **Revise** — By chapter, not by the date you wrote it.

ON GHISSU, THIS IS DONE FOR YOU — ON EVERY QUESTION.

Every question has a notes tab — and your notes gather into topic and chapter notes by themselves.



Build notes that find you →

ghissu.com

14 days free · no card

SCAN

P-IV
4(d)

Define nuclear energy. Discuss briefly on the mitigation of nuclear hazards and nuclear waste.

10

Introduction

Nuclear energy is energy released through controlled nuclear fission in reactors. It is critical to India's clean energy transition, and its low-emission, reliable nature suits the NDC goal of 40% clean capacity by 2030. The push for Small Modular Reactors makes managing hazards and waste urgent.

Institutional Framework

1. Regulatory Oversight: The Atomic Energy Regulatory Board enforces safety standards at mines, fuel plants and reactors through licensing, monitoring and emergency preparedness.
2. Liability Regime: The SHANTI Bill, 2025 sets a tiered liability framework based on reactor capacity, ensuring compensation and accountability.
3. Global Norms: As a founding IAEA member, India aligns with international safety standards.
4. Disaster Coordination: State disaster management authorities coordinate emergency response to protect people and ecosystems.

Mitigating Nuclear Hazards

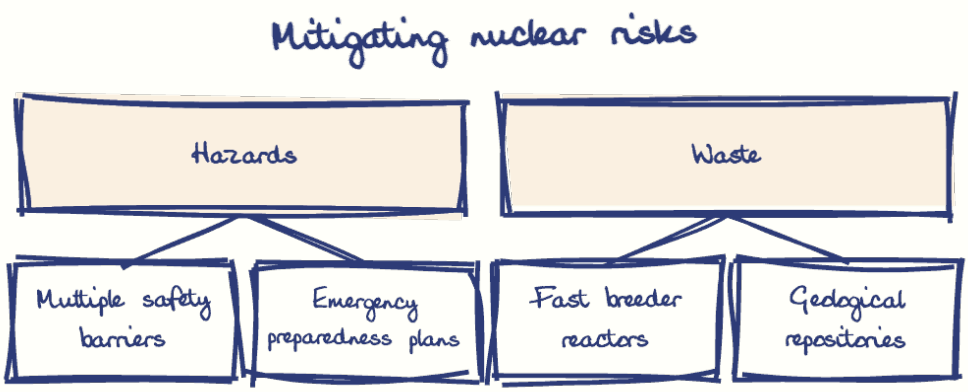
1. **Safety Layers:** Multiple barriers prevent core damage and radiation leaks, with real-time monitoring and rapid response.
2. **Emergency Preparedness:** On-site and off-site plans are regularly updated to limit public exposure.
3. **Public Trust:** Concerns persist near nuclear sites, including Andhra Pradesh's coastal belt, so transparent communication is needed to build social licence.

Managing Nuclear Waste

1. **Long-Lived Waste:** Spent fuel and radioactive waste stay radiotoxic for long periods, making safe handling a critical challenge.
2. **Advanced Reactors:** Fast Breeder Reactors and High Temperature Gas-Cooled Reactors reduce waste and use fuel better.
3. **SMR Research:** Waste disposal is costly and complex, Rs 20,000 crore under the Nuclear Energy Mission funds safer modular designs with easier waste handling.

Way Forward

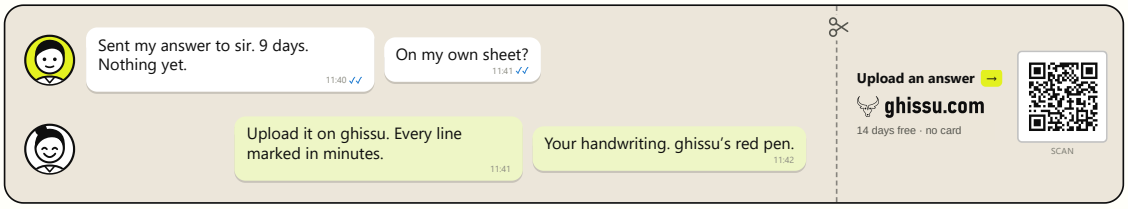
1. **Stronger AERB:** More technical and human resources.



2. Indigenous R&D: Faster SMR and waste recycling research.
3. Disposal: Long-term geological repositories and better interim storage.
4. Engagement: Awareness and participatory decisions.

Conclusion

Nuclear energy can meet India's growing demand and Paris commitments if hazards and waste are managed well. Institutional reform, innovation and inclusive governance will build a resilient sector supporting Atmanirbhar Bharat.



P-IV
4(a)

What is the difference between renewable energy and non-renewable energy? What are the different types of non-conventional energy? Add a note on their resources.

10

Introduction

Countries are trying to balance growth with sustainability. India, with rising demand and ambitious renewable targets, shows this transition. The renewable versus non-renewable divide shapes policy for energy security and lower emissions, including in Andhra Pradesh.

Difference Between Renewable and Non-Renewable Energy

Renewable vs non-renewable energy

Renewable	Non-renewable
Replenished by natural cycles	Finite and exhaustible
Low pollution	High carbon emissions
High initial, low running cost	Established but price volatile

1. **Replenishment:** Renewable energy (solar, wind, biomass) is renewed by natural cycles; coal, oil and gas cannot be replaced on a human timescale.

Practice 4(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

2. Environment: Renewables pollute little; fossil fuels drive air pollution and climate change.
3. Availability: Solar energy is inexhaustible on a human scale; fossil reserves face depletion, raising security concerns.
4. Economics: Renewables need high initial investment but save over time; fossil fuels dominate through established infrastructure but face price volatility, as in global oil markets.

The comparison shows a trade-off between sustainability and today's reliance on fossil fuels, which calls for a gradual shift to renewables.

Types and Resources of Non-Conventional Energy

1. Solar: Harnessed by photovoltaic cells or solar thermal systems; abundant in high-insolation Andhra Pradesh, whose solar parks add to India's 140 GW (January 2026).
2. Wind: About 54.65 GW installed, with strong potential along Andhra Pradesh's coast.
3. Biomass: Crop residues, animal waste and forest products; over 70% of people depend on biomass fuels, and Andhra Pradesh's farm economy supplies ample residue.
4. Hydropower: The Godavari and Krishna support

small and large projects.

5. Geothermal and Tidal: Earth's heat and ocean tides, still underdeveloped.

6. Fuel Cells and Hydrogen: Fuel cells give clean power with water as the only byproduct, the National Green Hydrogen Mission aims to scale hydrogen.

Conclusion

Renewables trade higher upfront cost for clean, lasting supply. These diverse non-conventional sources reduce fossil fuel dependence and strengthen energy security and sustainability.

FIVE-MINUTE LESSON EXAM-HALL THINKING

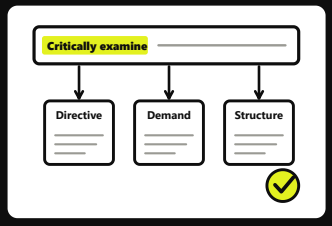
Decode the directive before you write

The directive word is an instruction. The marking scheme follows it — so should your answer.

1. **Discuss** — Both sides, then a verdict. The verdict carries marks.
2. **Critically examine** — The counter-view is compulsory — then commit to a judgement.
3. **Examine** — Causes, features and consequences, kept separate.
4. **Elucidate** — Make it clear with one example for every point.

ON GHISSU, THIS IS DONE FOR YOU — ON EVERY QUESTION.

Exam-hall thinking decodes the directive and the demand for every PYQ — Full for beginners, Crux for veterans.



See the thinking before the answer →

ghissu.com

14 days free · no card

SCAN

P-IV
4(b)

What are the factors for the increase in the worldwide energy demand? Discuss briefly to mitigate such demands.

10

Introduction

Global energy demand has surged in recent decades, driven by economic growth and urbanisation and technological advance. This threatens sustainable development and climate goals. India, with its expanding population and industrial base, shows these dynamics clearly.

Factors Behind Rising Energy Demand

1. **Growth and Urbanisation:** Rapid GDP growth and urban expansion, with urban population expected to double by 2030, raise industrial and household demand, as rising electricity use in Visakhapatnam shows.

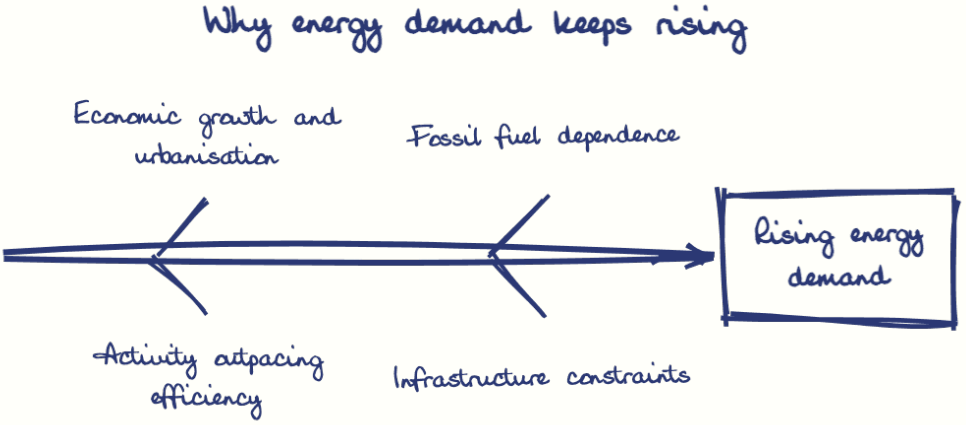
2. **Activity Effect:** Despite a 15% efficiency gain since 2000, India's final energy use nearly doubled by 2018 due to economic activity and appliance ownership, as activity outpaced efficiency.

3. **Fossil Fuel Dependence:** Coal supplies over half of India's electricity, slowing decarbonisation.

4. **Infrastructure Constraints:** Heatwave outages and water shortages at thermal plants expose bottlenecks, droughts in Andhra Pradesh have

hit power generation.

5. Climate Commitments: India's Nationally Determined Contributions under the Paris Agreement push non-fossil capacity, shaping how demand is met.



Strategies to Mitigate Demand

1. Energy Efficiency: Scale up the PAT scheme and efficient appliances under the Energy Conservation Act 2001 and the Bureau of Energy Efficiency; higher efficiency ambition can slow demand even as activity rises.
2. Renewables: Build on the PM Surya Ghar mission to expand capacity in states like Andhra Pradesh; its coastal location and port-led corridors aid renewable projects and energy trade. India already ranks fourth globally in renewable capacity, and the Electricity Act

2003 mandates renewable purchase obligations for discoms.

3. Resilient Infrastructure: Modernise grids and adopt water-efficient cooling for thermal plants.

4. Behavioural Change: Shift consumption patterns and promote energy-efficient buildings and sustainable transport to curb the activity effect.

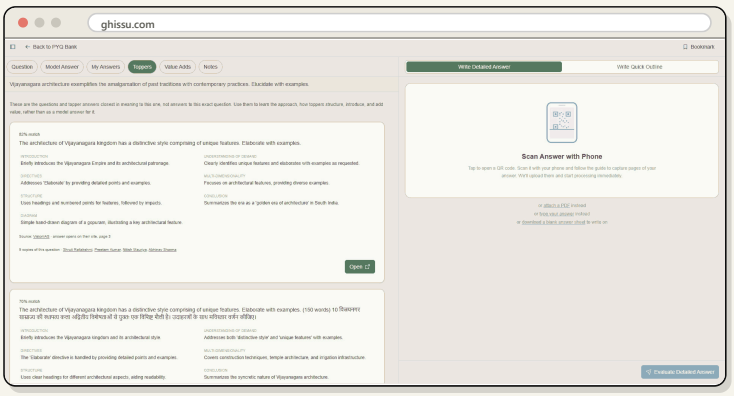
Conclusion

Rising demand comes from growth, urbanisation and fossil dependence. A coordinated approach of efficiency, renewables and resilient infrastructure can mitigate it and advance SDG 7 on affordable, reliable and sustainable energy.

THE REAL PRODUCT · NOT A MOCK-UP

TOPPERS' ANSWERS, PER QUESTION

Forty PDFs, and still no topper copy for this question?



See how toppers wrote it →

ghissu.com

14 days free · no card

SCAN

1 Topper answers matched by meaning to this exact question.

2 Each broken down: introduction, directives, structure.

3 Searched from 1.5 lakh+ topper answers.

P-IV
4(c)

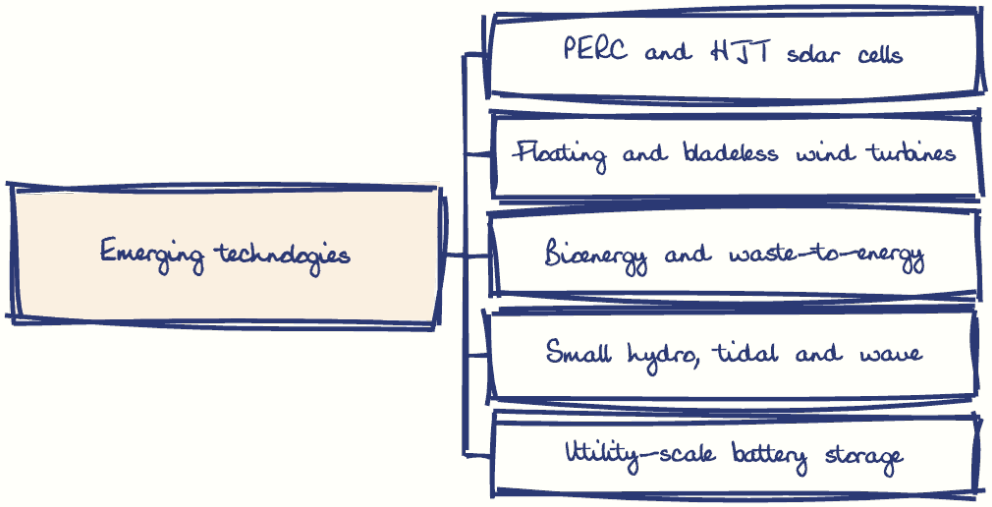
What are the new emerging renewable energy technologies?

10

Introduction

The climate crisis has made renewable technology critical to India's energy future. With a goal of 500 GW of non-fossil capacity by 2030, emerging renewable technologies are pivotal for energy security and lower emissions. Andhra Pradesh's coast and industrial corridors suit their deployment.

Emerging renewable technologies



Emerging Renewable Technologies

1. Advanced Solar: PERC and HJT cells raise panel efficiency by 6–12%, giving higher yield in adverse conditions, as in Andhra Pradesh's solar parks.

2. Next-Generation Wind: Taller turbines, longer

blades and bladeless designs cut maintenance and wildlife impact; floating offshore turbines and vertical axis turbines widen sites; AI uses real-time data to optimise output.

3. Bioenergy and Waste-to-Energy: Crop residues, municipal solid waste and biogas yield clean energy; Andhra Pradesh's horticulture and aquaculture offer biomass.
4. Small Hydro and Ocean Energy: Small hydro on the Krishna and Godavari has little ecological disruption; tidal and wave converters remain nascent.
5. Battery Storage: ReNew Power's utility-scale batteries tackle intermittency.

Potential Impact

1. Energy Security: Storage and the Green Energy Corridor give round-the-clock power and stable grids.
2. Environment and Economy: Lower emissions in line with Paris, plus jobs in manufacturing and installation.
3. Rural Benefits: Clean cooking fuel and power in Rayalaseema improve health and reduce deforestation.

Candidates must not write on this margin

Candidates must not write on this margin

4. Urban Use: Solar and wind power Visakhapatnam's industries and electric mobility.

Challenges

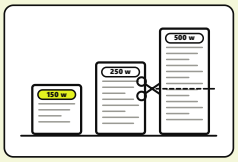
High capital costs, skill gaps, infrastructure gaps, regulatory uncertainty and R&D of only about 0.65% of GDP slow scaling, while land-use conflicts remain.

Way Forward

1. R&D: Fund locally adapted research in Andhra Pradesh's universities.
2. Skills: Train workers to run advanced systems.
3. Policy: Stable rules and faster approvals.
4. Decentralised Solutions: Mini-grids in Rayalaseema and coastal areas.

Conclusion

Emerging technologies, from efficient solar cells to floating turbines and storage, can transform India's clean energy future if R&D, skills and policy keep pace.



Writing 400 words for a 150-word question?

Model answers in three lengths

150, 250 and 500 words, in English and हिंदी. Learn what to cut before the examiner does.

Read one question three ways →

ghissu.com

14 days free · no card

SCAN

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 4(c) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
5(d)

Explain briefly on the role and responsibilities of National Disaster Management Authority (NDMA) in achieving its goal.

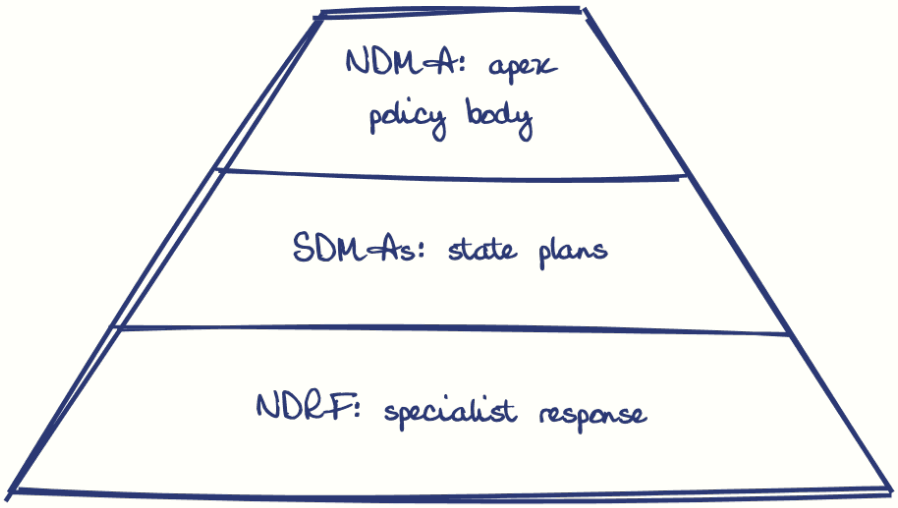
10

Introduction

The National Disaster Management Authority was established under the Disaster Management Act, 2005 for a coordinated, proactive approach to risk reduction and response. It is vital for states like Andhra Pradesh, which face cyclones along a 974 km coastline and droughts in Rayalaseema.

Role of the NDMA

NDMA's place in the system



1. Apex Policy Formulator: It is the highest policy body, laying down policies and guidelines for preparedness, mitigation and response.
2. Coordinator of Plans: It approves the National

Practice 5(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Plan and integrates disaster management into ministries' development plans.

3. Regulator and Enabler: Its guidelines help SDMAs, like Andhra Pradesh's, in planning and capacity building.
4. Promoter of Resilience: It builds a culture of prevention through knowledge, innovation and education.

Responsibilities of the NDMA

1. Guidelines: It guides SDMAs and departments to build risk reduction into development projects.
2. Coordination and Enforcement: It ensures timely implementation of policies, including during emergencies.
3. Funds: It recommends mitigation funds and supports use of the State Disaster Response Fund in Andhra Pradesh.
4. NDRF Oversight: It controls the NDRF, expanded to (16) battalions.
5. Relief Standards: It sets minimum relief standards, including concessional loans for victims.
6. Early Warning: It promotes IT-backed forecasting and early warning systems.

How These Achieve Its Goal

1. Holistic Risk Reduction: Mainstreaming disaster management into development reduces vulnerability.
2. Proactive Preparedness: Early warning and awareness shift management from reactive to proactive.
3. State Empowerment: Coordination with SDMAs enables local, context-specific action.
4. Technology-Driven Strategy: Seismic observatories and warning systems support timely, evidence-based action in hazard-prone places like Visakhapatnam.

Conclusion

Through policy coordination and technology the NDMA drives its vision of a safer India. Closer coordination with states and wider community preparedness will strengthen it further.

Correct, but average? One line from your answer

YOUR LINE
~~The 1857 revolt failed.~~
× true · forgettable

→

WITH A VALUE ADDITION
The 1857 revolt failed for want of a centre: no common plan, no modern arms, and the loyalty of the Sikhs, the Gurkhas and most princes to the Company.
✓ cause · evidence · specifics

ILLUSTRATIVE

Add value to your next answer →
ghissu.com
14 days free · no card
SCAN

Write your own answer to 5(d) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
5(a)

Explain various causes for the changes in climate and its effect on both the living and non-living bodies on the Earth.

10

Introduction

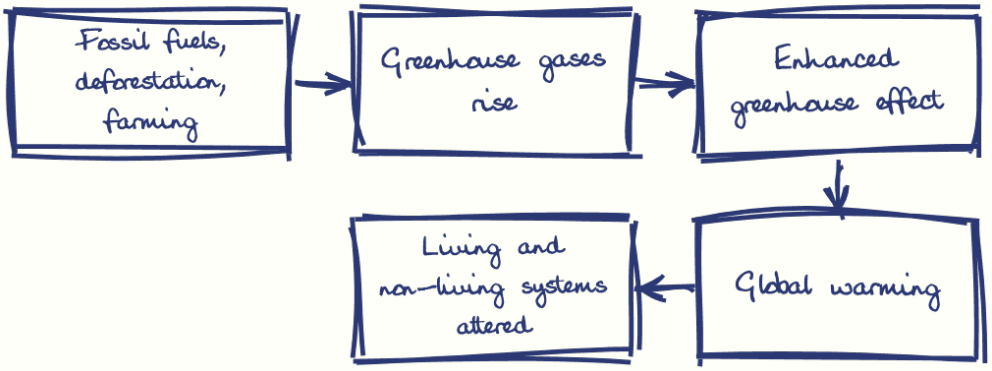
Climate change is a defining challenge of the 21st century. The rapid rise in greenhouse gas emissions from human activity has accelerated global warming, altering weather, sea levels and ecosystems. In India, including coastal Andhra Pradesh and the Rayalaseema drought belt, it threatens agriculture, water security and biodiversity.

Causes of Climate Change

1. Greenhouse gases: CO_2, methane, nitrous oxide and CFCs from fossil fuel combustion, deforestation and agriculture; in AP, aquaculture adds methane and Eastern Ghats deforestation raises CO_2.
2. Land use change: Deforestation for farming and cities shrinks carbon sinks, as forest loss in the Seshachalam Hills shows.
3. Agriculture and livestock: Livestock release methane and fertilisers emit nitrous oxide, with AP's chilli cultivation and livestock rearing adding to both.
4. Natural factors: Sunspot cycles, Milankovitch

orbital changes and volcanic eruptions shaped past climate but cannot explain warming since industrialisation.

From causes to effects



How the Causes Lead to Effects

Human activities raise greenhouse gases, which enhance the greenhouse effect and warm the planet, this warming reshapes both living and physical systems.

Effects on Living Bodies

1. Biodiversity loss: Species migrate or vanish, the Indian pangolin loses habitat in AP's mangroves and forests.
2. Agricultural stress: Changes in temperature and rainfall cut yields and spread pests, Rayalaseema faces more crop failures.
3. Health risks: Heatwaves, vector-borne disease

Practice 5(a) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V
Scan to practice

and malnutrition rise, heat illness grows in Visakhapatnam.

4. Livelihoods: Fisherfolk along AP's 974 km coastline face declining fish stocks.

Effects on Non-Living Bodies

1. Sea-level rise: Thermal expansion and glacial melt cause salinity ingress and land loss on AP's east coast.

2. Extreme weather: Stronger cyclones, floods and droughts, as Cyclone Hudhud showed.

3. Hydrological cycle: Altered monsoons make water availability in the Krishna and Godavari basins uncertain.

4. Ocean change: Warming and acidification in the Bay of Bengal harm corals, shellfish and fisheries.

Conclusion

Human activity, not nature, drives today's warming, and its effects run through living and non-living systems alike. Forest conservation under CAMPA and climate-resilient practices are the way forward.

Only ten minutes today?
Speak an outline. It gets evaluated too.
ghissu.com →

P-IV
5(b)

What are the different types of indicators of climatic changes? Write a short note on cryogenic indicators.

10

Introduction

India is seeing clear warming and altered rainfall that affect agriculture, water and livelihoods. Monitoring these changes needs reliable climatic indicators that reflect natural variability and human influence. They help assess past variations and predict trends, which matters for vulnerable states like Andhra Pradesh with its long coastline.

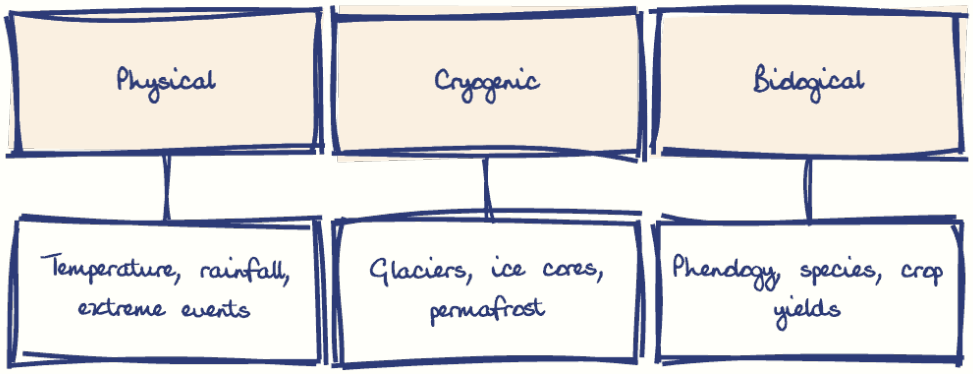
Types of Climatic Indicators

1. Physical indicators: Temperature records, rainfall patterns and extreme events such as droughts and cyclones; AP has seen more frequent cyclones like Titli.
2. Cryogenic indicators: Evidence from glaciers, ice caps and permafrost.
3. Biological indicators: Changes in phenology, species distribution and ecosystem health; shifting flowering times and yields in AP's horticulture reflect stress.

Note on Cryogenic Indicators

1. Meaning: They draw on frozen environments, which

Indicators of climatic change



preserve a record of past climate.

2. Methods: Ice core analysis, tracking glacial retreat and studying sediment deposits reveal historical climate.

3. Key evidence: The retreat of Himalayan glaciers is a key indicator of warming.

4. Value: They form a distinct but complementary category alongside physical and biological indicators.

5. Limits: Their spatial and temporal resolution is limited, and linking them to socio-economic impacts remains hard.

Monitoring Framework

1. INCCA: Over (120) institutions and (250) scientists track indicators nationally.

2. IMD: Observatories record temperature, pressure, rainfall and wind.

3. Reporting: National Communications and Biennial Update Reports under the UNFCCC include indicator data.

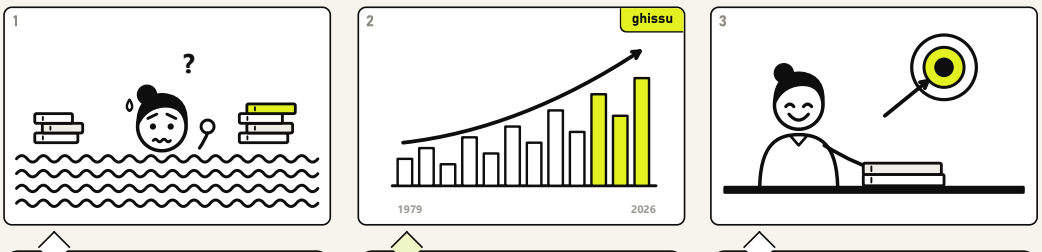
4. Gaps: Data gaps in remote areas of AP hinder precise monitoring.

Conclusion

Indicators already inform drought forecasting in Rayalaseema and state climate plans aligned with the N-APCC. Expanding cryogenic monitoring and closing data gaps will make adaptation sharper.

A STORY YOU MAY KNOW TREND ANALYSIS, PER SUB-TOPIC

Swimming in the syllabus



Every page might come. So I read every page.

Decades of papers say this sub-topic keeps rising.

So that's where I start.

— You have a feeling. The examiner has a pattern.

Find your next sub-topic →
ghissu.com
14 days free · no card

SCAN

P-IV
5(c)

What are the parameters considered to declare as a disaster? List out the various disasters which considered for declaration as national disasters.

10

Introduction

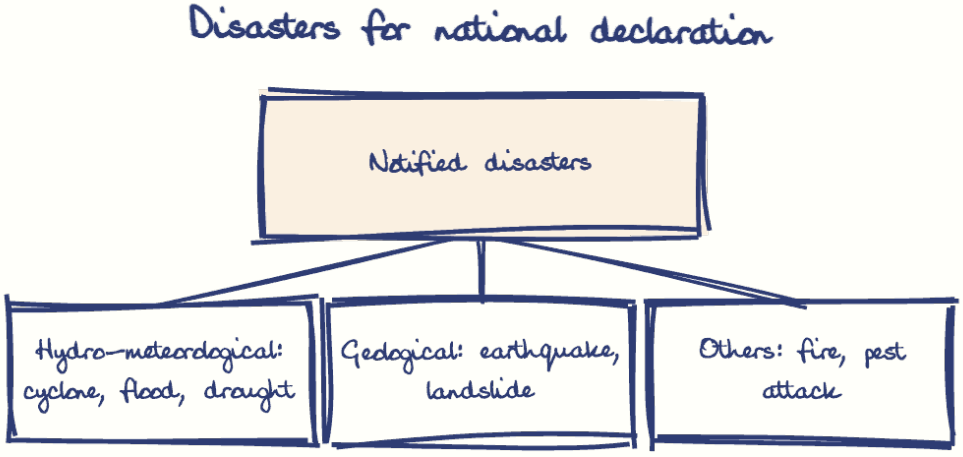
Declaring an event a disaster rests on parameters that judge its scale and severity, which decide the relief and mitigation resources mobilised.

Parameters for Declaring a Disaster

1. Scale of Damage: Destruction of life, property, infrastructure and services must overwhelm local capacity to cope without outside aid.
2. Casualties and Losses: Lives lost, injuries and economic losses, including crops and livestock.
3. Disruption of Normal Life: Social systems, livelihoods and public services are disrupted beyond normal coping.
4. Geographical Extent: The size of the area and the number of people affected.
5. Vulnerability and Capacity: Whether the event exceeds the resilience of local people and infrastructure.
6. Type and Frequency of Hazard: Its natural or human-made nature and recurrence.
7. Economic Impact: The relief and rehabilitation

burden on the state.

Disasters Considered for National Declaration



1. Cyclone: Severe storms damaging coastal and inland areas, frequent on Andhra Pradesh's coast.
2. Flood and Cloudburst: Riverine and flash floods, reported in Rayalaseema, and intense local rain causing landslides.
3. Draught: Dry spells causing water scarcity and crop failure, as in the Rayalaseema draught belt.
4. Earthquake and Tsunami: Tectonic shocks and seismic sea waves, a risk for ports like Visakhapatnam.
5. Landslide, Avalanche and Fire: Slides on

unstable terrain, forest or urban fires, hail, cold wave and pest attacks.

6. Others: Epidemics and pandemics, and industrial disasters such as chemical spills and nuclear accidents.

Implementation Challenges

1. Coordination: Central and Andhra Pradesh authorities, including the NDRF and SDRF, must work together.
2. Assessment and Funds: Damage assessment is often slow, and the National and State Disaster Response Funds sometimes fall short, as recent cyclones in AP showed.

Conclusion

Clear parameters and a defined list allow fair, prompt relief. Stronger early warning, quick assessment and adequate funds will make declarations more effective.

Improving — or just writing more?

My answers

Every attempt, every score, every enhanced version of your answer — on one timeline.

Watch your curve →

ghissu.com

14 days free · no card

SCAN

P-IV
6(d)

Differentiate between virus and bacteria. What are the harmful and useful effects caused by both in the humans and animals?

10

Introduction

The difference between viruses and bacteria is central to understanding infectious disease, as COVID-19 and zoonotic threats show. Both affect health and ecology, from harmful infections to useful symbiosis. In Andhra Pradesh, where livestock and aquaculture are vital, it aids animal health and prevention of zoonotic spillover.

Virus versus Bacteria

Virus vs Bacteria

Virus	Bacteria
Size	20–300 nanometres
Structure	Genes in protein coat
Reproduction	Only inside host cells
Living status	Non-living outside host
Treatment	Antivirals or vaccines

1. Why it matters: This contrast explains why COVID-19 needs a different approach from bacterial diseases like tuberculosis, prevalent in

Practice 6(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Andhra Pradesh.

Harmful Effects on Humans and Animals

1. Disease: Viruses cause influenza, dengue and COVID-19, spreading fast through mutation; bacteria cause typhoid, cholera and tuberculosis, still major challenges in the state's rural and urban areas.
2. Zoonotic spillover: Nipah and rabies are viral zoonoses from bats and wildlife; anthrax and leptospirosis are bacterial ones affecting livestock and humans.
3. Resistance: Antimicrobial resistance threatens bacterial treatment in Andhra Pradesh's animal husbandry sector, while viruses produce vaccine escape variants.
4. Economic impact: Pandemics disrupt economies, and bacterial diseases cause livestock losses that hurt Andhra Pradesh's agrarian communities.

Implication

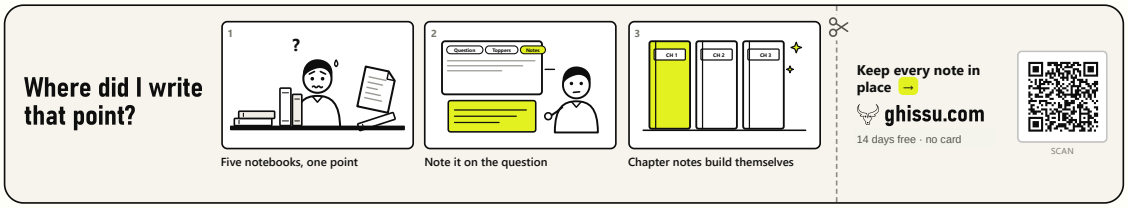
1. Right tools: Since antibiotics fail against viruses, their misuse in viral illness only feeds resistance; vaccines and antivirals are needed instead.

Useful Effects

1. Microbiome: Bacteria aid digestion, nutrient synthesis and immunity; gut bacteria in Andhra Pradesh's livestock improve feed efficiency, supporting the state's dairy and poultry industries.
2. Biotechnology: Bacteria produce antibiotics, enzymes and biofuels under the BioE3 Policy, and bacteriophages are explored as alternatives to antibiotics.
3. Ecological balance: Bacteria drive nutrient cycling and soil fertility for Andhra Pradesh's horticulture and aquaculture, while viruses regulate microbial populations.
4. Vaccines: Viruses are targets for vaccines, as in the COVID-19 vaccination drive in Andhra Pradesh, and bacterial components are used in typhoid vaccines.

Conclusion

Viruses and bacteria are both pathogens and allies. A One Health approach with integrated surveillance, biotechnology and vaccination will build resilience in Andhra Pradesh.



P-IV
6(a)

Differentiate between old and modern Biotechnology. Describe the discovery of Louis Pasteur on fermentation.

10

Introduction

Biotechnology, which harnesses living organisms for practical use, has evolved from ancient practice to cutting-edge genetic engineering, opening new paths to growth and sustainability. India, including Andhra Pradesh's growing biotech sector with biotech parks in Visakhapatnam, now uses modern biotechnology for agriculture, health and industry, and Pasteur's work on fermentation marks the turning point between the two eras.

Old versus Modern Biotechnology

1. Basis: Old biotechnology used natural processes like fermentation, selective breeding and cultivation since prehistoric times without understanding microbes; modern biotechnology, from the 20th century, uses recombinant DNA technology.
2. Control and consistency: Traditional yoghurt, cheese and wine making relied on natural microbes, giving variable quality and yield that limited scale; modern methods are precise, consistent and scalable.
3. Applications: Modern biotech yields

Practice 6(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

drought-tolerant SAATVIK chickpea, higher-yield genome-edited rice and vaccines, with India supplying a major global share.

Old vs Modern Biotechnology

Old Biotechnology	Modern Biotechnology
Control over process	Natural, empirical methods
Applications	Fermentation, selective breeding
Institutional support	Traditional knowledge systems
Outcome consistency	Variable quality and yield

4. Constraints: Low R&D spending of about 0.65% of GDP, regulatory complexity and infrastructure gaps limit modern biotechnology.

Pasteur's Discovery on Fermentation

1. Microbes cause fermentation: In the 1860s Pasteur showed that sterilised broths stayed free of life unless exposed to microorganisms in the air, disproving spontaneous generation.
2. Swan-neck flask: Its curved neck trapped airborne microbes, so no growth occurred unless the flask was tilted or broken.

Practice 6(a) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

3. Impact: It established germ theory, founded microbiology and enabled controlled fermentation, giving rise to industrial microbiology in food, pharmaceuticals and agriculture.

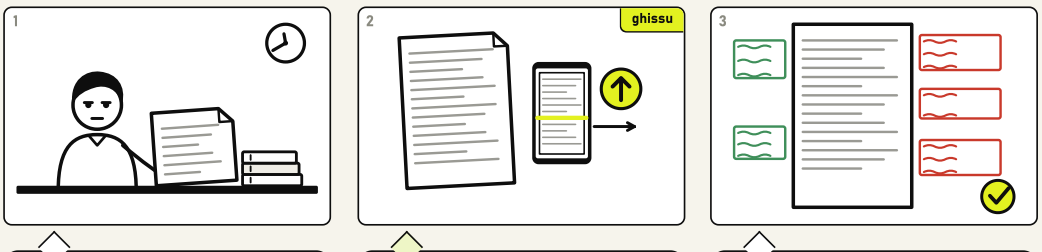
4. Today's link: Fermentation now uses genetically tailored yeast strains, and Andhra Pradesh's aquaculture and horticulture apply controlled fermentation for feed and product preservation.

Conclusion

Pasteur's discovery turned fermentation from an empirical craft into a controlled science, the base of modern biotechnology. Higher R&D funding, clear science-based regulation and skill development can carry it forward.

A STORY YOU MAY KNOW ANSWER EVALUATION

The answer nobody read



— Feedback you get tonight beats feedback you wait a week for.

Get your intro fixed on your sheet →

ghissu.com

14 days free · no card

SCAN

P-IV
6(b)

What are antibiotics? Write advantages and disadvantages of antibiotics use.

10

Introduction

Antibiotics are among the most transformative medical discoveries of the 20th century, drastically reducing mortality from bacterial infections. However, their widespread and often unregulated use has created antimicrobial resistance (AMR), a growing threat to health systems. Kerala's Operation AMRI TH, launched in 2024 to regulate antibiotic use, shows the need to balance benefits against misuse.

What Are Antibiotics

1. Nature: Chemical substances produced by microorganisms or synthesised artificially that kill bacteria (bactericidal) or inhibit their growth (bacteriostatic).

2. Spectrum: Broad-spectrum antibiotics like amoxicillin act on many bacteria, while narrow-spectrum ones target specific bacteria and need precise diagnosis.

3. History and production: Fleming's discovery of penicillin in 1928 began the antibiotic era, Hindustan Antibiotics Limited in Pimpri supports self-reliance.

Advantages vs Disadvantages of Antibiotics

Advantages	Disadvantages
Life-saving treatment	Risk of antimicrobial resistance
Broad disease control	Misuse through self-medication
Economic productivity	Used wrongly for viral illness
Support to livestock health	Weak drug regulation

Advantages

1. Lower mortality: Deaths from pneumonia, typhoid and meningitis have fallen, raising life expectancy.
2. Economic benefits: Preventing prolonged illness cuts health costs and productivity losses.
3. Veterinary use: Better livestock health supports rural livelihoods and food security in agrarian Andhra Pradesh.

Disadvantages

1. Resistance: Excessive use breeds resistant bacteria, India had an estimated 2,97,000 deaths attributable to AMR in 2019.
2. Self-medication: Over-the-counter sales without prescription, common in parts of Andhra Pradesh,

Candidates must not write on this margin

Candidates must not write on this margin

and incomplete courses fuel resistance.

3. Misconceptions: Many use antibiotics for viral illnesses like colds and dengue.

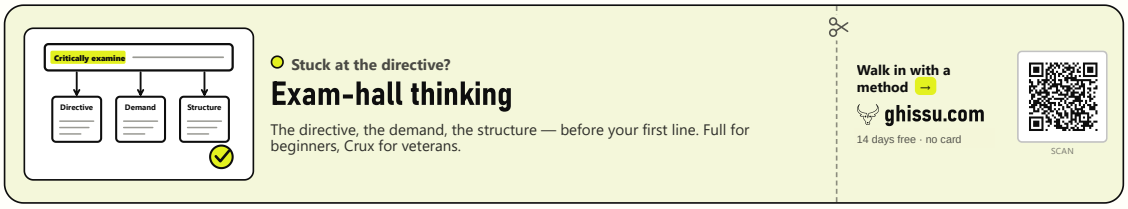
4. Weak regulation: Limited drug inspector capacity undermines prescription-only rules.

Way Forward

1. Regulation and awareness: Strengthen drug inspectors in Andhra Pradesh to enforce prescription-only sales and run education drives in the state's schools and health facilities.

Conclusion

Antibiotics remain life-saving. Stronger enforcement, training on ICMR guidelines and a One Health approach can keep them effective.



Write your own answer to 6(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
6(c)

What is the difference between Penicillin and Insulin? Describe the Industrial production of Penicillin.

10

Introduction

Antibiotics and recombinant DNA products have revolutionised medicine, tackling infections and chronic conditions like diabetes. Penicillin and Insulin are two landmark drugs, and their contrast, along with Penicillin's production, shows India's pharmaceutical capacity under schemes like the Production Linked Incentive.

Difference between Penicillin and Insulin

1. Nature and origin: Penicillin is a natural antibiotic from Penicillium mould, discovered accidentally by Alexander Fleming in 1928; Insulin is a pancreatic hormone now made by recombinant DNA technology.
2. Therapeutic use: Penicillin kills or inhibits infection-causing bacteria; Insulin regulates blood sugar in diabetics.
3. Production method: Penicillin comes from microbial fermentation; Insulin from engineered bacteria or yeast expressing the human gene, giving a pure product.
4. Specificity: Penicillin is narrow-spectrum, mainly

against Gram-positive bacteria; Insulin is a metabolic regulator with no antimicrobial action.

Penicillin vs Insulin

Penicillin	Insulin
Origin	Natural mould-derived antibiotic
Use	Antibacterial agent for infections
Production	Fermentation of Penicillium mould
Spectrum	Narrow spectrum antibiotic

Industrial Production of Penicillin

1. Submerged fermentation: Penicillium mould is cultured in large fermentors with nutrient media, aeration and pH control to maximise yield.
2. Downstream extraction: The antibiotic is extracted from the broth and purified for medical use.
3. Quality control: Good Manufacturing Practices ensure safety and efficacy.
4. Institutions: Hindustan Antibiotics Limited in Pimpri, with subsidiaries in Maharashtra, Karnataka and Manipur, has led public production of Penicillin and other antibiotics through this

Practice 6(c) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

decentralised design.

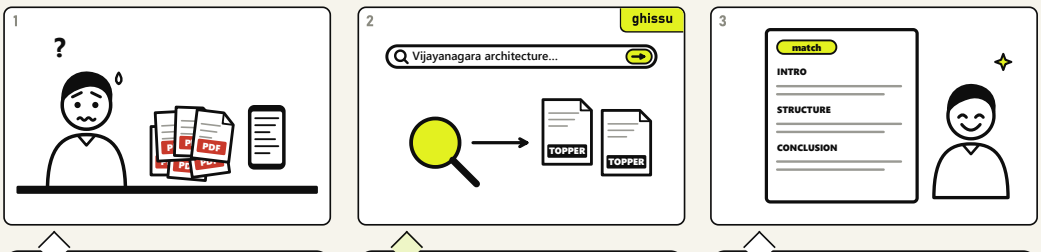
5. Revival: After closures of fermentation-based API plants, Bulk Drug Parks in Andhra Pradesh and other states under the PLI scheme aim to modernise production, as India's API share remains limited by competition from China.

Conclusion

Penicillin and Insulin differ in origin and use, but both reflect India's pharmaceutical strength and its role as a major global supplier of antibiotics. More R&D and modern fermentation facilities can reduce import dependence for critical antibiotics.

A STORY YOU MAY KNOW TOPPERS' ANSWERS, PER QUESTION

The topper copy hunt



Telegram, YouTube, forty PDFs... nothing on this question.

Type the question into ghissu.

Five topper answers to this exact one!

— Borrow the topper's approach, not the topper's sentences.

Try it on your question →

ghissu.com

14 days free · no card

SCAN

P-IV
7(d)

Write an essay on stem cells research and the significance of cloning in various fields.

10

Introduction

Stem cell research and cloning are frontier biotechnologies with transformative potential in medicine, agriculture and industry. India's regulatory frameworks and ethical guidelines show the need to harness them responsibly, including in states like Andhra Pradesh where biotech institutions are emerging.

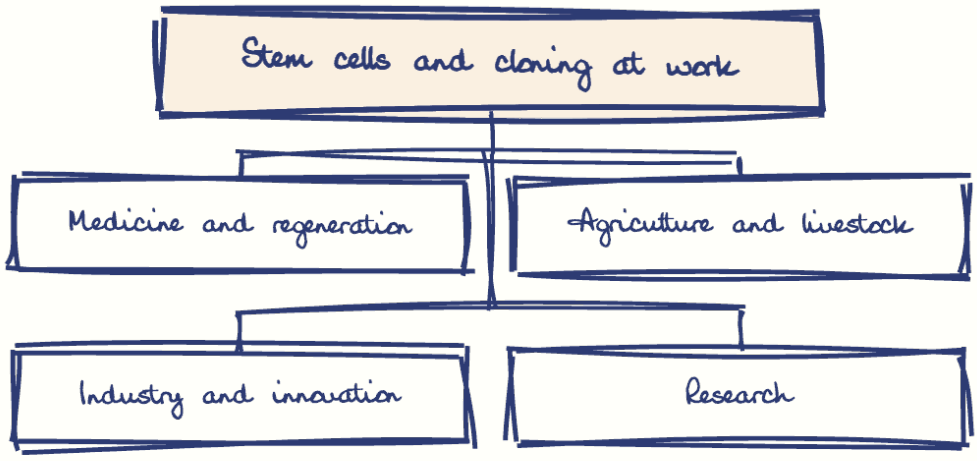
Regulatory Framework

1. **National Oversight:** The NAC-SCRT, set up by the Health Ministry, oversees stem cell research, with institutional committees (IC-SCR) at research centres ensuring ethical and scientific rigour.
2. **ICMR-DBT Guidelines:** They classify research as permitted, restricted or prohibited, and mandate registration and ethical review to prevent premature or fraudulent therapies.
3. **Clinical Trial Governance:** CDSCO regulates stem cell trials under the Drugs and Cosmetics Act, requiring registration in the Clinical Trials Registry-India and Institutional Ethics Committees.

Candidates must not write on this margin

Candidates must not write on this margin

Significance in Various Fields



1. Medicine: Pluripotency offers regenerative medicine for degenerative diseases, heart ailments and tissue repair, with potential to transform healthcare delivery.
2. Agriculture: Cloning and stem cell techniques propagate disease-resistant crops and livestock, raising yield and sustainability in AP's aquaculture and horticulture.
3. Industry and Innovation: Biotech startups and innovation hubs can reduce imports and build technological sovereignty.

Challenges

1. Ethical Concerns: Debate over embryonic stem cells and cloning the sanctity of human life and

How to use this book with ghissu.com
see page 4

Index · page 6

Practice 7(d) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

misuse demands public awareness and strict guidelines.

2. Technical Barriers: R&D at about 0.65% of GDP, brain drain and weak infrastructure limit scaling, so states like AP need more funding and capacity building.

Way Forward

1. Research Ecosystems: Raise funding and infrastructure for research in Andhra Pradesh, building on institutions near Visakhapatnam and public–private partnerships.
2. Ethical Enforcement: Run educational campaigns and enforce national guidelines rigorously to prevent exploitation in stem cell therapies.
3. Indigenous Innovation: Support biotech startups and innovation hubs in Andhra Pradesh.
4. Collaboration: Build national and international networks to share knowledge, standardise protocols and speed responsible clinical use.

Conclusion

Stem cell research and cloning hold immense promise for health, farming and industry. Aligning innovation with ethical governance will help India achieve self-reliance and serve humanity's well-being.

Write your own answer to 7(d) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V
3 / 3
Scan to practice

P-IV
7(a)

Define bioinsecticides, What are the different types of bioinsecticides?

10

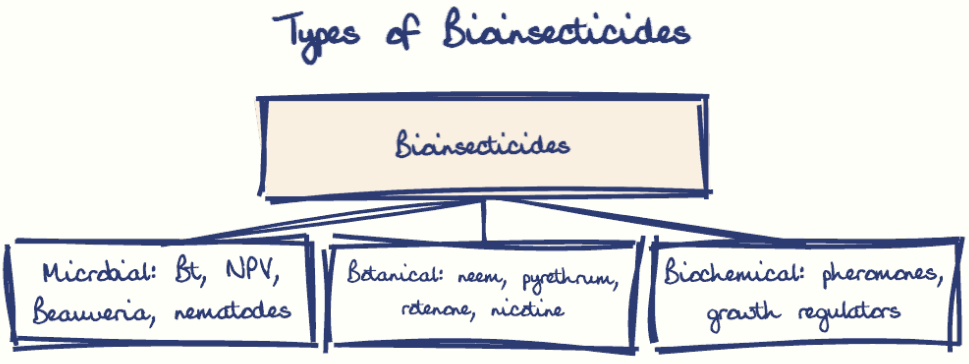
Introduction

Bioinsecticides have gained prominence as sustainable alternatives to chemical pesticides amid growing environmental concern. With India's bioeconomy projected to quadruple by 2030 and the BioE3 Policy promoting biotechnology, they are a critical part of integrated pest management, including in Andhra Pradesh's horticulture.

Definition

Bioinsecticides are insect-control agents derived from living microorganisms, plants or natural biochemicals. They are target-specific, less toxic and biodegradable, and are registered by the Central Insecticide Board and Registration Committee (CIB&RC) under the Insecticides Act, 1968.

Types of Bioinsecticides



1. Microbial Bioinsecticides: Bacillus thuringiensis (Bt) produces toxins lethal to specific insect larvae, and nuclear polyhedrosis virus infects and kills pests. Their specificity spares non-target species, suiting cotton, vegetables and chilli cultivation in Andhra Pradesh.

2. Botanical Bioinsecticides: Neem-based azadirachtin, pyrethrum from chrysanthemum, rotenone from Derris roots and nicotine from tobacco act as insecticides, repellents or growth disruptors.

3. Biochemical Bioinsecticides: Pheromones disrupt mating, and insect growth regulators interfere with pest development stages.

Advantages and Limitations

1. Advantages: Lower toxicity, reduced environmental persistence and target specificity make them biodegradable, safer alternatives that leave fewer chemical residues.

2. Limitations: They have shorter shelf life, slower action than chemicals and difficult production scale-up.

3. Adoption Barriers: Low awareness among farmers in Rayalaseema and coastal Andhra, and the need for cold chain infrastructure to keep microbes viable, constrain use.

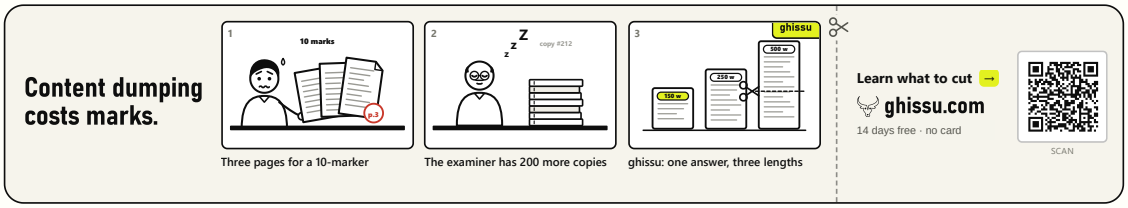
4. Compatibility: Their narrow pest range often requires integration with conventional pesticides, so farmer training and extension services are needed.

Policy Support

The Insecticides Act, 1968 framework ensures efficacy and safety, while the Bio-RIDE scheme with Rs 1,500 crore funds research and commercialisation, reducing import dependence.

Conclusion

With research on local pests at Andhra Pradesh's Acharya N.G. Ranga Agricultural University and Sri Venkateswara University, and alignment with Mission LiFE, bioinsecticides can reduce chemical residues and protect biodiversity.



P-IV
8(d)

What is a transgenic animal? List out the benefits from transgenic animals.

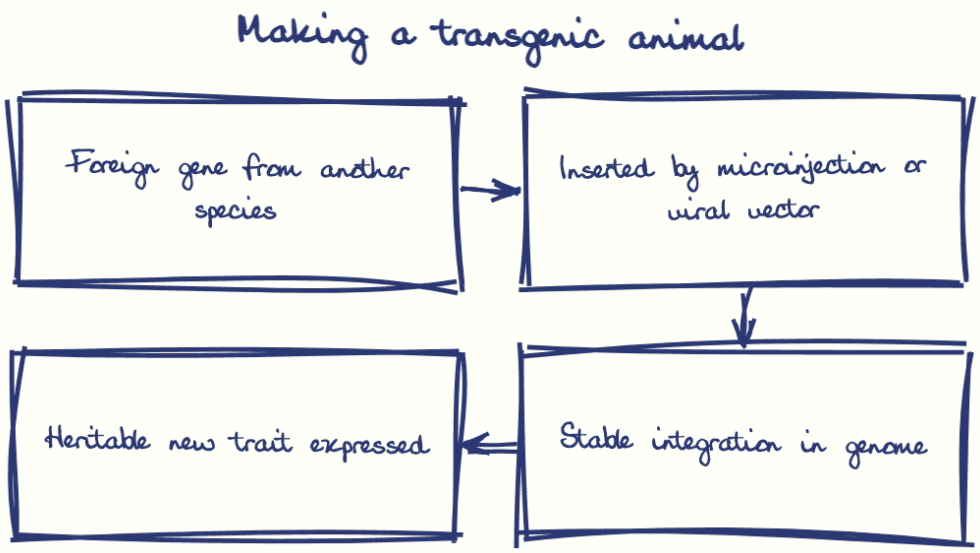
10

Introduction

Advances in biotechnology allow precise genetic changes that raise productivity and sustainability in animal husbandry. Transgenic animals are a major breakthrough with uses in medicine, agriculture and the environment, relevant to Andhra Pradesh with its strong agricultural base.

What Is a Transgenic Animal

1. **Definition:** A transgenic animal is one whose genome has been altered by inserting foreign DNA from another species using recombinant DNA technology, giving it new traits.
2. **Stable Integration:** The foreign gene integrates stably into the genome, so the trait is expressed and heritable.
3. **Methods:** They are created by microinjection, viral vectors or gene editing, bypassing traditional breeding barriers.
4. **Examples:** Animals engineered for disease resistance, faster growth or pharmaceutical proteins, such as goats producing human antithrombin.



Benefits of Transgenic Animals

1. **Disease Resistance:** Animals engineered to resist infections reduce mortality and losses in dairy and poultry, vital to Andhra Pradesh's agrarian economy.
2. **Enhanced Productivity:** Modifications can raise growth rates, milk yield or wool quality, making farming more profitable.
3. **Pharmaceutical Production:** As bioreactors, they produce therapeutic proteins, vaccines and enzymes, supporting affordable healthcare.
4. **Environmental Benefits:** Fish with better feed conversion ratios reduce aquaculture waste, relevant to the state's large aquaculture sector.
5. **Research Models:** They serve as valuable models for studying human diseases and developing treatments,

speeding biomedical research and innovation.

6. Conservation: Introducing beneficial genes or improving reproduction can help endangered species.

Way Forward

1. Regulation: Robust biosafety and ethical guidelines for transgenic animals, involving Andhra Pradesh research institutions, to ensure safe deployment and public trust.
2. Partnerships: Collaboration between government, academia and industry in Andhra Pradesh to speed research, commercialisation and adoption.
3. Farmer Training: Extension programmes in Andhra Pradesh to teach livestock farmers about benefits and safe use.
4. Indigenous R&D: Research on animals adapted to local agro-climatic conditions under Bio-RIDE.

Conclusion

Transgenic animals can transform animal health, productivity and biopharmaceutical output. Developed responsibly, they can advance sustainable agriculture and healthcare, supporting Aatmanirbhar Bharat in biotechnology.

Correct, but average?
A number, a committee, a case — for every question.
ghissu.com →

P-IV
8(a)

Write briefly on the types and uses of biopesticides.

10

Introduction

Biopesticides have gained prominence as sustainable alternatives to chemical pesticides, amid concern over environmental damage, health risks and pest resistance. The BioE3 Policy and focus on biotechnology-driven growth underline their importance for eco-friendly pest management.

Framework

The CIB&RC registers biopesticides for efficacy and safety, while Andhra Pradesh's Department of Agriculture builds them into state IPM programmes for horticulture and aquaculture, and Acharya N.G. Ranga Agricultural University runs field trials of local formulations.

Types and Uses

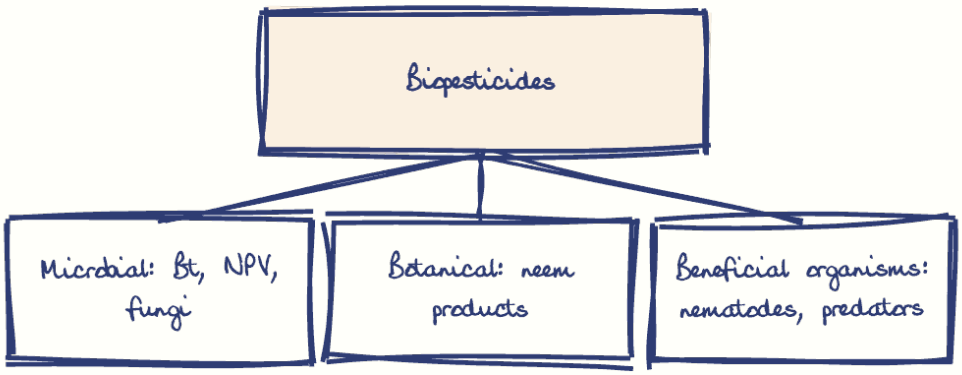
1. Microbial Biopesticides: Bacteria like Bacillus thuringiensis (Bt), viruses such as nuclear polyhedrosis virus (NPV) and fungi target specific pests. Bt products dominate, with 92% of biopesticide sales, controlling pests in cotton, vegetables and fruits.
2. Botanical Pesticides: Plant-derived products like

neem act as insecticides, repellents or growth regulators, with minimal environmental impact in Andhra Pradesh horticulture.

3. Beneficial Organisms: Protozoa, nematodes and predatory insects naturally suppress pests, forming part of integrated pest management.

4. Crops and Aquaculture: They control pests, fungal and bacterial diseases and weeds, and in Andhra Pradesh have been applied in chilli cultivation and shrimp farming, improving yield and reducing chemical residues.

Types of biopesticides



Challenges

1. Shelf Life: Many have limited shelf life because living agents must stay viable, complicating storage, distribution and farmer access.

2. Specificity: Narrow targets often require use

alongside chemical pesticides, which can inhibit biopesticide effectiveness.

3. Scale-Up: Agents needing in vivo production, like baculoviruses, are hard to scale, limiting availability and cost-effectiveness.

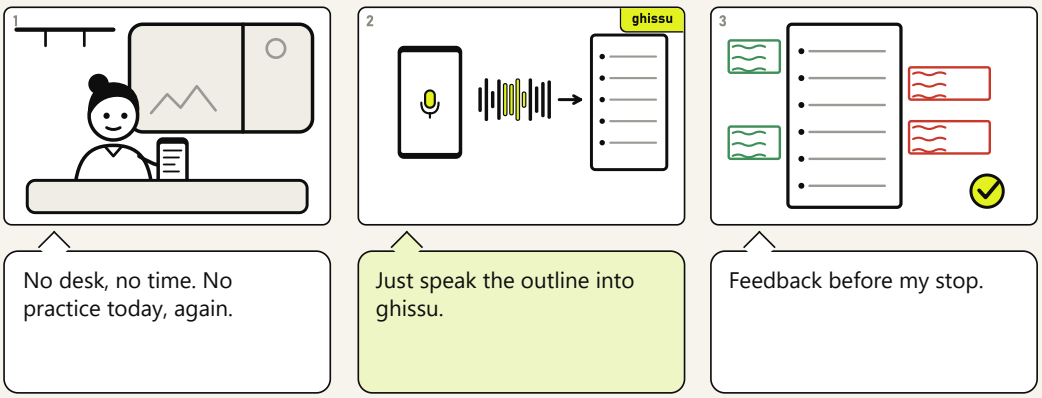
4. Positive Trend: Even so, sales in India are growing 10–25% a year, reflecting rising farmer awareness and demand.

Conclusion

Biopesticides reduce chemical load while protecting crops. More R&D under Bio-RIDE for longer shelf life, along with farmer awareness, can make biopesticides the backbone of sustainable pest control.

A STORY YOU MAY KNOW QUICK OUTLINE — TYPE, SPEAK OR SCAN

Practice on the bus



— An outline a day beats a full answer a week.

Try a quick outline →
ghissu.com
14 days free · no card
SCAN

P-IV
8(b)

Mention briefly the arguments in favour of and against GM crops.

10

Introduction

GM crops promise to address food security, climate resilience and nutritional deficiencies. With India's growing population, heavy imports of edible oils and pulses and the pending decision on GM mustard, the debate weighs their benefits against health, environmental and socio-economic concerns.

Arguments in Favour of GM Crops

1. **Higher Productivity:** Resistance to pests, diseases, drought and salinity raises yields, crucial for drought-prone Rayalaseema, and can cut import dependence for oilseeds and pulses.
2. **Less Pesticide Use:** Pest-resistant traits reduce chemical sprays; Bt cotton, widely adopted in India including Andhra Pradesh, has significantly decreased spraying, lowering pollution and conserving soil and water.
3. **Nutrition and Food Security:** Biofortified crops like Golden Rice, rich in vitamin A, address micronutrient deficiencies and malnutrition.
4. **Farmer Incomes:** Higher yields and lower input costs improve profits, as the Bt cotton experience

in Andhra Pradesh shows.

Arguments Against GM Crops

1. **Health Concerns:** Long-term effects remain uncertain, with fears of allergenicity and antibiotic resistance markers; though no conclusive evidence has emerged, the precautionary principle fuels apprehension.
2. **Environmental Risks:** Gene flow to wild relatives may create herbicide-resistant superweeds and affect native flora in biodiverse regions of Andhra Pradesh.
3. **Corporate Dependence:** Multinational dominance of seed markets raises fears of costly patented seeds, threatening traditional practices and seed sovereignty. Andhra Pradesh smallholders may face legal trouble if crops are contaminated.
4. **Regulatory Gaps:** The Bt brinjal moratorium and GM mustard litigation show the need for rigorous, transparent and independent safety assessments.

Balanced View

1. **Science-Based Regulation:** Robust, transparent mechanisms with independent experts and farmer representatives can build public trust and ensure safety.

Practice 8(b) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

The GM crops debate

For	Against
Higher yields, fewer imports	Uncertain long-term health effects
Less spraying with Bt cotton	Superweeds through gene flow
Golden Rice for nutrition	Costly patented seeds
Better farmer incomes	Weak, delayed regulation

2. Indigenous R&D: Research by Indian institutions can reduce reliance on multinational corporations and tailor crops to local agro-climatic conditions.
3. Farmer Awareness: Outreach and capacity building in states like Andhra Pradesh can help farmers weigh benefits and risks.

Conclusion

GM crops can boost productivity, nutrition and incomes, especially in climate-stressed regions like Andhra Pradesh, but concerns about health, environment and equity demand cautious, well-regulated adoption aligned with Atmanirbhar Bharat and sustainable development.

Which sub-topic comes next?
Trend analysis for every syllabus topic.
ghissu.com →

P-IV
8(c)

What are the different types of sterilization methods adopted for different types of materials?

10

Introduction

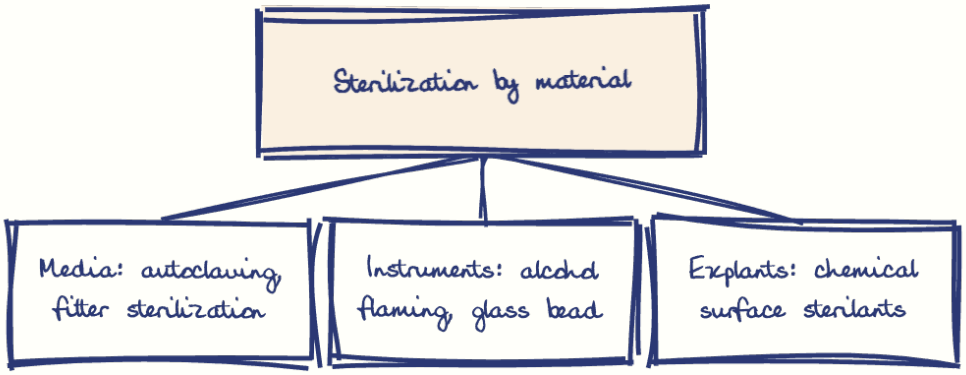
Sterilization is the elimination of all microorganisms to maintain aseptic conditions. It is vital in tissue culture, medical and laboratory settings, where contamination compromises results or safety. Materials need tailored methods based on their physical and chemical properties.

Sterilization Methods for Different Materials

1. Nutrient media and glassware: Autoclaving with steam under pressure at 121°C and (15) psi for (20) minutes is the most widely used method, also suiting heat-resistant plastic labware.
2. Heat-sensitive materials: Thermolabile media components and certain plastics need filter sterilization or UV radiation, though UV is less common due to cost and damage to plastics.
3. Instruments: Forceps and scalpels are dipped in 70–95% alcohol and flamed, or sterilised in glass bead and infrared sterilizers, which are safer than Bunsen burner flaming.
4. Plant explants: Surface sterilization uses sodium hypochlorite, calcium hypochlorite, hydrogen peroxide

or mercuric chloride, with time and concentration balancing efficacy and tissue safety.

Sterilization Methods by Material Type



Maintaining Aseptic Conditions

1. Work area: In Andhra Pradesh, tissue culture labs follow national-standard protocols, using laminar airflow cabinets and separate pathological and tissue culture spaces.
2. Discipline: Avoiding talking in the hood and placing materials so airflow is not disrupted prevent contamination, critical in Andhra Pradesh labs serving horticulture and aquaculture.
3. Chemical safety: Toxic mercuric chloride needs careful rinsing and disposal to avoid environmental and health hazards.
4. Equipment gaps: Some Andhra Pradesh laboratories struggle to acquire and maintain

advanced sterilization equipment.

Way Forward

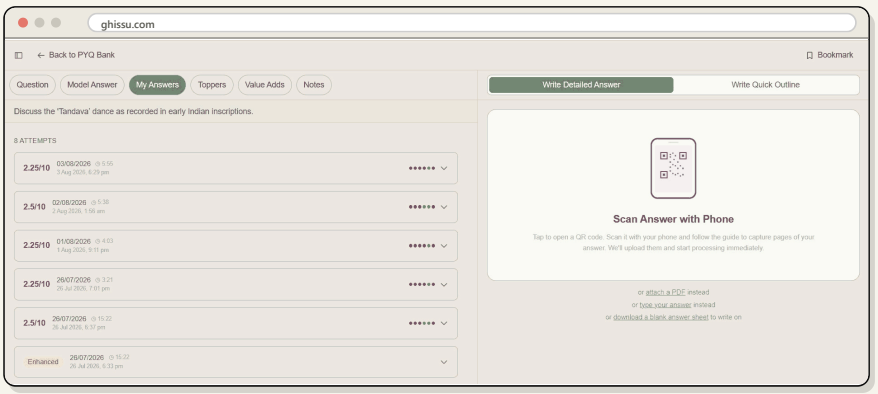
1. Capacity: Promote cost-effective indigenous glass bead and infrared sterilizers, train lab staff and fund equipment maintenance in Andhra Pradesh.
2. Safer sterilants: Research less toxic, eco-friendly sterilants for plant explants.

Conclusion

Matching the method to the material balances efficacy with compatibility. Stronger capacity will improve tissue culture and biotechnology, in line with SDG (9).

THE REAL PRODUCT · NOT A MOCK-UP MY ANSWERS — YOUR PROGRESS

Fifty answers written. Are you any better?



1 Eight attempts at one question, first to latest.

2 Every one scored — the curve is visible.

3 Enhanced versions kept beside your own.

Start your timeline →

ghissu.com

14 days free · no card

SCAN

P-IV
9(d)

Discuss briefly on the morphology and biological characters of measles virus.

10

Introduction

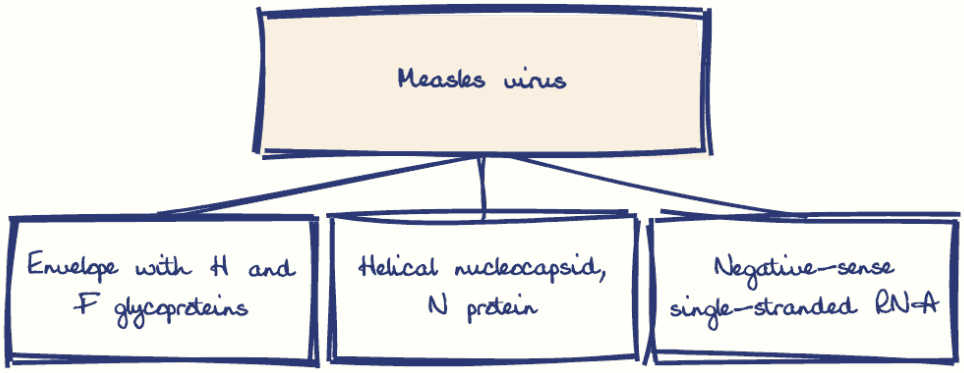
Measles remains a significant public health concern in India, including Andhra Pradesh, where outbreaks continue despite vaccination efforts. Understanding the morphology and biological characters of the measles virus is crucial for vaccine development, surveillance and outbreak control.

Morphology of the Measles Virus

1. **Enveloped Spherical Virus:** It is an enveloped virus with a roughly spherical shape, about 100 to 300 nanometres in diameter, which helps it enter host cells through membrane fusion.
2. **RNA Genome:** It contains a single-stranded, negative-sense RNA genome of about 15,894 nucleotides, encoding six structural proteins needed for replication and assembly.
3. **Surface Glycoproteins:** The envelope is studded with the hemagglutinin (H) protein, which attaches to host cells, and the fusion (F) protein, which fuses viral and cell membranes.
4. **Helical Nucleocapsid:** Inside the envelope, the nucleoprotein (N) is tightly bound to the RNA,

protecting the genome and aiding replication.

Structure of the measles virus



Biological Characters

1. Airborne Transmission: It spreads through respiratory droplets and aerosols, with a high basic reproduction number (R0), making it one of the most contagious human viruses.
2. Host Range and Tropism: It infects only humans, targeting respiratory epithelial cells and immune cells such as lymphocytes and dendritic cells, which spreads it through the body.
3. Immune Evasion: It suppresses host immunity by inhibiting interferon signalling, escaping early antiviral defences.
4. Clinical Effects: Infection causes a characteristic rash and fever, but can lead to pneumonia and encephalitis in malnourished or

immunocompromised children, a concern for vulnerable groups in Andhra Pradesh.

Public Health Implications

1. Vaccine Design: The live attenuated vaccine induces neutralising antibodies against the H and F proteins, the very glycoproteins the virus needs for infection.
2. Molecular Typing: The virus's genetic and phenotypic stability aids molecular epidemiology, helping track outbreaks and vaccine effectiveness, which is essential for states like Andhra Pradesh with reported cases.
3. Eradication Challenge: Its high transmissibility causes outbreaks wherever vaccination coverage is low, so immunisation must be sustained and monitored.

Conclusion

The structure and biology of the measles virus underpin its high infectivity and pathogenic potential, and a clear grasp of these features enables effective vaccine design and outbreak control. Stronger molecular surveillance and wider vaccination coverage are essential for elimination and for SDG (3.)

Write your own answer to 9(d) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
9(a)

What is meant by infection? Explain the terms Acute, chronic, fulminating, localized, generalized, primary and secondary used for infections with examples.

10

Introduction

Infections remain a critical public health challenge in Andhra Pradesh, where climate and socio-economic factors shape disease patterns. Classifying infections as acute, chronic, fulminating, localized, generalized, primary or secondary guides clinical and public health responses.

Meaning of Infection

1. **Definition:** Infection occurs when pathogenic microorganisms such as bacteria, viruses, fungi or parasites invade and multiply within a host, causing tissue damage and an immune response. Its severity depends on the pathogen, the site and host immunity.

Types of Infection with Examples

1. **Acute:** Rapid onset and short duration, often resolving with or without treatment. Acute respiratory infections in Visakhapatnam appear suddenly with cough and fever and last days to weeks.

2. **Chronic:** Develops slowly and persists long,

sometimes lifelong. In tuberculosis, prevalent in parts of Andhra Pradesh, the bacteria stay dormant or active in lung tissue.

3. Fulminating: Severe, rapidly progressing and life-threatening. Meningococcal meningitis outbreaks in Andhra Pradesh districts can escalate quickly with high mortality if untreated.
4. Localized: Confined to one tissue or organ, like a skin abscess by Staphylococcus aureus in rural villages, with redness, swelling and pain.
5. Generalized: A systemic infection spreading through blood or lymph. Septicaemia, arising from untreated localized infection, causes widespread inflammation and organ dysfunction.
6. Primary: The initial infection of a healthy host, such as primary dengue transmitted by Aedes mosquitoes in coastal Andhra Pradesh.
7. Secondary: Occurs when a primary infection weakens immunity or damages tissue, letting opportunists invade, as in bacterial pneumonia after viral influenza in Andhra Pradesh.

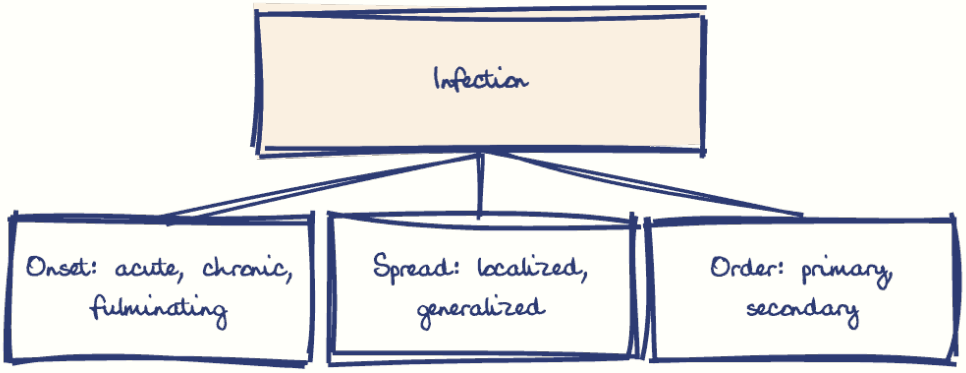
Challenges

1. Diagnosis and access: Weak laboratories in Andhra Pradesh's district hospitals delay

Candidates must not write on this margin

Candidates must not write on this margin

Types of infection



diagnosis, and poor rural access in Rayalaseema raises deaths from fulminating infections.

2. Surveillance: Localized infections are easier to contain, but generalized ones strain disease monitoring and need integrated surveillance.

3. Resistance: Secondary infections need comprehensive care and careful antibiotic use to avoid antimicrobial resistance, a growing concern in state hospitals.

Conclusion

Infections differ in onset, duration, severity and spread. Better diagnostics, digital tracking through ABHA and antimicrobial stewardship will improve infection control in Andhra Pradesh.

Where did you write that point?
Notes that live with the question.
ghissu.com

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 9(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
9(b)

Write a short note on Pathogenesis.

10

Introduction

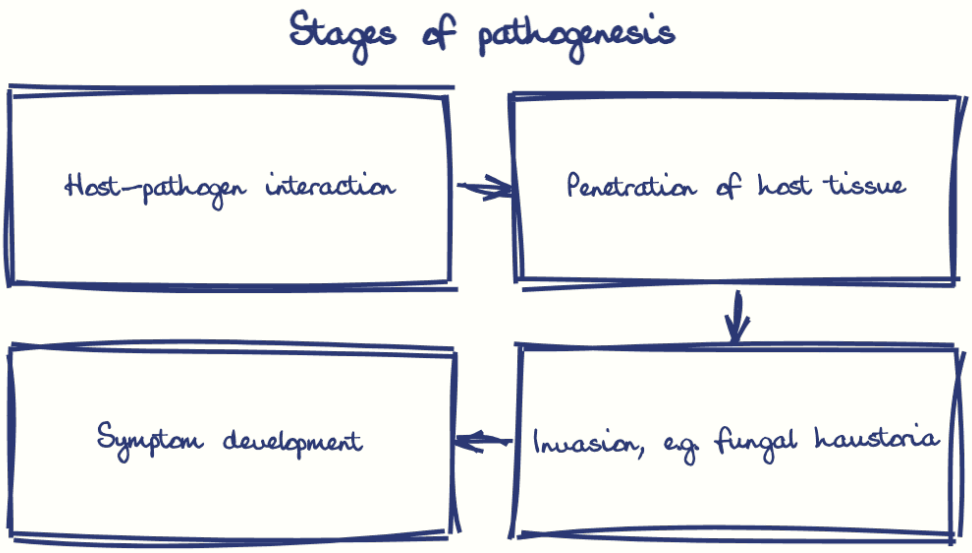
Pathogenesis is a fundamental concept in plant pathology explaining the sequence of events that leads to disease in plants. It underpins disease management, vital for agricultural productivity and food security in Andhra Pradesh.

Process of Pathogenesis

1. Host-pathogen interaction: Pathogenesis unfolds in stages of penetration, invasion and symptom development as the pathogen interacts with the host plant.
2. Invasion structures: Pathogens have structural and physiological traits for attack, such as fungi developing haustoria to invade host tissues.
3. Growth strategies: Pathogens grow ectophytically or endophytically, which complicates control of diseases like downy mildew and rust in Andhra Pradesh.
4. Environmental influence: Environmental factors shape the progress of pathogenesis, so incidence and severity vary unpredictably across Andhra Pradesh's regions.

Practice 9(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice



Study and Institutional Base

1. Pioneers: E. J. Butler laid the groundwork for understanding disease causation and management in India, a legacy continued by Andhra Pradesh's agricultural universities.
2. Diagnosis and quarantine: Pathogen identification enables precise diagnosis, while quarantine protocols keep pathogens out of disease-free areas.

Significance

1. Stage-wise control: Because disease develops in stages, blocking penetration through resistant varieties or quarantine stops it before invasion begins.

Challenges

1. Resource gaps: Limited access to diagnostic tools

and fungicides in rural Andhra Pradesh delays detection and causes crop losses.

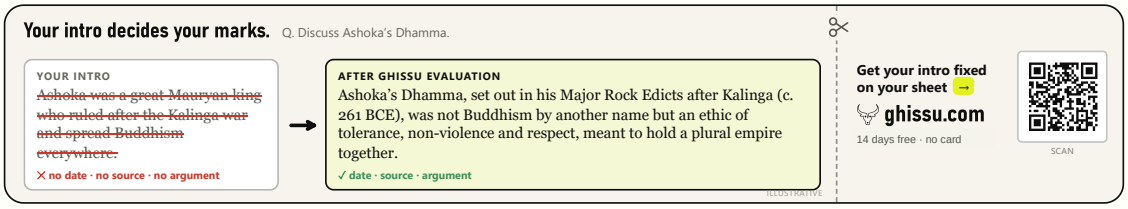
2. Knowledge gaps: Farmers often rely on traditional practices that may miss the nuances of pathogenesis, so extension services must bridge the gap.

Way Forward

1. Research: Fund plant pathology research in Andhra Pradesh on local pathogen dynamics.
2. Resistant varieties: Breeding informed by pathogenesis can yield crops that resist prevalent pathogens.
3. Integrated management: Combine chemical, biological and cultural controls with farmer training for early detection.

Conclusion

Pathogenesis explains the chain of events by which pathogens cause plant disease. Applying it through research and extension can enhance crop resilience and food security in the spirit of Sabka Saath Sabka Vikas.



Write your own answer to 9(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
9(c)

What is hydrophobia? how is it caused? What remedial or precautionary measures are to be taken?

10

Introduction

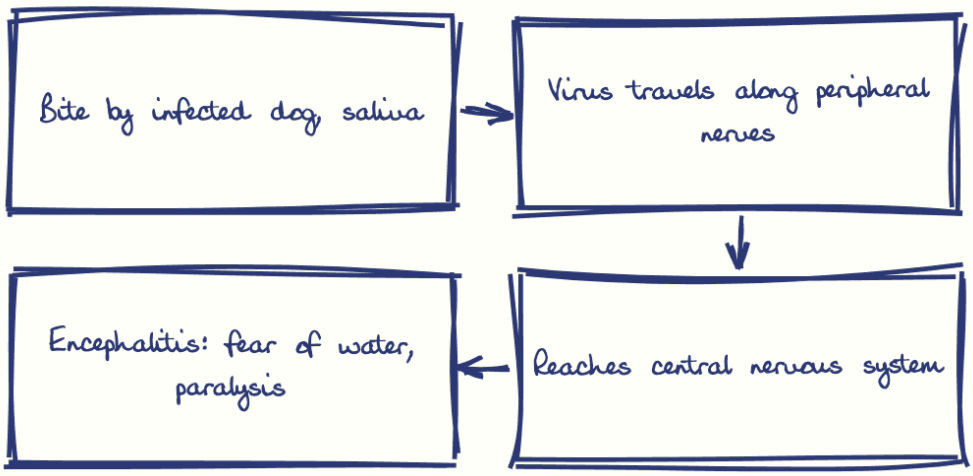
Hydrophobia, commonly known as rabies, is a fatal viral disease that attacks the central nervous system. Once clinical signs appear, death is almost certain. In Andhra Pradesh, where dog bites are a common route of transmission, it remains a serious public health concern.

How Hydrophobia Is Caused

1. Viral Etiology: It is caused by the rabies virus, a neurotropic virus transmitted mainly through the saliva of infected animals, most commonly dogs in India.
2. Pathophysiology: After an infected bite, the virus travels via peripheral nerves to the central nervous system, causing encephalitis and symptoms such as fear of water, agitation and paralysis.
3. Animal Reservoirs: Stray and unvaccinated dogs are the main reservoirs in Andhra Pradesh, though bats and wild carnivores can also transmit the virus.
4. Risk Factors: Lack of awareness, poor access to

post-exposure prophylaxis and inadequate animal control sustain transmission in rural and peri-urban areas of Rayalaseema and coastal districts.

How hydrophobia develops



Remedial and Precautionary Measures

1. Post-Exposure Prophylaxis (PEP): Immediate, thorough wound washing with soap and water, followed by timely rabies vaccine and immunoglobulin, is the most effective way to prevent the disease.
2. Animal Vaccination and Control: Mass dog vaccination and sterilisation programmes reduce stray populations and interrupt virus transmission, as municipal bodies in urban areas like Visakhapatnam do.
3. Public Education: Through state health

campaigns in Andhra Pradesh, communities must learn to avoid stray animals, seek prompt medical care after bites and complete vaccination schedules.

4. Integrated One Health Approach: Collaboration between human health, veterinary services and local bodies ensures coordinated surveillance, vaccination and outbreak response.
5. Institutional Support: The National Rabies Control Programme of the Health Ministry guides vaccination and PEP, implemented through Andhra Pradesh's Directorate of Public Health and Family Welfare.
6. Surveillance: Tracking animal bites and suspected cases needs coordination between veterinary and human health sectors, so that hotspots get targeted vaccination drives.

Conclusion

Hydrophobia is deadly yet preventable through timely intervention. Stronger surveillance, wider dog vaccination and community awareness can sharply reduce rabies in Andhra Pradesh, in line with SDG (3) on good health and well-being.

Stuck at "Critically examine"?
See the thinking before you write.
ghissu.com →

P-IV
10(d)

Write a short note on the adverse consequences of vaccination.

10

Introduction

Vaccination is one of the most effective public health interventions against infectious disease. In India, the Universal Immunisation Programme and Pulse Polio Programme have protected millions. Yet vaccination also has adverse consequences that can affect public trust and programme effectiveness.

Adverse Consequences of Vaccination

1. Vaccine-associated adverse events: Effects range from mild fever and soreness to rare allergic reactions or neurological complications. Studies from Delhi in 1989–94 reported paralytic poliomyelitis in some children after multiple doses of Oral Polio Vaccine.

2. Hesitancy and resistance: Misinformation, cultural beliefs and mistrust breed hesitancy; gendered mistrust slows HPV vaccination.

3. Coercion: Cancelling ration cards or cutting power supply for refusal, as in Uttar Pradesh polio campaigns, alienates communities.

4. Surveillance gaps: India's fragmented surveillance gives patchy adverse event data and delayed responses, so safety concerns are not addressed

Practice 10(d) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

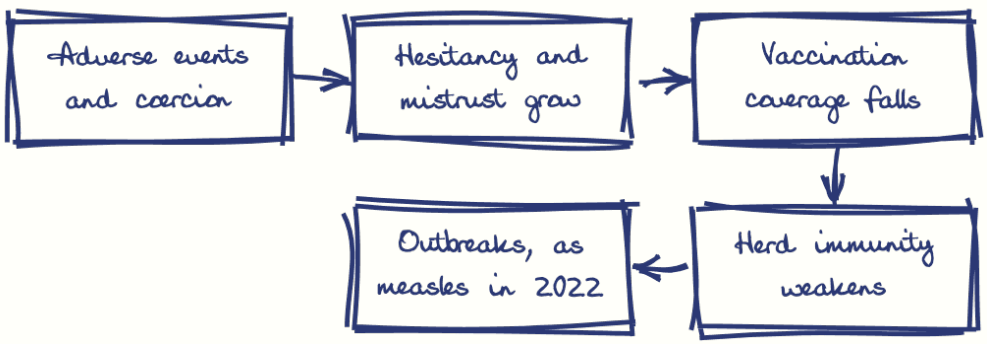
Scan to practice

promptly.

Implications

1. Erosion of trust: Lower coverage threatens herd immunity and invites outbreaks, as with measles in Maharashtra in 2022.
2. Health system strain: Even rare complications need medical attention and resources, burdening regions like Rayalaseema in Andhra Pradesh.
3. Social inequity: Hesitancy clusters in vulnerable populations, widening health gaps; gendered mistrust in HPV vaccination shows how social factors shape access.
4. Ethical dilemmas: The debate over mandatory versus voluntary vaccination pits public health goals against individual rights, needing ethical frameworks.

How adverse consequences hurt coverage



Way Forward

1. Surveillance: Integrate adverse event monitoring with adequate human resources for timely detection and response, especially in Andhra Pradesh's rural districts.

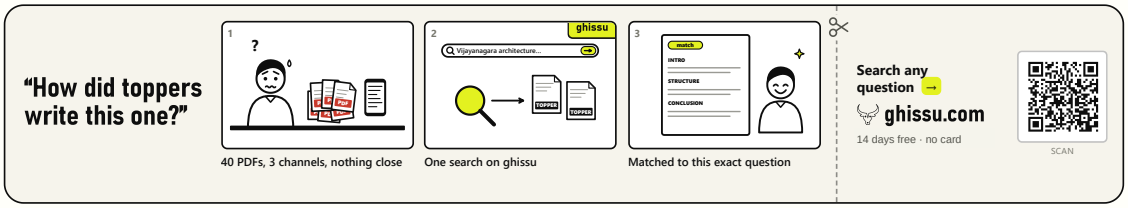
2. Community engagement: Culturally sensitive communication through local leaders and health workers can build trust.

3. Ethical policy: Avoid coercion and promote informed consent and transparency, with proactive disclosure of safety information under the RTI Act 2005.

4. Equity: Prioritise vulnerable groups with gender-sensitive approaches in campaigns like HPV vaccination.

Conclusion

Vaccination is indispensable, but addressing its adverse consequences sustains trust and coverage, supporting SDG (3) on healthy lives for all.



P-IV
10(a)

Enumerate the mechanisms of humoral immunity system and the usefulness of the antibodies.

10

Introduction

Humoral immunity is a critical part of the adaptive immune system that protects the body by producing antibodies. This antibody-mediated response identifies and neutralises foreign antigens, prevents infections and builds immune memory, making it central to vaccine development and disease management.

Mechanisms of Humoral Immunity

1. B-cell activation: The response begins when B-lymphocytes recognise specific antigens through their surface receptors, leading to activation and proliferation.
2. Antibody production: Activated B-cells differentiate into plasma cells that secrete antibodies (immunoglobulins) specific to the antigen.
3. Antibody structure: Each antibody has two identical heavy and two identical light chains forming a Y-shaped structure that binds antigens.
4. Types of antibodies: IgM (primary response), IgG (secondary response, most abundant), IgA (mucosal immunity), IgE (allergy and parasite defence) and IgD (early B-cell receptor).

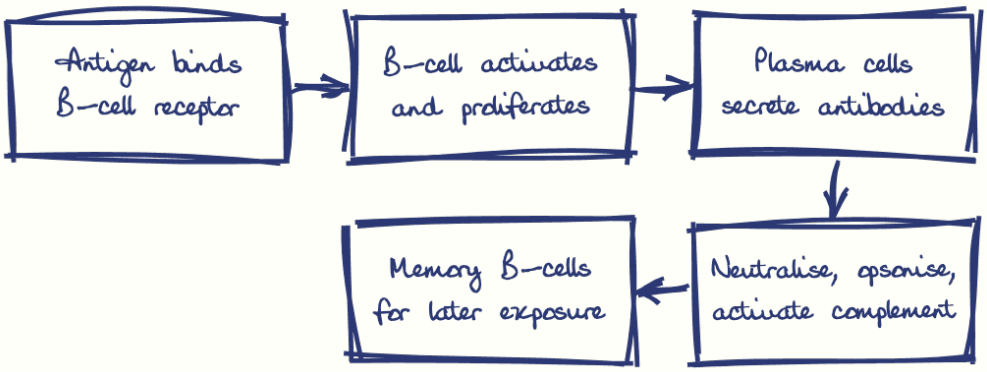
Practice 10(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

5. Memory cell formation: Some B-cells become memory cells that give a faster and stronger response on later exposure.

6. Antigen-antibody interaction: Antibodies bind antigens specifically, neutralising pathogens, marking them for phagocytes or activating complement.

How humoral immunity works



Usefulness of Antibodies

1. Pathogen neutralisation: Antibodies block viruses and bacteria from infecting host cells, as seen in responses to COVID-19.

2. Opsonisation: They coat pathogens so that macrophages recognise and ingest them easily.

3. Complement activation: Binding triggers the complement cascade, lysing pathogens and recruiting immune cells.

4. Immune memory: Long-lasting immunity forms the

basis of vaccination programmes, including those in Andhra Pradesh.

5. Diagnostic and therapeutic use: Serological tests detect past infections and monoclonal antibody treatments fight infectious diseases.
6. Mucosal protection: IgA guards the respiratory and gastrointestinal tracts at the entry points of infection.

Challenges

1. Testing accuracy: False positives and cross-reactivity weaken serological tests used in Andhra Pradesh to assess immunity.
2. Immunity uncertainty: Evidence on how long antibody protection lasts against novel pathogens like SARS-CoV-2 is limited.

Conclusion

Humoral immunity protects the body and enables long-term memory. Antibodies extend beyond natural defence to diagnostics and therapy, forming the backbone of vaccination. Strengthening antibody research and use in Andhra Pradesh will improve disease surveillance.

150 · 250 · 500 words.
Same question, three model answers.
ghissu.com →

P-IV
10(b)

Give an account of the adverse effects of passive immunization.

10

Introduction

Passive immunisation is the direct transfer of preformed antibodies to give immediate protection against infections and toxins, as in snakebites, tetanus and some viral exposures. Despite its life-saving role, it carries adverse effects at the medical, implementation and social levels.

Medical and Safety Effects

1. Short-lived protection: Transferred antibodies degrade over weeks, so repeat doses or active immunisation are needed, straining Andhra Pradesh's public health system.
2. Hypersensitivity reactions: Heterologous antibodies such as equine-derived antivenoms used for snakebite in Visakhapatnam can cause reactions from mild rashes to severe anaphylaxis.
3. Serum sickness: Some recipients develop a delayed immune complex reaction with fever, rash and joint pain, needing extra care.
4. Quality concerns: Impure preparations threaten safety where regulation and quality assurance are uneven, as in many parts of Andhra Pradesh.

Practice 10(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Implementation Problems

1. High cost: Immunoglobulins are expensive and scarce in underserved areas like Rayalaseema, limiting emergency access.
2. Cold chain: Strict storage needs in Andhra Pradesh's remote districts risk loss of potency.
3. Low awareness: Myths and mistrust in some Andhra Pradesh communities delay timely treatment.
4. Weak integration: Poor coordination with active immunisation leaves gaps in long-term immunity.

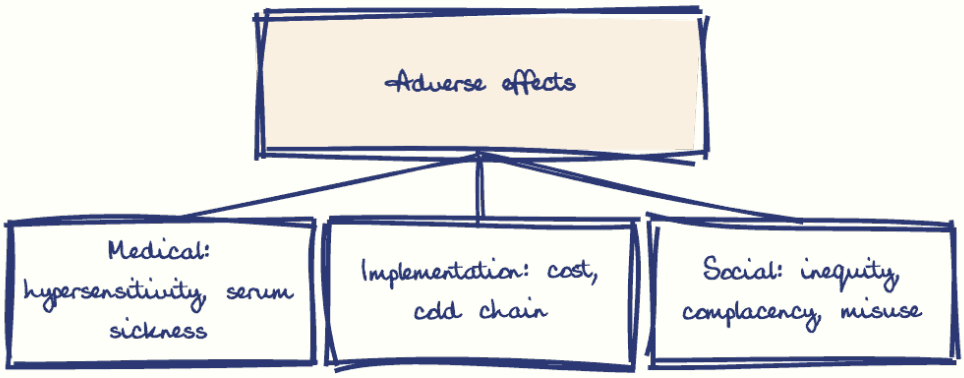
Social and Ethical Effects

1. Equity: Uneven distribution raises fairness concerns for vulnerable groups in drought-prone Rayalaseema.
2. Complacency: Overreliance may reduce vaccination uptake, undermining herd immunity under UIP as implemented in Andhra Pradesh.
3. Consent and misuse: Low literacy hampers informed consent, and inappropriate use wastes resources and exposes patients to risk.

Way Forward

1. Regulation: Strengthen quality control and pharmacovigilance to monitor adverse reactions

Adverse effects of passive immunisation



and ensure safe antibody products.

2. Cold chain: Invest in cold storage and transport in rural and tribal areas to maintain antibody efficacy.
3. Awareness: Targeted campaigns can dispel myths and promote timely acceptance alongside active vaccination.
4. Integration: Coordinated protocols should link passive and active immunisation to optimise coverage and resource use.

Conclusion

Passive immunisation remains indispensable for immediate protection. Stronger pharmacovigilance, cold chains, awareness and integration with active immunisation can contain its adverse effects.

Write your own answer to 10(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V
Scan to practice

P-IV
10(c)

What are the organs of immune system? Give a schedule of immunization for newborn children.

10

Introduction

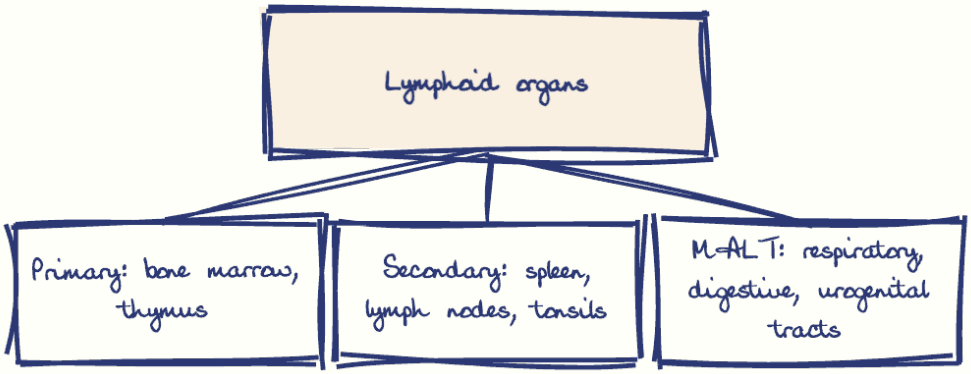
Immunisation is a cornerstone of public health, protecting newborns from infectious diseases that cause much child mortality in India. The Universal Immunisation Programme (UIP) and Mission Indradhanush have widened coverage, and knowing the immune organs and the newborn schedule helps sustain these gains.

Organs of the Immune System

1. Primary Lymphoid Organs: Bone marrow produces all blood cells including lymphocytes, while the thymus, near the heart, matures T-lymphocytes and is prominent in newborns and children.
2. Secondary Lymphoid Organs: Mature lymphocytes move to the spleen, lymph nodes, tonsils, Peyer's patches in the small intestine and the appendix, where they meet antigens and mount responses.
3. Mucosa-Associated Lymphoid Tissue: MALT, about 50% of lymphoid tissue, lines the respiratory, digestive and urogenital tracts where pathogens often enter.
4. Functions: The spleen filters blood-borne pathogens

and stores erythrocytes, while lymph nodes trap antigens from lymph and activate lymphocytes, enabling innate and adaptive immunity.

Organs of the immune system



Immunisation Schedule for Newborns

1. At Birth: BCG against tuberculosis, the oral polio vaccine (OPV) and Hepatitis B vaccine.
2. Through the First Year: Further OPV doses and the Pentavalent vaccine protecting against diphtheria, pertussis, tetanus, hepatitis B and Haemophilus influenzae type b.
3. Why Timing Matters: Timely doses build active immunity by generating memory B and T cells.
4. Coverage: UIP gives free vaccines against (12) diseases to over (2.54 crore) newborns a year, including in Andhra Pradesh.

Candidates must not write on this margin

Candidates must not write on this margin

Challenges

1. Vaccine Hesitancy: Misinformation keeps some parents away despite high coverage.
2. Logistics: Remote areas and rural–urban gaps make timely doses harder; Andhra Pradesh prioritises cold chain and health worker training.
3. Surveillance: Integrating systems to track coverage and adverse events remains a challenge, which Andhra Pradesh is tackling under Mission Indradhanush 4.0.

Way Forward

1. Cold Chain: Upgrade storage and transport in Andhra Pradesh to keep vaccines potent and reduce wastage.
2. Capacity Building: Train frontline workers in protocols and adverse event management.

Conclusion

Knowing the immune organs and keeping to the newborn schedule safeguard child health. Anganwadi-led awareness and a unified digital immunisation platform in Andhra Pradesh will protect every newborn and advance SDG 3.

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 10(c) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
11(b)

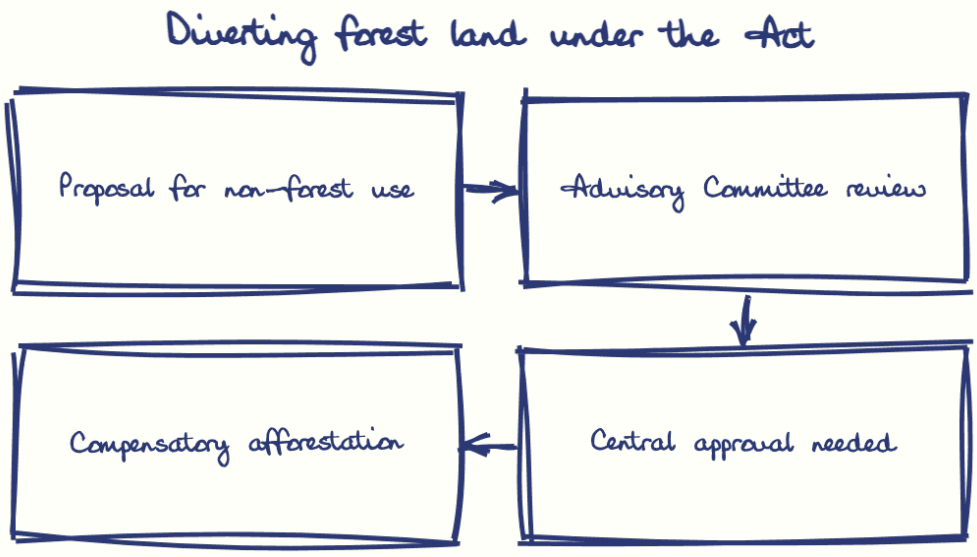
What are the salient features of Forest Act of 1980?

10

Introduction

The Forest (Conservation) Act, 1980 was enacted in response to alarming deforestation and diversion of forest land for non-forest uses. It marked a shift by centralising forest conservation authority, and remains relevant in states like Andhra Pradesh, where forests sustain tribal livelihoods.

Salient Features



1. **Restriction on Diversion:** Dereservation or use of forest land for non-forest purposes requires prior Central approval.
2. **Wide Coverage:** It covers reserved, protected, village and private forests.

Practice 11(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

3. Advisory Committees: The Centre can constitute committees for expert guidance on diversion cases.
4. Penalties: Unauthorised forest land use is penalised, deterring encroachment.
5. Regulated Non-Forest Activities: Certain activities are allowed with prior approval, balancing ecology with development where Andhra Pradesh's industrial corridors and ports meet forests.
6. Compensatory Afforestation: Developers diverting forest land must undertake afforestation elsewhere.

Implementation Issues

1. Centralised Decisions: Shifting approvals to the Centre curbed unregulated diversion but caused delays and Centre-state conflicts, as with Polavaram in Andhra Pradesh.
2. Judicial Expansion: The (1996) Godavarman judgment extended the Act to all forest lands, even unnotified ones, though unclear boundaries complicate it.
3. Tribal Rights: The Act overlooked forest dwellers, a gap partly addressed by the Forest Rights Act, (2006).
4. Recent Changes: The Forest Conservation Rules, (2022) and the (2023) Amendment streamline approvals

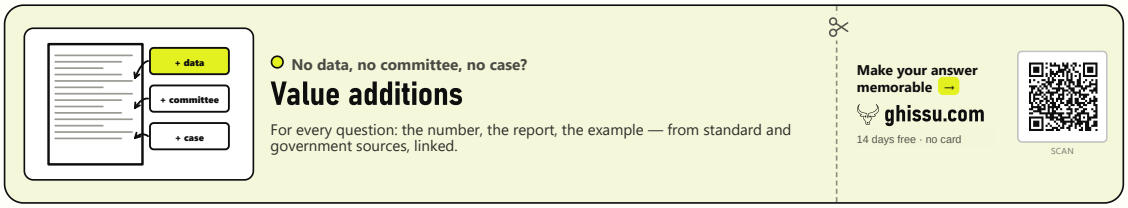
and allow compensatory afforestation in less forested states, but critics fear easier diversion will dilute conservation.

Way Forward

1. Community Participation: Use joint forest management and forest rights for forest-dependent communities in Andhra Pradesh.
2. Transparency: Digitise forest land records and streamline approvals.
3. Restoration: Invest in scientifically guided compensatory afforestation suited to local forest types.
4. Planning: Build rigorous environmental impact assessments into projects from the start.

Conclusion

The Act remains a cornerstone of India's forest governance. Aligning it with community rights and ecological restoration will help balance conservation with development.



P-IV
11(c)

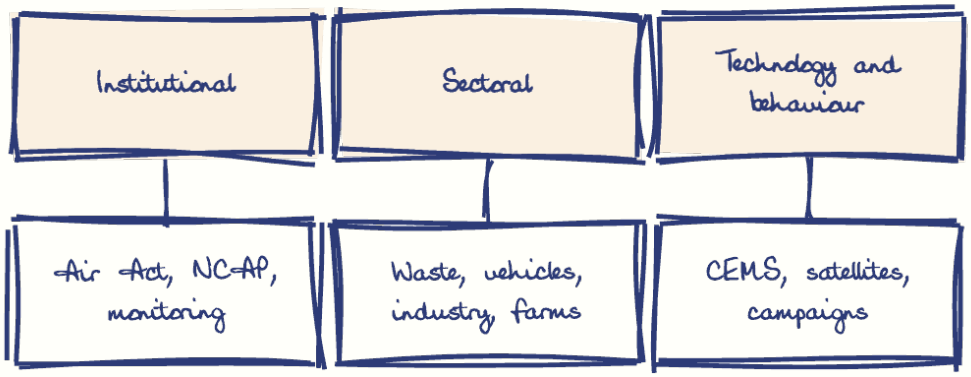
What are the measures to be taken in prevention and control of air pollution?

10

Introduction

Air pollution threatens public health, economic growth and the environment, causing over 2 million deaths a year in India. The challenge is acute in urban and industrial regions. In Andhra Pradesh, solid waste management gaps in urban local bodies worsen air quality, so targeted prevention and control are needed.

Measures to prevent air pollution



Institutional Measures

1. Stronger Laws: Enforce the Air (Prevention & Control of Pollution) Act strictly and align NAAQS with WHO guidelines; the Commission on Air Quality Management in Delhi–NCR is a model for coordinated governance.

2. Decentralised Governance: Empower municipal corporations and urban development authorities in Andhra Pradesh to implement city plans under the National Clean Air Programme.

3. Monitoring and Transparency: Expand continuous ambient monitoring stations and adopt low-cost sensors for real-time data, while transparency under the RTI Act 2005 aids public participation and policy response.

4. Capacity and Awareness: Train officials and promote clean cooking fuel under Pradhan Mantri Ujjwala Yojana.

Sectoral Measures

1. Waste Management: Build sanitary landfills and wet waste plants and enforce segregation, since their absence in most local bodies leads to open burning.

2. Vehicles: Build on BS-VI fuel and electric vehicles by phasing out older vehicles and easing traffic congestion, which add heavily to urban air pollution.

3. Industry and Agriculture: Relocate polluting units, promote cleaner technology and support in-situ crop residue management to curb stubble burning.

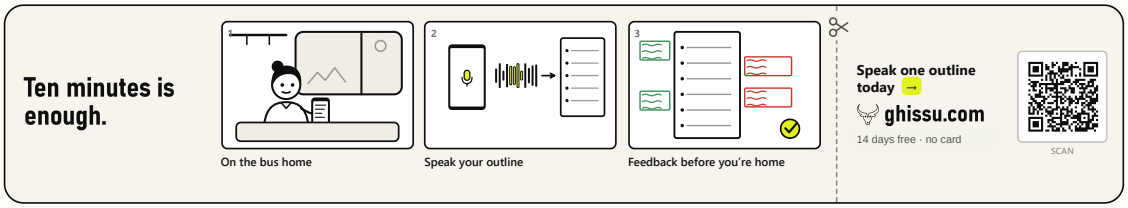
4. Finance: Raise NCAP's limited budget through green bonds, environment funds and public-private partnerships.

Technology and Behaviour

1. Emission Monitoring: Continuous emission monitoring systems (CEMS) and satellite-based air quality surveillance enable proactive management and rapid response to pollution spikes.
2. Integrated City Plans: Combine air pollution control with waste management, transport planning and industrial regulation in Andhra Pradesh's cities.
3. Behavioural Change: Targeted campaigns and incentives for clean fuels and waste segregation reduce household and municipal pollution.

Conclusion

Preventing and controlling air pollution needs a multi-dimensional approach: strong institutions, technological innovation and community participation working together. Aligned with NCAP and the SDGs, this will improve health and support sustainable development.



Write your own answer to 11(c) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

P-IV
13(d)

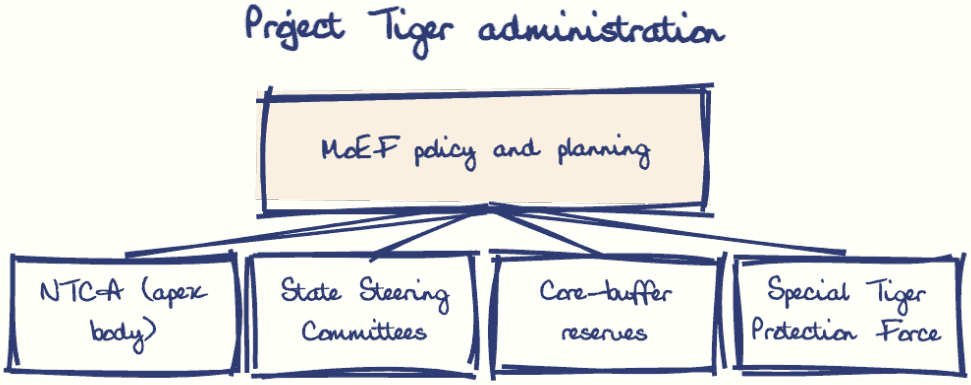
In the Project Tiger, what are the administrative measures taken by the Ministry of Environment and Forests.

10

Introduction

Project Tiger, launched in 1973, remains the flagship conservation initiative of the Ministry of Environment, Forest and Climate Change. India hosts around 70% of the global tiger population, and the Ministry's administrative measures have evolved to address poaching, habitat loss and human-wildlife conflict.

Administrative Measures under Project Tiger



1. **Core-buffer strategy:** The core is strictly protected while the buffer allows regulated activity, reducing fragmentation. The Nagarjunsagar-Srisailam Tiger Reserve in Andhra Pradesh, the largest, covers 3,538 sq. km.
2. **Special Tiger Protection Force:** A dedicated anti-poaching unit conducts patrols, intelligence

gathering and enforcement, backed by watchtowers, camps and communication networks.

3. Financial support to states: Centrally Sponsored Schemes fund habitat management, anti-poaching work and relocation of communities to reduce conflict, in 2022-23 significant funds went to anti-poaching infrastructure and water bodies in reserves.

4. Conflict SOPs: The National Tiger Conservation Authority has issued SOPs for rescue, compensation and awareness.

Role of the Ministry in Administration

1. Policy and planning: It framed the National Wildlife Action Plan, created the NTCA as apex body and works through State-level Steering Committees chaired by Chief Ministers.

2. Implementation and monitoring: It disburses funds and evaluates reserves, using the Wildlife Institute of India for data-driven decisions.

3. Coordination and capacity: Agreements with Nepal, Bangladesh and China support trans-boundary conservation, and training of forest staff and research through the Indian Council of Forestry Research and Education build capacity.

4. Challenges: Balancing the Forest Rights Act with conservation, habitat saturation and funding gaps strain administration.

Way Forward

1. Community participation: Livelihood support and participatory management.
2. Technology: GIS, camera traps and AI-based analytics for monitoring.
3. Funding: More resources for habitat restoration and anti-poaching, including in Andhra Pradesh.
4. Legal harmony: Refine policies to align forest rights with conservation.

Conclusion

These measures have been pivotal in stabilising and increasing India's tiger population. Scientific management, community engagement and inter-governmental coordination can secure tigers' future, in line with SDG (15).

STUDY TIP · BEFORE YOU OPEN A CHAPTER
List what was asked from it in the last ten years. Read that first; the rest second.

ghissu
ghissu has already done this for every topic — with the trend drawn out.

See the trend for your chapter →
ghissu.com
14 days free · no card
SCAN

Write your own answer to 13(d) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
13(a)

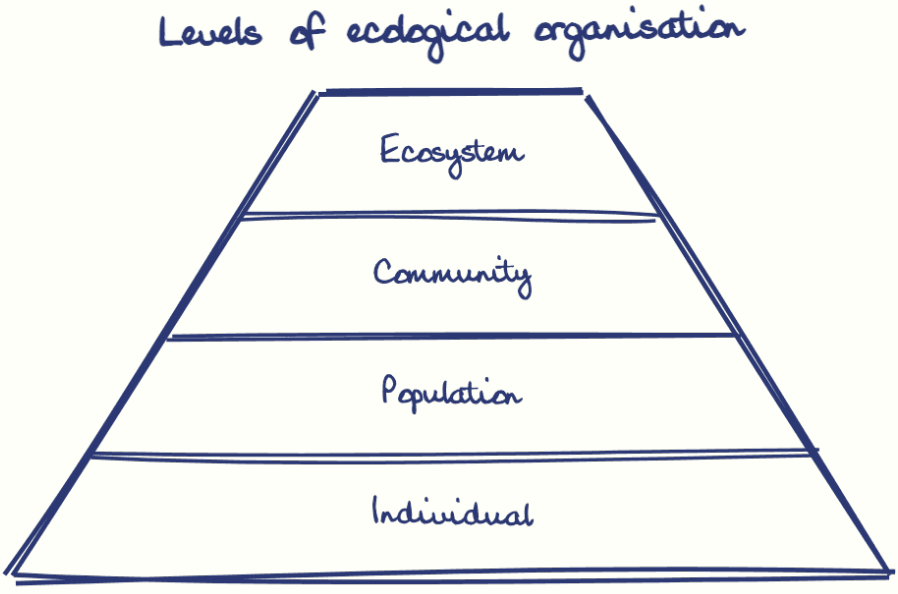
What are the Principles of Ecology?

10

Introduction

Ecology, from the Greek "oikos" (house) and "logos" (study), is the science of relations between living organisms and their physical environment. Amid biodiversity loss, climate change and ecosystem degradation, its principles are crucial for sustainable development, including in Andhra Pradesh, where diverse ecosystems meet development pressures.

Core Principles of Ecology



1. **Ecosystem as a Functional Unit:** Biotic communities and their abiotic environment interact continuously through energy flow and nutrient cycling, as in the interdependence of plants and decomposers.

2. Hierarchical Organisation: Ecology spans individuals, populations, communities and ecosystems, each level with distinct properties.
3. Interdependence and Connectivity: Organisms rely on each other and on their environment, so a change in one part affects the whole system, critical in managing the Krishna and Godavari basins.
4. Balance and Dynamic Change: Ecology once stressed a "balance of nature"; modern understanding adds dynamic change and succession, as seen in the Pichavaram mangroves.

Challenges in Applying the Principles

1. Complexity: Non-linear interactions make prediction difficult, as in Rayalaseema's drought-prone agro-ecosystems.
2. Human Impact: Deforestation, pollution and urbanisation disrupt balance; coastal aquaculture in AP has altered native habitats.
3. Governance Gaps: Weak enforcement and coordination among bodies like the AP Pollution Control Board.
4. Development Pressures: Industrial corridors like Visakhapatnam make it hard to balance growth and conservation.

Integrating Ecology into Development

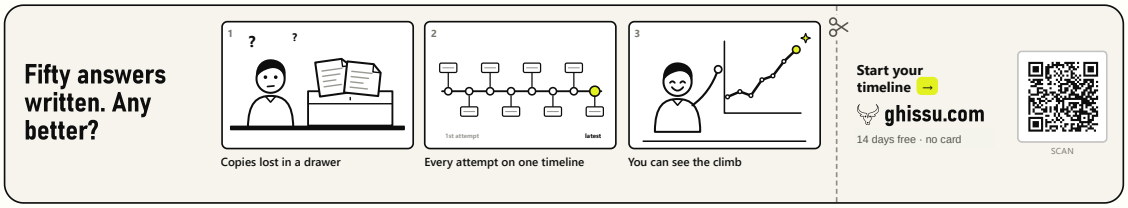
1. Policy: The National Environment Policy calls for ecosystem-sensitive planning, as in the Polavaram project area.
2. Community Knowledge: Participatory forest management in Eastern Ghats tribal areas strengthens stewardship.
3. Restoration: Mangrove reforestation at Pichavaram recovers biodiversity and ecosystem services.
4. Awareness: Ecological literacy builds support for campaigns like Ek Ped Maa Ke Naam.

Way Forward

Strengthen the AP Forest Department, adopt ecosystem-based management in coastal zones and mandate ecological impact assessment in growing districts like Amaravati.

Conclusion

The principles of ecology show that all life is interconnected. Applying them through ecosystem-based management and community participation is the foundation of sustainable development.



P-IV
13(b)

Define biodiversity. Discuss briefly on the consumptive use of biodiversity.

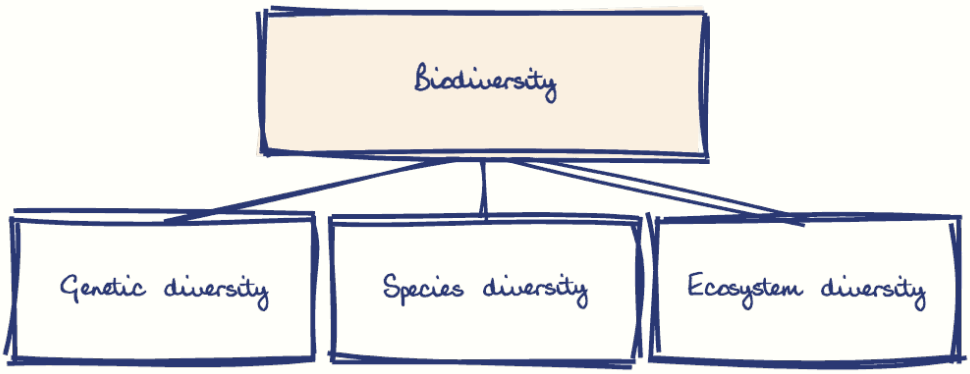
10

Introduction

Biodiversity is critical for ecological balance and human well-being. India, including Andhra Pradesh's diverse agroecosystems and forest landscapes, is rich in biodiversity, so its understanding and sustainable use are vital for ecological security.

Defining Biodiversity

Levels of biodiversity



1. Comprehensive biological variety: Biodiversity is the variety of life forms on Earth, covering genetic, species and ecosystem diversity, which are interdependent and collectively sustain ecological processes.
2. Genetic diversity: Multiple crop strains in Andhra Pradesh's horticulture sector provide resilience

against pests, diseases and environmental stress, underpinning food security and agricultural sustainability.

3. Species and ecosystem diversity: Species diversity covers plants, animals and microorganisms, while ecosystem diversity covers habitats like forests, wetlands and coasts, such as the Eastern Ghats and Krishna delta.

Consumptive Use of Biodiversity

1. Definition and scope: It is the direct harvest of biological resources such as fuelwood, fodder, medicinal plants and food for local consumption; these are not traded commercially but are essential for rural livelihoods.

2. Livelihoods and culture: In Andhra Pradesh's tribal and rural communities, consumptive use meets subsistence needs and sustains traditional knowledge, cultural identity and a close bond with nature.

3. Sustainability concerns: While it gives immediate benefits, overexploitation without management degrades habitats and causes biodiversity loss, threatening long-term resource availability.

4. Policy linkages: Biodiversity Management Committees and People's Biodiversity Registers encourage community

participation, so that consumptive use aligns with conservation goals.

Way Forward

1. Community-based management: Empower local communities through capacity building and participatory governance to manage resources sustainably.
2. Knowledge and enforcement: Blend documented indigenous practice with science, and enforce biodiversity laws at the state level.
3. Alternative livelihoods: Develop eco-friendly enterprises and value chains around biodiversity products to reduce pressure on wild resources and raise incomes.

Conclusion

Biodiversity must be conserved while recognising local consumptive needs. Governance that balances ecology with socio-economic realities, as in the National Biodiversity Strategy and Action Plan and community conservation in Andhra Pradesh, advances SDG (15) for present and future generations.

STUDY TIP · WHERE TO WRITE A NOTE
Write the note against the question that taught it. That is the only place revision will look for it.

ghissu
On ghissu every question has a notes tab — and your notes gather into chapter notes by themselves.

Open the notes tab →
ghissu.com
14 days free · no card
SCAN

Write your own answer to 13(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
13(c)

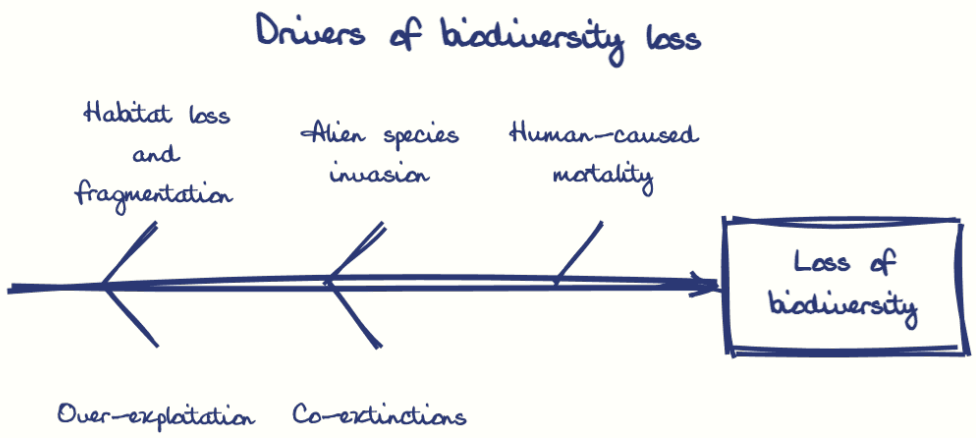
Discuss briefly various reasons leading to the loss of bio-diversity.

10

Introduction

Biodiversity loss has become a critical environmental challenge globally and in India, including Andhra Pradesh, where rich ecosystems face increasing threats. Rapid urbanisation, habitat destruction and unsustainable resource use have accelerated species extinction rates, undermining ecological balance and human well-being. Understanding these causes is essential for effective conservation.

Reasons for Biodiversity Loss



1. **Habitat loss and fragmentation:** The primary driver is deforestation, agricultural expansion and urban development. Conversion of forest land in Andhra Pradesh for aquaculture and agriculture has fragmented habitats, threatening the Asian elephant,

Practice 13(c) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V
1 / 3
Scan to practice

with only about 142 in the state.

2. Over-exploitation of resources: Unsustainable hunting, fishing and plant harvesting cause declines; overfishing along Andhra Pradesh's coast and excess extraction of forest products have endangered several species.

3. Invasion of alien species: Water hyacinth and Lantana have altered native ecosystems, and illegally introduced African catfish threaten native fish, including in Andhra Pradesh's waterways.

4. Co-extinctions: Extinction of one species triggers loss of others dependent on it, such as specialised pollinators or parasites, a cascade that weakens resilience.

5. Human-caused mortality: In South India, electrocution and habitat loss drive elephant deaths, with over 80% of elephant deaths attributed to human activity.

Impact of These Causes

1. Ecosystem services lost: Biodiversity underpins pollination, seed dispersal and water purification; its loss threatens farm productivity and food security.

2. Greater vulnerability: Reduced biodiversity lowers

stability and fragmented forests in Andhra Pradesh cut species' ability to migrate and adapt to climate change.

3. Socioeconomic consequences: Communities reliant on forest and aquatic resources face poverty, migration and conflict over dwindling resources.

Way Forward

1. Habitat protection: Enforce stricter land-use rules in Andhra Pradesh and expand protected areas and ecological corridors for species like elephants.
2. Sustainable use: Promote community-based conservation that draws on traditional knowledge.
3. Control invasives: Run targeted programmes to monitor and eradicate invasive species.
4. Better monitoring: Use DNA-based censuses to track populations and guide adaptive conservation.

Conclusion

Biodiversity loss driven by habitat destruction, over-exploitation, invasive species and co-extinctions poses severe ecological and socioeconomic risks in regions like Andhra Pradesh. Integrated conservation aligned with sustainable development will safeguard ecosystem services, in the spirit of "Sabka Saath Sabka Vikas".

Write your own answer to 13(c) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V
3 / 3
Scan to practice

P-IV
14(d)

Write a short note on Eutrophication.

10

Introduction

Eutrophication is the excessive growth of algae and aquatic plants in water bodies, driven by nutrient pollution from agricultural runoff, sewage discharge and industrial effluents. It severely degrades aquatic ecosystems and water quality. In Andhra Pradesh, with the Krishna and Godavari river systems and a long coastline, it threatens freshwater and coastal resources, fisheries and livelihoods.

Causes

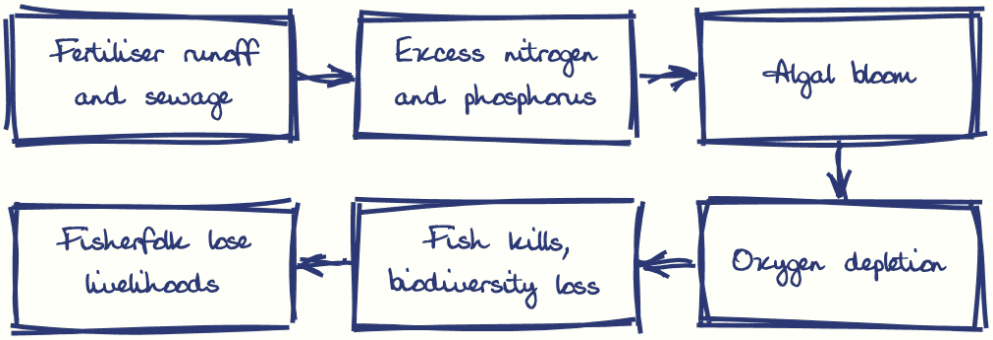
1. Fertiliser runoff: Intensive chilli and horticulture cultivation uses heavy fertiliser, and excess nitrogen and phosphorus leach into rivers and reservoirs.

2. Inadequate sewage treatment: Many sewage treatment plants operate below capacity or are outdated, so untreated or partly treated effluent enters water bodies.

Consequences

1. Oxygen depletion: Algal blooms cause oxygen depletion, fish kills and shifts in species composition in freshwater lakes and reservoirs.

How eutrophication sets in



2. Biodiversity loss: Aquatic ecosystems vital for local communities and fisheries are threatened.

3. Livelihood effects: Fisherfolk and rural populations in coastal Andhra Pradesh and Rayalaseema lose income and food security.

Institutional Framework

1. Pollution Control Boards: The Central and Andhra Pradesh Boards monitor water under the National Water Quality Monitoring Programme, identify eutrophic water bodies and enforce norms.

2. River Rejuvenation Committees: The Central Monitoring Committee and state committees coordinate river restoration by reducing nutrient inflows.

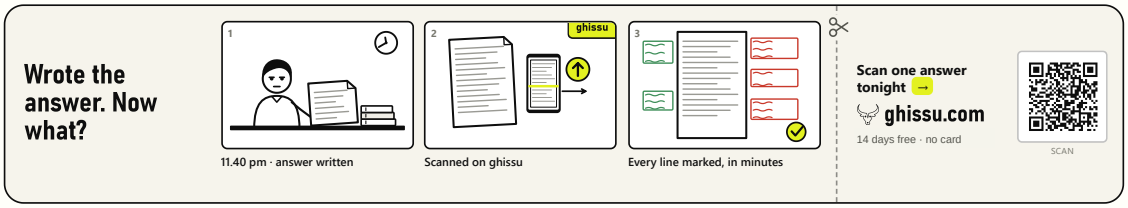
3. Balanced fertiliser use: ICAR promotes the Soil Health Card Scheme, actively implemented in the state to cut runoff.

Way Forward

1. Integrated nutrient management: Promote neem-coated urea, precision fertiliser application and organic farming.
2. Modern sewage infrastructure: Build decentralised, efficient treatment plants in urban and peri-urban areas, building on AMRUT.
3. Wetland restoration: Rehabilitate wetlands and riparian buffers as natural nutrient filters, with community monitoring of water bodies.

Conclusion

Eutrophication links environmental damage with economic and social loss in a water-rich state. Coordinated institutional action, sustainable agriculture and robust wastewater management can restore aquatic ecosystems, in line with SDG (6) on clean water and sanitation.



P-IV
14(a)

What are the major Air Pollutants? Add a note on their effects.

10

Introduction

Air pollution remains a critical environmental and public health challenge in India, acute in urban and industrial regions like Visakhapatnam's industrial corridor, where rapid growth has raised emissions. Understanding the major pollutants and their effects is essential for mitigation under the National Clean Air Programme (NCAP).

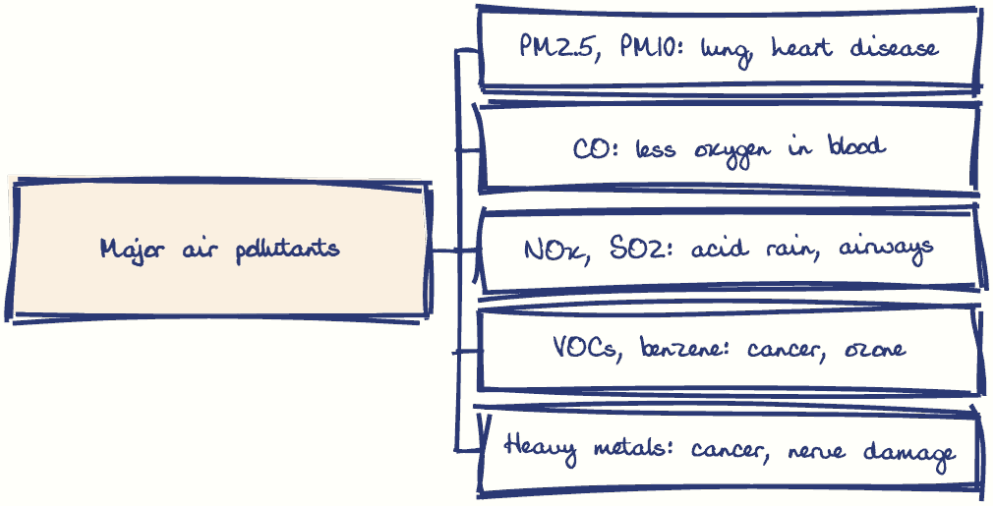
Major Air Pollutants and Their Effects

1. Particulate Matter (PM2.5 and PM10): Fine particles from vehicles, industry, construction dust and biomass burning; elevated levels in cities like Visakhapatnam cause respiratory diseases and aggravate cardiovascular conditions.
2. Carbon Monoxide (CO): From incomplete combustion of fossil fuels; it reduces oxygen delivery in the blood, causing headaches, dizziness and, in severe cases, death.
3. Nitrogen Oxides and Sulphur Dioxide: From thermal power plants and smelters; they cause acid rain that damages crops and forests and worsen respiratory illness.
4. VOCs and Benzene: From refineries and chemical

industries; they are carcinogenic and form ground-level ozone, which impairs photosynthesis and crop yields.

5. Heavy Metals: Arsenic, cadmium and mercury from coal combustion and effluents accumulate in soil and water, posing cancer and neurological risks.

Major air pollutants and effects



Wider Environmental and Economic Effects

Pollution weakens vegetation, reduces solar panel efficiency and raises healthcare costs, slowing industrial and agricultural growth. Outdoor workers and children in urban slums bear the heaviest health burden.

Regulatory Response

1. NCAP: Launched in 2019, it targets 131 cities, including Vijayawada and Visakhapatnam, to cut PM10 by 40% by 2025–26.
2. Vehicle Norms: BS-VI fuel since 2020 and EVs under FAME-II reduce CO and NOx.
3. Enforcement Gaps: APPCB faces capacity limits in industrial hubs of Krishna and Godavari districts, and norms are enforced inconsistently in the Visakhapatnam–Chennai corridor.
4. Monitoring Gaps: Uneven data coverage and weak inter-agency coordination limit real-time tracking.

Way Forward

Expand real-time monitoring, give APPCB more authority and resources, and promote clean energy and transport with public awareness.

Conclusion

Each major pollutant harms health, crops and the economy in its own way, so a multi-pronged response is needed to protect public health and growth.

Stuck at "Critically examine"?
See the thinking before you write.
ghissu.com →

Write your own answer to 14(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
14(b)

What is meant by Electrostatic precipitator? Explain its working principle.

10

Introduction

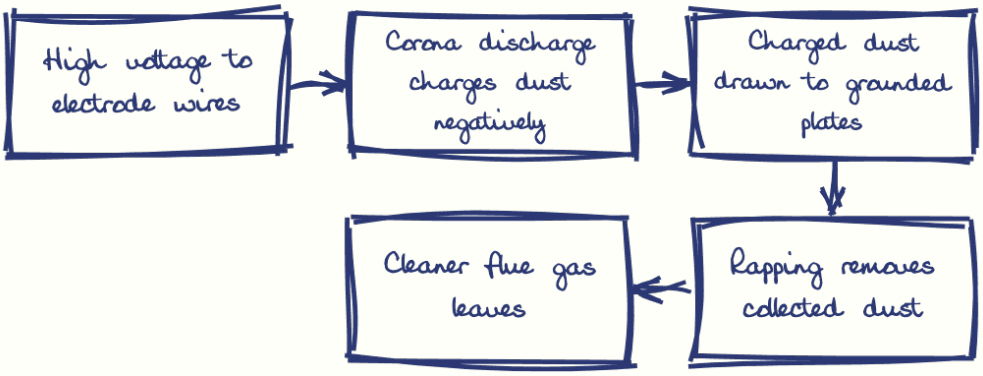
Air pollution control is a critical concern in industrial areas, especially in Andhra Pradesh's growing hubs like Visakhapatnam. An electrostatic precipitator (ESP) is a pollution control device that removes particulate matter from flue gas using electric charge, used in thermal power plants, cement factories and steel plants.

Working Principle of an ESP

1. **Charging Particulate Matter:** A high voltage applied to electrode wires creates a corona discharge that releases electrons, which attach to dust particles in the flue gas and give them a negative charge.
2. **Collection of Charged Particles:** The negatively charged particles are attracted to grounded collection plates of opposite charge, where they accumulate.
3. **Removal of Dust:** Collected dust is periodically removed by mechanical rapping or shaking to maintain efficiency.
4. **Gas Flow Control:** Air velocity between the plates must be controlled so particles have

enough time to settle.

How an electrostatic precipitator works



Operational Challenges

1. Ash Resistivity: Indian coals with high silica and alumina produce high-resistivity ash that reduces ESP performance, a challenge for plants across Andhra Pradesh.
2. Operational Lapses: Fluctuating gas flow and maintenance lapses reduce effectiveness.
3. High Capital Cost: Running and maintenance costs are low, but initial capital cost is high.
4. Coal Quality: High ash content in coal used in the state's thermal plants further lowers efficiency.

Regulatory Framework

1. Central Mandate: MoEF&CC mandates installing and upgrading ESPs in thermal power plants

to meet particulate norms, the Visakhapatnam Thermal Power Station is implementing these upgrades.

2. State Enforcement: APPCB ensures industries install ESPs under the Air Act.

Way Forward

1. Integrated Controls: Combine ESPs with scrubbers and bag filters in cement and steel plants that emit diverse pollutants.
2. Better Coal: Adopt coal washing to reduce high ash content.
3. Finance and Research: Subsidies and low-interest loans for retrofitting, and ESP designs suited to local coal, drawing on Andhra University and Sri Venkateswara University.

Conclusion

By charging and collecting dust particles, ESPs remove particulate matter at low running cost. Backed by APPCB monitoring, regular upgrades and better coal, they help industrial hubs like Visakhapatnam meet air quality standards.

How did toppers write this one?
1.5 lakh+ topper answers, matched to each question.
ghissu.com →

Write your own answer to 14(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
14(c)

Differentiate between BOD and COD and discuss their importance.

10

Introduction

Water quality assessment is critical for environmental management and public health, especially in Andhra Pradesh where agriculture and aquaculture depend on water. Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) are two key parameters of water pollution.

Differences

BOD vs COD

BOD	COD
Oxygen demand by microbes	Oxygen demand by chemical oxidants
Only biodegradable organic matter	Biodegradable and non-biodegradable organics
Requires 5-day incubation	Results available within hours
Shows biological treatment efficiency	Shows total organic pollution load

1. Measurement principle: BOD is the oxygen used by aerobic bacteria to decompose organic matter, typically over five days at 20°C, COD uses strong chemical reagents to oxidise

Practice 14(c) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

organics rapidly.

2. Scope: COD captures pollutants resistant to biodegradation, so it is broader and quicker, while BOD is slower but tied to biology and so shows how well biological treatment is working.

Importance

1. Aquatic life: High BOD indicates oxygen depletion risk, leading to fish kills and ecosystem damage.
2. Industrial effluents: Where effluents carry non-biodegradable chemicals, BOD and COD diverge; COD is crucial in Visakhapatnam industrial zones.
3. Targeting sources: Together they help regulators act, such as on tannery effluents near Chittoor and agricultural runoff in Nellore.
4. Regulation: The APPCB tests the Godavari and Krishna with both and builds them into discharge norms under the Water (Prevention and Control of Pollution) Act to protect water bodies like the Pulavaram reservoir.

Practice 14(c) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Way Forward

1. Laboratory networks: Rural areas, including drought-prone Rayalaseema and coastal aquaculture zones, lack testing labs, so accredited labs with rapid testing are needed for timely monitoring.
2. Integrated control: Combined BOD and COD data should design targeted treatment for industrial and agricultural pollutants in high-risk zones like the Krishna delta.

Conclusion

BOD and COD are complementary: one shows the biological stress on water, the other the total organic load. Together they guide treatment design, enforcement and public awareness, as seen in better compliance in urban centres like Amaravati.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 14(c) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
15(d)

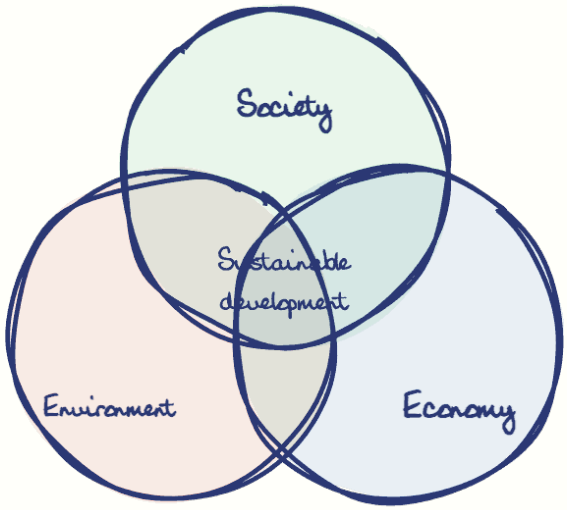
What is Brundtland Report? Enumerate.

10

Introduction

The Brundtland Report, formally titled Our Common Future, was published in 1987 by the World Commission on Environment and Development (WCED), chaired by Gro Harlem Brundtland, then Prime Minister of Norway. This landmark document brought sustainable development to the forefront of global policy, linking environmental, social and economic issues.

Our Common Future (1987)



Key Features of the Report

1. WCED: An international body set up by the UN amid concern over the accelerating deterioration of the environment and natural

resources.

2. Definition: Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

3. Social and economic links: It tied poverty, social equity and environmental degradation together and sought strategies that address all three.

4. Global influence: It laid the ground for the 1992 Earth Summit in Rio and Agenda 21.

Challenges and Outcomes

1. Poverty and environment: The report showed how poverty drives degradation, visible in AP where agrarian distress and resource depletion coexist and complicate sustainable development.

2. Policy integration: Turning its holistic vision into action demands coordination across forestry, water management and urban planning in AP.

3. Non-binding nature: Its recommendations lacked legal force, limiting enforcement and accountability.

4. Catalyst for SDGs: Its principles shaped the UN Sustainable Development Goals, which guide

planning in India and AP through NITI Aayog.

Legacy

1. Intersectoral approach: It encouraged integrated approaches that combine economic growth with environmental protection, as in AP's sustainable agriculture and coastal zone management.
2. Awareness: It raised global awareness of sustainability, shaping curricula and public discourse on conserving resources for future generations.
3. Cooperation: It fostered international dialogue, letting India carry a sustainability agenda to forums like BRICS and the UNFCCC.

Conclusion

The Brundtland Report redefined development by placing intergenerational equity at its core. Its enduring legacy is the recognition that economic progress must harmonise with environmental protection and social equity, a vision that can guide Andhra Pradesh and India towards the SDGs.

REVIEWER'S NOTE · VALUE ADDITION

Examiners read hundreds of answers a day. The one with a number, a committee and a case is the one they remember.

ghissu

ghissu gives you the data point, the report and the example for every question — each linked to its source.

Get the value adds for any question →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 15(d) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
15(a)

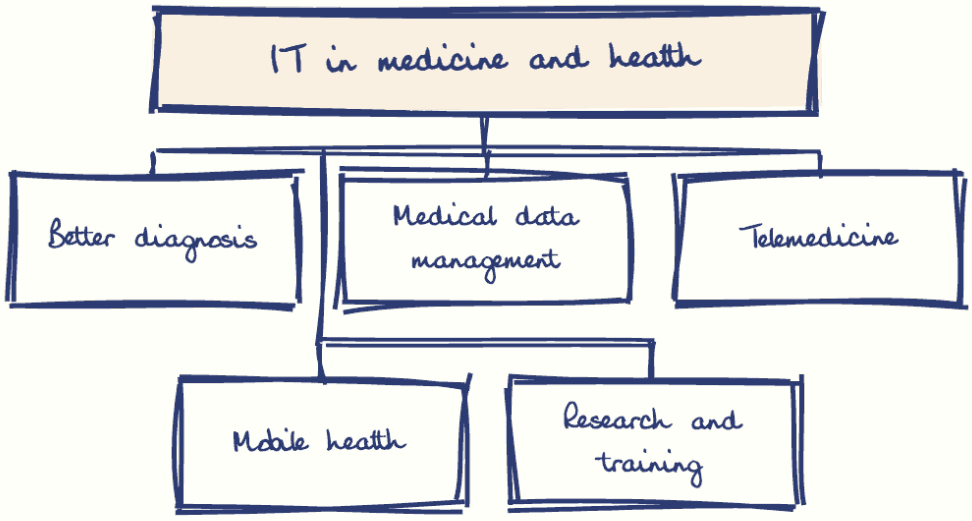
What are the salient benefits of the application of Information Technology in the field of medicine and medical health?

10

Introduction

The rapid advance of Information Technology (IT) has revolutionised medicine, and the COVID-19 pandemic underlined the need for accessible, efficient care. In India, the Ayushman Bharat Digital Mission (ABDM) has accelerated IT in healthcare to bridge urban-rural divides, improve care and enable data-driven policy.

Salient Benefits in Medicine



1. **Improved diagnosis:** AI tools give doctors unbiased second opinions and predictive analytics; AI screening for TB and diabetic

retinopathy brought a 27% reduction in adverse TB outcomes and more case detection.

2. Medical data management: Electronic Health Records under the National Digital Health Blueprint enable interoperable data, less duplication and continuity of care; ABDM has issued over 799 million digital health IDs.
3. Telemedicine: eSanjeevani completed over 183 million consultations by December 2023, serving about half a million patients daily, cutting hospital visits and infection risk and widening specialist reach.
4. Research and education: VR and AR give immersive training, while big data and bioinformatics speed drug discovery, genome sequencing and personalised medicine.

Benefits in Public Health

1. Mobile health: The ANMOL app helps frontline workers register pregnant women, monitor immunisation and promote institutional deliveries.
2. AI in public health: The IndiaAI Mission links AI with ABDM for disease management and surveillance.
3. Health informatics: The National Health Stack

helps policymakers detect fraud, measure outcomes and allocate resources.

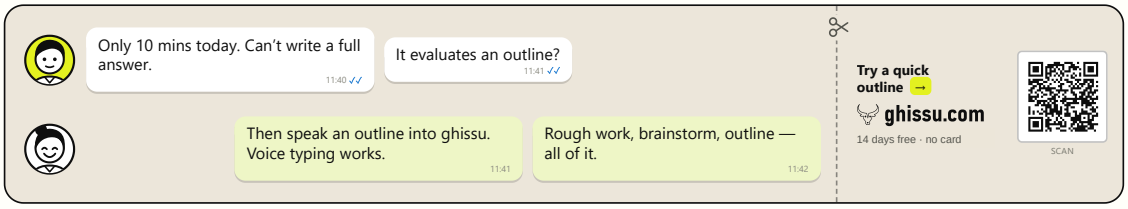
4. Wearables and IoMT: Devices track vital signs and alert users and doctors to emergencies, aiding chronic disease care.

Way Forward

1. Rural infrastructure: Expand high-speed internet and mobile connectivity in underserved parts of Andhra Pradesh for equitable telemedicine.
2. Training and privacy: Train providers in digital tools and enforce health data protection to build trust.
3. Scale up: Bring AI diagnostics and teleconsultation to Andhra Pradesh's primary health centres for early detection.

Conclusion

IT makes healthcare more accessible, efficient and patient-centric, and initiatives like ABDM show how digital health can advance universal health coverage.



P-IV
15(b)

What is meant by watershed? How can watershed management be done effectively with modern Information Technology?

10

Introduction

Water scarcity and erratic monsoons, worsened by climate change, make sustainable water management critical for India's farm and ecological security. Watershed management treats a natural hydrological unit as the planning entity to conserve soil and water, vital in Andhra Pradesh where rainfed farming and drought-prone Rayalaseema need efficient water use.

What Is a Watershed?

1. **Meaning:** A watershed is a geographic area draining runoff to a common point. It transcends administrative boundaries, making it ideal for integrated management; AISLUS delineates sub-watersheds and micro-watersheds.
2. **Institutions:** National and State Steering, Technical and Implementation Committees coordinate, while village watershed committees and Panchayati Raj bodies ensure participation under the IWMP guidelines revised in 2011. Andhra Pradesh adapts IWMP projects to local drought and groundwater depletion.

Practice 15(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Challenges Today

1. **Funding Cuts:** Reduced central funding under PMKSY and the sidelining of NGOs weaken participation.
2. **Technocratic Fixes:** Engineering quick-fixes that ignore hydrological links, as critiqued in Mission Kakatiya, hurt long-term health.
3. **IT Gaps:** Many projects still lack modern monitoring tools.

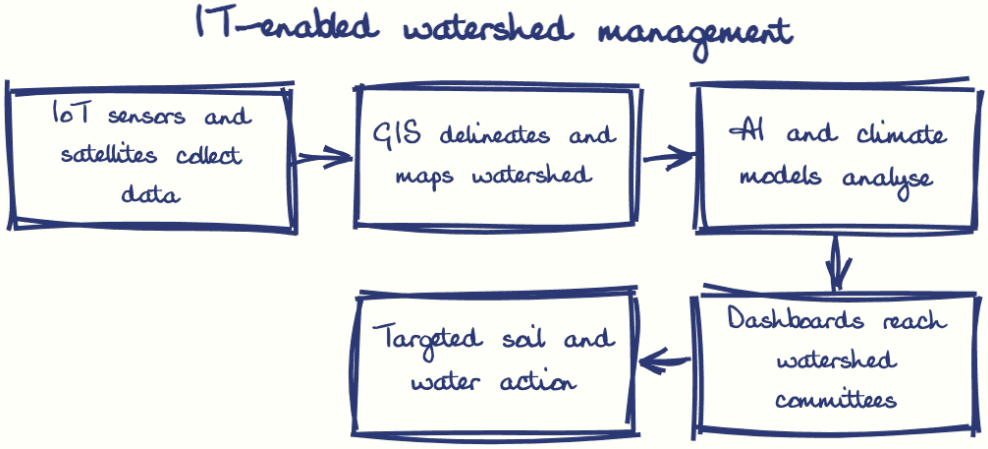
Effective Management Using Modern IT

1. **Real-Time Monitoring:** IoT and AI-enabled sensors track soil moisture, rainfall and water quality for precise irrigation scheduling and early warning of drought or floods.
2. **Remote Sensing and GIS:** Satellite imagery and GIS mapping support delineation, land use classification and monitoring of vegetation change for targeted interventions.
3. **Open Water Data:** Nodal agencies like the National Water Informatics Centre share unclassified hydrological data for informed decisions.
4. **Digital Community Platforms:** Mobile apps and dashboards give watershed committees forecasts, best practices and participatory monitoring tools, building ownership.

Candidates must not write on this margin

Candidates must not write on this margin

5. Climate Modelling: IT-enabled climate models and water budgeting align interventions with climate resilience in drought-prone areas.

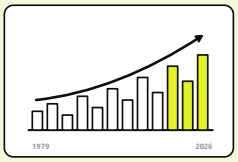


Way Forward

Better coordination between national and Andhra Pradesh watershed committees for data sharing and funding and investment in IoT networks and AI analytics suited to regional watersheds.

Conclusion

Where managed holistically, watersheds have improved soil conservation, water retention and livelihoods. Combining IT with community participation can make Andhra Pradesh's rainfed zones drought-resilient.



Which sub-topic will they ask next?

Trend analysis

Every syllabus topic: the direction of travel, the most tested sub-topic, the probable next theme.

Prepare the pattern →

ghissu.com

14 days free · no card

SCAN

P-IV
15(c)

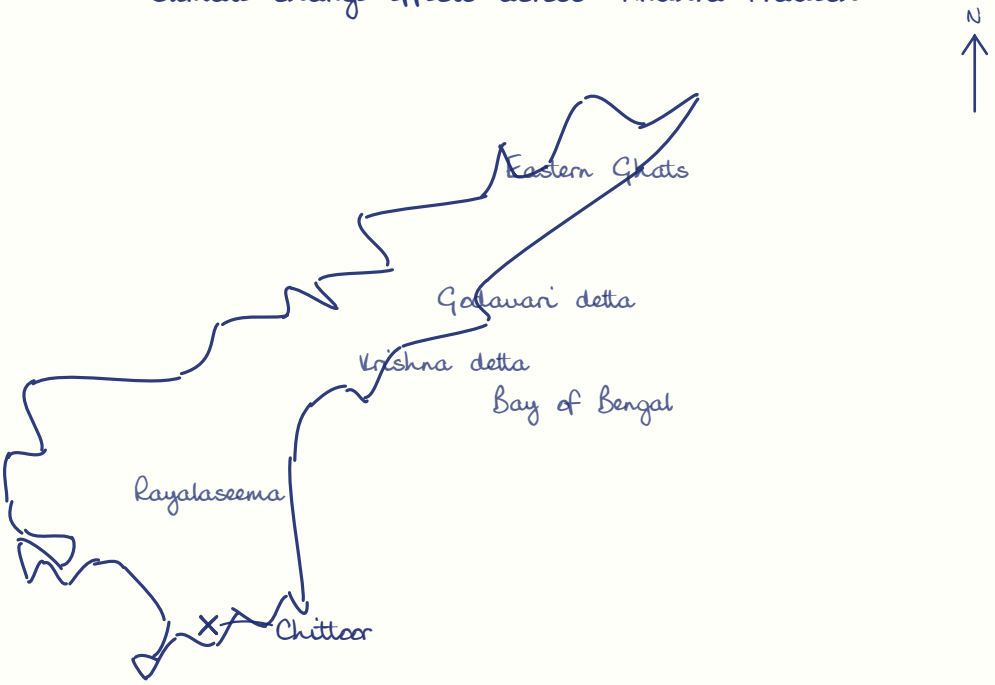
What are the effects of climate change on nature, sea level fluctuations, plants and animal migration, changes in insect pests?

10

Introduction

Climate change has profound effects on natural systems and societies. In India, rising temperatures, altered rainfall and extreme events already affect ecosystems, biodiversity and livelihoods. Andhra Pradesh, with its 974 km coastline and rich biodiversity, shows the vulnerability to sea level rise and shifting species.

Climate change effects across Andhra Pradesh



Effects on Nature

1. Biodiversity disruption: Altered temperature and rainfall upset life cycles, pollination and food availability; endemic species of the Eastern Ghats are affected.
2. Extinction risk: Droughts and heatwaves have caused declines in amphibians and reptiles in Rayalaseema.
3. Invasive species: Changing climates help invasive plants encroach on mangroves in the Krishna and Godavari deltas, weakening these biodiversity hotspots and natural buffers.

Sea Level Fluctuations

1. Faster rise: The Indian Ocean region is rising faster than the global average, with about 300 mm projected, causing erosion and saline intrusion into aquifers.
2. Coastal flooding: Storm surges during cyclones like Titli flood farmland and damage infrastructure; salinity threatens aquaculture.

Plant and Animal Migration

1. Range shifts: Rising temperatures push species towards cooler habitats; some bird and butterfly species in AP's forests have shifted attitudinal

Practice 15(c) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2012 · printed in old-syllabus Paper IV, now Paper V
Scan to practice

ranges.

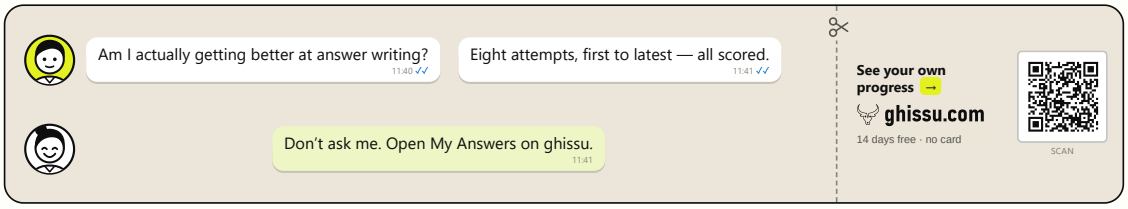
2. Phenology: Altered flowering of chillies disrupts pollinators and yields in Rayalaseema.
3. Human–wildlife conflict: Animals moving near settlements raise human–wildlife conflict in Chittoor.

Changes in Insect Pests

1. Higher incidence: Warmer temperatures and humidity intensify pest incidence in chillies and paddy, raising pesticide use and costs.
2. Wider ranges: Pests invade new areas that were earlier unsuitable for their survival, threatening fresh regions and crops.

Conclusion

Climate change acts at every level of nature: it shifts coastlines, moves species, breaks ecological timing and multiplies pests. Adaptive strategies under the NAPCC and Panchamrit targets are essential to protect ecosystems and livelihoods.



APPSC GROUP-I MAINS · PAPER V

2011

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

P-IV
1(a)

(i) Illustrate the National Policy on Science and Technology. (ii) Discuss the role of technology missions in India with a suitable example.

10

Introduction

Science and technology are pivotal to India's development, given global competition and the need for self-reliance. The National Policy on Science and Technology has evolved to align scientific capability with socio-economic goals. Technology missions, as focused interventions, translate policy into outcomes, as ISRO's missions from Sriharikota and the India Semiconductor Mission show.

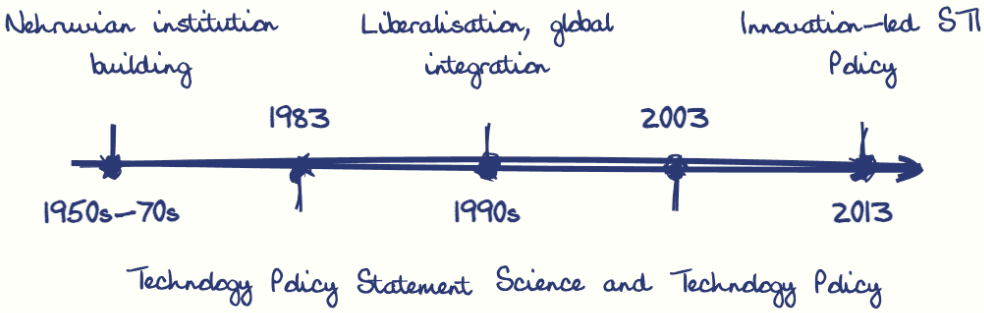
(i) National Policy on Science and Technology

1. Historical phases: Policy moved from Nehruvian institution-building in the 1950s–70s, to the 1983 Technology Policy Statement promoting indigenous development, to 1990s liberalisation and global integration.
2. Policies of 2003 and 2013: They emphasised innovation-led growth, higher R&D funding and academia-industry-government collaboration in biotechnology, electronics and pharmaceuticals.
3. Institutional capacity: IIT Tirupati and the Sriharikota spaceport support national missions, while technology aids aquaculture and horticulture in

Andhra Pradesh.

4. Governance: DST and the new National Research Foundation improve funding and coordination, addressing past fragmentation.

Evolution of S&T policy



(ii) Role of Technology Missions

1. Sectoral focus: Missions deploy dedicated resources in agriculture, health and energy. The Technology Mission on Oilseeds and Pulses improved yields and farmer incomes in Rayalaseema.

2. Lab to land: Despite CSIR and DRDO, transfer to industry is slow; missions speed it through PPPs and incubation.

3. Funding limits: R&D spending of about 0.65% of GDP and weak venture capital restrict missions.

4. Example: The India Semiconductor Mission builds domestic chip manufacturing to reduce import dependence, with projects near Visakhapatnam,

linking growth with technological sovereignty.

Policy to Mission Link

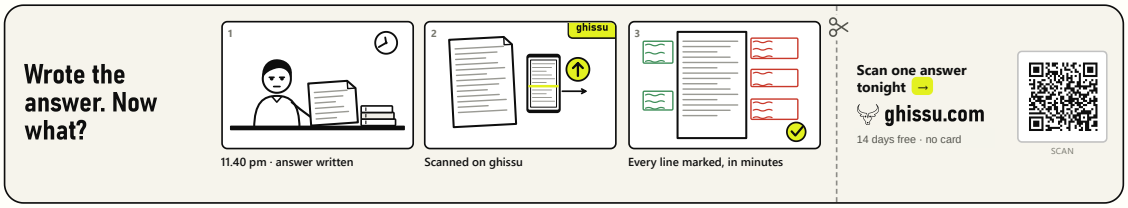
The policy sets goals of indigenisation and innovation; missions then build sectoral capacity which produces socio-economic impact and strategic autonomy.

Way Forward

1. State ecosystems: Raise funding and support for S&T in states like Andhra Pradesh to use local strengths in farming and industry.
2. Public-private partnerships: Give clear incentives for private participation so missions stay demand-driven.
3. Commercialisation: Set up value addition centres in public research institutions to improve technology readiness.

Conclusion

The policy and its missions form the backbone of India's innovation-led growth and strategic autonomy.



P-IV
1(b)

Discuss on (i) Intellectual Property Rights. (ii) National Policy on Technology Missions.

10

Introduction

In an era where innovation drives growth and competitiveness, Intellectual Property Rights (IPR) and technology missions are crucial to India's development. The National IPR Policy (2016) and various technology missions aim to foster creativity, protect innovations and promote technological self-reliance in a knowledge-driven economy, as ISRO's Sriharikota facility in Andhra Pradesh shows.

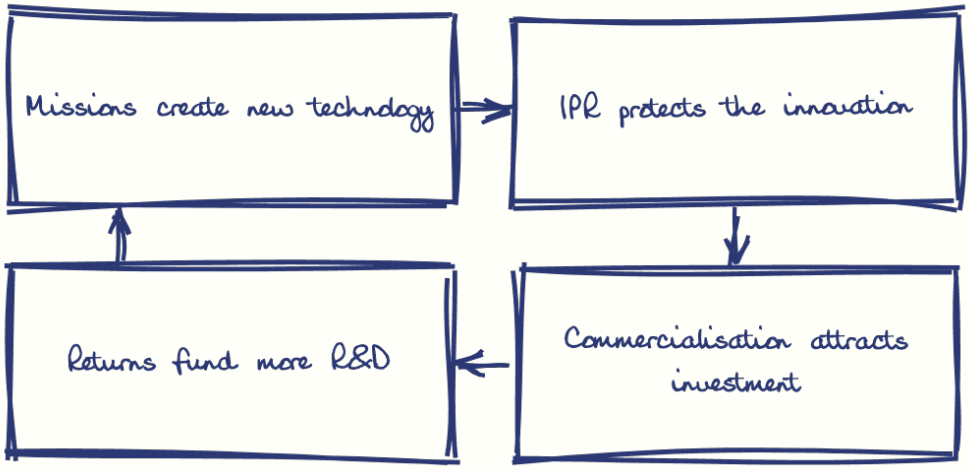
(i) Intellectual Property Rights

1. Meaning and importance: IPR protects creations of the mind such as inventions, literary works and designs, giving innovators recognition and financial reward that encourages more innovation.
2. Components: Patents, copyrights, trademarks, geographical indications, semiconductor layout designs and plant variety protection.
3. Economic and social role: IPR encourages R&D and start-ups, as with the Start-Up Intellectual Property Protection Scheme extended till 2026.
4. Balance: Monopoly rights must not stifle access; compulsory licensing safeguards affordable medicines.

(ii) National Policy on Technology Missions

1. Objectives: Promote indigenous technology, reduce import dependence and build innovation ecosystems in semiconductors, biotechnology and digital technologies.
2. Features: Stronger R&D infrastructure, PPPs and monitoring mechanisms, with the Visakhapatnam–Chennai corridor as a technology hub.
3. Impact: India Semiconductor Mission 2.0 has approved projects worth 1.60 lakh crore across six states, aiming to meet 70–75% of chip demand by 2029.
4. Limitations: R&D at about 0.65% of GDP, talent shortages and uneven diffusion to regions like Rayalaseema.

IPR and technology missions



Interlinkages

1. Mutual reinforcement: IPR lets mission innovations be commercialised and attract investment.
2. Policy synergy: Copyright and semiconductor layout design administration sit together under DPIIT.
3. Access and incentives: Both must prevent monopolistic practices that block essential technologies.

Way Forward

1. R&D funding: Raise public and private R&D, especially in Andhra Pradesh's technology clusters.
2. IPR infrastructure: Modernise administration with digital tools and skilled staff to speed patents and trademarks.
3. Inclusive innovation: Use frugal models for drought in Rayalaseema and coastal aquaculture.
4. Industry–academia links: Turn mission research into market-ready solutions.

Conclusion

Together, IPR and technology missions can protect and scale Indian innovation, driving innovation-led growth and self-reliance.

Stuck at "Critically examine"?
See the thinking before you write.
ghissu.com →

Write your own answer to 1(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
2(a)

(i) Comment on "Chandrayan" Programme. (ii) Discuss two salient features of IRS-5 Satellite.

10

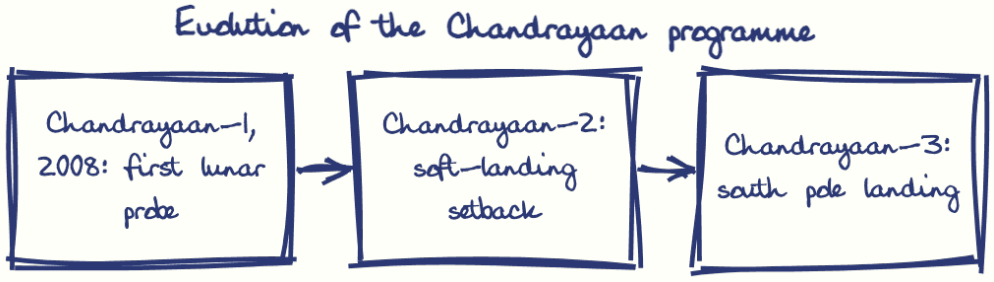
Introduction

India's Chandrayaan programme, begun with Chandrayaan-1 in 2008, has expanded the country's lunar exploration and global standing in space science. Alongside it, the Indian Remote Sensing (IRS) series supports earth observation for agriculture and disaster management, vital for states like Andhra Pradesh.

(i) Comment on the Chandrayaan Programme

1. **Indigenous Development:** The missions rest on ISRO's capacity built since the 1970s, including the PSLV and GSLV Mk-III, allowing complex missions on limited budgets.
2. **Scientific Exploration:** They have advanced lunar science while matching India's dual goals of exploration and practical applications.
3. **Learning from Setbacks:** Chandrayaan-2's landing faced setbacks in soft-landing technology, but Chandrayaan-3 landed near the lunar south pole in 2023, showing resilience.
4. **Strategic Gains:** The evolution from Chandrayaan-1 to Chandrayaan-3 boosted India's credibility and reduced dependence on foreign

systems.



(ii) Two Salient Features of IRS-5

1. **High-Resolution Imaging:** It provides detailed images for monitoring land use, crop health and water bodies, supporting resource planning.
2. **Sun-Synchronous Orbit:** It allows consistent data acquisition, crucial for precision agriculture and environmental monitoring in Rayalaseema and coastal districts.

Applications and Gaps

1. **Scheme Support:** IRS data supports IWMP, PMFBY, AIBP and the Soil Health Card Scheme in Andhra Pradesh.
2. **Data Dissemination:** The Bhuvan geo-portal and Village Resource Centres carry data to users, though capacity in rural Andhra Pradesh needs strengthening.
3. **Coordination:** Bodies like the INSAT Coordination Committee and the National Natural Resources

Management System ensure the data is used effectively across ministries.

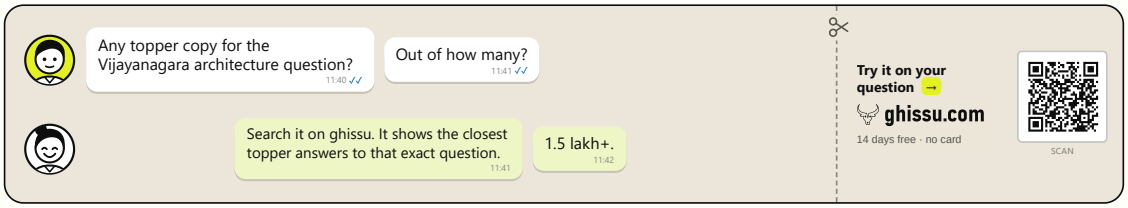
4. Why It Matters: The state's agrarian economy and coastal vulnerability make regular satellite data a direct support to farmers and disaster managers.

Way Forward

Continued investment in indigenous payloads, offering IRS data services globally and better digital infrastructure in rural areas will widen the impact and improve disaster preparedness.

Conclusion

Chandrayaan and IRS-5 show India balancing scientific exploration with development needs. Training local officials and farmers will turn this technological sovereignty into better agriculture and disaster management.



P-IV
2(b)

Illustrate the role of Indian Space Programmes on (i) Rural India (ii) Natural Resources

10

Introduction

India's space programme has become a critical enabler of rural development and natural resource management. With over 65% of the population in rural areas, space applications improve productivity, governance and resilience. ISRO's Bhuvan geoportal and Village Resource Centres show this, including in Andhra Pradesh, home to the Sriharikota launch facility.

(i) Role in Rural India

1. **Precision Agriculture:** RESOURCESAT imagery monitors crop health, pests and soil moisture, enabling transparent and timely insurance settlements under PMFBY.

2. **Water Management:** Mapping groundwater and surface water aids irrigation planning under PMKSY and watershed management in drought-prone Rayalaseema, complementing the Polavaram scheme.

3. **Disaster Response:** Bhuvan provides near real-time flood and drought imagery, vital along Andhra Pradesh's 974 km cyclone-prone coastline.

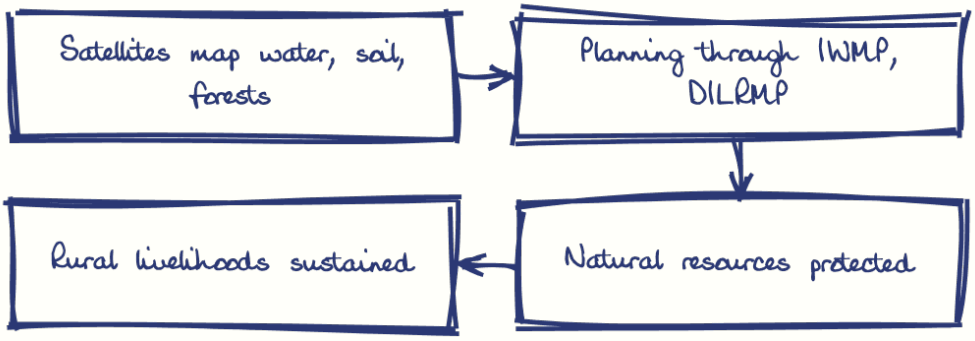
4. **Infrastructure and Governance:** GIS helps plan and

monitor PMGSY roads and electrification, while SVAMITVA uses drone and satellite data for property records, empowering gram panchayats.

(ii) Role in Natural Resources

1. Resource Inventory: Remote sensing databases on land use, forest cover and minerals support conservation, addressing gaps flagged in Andhra Pradesh's mining audits.
2. Environment and Biodiversity: Monitoring of coastal zones and forest health allows early detection of degradation, critical for the state's diverse ecosystems.
3. Weather Forecasting: Meteorological satellites improve prediction, helping farmers adapt to erratic monsoons.
4. Land and Water Planning: Imagery underpins DILRMP and IWMP watersheds, optimising land use and water harvesting.

From space data to rural livelihoods



Candidates must not write on this margin

Candidates must not write on this margin

Way Forward

1. Panchayat Capacity: Train local bodies to use space data in planning.
2. Extension Services: Feed satellite data into Kisan Call Centres.
3. Private Innovation: Encourage startups to build localised space solutions.
4. Data Access: Expand BharatNet so villages can reliably use space-enabled services.

Conclusion

Mapping water, soil and forests protects natural resources, which in turn sustain rural livelihoods. Space programmes thus bridge gaps in agriculture, water security, disaster resilience and environmental sustainability, making indigenous space technology a tool of rural transformation.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 2(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
3(a)

Comment on the impact of the growth of Computer industry on the economic growth of India.

10

Introduction

The rapid expansion of the computer industry has been a cornerstone of India's economic transformation this century. Contributing nearly 9% of GDP and millions of jobs, it has spurred services, manufacturing and innovation, and it now underpins Digital India and the India-AI Mission.

Contribution to Economic Growth

1. Export-Led Growth: Software exports rose from under US$100 million in 1991 to about US$194 billion by 2023, making India a global IT leader.
2. Employment: Over 5.4 million direct jobs, nurtured by IITs and engineering colleges; India holds 23% of the world's software engineering talent.
3. Innovation Spillovers: Over 159,000 startups and more than 100 unicorns as of 2025 drive investment and SME technology adoption.
4. Regional Development: AP's GSDP grew 9.24% in 2024-25, ahead of the national 6.4%, helped by IT hubs in Visakhapatnam and Amaravati.

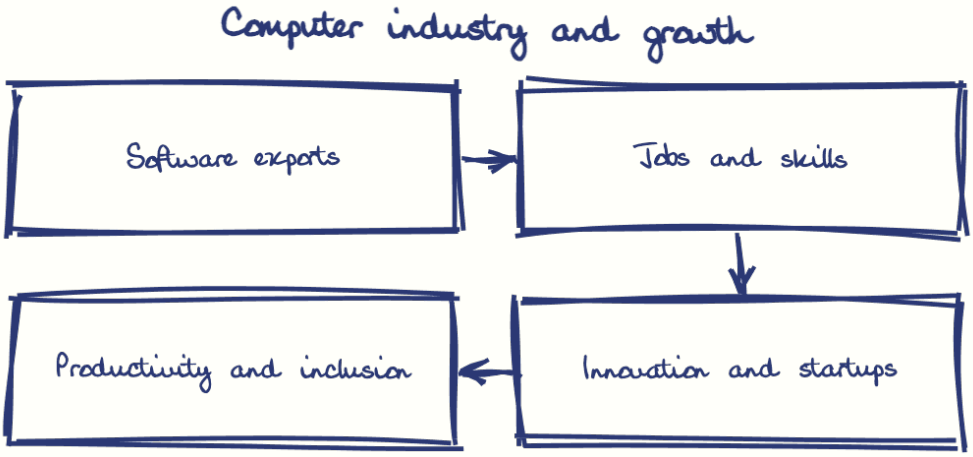
Wider Benefits

1. Productivity: Digital services, automation and data-driven decisions raise productivity across sectors.
2. Financial Inclusion: Jan Dhan Yojana and e-governance widen access to banking and services in underserved areas.

Challenges

1. Infrastructure and Costs: Rural IT infrastructure is thin, and ICT adoption, maintenance and customisation are costly for SMEs.
2. Automation Risk: Much of the workforce does low-value services vulnerable to automation, as the Economic Survey 2024-25 warns.

The impact flows from exports to jobs and skills, then to innovation, and on to productivity and inclusion.



Way Forward

1. Rural Connectivity: Broadband in rural and underserved areas through public-private partnerships to reduce the digital divide.
2. Reskilling: Targeted training in AI and Industry 4.0 to counter automation risks and protect job quality.
3. SME Support: Incentives and customised solutions to lower adoption costs, in line with Make in India.
4. Innovation: Academia-industry-government collaboration for R&D under the India AI Mission.

Conclusion

The computer industry has driven exports, jobs, innovation and inclusion. Closing infrastructure and skill gaps through reskilling will sustain its momentum.

Correct, but average? One line from your answer

YOUR LINE
~~The 1857 revolt failed.~~
× true · forgettable

→

WITH A VALUE ADDITION
The 1857 revolt failed for want of a centre: no common plan, no modern arms, and the loyalty of the Sikhs, the Gurkhas and most princes to the Company.
✓ cause · evidence · specifics

ILLUSTRATIVE

Add value to your next answer →
ghissu.com
14 days free · no card
SCAN

P-IV
3(b)

Write on (i) Communications and remote sensing (ii) Influence of Information Technology on Indian Rural Sector.

10

Introduction

Over 65% of Indians live in rural areas. Communications, remote sensing and information technology, backed by ISRO's Sriharikota launch facility and Digital India, are bridging the urban–rural divide, raising farm productivity, improving governance and delivering essential services.

(i) Communications and Remote Sensing

1. **ISRO's Remote Sensing:** Satellite data supports flood damage assessment through the Bhuvan portal and precision agriculture for PM-FBY.
2. **Village Resource Centres:** ISRO's 473 VRCs are local hubs for geospatial data, helping villagers in Andhra Pradesh decide on farming and resources.
3. **Communication Backbone:** BharatNet links over 2.15 lakh Gram Panchayats by optical fibre for e-governance.
4. **Resource Management:** Soil Health Card, IWMP and Jal Shakti Abhiyan use remote sensing for sustainable, climate-resilient farming, crucial for drought-prone Rayalaseema.

Practice 3(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V
1 / 3
Scan to practice

(ii) Influence of IT on the Rural Sector

1. Market Access: eNAM gives farmers in Andhra Pradesh real-time prices and direct urban buyers, raising incomes.
2. Health and Education: ABDM extends digital health records and telemedicine, and PM eVIDYA delivers multi-mode education.
3. E-Governance: Digital land records and the SVAMITVA scheme reduce corruption and disputes, benefiting farmers in Andhra Pradesh.
4. Social Change: With 85.5% of households owning smartphones and 96.8% of rural youth using mobiles, banking and e-commerce reach villages.
5. Employment: Agri-tech startups create jobs, and PMGDISHA builds digital skills.

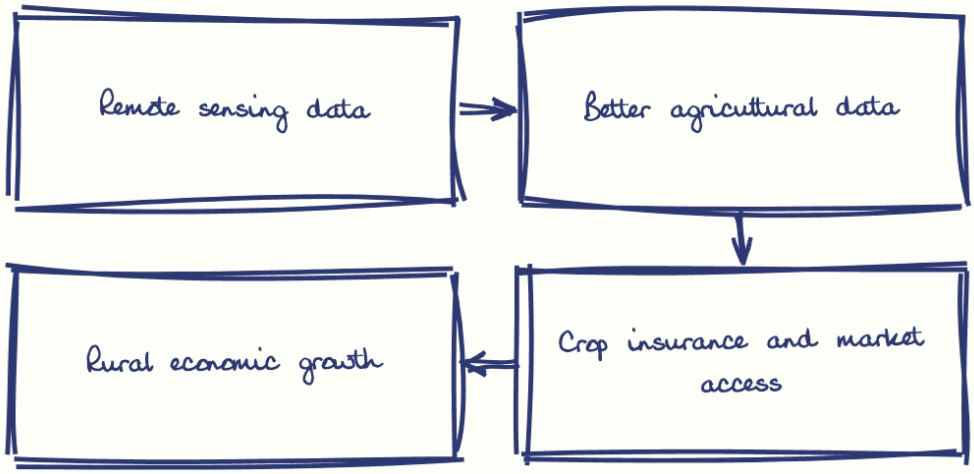
Challenges

1. Digital Divide: Rural internet penetration is 60.46% nationally and only 28% of rural women have internet access, revealing gender and regional disparities.
2. Infrastructure Gaps: Erratic power, low literacy and weak last-mile links in rural Andhra Pradesh.

Linking the Two

Remote sensing yields better agricultural data, which improves crop insurance and market access, and in turn drives rural economic growth.

From satellites to rural growth



Way Forward

1. Infrastructure: Faster BharatNet and 5G rollout.
2. Inclusion: Digital literacy for women and marginalised groups.

Conclusion

Remote sensing and IT are steadily turning data into better crops, fairer markets and closer services for rural India. Closing access and skill gaps will let every village share in this change.

Only ten minutes today?
Speak an outline. It gets evaluated too.
ghissu.com

P-IV
4(a)

(i) Of late there is lot of public resistance towards the establishment of coal based power plants. Comment. (ii) Discuss the reasons for the sparse use of renewable resources in India

10

Introduction

India is balancing development needs with climate commitments. Coal plants have been the backbone of its electricity, but public resistance has intensified. At the same time, despite ambitious targets and fast capacity additions, renewables remain limited by structural and economic barriers.

(i) Public Resistance to Coal Power

1. **Pollution and Health:** Coal plants are major sources of air pollution and greenhouse gases, causing health crises that fuel protest near clusters like Visakhapatnam.
2. **Land and Displacement:** Large coal projects need extensive land, displacing people and destroying livelihoods of local communities.
3. **Policy Shift:** Renewable Purchase Obligations and incentives for cleaner technology discourage coal expansion.
4. **Awareness:** Community mobilisation and activism demand cleaner alternatives and accountability.

Practice 4(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

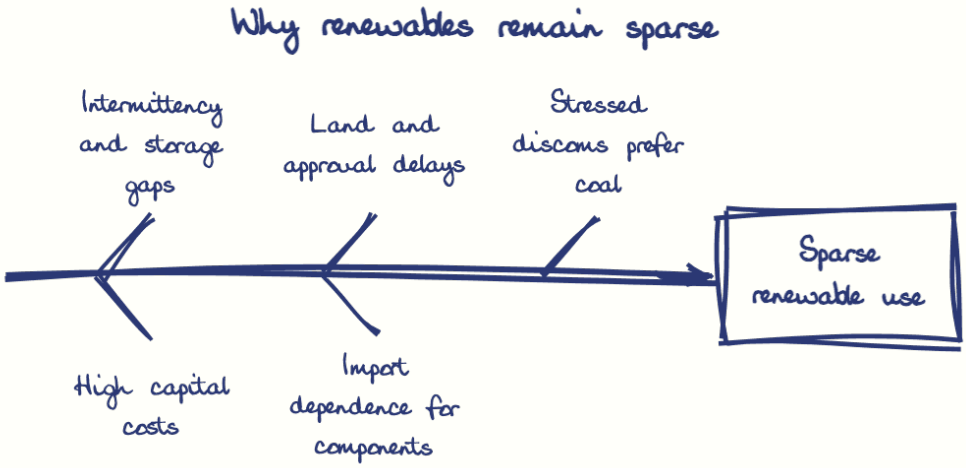
5. The Other Side: Coal still ensures reliable supply and millions of jobs, so a rapid phase-out is hard.

(ii) Reasons for Sparse Use of Renewables

1. Intermittency: Solar and wind threaten grid stability and need costly storage, with limited battery manufacturing.
2. High Capital Costs: Despite the PM Surya Ghar mission, rooftop solar lacks affordable credit; rural areas adopt more, as Andhra Pradesh's rural solar initiatives show, but urban uptake lags.
3. Land and Regulation: Contiguous land for solar parks is hard to acquire, and approvals are slow.
4. Import Dependence: Components come mainly from China, creating supply chain risks.
5. Discom Finances: Stressed DISCOMs prefer cheaper coal contracts.

Way Forward

1. Grid and Storage: Large-scale storage and better grid management.
2. Faster Approvals: Simpler land acquisition and



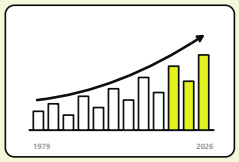
time-bound regulation.

3. Domestic Manufacturing: Local components and batteries to cut imports.

4. Finance: ~~Affordable~~ credit and stronger carbon credit markets.

Conclusion

Public resistance to coal reflects real environmental, social and health costs, while sparse renewable use stems from technical, economic and institutional gaps. Addressing both through infrastructure and policy reform advances SDG 7.



Which sub-topic will they ask next?

Trend analysis

Every syllabus topic: the direction of travel, the most tested sub-topic, the probable next theme.

Prepare the pattern →

ghissu.com

14 days free · no card

SCAN

P-IV
4(b)

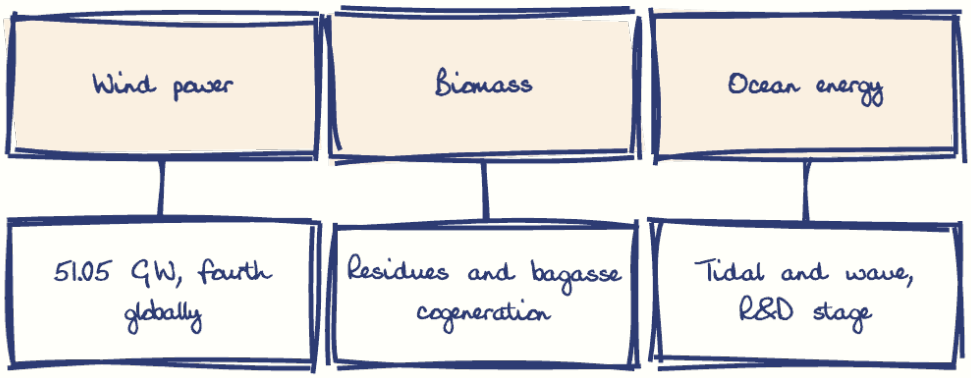
Deliberate on (i) Wind power in India (ii) Biomass management in India (iii) Ocean energy

10

Introduction

Expanding renewable energy is critical for climate action and energy security. Under the National Action Plan on Climate Change and the Paris Agreement, wind, biomass and ocean energy support India's goal of 40% non-fossil installed capacity by 2030.

Three renewable options for India



(i) Wind Power in India

1. **Capacity:** India ranks fourth globally, with 51.05 GW by June 2025, up from 21.04 GW in 2014, backed by about 10,000 MW of annual manufacturing.
2. **Offshore Potential:** A 7,500 km coastline

supports a 30 GW offshore target by 2030, with Denmark and the UK as partners.

3. Challenges: High maintenance costs, multi-ministry clearances and noise; circular economy methods can recycle 85–95% of turbine materials.

4. Andhra Pradesh: Its coast and Visakhapatnam port suit offshore wind.

(ii) Biomass Management in India

1. Contribution: A 10,000 MW target, with support for grid biomass power and bagasse cogeneration.

2. Sustainable Use: Converting crop residues and waste cuts open burning; Swachh Bharat gives viability gap funding.

3. Challenges: Seasonal supply, transport logistics and inefficient conversion limit scale.

4. Andhra Pradesh: Crop and aquaculture waste offer potential, but need better collection.

(iii) Ocean Energy

1. Status: Tidal and wave energy remain at the R&D stage due to high cost and few sites.

2. Legal Framework: Article 297 vests seabed

resources in the Union, and the Territorial Waters Act regulates offshore projects, so Centre–state coordination matters.

3. Challenges: Capital cost, complexity and harm to marine life and fisherfolk.

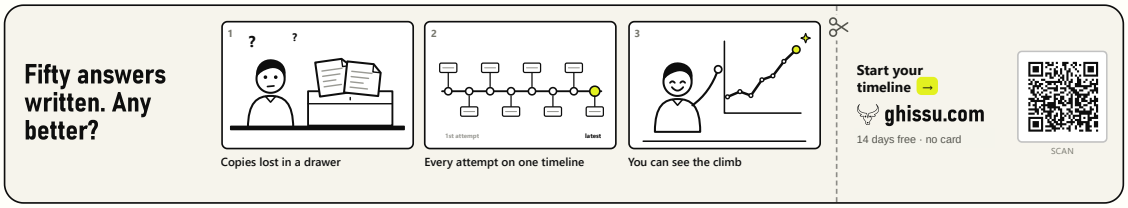
4. Andhra Pradesh: Its 974 km coast, near Visakhapatnam, suits pilots.

Way Forward

1. Ports and Clearances: Stronger ports and faster approvals for offshore projects.
2. Recycling: Frameworks for decommissioned turbines.
3. Biomass Logistics: Efficient conversion and year-round feedstock.

Conclusion

Together, wind, biomass and ocean energy can diversify India's energy mix and cut fossil dependence, with Andhra Pradesh's coast as a hub.



P-IV
5(a)

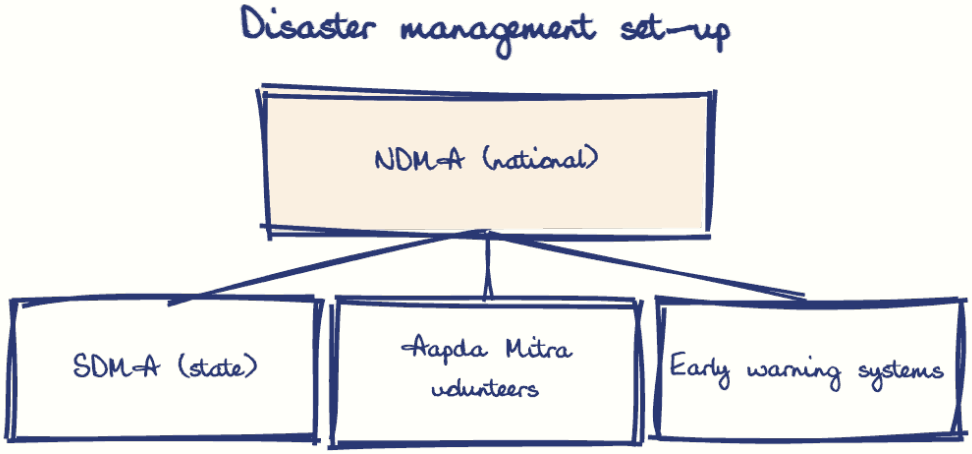
Write on (i) Disaster Management Programmes in India (ii) Mitigation methods to address Global warming.

10

Introduction

India faces growing disasters worsened by climate change, as the 2023 cyclone season and heatwaves showed. Its diverse geography and socio-economic vulnerability demand robust disaster management and global warming mitigation. Andhra Pradesh, with its long coastline and agrarian economy, shows this need clearly.

(i) Disaster Management Programmes in India



1. **National and State Coordination:** The NDMA provides a framework for prevention, preparedness and response, which Andhra Pradesh's SDMA adapts to local risks such as cyclones and Rayalaseema droughts.

2. Aapda Mitra Scheme: It has trained over 100,000 community volunteers nationwide, including in AP, building local capacity for rescue.
3. Early Warning Systems: AP's cyclone warnings, linked to the National Cyclone Risk Mitigation Project, improved evacuation during Cyclone Titli.
4. Climate Adaptation: NAPCC missions like the Green India Mission support afforestation and ecosystem restoration, though progress is uneven.

Challenges

1. Project Delays: Delays in Polavaram affect flood control and rehabilitation.
2. Urban Resilience: 82 of 123 AP urban local bodies lack wet waste processing plants, adding to hazards during disasters.
3. Livelihood Shocks: Droughts and cyclones disrupt farming, needing diversified risk financing and safety nets.
4. Knowledge Gaps: India has advanced warning and GIS-based risk mapping, but AP needs better data integration and community awareness to turn science into local action.

Candidates must not write on this margin

Candidates must not write on this margin

(ii) Mitigation Methods for Global Warming

1. **Renewable Energy:** Under the Paris Agreement, India is raising non-fossil fuel capacity, and AP is investing in solar and wind projects.
2. **Ecosystem-Based Approaches:** Protecting wetlands and forests creates natural carbon sinks and flood buffers, as promoted under the S-APCC.
3. **Sustainable Urban Planning:** Green infrastructure in Visakhapatnam under AMRUT improves drainage and reduces heat island effects.
4. **Policy and Capacity:** Stronger enforcement and local institutional capacity keep mitigation sustained.

Way Forward

Integrated risk-informed planning, timely completion of projects like Polavaram and wider Aapda Mitra training will build resilience.

Conclusion

Anchored in the Sendai Framework and N-APCC, these programmes reduce disaster risk, protect livelihoods and contribute to global climate goals.

STUDY TIP · WHERE TO WRITE A NOTE

Write the note against the question that taught it. That is the only place revision will look for it.

ghissu

On ghissu every question has a notes tab — and your notes gather into chapter notes by themselves.

Open the notes tab →

ghissu.com

14 days free · no card

SCAN

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 5(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

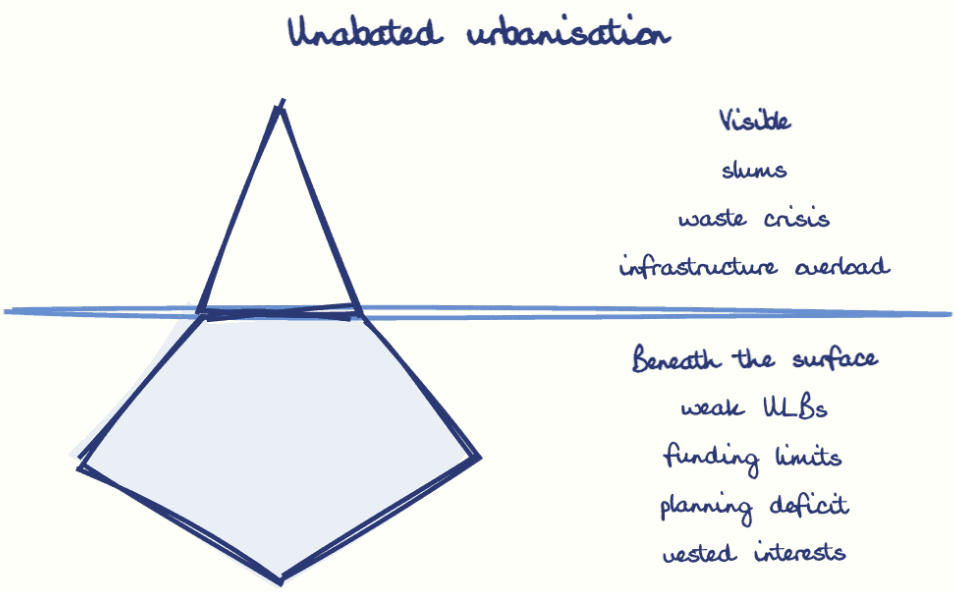
P-IV
5(b)

Discuss the problems associated with unabated urbanization and suggest the ways and means to arrest this.

10

Introduction

India is urbanising fast, with its urban population projected to reach 675 million by 2035, second only to China. Driven by jobs and rural–urban migration, unplanned growth strains infrastructure, environment and social equity, demanding urgent action.



Governance Constraints

1. **Weak Local Bodies:** Despite the 74th Amendment, many ULBs lack the capacity and autonomy to plan and manage growth, leading to poor service delivery.
2. **Funding Limits:** State governments restrict ULBs

from raising adequate funds, so infrastructure and waste gaps persist.

3. Planning Deficit: Uncontrolled expansion is accepted fatalistically, with inadequate town planning and no second National Commission on Urbanisation since the 1980s.

4. Vested Interests: Powerful elites push commercial interests, leading to unauthorised construction and neglect of the urban poor.

Urban Problems

1. Infrastructure Overload: Water, sewage, transport and power fail, as Mumbai shows.

2. Slums: The inability to provide affordable housing creates sprawling, overcrowded slums with poor sanitation, as in Visakhapatnam.

3. Waste Crisis: Andhra Pradesh's (123) ULBs generated about (6,890) tons of solid waste daily in (2022); only (4) had operational sanitary landfills and (82) lacked wet waste plants, showing severe gaps.

4. Social Strain: Many migrants stay in informal jobs without social security, while exclusion and alienation raise poverty and crime.

5. Environment and Health: Lost green spaces,

pollution, overcrowding and poor sanitation harm health and the quality of urban life.

Ways to Arrest Unplanned Urbanisation

1. Empower ULBs: Greater financial and administrative autonomy with capacity building.
2. Planned Development: Strategic densification and multiple urban centres to ease megacities.
3. Affordable Housing: Scale up PM-AY and slum upgrading, with focused implementation in Andhra Pradesh.
4. Waste Management: Sanitary landfills, wet waste processing and strict enforcement of solid waste rules.
5. Inclusive Services: Extend basic services and social security to slum dwellers and informal workers.

Conclusion

Unabated urbanisation threatens sustainable development and liveability. Integrated governance reform, planned infrastructure and social inclusion can make cities equitable and resilient, in line with SDG (11) on sustainable cities.

Wrote the answer. Now what?
Upload it — every line marked, in minutes.
ghissu.com →

P-IV
6(a)

Discuss the Agricultural research in India mentioning its linkages. Describe the research and educational activities in the field of agriculture, animal husbandry and fisheries.

10

Introduction

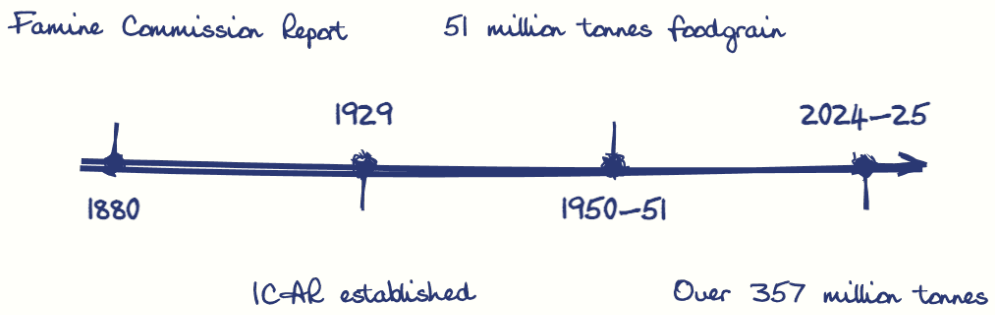
Agricultural research in India is crucial for food security, rural livelihoods and economic growth amid a rising population and climate challenges. The Green Revolution showed how continuous innovation in agriculture, animal husbandry and fisheries raises productivity and resilience.

Agricultural Research in India

1. Foundation and Growth: Research began with the Famine Commission Report of 1880 and grew through IARI and ICAR, established in 1929. Today over 70 Agricultural Universities make it one of the largest systems globally.
2. Productivity: Foodgrain output rose from 51 million tonnes in 1950–51 to over 357 million tonnes in 2024–25.
3. Economic Returns: Research has reduced poverty and raised rural employment and exports. Every rupee invested in research yields far higher returns than subsidies on fertilisers or power.
4. Challenges: A 50% gap between demand and

supply of skilled scientists, while the scope now extends to agribusiness and value chains.

Growth of agricultural research



Linkages

1. Allied Sectors: Animal husbandry and fisheries show stable growth and support rural livelihoods. Livestock genetics and health research support India's nearly 25% share of global milk production.
2. Extension: Krishi Vigyan Kendras and ATMAs link research to farmers through on-farm testing and demonstrations, as in Andhra Pradesh.
3. Partnerships: Private participation and collaboration with over 100 countries, including the USA and Germany.
4. Policy: ICAR aligns research with schemes like PM Fasal Bima Yojana and the Soil Health

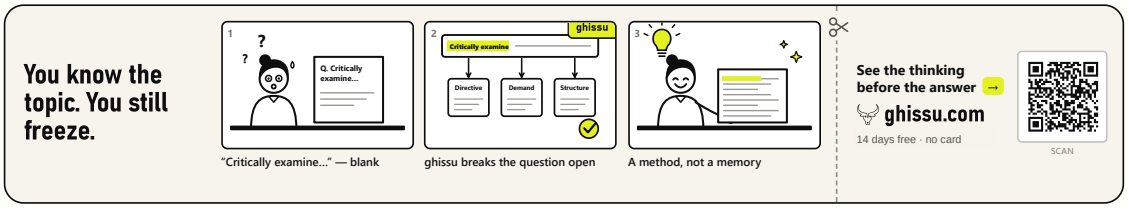
Card Scheme.

Research and Educational Activities

1. **Institutional Network:** ICAR and SAUs teach crop sciences, veterinary sciences, fisheries and food technology.
2. **Livestock and Fisheries:** With over 535 million livestock, research focuses on indigenous breed conservation, diagnostics and vaccines; fisheries research expands aquaculture along Andhra Pradesh's 974 km coastline.
3. **Education:** Courses now include agribusiness, food processing and sustainability, and the output of graduates is rising to meet demand in research, extension and industry.
4. **Outreach:** KVKs run on-farm trials and supply quality seeds and livestock strains.

Conclusion

Raising investment in biotechnology, GIS and climate-resilient crops, while strengthening extension, will keep research serving India's farmers.



Write your own answer to 6(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
6(b)

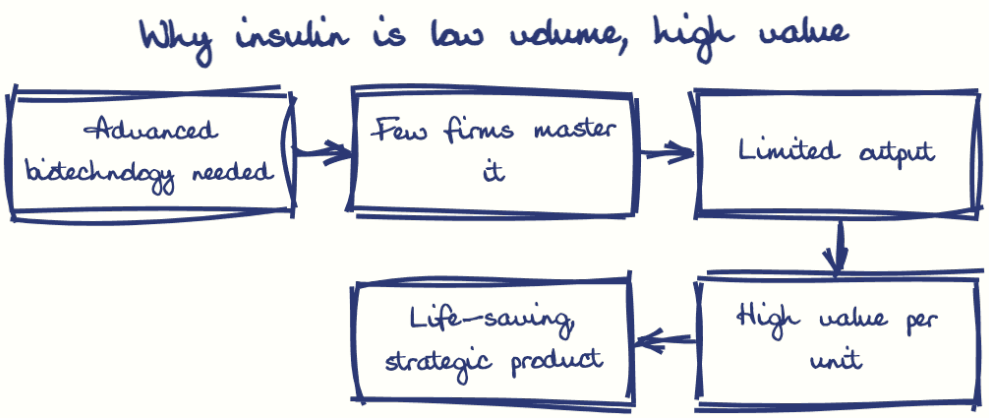
Elucidate a low volume high value product giving the example of insulin. Mention the role of Indian companies for its production.

10

Introduction

Low volume high value products are specialised goods made in limited quantities but with great economic and strategic value due to their complexity and critical use. Insulin exemplifies this, and rising diabetes in India, including Andhra Pradesh, makes domestic production vital for affordable access.

Insulin as a Low Volume High Value Product



1. Complex production: It requires advanced biotechnology, so output is limited while market value is high.
2. Critical use: As a life-saving hormone for diabetes, supply disruption is costly, giving it strategic value for self-reliance and reduced

import dependence.

Role of Indian Companies

1. **Biotech innovation:** Firms such as Biocon pioneered affordable biosimilar insulin, combining biotech expertise with cost-effective manufacturing.
2. **Scaling capacity:** Expanded cGMP-compliant facilities give stable, consistent production.
3. **Collaborations:** Partnerships with multinationals and exports make India a growing global insulin supplier, showing rising technological sovereignty.
4. **Affordability:** Companies work with Jan Aushadhi Kendras to supply insulin at lower cost, improving access in Andhra Pradesh and across India.

Government Support

State policy creates the conditions in which companies can scale.

1. **PLI Scheme:** A Rs 15,000 crore outlay boosts domestic manufacture of complex drugs including insulin.
2. **Bulk Drug Park:** Andhra Pradesh hosts one, with up to Rs 1,000 crore central assistance for infrastructure and logistics.

3. Regulatory reform: A new Drugs, Medical Devices and Cosmetics Bill is to replace the 1940 Act, streamlining quality standards for biologics.

Challenges

1. Imported APIs: Heavy reliance on China hurts cost and supply chain resilience.
2. Low R&D: Firms have focused on generics rather than discovery.
3. Quality compliance: GMP gaps affect international acceptance.
4. Affordability: Complexity limits cheap availability, so schemes like Pradhan Mantri Bhartiya Janaushadhi Pariyojana are needed.

Conclusion

Insulin shows how a low volume high value product underpins healthcare security. Indigenous API capacity, particularly in Andhra Pradesh, more R&D, regulatory harmonisation and public–private partnerships will strengthen Indian companies' role.

1.5 lakh+

Borrow the approach, not the answer.

Topper answers matched by meaning to your question — with how each one opened, structured and concluded.

See how toppers wrote it → ghissu.com

14 days free · no card

SCAN

P-IV
7(a)

Discuss the applications of plants in different areas with examples.

10

Introduction

Plants have been integral to human civilisation, serving diverse roles from food and medicine to industry and culture. With scientific advances and policy support their applications have widened. India's rich biodiversity and traditional knowledge make plant resources crucial for sustainable development, especially in Andhra Pradesh where agriculture and horticulture are key economic pillars.

Applications of Plants in Different Areas

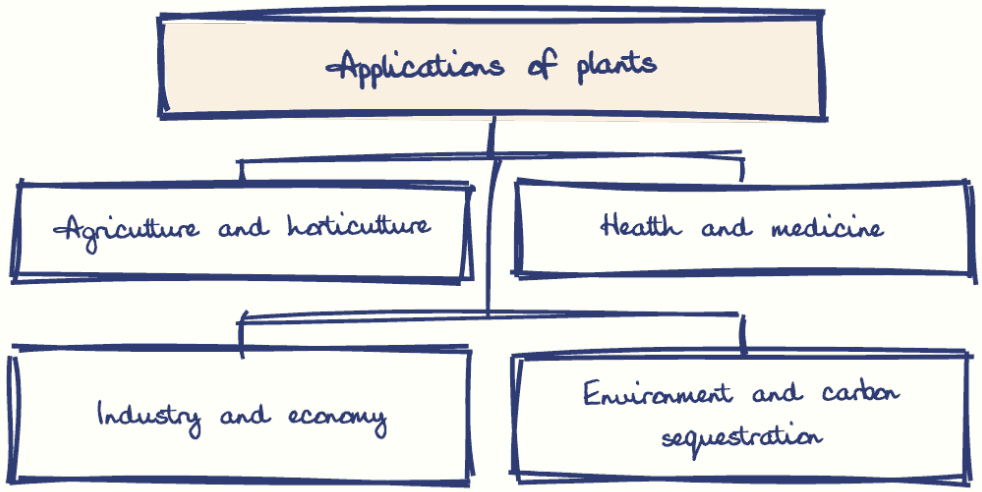
1. **Agriculture and horticulture:** Plants such as eggplant, mango and turmeric are cultivated for food and spices. Andhra Pradesh leads with 8.8% CAGR in agricultural growth, driven by diversification into high-value crops like chillies and fruits.

2. **Health and medicine:** Over 7,500 plant species are used in traditional medicine, and India's herbal medicine market is growing fast. Standardisation and reproducibility of plant-based drugs remain challenges.

3. **Industry and economy:** Plants provide raw material for textiles (cotton, jute), timber and bio-based

products. The rise of floriculture exports and the coconut trade shows their export potential.

4. Environment: Plants mitigate climate change through carbon sequestration and protect soil, though sustainable cultivation practices need wider adoption.



Institutional Support

1. Conservation: India has set up (115) Medicinal Plants Conservation Areas to protect medicinal plant diversity.

2. Quality planting material: The Clean Plant Programme, backed by the Asian Development Bank, supplies disease-free planting material to raise horticulture productivity, benefiting states including Andhra Pradesh.

3. Traditional knowledge: Indigenous women healers conserve medicinal plant diversity, showing the value of participatory breeding.

Way Forward

1. R&D and breeding: Andhra Pradesh's agricultural research centres should use genome editing and biotechnology to develop climate-smart, high-yield and disease-resistant varieties.
2. Traditional knowledge: Involve local communities in participatory breeding to conserve diversity and find novel uses.
3. Sustainable cultivation: Good Agricultural Practices and a wider Clean Plant Programme can improve planting material and cut pesticide use.
4. Market linkages: Infrastructure and digital platforms like e-NAM can connect farmers to markets, boosting income from high-value and medicinal crops.

Conclusion

Plants are foundational to food security, health, industry and environmental sustainability. Leveraging India's plant wealth through innovation and inclusive policy will advance Atmanirbhar Bharat and SDG (9.)

Write your own answer to 7(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
7(b)

(i) Differentiate between wild and domestic animals. (ii) Give an overview of how animals are exploited in medical research.

10

Introduction

The distinction between wild and domestic animals is basic to biodiversity, ecological balance and human–animal interaction in India, which has rich wildlife and leads the world in milk production. It also shapes regulation and the ethics of using animals in medical research.

(i) Wild and Domestic Animals

Wild vs Domestic Animals

Wild Animals	Domestic Animals
Natural ecosystems	Human-controlled environment
Natural breeding	Bred by humans
Independent of humans	Depend on humans
Wildlife laws	Husbandry, welfare laws
Biodiversity, ecosystem services	Agriculture, food, companionship

1. **Habitat and Behaviour:** Wild animals live and reproduce in natural ecosystems without human control, domestic animals are raised in human-controlled environments for food, labour

or companionship.

2. Dependency: Domestic animals depend on humans for food, shelter and breeding, while wild animals avoid humans, though encroachment affects them.

3. Legal Status: Wild animals are protected under wildlife laws and programmes like Project Elephant in Andhra Pradesh; domestic animals come under husbandry and welfare laws.

4. Roles: Wild animals provide ecosystem services; domestic animals support rural economies, with milk output of 239.30 million tonnes in 2023–24. In AP, growing livestock and aquaculture encroach on elephant corridors.

(ii) Animals in Medical Research

1. Necessity of Testing: Animals test drug safety and efficacy before human trials, since early human testing is unethical and risky.

2. Pain and Ethics: Higher mammals feel pain and distress like humans, so anaesthesia and humane endpoints are required, and painless experiments are condemned.

3. Regulation: The CPCSEA enforces standards of animal care, though enforcement is weak in

resource-limited settings.

4. Alternatives: In vitro methods reduce animal use, but full replacement is not yet feasible.

Way Forward

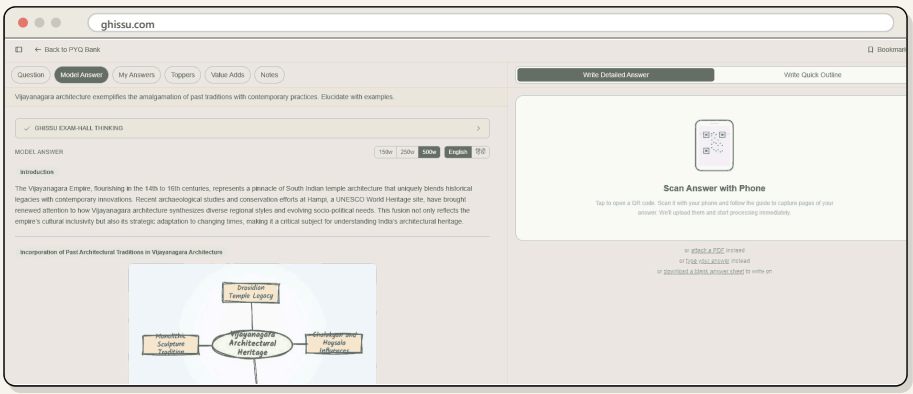
Strengthen CPCSEA oversight in Andhra Pradesh, invest in alternatives and sensitise researchers and farmers to humane treatment.

Conclusion

Recognising the distinct roles of wild and domestic animals, and limiting their use in research to what is necessary and humane, balances science with welfare.

THE REAL PRODUCT · NOT A MOCK-UP MODEL ANSWERS IN 150 · 250 · 500

Writing 400 words for a 150-word question?



1 One question, three lengths: 150 · 250 · 500 words.

2 Two languages at a tap: English and हिंदी.

3 Keywords, structure and diagrams the examiner rewards.

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

P-IV
8(a)

Give a brief description of GM crops and explain the regulations and concerns associated with the release of a Genetically Modified crop.

10

Introduction

Genetically Modified (GM) crops are a major advance in agricultural biotechnology. With India's growing population and food security needs, their release has become crucial, but it involves complex regulation and concerns about environment, health and biodiversity.

Brief Description of GM Crops

GM crops are plants whose genetic material is altered through biotechnology to give traits such as pest resistance, herbicide tolerance and climate adaptability, as in Bt-cotton.

Regulatory Framework

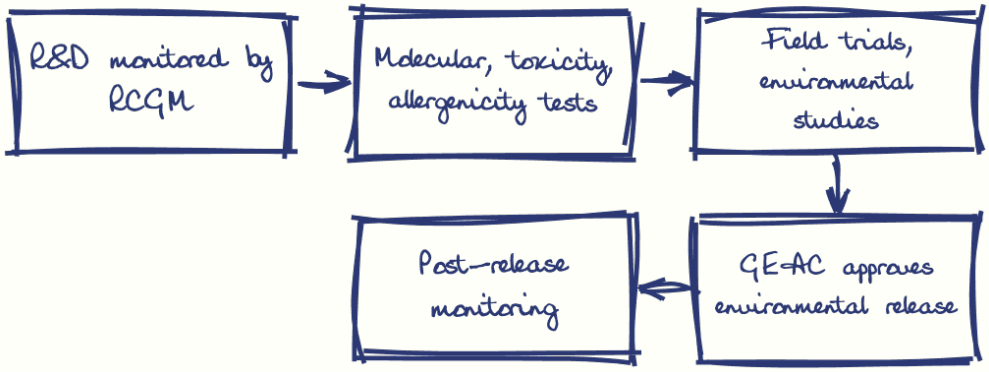
1. **Legal Structure:** The Environment Protection Act, 1986, through its 1989 Rules on hazardous microorganisms and genetically engineered organisms, governs everything from research to commercial release.
2. **Key Bodies:** The Genetic Engineering Appraisal Committee (GEAC) under the Environment Ministry approves environmental release, while the

RCGM under DBT monitors R&D.

3. Safety Protocols: GE-AC requires molecular characterisation, food safety tests for toxicity and allergenicity, and environmental studies such as field trials and pollinator effects.

4. Gene-Edited Crops: Those without foreign DNA are exempt from biosafety assessment and can be released as new varieties.

Path to GM crop release



Concerns about Release

1. Environmental Risks: Transgene flow through pollen or seed may create herbicide-resistant weeds or reduce genetic diversity.

2. Health and Biodiversity: Despite scientific consensus on safety, fears persist about health effects and harm to honeybees from herbicide overuse.

3. Regulatory Delays: The Bt-brinjal moratorium and

Supreme Court challenges to GM mustard delay farmers' access.

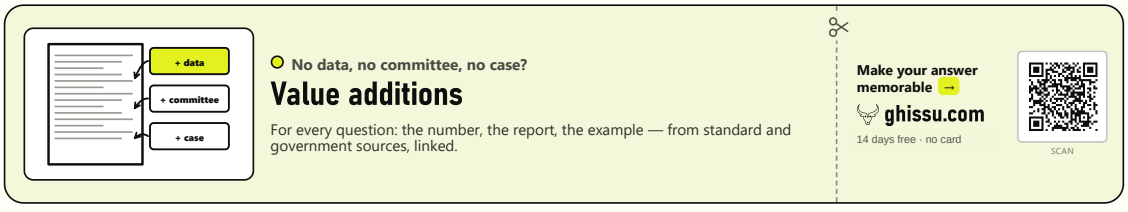
4. Social Acceptance: Public scepticism and trade barriers like labelling rules complicate adoption, even in cotton-growing Andhra Pradesh.

Way Forward

1. Regulatory Capacity: Strengthen GE-AC and RCGM to speed evaluations while keeping standards rigorous.
2. Post-Release Monitoring: Track environmental and health impacts after release and adapt policy on evidence.
3. Public Awareness: Outreach to farmers and consumers in Andhra Pradesh to build trust.

Conclusion

Balancing innovation with stringent regulation, public engagement and indigenous research at Andhra Pradesh's agricultural universities on locally adapted crops can let India harness GM crops responsibly, advancing SDG (2) and Atmanirbhar Bharat.



P-IV
8(b)

Define biofertilizers. Give an account of technology involved for the mass production of different biofertilizers, with their application.

10

Introduction

The rising cost and environmental impact of chemical fertilizers have intensified the search for sustainable alternatives. Biofertilizers, which harness beneficial microbes to enhance soil fertility, are eco-friendly inputs that suit Andhra Pradesh's push for soil health and the goal of Atmanirbhar Bharat in farming.

Definition and Role

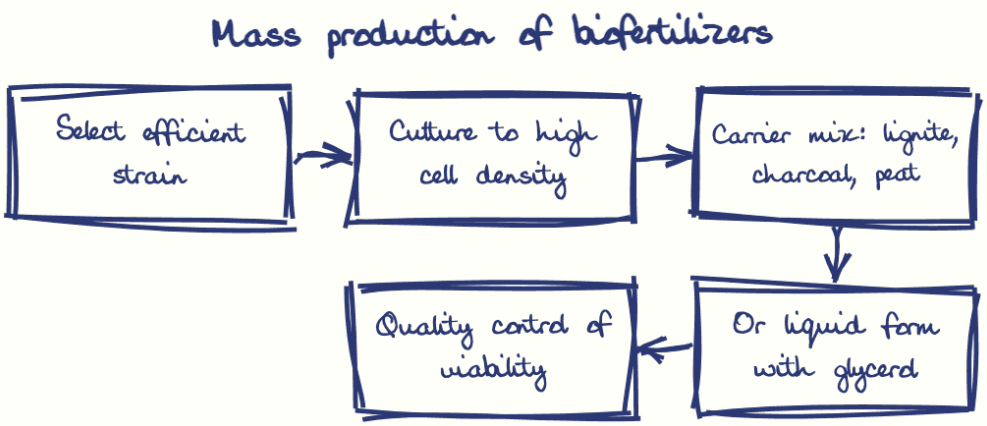
1. **Definition:** Biofertilizers are products containing living microorganisms that, applied to seeds, soil or plants, colonise the rhizosphere and fix nitrogen, solubilise phosphorus or mobilise other nutrients.
2. **Distinct from Organic Fertilizers:** They are carrier-based microbial inoculants such as Rhizobium, Azotobacter and Mycorrhiza, not decomposed organic matter.
3. **Benefits:** They reduce chemical fertilizer use, lower production costs and limit pollution, which matters in Rayalaseema with its soil degradation and drought stress.

Practice 8(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Technology for Mass Production

1. **Strain Selection and Cultivation:** Efficient strains, like Rhizobium for legumes or phosphate-solubilising bacteria, are isolated and cultured in nutrient-rich liquid media under controlled conditions to a high cell density.
2. **Carrier-Based Formulation:** Cultures are mixed with sterile carriers such as lignite, charcoal or peat to make solid biofertilizers.
3. **Liquid Inoculants:** Liquid formulations with protectants like glycerol and polyvinyl pyrrolidone improve viability and extend shelf life up to one year.
4. **Quality and Scale:** Poor quality and short shelf life still deter farmers. In 2019-20, India produced over 79,000 metric tonnes of solid and 30,000 kL of liquid biofertilizers from 537 units, including facilities in Andhra Pradesh serving local crops.

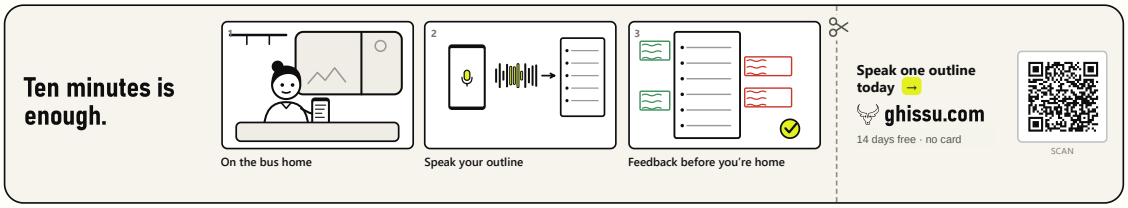


Applications

1. **Nitrogen Fixers:** Rhizobium serves pulses like gram and lentil, while Azospirillum and Azotobacter help rice and maize, reducing urea use in Andhra Pradesh's paddy fields.
2. **Phosphorus Solubilisers:** Bacillus and Aspergillus free insoluble phosphates for sugarcane and finger millet in coastal Andhra Pradesh.
3. **Mycorrhizal Fungi:** Endomycorrhizae improve nutrient and water uptake in horticultural crops of Chittoor district.
4. **Micronutrient Solubilisers:** Bacteria mobilise potassium and zinc in Rayalaseema's degraded, drought-prone soils.

Conclusion

Regional mother culture facilities and farmer training through Krishi Vigyan Kendras can make biofertilizers a mainstay of sustainable farming in Andhra Pradesh.



P-IV
9(a)

Discuss in detail about HIV virus; also mention the symptoms associated with the infection, preventive measures and diagnostic test for disease confirmation.

10

Introduction

HIV/AIDS remains a critical public health challenge globally and in India despite progress in control and treatment. By weakening the immune system, the virus exposes people to opportunistic infections, so early diagnosis and prevention matter. India aims to end the epidemic as a public health threat by 2030 under SDG 3.3.

The HIV Virus

1. Nature: HIV is a retrovirus of the lentivirus genus that mainly infects CD4+ T-helper lymphocytes and macrophages.
2. Mechanism: Using reverse transcriptase, it integrates its DNA into host cells, leading to progressive immune failure.
3. Transmission: Unprotected sexual contact, contaminated blood transfusions, shared needles, and mother-to-child spread during pregnancy or breastfeeding.

Symptoms

1. Early stage: Flu-like fever, weight loss, malaise

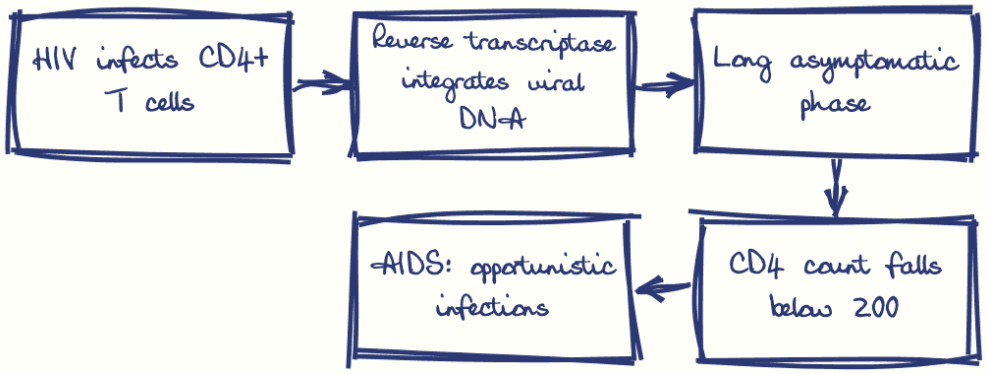
and lymphadenopathy, followed by a long asymptomatic phase.

2. AIDS stage: Opportunistic infections such as tuberculosis, fungal infections and neurological disorders.

Diagnostic Tests

1. Screening: ELISA detects antibodies against the virus.
2. Confirmation: Western blot or nucleic acid tests ensure accuracy.
3. Monitoring: CD4 counts below 200 cells/mm^3 indicate advanced immunodeficiency, and viral load shows treatment response.

How HIV leads to AIDS



Preventive Measures

1. Safe practices: Safe sex, blood safety and harm reduction among intravenous drug users.

2. Antiretroviral therapy: ART, free in India including Andhra Pradesh, cannot cure HIV but prolongs life and cuts transmission.

3. Targeted interventions: Sentinel surveillance and outreach to sex workers and migrants in Andhra Pradesh, with awareness drives like Mission Sampark in the state.

4. Institutions and law: NACO leads the response, NACP Phase V (2021–26) has an outlay of Rs. 15,471.94 crore for 95–95–95 targets, and the HIV/AIDS Act, 2017 bars discrimination.

Challenges and Way Forward

1. Barriers: Social stigma, low awareness and gaps in blood safety and needle exchange hinder prevention in rural and marginalised communities of Andhra Pradesh.

2. Remedies: Improve access to rapid and confirmatory testing in remote areas of Andhra Pradesh and link patients to care quickly.

Conclusion

Wider testing, ART adherence and community education can end HIV as a public health threat by 2030.

Write your own answer to 9(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
9(b)

Describe the causative agent for malaria, emphasize its life cycle in different hosts and discuss the preventive measures for the disease.

10

Introduction

Malaria remains a critical public health challenge in tropical regions like Andhra Pradesh, where the climate favours mosquito breeding. Despite a global rise in 2023, India cut malaria cases by 69% and deaths by 68% between 2017 and 2023, and aims for elimination by 2030.

Causative Agent

1. **Plasmodium:** Malaria is caused by protozoan parasites of the genus Plasmodium, mainly P. falciparum and P. vivax in India, with P. falciparum the most lethal.
2. **Two hosts:** The parasite needs humans and female Anopheles mosquitoes, which act as vectors.

Life Cycle in Different Hosts

1. **Entry into humans:** An infected mosquito bite injects sporozoites into the bloodstream.
2. **Liver stage:** Sporozoites migrate to the liver and multiply into merozoites.
3. **Blood stage:** Merozoites infect red blood cells

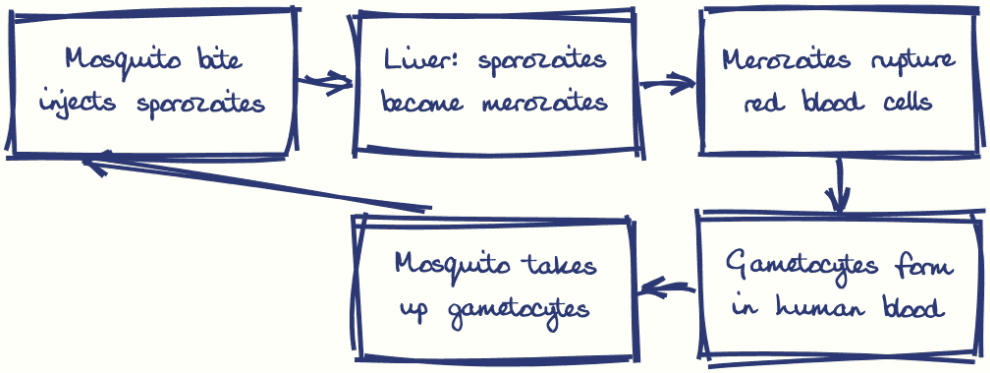
Practice 9(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

and rupture them, causing clinical symptoms.

4. Mosquito stage: Sexual forms called gametocytes develop in humans, are taken up by a biting mosquito and continue the cycle inside the insect.

Life cycle of Plasmodium



Preventive Measures

1. Vector control: Insecticide-treated nets, indoor residual spraying and removal of breeding sites such as stagnant water in coastal and canal areas of Andhra Pradesh.

2. Vaccines: WHO-approved RTS,S/AS01 and R21/Matrix-M can protect high-risk groups.

3. Surveillance and early diagnosis: The Test, Treat, Track strategy with health workers and mobile units in remote areas of Andhra Pradesh ensures prompt treatment.

4. Awareness: Campaigns in local languages promote nets and early care.

Programmes and Challenges

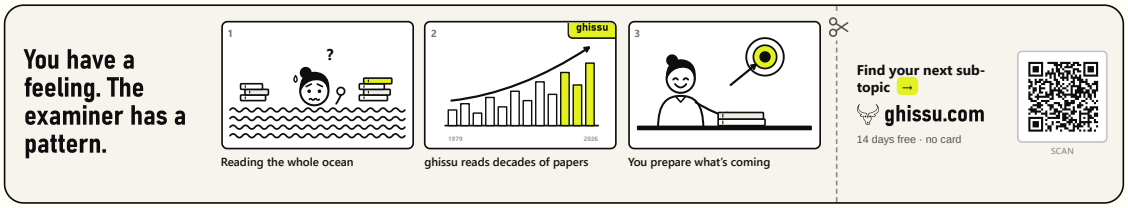
1. Plans: The National Strategic Plan for Malaria Elimination (2023–27) and IMEP–3 guide action; 122 districts reported zero cases in 2023, and Andhra Pradesh contributed through targeted interventions in endemic districts.

2. Challenges: Drug-resistant strains need constant monitoring and access to antimalarials is uneven in rural and tribal areas of Andhra Pradesh.

3. Social barriers: Poverty, migration and low health literacy limit use of nets and timely treatment.

Conclusion

India's 2024 exit from the WHO High Burden to High Impact group shows progress. Sustained prevention can achieve elimination by 2030.



P-IV
10(a)

Define immunity. Discuss the concept of vaccination and explain how the vaccine for OPT is produced conventionally.

10

Introduction

Vaccination has emerged as a critical tool to enhance immunity and prevent infectious disease, especially since the COVID-19 pandemic. The success of India's Universal Immunisation Programme (UIP), which reaches about 2.54 crore newborns a year, shows why immunity and vaccine production matter.

Defining Immunity

Immunity is the body's natural ability to resist infections caused by pathogens such as bacteria, viruses and parasites. Active immunity develops when the immune system encounters antigens and forms memory cells that respond rapidly on future exposure.

Concept of Vaccination

1. Principle: Vaccination mimics infection by introducing weakened or inactivated pathogens, stimulating immunity without causing disease.
2. Regulation: CDSCO regulates vaccine approval and quality, and the National Vaccine Policy,

2011 guides procurement and distribution.

3. Delivery: UIP provides vaccines free of cost, and Mission Intensified Indradhanush 4.0 targets immunisation gaps, implemented in Andhra Pradesh through state health services, even in remote areas.

Conventional Production of OPV

1. Virus Culture: Attenuated poliovirus strains are grown in monkey kidney cell cultures.
2. Purification and Formulation: The virus is harvested, purified and formulated into oral drops.
3. Quality Control: Mass production needs stringent checks to prevent reversion to virulence.
4. Mode of Action: As a live attenuated vaccine, it induces strong mucosal immunity in the gut, preventing poliovirus replication and transmission.

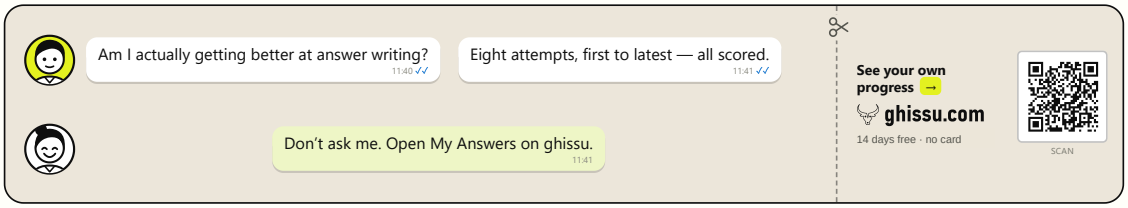
Conventional production of OPV

Significance and Risks

OPV is cost-effective and easy to give, making it ideal for mass campaigns like the Pulse Polio Programme. Its wide use cut polio incidence worldwide and in India, including Andhra Pradesh's districts, and was central to India's polio-free certification. Rare vaccine-derived poliovirus cases require monitoring and an eventual shift to inactivated polio vaccine.

Conclusion

Immunity is the body's shield, and vaccines like OPV strengthen it at population scale. Vaccine hesitancy from misinformation and cold chain gaps in remote areas such as the Rayalaseema drought belt remain challenges. Strong cold chains, community engagement and continued surveillance will sustain immunisation gains in Andhra Pradesh.



P-IV
10(b)

What do you understand by a vaccine? Describe the journey of development from conventional to modern day (Recombinant vaccines).

10

Introduction

Vaccines are a cornerstone of public health, as the rapid development of COVID-19 vaccines showed. The journey from conventional to recombinant vaccines reflects scientific progress towards faster, safer and more targeted immunisation, which the Universal Immunisation Programme now relies on.

What is a Vaccine

1. **Definition:** A vaccine is a biological preparation that stimulates adaptive immunity against a specific pathogen by exposing the immune system to a weakened, inactivated or partial form of it, or its genetic material.

Conventional Vaccines

1. **Live attenuated:** Weakened whole pathogens trigger immunity, as in BCG.
2. **Inactivated:** Killed whole pathogens are used, as in Polio vaccine and COVAXIN.

Practice 10(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Modern Recombinant Vaccines

1. Sub-unit vaccines: Only a part of the pathogen is used, as in the HPV vaccine.
2. DNA and mRNA vaccines: Zydus Cadila's DNA vaccine and the mRNA vaccines of Pfizer and Moderna deliver genetic instructions so human cells make pathogen proteins.
3. Viral vectored vaccines: AstraZeneca's COVID vaccine uses a viral vector to carry these instructions.

Conventional vs recombinant vaccines

Conventional	Recombinant
Whole pathogen, weakened or killed	Part of pathogen or genes
BCG, Polio vaccine, COVAXIN	HPV, DNA, mRNA, viral vector
Fixed preparation	Platform adapts to new pathogens

Development and Challenges

1. Clinical trials: Phases I to IV test safety and efficacy with regulators requiring over 50% efficacy. India used emergency use authorisation in the pandemic.
2. Indian capacity: Serum Institute and Bharat

Biotech are among the world's largest manufacturers, backed by the COVID Suraksha Mission.

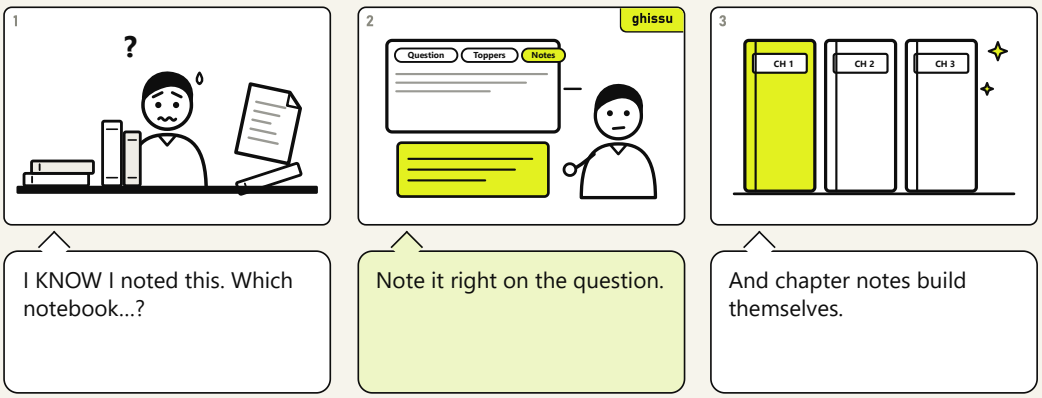
3. R&D gap: Vaccine R&D is costly and slow, and low R&D intensity keeps most firms on traditional vaccines.

Conclusion

Vaccines have moved from whole pathogens to precise, scalable recombinant platforms. Stronger indigenous R&D, surveillance and public acceptance, especially for HPV vaccines in Andhra Pradesh's rural districts, will advance health security and SDG 3.

A STORY YOU MAY KNOW NOTES THAT LIVE WITH THE QUESTION

Where did I write that?



— Note it where you learnt it.

Keep every note in place →
ghissu.com
14 days free · no card

SCAN

P-IV
11(a)

In your own words explain the steps to bring awareness and in educating towards Environmental Protection in India

10

Introduction

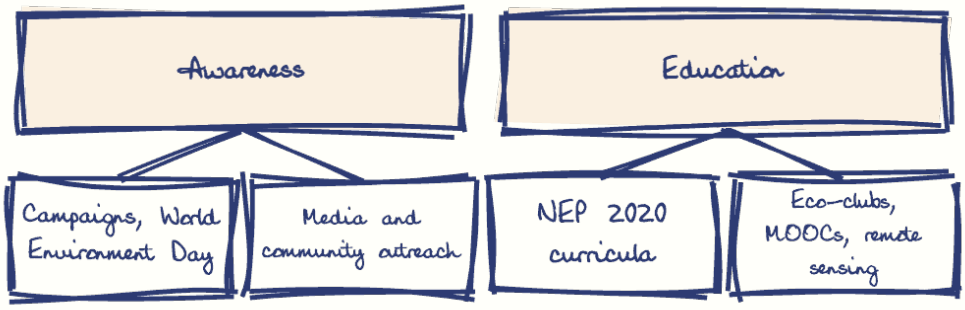
With erratic monsoons and rising temperatures, environmental protection has become a critical priority for India. Its vast population and diverse ecosystems face pollution, deforestation and resource depletion, making awareness and education essential. Educated citizens join conservation and support India's commitments under the Paris Agreement and the SDGs.

Steps to Create Awareness

1. Government campaigns: Environment and agricutture ministries promote plantation drives, and World Environment Day on June 5th is celebrated nationwide.
2. Multimedia and print: Television, documentaries, films, the internet and newspapers carry environmental issues to urban and rural audiences.
3. Community engagement: Exhibitions, posters, face-to-face contact, SMS and email spread the message, while individual action inspires collective change.
4. Countering misinformation: Expert-driven communication keeps issues from becoming

propaganda, as in the Enron (Ratnagiri) controversy.

Building environmental awareness



Steps in Environmental Education

1. Formal education: NEP 2020 embeds climate change, pollution control and biodiversity in school curricula.
2. MOOCs and distance learning: Online platforms overcome geographical barriers, in line with NITI Aayog's Strategy for New India @75.
3. Scientific tools: Remote sensing and satellites, which monitor resources and predict forest fires, strengthen content.
4. Experiential learning: Eco-clubs and participatory forest management, AP's ULBs use awareness to improve waste management.

Outcomes

Awareness drives waste segregation and water conservation, and in AP it has begun to improve

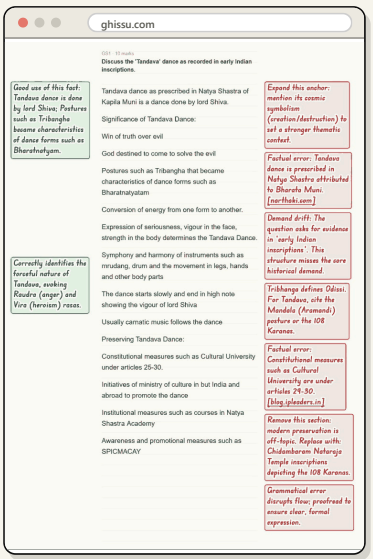
waste handling; state campaigns can target local issues like coastal ecosystem protection. Meanwhile educated citizens comply with regulations, join public hearings, pass conservation values to future generations and support the LiFE movement launched at COP26.

Conclusion

Awareness and education turn citizens from bystanders into partners in protection. Campaigns, media, schools and community action together can build the environmental consciousness India needs for a sustainable future.

THE REAL PRODUCT · NOT A MOCK-UP ANSWER EVALUATION

Nine days waiting for feedback on one answer?



1. Green notes: what you did well — keep doing it.
2. Red notes: the missing dimension, the better data, the sharper intro.
3. Handwritten or typed — scanned with your phone, marked in minutes.

Upload an answer →

ghissu.com

14 days free · no card

SCAN

P-IV
11(b)

Write the salient features of Water Act and Water pollution Cess Act.

10

Introduction

Water pollution remains a critical challenge in India, affecting public health, agriculture and biodiversity. The Water Act, 1974 and the Water Cess Act, 1977 were enacted to regulate pollutant discharge and to generate resources for pollution control. The 2024 amendment seeks to modernise them.

Salient Features of the Water Act

Water Act vs Water Cess Act

Water Act	Water Cess Act
Creates Pollution Control Boards	Levies cess on water use
Consent for industries	Funds the Boards
2024 amendment	Fund utilisation issues

1. **Pollution Control Boards:** The Act created the CPCB and SPCBs. The CPCB sets standards and provides technical guidance, while SPCBs enforce regulations and take legal action against violators.

2. Consent Mechanism: Industries must obtain prior consent from SPCBs before discharging effluents into water bodies, ensuring compliance with standards.

3. 2024 Amendment: It decriminalises minor offences by replacing imprisonment for technical violations with fines, and allows the Centre to exempt certain industries from prior consent, balancing protection with ease of doing business.

Salient Features of the Water Cess Act

1. Cess on Water Use: It imposes a cess on water consumed by industries and local authorities.

2. Funding the Boards: The proceeds support the CPCB and SPCBs, linking economic instruments with environmental governance.

Implementation Challenges

1. Enforcement Capacity: SPCBs face capacity constraints and low prosecution rates; in Andhra Pradesh, urban local bodies struggle with waste management infrastructure, indirectly affecting water quality.

2. Industrial Compliance: The CPCB's National Water Quality Monitoring Programme reports better compliance among Grossly Polluting

Practice 11(b) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Industries, yet significant non-compliance persists.

3. Fund Utilisation and State Adoption: Cess funds need efficient, transparent use, and as water is a state subject, the 2024 amendment applies only where states adopt it.

Way Forward

1. Stronger SPCBs: Enhance their technical, financial and human resources for timely action.
2. Partnerships and Data: Use cess funds with industries for cleaner technology and wastewater treatment, backed by open data and better monitoring networks.
3. Wider Adoption: Encourage all states, including Andhra Pradesh, to adopt the 2024 amendments so that regulatory practice is uniform and legal ambiguities reduce.

Conclusion

The two Acts form the backbone of India's legal fight against water pollution. Effectively implemented, they support cleaner water bodies and the constitutional duty to protect the environment.

Stuck at "Critically examine"?
See the thinking before you write.
ghissu.com →

Write your own answer to 11(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
12(a)

Give an account on mineral resources of India. Explain the impacts of mining activity on the environment.

10

Introduction

India's vast and varied geological formations have endowed it with a rich array of minerals crucial for industrial and economic development. As infrastructure and industrialisation accelerate, demand has surged, making sustainable management and the ecological cost of mining a pressing concern. Understanding both is essential to balance development with environmental stewardship.

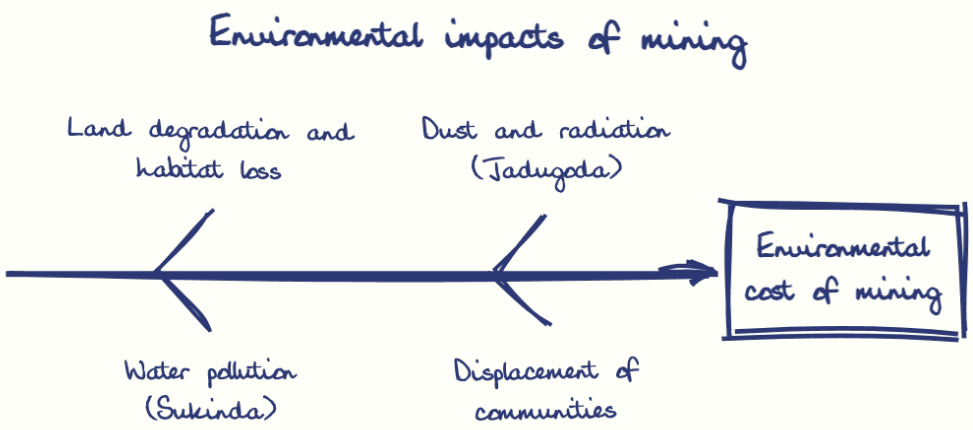
Mineral Resources of India

1. Diverse wealth: India has about (84) minerals, metallic ones like iron ore, bauxite and manganese, and non-metallic ones like graphite and mica, supporting industry and exports.
2. Key reserves: Significant coal, iron ore and bauxite; Andhra Pradesh holds (86.11%) of India's barytes and (18.49%) of its road metal reserves.
3. Import dependence: Lithium, cobalt and nickel are imported, prompting the National Critical Mineral Mission to boost domestic exploration.
4. Underutilisation: Mining contributes only about (1.4%) to GDP, far below countries like Australia,

underscoring the need for better exploration and sustainable mining.

Environmental Impacts of Mining

1. **Land degradation:** Removal of vegetation and topsoil causes soil erosion and habitat loss, especially in forests and sensitive zones.
2. **Water pollution:** Effluents and leaching contaminate surface and groundwater, as with hexavalent chromium from the Sukinda chromite mines.
3. **Air pollution and health:** Dust and harmful gases cause respiratory illness; radioactive hazards arise near the Jaduguda uranium mines.
4. **Social disruption:** Mining-induced displacement and land encroachment disrupt communities and demand effective rehabilitation; audits flagged compliance gaps in Andhra Pradesh's mineral-rich districts.

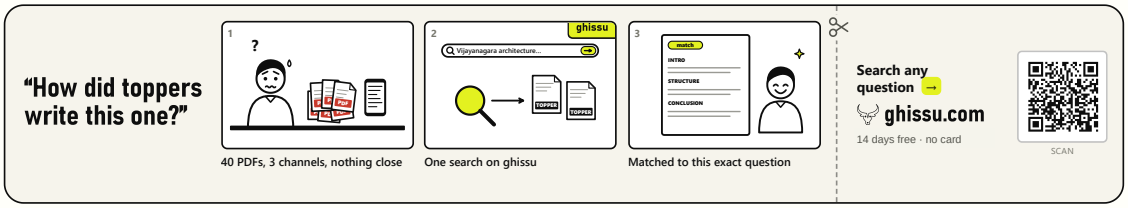


Way Forward

1. Environment Management Plans: Enforce comprehensive EMPs at all mining stages, prioritise eco-friendly technologies and avoid ecologically fragile zones.
2. Recycling: Recycle aluminium and copper to reduce extraction pressure, backed by strict laws and incentives for sustainable use.
3. Regulatory coordination: A unified regulatory authority at state and central levels can streamline approvals, ensure compliance and bring transparency to mining operations.
4. Rehabilitation: Secure livelihoods through community participation and benefit-sharing.

Conclusion

Minerals are a cornerstone of India's growth, but mining carries serious environmental and social costs. Stringent safeguards, resource efficiency and inclusive development can deliver sustainable mineral use that supports India's climate commitments without compromising ecological integrity.



P-IV
12(b)

Explain Integrated Water Resource Management. Give the importance of Water user Associations.

10

Introduction

Water scarcity, driven by climate change, population growth and urbanisation, makes sustainable management critical for India. Integrated Water Resource Management (IWRM) offers a holistic approach balancing agriculture, ecosystems and human use, while Water User Associations (WUAs) let local communities take part in equitable and efficient management, vital in agrarian Andhra Pradesh.

Integrated Water Resource Management

1. Meaning: A process of coordinated development and management of water, land and related resources to maximise economic and social welfare equitably without compromising ecosystems. It looks at water's many uses on a basin-wide scale rather than in sectoral silos.

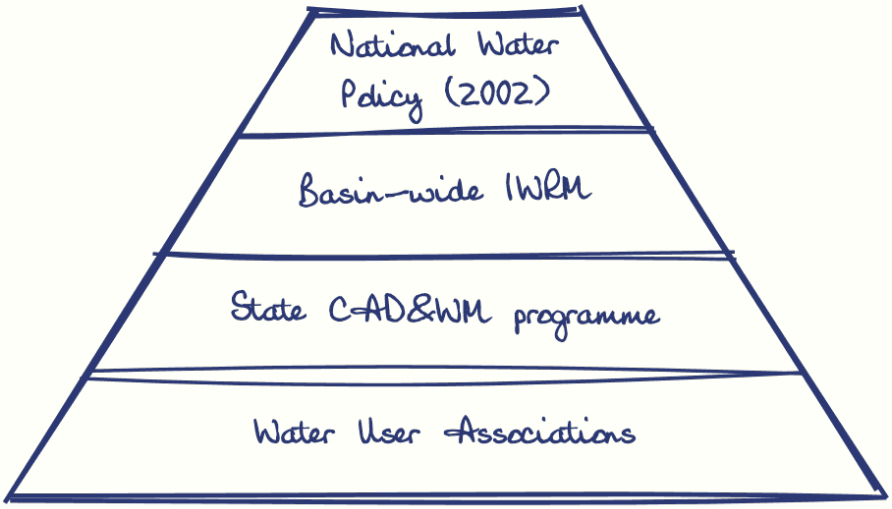
2. Key Components: Participatory decision-making, cross-sectoral coordination, demand and supply management and adaptive governance, with allocation for farming, homes, industry and environment.

3. Policy Framework: The National Water Policy

(2002) invdves Panchayats, WUAs and local bodies in planning, operation and maintenance, creating ownership.

4. Andhra Pradesh: The CAD&WM programme mandates WUAs for water distribution and recovery of charges, reflecting participatory irrigation management.

Layers of participatory water management



Importance of Water User Associations

1. Local Empowerment: Farmer-led cooperatives decide on distribution, maintenance and conflict resolution, building ownership and efficiency.
2. Equitable Distribution: They ensure fair shares for small and marginal farmers, crucial in drought-prone Rayalaseema.
3. Financial Sustainability: Recovery of water charges

funds repairs and reduces dependence on subsidies.

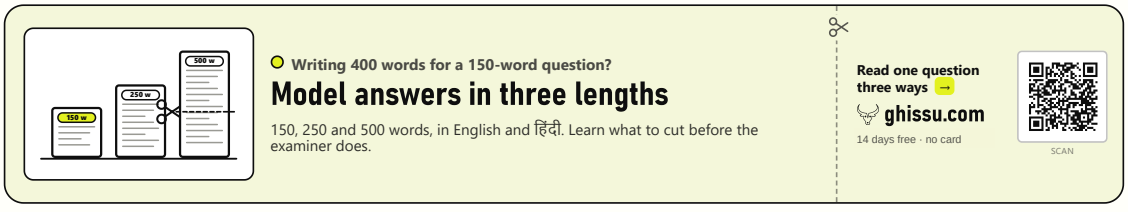
4. Interface with Government: As intermediaries with irrigation authorities, they help implement PMKSY and promote water-saving technology and cropping suited to local water availability.

Way Forward

1. Legal Backing: States like Andhra Pradesh should update and enforce WUA laws with clear rights and duties, so the associations last.
2. Capacity Building: Train members in water budgeting, conflict resolution and modern irrigation.
3. Gender Inclusion: Bring in more women to address gender-specific water needs.
4. Technology: Digital metering, monitoring and data sharing for transparency and better decisions.

Conclusion

IWRM gives the framework and WUAs make it work at the grassroots, ensuring participatory, equitable and efficient water use in line with SDG 6.



P-IV
13(a)

What are the natural and human induces threats to biodiversity? Give an example of each.

10

Introduction

Biodiversity, the variety of life on Earth, is crucial for ecosystem stability, human well-being and sustainable development. It faces natural and human-induced threats that have accelerated species decline and habitat degradation globally and in India.

Natural Threats

Natural vs human-induced threats

Natural	Human-induced
Natural disasters	Habitat loss
Climate variability	Overexploitation
Disease outbreaks	Pollution
Natural dispersal	Exotic species

1. Natural disasters: Earthquakes, floods, droughts and wildfires cause sudden, extensive damage to habitats and populations. Example: the Nilgiris wildfires showed how seasonal disturbances can severely hit forests and wildlife in southern India.

2. Climate variability: Natural shifts in temperature and rainfall alter habitats, forcing species to migrate or adapt.
3. Disease outbreaks: Epidemics can decimate wildlife with limited genetic diversity, causing local extinctions and affecting ecosystem functions.
4. Natural dispersal: Species occasionally migrate to new areas on their own and may outcompete native species.

Human-Induced Threats

1. Habitat loss and fragmentation: The most significant threat, driven by agriculture, urbanisation, mining and infrastructure. Example: rapid coastal development and aquaculture in Andhra Pradesh have fragmented habitats.
2. Overexploitation: Unsustainable hunting, fishing and harvesting cause declines, illegal wildlife trade, including in parts of Andhra Pradesh, targets species for ivory and horns.
3. Pollution: Industrial effluents, pesticides and plastics poison species. Example: pollution in the Godavari river basin harms aquatic biodiversity and fisheries vital to local livelihoods.
4. Exotic species: Introduced species turn invasive

and outcompete or prey on natives, a form of biological pollution.

5. Climate change: It intensifies natural variability and extinction risk; even renewable projects have shrunk the Great Indian Bustard's habitat in Rajasthan, showing complex trade-offs.

Way Forward

1. Habitat protection: Protect Key Biodiversity Areas and restore fragmented habitats to keep ecological connectivity.
2. Sustainable resource use: Promote community-based management using local knowledge.
3. Pollution control: Tighten monitoring of pollutants in river basins and coastal zones of Andhra Pradesh.
4. Nature-based solutions: Align climate action with biodiversity through resilient ecosystems and carbon sequestration.

Conclusion

Integrated conservation and sustainable development will preserve India's biological heritage, including Andhra Pradesh's, in line with SDG (15.)

Correct, but average?
A number, a committee, a case — for every question.
ghissu.com →

P-IV
13(b)

What is meant by the term natural Selection? Draw a food chain that has you as the top consumer.

10

Introduction

Natural selection is a foundational concept of biological evolution and human adaptation. Research in human genetics shows populations still evolve in response to changing ecological and social conditions.

Meaning of Natural Selection

1. Darwin's Five Facts: Super fecundity, steady populations, limited resources, unique individuals and heritability explain why organisms compete and only the fittest survive to reproduce.
2. Genetic Variation: Individual uniqueness comes from heritable genetic variation passed to offspring, letting populations adapt over generations.
3. Darwinian Fitness: Fitness is reproductive success relative to others, traits giving a survival advantage spread.
4. Types of Selection: Stabilising selection favours average traits, directional favours one extreme and disruptive favours both, as seen in human birth weight.

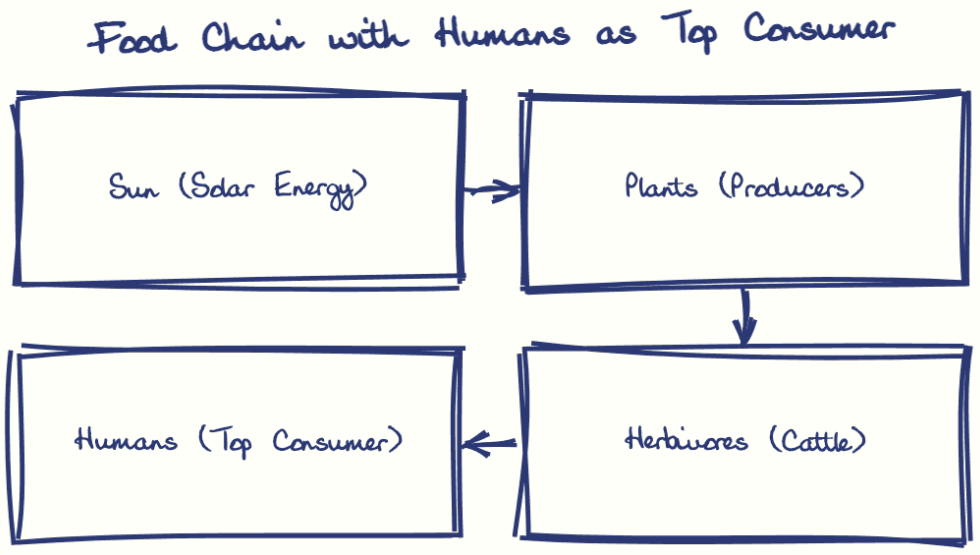
Practice 13(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Natural Selection in Humans Today

1. Culture and Technology: Better healthcare, as in Andhra Pradesh, raises survival regardless of genes, reducing selection's role.
2. Environmental Variability: Lighter skin at high latitudes, where sunlight is weaker, aids vitamin D synthesis, an adaptation to local conditions.
3. Genetic Disease: Huntington's chorea shows selection lowering allele frequency when reproductive success falls.
4. Intra-Group Selection: Selection can favour traits that raise growth or survival relative to other groups.

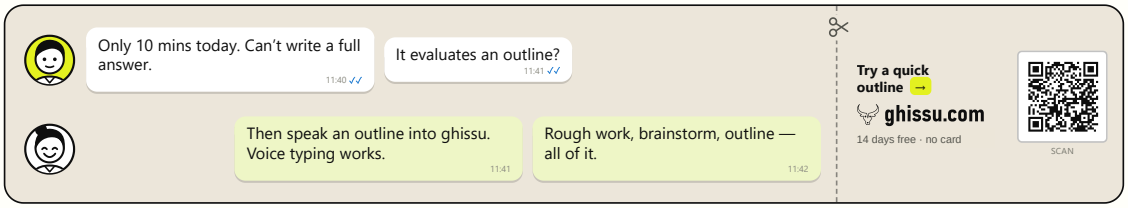
Food Chain with Me as Top Consumer



1. Energy Source: Green plants convert solar energy into chemical energy by photosynthesis, forming the base.
2. Consumers: Cattle as herbivores are primary consumers; humans, being omnivores, sit at the top.
3. Energy Loss: Much energy is lost as heat at each transfer, limiting chains to four or five levels.

Conclusion

Natural selection explains how organisms adapt and evolve, and remains relevant despite technology. The food chain shows humans as apex consumers who depend on every level below, so protecting ecological balance serves our own survival.



Write your own answer to 13(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2011 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
14(a)

Explain Urban Governance and role of community in Municipal Solid Waste Management.

10

Introduction

Rapid urbanisation has made municipal solid waste management (MSWM) a critical urban governance issue, as growing waste and weak infrastructure threaten health and the environment. In Andhra Pradesh, ULBs generate about 6,890 tons of waste daily (2022), so governance and community involvement are pivotal to cleaner cities.

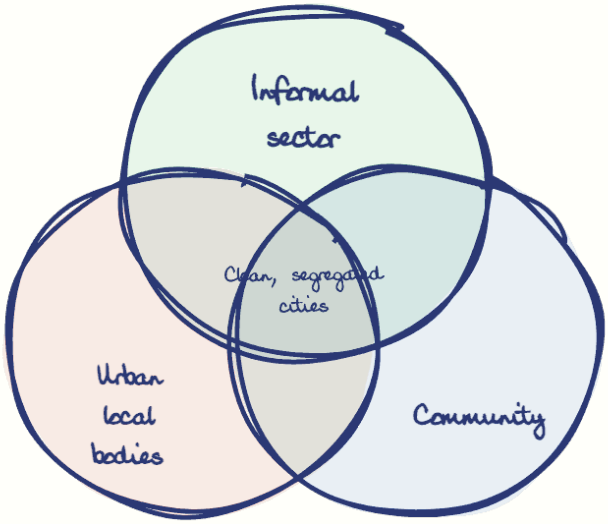
Urban Governance in MSWM

1. **Decentralised responsibility:** The 74th Constitutional Amendment (1992), through the 12th Schedule, assigns cleanliness and waste disposal to ULBs, bringing governance closer to citizens.
2. **Institutional coordination and gaps:** Municipal sanitation wings, the CPCB and state committees must coordinate, but ULBs face financial, technical and infrastructural deficits, only 4 of 123 had operational sanitary landfills in 2022.
3. **Policy framework:** The SWM Rules, 2016, revising the 2000 rules, cover segregation to disposal, yet unclear roles, poor monitoring and few bye-laws weaken enforcement.

4. Partnerships: Private firms and informal workers can raise efficiency, but weak municipal finances and a lack of bankable projects deter investment.

Role of the Community

Partners in MSW management



1. Source segregation: ULBs claim 98.17% segregation but audits show about 75%, so citizen engagement is essential to cut landfill burden.
2. Informal sector: Waste pickers and recyclers drive much of the recovery, so formal recognition, training and health support for waste pickers improve recycling and livelihood security.
3. Awareness and behaviour change: Public campaigns, education, cleanliness drives and local composting initiatives build responsible disposal habits, ownership and accountability.

4. Community monitoring: Active citizen involvement in monitoring services, with feedback mechanisms, keeps ULBs transparent, exposes gaps early and makes them responsive.

Way Forward

1. Stronger ULBs: Enhance the financial, technical and infrastructural capacity of ULBs through targeted training, increased budgetary allocations and technology adoption for efficient MSWM.
2. Inclusion and IEC: Give waste pickers social security and skills, and run sustained IEC campaigns with incentives for segregation.
3. Monitoring: Set up transparent grievance platforms with community representatives.

Conclusion

Governance that joins strong institutions with active communities is essential for sustainable MSWM. It can give Andhra Pradesh's cities cleaner environments and better health, in line with Swachh Bharat Mission and the SDGs.

STUDY TIP · BEFORE YOU OPEN A CHAPTER
List what was asked from it in the last ten years. Read that first; the rest second.

ghissu
ghissu has already done this for every topic — with the trend drawn out.

See the trend for your chapter →
ghissu.com
14 days free · no card
SCAN

P-IV
14(b)

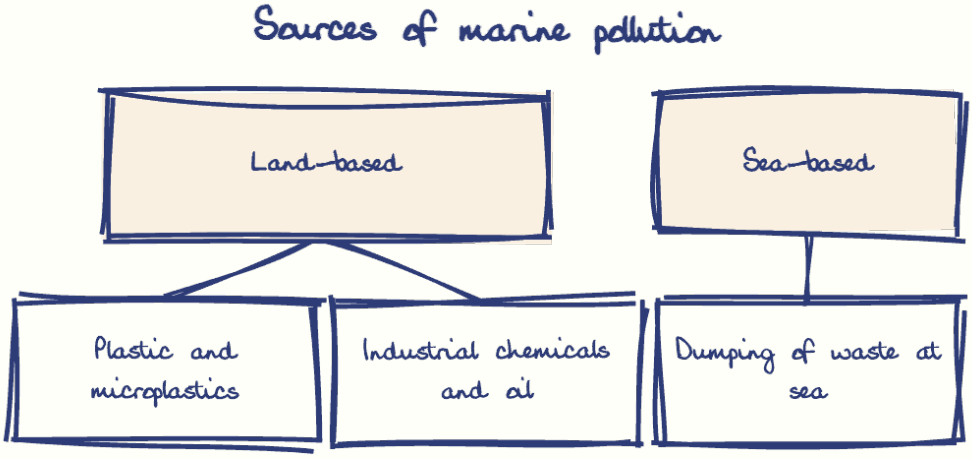
Discuss the sources, impacts and control measures of Marine pollution.

10

Introduction

Marine pollution has emerged as a critical challenge for countries with long coastlines like India, which has a 7,500 km coast. Andhra Pradesh's 974 km coastline is particularly vulnerable to land-based and sea-based pollution. Plastic waste, chemical discharges and oil spills threaten marine biodiversity, fisheries and coastal livelihoods.

Sources



1. Plastic and microplastics: Despite bans, plastic keeps entering the sea; a 71.67% increase in plastic discharge from nearby Chennai signals similar risks for the state's ports.
2. Chemicals and oil: Industrial effluents with

mercury, pesticides and hydrocarbons reach the sea through river runoff and coastal industries, as in Visakhapatnam's industrial corridor.

3. Sea-based dumping: Waste dumped at sea adds to land-based pollution, which is why international conventions restrict it.

Impacts

1. Ecological damage: Pollution causes coral bleaching, loss of fish stocks and low-oxygen dead zones, disturbing ecological balance.
2. Health risks: Microplastics have been detected in seafood, and toxic chemicals such as mercury pass into the catch and harm human health.
3. Livelihoods: Over four million fisherfolk nationally, including the state's coastal communities, depend on marine resources.

Control Measures

1. Legal framework: CRZ Notifications under the Environment Protection Act, 1986, regulate activity along sensitive coasts, including Andhra Pradesh's coastal districts.
2. International conventions: The London Convention 1972 and its 1996 Protocol restrict dumping, and UNCLOS lets coastal states regulate their

EEZs.

3. Plastic ban: The 2022 Single-Use Plastic Ban, backed by UNEP's Clean Seas Campaign, cuts marine debris.

4. Monitoring and coordination: ICM-AM and the National Centre for Ocean Research collect data along Andhra Pradesh's coast, and expanding COM-APS there would give real-time data.

5. Enforcement and blue economy: Weak coordination and inconsistent enforcement are key gaps. A dedicated coastal pollution task force in Andhra Pradesh, recycling infrastructure in coastal cities and eco-friendly aquaculture and port operations at Visakhapatnam and other ports can close these governance gaps.

Conclusion

Effective control needs robust institutions, community participation and technology aligned with SDG 14, securing a sustainable blue economy for coastal people and preserving marine biodiversity for future generations.

Improving — or just writing more?

My answers

Every attempt, every score, every enhanced version of your answer — on one timeline.

Watch your curve → ghissu.com

14 days free · no card

SCAN

P-IV
15(a)

Explain National Action Plan on Climate Change (NAPCC), objectives and recommendations.

10

Introduction

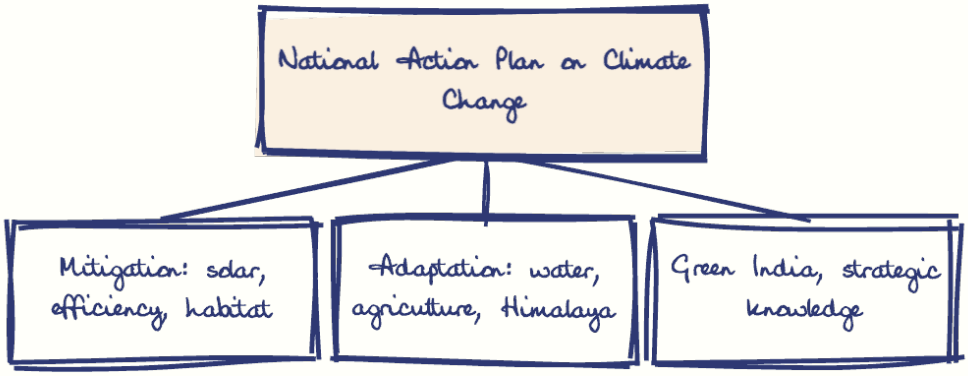
With erratic monsoons, rising temperatures and coastal risks, India must integrate climate action with development. Launched in 2008, the National Action Plan on Climate Change (NAPCC) is India's pioneering framework of focused missions. It aims to reduce the emission intensity of the economy while enhancing adaptive capacity, reflecting India's Paris commitments and NDCs.

Structure and Objectives

1. Nodal coordination: The MoEFCC coordinates eight missions, each anchored in a relevant ministry for accountability.
2. Eight missions: Solar energy, enhanced energy efficiency, sustainable habitat, water, agriculture, Himalayan ecosystem, Green India and strategic knowledge, covering mitigation and adaptation.
3. State integration: States like AP localise missions, managing risks along the 974 km coastline and promoting renewables in corridors like Visakhapatnam–Chennai.
4. Resources: Budgets, public–private partnerships and

international climate finance.

N-APCC (2008): eight missions



Implementation and Outcomes

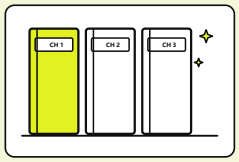
1. Solar: The National Solar Mission helped install over (49) GW nationally by (2021), with AP's solar projects contributing through grid integration.
2. Efficiency: The PAT mechanism under NMEEE cut over (31 million) tonnes of CO_2 in its first cycle.
3. Green India Mission: Afforestation and ecosystem restoration enhance carbon sinks and biodiversity. AP's forest programmes improve resilience in Rayalaseema.
4. Gaps: Inadequate finance, limited technology diffusion and poor centre–state coordination have constrained the plan, requiring revisions aligned with updated NDCs.

Recommendations

1. State capacity: Strengthen institutions in climate-sensitive sectors like agriculture and aquaculture in AP for better local adaptation.
2. Innovative finance: Expand the National Clean Energy Fund and incentivise private participation to close funding gaps.
3. Technology: Promote environment-friendly technologies and use ISRO satellite data for climate monitoring and evidence-based decisions.
4. Communities: Blend traditional conservation practices with modern strategies to build sustainable livelihoods in Rayalaseema and coastal areas.

Conclusion

The N-APCC laid the foundation of India's climate strategy by joining mitigation and adaptation through targeted missions. Revised to match updated NDCs and backed by strong state action in regions like Andhra Pradesh, it can advance SDG (13), SDG (15) and the Panchamrit commitments.



Notes in five notebooks, none at hand?

Notes that live with the question

Note from toppers' answers, exam-hall thinking and model answers — your notes gather into topic and chapter notes on their own.

Build notes that find you →

ghissu.com

14 days free · no card

SCAN

P-IV
15(b)

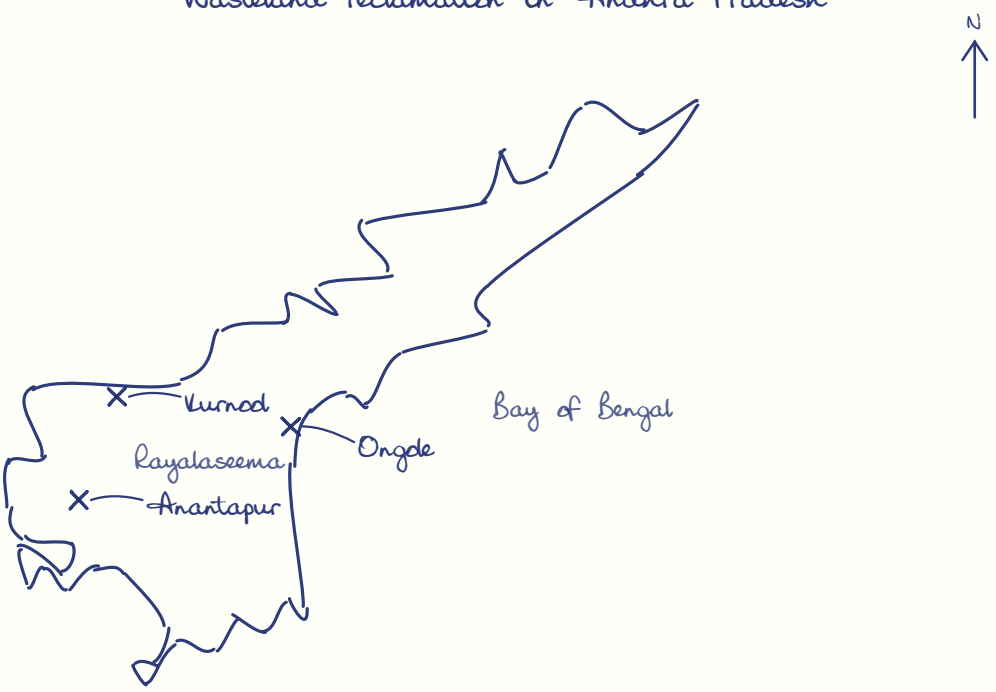
Discuss the strategies for wasteland reclamation.

10

Introduction

Industrial activity, unscientific farming and natural factors have caused extensive land degradation. The 2019 Wastelands Atlas classes about 17% of India's area as wasteland, much of it arid and semi-arid, threatening ecology, farm output and livelihoods. In Andhra Pradesh, with the Rayalaseema drought belt, effective reclamation is vital.

Wasteland reclamation in Andhra Pradesh



Institutional and Policy Strategies

1. **Mapping and Coordination:** The Department of Land

Resources leads mapping with geospatial data from the National Remote Sensing Centre, and Andhra Pradesh's Department of Agriculture and Cooperation aligns state programmes with it.

2. Legal Protection: The Forest Conservation Act and Soil Conservation Acts prevent further degradation.

3. Community Involvement: Local self-government bodies manage wasteland development, integrating customary land tenure in tribal areas.

4. Financial Support: MGNREGA funds soil and water conservation, including watershed projects in Anantapur and Kurnool.

Technical Strategies

1. Tailored Treatment: Wastelands range from saline and waterlogged to ravinous and mining-affected; saline soils in coastal Prakasam need gypsum, while Rayalaseema's ravines need structural soil conservation.

2. Agroforestry: Combining trees with annual crops under the National Agroforestry Policy (2014) improves soil health and livelihoods; multipurpose tree species suit degraded land.

3. Monitoring: Remote sensing and GIS allow continuous tracking; atlas updates show a gradual national

decline in wasteland area, though district-level use is still limited in Andhra Pradesh.

Challenges

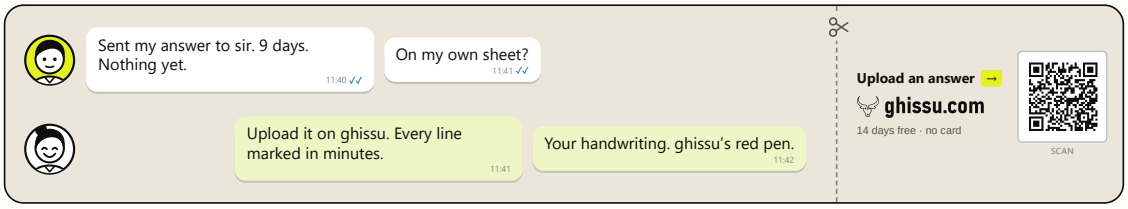
1. Cost and Expertise: High capital and technical needs put reclamation beyond many small and marginal farmers in drought-affected regions.
2. Social Issues: Unclear land ownership and encroachments complicate restoration.

Way Forward

1. Community-Based Approach: Clear land rights and financial incentives for gram panchayats and SHGs, ensuring social inclusion and sustainability.
2. Technology: Soil health cards and precision agroforestry for each landscape.
3. Climate Resilience: Link reclamation with drought mitigation and water conservation in Rayalaseema.

Conclusion

No single method fits all wastelands. A mix of institutional, technical and community-led strategies can restore them to productivity and ecological resilience.



APPSC GROUP-I MAINS · PAPER V

2008

This book is arranged year by year. For subject-wise, topic-wise and sub-topic-wise PYQs, with UPSC questions from 1979 to 2026 mapped to the APPSC syllabus, sign in at **ghissu.com** and join the waitlist (usually approved in 1 to 2 days).

P-IV
1(a)

What are the strategies of development, transfer and diffusion of technology in our country ?

10

Introduction

Technology is a critical driver of India's growth and strategic autonomy, as the India Semiconductor Mission and the rapid growth of digital infrastructure show. Development, transfer and diffusion of technology raise productivity, innovation and inclusive growth. Andhra Pradesh, with ISRO's Sriharikota facility and industrial corridors, reflects these national strategies.

Strategies of Development

1. Research ecosystem: National laboratories and Technology Transfer Offices at central and university levels commercialise innovations.
2. Policy evolution: Policy moved from Nehruvian institution-building to the 1983 Technology Policy Statement on indigenous development, and now Make in India and Digital India.
3. Public-private partnerships: The Technology Development Board and innovation clusters bridge research and market.
4. State initiatives: The Andhra Pradesh MedTech Zone and the Visakhapatnam-Chennai corridor

support incubation.

Strategies of Transfer

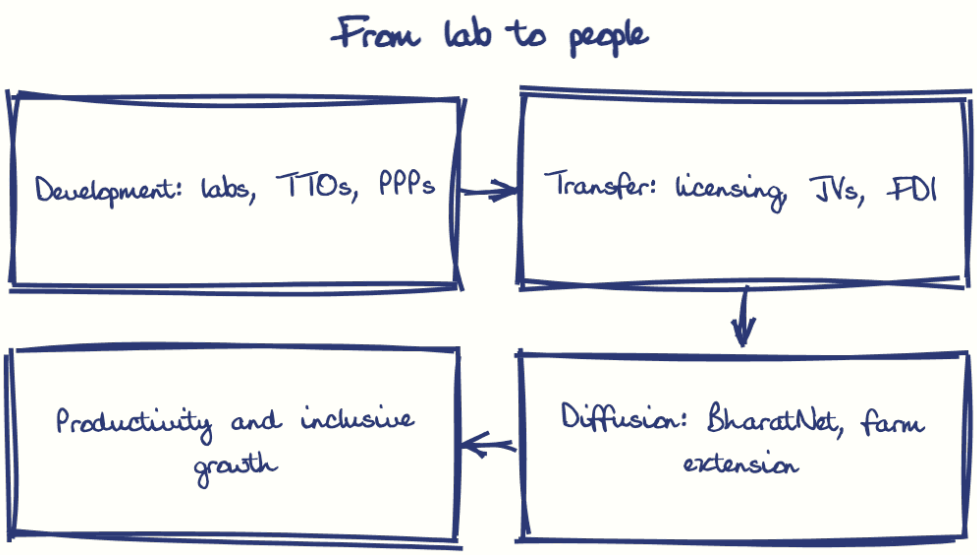
1. Channels: Licensing, joint ventures, FDI and reverse engineering, though foreign firms hesitate to share cutting-edge technology.
2. Flexible models: India promotes co-development and co-ownership, adapting technology to local needs.
3. Global partnerships: Foreign research ties and the NITI Frontier Tech Hub link global and local knowledge.

Strategies of Diffusion

1. Digital public infrastructure: BharatNet connects over 2.15 lakh Gram Panchayats and 5G reaches 99.9% of districts.
2. Sectoral diffusion: Precision farming and aquaculture spread via extension; India supplied nearly 60% of global COVID-19 vaccines.
3. Skills: Local upskilling enables effective technology use.

Challenges

R&D at about 0.65% of GDP, brain drain, subsidy delays and only 28% of rural women online slow



adoption.

Way Forward

1. R&D funding: Raise R&D to at least 2% of GDP in semiconductors and clean energy.
2. Stronger TTOs: Give them resources for faster licensing.
3. Digital access: Expand affordable connectivity and digital literacy in rural areas.

Conclusion

Integrated strategies of development, transfer and diffusion can make technology an engine of inclusive growth.

How did toppers write this one?
1.5 lakh+ topper answers, matched to each question.
ghissu.com →

P-IV
1(b)

Discuss our national policy on generation and management of intellectual property.

10

Introduction

India's National IPR Policy, 2016 is a strategic framework to stimulate creativity, innovation and entrepreneurship. As knowledge and intangible assets drive growth, generating and managing IP is vital for development and technological sovereignty. Aligned with TRIPS, the policy balances rights holders' interests with public welfare.

Institutional Framework for IP Management

1. Integrated administration: Copyright and semiconductor layout design were brought under DPIIT, creating synergy among IP offices and sharply reducing pendency.
2. Balanced laws: IP laws were modernised, and Novartis AG v. Union of India (2013) upheld access to affordable medicines while protecting genuine innovation.
3. Human capital: The policy expands IP education, training and research to build a skilled workforce able to generate and manage IP.
4. Enforcement: Merging the Copyright Board with the IPAB, along with digitalisation, improved dispute resolution and protection against

infringement.

Generation of IP: Outcomes and Challenges

1. Simplification and digitalisation: Expedited examination for start-ups, reduced MSME fees and AI-enabled IP search accelerated filings, placing India in the global top 10 for patents and trademarks.
2. R&D constraint: R&D spending of about 0.65% of GDP limits patentable innovation in biotechnology and pharmaceuticals.
3. Awareness gaps: Awareness of IP benefits stays uneven in rural and semi-urban areas of Andhra Pradesh, holding back grassroots innovators.
4. Commercialisation: The Start-Up Intellectual Property Protection Scheme supports innovators, but IP valuation and financing remain weak.

National IPR Policy 2016 scorecard

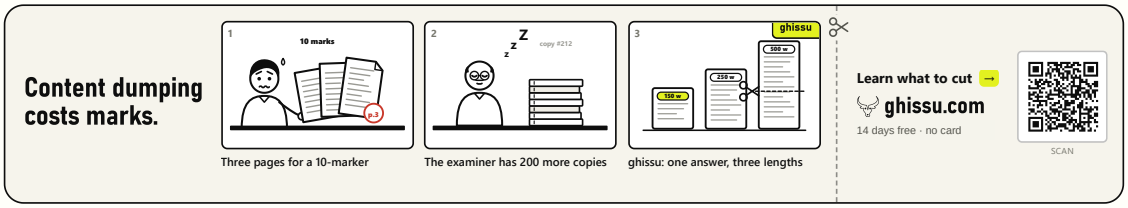
Achievements	Gaps
Integrated IP offices under DPIIT	R&D only 0.65% of GDP
Faster filings, top 10 globally	Uneven awareness in Andhra Pradesh
IP-AB merger, better enforcement	Weak IP valuation and finance

Way Forward

1. Outreach: Targeted programmes in Rayalaseema and coastal Andhra can educate innovators and entrepreneurs about IP benefits and processes.
2. R&D funding: Higher public and private R&D, especially in aquaculture and horticulture prominent in Andhra Pradesh, with academia–industry partnerships, can raise IP generation.
3. IP finance: Valuation frameworks and credit facilities can help MSMEs and start-ups raise capital on IP assets.
4. Digital services: AI-driven IP tools and virtual hearings can cut pendency further.

Conclusion

The National IPR Policy has laid a strong foundation for innovation and IP protection. Closing gaps in awareness, R&D and finance can make India a global innovation leader, in line with SDG 9.



P-IV
2(a)

What are the salient features of IRS-P5 ?

10

Introduction

IRS-P5, better known as Cartosat-1, is an Indian Remote Sensing satellite built by ISRO for cartography, that is, accurate maps and terrain models. It was launched on 5 May 2005 by PSLV-C6 from Sriharikota in Andhra Pradesh, and was the first Indian satellite able to take along-track stereo images in orbit.

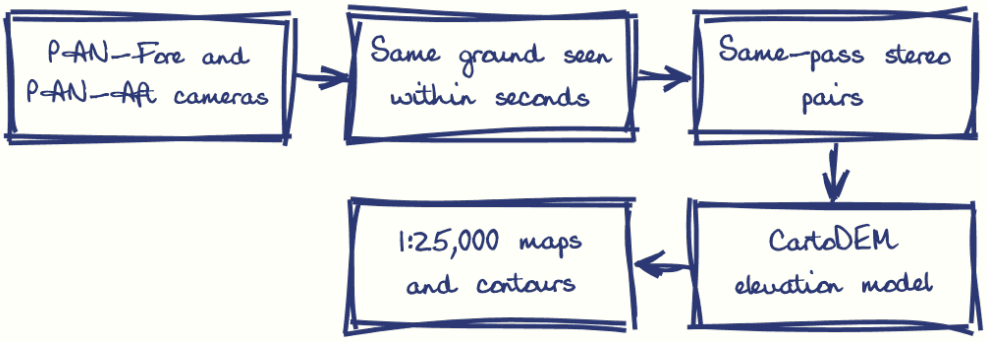
Orbit and Platform

1. Sun-synchronous orbit: A near-polar orbit at about 618 km lets it pass over any place at the same local time, giving uniform lighting.
2. Mass and launch: It weighed about 1,560 kg and was launched with HAMSAT, an amateur radio satellite.
3. Revisit: Steerable cameras gave a revisit of about 5 days over an area.
4. Design life: Designed for about five years, it served well beyond it.

Payload: Twin Panchromatic Cameras

1. Two PAN cameras: PAN-Fore and PAN-Aft are tilted at about +26 and −5 degrees along the track.
2. High resolution: Each has a resolution of about 2.5 metres, enough to see buildings, roads and field boundaries.
3. Wide swath: A 30 km swath maps large areas quickly.
4. Same-pass stereo: Both cameras see the same ground within seconds, avoiding errors from changing season or light.
5. On-board storage: A solid state recorder stores images out of ground station range.

How Cartosat-1 makes maps



Applications

1. Topographic mapping: Stereo pairs update maps at 1:25,000 and draw contours.

Candidates must not write on this margin

Candidates must not write on this margin

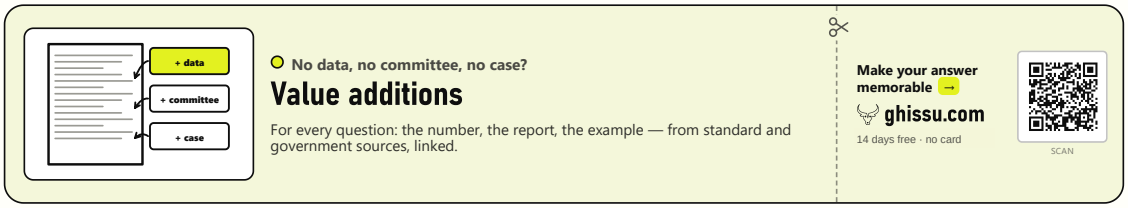
2. CartoDEM: The national elevation model aids watershed planning, flood modelling and canal alignment.
3. Infrastructure: Images support town planning and new capital regions and irrigation command areas in states like Andhra Pradesh.

Significance

1. Self-reliance: It reduced dependence on foreign high-resolution imagery.
2. Foundation: It led to the Cartosat series up to Cartosat-3.
3. State use: The Andhra Pradesh Space Applications Centre uses its data for planning and disaster management.

Conclusion

Combining high resolution, a wide swath and same-pass stereo, Cartosat-1 gave India its own mapping capability and remains a landmark in using space technology for governance.



P-IV
2(b)

What is the role of Indian Space Program in agricultural and rural developmental activities ?

10

Introduction

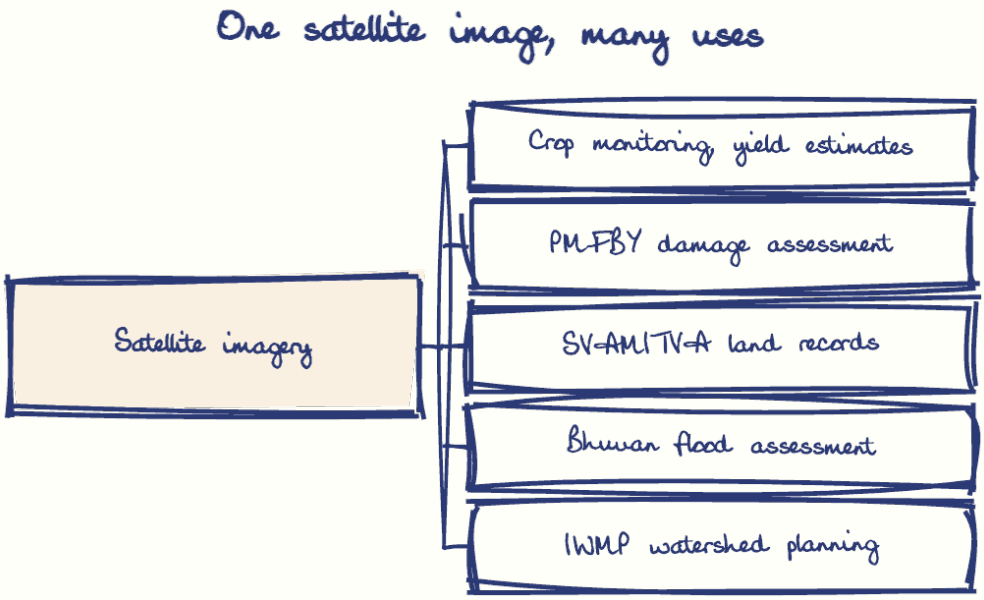
The Indian Space Program, led by ISRO, has become a pivotal force in transforming agriculture and rural development, as over 65% of India's population lives in rural areas. Satellite and geospatial technologies support food security, resource management and sustainable growth, especially in states like Andhra Pradesh.

Role in Agricultural Development

1. **Crop Monitoring and Yield Estimation:** Remote sensing satellites track crop health, pests and water stress in real time, enabling early action and accurate yield forecasts.
2. **Soil Health and Land Use:** Mapping soil moisture and fertility supports precision agriculture, crucial for drought-prone Rayalaseema.
3. **Crop Insurance:** Imagery assesses flood and drought damage, speeding claims under PMFBY.
4. **Market Decisions:** Surplus estimates guide export potential and market interventions, benefiting Andhra Pradesh's horticulture and aquaculture exports.

Role in Rural Development

1. **Natural Resource Management:** Geospatial planning of water and land through IWMP and On Farm Water Management.
2. **Connectivity:** NavIC supports rural road projects under PMGSY, improving market access.
3. **Disaster and Health:** The Bhuvan portal and Village Resource Centres help flood damage assessment and health resource mapping.
4. **Land Records:** DILRMP and SVAMITVA use satellite imagery to regularise ownership, easing credit and reducing disputes in rural Andhra Pradesh.



Impact and Challenges

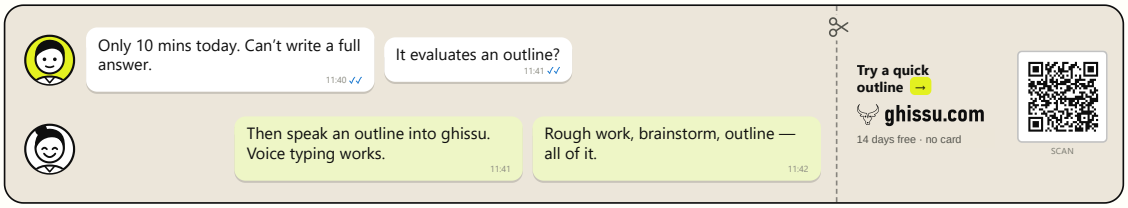
1. Socio-Economic Gains: Better productivity and resource use contributed to 10.7% agriculture GVA growth in Andhra Pradesh (2024-25).
2. Digital Divide: Gaps in rural digital literacy and infrastructure limit use, needing capacity building.
3. Cost: Costly satellite data restricts smallholder adoption.
4. Integration: Satellite data must be linked to ground-level schemes like MGNREGA and AIBP under robust governance for transparency.

Way Forward

Training for farmers and officials, low-cost mobile advisories for small farmers and alignment with Jal Shakti Abhiyan and IWMP will widen the benefits.

Conclusion

Training farmers, affordable services and better data governance will let space technology fully drive agricultural productivity and rural development.



P-IV
3(a)

Discuss the role of software development in the economic growth of our country.

10

Introduction

Software development has been a critical driver of India's economic growth in the post-liberalisation era, contributing to GDP, employment and innovation. Ongoing digital transformation under Digital India and the India Semiconductor Mission 2.0 makes it even more pivotal for sustained progress.

Contribution to Economic Growth

1. **Exports and GDP:** The IT and software industry is projected to reach USD 194 billion by 2025, about 5% of GDP; its exports strengthen forex reserves and the trade balance.
2. **Employment and Skills:** The sector employs over 5 million people, including a significant share of women, with high labour productivity that lifts incomes.
3. **Startup Ecosystem:** Over 159,000 startups and more than 100 unicorns drive innovation and attract venture capital, adding an estimated USD 140 billion by 2030.
4. **Clusters and Spillovers:** Bengaluru and Hyderabad generate knowledge spillovers and

draw global capability centres into India.

How Software Drives Growth

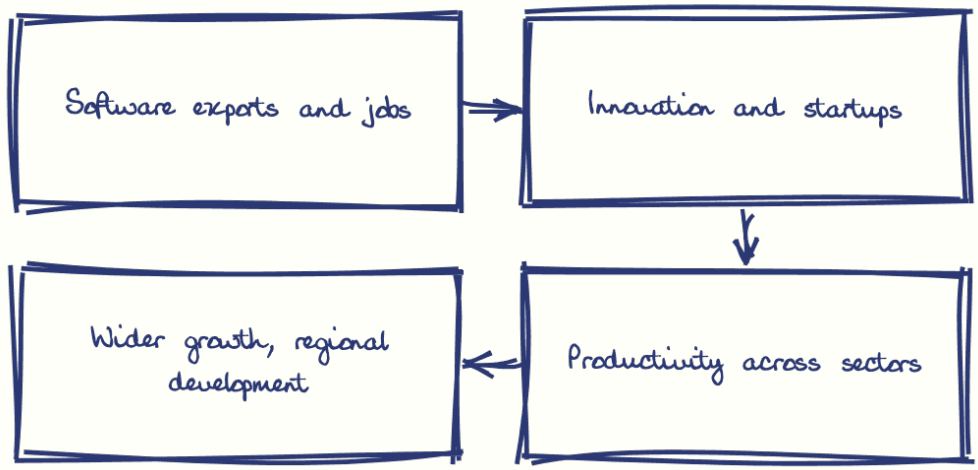
1. Growth Multiplier: As a general-purpose technology, software lifts productivity from agriculture to manufacturing; cloud and AI in SMEs show this.
2. Domestic Market: Exports drove early growth, but domestic adoption, aided by e-governance as in Andhra Pradesh, is key to sustainability.
3. Skill and Infrastructure Gaps: Advanced skills sit in a few firms, and smaller cities lack infrastructure.
4. Global Vulnerability: The 2008 crisis and COVID-19 exposed dependence on global markets, though quick recovery showed resilience.

Thus software development leads to exports and jobs, then to innovation and startups, and finally to wider growth and regional development.

Way Forward

1. Skills: Expand the Skill India Digital Platform for AI and blockchain.

How software drives growth



2. R&D: Use the IT Venture Capital Fund to move up the value chain.
3. Domestic Adoption: Fiscal incentives for SMEs to adopt software, using Andhra Pradesh's digital governance models as best practice.
4. Regional Hubs: Build IT infrastructure beyond metros for balanced growth.

Conclusion

Software development has become a cornerstone of growth through exports, jobs and innovation. Linked with domestic demand and skills, it supports SDG (9) and a knowledge-driven economy.

Which sub-topic comes next?
Trend analysis for every syllabus topic.
ghissu.com →

Write your own answer to 3(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
3(b)

Discuss the various information technology applications with special reference to rural India.

10

Introduction

Information technology has become a cornerstone of India's development, with rural internet users reaching 293 million. Since over 65% of Indians live in villages dependent on agriculture, IT can bridge the urban-rural divide by raising productivity and improving governance, education and healthcare.

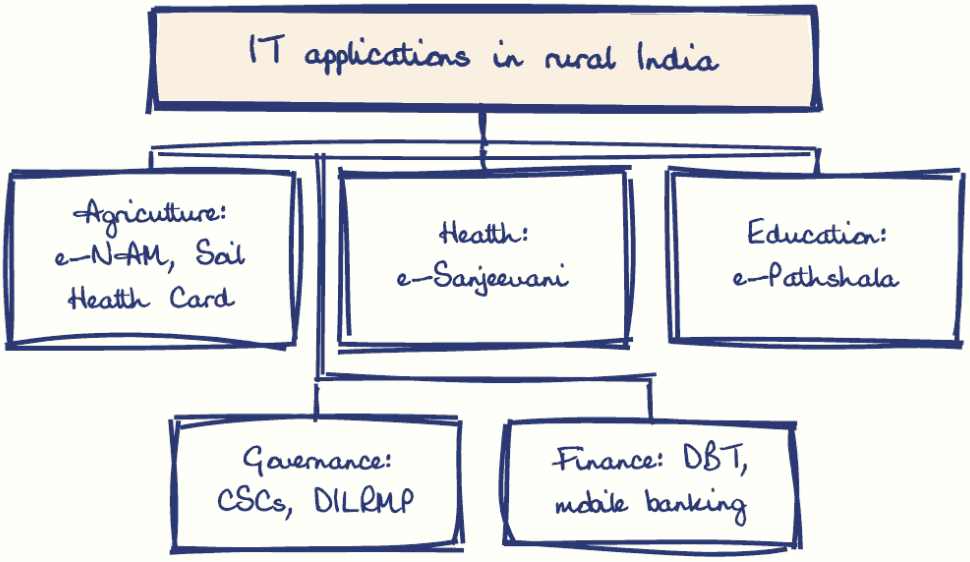
IT Applications in Rural India

1. Agriculture: e-NAM unifies fragmented markets for better prices, while the Soil Health Card and PM-FBY give data on soil quality and crop insurance.
2. Healthcare: e-Sanjeevani telemedicine and the NHRR project deliver remote care, offsetting the shortage of rural doctors and long journeys.
3. Education: e-Pathshala and digital classrooms bring interactive learning and skills to remote villages.
4. E-Governance: e-Panchayat, e-Gram and DILRMP digitise processes, and Common Services Centres deliver services, DBT and grievance redressal.

Practice 3(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

5. Financial Inclusion: Mobile banking and Direct Benefit Transfer, as in the PAHAL LPG subsidy, ensure timely, leak-free transfers.



Impact on Rural India

1. Farmers: Real-time prices, weather forecasts and best practices raise yields and incomes; e-NAM and soil health cards helped Andhra Pradesh farmers diversify.
2. Health and Education: Remote specialist consultations and online learning, vital during COVID-19, widen access.
3. Governance: Digital platforms raise transparency, and BharatNet, covering over 2.15 lakh Gram Panchayats, aids efficient services and participatory governance.

Candidates must not write on this margin

Candidates must not write on this margin

Challenges

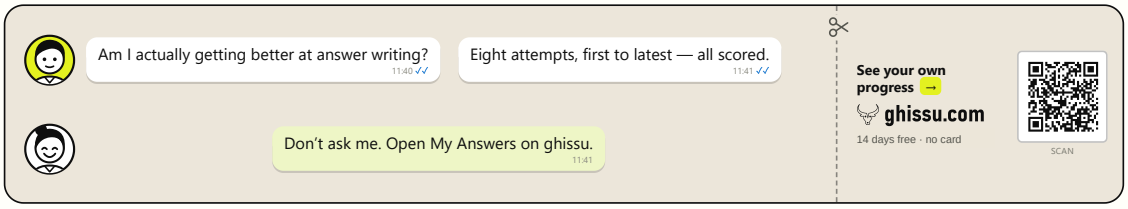
1. Digital Divide: Only 28% of rural women have internet access.
2. Infrastructure: Weak connectivity and low digital literacy, as in drought-prone Rayalaseema, limit the use of telemedicine and e-learning by rural students and patients.

Way Forward

1. Infrastructure: Complete BharatNet Phase-II.
2. Literacy: Training focused on women and marginalised groups.
3. Local Content: Applications in regional languages and culturally relevant formats.
4. Partnerships: Government, private sector and civil society together to scale solutions.

Conclusion

IT applications are reshaping rural livelihoods; closing the access gap will make them an engine of inclusive growth.



P-IV
4(a)

What are the various renewable energy resources ? What is their potential in our country ?

10

Introduction

India's renewable sector has surged on national targets and climate commitments. Installed capacity crossed 203 GW in 2024, with a goal of 500 GW by 2030. Renewables are central to energy security, cutting fossil dependence and inclusive rural growth.

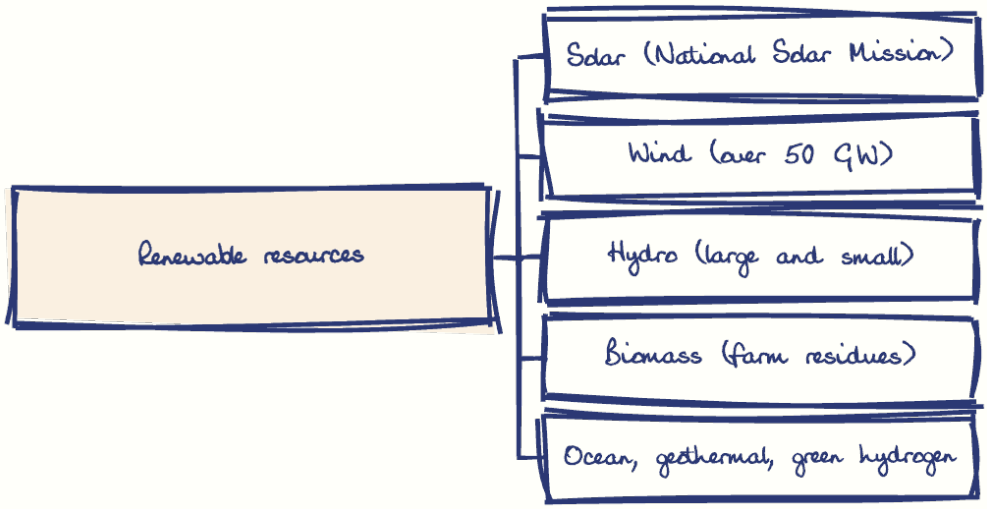
Renewable Energy Resources

1. **Solar:** Photovoltaic and solar thermal, expanded by the National Solar Mission.
2. **Wind:** Over 50 GW installed; Andhra Pradesh's 974 km coastline offers strong wind and solar potential.
3. **Hydro:** Large and small hydro projects add clean, steady capacity.
4. **Biomass:** Agricultural and organic residues meet rural energy needs.
5. **Emerging Sources:** Ocean, geothermal and green hydrogen.

Practice 4(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Renewable energy resources of India



Potential in Our Country

1. Institutions: MNRE leads policy and IREDA finances projects, while Andhra Pradesh handles land, grid links and tariffs for its solar parks and wind projects.
2. Agriculture: PM-KUSUM solar pumps raise farm output in drought-prone Rayalaseema and cut diesel use.
3. Households: Rooftop solar under PM Surya Ghar: Muft Bijli Yojana is spreading in Andhra Pradesh.
4. Grid: The Green Energy Corridor helps integrate variable power, though transmission capacity remains a concern.
5. Environment: Renewables offer low-carbon power,

cutting greenhouse gases and air pollution in line with climate commitments.

Constraints

1. **Finance:** About $189 billion is needed for 2022–30 despite falling solar tariffs and PLI support.
2. **Variability:** Solar and wind need storage and demand-side management for reliable supply.

Way Forward

1. **States:** Faster approvals, grid links and tariff rationalisation in Andhra Pradesh.
2. **Finance:** Affordable credit and risk cover through PLI and IREDA.
3. **R&D:** Indigenous solar cells, turbines and storage for strategic autonomy.

Conclusion

India's renewable potential is vast and diverse, from solar and wind to biomass and green hydrogen. It can be realised through finance, storage and stronger state implementation.

STUDY TIP · WHERE TO WRITE A NOTE

Write the note against the question that taught it. That is the only place revision will look for it.

ghissu

On ghissu every question has a notes tab — and your notes gather into chapter notes by themselves.

Open the notes tab →

ghissu.com

14 days free · no card

SCAN

P-IV
4(b)

What are the challenges faced by energy sector in our country ?

10

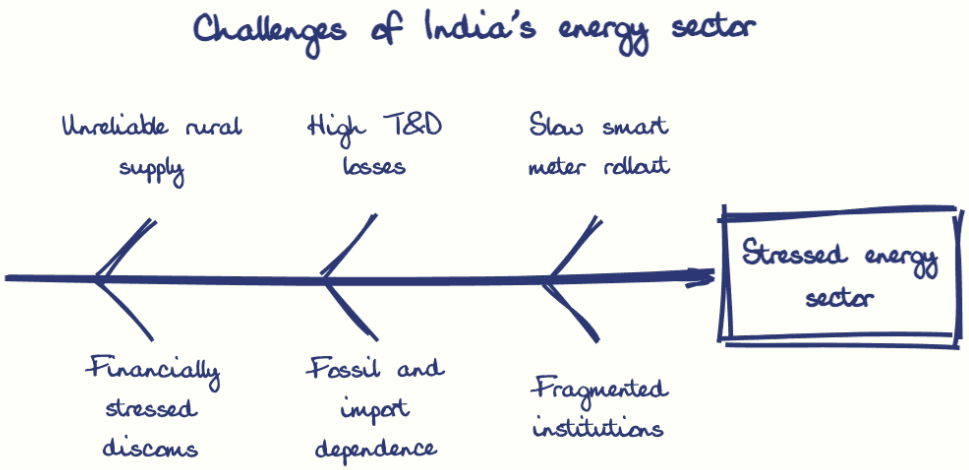
Introduction

India's energy sector is at a critical juncture as rapid industrialisation and urbanisation push demand sharply upward. Despite near-universal electrification, unreliable supply, discom finances and infrastructure gaps persist, affecting growth, welfare and Paris Agreement commitments.

Key Challenges

1. **Unreliable Supply:** Though India is declared power surplus, about (53%) of villages get electricity for under (12) hours a day, so poor quality and continuity of supply undermine the gains of electrification.
2. **Discom Financial Distress:** Financially stressed discoms cannot procure power or upgrade networks, even after UDAY, dues to producers remain, and (34) thermal plants are stressed with Rs (1.74 lakh) crore involved.
3. **Infrastructure Deficits:** Inadequate networks with high T&D losses and poor grid connectivity in eastern and rural areas impede renewable integration and reliable supply.
4. **Fossil Dependence:** Reliance on coal and imported

crude exposes the economy to price volatility and supply shocks, and clashes with climate goals.



Other Dimensions

1. Economic: Delayed payments to producers discourage investment and burden state budgets; in Andhra Pradesh, power SPSEs matter to the economy but face governance and financial gaps.
2. Technological: Slow adoption of smart grids; the Smart Meter National Programme struggles in remote areas.
3. Institutional: Overlapping central and state roles, inconsistent regulation and no unified energy data repository hinder coordinated planning.
4. Social and Environmental: Access gaps hurt rural and vulnerable groups, while continued fossil use raises carbon emissions and degrades

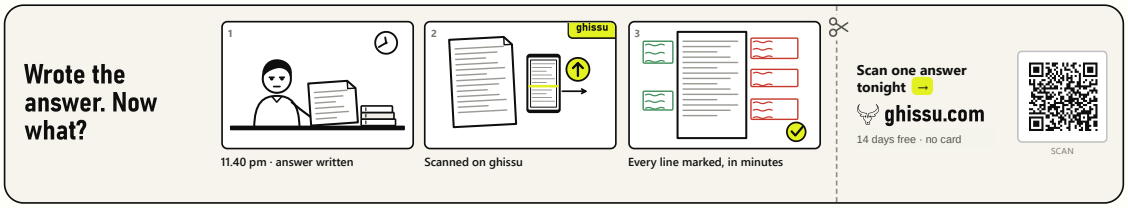
the environment.

Way Forward

1. Discom Health: Improve operational efficiency through tariff rationalisation and fewer regulatory assets.
2. Modern Infrastructure: Upgrade T&D networks, expand smart grids and integrate renewables for reliable supply.
3. Energy Security: Scale up renewables and develop indigenous resources to cut fuel imports.
4. Coordination: A centralised energy data platform and streamlined regulation for transparency.

Conclusion

Overcoming these challenges is essential for growth and energy security. Fixing finances, infrastructure and institutions will give India reliable, affordable and clean energy, in line with the National Action Plan on Climate Change and SDG (7.)



P-IV
5(a)

What are the potential impacts of climate change in India ?

10

Introduction

With 2024 the warmest year on record and 2016–2025 the hottest decade, India faces an urgent climate crisis of rising temperatures, erratic monsoons and extreme weather. Around 75 percent of districts are climate hotspots, threatening people, ecosystems and economic progress.

Physical Impacts

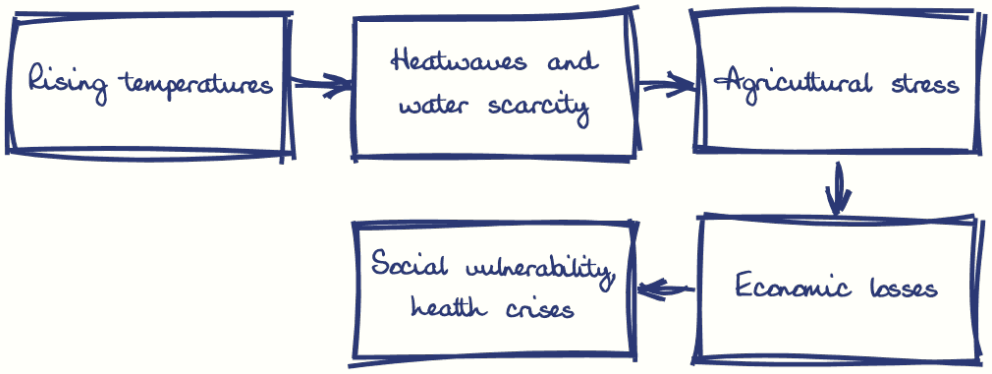
1. Rising temperatures: Land surface temperatures are rising 0.68°C per century, intensifying heatwaves; a 2010 heatwave killed over 1,300 in Ahmedabad.

2. Erratic monsoon: Greater rainfall variability brings droughts and floods, threatening water security in northern India and drought-prone Rayalaseema.

3. Sea-level rise: A rise projected at 300 mm threatens coasts, including AP's 974 km, through salinity ingress and storm surges.

4. Extreme events: Cyclones Titli and Gaja in 2018 caused damage and displacement on the east coast.

The chain of climate impacts



Ecological, Economic and Social Impacts

1. **Ecological disruption:** Altered ecosystems threaten species diversity and habitat connectivity, hurting fisheries and farming.
2. **Economic strain:** Agriculture, growing at 10.7% GVA in AP, is highly vulnerable; crop failures deepen farmer distress, and extreme weather damages industry.
3. **Social and health risks:** Heat stress, waterborne disease and air pollution hit marginalised groups hardest.
4. **Urban vulnerability:** Coastal cities like Visakhapatnam face flooding and heat, worsened by poor drainage.

The Chain of Impacts

Rising temperatures bring heatwaves and water scarcity, which cause agricultural stress and

economic losses, which finally show up as social vulnerability and health crises.

Way Forward

1. Resilient agriculture: Drought-resistant crops and efficient irrigation under the NMSA, focused on Rayalaseema.
2. Coastal protection: Mangrove restoration and cyclone-resilient infrastructure along AP's coast.
3. Urban adaptation: Better drainage through the Smart Cities Mission and AMRUT.
4. Health systems: Stronger capacity for heat-related and climate-induced illness, prioritising vulnerable groups.

Conclusion

India's climate challenges demand integrated responses. Guided by the Panchamrit commitments and the NAPCC, India can protect livelihoods and move towards net zero by 2070.

REVIEWER'S NOTE · CRITICALLY EXAMINE

"Critically examine" pays for the counter-view. An answer that stops at the first view has stopped at half the marks.

ghissu

ghissu exam-hall thinking decodes every directive — Discuss, Examine, Elucidate — before you write a line.

Decode your next question →

ghissu.com

14 days free · no card

SCAN

Write your own answer to 5(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

Scan to practice

P-IV
5(b)

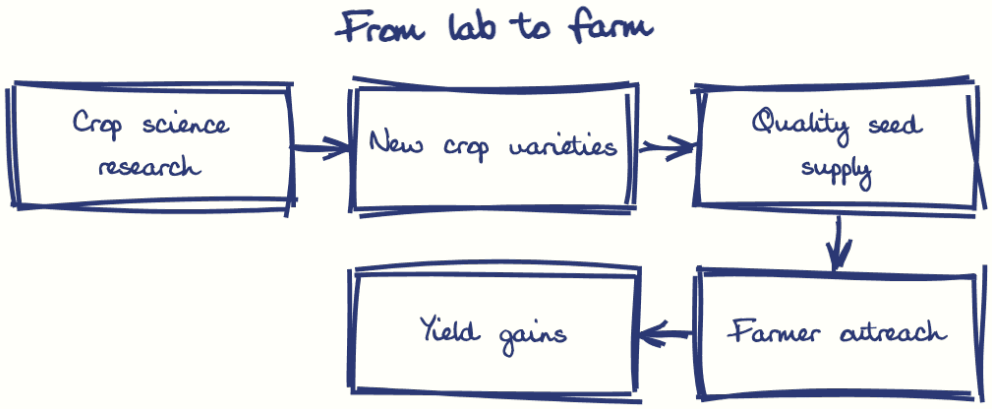
What are the outcomes of crop science research in India ?

10

Introduction

Crop science research has become vital as India seeks food security, higher farmer incomes and climate adaptation. With vast agro-ecological diversity and rising biotic and abiotic stresses, innovations in breeding, biotechnology and resource management drive productivity, resilience and quality including for farmers in Andhra Pradesh.

Institutional Outcomes



1. New Crop Varieties: ICAR has developed over 100 varieties suited to diverse zones, raising yield and stress tolerance, AP benefits from location-specific breeding.
2. Quality Seed Supply: Over 11,800 tonnes of breeder seed ensure high-quality seed and faster

adoption of new varieties.

3. Regulatory Backbone: The Environment (Protection) Act, 1986 and guidelines for gene-edited products balance innovation with biosafety, enabling molecular breeding in rice, wheat and cotton.
4. Collaboration: Partnerships with the private sector and access to global technology help bridge innovation gaps.

Tangible Outcomes in the Field

1. Yield Gains: The drought-tolerant chickpea SAATVIK and higher-yielding gene-edited rice lines have been commercialised.
2. Technology in Crop Management: Micro-irrigation, precision farming and drones for locust control in 2020 improve resource efficiency and reduce losses.
3. Farmer Outreach: Over 10,000 farmers have been trained in and exposed to new crop science technologies.

Limitations

1. Modest Adoption: Weak extension and awareness slow uptake in rural AP.
2. Resource Constraints: Low R&D funding,

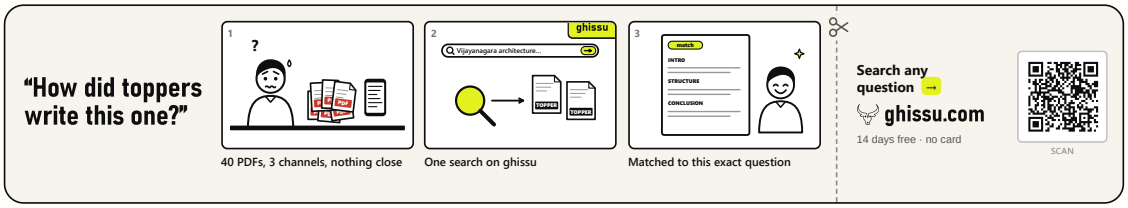
fragmented disciplinary efforts and IPR issues hinder diffusion, while infrastructural and administrative bottlenecks in AP limit benefits.

Way Forward

1. More R&D: Raise funding and build centres of excellence akin to IITs and IISc.
2. Extension: Use ICT and social media for farmer engagement.
3. Partnerships: Ease access to global germplasm through supportive IPR policies.
4. Sustainability: Combine crop science with resource conservation and integrated crop management.

Conclusion

Crop science has delivered better varieties, higher productivity and sustainable technologies, strengthening food and nutritional security. Stronger research, extension and policy will make it a cornerstone of rural prosperity.



Write your own answer to 5(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
6(a)

Define fermentation. Discuss the general procedure for the production of alcohol in the industry.

10

Introduction

Fermentation is a time-honoured biochemical process, from ancient winemaking to modern industry. India's emphasis on biofuels has renewed interest in it for alcohol production, which supports agro-industrial economies, rural livelihoods and the goal of ethanol blending to cut fossil fuel dependence.

Definition of Fermentation

Fermentation is the enzymatic breakdown of organic substances by microorganisms, mainly yeasts like Saccharomyces cerevisiae, converting sugars into alcohol and carbon dioxide under anaerobic conditions.

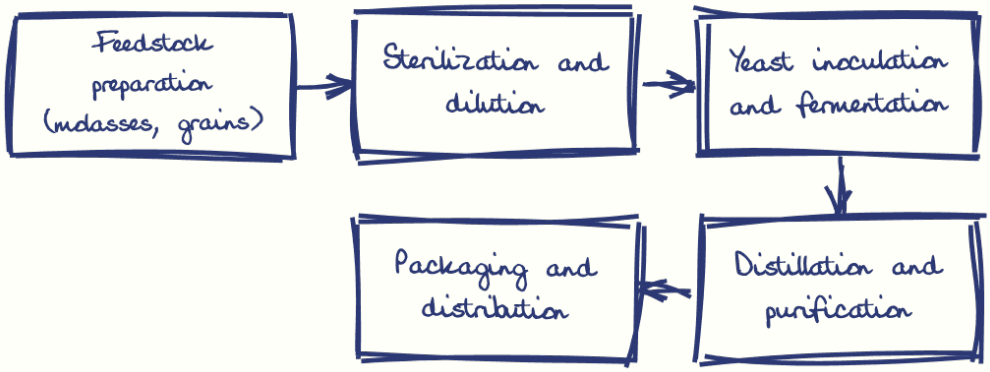
General Procedure for Industrial Alcohol

1. Feedstock selection: Sugarcane molasses, grains or damaged food grains are used.
2. Preparation: Substrates are diluted and sterilised to remove unwanted microbes.
3. Yeast inoculation and fermentation: Grain starch is first turned to sugar by amylase, while molasses is diluted to about 10–18% sugar. Yeast then ferments the sugars through glycolysis at about

30–35°C and pH 4–5 for 48–72 hours, giving a wash of 8–10% alcohol.

4. Distillation and purification: The wash is distilled to rectified spirit of about (95%) alcohol, then purified and stored.

Industrial Alcohol Production Process



Process Challenges

1. Optimal conditions: Temperature, pH and oxygen levels decide yeast viability and efficiency.
2. Alcohol toxicity: Yeast cannot survive beyond certain alcohol levels, so fermentation must be controlled.
3. Local constraints: In Andhra Pradesh, climate and feedstock availability need adaptive management.

Policy and Institutional Support

1. Pricing and finance: Administered pricing

reintroduced in 2014 and interest subvention by the DFPD have expanded distilleries, including molasses and grain units in Andhra Pradesh.

2. Research: The Central Food Technological Research Institute and agricultural universities in Andhra Pradesh improve yeast strains and yield.

3. Social balance: Production must be balanced with public health concerns and regulatory compliance.

Conclusion

Fermentation underpins industrial alcohol, economic development and energy security. Better yeast strains and non-food feedstocks can expand production sustainably, supporting SDG 9.

FIVE-MINUTE LESSON MODEL ANSWERS IN 150 · 250 · 500

Write to the word limit, not past it

In UPSC GS Mains, a 10-marker takes about 150 words and a 15-marker about 250. Twenty questions in three hours leaves roughly 7 and 11 minutes each.

1. **150 words** — A definition, two dimensions, a one-line verdict.
2. **250 words** — Add a line of context, a third dimension, one example each.
3. **500 words** — Add a diagram, a second example each, a way forward.
4. **The rule** — Draft the 150 first. Expand only where marks are.

ON GHISSU, THIS IS DONE FOR YOU — ON EVERY QUESTION.

Every model answer comes in 150, 250 and 500 words, in English and हिंदी — so you see what survives the cut.

Read one question three ways →

ghissu.com

14 days free · no card

SCAN

P-IV
6(b)

(i) What is a low volume, high value product in biotechnology ? (ii) Write about interferons, their applications and a general scheme for the industrial production of interferons.

10

Introduction

Biotechnology is a cornerstone of India's bioeconomy, targeted to reach USD 300 billion by 2030 under the BioE3 Policy. Low volume, high value products give targeted, high-impact solutions in health and industry, supporting Aatmanirbhar Bharat, with Andhra Pradesh's Bio-NEST incubation centres fostering such enterprises.

(i) Low Volume, High Value Products

1. Concept: Specialised biopharmaceuticals or biomolecules produced in small quantities but commanding high prices due to their complexity and therapeutic importance.
2. Examples: Recombinant proteins, monoclonal antibodies and synthetic vaccines serve niche medical needs and earn significant revenue despite limited production scale.
3. Production limits: High R&D cost, strict regulation and sophisticated plants prevent mass production, so focused innovation hubs are needed to bridge gaps in technology and scale.

Practice 6(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

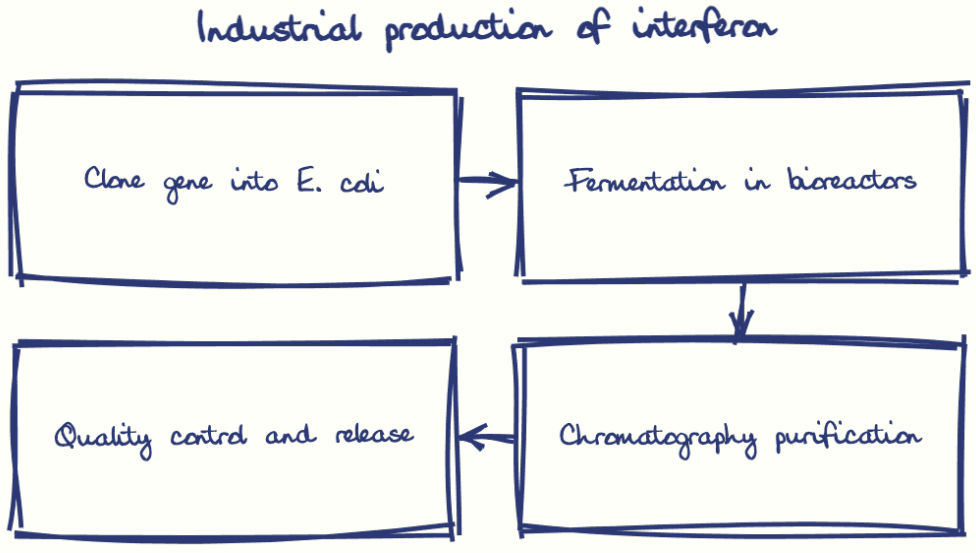
Scan to practice

4. Strategic value: They reduce import dependence, as with the indigenous DNA vaccine ZyCoV-D under BIRAC.

(ii) Interferons and Their Applications

1. Nature and types: Signalling proteins produced by host cells in response to pathogens; alpha, beta and gamma types modulate immunity differently.
2. Therapeutic uses: They treat hepatitis B and C, cancers like melanoma and lymphoma, and autoimmune diseases through antiviral, antiproliferative and immunoregulatory action.
3. Limitations: Flu-like side effects demand careful dosing and monitoring, and better formulations and delivery methods.

General Scheme of Industrial Production



Practice 6(b) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V
2 / 3
Scan to practice

Candidates must not write on this margin

Candidates must not write on this margin

1. Recombinant DNA step: Interferon genes are cloned into host cells like E. coli or mammalian cell lines.
2. Fermentation: Host cells are cultured in bioreactors under controlled conditions.
3. Downstream purification: Chromatography isolates pure, therapeutic-grade interferon.
4. Quality control: Checks for safety, potency and consistency under the Central Drugs Standard Control Organization, which also monitors units in Visakhapatnam's pharmaceutical clusters.
5. Scale and innovation: Advances in bioprocess engineering and cell culture raise yield and cut costs.

Conclusion

Low volume, high value products like interferons strengthen India's health security and strategic autonomy in critical medicines. Incubation hubs, BIRAC partnerships and the PLI Scheme for Pharmaceuticals can scale their domestic production.

REVIEWER'S NOTE · VALUE ADDITION

Examiners read hundreds of answers a day. The one with a number, a committee and a case is the one they remember.

ghissu

ghissu gives you the data point, the report and the example for every question — each linked to its source.

Get the value adds for any question →

ghissu.com

14 days free · no card

SCAN

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 6(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
7(a)

Write the general characteristics of terrestrial plants. Write about the uses of plants to mankind.

10

Introduction

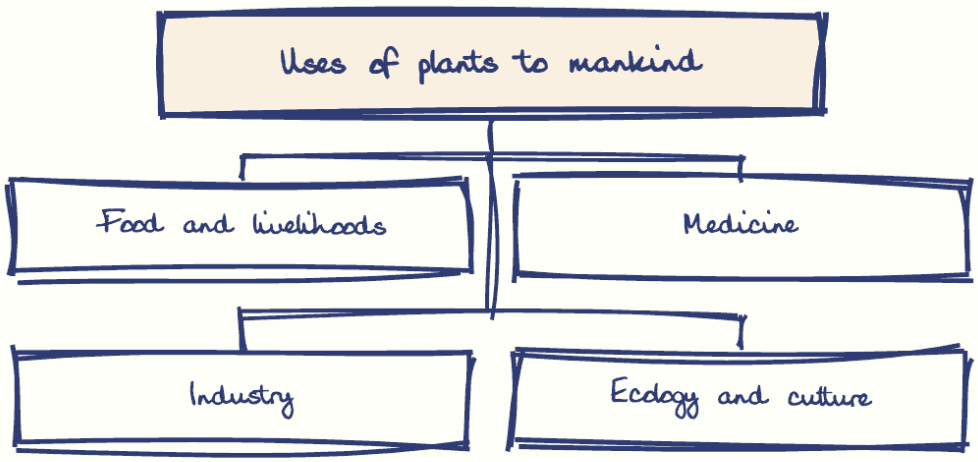
Terrestrial plants are the backbone of life on land, sustaining ecosystems and human civilisation. Millions of years of evolution have equipped them to survive diverse land environments. In Andhra Pradesh, with varied agro-climatic zones, they are central to agriculture, medicine and industry.

General Characteristics of Terrestrial Plants

1. Adaptation to land: Differentiated roots, stems and leaves absorb water, provide support and perform photosynthesis efficiently in air rather than water.
2. Cell features: Cell walls contain cellulose, and chloroplasts carry out photosynthesis to meet energy needs.
3. Reproduction: They reproduce sexually through seeds or spores; vascular plants such as gymnosperms and angiosperms bear cones or flowers, helping them colonise diverse habitats.
4. Diversity: Angiosperms include monocots and dicots, seen in mango, tamarind, turmeric and pepper in Andhra Pradesh.

5. Physiology: Stomata regulate water loss, and secondary metabolites aid defence and interaction with the environment.

Uses of Plants to Mankind



1. Food and livelihoods: Plants provide staples, fruits, vegetables and spices; chilli and horticultural produce support millions of farmers in Andhra Pradesh.
2. Medicine: Used in Indian medicine for over 4,000 years, with around 7,500 species documented for therapeutic use.
3. Industry: Raw materials for textiles like cotton and jute, timber, oils and bioactive compounds for pharmaceuticals and agrochemicals.
4. Ecology and culture: They maintain soil fertility,

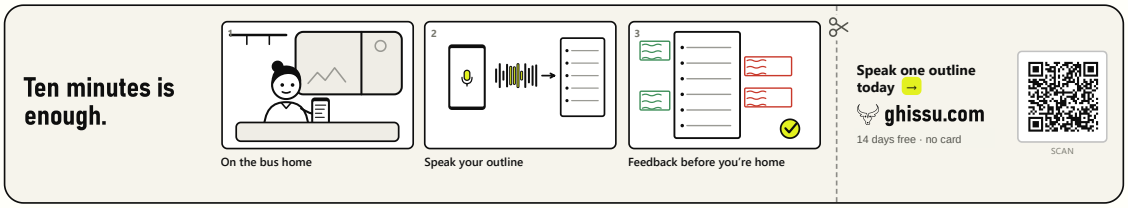
prevent erosion, support biodiversity and hold ritual significance.

Challenges

1. Variability: Plant chemistry varies with genetics and growing conditions, complicating standardisation for medicinal and industrial uses.
2. Biodiversity loss: Urbanisation and land-use change threaten plant diversity in Andhra Pradesh's sensitive zones like the Eastern Ghats.
3. Management: Sustainable harvesting needs institutions that integrate traditional knowledge with science.

Conclusion

Terrestrial plants underpin ecological stability, economic prosperity and cultural heritage, especially in a biodiverse state like Andhra Pradesh. Expanding Medicinal Plants Conservation Areas, agroforestry in drought-prone Rayalaseema and climate-resilient farming will sustain them, in line with the Green India Mission.



P-IV
7(b)

Write about the definition, criteria, history and process of domestication of some common animals by humans.

10

Introduction

Domestication of animals is a foundational process that shaped human civilisation by enabling settled agriculture, food security and socio-economic development. It involves selective breeding and taming of wild species, a practice over 10,000 years old and deeply embedded in Indian agrarian culture, including in Andhra Pradesh.

Definition and Criteria

1. Definition: Domestication is the intentional control over the breeding, distribution and biological traits of animals to increase their utility for food, labour and companionship.

2. Criteria: Key markers are genetic changes, behavioural tameness and morphological differences from wild ancestors, such as smaller size or altered colouration.

History of Domestication

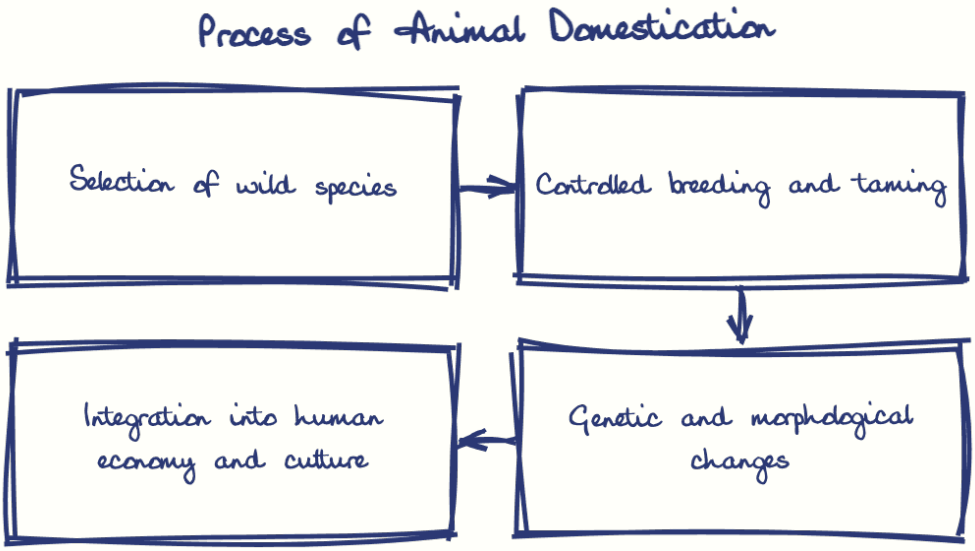
1. Origins: Cattle, sheep and goats were domesticated around 10,000 years ago in the Fertile Crescent, with parallel developments in the Indian

subcontinent.

2. Archaeological evidence: Domesticated animal bones from Andhra Pradesh and nearby regions show morphological changes, proof of long-term human intervention.

3. Cultural embedding: In Andhra Pradesh, animals are part of religious ceremonies and social status, sustaining rearing traditions.

Process of Domestication



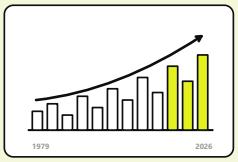
1. From wild to domestic: Suitable wild animals are selected, controlled breeding produces genetic adaptations, and the animals become part of human livelihoods.

Present Status

1. **Livelihoods:** Livestock gives income and security to small farmers in Andhra Pradesh, with women central to household animal care.
2. **Challenges:** Declining indigenous cattle, mechanisation and chemical fertilisers replacing dung reduce traditional benefits, especially in rural Andhra Pradesh.
3. **Data and demand:** The 21st Livestock Census and eLISS guide breeding, while rising demand for milk and eggs, with India the largest milk producer, calls for better practice.

Conclusion

Domestication has given humans food, labour and cultural value. Indigenous breeding, data-driven management and support for women rearers can sustain these benefits.



Which sub-topic will they ask next?

Trend analysis

Every syllabus topic: the direction of travel, the most tested sub-topic, the probable next theme.

Prepare the pattern →

ghissu.com

14 days free · no card

SCAN

P-IV
8(a)

What is meant by genetically modified crops ? Write their role in agriculture with a note on the legitimate concerns about transgenic agriculture.

10

Introduction

Genetically modified (GM) crops have become a pivotal innovation as India and Andhra Pradesh face climate change, water scarcity and food insecurity. Recent approval of climate-smart varieties like SAATVIK chickpea and genome-edited rice lines makes them more relevant than ever.

Meaning of GM Crops

1. **Definition:** GM crops are plants into which specific genes are introduced to enhance traits such as pest resistance and drought tolerance, raising productivity and sustainability.
2. **Regulation:** India regulates them under the Environment Protection Act, 1986, with DBT guidelines since 1989 requiring case-by-case risk assessment.

Role in Agriculture

1. **Productivity and Sustainability:** Bt cotton has raised yields and reduced pesticide use, benefiting farmers in Rayalaseema and other regions of Andhra Pradesh.

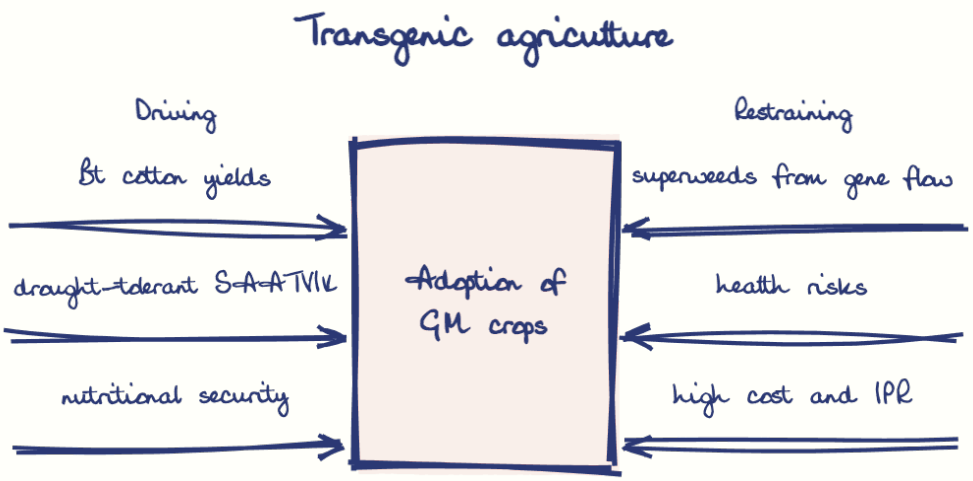
2. Climate Resilience: Drought-tolerant SAATVIK chickpea and genome-edited rice address water scarcity, crucial for Andhra Pradesh's agrarian economy.
3. Nutritional Security: Improved varieties support nutrition alongside food supply.
4. Local Research: Agricultural universities in Andhra Pradesh develop and test crops suited to local conditions.

Legitimate Concerns

1. Gene Flow: Transfer of genes to wild relatives may create superweeds.
2. Health Risks: Potential allergenicity and antibiotic resistance transfer need long-term monitoring.
3. Socio-Economic Barriers: High initial investment, intellectual property restrictions and limited awareness can widen rural disparities.
4. Farmers' Rights and Biodiversity: The Protection of Plant Varieties and Farmers' Rights Act, (2001) and the National Biodiversity Act, (2002) seek to balance innovation with these rights.

Way Forward

1. State R&D: Fund agricultural biotechnology in



Andhra Pradesh to develop locally adapted GM crops with farmer participation.

2. Farmer Training: Extension programmes on benefits and safe use, blending traditional knowledge with modern biotechnology.
3. Biosafety Monitoring: Robust, transparent and continuous biosafety monitoring at the state level.
4. Inclusive Policy: Balance intellectual property rights with farmers' access and biodiversity conservation.

Conclusion

Governed by stringent regulation and inclusive policy, transgenic agriculture can advance SDG (2): Zero Hunger and equitable growth.

Improving — or just writing more?

My answers

Every attempt, every score, every enhanced version of your answer — on one timeline.

Watch your curve →

ghissu.com

14 days free · no card

SCAN

P-IV
8(b)

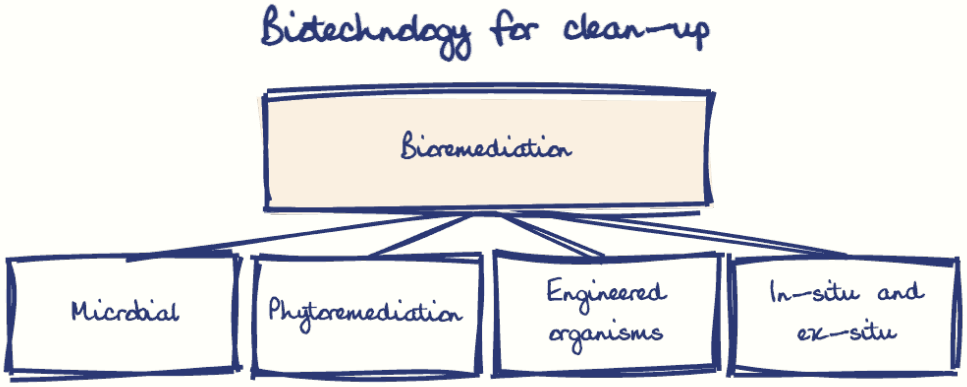
Write about the role of biotechnology in environmental clean-up processes.

10

Introduction

Biotechnology has become a transformative tool against pollution and waste. As industrialisation and urbanisation intensify environmental degradation in India, biological solutions are crucial for sustainable development. The BioE3 Policy and Bio-RIDE scheme show the government's commitment to them.

Biotechnology in Clean-Up Processes



1. Microbial Bioremediation: Bacteria and fungi degrade pollutants into less harmful substances in soil and water. TERI's bacterial mixture Oilzapper remediates oil-contaminated sites without toxic residues.
2. Phytoremediation: Plants absorb, transform or stabilise contaminants through phytoextraction and phytostabilisation, useful against heavy metals in

industrial areas.

3. Genetic Engineering: Recombinant DNA creates organisms that target specific pollutants, and bacteria that glow in response to toxins act as bioindicators for real-time monitoring.

4. In-situ and Ex-situ Methods: Biopiles, bioreactors and landfarming optimise microbial activity to speed up degradation.

Applications Across Sectors

1. Waste Management: Anaerobic digestion turns biodegradable waste into biogas and manure; portable biogas plants convert food waste into energy.

2. Water and Soil: Bioremediation treats industrial effluents, and in AP's coastal and aquaculture zones it maintains water quality for livelihoods.

3. Plastic Recycling: Enzymatic and microbial processes convert plastics into chemicals or fuels, supporting a circular economy.

4. Bio-Based Materials: Biodegradable polymers replace petroleum-derived products.

Challenges

1. Scalability: Site variability and slow degradation limit large clean-ups.

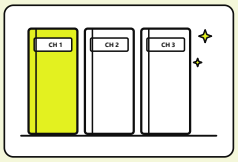
2. Biosafety: Releasing GMOs in the open needs robust regulation.
3. Costs: High initial costs, weak infrastructure and little private participation hinder use in rural AP.
4. Opportunities: Bio-RIDE and BioE3 foster R&D, entrepreneurship and skilled jobs, speeding indigenous technology.

Way Forward

1. State R&D: Strengthen AP's biotechnology institutions for locally adapted solutions.
2. Partnerships: Link government, academia and industry to scale clean-up and cut costs.
3. Biosafety Rules: Build clear regulation so that innovation does not compromise public health.

Conclusion

Biotechnology offers cost-effective, eco-friendly clean-up. With Bio-RIDE and BioE3 fostering R&D and skilled jobs, it can drive a sustainable bioeconomy.



Notes in five notebooks, none at hand?

Notes that live with the question

Note from toppers' answers, exam-hall thinking and model answers — your notes gather into topic and chapter notes on their own.

Build notes that find you →

ghissu.com

14 days free · no card

SCAN

P-IV
9(a)

Write about the causative agent for diarrhoea, its symptoms, modes of infection and the required preventive measures by the government during the outbreak.

10

Introduction

Diarrhoeal diseases remain a major public health challenge in India, especially among children under five, contributing significantly to illness and death. Persistent outbreaks, including in Andhra Pradesh, show the need to understand their causes, spread and control.

Causative Agents

1. Pathogens: Diarrhoea is caused by bacterial, viral and parasitic organisms. Vibrio cholerae causes cholera, while rotavirus, Escherichia coli and Entamoeba histolytica, the cause of waterborne amoebiasis, are other agents.

Symptoms

1. Clinical signs: Frequent loose or watery stools with dehydration, abdominal cramps, nausea and vomiting. Severe cholera causes rapid fluid loss, kidney failure and death if untreated.

Modes of Infection

1. Contaminated water and food: Ingestion is the main route, aided by polluted rivers, ponds and

Practice 9(a) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

inadequate sanitation.

2. Social factors: Overcrowding at mass gatherings like Kumbha Mela and poor hygiene in slums raise risk.

3. Flies and pests: They contaminate food and water sources.

Government Measures During an Outbreak

1. Surveillance: The Integrated Disease Surveillance Programme enables early detection, and Andhra Pradesh coordinates with central authorities, confirming diagnosis by bacteriological stool tests.

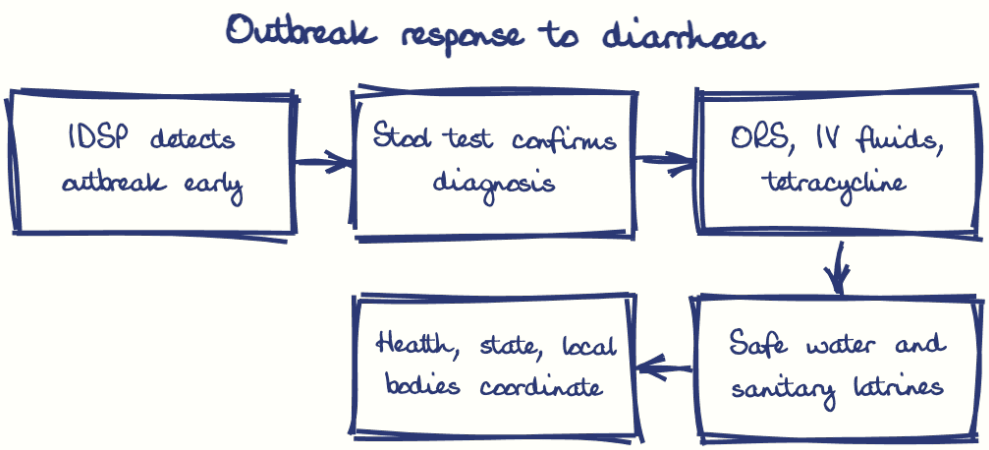
2. Treatment: Health facilities give oral rehydration salts, or IV fluids in severe cases, and tetracycline for cholera, backed by Andhra Pradesh's district-level preparedness.

3. Water and sanitation: Sanitary latrines and safe drinking water under Swachh Bharat Mission Gramin cut spread.

4. Coordination: The health ministry, state departments and local bodies act together, with the State Disaster Response Fund mobilising resources.

Challenges

1. Infrastructure gaps: Informal settlements and



unauthorised colonies in Andhra Pradesh lack piped water and sewers.

2. Behaviour and season: Low hygiene awareness and open defecation persist, and monsoon flooding contaminates water in coastal and riverine districts of Andhra Pradesh, causing seasonal spikes.

Way Forward

1. Prevention: Expand water and sewage networks and run behaviour change campaigns on handwashing, food safety and latrine use.

Conclusion

Diarrhoea is preventable. Safe water, sanitation, awareness and rapid response can contain outbreaks and protect children.

Wrote the answer. Now what?
Upload it — every line marked, in minutes.
ghissu.com →

P-IV
9(b)

Write about different types of tuberculosis, modes of infection, symptoms, common diagnostic tests and causes for its prevalence in India.

10

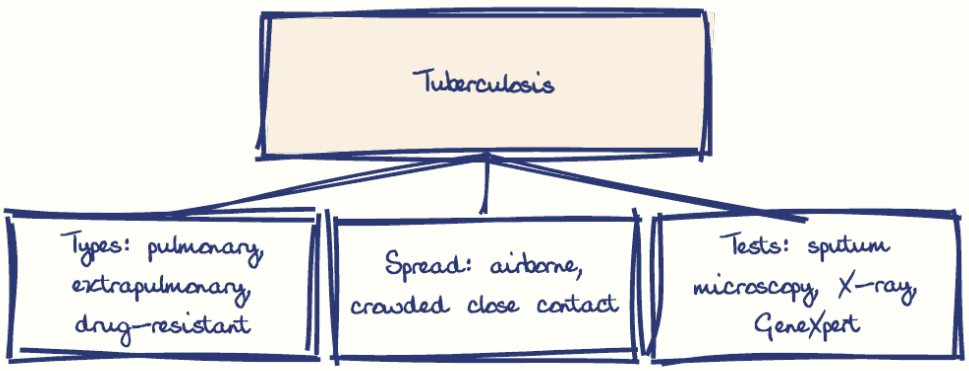
Introduction

Tuberculosis (TB) remains a critical public health challenge in India, which accounts for about 28 percent of the global TB burden. Despite progress under the National Tuberculosis Elimination Programme (NTEP) and the target to end TB by 2025, it still causes heavy illness and death.

Types of TB

1. Pulmonary and extrapulmonary: Pulmonary TB affects the lungs, while extrapulmonary TB involves lymph nodes, bones or the central nervous system.
2. Drug-resistant TB: MDR-TB and XDR-TB complicate treatment.

Tuberculosis Overview



Mode of Infection

1. Airborne transmission: Mycobacterium tuberculosis spreads when people inhale droplets from an infected person's cough or sneeze, especially in crowded, poorly ventilated settings.

Symptoms

1. Common signs: Cough beyond two weeks, chest pain, haemoptysis, fever, night sweats, weight loss and fatigue, extrapulmonary symptoms depend on the organ.

Diagnostic Tests

1. Frontline tests: Sputum smear microscopy and chest X-ray.
2. Molecular tests: GeneXpert rapidly detects TB DNA and rifampicin resistance.

Causes of Prevalence in India

1. Socio-economic factors: Poverty, malnutrition, overcrowding, diabetes and HIV impair immunity.
2. Private sector: Over 70 percent of care is private, often lacking standard protocols, causing underreporting and broken treatment in rural areas such as Andhra Pradesh.
3. Treatment gaps: Stigma, low awareness and

delays fuel spread and drug resistance.

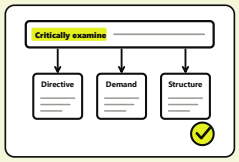
4. COVID-19: The pandemic disrupted services, though the TB Mukt Bharat Abhiyaan and the 100-Day TB Campaign revived case detection, diagnosing over 28 lakh patients including in Andhra Pradesh.

Way Forward

1. Integrated care: Partner with private providers in Andhra Pradesh, use active case finding under the National Strategic Plan 2017-2025, and add nutritional support.

Conclusion

Free diagnosis, the Nikshay portal, nutritional support and stronger private partnership can bring India closer to TB elimination.



Stuck at the directive?

Exam-hall thinking

The directive, the demand, the structure — before your first line. Full for beginners, Crux for veterans.

Walk in with a method →

ghissu.com

14 days free · no card

SCAN

P-IV
10(a)

What are vaccines ? Write the fundamental concept of vaccination and list some of the compulsory vaccines for children.

10

Introduction

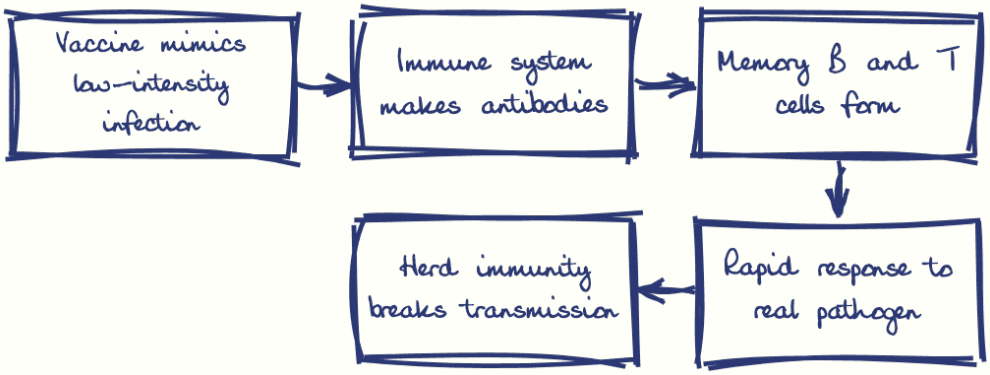
Vaccines are a cornerstone of public health, preventing infectious diseases that cause widespread illness and death. In India, vaccination has sharply reduced polio and measles. Its core idea is to train the immune system to recognise and fight pathogens without causing the disease.

What Vaccines Are and How They Work

1. Definition: Vaccines are biological preparations that mimic a low-intensity infection by introducing antigenic components of pathogens, stimulating immunity without illness.
2. Immune memory: Vaccination induces antibodies and memory B and T cells, so the body responds rapidly and effectively to the real pathogen later.
3. Active and passive immunity: Vaccines mainly induce active immunity through the body's own response, while passive immunity is a direct transfer of antibodies, as through maternal colostrum.
4. Herd immunity: Mass vaccination breaks the chain of transmission, protecting even those who

cannot be vaccinated.

How a vaccine protects



Compulsory Vaccines for Children

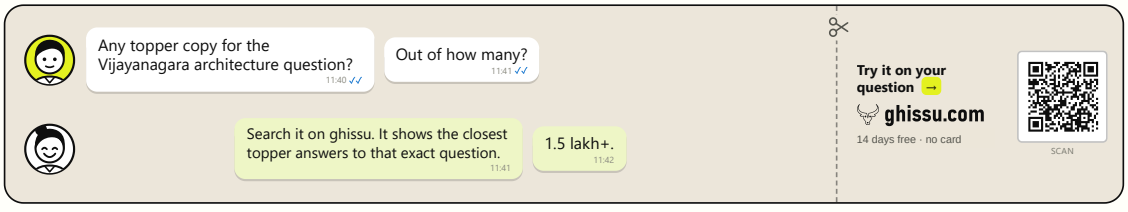
1. **Universal Immunisation Programme:** The UIP provides free vaccines against (12) diseases to children and pregnant women across India, including Andhra Pradesh, which has actively run UIP campaigns.
2. **Key vaccines:** These include BCG (tuberculosis), DPT (diphtheria, pertussis, tetanus), polio, measles, rubella and hepatitis B.
3. **Newer additions:** Vaccines against meningitis, rotavirus diarrhoea and pneumococcal pneumonia widen protection.
4. **HPV vaccine:** The HPV vaccine for adolescent girls aims to prevent cervical cancer and reduce cancer mortality.

Challenges

1. Coverage gaps: Despite progress, coverage varies across states and districts, with areas like Rayalaseema and tribal regions of Andhra Pradesh lagging behind.
2. Vaccine hesitancy: Misinformation and social resistance, seen in HPV drives, need community engagement and trust-building.
3. Infrastructure: Cold chain logistics, health worker training and public health infrastructure in Andhra Pradesh must be strengthened to sustain success.
4. Safety monitoring: Continuous monitoring of vaccine safety and pathogen evolution maintains public confidence and lets strategies adapt.

Conclusion

Vaccination is a scientifically proven, cost-effective intervention. Strengthening coverage, infrastructure and public trust will protect children and support Sustainable Development Goal (3) on good health and well-being.



P-IV
10(b)

Write about the different types of modern vaccines available today. How are the modern vaccines better than the conventional vaccines ?

10

Introduction

The rapid development of vaccines during COVID-19 highlighted the transformative potential of modern vaccine technologies. Compared with conventional vaccines, they offer faster development, adaptability and better efficacy, making them vital for universal immunisation and pandemic preparedness.

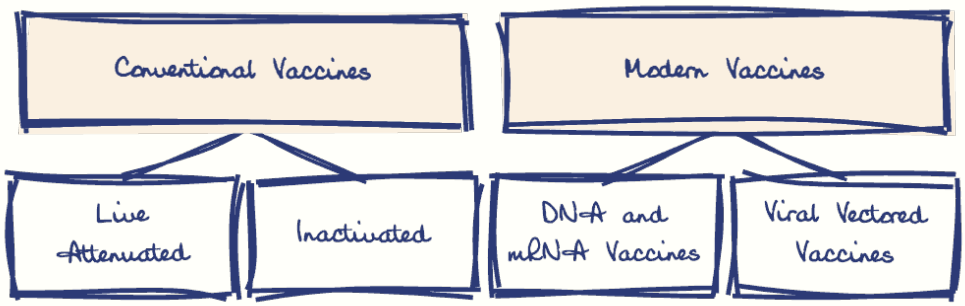
Types of Modern Vaccines

1. DNA vaccines: Genetic material instructs human cells to produce pathogen proteins, as in Zydus Cadila's indigenous COVID-19 vaccine.
2. mRNA vaccines: mRNA vaccines showed high efficacy against COVID-19, aided by extensive public and charitable funding.
3. Viral vectored vaccines: A viral vector delivers the genetic instructions that stimulate immunity.
4. Contrast with conventional: Conventional vaccines rely on weakened or inactivated whole pathogens, while these platforms use genetic instructions and can be modified quickly.

Practice 10(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

Vaccine Technology Types



Why Modern Vaccines are Better

1. **Speed and adaptability:** They can be developed and modified quickly as pathogens mutate, as with COVID-19 variants, unlike conventional vaccines with longer production cycles.
2. **Targeted immune response:** By instructing cells to make specific antigens, they induce strong cellular and humoral immunity, with fewer side effects than whole-pathogen vaccines.
3. **Scalability:** Platform technologies need no live pathogen culture, enabling faster scale-up in pandemics, critical for populous states like Andhra Pradesh.
4. **Wider scope:** They pave the way for universal and therapeutic vaccines, such as against cancer.

Limitations

1. **Cold chain:** mRNA vaccines need demanding

storage, while Covaxin is easier to deploy across Andhra Pradesh's diverse regions.

2. Surveillance: Long-term safety and new variants need robust post-approval monitoring, a challenge for India's fragmented system.
3. Hesitancy: Gendered mistrust and misinformation slow HPV vaccine uptake in rural India, including Andhra Pradesh, despite campaigns for 11.5 million adolescent girls.
4. R&D gap: India's low R&D intensity favours adaptive improvements over breakthroughs.

Conclusion

Modern vaccines shift design from whole pathogens to genetic platforms. Indigenous R&D and stronger surveillance, in line with Aatmanirbhar Bharat, will help India use them fully.

Same question. Three word limits.

500 words — context · three dimensions · two examples each · a diagram · a conclusion

250 words — one-line context · three dimensions · one example each · a conclusion

150 words — a definition · two dimensions · one line each · a one-line verdict

See every model answer at 3 lengths →

ghissu.com

14 days free · no card

SCAN

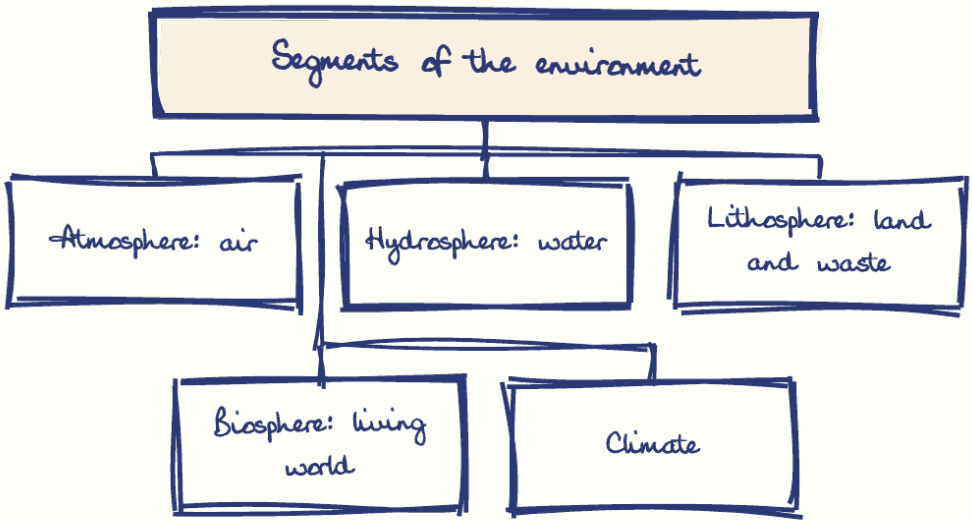
P-IV
11(a)

Write in detail about various segments of environment.

10

Introduction

The environment is the complex interplay of natural and human-made elements that sustain life on Earth. With climate change, biodiversity loss and pollution escalating, understanding its segments is crucial for conservation. In Andhra Pradesh, with rich biodiversity, a long coastline and rapid urbanisation, this is vital for livelihoods and ecological balance.



Segments of the Environment

1. Air (atmosphere): Industrial growth and vehicular emissions in cities like Visakhapatnam deteriorate air quality.

2. Water (hydrosphere): Agricultural runoff and untreated sewage threaten aquatic ecosystems and public health.

3. Land and waste: AP's (123) ULBs generate about (6,890) tons of solid waste a day, yet in (2022) only (4) had sanitary landfills and (82) lacked wet waste plants.

4. Living world (biosphere): AP's flora and fauna face habitat fragmentation and invasive species, needing support from Project Tiger and MISHTI.

5. Climate: Coastal and drought-prone regions face sea-level rise and erratic rainfall, hurting farming and water security.

Framework Protecting the Segments

1. Legislation: The Environment Protection Act (1986), Forest Conservation Act (1980) and Wildlife Protection Act (1972) cover air, water, forests and biodiversity.

2. Authorities: CPCB and the National Green Tribunal act nationally, and the AP Pollution Control Board enforces compliance.

3. EIA: Projects such as infrastructure and aquaculture must assess ecological impact before approval.

4. Public participation: Mission LiFE and the Green Credit Programme reward sustainable practices.

Way Forward

1. Local bodies: Give ULBs technical and financial resources to segregate and process waste under the Solid Waste Management Rules.
2. Monitoring: Real-time air and water monitoring with GIS mapping in industrial corridors and fragile ecosystems.
3. Community action: Programmes like Plant4Mother and Ek Ped Maa Ke Naam foster stewardship of forests and biodiversity.
4. Green industry: Cleaner technology through ADEETIE and FAME-II to cut emissions.

Conclusion

The segments of the environment are interlinked, so damage to one spreads to others. Coordinated policy and public participation can protect them together.

Correct, but average? One line from your answer

YOUR LINE
~~The 1857 revolt failed.~~
✕ true · forgettable

→

WITH A VALUE ADDITION
The 1857 revolt failed for want of a centre: no common plan, no modern arms, and the loyalty of the Sikhs, the Gurkhas and most princes to the Company.
✓ cause · evidence · specifics

ILLUSTRATIVE

Add value to your next answer →
ghissu.com
14 days free · no card
SCAN

P-IV
11(b)

(i) Write in detail about the Water (Prevention and Control of Pollution) Act. (ii) What are the issues involved in the enforcement of environmental legislation and law ?

10

Introduction

Water pollution remains a critical challenge to public health, biodiversity and sustainable development in India. The Water (Prevention and Control of Pollution) Act, 1974 established regulatory bodies and standards to control it. Yet enforcement gaps persist, and the 2024 amendment shows a shift towards balancing regulation with economic realities.

(i) The Water Act

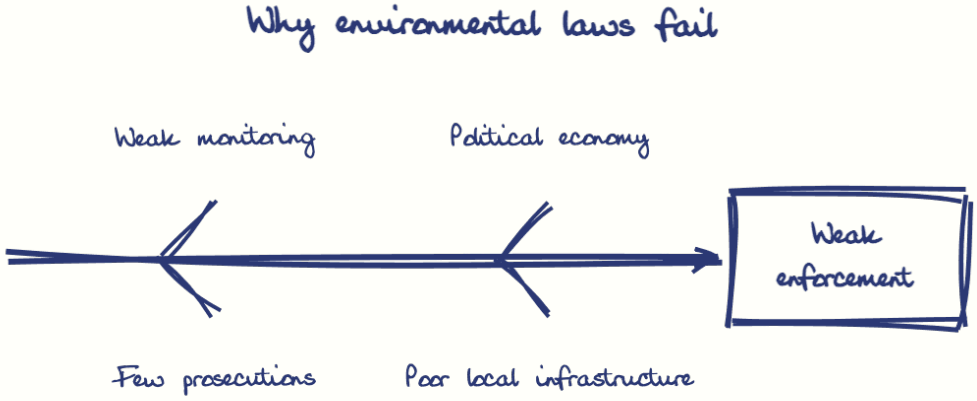
1. **Pollution Control Boards:** The Act created the CPCB and the State Pollution Control Boards as statutory authorities, the CPCB sets standards and SPCBs enforce them.
2. **Consent mechanism:** Industries must get prior SPCB consent before establishment and operation, which is crucial to prevent untreated effluent discharge.
3. **Rationalised penalties:** The 2024 amendment decriminalises minor violations, replacing imprisonment with fines to reduce hardship on business while keeping safeguards.

Practice 11(b) on Ghissu →
Exam-hall thinking for this question · relevant toppers' answers · related questions from other Mains exams
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

4. Central exemptions: The amendment lets the Centre exempt certain industries from prior consent, streamlining approvals but raising fears of regulatory dilution.

(ii) Issues in Enforcing Environmental Law



1. Weak monitoring: Despite networks like the National Water Quality Monitoring Programme, data gaps and inconsistent surveillance delay action against polluters.
2. Few prosecutions: Very few complaints and prosecutions are recorded under the Act, reflecting enforcement inertia and weak SPCB capacity.
3. Political economy: Bureaucratic collusion, political interference and economic pressure help industries evade compliance, slowing progress in industrial clusters.
4. Local infrastructure: In Andhra Pradesh,

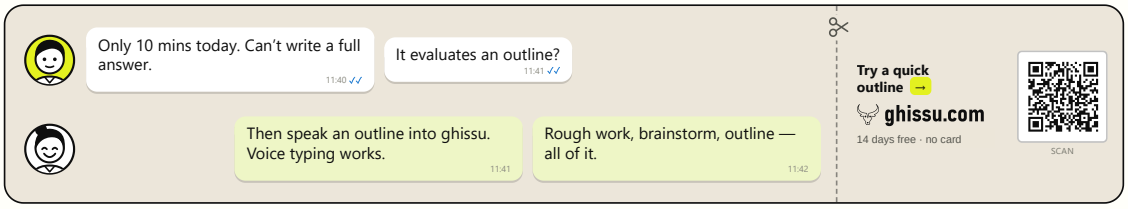
inadequate waste treatment infrastructure and few operational sanitary landfills, flagged by the 2022 CAG report on ULBs, hamper local pollution control.

Way Forward

1. Stronger SPCBs: More technical, financial and human resources for robust monitoring, enforcement and timely legal action against violators.
2. Technology: Real-time sensors and open data platforms under the National Hydrology Project.
3. Community engagement: Awareness campaigns and participatory monitoring create bottom-up pressure for compliance.
4. Judicial reform: Swift adjudication of pollution cases and clear penalty guidelines will deter violations.

Conclusion

The Act and its 2024 amendment form a vital framework, but fixing institutional weaknesses is essential to secure the right to a clean environment.



P-IV
12(a)

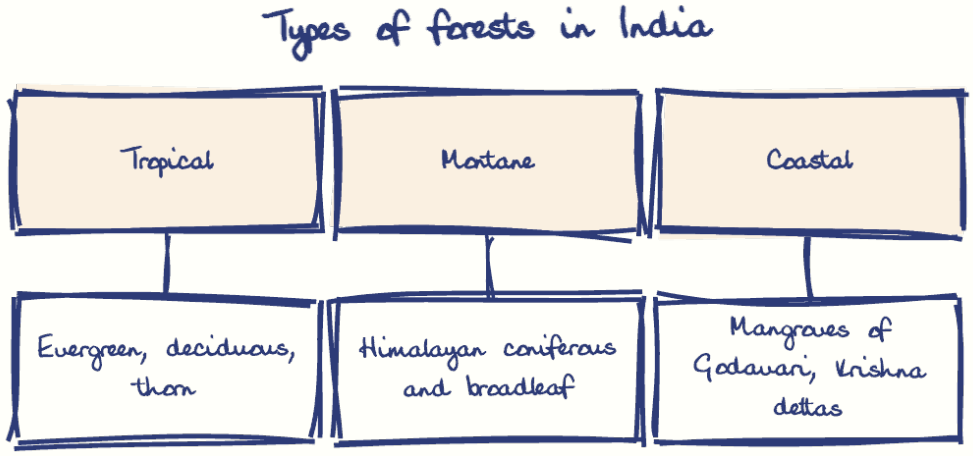
(i) Explain about the types of forests present in India. (ii) How are forests helping the human society ? Explain.

10

Introduction

India's forests cover about 25.17% of its geographical area, according to ISFR 2023. They sustain rich biodiversity and provide ecosystem services that support livelihoods and environmental stability, making them central to sustainable development.

(i) Types of Forests in India



1. **Tropical Evergreen:** Dense, multi-layered forests of heavy-rainfall areas like the Western Ghats and northeast, rich in endemic species.
2. **Tropical Deciduous:** Monsoon forests covering the largest area, including the Eastern Ghats of

Andhra Pradesh, they shed leaves seasonally and yield timber and minor forest produce.

3. Tropical Thorn: Drought-resistant forests of arid zones like Rayalaseema that conserve soil and give fodder and fuelwood.

4. Montane: Himalayan coniferous and broadleaf forests that vary with attitude, regulate water cycles and check erosion.

5. Mangroves: Coastal forests of the Godavari and Krishna deltas that protect shores from erosion and cyclones and support fisheries.

Forest types of Andhra Pradesh

Candidates must not write on this margin

Candidates must not write on this margin

(ii) How Forests Help Human Society

1. Ecological Services: Forests regulate rainfall and temperature and prevent soil erosion; the Eastern Ghats conserve soil and water for farming.
2. Livelihood Support: Millions depend on fuelwood, fodder, timber and medicinal plants; MGNREGS in Andhra Pradesh links conservation with rural jobs.
3. Biodiversity: The Nallamala forests sustain ecological balance and eco-tourism.
4. Climate Mitigation: Forests are carbon sinks, helped by Andhra Pradesh's afforestation and mangrove restoration under MISHTI.

Way Forward

1. Mangrove Conservation: Scale up MISHTI along Andhra Pradesh's coastline to protect biodiversity and livelihoods.

Conclusion

Scientific management, community participation and agroforestry can make India's forests a lasting base for ecology, livelihoods and climate resilience, in line with SDG (15.)

Which sub-topic comes next?
Trend analysis for every syllabus topic.
ghissu.com →

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 12(a) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
12(b)

Explain about the mineral resources present in India.

10

Introduction

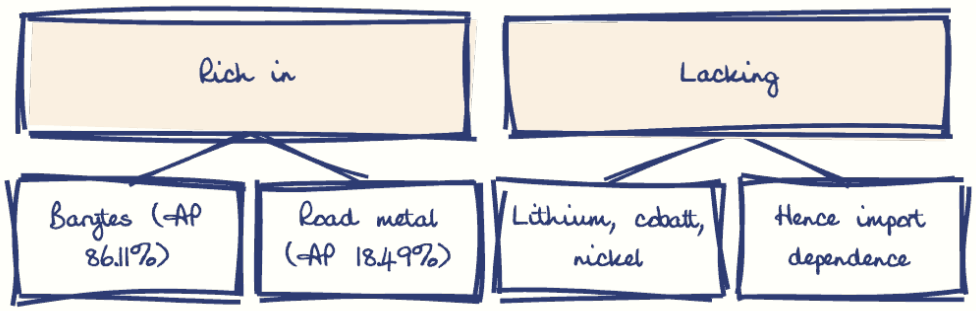
India's mineral resources are a cornerstone of economic development, underpinning sectors from steel to the energy transition. A vast, geologically diverse landmass gives India a rich variety of minerals, unevenly distributed. Rising demand from urbanisation, infrastructure and clean energy goals makes managing this wealth crucial for strategic autonomy.

Mineral Resources and Their Governance

1. **Exploration agencies:** The Geological Survey of India, MECL and NMDC lead systematic surveys that identify and develop deposits.
2. **Regulation:** The MMDR Act, after recent amendments, includes (30) critical minerals, such as rare earths and energy transition metals, in Schedule I.
3. **State role:** Andhra Pradesh holds (86.11%) of India's barytes and (18.49%) of its road metal reserves.
4. **Public and private roles:** The private sector accounts for about (60%) of mineral production value, especially non-ferrous metals and

industrial minerals.

India's mineral position



Challenges in Development

1. Uneven distribution: Metallic minerals concentrate in the peninsular plateau and eastern belts, while lithium, cobalt and nickel are largely absent, causing high import dependence.
2. Social licence: Poor regulatory governance has allowed illegal mining and weakened the social licence to operate, causing conflicts with communities in tribal and ecologically sensitive regions.
3. Infrastructure and skills: Weak evacuation infrastructure and skill gaps limit the sector's GDP share.
4. State finances: Andhra Pradesh has shown mineral revenue shortfalls against budget estimates and gaps in remittances to the

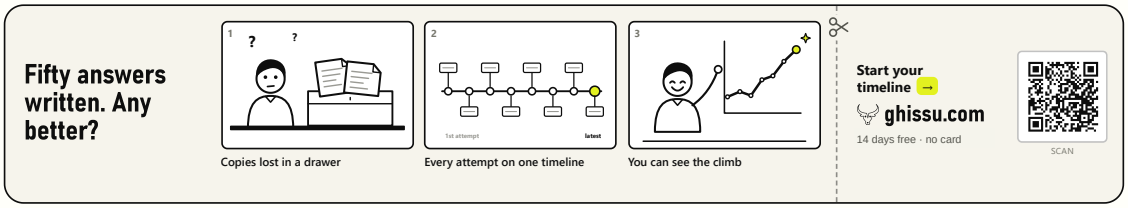
exploration trust fund.

Strategies

1. Explore in India: Encourage private participation and better geological data.
2. Sustainable mining: Free, prior and informed consent in tribal and sensitive areas.
3. Infrastructure and skills: Invest in transport and processing infrastructure, and align vocational training with mining needs.
4. Critical mineral security: The National Critical Mineral Mission secures supply chains for clean energy minerals and cuts import dependence.

Conclusion

India's mineral wealth is pivotal for its industrial and clean energy ambitions. Addressing governance, technological and social gaps through better exploration, stronger state management and sustainable practices can deliver resource security, inclusive growth and climate resilience.



P-IV
13(a)

(i) What do you understand by the term ecosystem ? (ii) Write the structure of ecosystem. (iii) List out different types of aquatic ecosystems existing in India. (iv) What are the threats to aquatic ecosystems and how can they be conserved ?

10

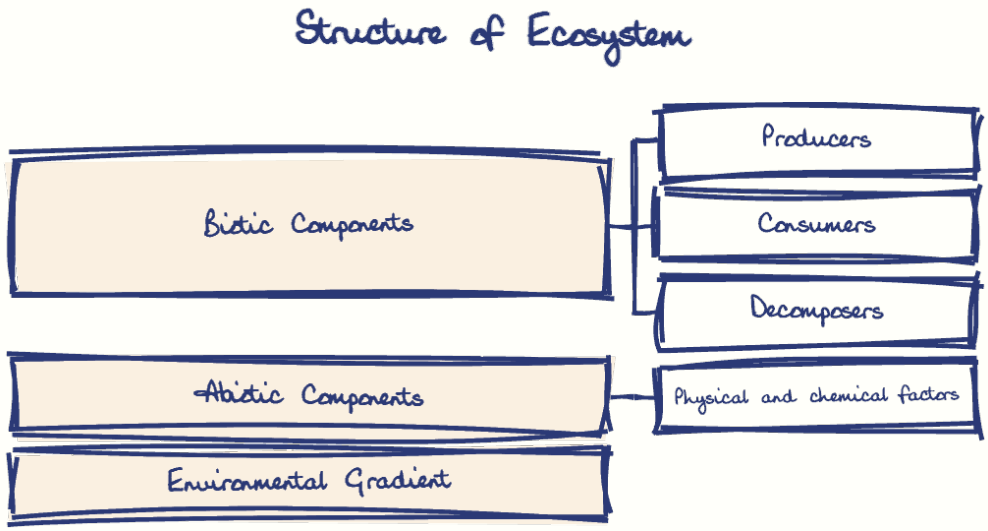
Introduction

Ecosystems are Earth's basic life-support systems, linking living organisms with their physical environment. In diverse India, aquatic ecosystems support livelihoods, maintain ecological balance and sustain the growing Blue Economy.

(i) Meaning of Ecosystem

An ecosystem is a dynamic complex of plants, animals and microbes interacting with each other and with air, water and soil as a functional unit.

(ii) Structure of Ecosystem



1. Biotic: Producers (plants, algae), consumers (herbivores, carnivores) and decomposers (bacteria, fungi).
2. Abiotic: Sunlight, temperature, water, minerals and nutrients.
3. Environmental Gradient: Ranges of temperature, light and moisture that shape the distribution of organisms.

(iii) Aquatic Ecosystems in India

1. Freshwater: Salt below (5) ppt; rivers, lakes, ponds and springs, as in the Krishna and Godavari basins.
2. Marine: Salt of (35) ppt or more, shallow seas and open ocean along a (974 km) coastline, like the Visakhapatnam coast.
3. Brackish Water: (5) to (35) ppt; estuaries, salt marshes and mangroves such as the Godavari delta.
4. Coral Reefs: Gulf of Mannar and Andaman and Nicobar Islands.

(iv) Threats and Conservation

1. Pollution: Effluents, farm runoff and sewage degrade water.

Practice 13(a) on Ghissu →
Andhra Pradesh and national value addition · quick outline practice · friends' and public answers
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

2. Habitat Loss and Overexploitation: Encroachment, unsustainable aquaculture and overfishing destroy wetlands and reduce biodiversity.

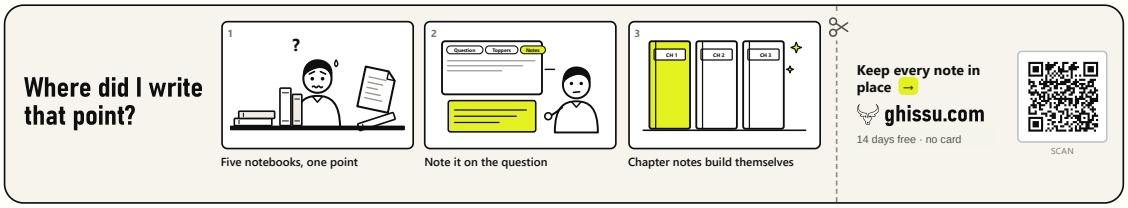
3. Climate Change: Warming and sea-level rise alter habitats.

4. Conservation: The NPCA and Wetlands Rules 2017 protect habitats, CRZ rules and the Ramsar Convention add safeguards, and BMCs involve communities.

5. Restoration: Mangrove afforestation in the Godavari-Krishna delta builds resilience.

Conclusion

Rising mangrove cover and more Ramsar sites show progress. Better funding, coordination and awareness will secure aquatic ecosystems.



P-IV
13(b)

(i) What are the threats to Indian biodiversity ? (ii) How can Indian biodiversity be conserved ?

10

Introduction

India is one of the world's mega-diverse countries, hosting about 4–5% of all known species due to its varied ecosystems. Rapid industrialisation, urbanisation and unsustainable resource use have intensified threats to it. Conservation is critical because biodiversity sustains ecological balance, livelihoods and cultural heritage, especially for forest-dependent communities.

(i) Threats to Indian Biodiversity

1. Habitat destruction and fragmentation: Agriculture, infrastructure, mining and settlements cause large-scale habitat loss; encroachment in hotspots like the Western Ghats has reduced forest cover.

2. Overexploitation and poaching: Commercial hunting and illegal trade, driven by demand for animal parts and traditional medicines, threaten tigers, elephants and medicinal plants.

3. Pollution and degradation: Polluted water, degraded soil and forest fires reduce species viability. India has lost over half its natural forests and (40%) of mangroves, and (70%) of surface water bodies are

polluted.

4. Climate change and invasives: Changing climate shifts species distribution while invasives outcompete natives, leaving the Himalayan and Indo-Burma hotspots especially vulnerable.

(ii) Conserving Indian Biodiversity

Threats and their remedies

Threat	Conservation response
Habitat destruction	In situ protected areas
Poaching	Targeted species projects
Pollution and degradation	Community governance
Loss of rare species	Ex situ conservation

1. In situ conservation: (14) biosphere reserves, (90) national parks and over (450) wildlife sanctuaries protect species in natural habitats; the Nagarjuna Sagar-Srisailam Tiger Reserve in Andhra Pradesh safeguards tigers and forests.
2. Ex situ conservation: Zoos, botanical gardens, seed banks and tissue culture preserve endangered species outside their habitats, enabling breeding and genetic conservation.

3. Community governance: The three-tier model with over 2,76,653 Biodiversity Management Committees and People's Biodiversity Registers empowers locals; sacred groves and wetland efforts in Andhra Pradesh show results.

4. Targeted species projects: Project Tiger, Project Elephant and Project Snow Leopard conserve keystone species; tigers rose from 2,226 in 2014 to 3,682.

Way Forward

1. Habitat connectivity: Enhance ecological corridors between protected areas, especially in the Eastern Ghats.

2. Sustainable livelihoods: Support forest communities through eco-tourism and sustainable harvesting.

3. Technology: Use GIS, remote sensing and AI-based tools for real-time monitoring.

4. Enforcement and awareness: Strengthen wildlife laws and public awareness.

Conclusion

Indian biodiversity faces threats from habitat loss, exploitation, pollution and climate change. A comprehensive approach combining in situ and ex situ methods, community involvement and technology will sustain it, in line with SDG 15.

Write your own answer to 13(b) and get it evaluated →
Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed
ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
14(a)

As an individual what is your role in the prevention of environmental pollution ? Explain.

10

Introduction

Environmental pollution threatens public health, biodiversity and sustainable development, especially in fast-urbanising regions like Andhra Pradesh. Individual actions, added together, can sharply cut emissions, waste and resource misuse. Since pollution prevention begins at home, each person complements government efforts.

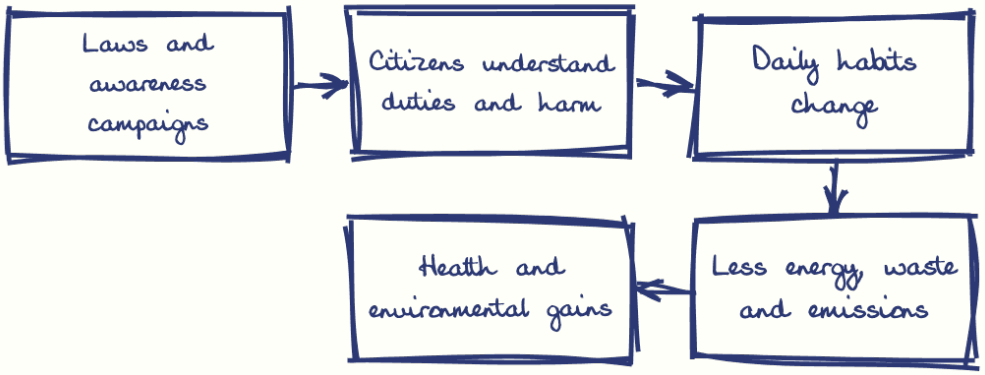
Basis of Individual Responsibility

1. Constitutional duty: Every citizen must protect and improve the natural environment, including forests, rivers and wildlife.
2. Environmental laws: Under the Water Act, 1974 and Air Act, 1981, State Pollution Control Boards, including AP's, regulate pollution, and citizens help by complying and reporting violations.
3. Local governance: AP's Urban Local Bodies run waste drives where citizen vigilance can fill gaps that audits reveal.
4. Awareness: Campaigns by government and NGOs enable informed choices.

My Role as an Individual

1. **Save energy:** Switching off unused lights and appliances reduces demand on power plants, directly cutting fuel use and the pollution it causes.
2. **Choose clean transport:** Walking, cycling, carpooling and public transport lower emissions in cities like Visakhapatnam and Amaravati.
3. **Segregate waste:** Segregation compliance in AP is lower than claimed, so I must sort waste at home and prefer reusable products.
4. **Join conservation drives:** Tree planting and water conservation restore ecological balance and soften local pollution.
5. **Use water wisely:** Efficient use at home supports community programmes.

How individual action cuts pollution



Candidates must not write on this margin

Candidates must not write on this margin

How Individual Action Works

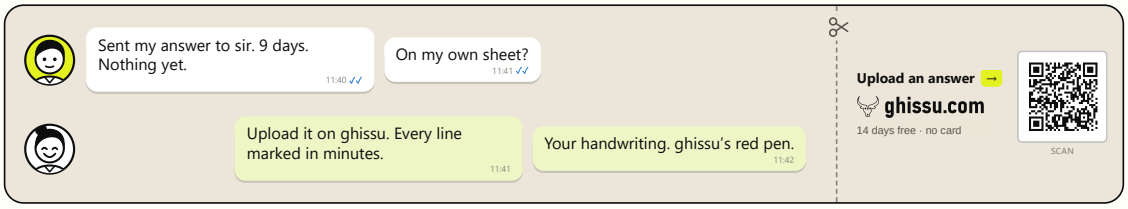
The chain runs from law and awareness to behaviour change: once citizens understand their duties and the harm pollution causes, they change daily habits, which reduces pollution and brings health and environmental benefits.

Why Individual Action Matters

1. Collective impact: Millions of small changes in energy, transport and waste habits add up to large reductions at the grassroots.
2. Accountability: Informed and vigilant citizens push Pollution Control Boards and local bodies to enforce laws.

Conclusion

Individual responsibility is indispensable for public health and sustainable development. By conserving resources, cutting waste and supporting pollution control laws, citizens help achieve cleaner air and water, advance SDG 13 and SDG 15, and help Andhra Pradesh meet its environmental goals.



How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 14(a) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

P-IV
14(b)

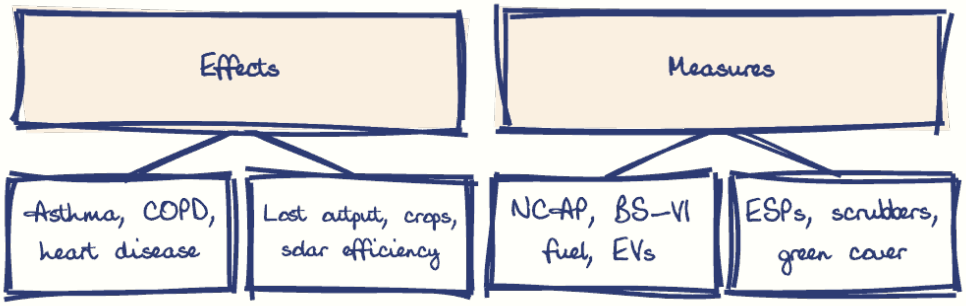
(i) Explain the effects of air pollution. (ii) Suggest various measures to control air pollution.

10

Introduction

Air pollution has become a critical environmental and public health challenge in India, which accounts for a large share of global pollution-related deaths, many in urban centres. Andhra Pradesh, with growing urbanisation and industrial corridors like Visakhapatnam, reflects these trends and needs targeted action.

Air pollution: effects and control



(i) Effects of Air Pollution

1. **Health Burden:** It causes asthma, COPD and cardiovascular disease, in India it caused about 2.1 million deaths in 2021, with people in industrial areas like Visakhapatnam most at risk.

2. **Economic Impact:** Pollution-related illness lowers

worker productivity and raises healthcare costs; in Andhra Pradesh it can cut labour in agriculture and aquaculture and raise public health spending.

3. Environmental Degradation: Acid rain damages ecosystems, and impaired photosynthesis cuts crop yields, worsening vulnerability in drought-prone Rayalaseema.

4. Renewable Energy Loss: Pollutants block sunlight and raise ambient temperatures, cutting solar panel efficiency. India could lose significant solar generation by 2050, threatening energy transition goals.

(ii) Measures to Control Air Pollution

1. National Clean Air Programme: NCAP aims to cut particulate matter by 40% by 2025-26 through city action plans; Vijayawada and Tirupati target vehicle and industrial pollution.

2. Vehicular Controls: BS-VI fuel standards since 2020 and FAME-II, while Andhra Pradesh expands EV infrastructure and compressed biogas under SATAT.

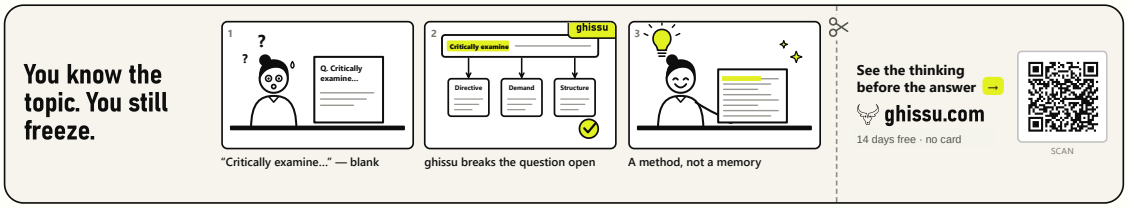
3. Industrial Controls: Electrostatic precipitators and scrubbers in factories, and zoning that separates industry from homes, as in Visakhapatnam's

steel and chemical plants.

4. Public Awareness: Education and community monitoring foster accountability, schools in Amaravati now teach air quality, and NGOs work alongside government.
5. Green Cover: Afforestation and wetland restoration in urban and rural areas act as natural air filters that reduce particulate matter.
6. Inter-Agency Coordination: State, district and municipal authorities must implement air quality plans together rather than in isolation.
7. Technology and Clean Energy: Real-time sensors and analytics enable timely advisories, while subsidies support renewable energy and clean fuels in transport and industry.

Conclusion

Air pollution harms health, the economy and the environment. Strong NCAP implementation, technology and public participation can deliver cleaner air and sustainable growth in line with the SDGs.



Write your own answer to 14(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

P-IV
15(a)

Write about the following : (i) Climate change (ii) Acid rain

10

Introduction

Climate change and acid rain are critical environmental challenges that threaten ecological balance, human health and economic development. Rising greenhouse gas emissions and industrial pollutants have intensified both, including in Andhra Pradesh, where coastal vulnerability and industrial growth intersect.

(i) Climate Change

1. Causes: Rapid urban growth in Amaravati and Visakhapatnam raises energy demand and vehicular emissions, adding to greenhouse gases.
2. Impacts: Extreme weather events such as Cyclone Titli have become more frequent, causing socio-economic distress.
3. Framework: The N-APCC (2008), with the National Solar Mission and National Mission on Sustainable Habitat, guides mitigation, including in AP's cities.
4. Monitoring: ISRO's Sriharikota centre provides satellite data on climate variables.
5. Urban resilience: The Climate-Smart Cities

Assessment Framework in Visakhapatnam and Vijayawada promotes energy efficiency.

(ii) Acid Rain

1. Causes: SO_2 and NO_x from fossil fuels, including thermal plants near the Krishna and Godavari basins, form acid rain.
2. Impacts: It lowers soil pH and harms agriculture and aquatic ecosystems, and with them the biodiversity and fisheries that support livelihoods.
3. Regulation: The Air Act, 1981 and Environment Protection Act, 1986 empower agencies; the APPCB monitors emissions in the Visakhapatnam port area.
4. Gaps: Weak enforcement, low awareness and poor centre–state coordination.

Climate change vs acid rain

Climate change	Acid rain
Greenhouse gases	SO_2 and NO_x
Extreme weather, Cyclone Titli	Lower soil pH
NAPCC (2008)	Air Act, 1981

Common Solutions

1. Clean energy: AP's solar parks and wind farms cut coal use.
2. Afforestation: The Green India Mission in the Eastern Ghats aids carbon sequestration and buffers soil and water.
3. Stricter norms: Real-time data and stricter penalties for SO_2 and NO_x violations in industrial zones.
4. Community engagement: Educating people to reduce fossil fuel use and protect water bodies from acidification.

Conclusion

Climate change and acid rain share a root in fossil fuel emissions, so they share a cure. Clean energy, green cover, strict norms and community engagement can protect Andhra Pradesh's ecosystems and build climate resilience.

1.5 lakh+

Borrow the approach, not the answer.

Topper answers matched by meaning to your question — with how each one opened, structured and concluded.

See how toppers wrote it → ghissu.com

14 days free · no card

SCAN

P-IV
15(b)

(i) What is the need for the development of wastelands ? (ii) Write about reclamation of wastelands.

10

Introduction

With rising pressure on land from population growth, urbanisation and climate change, wasteland development has become an environmental and economic priority. Wastelands are underused lands of degraded productivity; the Department of Land Resources estimated about 16.96% of India's land as wasteland in 2015–16. In Andhra Pradesh, reclaiming them can raise land availability, especially in drought-prone Rayalaseema.

(i) Need for Wasteland Development

1. **Land Availability:** Reclaimed land eases pressure from population growth and urbanisation on farmland.
2. **Ecological Balance:** Degraded land threatens ecological balance; restoration supports the state's commitment to land degradation neutrality.
3. **Rural Livelihoods:** Evidence from Andhra Pradesh shows agroforestry on degraded land improves soil health and provides livelihoods for small and marginal farmers.
4. **Climate Resilience:** Restored land increases carbon sequestration.

5. Urban Legacy Waste: Andhra Pradesh had 85.89 lakh metric tons of waste in dump yards in 2022, degrading land that must be remediated.

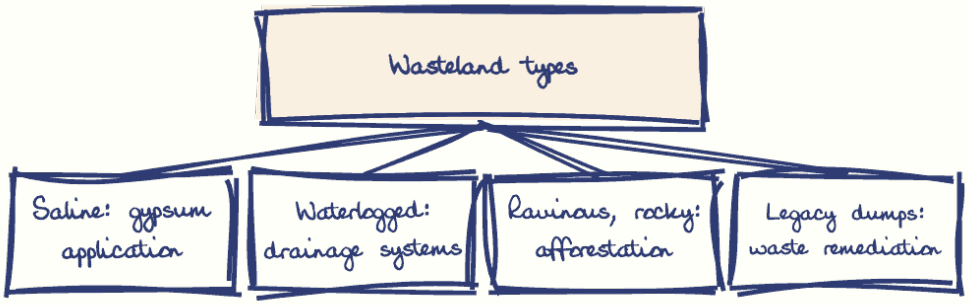
(ii) Reclamation of Wastelands

1. Tailored Techniques: Saline, waterlogged, ravinous or rocky lands need specific methods such as soil amendments like gypsum application, drainage systems and afforestation.
2. Agroforestry: The National Agroforestry Policy (2014) promotes tree-based use of wastelands for economic and ecological returns, which Andhra Pradesh uses to restore degraded land.
3. Scheme Convergence: MGNREGA and Andhra Pradesh's watershed programmes give financial and technical support.
4. Community-Led Management: Panchayats can give long-term leases to marginal farmers and SHGs, resolving complex common or absentee ownership.
5. Technology: Remote sensing and GIS identify and monitor wastelands precisely.
6. Integrated Waste and Land Strategy: Linking urban solid waste reform with rural reclamation clears legacy dumps and prevents new degradation.

Candidates must not write on this margin

Candidates must not write on this margin

Matching treatment to wasteland

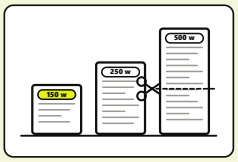


Challenges

1. Technical Complexity: Tailored methods need significant investment and expertise, limiting large-scale reclamation.
2. Social Barriers: Small and marginal farmers lack capital and technical knowledge, so subsidies and capacity building are needed for fair participation.

Conclusion

Wasteland reclamation restores ecology, raises productivity and supports rural livelihoods. With institutional reform, technology and community participation, it can drive sustainable development and climate resilience.



Writing 400 words for a 150-word question?

Model answers in three lengths

150, 250 and 500 words, in English and हिंदी. Learn what to cut before the examiner does.

Read one question three ways →

ghissu.com

14 days free · no card

SCAN

How to use this book with ghissu.com
see page 4

Index · page 6

Write your own answer to 15(b) and get it evaluated →

Detailed, examiner-style feedback marked on your handwritten sheet · first 14 days of Pro free, no card needed

ghissu.com · APPSC Group-I Mains 2008 · printed in old-syllabus Paper IV, now Paper V

Scan to practice

APPSC GROUP-I MAINS · PAPER V

Read an answer here. Write yours on ghissu.com.

This book collects 292 previous year questions of Paper V, Science, Technology and Environmental Issues, from eight sittings between 2008 and 2025. Each answer is written to the length an aspirant can produce in the exam hall: three pages, an introduction that meets the demand, structured body points, a diagram or map where it earns its place, and a conclusion.

IN THIS BOOK

- Questions numbered exactly as in the official papers
- Arranged paper-wise and year-wise under the latest syllabus
- Old-syllabus questions placed in the paper they belong to today
- A clickable index and a link to practice every question

ON GHISSU.COM

- Subject, topic and sub-topic-wise PYQs with UPSC questions (1979 to 2026)
- Answer evaluation with detailed feedback in minutes
- Exam-hall thinking, toppers' answers and value additions
- Ask Ghissu and the Ghissu bull, in your own language

FOLLOW GHISSU @tryghissu

WhatsApp / Telegram: 85228 64298

TELEGRAM

ABOUT THE AUTHOR

Sri Charan Vegireddi studied Mathematics and Computing Engineering at DTU, New Delhi, and has written two UPSC Civil Services Mains, two APPSC Group-I Mains and one APPSC Group-II Mains. As Mentor-cum-Head Reviewer at Vajiram & Ravi, New Delhi, he headed the evaluation of its Mains Test Series, reviewed more than 10,000 answer copies, trained over 600 evaluators and mentored more than 100 aspirants.

Continue on ghissu.com

1. Open ghissu.com and sign in.
2. Join the waitlist. Requests are usually approved within one or two days.
3. Open any question from this book and start writing.

SCAN TO BEGIN

Ghissu

A free resource. Share it with an aspirant.

Questions are reproduced from the official APPSC question papers for educational use.
© 2026 Ghissu · hello@ghissu.com